Chapter

1

IDEAL FLUIDS AND PRESSURE LOSSES

LEARNING OUTCOMES

- This is a self-study chapter
- Have knowledge and understanding of Bernoulli's law for ideal fluids
- Understand the instruments and the means of measuring static, dynamic and total pressure
- Apply Bernoulli's Equation to fans
- Describe and calculate the effects of an evaseè.

REFERENCES

Brake, DJ, 'Fan total pressure or fan static pressure: which is correct when solving ventilation problems?', *Journal of the Mine Ventilation Society of South Africa*, January/March 2002, p6-11.

Hemp, R, 'The effect of changes in barometer pressure on mines in the Highveld of South Africa, *Journal of the Mine Ventilation Society of South Africa,* April/June 1998, p41-52.

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1.1 Ideal Fluids

This chapter contains information about the application of various thermodynamic and fluid flow principles for ideal situations as well as where we consider the effect of pressure losses. It is almost important to note that fluids can be for example:

- Water
- Petrol
- Diesel
- Air
- Gases etc.

Fluid flow therefore relates to different substances as mentioned before, but that the principles applied to each remains the same with in most cases only the density (and also indirectly then the compressibility of the substance) that changes. It is important for the student to be able to understand and apply these principles in the context of the mining engineering discipline. Many of the aspects mentioned in this chapter such as Bernoulli's law, potential and kinetic energy, compressibility, incompressibility, different types of pressure etc. have been covered in previous courses such as thermodynamics (MTX 221) and Thermo fluids (MTV 310). The purpose of this subject is therefore to introduce its application in the mining industry.

1.1.1 Bernoulli's Application in the Mining Context

The simplest form of the Bernoulli Equation for ideal fluids can be described through the use of a very simple explanation. Figure 1.1 represents a pipe full of water or an airway underground with air flowing through it.

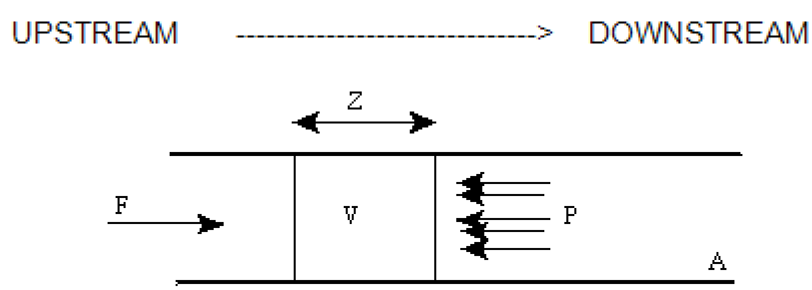


Figure 1.1 Fluid Flow in a controlled 'film'

NB

Assume that this horizontal pipe or tunnel is open on one end with a cross-sectional area A. If a fluid with volume V and mass m flows through, the pipe, we must remember that, even in the absence of friction, there is also a pressure P_2 because the fluid is already in the pipe/tunnel. Thus, there must be a force F applied to the fluid to overcome the apposed pressure. The purpose is to determine the amount of work that must be done to displace the fluid a distance z.

The force F must be in balance with the pressure of the liquid over the area A in the pipe:

Work done = force (F) x distance (z)

= PAz

But:

Az is the volume displaced, therefore:

Work done = P x volume, but ρ = m/volume

Thus:

Work done = Pm/ρ

We are now in a position to quantify the total amount of mechanical energy:

Total Mechanical Energy

=	KE	+	PE	+	Flow
=	(Energy)		(Energy)		(work done)
=	$mU^2/2$	+	mgh	+	Pm/ρ

If no mechanical energy is added or removed whilst the fluid is in the pipe and with friction ignored the total amount of mechanical energy must be constant, therefore:

$$m\,(U^2/2 + gh + P/\rho) = \text{constant}$$

If we now consider two points in the pipe, the following Equation can be derived:

$$m \ x \left(\frac{U_1^2}{2} + gh_1 + \frac{P_1}{\rho}\right) = m \ x \left(\frac{U_2^2}{2} + gh_2 + \frac{P_2}{\rho}\right)$$

If the fluid is incompressible, $\rho_2 = \rho_1 = \rho$ (so assumed liquid or gas) then:

$$\frac{U_1^2 - U_2^2}{2} + g(h_1 - h_2) + \frac{P_1 - P_2}{\rho} = 0 \text{ . J/kg} \qquad \text{………... (1.1)}$$

Remember that Equation1.1 is done for:

- Frictionless conditions
- Constant density
- Steady flow.

NB

Equation1.1 will be adapted later for real compressible fluids. It is also important to note that '1' and '2' as footnote shown in Equation 1.1 always refer to upstream and downstream respectively.

If density ρ is multiplied right through for Equation 1.1 we then get Bernoulli's Equation in terms of pressure (Pa):

$$\frac{\rho U_1^2}{2} + \rho g h_1 + P_1 = \frac{\rho U_2^2}{2} + \rho g h_2 + P_2 \quad \text{......... (1.2)}$$

Where:

P	=	Static pressure (pressure as a result of work done) in Pa
ρ	=	density (kg/m^3)
h	=	elevation (m)
U	=	velocity (m/s).

If the Equation is divided by ρg, we get Bernoulli's Equation as follows:

$$\frac{U_1^2}{2g} + h_1 + \frac{P_1}{\rho g} = \frac{U_2^2}{2g} + h_2 + \frac{P_2}{\rho g} \quad \text{.................. (1.3)}$$

This is Bernoulli's Equation in terms of pressure head expressed in metre (m). P is the static pressure in all directions and is **independent** of flow direction. If we want to bring the fluid to rest at the **end** of the column a **distance x** away from the position 1 (upstream), a specific pressure in the opposite direction will be needed to do this (assume column horizontal).

Let us consider the Bernoulli Equation:

$$
\begin{aligned}
\frac{\rho U_1^2}{2} + \rho g h_1 + P_1 &= \frac{\rho U_2^2}{2} + \rho g h_2 + P_x \\
\frac{\rho U_1^2}{2} + 0 + P_1 &= \frac{\rho U_2^2}{2} + 0 + P_x \\
\frac{\rho U_1^2}{2} + P_1 &= 0 + 0 + P_x \quad (U_2 = 0, \text{ fluid stopped}) \\
VP + SP &= P_T \text{(total pressure needed to stop fluid (TP))}
\end{aligned}
$$

Photos

The various pressures mentioned previously, namely velocity pressure, static pressure and total pressure can all be measured with a vertical manometer and pitot tube (metal tube facing oncoming fluid and also has side holes) combination. The following explanations will clarify the definitions.

1.1.1.1 Static Pressure (SP)

Definition

Animation

As mentioned above, static pressure indicates the potential energy of the air (i.e. the energy a body has by virtue of its position). Static pressure acts in all directions, regardless of the direction of flow and is often called 'bursting pressure' (i.e. static pressure is the pressure exerted by the air on the walls of a duct, tending to force it apart). Static pressure is measured with a vertical manometer (side gauge) at right angles to the flow of air as illustrated in Figure 1.2 (there is a small hole in the pipe and a rubber tube attached to the manometer is held against the pipe to measure the pressure). An important note is that static pressure can be positive or negative (suction) and this pressure measurement is normally referred to as a side gauge pressure measurement. The static pressure measured at the end or delivery side of an air column, is always 'zero'.

NB

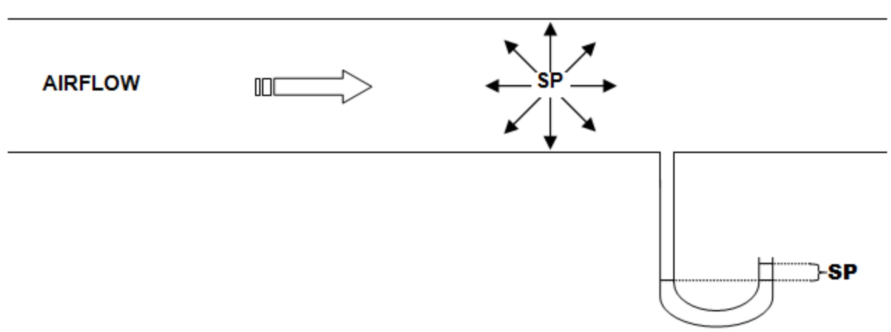
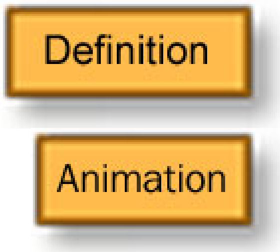


Figure 1.2 Graphical Representation measuring static pressure



1.1.1.2 Velocity Pressure (VP)

Velocity pressure indicates the kinetic energy of the air (i.e. the energy a body has by virtue of its motion) and hence can be defined as the pressure due to movement of the fluid. Thus, the higher the fluid velocity, the higher the velocity pressure will be, and vice versa. Velocity pressure is measured with a Pitot tube (pushed inside the pipe), measuring fluid (moving) pressure in the middle of the pipe (facing the fluid flow) and connected to a side gauge (vertical manometer) and is always 'positive'. Figure 1.3 illustrates.

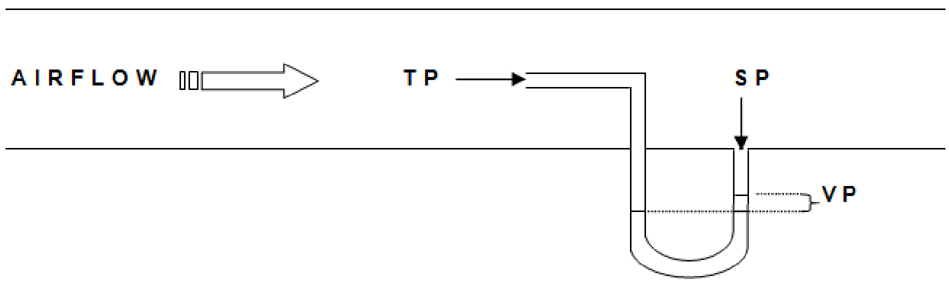
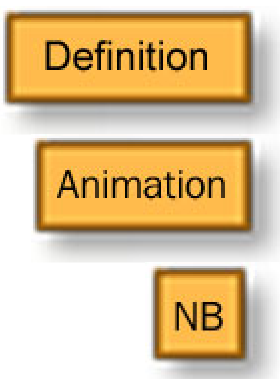


Figure 1.3 Graphical Representation measuring velocity pressure



1.1.1.3 Total Pressure (TP)

Total pressure is defined as the algebraic sum of velocity and static pressure and can be expressed as:

TP = SP + VP

Total pressure is measured with a facing gauge, which is illustrated in Figure 1.4. It is also important to note that total pressure can be positive or negative.

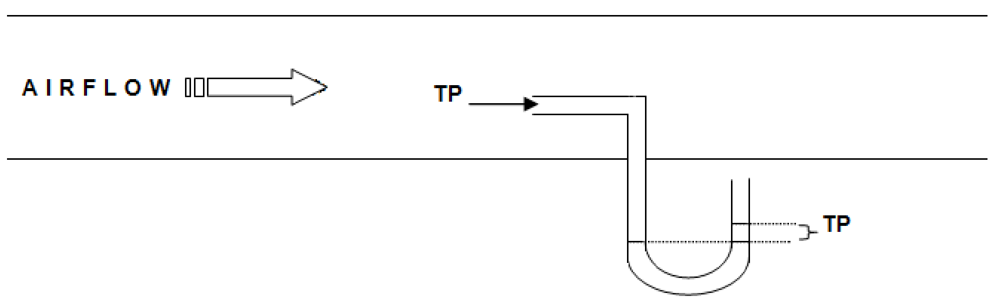


Figure 1.4 Graphical Representation measuring total pressure (TP)

WORKED EXAMPLE 1

The following example will explain the explanation of the Bernoulli Equation for ideal fluid flow more clearly. The drawing represents a portion of a column of a pipe that is fitted with a venturi (narrowing and widening portion of a pipe column to enable change in pressure). Attached to this pipe are three sets of (mercury) manometers. Each manometer measuring a unique pressure associated with the flow of the substance within the pipe.

From Figure 1.5 the bold dark shaded areas represent mercury in the respective tubes. The difference in mercury level for the lower tube is 20 mm. Determine the quantity of water flowing through the venturi, if the diameter of the column at point 1 is 100 mm and 50 mm at point 2 (use Equation 1.2 to solve the problem). The density of water to be assumed as 1 000 kg/m^3 and the density of the mercury as 13 500 kg/m^3.

Excel

Answer

4.7 litres/second

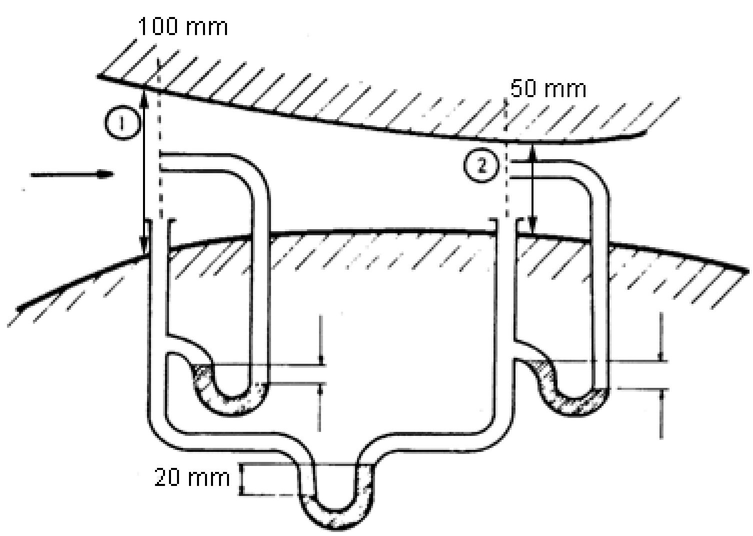


Figure 1.5 Graphical representation of pressure measurements

NB

An important part of understanding the application of Bernoulli's Equation is to know when we refer to moving fluids and when we refer to stationary fluids within the Equation.

1.2 Pressure Losses in Mine Fluid Flow

NB

It is important to understand the various types of pressure when looking at Bernoulli's Equation. The aspects related to the various types of pressure were discussed in the previous section (1.1.1). Because we deal with actual conditions in mine fluid flow it is necessary to consider the effect of losses in the whole process.

1.2.1 The Effect of Pressure Losses

Bernoulli's Equation as derived before is for ideal fluids. In our case the fluid is air and there are losses due to friction. The effect of losses is thus added to the 'outflow' side. Bernoulli's Equation with losses is:

$$\frac{\rho U_1^2}{2} + \rho g h_1 + P_1 = \frac{\rho U_2^2}{2} + \rho g h_2 + P_2 + p_L \quad \text{...(1.4)}$$

If there is no elevation difference (as in a ventilation column), the h component will be eliminated. Pressures in Bernoulli's Equation refer to absolute pressures.

Definitions

In the examples that follow, the pressures will be expressed as **gauge pressures**. That is pressures that are measured relative to some reference base pressure. The '**base pressure'** is the air pressure outside the ventilation column. We also assume that the 'base pressure' (normally barometric pressure) is the same at all points outside the ventilation column.

The relationship between absolute, atmospheric and gauge pressure is as follows:

Absolute Pressure = Atmospheric pressure + Gauge pressure

Consider the following situation in Figure 1.6. The line in the sketch refers to a ventilation pipe or column in an underground tunnel, but can also represent a water pipe in the same area. Let us assume for this example the fluid in the pipe is air (fan driven).

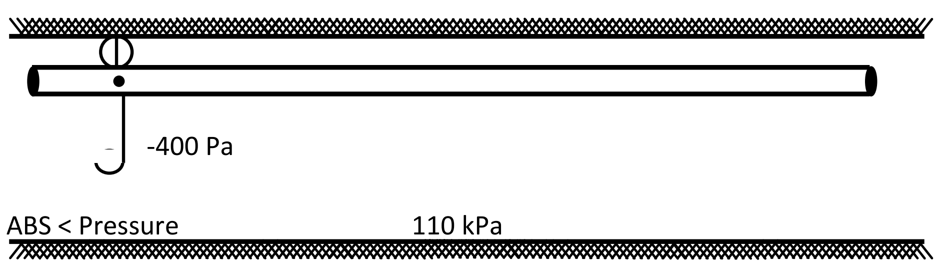


Figure 1.6 Measuring pressure in ventilation pipe

Relationship between atmospheric, total and metre pressure is as follows. The vertical manometer in Figure 1.6 shows a reading of – 400 Pa (suction pressure), which means that the pressure in the column is 400 Pa less than the pressure outside the column. The absolute pressure for air inside the column is:

Absolute pressure in column = atmospheric pressure + gauge pressure
= 110 + (- 0.4)
= 109.6 kPa

To get an expression for the pressure loss, Equation 1.4 can be rewritten as: (elevation is horizontal)

$$p_L = [P_1 + \frac{\rho U_1^2}{2}] - [P_2 + \frac{\rho U_2^2}{2}]$$

Or, written in more simple terms:

$$p_L = (SP_1 + VP_1) - (SP_2 + VP_2)$$
$$= TP_1 - TP_2 \quad \text{...} \quad (1.5)$$

If the velocity pressure in the column remains constant (because of the diameter of the pipe not changing), the Equation can be simplified as follows:

$$p_L = SP_1 - SP_2 \text{ (for column with constant diameter)}$$

Figure1.7 shows two static pressure readings some distance apart in a pipe with constant diameter (D) and it will be used to determine the pressure loss between the two points ‘1’ and ‘2’.

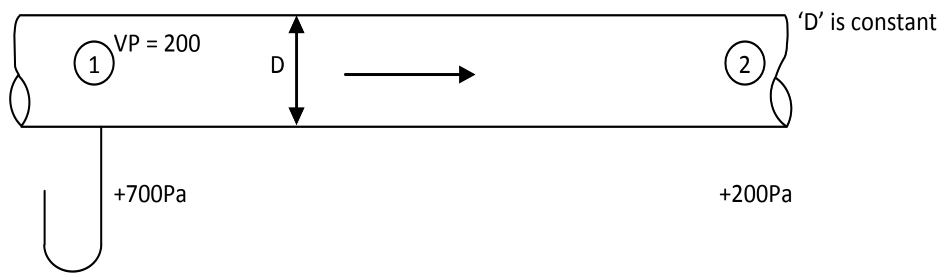


Figure 1.7 Determining pressure loss within ventilation column (diameter constant)

From Figure 1.7 the pressure loss can be expressed as:

$$\begin{aligned} p_L &= SP_1 - SP_2 \\ &= (+700) - (+200) \\ &= +500 \text{ Pa} \end{aligned}$$

NB

In a uniform column with diameter D the pressure loss is the difference in static pressure at the two points and is measured with a vertical manometer (side gauge). When the diameters at the beginning and end of the column differ, the total pressure (facing gauge) **must** be measured, as the velocity pressure at the two points will differ (change in diameters).

The pressure loss is then expressed as:

$$p_L = TP_1 - TP_2$$

Figure 1.8 illustrates the situation better.

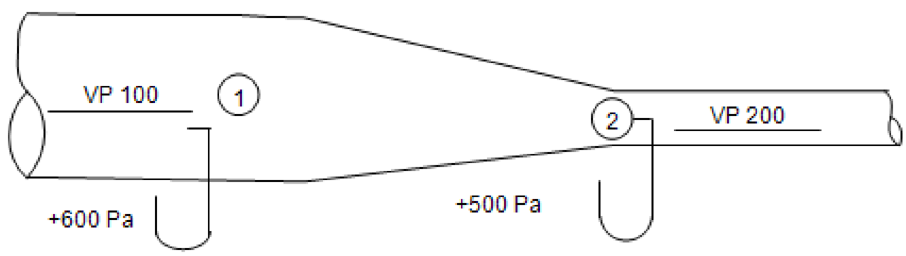


Figure 1.8 Determining pressure loss within ventilation column changing diameter

$$\begin{aligned} p_L &= TP_1 - TP_2 \\ &= (+\,600) - (+\,500) \\ &= 100 \text{ Pa} \end{aligned}$$

Self study

Read the article by S Ambrosio: ‘Steel spiral ducting friction factor determination’.

1.2.2 Determining Column Inlet and Outlet Pressures

There is a specific procedure to be followed when pressures are to be measured at the inlet and outlet of a column. There are two scenarios, one where shock losses are ignored and the other where these losses (which are a reality in mine ventilation airflow calculations) need to be considered. We first need to establish what is meant by shock losses applicable to the airflow calculations in the mining industry.

Definitions

Animation

1.2.2.1 Shock Losses

When the direction of airflow changes direction or the velocity changes due to changes in airway sizes, this can result in pressure loss, which is called **shock losses**.

Shock losses can occur in the following situations (**airway** normally refers to a tunnel through which air is flowing in underground and **air column** normally refers to a ventilation pipe underground):

- Entrance to an airway or air column
- Obstructions in an airway or column
- Turns in an airway or column
- Sudden changes in the size of an airway or column
- The outlet and, or inlet of the airway or column.

Good planning or small changes in the mining systems can overcome or reduce shock losses. The losses are also sometimes referred to as **entry** or **exit losses**, depending on the situation.

1.3 Bernoulli's Equation and Fans

Fans are used to make the flow of air possible in the underground working environment. Air is drawn down a down-cast shaft and up the up-cast shaft by a fan that is normally situated at the top of the up-cast shaft. Figure 1.9 shows a basic layout for airflow through a mine. Various conventional basic plan symbols are used for underground mine plans (with specific reference to ventilation items) and these are shown in the image attached.

Image

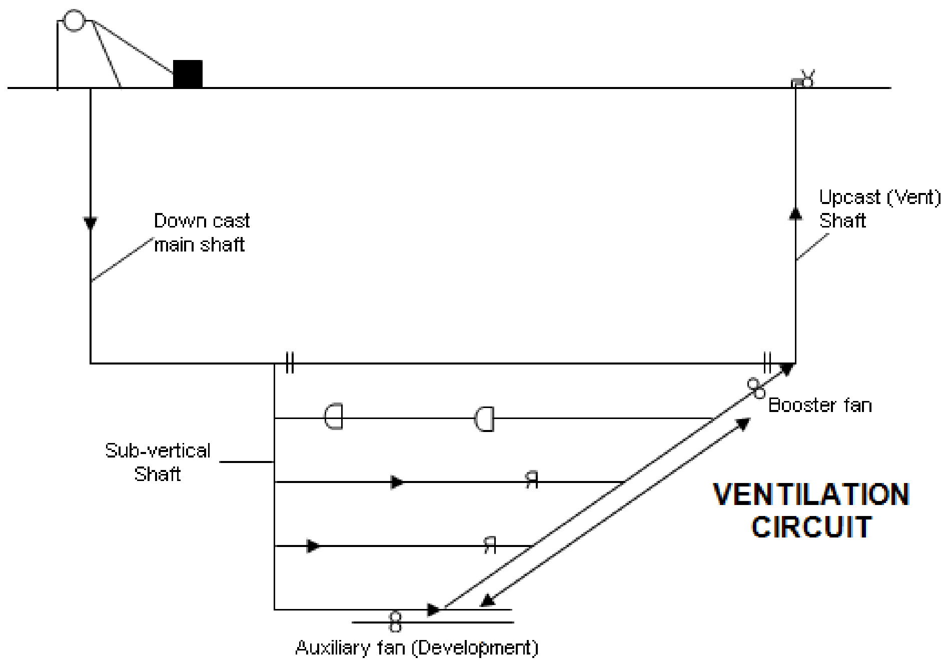


Figure 1.9 Basic flow of air through a mine

Definition

Fans are also installed in ventilation columns or pipes underground and is used to get fresh air to working places where there is not through ventilation (dead ends). The definition of a **dead end** is a tunnel or airway underground under construction of which the heading (or advance) is more than twice the width. Under these conditions a fan must be installed with a pipe column so as to ensure that healthy productive working conditions will be created.

Later in the course different fan combinations and types of air supply and air extraction layouts will be discussed, such as:

- Force columns
- Exhaust columns
- Combination of force and exhaust columns.

In solving airflow calculations underground, Bernoulli's Equation can be applied to fans as well and are approached in a simple manner in this section as was done before.

We first need to address the concept of inlet and outlet column pressure measurements applicable to fan columns (columns with fans to ventilate dead ends).

1.3.1 Determining Column Inlet Pressures

NB

Consider Figure 1.10 (let us ignore the position of the fan in the pipe for the moment). It is irrelevant now whether the fan is in the beginning, middle or end of the column. It is important to understand the principles applicable to pressure measurements in a pipe or ventilation column underground.

Rule

The points ‘1’ and ‘2’ are two points where point ‘1’ is just outside the ventilation pipe/column and point ‘2’ at the entrance to the pipe. Because points ‘1’ and ‘2’ are in the atmosphere (open space) the pressure loss between the two points is ‘zero’ (no shock losses at inlet and condition described as ideal). The velocity pressure (VP) in the pipe is measured as 200 Pa (and because the pipe has a uniform diameter, the VP will remain constant). The only time that the velocity pressure will change is when there is leakage into or out of the pipe (velocity pressure therefore stays constant for a leak-less column). From Equation 1.5 as follows:

$p_L = TP_1 - TP_2 = 0$, but $TP_1 = 0$

$TP_2 = 0$ and $VP_2 = 200$ Pa

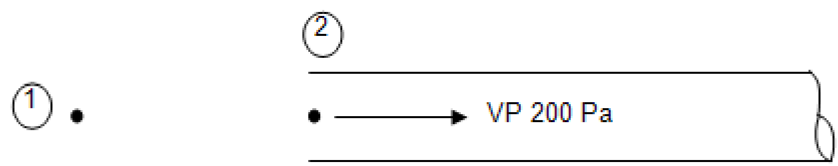


Figure 1.10 Column inlet pressures

From the Equation 1.5:

$TP_2 = VP_2 + SP_2$

We then get:

NB

$$
\begin{aligned}
SP_2 &= TP_2 - VP_2 \\
&= (0) - (+200) \\
&= -200 \text{ Pa (not ‘zero’ as at end of column)}
\end{aligned}
$$

For the inlet column we can say:

1. $TP_2 = 0$ (no ‘shock’ losses assumed)
2. $SP_2 =$ negative value from the VP

NB

If there are 'shock' losses ('real life' scenario), the situation can be handled as if there are friction losses between points '1' and '2' (as per Figure 1.10). For example given that inlet 'shock' losses are 75% of the velocity pressure, then the inlet shock loss can be determined as follows:

$$
\begin{aligned}
\text{Inlet shock loss} &= 0.75 \times 200 \\
&= 150 \text{ Pa and then} \\
p_L &= TP_1 - TP_2 \\
TP_2 &= TP_1 - p_L \\
&= 0 - 150 \\
&= -150 \text{ Pa} \\
SP_2 &= TP_2 - VP_2 \\
&= (-150) - (+200) \\
&= -350 \text{ Pa}
\end{aligned}
$$

NB

Compared with static pressure as '- 200 Pa' when no shock losses were considered!

1.3.2 Determining Column Outlet Pressure

At the outlet of the column, the only energy that remains in the air is the energy as a result of air velocity, because all the internal energy is used to overcome friction, the following can therefore be said:

$$
\begin{aligned}
SP &= 0 \\
VP &= \frac{\rho U^2}{2} \\
TP &= SP + VP \\
&= 0 + VP \\
TP &= VP \text{ (at the end of the column)}
\end{aligned}
$$

In summary therefore:

1. $TP_{end} = VP_{end}$
2. $SP = 0$ (always 'zero' at the end of a delivery column)

1.3.3 Bernoulli's Equation for Fans

From previous work, we can use Bernoulli's Equation in the application of fans. The Equation will look as follows:

$$\frac{\rho U_1^2}{2} + \rho g h_1 + P_1 + FTP = \frac{\rho U_2^2}{2} + \rho g h_2 + P_2 + p_L \quad .(1.6)$$

Where:

FTP = Fan Total Pressure and the other symbols the same as was used before.

From Equation 1.6 and with elevation constant (pipe horizontal) the following Equation can be derived:

$$FTP = \left(\frac{\rho U_2^2}{2} + P_2\right) - \left(\frac{\rho U_1^2}{2} + P_1\right) + p_L$$

Or in simpler terms:

$$FTP = (SP_2 + VP_2) - (SP_1 + VP_1) + p_L$$

In terms of total pressure, the Equation can be expressed as:

$$FTP = TP_2 - TP_1 + p_L$$

We can therefore also say as follows:

$$FTP_{\text{fan total pressure}} = FSP_{\text{fan static pressure}} + FVP_{\text{fan velocity pressure}} + p_L$$

NB

Fan static pressure (FSP) relates to the ability of the fan to do work. In many cases when a fan curve is drawn (pressure versus quantity), the pressure on the graph relates to fan static pressure unless otherwise stated.

Animation

1.3.3.1 Fan Pressure Readings

Fans, as was mentioned before can be installed on surface but also used in columns underground to ensure the proper distribution of air. When we look at fans for columns; they can be installed at the beginning of the column (force fan), in the middle of the column, normally in conjunction with a fan at the end or beginning of a column to act as a booster fan and fans installed at the end of the ventilation column (exhaust fan – fan sucking air out).

FAN STATIC PRESSURE

The fan static pressure (FSP) can be determined as follows (using first principles):

NB

FTP = FSP + FVP + p_L (assume losses across fan as nil)

FSP = FTP – FVP

= $(TP_2 – TP_1)$ - FVP

FSP = $(SP_2 + VP_2) - TP_1$ – FVP (FVP = VP_2 = VP_1)

= $SP_2 – TP_1$

MEASUREMENT OF PRESSURES FOR FORCE FANS (FAN AT COLUMN INLET)

If points ‘1’ and ‘2’ are the inlet and outlet of the fan (not column outlet) respectively (see Figure 1.11) where the fan is used as a force fan, then:

TP_1 = 0 (from previous example)

TP_2 = $VP_2 + SP_2$

= 200 + 600

= 800

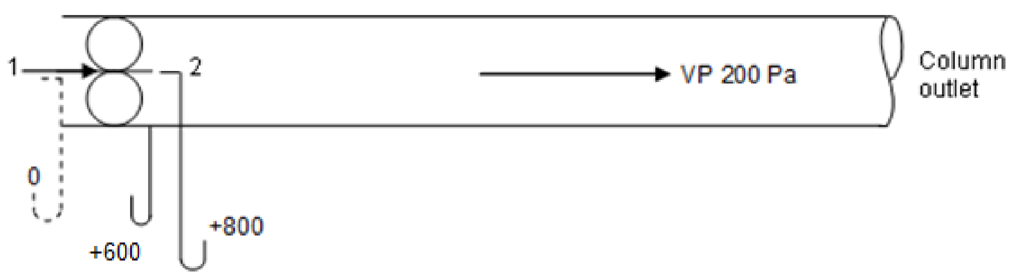


Figure 1.11 Fan at inlet of column (force fan)

We can therefore say that (600 Pa refers to the static pressure measurement on the delivery side of the fan the total pressure on the inlet side of the fan is 0 Pa):

FTP = $TP_2 - TP_1 + p_L$

= $800 - 0 + p_L$

= $800 + p_L$

NB

With taking losses into account at the intake side of the fan, a static pressure reading would have read as ‘- 200 Pa’, the negative of the velocity pressure.

The FTP could then also have been calculated as follows:

$$\begin{aligned} FTP &= SP_2 - SP_1 \\ &= 600 - (-200) \\ &= 800 \text{ Pa} \end{aligned}$$

$$\begin{aligned} FSP &= SP_2 - TP_1 \\ &= 600 - 0 \\ &= 600 \text{ Pa} \end{aligned}$$

MEASUREMENT OF PRESSURES FOR BOOSTER FANS

Consider Figure 1.12 ('D' upstream and 'E' downstream).

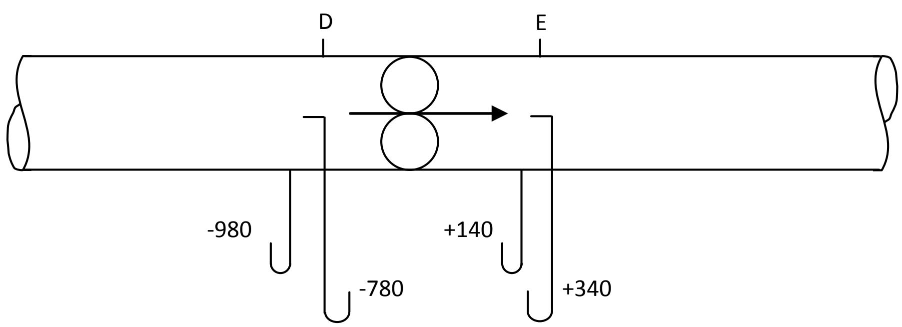


Figure 1.12 Fan in the middle of a column (booster fans)

The total fan pressure (FTP) can be calculated by (assume pressure losses are nil):

$$\begin{aligned} FTP &= TP_E - TP_D + p_L \\ &= 340 - (-780) + 0 \\ &= 1\,120 \text{ Pa} \end{aligned}$$

Or as follows (only because the diameter of the column is uniform and the VP is therefore the same in and out of the fan)

$$\begin{aligned} FTP &= SP_E - SP_D \\ &= +140 - (-980) \\ &= 1\,120 \text{ Pa} \end{aligned}$$

$$\begin{aligned} FSP &= SP_2 - TP_1 \\ &= +140 - (-\,780) \\ &= 920 \text{ Pa} \end{aligned}$$

These definitions of FTP and FSP stand for any position in the column.

MEASUREMENT OF PRESSURES FOR FANS AT COLUMN OUTLET (EXHAUST FANS)

Consider the values for the pressures shown in Figure 1.13.

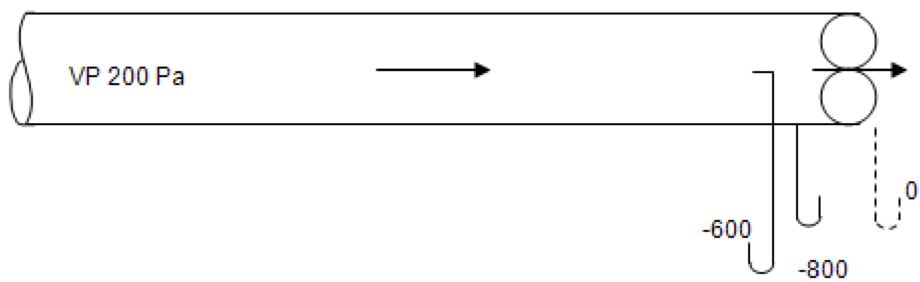


Figure 1.13 Fan in the middle of a column (booster fans)

It has been mentioned before that the static pressure reading at the outlet is zero. Thus, the total fan pressure can be calculated as follows:

$$
\begin{aligned}
FTP &= SP_2 - SP_1 \\
&= 0 - (-800) \\
&= 800 \text{ Pa}
\end{aligned}
$$

To get the total fan pressure at the end of a column, we just need to measure the static pressure at the inlet of the fan. We can also calculate the values:

$$
\begin{aligned}
FTP &= TP_2 - TP_1 \\
&= 200 - (-600) \\
&= 800 \text{ Pa}
\end{aligned}
$$

NB

It must be remembered that the static pressure readings can **only be used when the in and outlet areas of the fan are the same**. The static pressure of the fan (FSP) can be expressed as:

$$
\begin{aligned}
FSP &= SP_2 - TP_1 \\
&= 0 - (-600) \\
&= 600 \text{ Pa}
\end{aligned}
$$

WORKED EXAMPLE 2

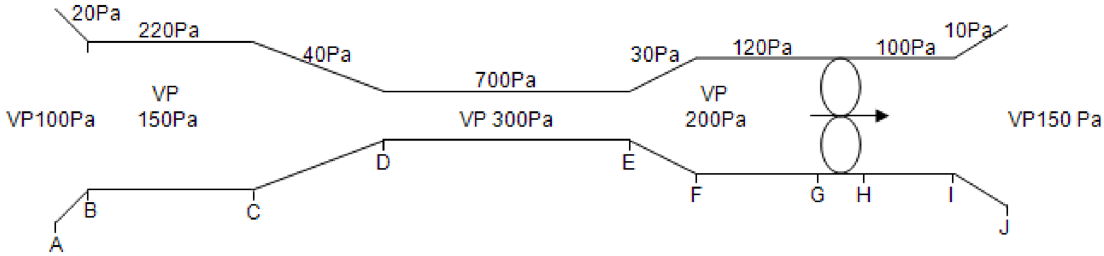


Figure 1.14 Fan installed in ventilation column

Figure 1.14 shows a fan installed in a ventilation column. Velocity pressures and pressure losses in the various sections are indicated. Determine the total, static and velocity pressure at each point and also the fan pressure, fan static pressure and fan velocity pressure.

Answer

A: $TP_A = 0$, $VP_A = 100$ Pa

$SP_A = TP_A - VP_A$

$= -100$ Pa

The pressure loss A - B = 20 Pa

$PL = TP_1 - TP_2$

$TP_2 = TP_1 - P_L$

So:

$TP_B = TP_A - P_L$

$= 0 - 20$

$= -20$ Pa

And

$VP_B = 150$ Pa

$SP_B = TP_B - VP_B$

$= (-20) - (+150)$

$= -170$ Pa

C: The pressure loss B-C = 220 Pa

Now:

$$TP_2 = TP_1 - P_L$$
$$TP_C = TP_B - P_L$$
$$= (-20) - (+220)$$
$$= -240 \text{ Pa}$$

And

$$VP_C = 150 \text{ Pa}$$

So:

$$SP_C = TP_C - VP_C$$
$$= (-240) - (+150)$$
$$= -390 \text{ Pa}$$

The pressures at D, E, F and G can be obtained in the same manner. No SP or TP on the outlet side of the fan is given. But we know that SP at the end of the column equals zero.

J: $SP_J = 0, \quad VP_J = 150 \text{ Pa}$

$$TP_J = VP_J + SP_J$$
$$= 150 + 0$$
$$= 150 \text{ Pa}$$

I: The pressure loss I - J = 10 Pa

And

$$P_L = TP_1 - TP_2$$

(Remember that '1' indicates the upstream point and '2' indicates the downstream point).

$$TP_1 = TP_2 + P_L$$

So:

$$TP_1 = TP_J + P_L$$
$$= 150 + 10$$
$$= 160 \text{ Pa}$$

And

$VP_1 = 200$ Pa

$SP_1 = TP_1 - VP_1$

$= (+160) - (+200)$

$= -40$ Pa

H The pressure loss H - I = 100 Pa

And

$TP_1 = TP_2 + P_L$

$TP_H = TP_1 + P_L$

$= 160 + 100$

$= 260$ Pa

And

$VP_H = 200$ Pa

$SP_H = TP_H - VP_H$

$= (+260) - (+200)$

$= 60$ Pa

Table 1.1 Summery of results

	A	B	C	D	E	F	G	Fan	H	I	J
TP	0	-20	-240	-280	-980	-1 010	-1 130	1 390	260	160	150
SP	-100	-170	-390	-580	-1 280	-1 210	-1 330	1 190	60	-40	0
VP	100	150	150	300	300	200	200	200	200	200	150

Now:

FTP = (+260) - (-1130) (difference between 2 facing gauges)

= 1 390 Pa

Or:

FTP = (+60) - (-1330) (difference between 2 side gauges, equal areas)

= 1 390 Pa

And

FSP = (+60) - (-1130) (diff. Facing gauge inlet, side gauge Delivery)

= 1 190 Pa

FVP = 200 Pa

(Check: FTP = FSP + FVP = 1 190 + 200 = 1 390 Pa)

These results could also have been checked by incorporating the pressure losses which states:

FTP = $VP_{outlet} + P_L$

= (150) + (20 + 220 + 40 + 700 + 30 + 120 + 100 + 10)

= 150 + 1 240

= 1 390 Pa

1.4 Evasee's

Evasee's are usually installed at the outlet of a column to thus convert the loss in velocity pressure to static pressure. This is done by increasing the outlet area and thereby decreasing the velocity of the air. The decrease in velocity pressure thus has, as a result, an increase in the static pressure. Consider the following condition shown in Figure 1.15.

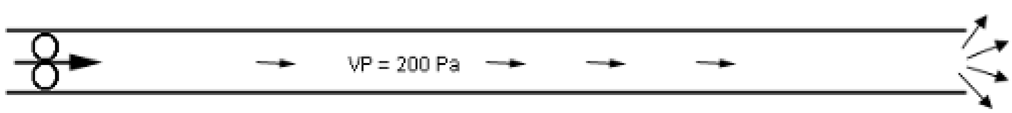


Figure 1.15 The effect of having no evaseè at the end of a column

At the outlet the 200 Pa is 'wasted' because no additional useful work is done. An evaseè at the outlet will result in the velocity pressure being decreased from 200 Pa to 50 Pa (50 Pa used as an example and dependent on the area at the outlet of the evaseè). This is clearly shown in Figure 1.16.

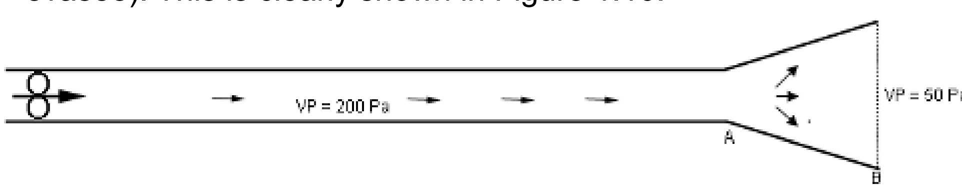


Figure 1.16 The effect of having an evaseè at the end of a column

A static pressure measurement (made with a 'side gauge') at the discharge end of any column or system will read zero.

NB

This is because all of the static pressure has been used up in **overcoming the friction** of the system as shown in Figure 1.16 (SP_B=0). But between A and B there is a gain in SP.

Rule

This is a very important aspect to grasp. The SP at B is greater than the SP at A; since the SP at B is 0, the SP at A must be negative (and therefore a gain in static pressure from a high negative to 'zero'). This will always be the case when an evaseè is fitted at the extreme end of a system.

For example, let us assume a pressure loss of 30 Pa between points A and B mentioned in Figure 1.16. From the sketch it is clear that the change in velocity pressure (VP) is 150 Pa (from inside the column at position A to the end of the column position B).

This is also the increase in static pressure (SP) as mentioned before. There is therefore 150 Pa available to overcome friction in the system (if evaseè is 100% efficient).

This is just the theoretical increase in static pressure, because a certain amount of pressure is used to overcome friction (30 Pa), we can therefore calculate the actual increase in static pressure as follows:

Actual increase in SP = theoretical increase in SP-friction losses
= 150 - 30
= 120 Pa

The efficiency of the evaseè can be expressed as:

Evaseè efficiency = actual increase SP / Theoretical increase in SP
= 120/150 x 100
= 80 %

NB

The static pressure at point B will be zero, but higher than A (Increase in SP from A to B). The **static pressure at A will thus be a negative value.**

The increase in SP from A to B = $SP_B - SP_A$

Therefore the:

SP_A = SP_B – actual increase in SP
= 0 - 120
= -120 Pa

From this example we can therefore say that:

$$TP_A = VP_A + SP_A$$
$$= 200 + (-120)$$
$$= 80$$

$$TP_B = VP_B + SP_B$$
$$= 50 + 0$$
$$= 50 \text{ Pa}$$

Pressure loss

$$= TP_A - TP_B$$
$$= 80 – 50$$
$$= 30 \text{ Pa}$$

NB

Note that the installation of an evaseè would result in the **fan pressure being reduced** to an amount equal to the **actual** (not theoretical) increase in static pressure at the evaseè and therefore more air will flow through the system. This lowers the system resistance and this aspect will be explored in detail in Chapter 2 of this course (lowering of the system resistance curve). Figure 1.17 and Figure 1.18 illustrate graphically what happens to the SP and VP in a system with and without an evaseè.

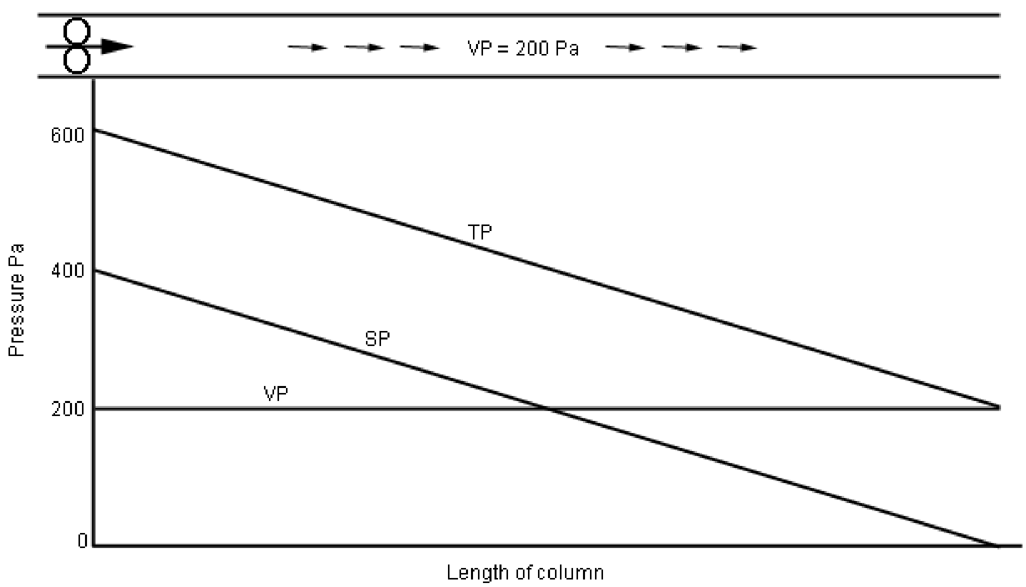


Figure 1.17 System without an evaseè at the end of a column

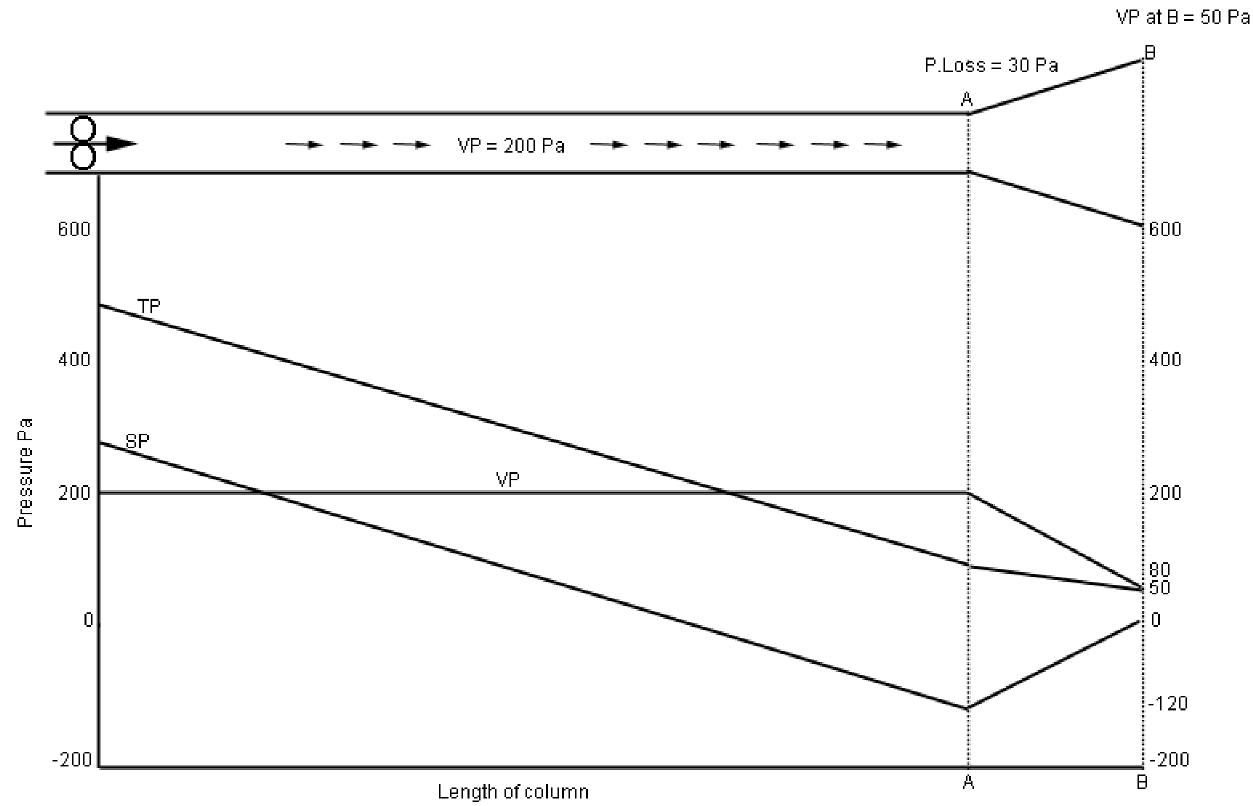


Figure 1.18 System with an evaseè at the end of a column

Take note of the following:

- VP is constant as long as the column diameter is constant and decreases as soon as the diameter increases in the evaseè.
- A negative static pressure is obtained at the evaseè inlet (A) of -120 Pa. There is then a gain in SP of 120 Pa and the SP at the end of the column equals zero.
- The friction losses between A and B is 30 Pa. In order to measure this pressure loss due to friction in the evaseè it would be necessary to measure TP at A and TP at B and obtain the difference: = (80 - 50) Pa = 30 Pa.
- TP would be measured with a facing gauge.
- By fitting the evaseè the fan pressures have decreased by an amount equal to the **regain in SP** at the evaseè. This means that a larger quantity of air will flow through the system (making available more air at lower pressure due to the reduction in the system resistance).

WORKED EXAMPLE 3

An evaseè with an inlet area of 0.5 m^2 and an outlet area of 1.2 m^2 is 80% effective and is installed at the end of a ventilation column. The air density is 1.2 kg/m^3. The volume flow through the system is 8 m^3/s. Calculate the **actual** increase in static pressure. The Velocity pressure at evaseè inlet must be determined:

$$VP_{in} = \frac{\rho U_{in}^2}{2}$$

But:

$$U_{velocity} = \frac{Q_{\text{quantity of air}}}{A_{\text{area of colum}}}$$

$$VP_{in} = \frac{\rho U_{in}^2}{2}$$

$$= \frac{1.2 \times (\frac{8.0}{0.5})^2}{2}$$

$$= 154 \text{ Pa}$$

Theoretical Increase in static pressure	=	$VP_{in} - VP_{uit}$
	=	154 - 27
	=	127 Pa
Actual increase in static pressure	=	127 x 80%
	=	102 Pa

WORKED EXAMPLE 4

Figure 1.19 shows 28.6 m^3/s of air at a density of 1.16 kg/m^3 that flows through an evaseè. The evaseè is 75% efficient. In Figure 1.19 manometers connected to side gauges measure static pressure and the manometers connected to facing gauges measure total pressure. All pressures are in pascals.

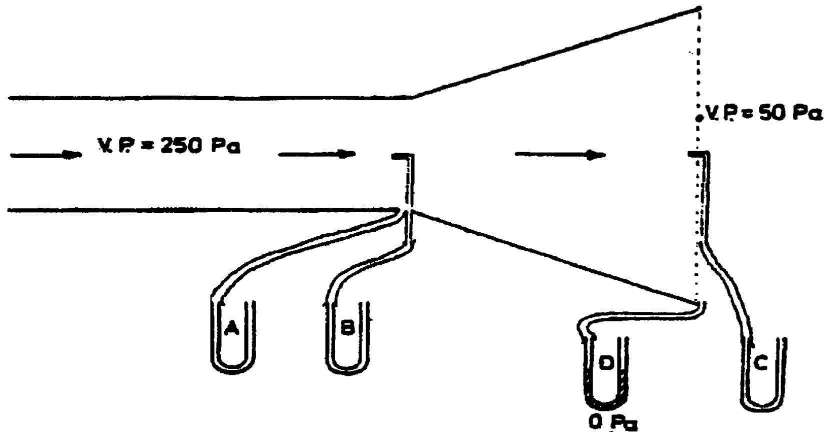


Figure 1.19 Airflow through an evaseè

a) Determine the diameter of the inlet and the outlet of the evaseè
b) Determine the pressure loss due to friction and shock in the evaseè.
c) Determine the pressure measured by the manometers A, B and C.
d) Indicate which two manometers could be used to measure the pressure loss in the evaseè and indicate their respective values.

Answer

(a) Determine the diameter of the inlet and the outlet of the evaseè by using the following approach:

$$VP = \frac{\rho U^2}{2}$$

$$U_{in} = \sqrt{\frac{2 \text{ x } VP_{in}}{\rho}}$$

$$= \sqrt{\frac{2 \text{ x } 250}{1.16}}$$

$$= 20.76 \text{ say } 20.8 \text{ m/s}$$

$$U_{out} = \sqrt{\frac{2 \text{ x } VP_{in}}{\rho}}$$

$$= \sqrt{\frac{2 \text{ x } 50}{1.16}}$$

$$= 9.28 \text{ say } 9.3 \text{ m/s}$$

$$Q = U_1A_1 = U_2A_2$$

(**Note**: If the density of the air remains constant)

$$A_1 = \frac{Q}{U_1} \qquad A_2 = \frac{Q}{U_2}$$

$$= \frac{28.6}{20.8} \qquad = \frac{28.6}{9.3}$$

$$= 1.375 \text{ m}^2 \qquad = 3.075 \text{ m}^2$$

$$A_1 = \frac{\pi D^2}{4}$$

$$D_1 = \sqrt{\frac{4 \text{ x } A_1}{\pi}}$$

$$= \sqrt{\frac{4 \text{ x } 1.375}{\pi}}$$

$$= 1.3 \text{ m}$$

$$A_1 = \frac{\pi D^2}{4}$$

$$D_1 = \sqrt{\frac{4 \text{ x } A_1}{\pi}}$$

$$= \sqrt{\frac{4 \text{ x } 3.075}{\pi}}$$

$$= 1.978 \text{ say } 2.0 \text{ m}$$

(b) Determine the pressure loss due to friction and shock in the evaseè.

Theoretical increase in SP	=	$VP_1 – VP_2$
	=	250 – 50
	=	200 Pa
Evaseè efficiency	=	$\frac{\text{actual SP increase}}{\text{theoretical SP increase}}$
Actual SP increase Theoretical	=	Evaseè efficiency x SP increase
	=	0.75 x 200
	=	150 Pa
Frictional pressure losses	=	Theoretical increase in SP - Actual increase in SP
	=	200 - 150
	=	50 Pa

(c) Determine the pressure measured by the manometers A, B and C.

NB

The static pressure at point B will be zero, but higher than A (Increase in SP from 1 to 2). The **static pressure at 1 will thus be a negative value.**

The increase in SP from 1 to 2 = $SP_{2\,(D)} - SP_{1\,(A)}$

Therefore the:

$$
\begin{aligned}
SP_{1\,(A)} &= SP_B - \text{actual increase in SP} \\
&= 0-150 \\
&= -150 \text{ Pa } (\textbf{manometer A})
\end{aligned}
$$

From the example we thus deduce that:

$$
\begin{aligned}
TP_{1\,(B)} &= VP_1 + SP_{1\,(A)} \\
&= 250 + (-150) \\
&= 100 \text{ Pa } \textbf{(manometer B)}
\end{aligned}
$$

$$
\begin{aligned}
TP_{2\,(C)} &= VP_2 + SP_{2\,(D)} \\
&= 50 + 0 \\
&= 50 \text{ Pa } (\textbf{manometer C)}
\end{aligned}
$$

(d) Indicate which two manometers could be used to measure the pressure loss in the evaseè.

$$
\begin{aligned}
\text{Pressure loss} &= TP_{1\,(B)} - TP_{2\,(C)} \\
&= 100 - 50 \\
&= 50 \text{ Pa (same as before)}
\end{aligned}
$$

SELF STUDY

References

Read the article: 'Fan total pressure or fan static pressure: which is correct when solving ventilation problems', D.J. Brake - this for examination purposes as well, what is the main message from this article?

Also read the article: 'The effect of changes in barometer pressure on mines in the Highveld of South Africa' by R Hemp.

Chapter 2

BASIC AIRFLOW PRINCIPLES AND AIRFLOW CIRCUITS

LEARNING OUTCOMES

2.1 Laminar and Turbulent Flow and Basic Airflow Principles

- Limits of incompressible fluids
- Friction losses in turbulent flow
- Describe real fluids in terms of Bernoulli
- Explain Reynold's number
- Differentiate turbulent flow
- Describe the following terms: Streamlines, boundy layers and separation
- Airpower
- Airways in series and parallel (graphical representation)
- Booster fans, regulators, door pressures.

2.2 Measuring of Airflow

Know the following measuring techniques:

- Anemometer
- Velometer.

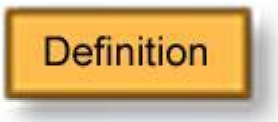
REFERENCES

Marks, J, 'Back to basics - why we ventilate mines, or what is adequate ventilation?', Journal of the Mine Ventilation Society of South Africa, April/June 2002, p54-56.

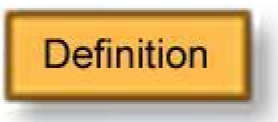
Mine Environmental Control (MEC) Certificate, Chamber of Mines, South Africa, Workbook 1.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book)

2.1 Laminar (Streamline) and Turbulent Flow



A **fluid**, which is defined as that which flows and can be either a liquid or a gas, flows in two fundamentally different ways. The nature of these ways was demonstrated by Reynolds in a series of experiments in which he released a coloured liquid into a stream of water flowing through a fine tube. Reynolds shows that at low velocities the flow pattern is regular and that the colouring matter remained in distinct layers indicating that the flow pattern was laminar or streamline. Reynolds also showed that as the velocity increases a point is reached where the flow pattern becomes disturbed and that the colouring matter does not flow in distinct lines but, that irregularities are formed and that the flow pattern is turbulent. The velocity at which this change in flow pattern takes place is called the **critical velocity**.



A practical example of these two different types of flow arises in a river flowing over narrow rapids; here the water is fast - moving and the flow is turbulent, while when this same river flows between two small river banks and the river is deep the river flows quietly and the flow pattern is streamline.

According to Reynolds, the nature of the fluid flow depends upon several factors, namely:

- Fluid velocity and density
- Diameter of the duct
- Viscosity of the fluid.

These factors could be combined and identified by a dimensionless number known as the Reynolds number (R_e).

The Equation for defining the Reynolds number of fluid flowing in a circular pipe or duct is:

$$R_e = \frac{\rho D U}{\mu} \quad \text{.......} \quad (2.1)$$

Where:

R_e	=	Reynolds number and is dimensionless
ρ	=	the fluid density (kg/m^3)
U	=	the fluid velocity (m/s)
D	=	the duct diameter (m)
μ	=	the fluids dynamic viscosity (Ns/m^2)

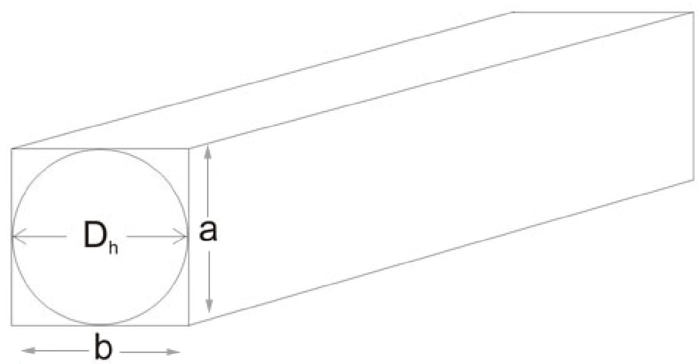


Figure 2.1 Graphical explanation of the hydraulic diameter

If the airway or duct is not circular, but is a rectangular shape of sides 'a' and 'b', then 'D' in the Equation is replaced by the equivalent diameter D_h shown below:

$$D_h = \frac{4\,A}{C}$$

Where:

A	=	Area of the duct ('a' x 'b')
C	=	Circumference of the duct (2 x 'a' + 2 x 'b')

When the Reynolds number is below 2 000 the flow pattern is laminar (streamline). When the Reynolds number is between 2 000 and 5 000 the flow pattern is neither streamline nor fully turbulent, and when the Reynolds number is greater than 5 000 the flow pattern is deemed to be fully turbulent.

The Reynolds number is significant because it gives an indication

Rule

If the Reynolds number has the same numerical value for two different incompressible flow systems and if geometric and surface similarly are maintained, the flow pattern remains the same.

This is a very important point when scale models are used to represent larger networks, although with the latest technology now available, air and water circuits are nowadays simulated through computer programmes, which make the use of scale models almost irrelevant but still useful to understand basic airflow concepts.

Scale Model Testing

Animation

In planning the layout of a new mine, or part of a mine, mining and ventilation engineers may make use of models. There are many reasons for doing this. Basically, models enable information to be obtained comparatively cheaply and without the difficulties which would arise in an in-situ case. Imagine, for instance, how difficult and costly it would be to determine the effects on resistance of altering shaft steel work spacing and design in a main hoisting shaft.

In order to compare measurements made in shafts and their scale models (the same principle applies for simulation models), it is necessary that:

- Geometric similarity can be obtained i.e. the model shafts must be exactly scaled replicas of the actual shafts.
- Dynamic similarity must be obtained, that is the flow patterns in the scale model and in the in-situ shaft must be similar. If the Reynolds number is the same in both cases, it implies that dynamic similarity had been obtained and that the flow patterns in the two systems were the same.

This need for geometric and dynamic similarity applies in all scale model testing. If the scale model of a shaft is thirty times smaller than the actual shaft and assuming that air is used as fluid in the test situation, to achieve the same Reynolds number the air velocity in the model would have to be 30 times that in the actual case, because the other term in the Equation (density and viscosity) cannot be altered appreciably. This could mean having air velocities in the scale model as high as 200 m/s and in fact is very difficult to obtain.

Fortunately it is often found that a smaller Reynolds number will suffice in the scale model to achieve ‘dynamic similarity’.

Rule

The co-efficient of friction value does not change appreciably once a certain value of the Reynolds number is reached; this is shown in the Figure 2.2.

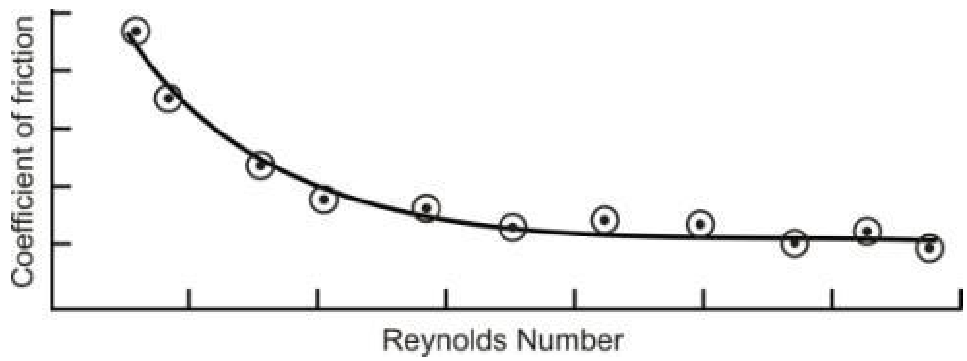


Figure 2.2 Impact of co-efficient of friction with increased Reynolds number

Scale model testing is rarely, if ever conducted today because of the computerised modeling mentioned before, but it is important to understand the principles of scale.

Calculation using the Reynolds Number

Calculations involving Reynolds number are usually very simple. The following example shows how this can be done.

WORKED EXAMPLE 1

Calculate the density in a shaft given the following information:

$R_e = 37\,000$

$U = 7$ m/s

$D = 6.9$ m

$\mu = 0.0016$ Ns/m^2

$$R_e = \frac{\rho D U}{\mu}$$

$$\rho = \frac{R_e\ \mu}{UD}$$

$$= \frac{37\,000 \times 0.0016}{7.0 \times 6.9}$$

$$= 1.23 \text{ kg/m}^3$$

NB

We must know what the limits to incompressible flow are. This aspect is fully covered in Environmental Engineering for South African mines Chapter 1 Page 10, paragraph 9.

2.1.1 Friction Losses in Laminar and Turbulent Flow

For the purposes of this course we will accept that airflow underground will in all likely circumstances be turbulent and therefore laminar flow of air is not considered for underground (remembering that a R_e number more than 5 000 is an indication of fully turbulent flow). As a result of the complexity of turbulent flow, analytical derivation is not really possible. The construction of physical models and observations in wind tunnels and other ‘Flow’ test facilities are the primary means of predicting the results of turbulent flow.

2.1.1.1 The Chezy-Darcy Equation for Turbulent Flow

Chezy found the average velocity of water in a duct can be represented by the following Equation and this will be used to show how the Equation applicable to airflow can be determined:

$$u = c\sqrt{\frac{A}{\text{'per'}}\left(\frac{h}{L}\right)} \quad \text{m/s} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots(2.2)$$

Where:

U	=	Velocity (m/s)
c	=	proportionality constant (Chezy coefficient), note not circumference
‘per’	=	perimeter or circumference (m)
A	=	cross sectional area (m^2)
h	=	vertical height difference (m)
L	=	length (m)
$\left(\frac{h}{L}\right)$	=	hydraulic gradient (dimensionless)

Equation 2.2 is also known as Chezy’s Equation for channel flow. If a fluid flows in a channel, there are shear forces, which act on the edges of the channel.

The drag force on the channel edges is thus:

Drag Force = $\tau\, C\, L$

Where:

τ	=	shear force (tau)
C	=	circumference that is covered by air or wetted by water
L	=	length of channel

This force must be equal to the pressure force, which is required to set the fluid into motion (from Pressure = $\frac{\text{Force}}{\text{Area}}$ in N/m^2 or Pa we get that Force (F) = P x A), where P is the pressure difference over the length of the duct or tunnel. We can also say that the pressure associated with a static column of fluid, say water (vertical) is related to the density of the fluid (ρ), the height (h) of the column of fluid and the gravitational acceleration component (g) and can be expressed as follows:

$$\text{Pressure}_{\text{(vertical column)}} = \rho g h \qquad (2.3)$$

From the above the following can therefore be derived:

$$\tau.C.L = pA$$

$$\text{But } p = \rho g h$$

We can therefore conclude that:

$$\tau = \frac{A\rho g h}{C \times L} \qquad (2.4)$$

If the flow is fully turbulent, the drag-force (τ) experienced on the channel edges is directly proportional to the kinetic energy for the flow, expressed in J/m^3 (Chezy).

$\tau \propto \frac{\rho U^2}{2}$ $\left(\frac{J}{m^3} = \frac{Nm}{m^3} = \frac{N}{m^2}\right)$, velocity pressure (VP) dealt with in Chapter 1.

From the above the following can be derived:

$$\tau = \frac{f\,\rho\,U^2}{2} \quad N/m^2 \qquad (2.5)$$

Where:

f	=	dimensionless coefficient which is for full turbulent flow, and is dependent on the roughness of the channel.

If Equations 2.4 and Equation 2.5 are combined we get the following Equation:

$$\frac{fu^2}{2} = \frac{Agh}{C\ L} \quad \text{...} \quad (2.6)$$

And from this Equation 2.7:

$$U = \sqrt{\frac{2g}{f}} \sqrt{\left(\frac{h}{L}\right)\frac{A}{C}} \quad m/s \quad \text{.......................................} \quad (2.7)$$

With the symbols being the same as those previously defined. If we compare the above Equation with Equation 2.1, we can see that Chezy's coefficient f holds a relationship with the roughness of the channel.

$$c = \sqrt{\frac{2g}{f}} \quad m^{1/2}/s$$

Darcy was interested in turbulent flow in pipes, and adapted Chezy's Equation for this purpose. The area of the pipe is $\pi d^2/4$, the circumference πd, h is the pressure head or pressure head loss and L is the length of the pipe.

We can rewrite Equation 2.7 as follows:

$$U^2 = \frac{2g}{f} \frac{\pi d^2}{4} \frac{1}{\pi d} \frac{h}{L}$$

Or in terms of pressure head (m) Equation 2.8:

$$h = \frac{4fLU^2}{2gd} \quad m\ \text{fluid} \quad \text{...} \quad (2.8)$$

The above Equation is also known as the Darcy-Weisbach Equation. The pressure head loss h can be substituted by the pressure head Equation, $p = \rho gh$, to get Equation 2.9:

$$p = \frac{4\ f\ L}{d} \frac{\rho U^2}{2} \quad Pa \quad \text{.......................................} \quad (2.9)$$

The above Equation can also be expressed in work done against friction:

$$F_{12} = \frac{p}{\rho} = \frac{4fL}{d} \frac{U^2}{2} \quad (J/kg)$$

The Bernoulli Equation for friction and turbulent flow can therefore be expressed as:

$$\frac{U_1^2-U_2^2}{2} + (h_2 - h_1)g + \frac{P_1-P_2}{\rho} = \frac{4\,f\,L}{d}\,\frac{U^2}{2}\ \text{(J/kg)} \qquad \ldots\ldots\ldots(2.10)$$

Where:

u	=	average velocity
f	=	Chezy-Darcy coefficient

As was previous shown to get to a general term for airways etc., which are not circular, we define the hydraulic diameter - similar to the equivalent diameter as follows:

$$r_h = \frac{A}{C} = \left(\frac{\frac{\pi d^2}{4}}{\pi d}\right) = \frac{d}{4}$$

The hydraulic diameter is now:

$$r_h = 4\,\frac{A}{C}$$

If we substitute the above Equation into Equation 2.10, the following Equation expresses frictional pressure loss:

$$p = f\,L\,\frac{C}{A}\,\frac{\rho U^2}{2} \quad \text{Pa}$$

This Equation can also be written in terms of $u = \frac{Q}{A}$ and will look as follows:

$$p = \frac{f\,L}{2}\,\frac{C}{A^3}\,\rho\,Q^2 \quad \text{Pa}$$

Or as:

$$P = R_t\,Q^2 \quad \text{Pa} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2.11)$$

Where:

$$R_t = \frac{f\,L}{2}\,\frac{C}{A^3} \quad m^{-4}$$

And R_t is the turbulent resistance of the pipe.

2.1.1.2 The Friction Coefficient f

The important question to answer here is how to calculate the value for f for any pipe. Further it is found that f is not really constant, but changes with Re value for smooth pipes and specifically with low Re values. The previous mentioned is not a surprise as f was derived specifically for full turbulent flow. By making reference to Figure 2.3 we can see typical values.

From the sketch it is clear that values for f for a smooth pipe decrease drastically as the R_e value increases, but there is a very high value for f if the pipe is not smooth. These values were attained by a number of experiments done by Stanton and Nikuradse. He painted the inside of a number of smooth pipes uniformly with grains of sand. The relationship between f and the R_e value is also shown in Figure 2.3.

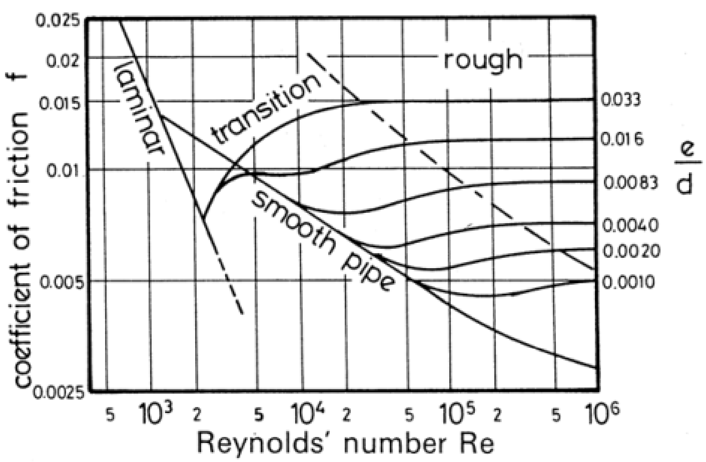


Figure 2.3 Relationship between co-efficient of friction and R_e and e/d (roughness) (Stanton and Nikuradse diagrams)

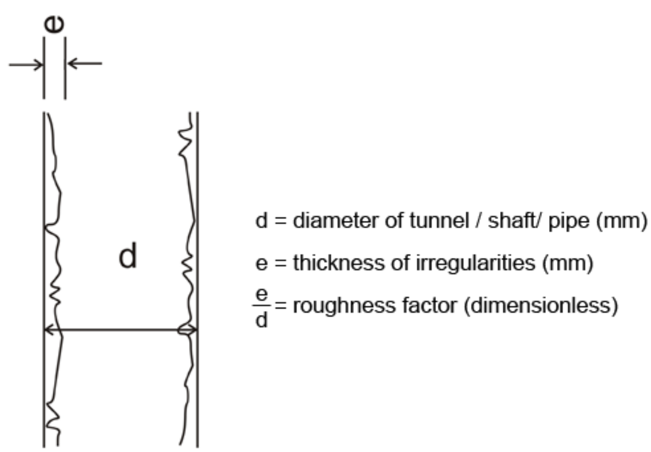


Figure 2.4 Graphical explanation of roughness factor

In the practical environment for pipes and underground airways the sides are not uniform (Figure 2.4). Specifically mine airways vary greatly in roughness. A cement ventilation shaft might have a reasonably constant value for e/d (representing the roughness factor of the side walls), as low as 0.001.

An American engineer, Moody, further developed the concept of equivalent sand roughness. He designed the moody diagram, shown in Figure 2.5 This graph also shows a curve for laminar flow and is included for completeness. A very important aspect that is also shown on this graph is the relationship with the friction factor k of Atkinson.

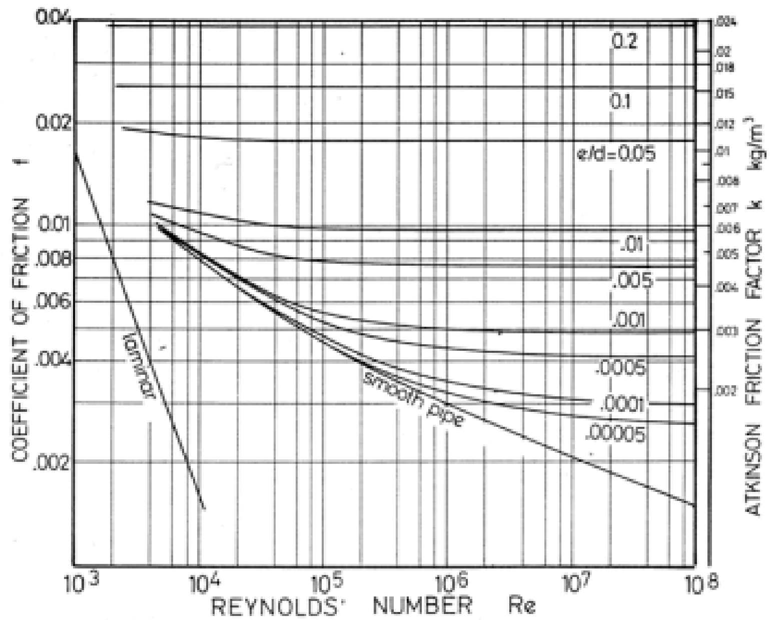


Figure 2.5 Relationship between co-efficient and R_e and Atkinson's friction factor k

2.1.1.3 Smooth Pipes and Turbulent Flow

The general Equation for turbulent flow in smooth pipes, which was developed by Nikuradse Von Karman, is given as the following Equation 2.12:

$$\frac{1}{\sqrt{f}} = 4 \log_{10} (R_e \sqrt{f}) - 0.4 \qquad \text{(2.12)}$$

This simply gives:

$$f = \frac{0.0791}{R_e^{0.25}}$$

This is the Equation for f for turbulent flow (R_e = 3 000 to 10^5) in smooth pipes.

2.1.1.4 Rough Pipes and Turbulent Flow

When rough pipes hinder the development of full turbulent flow, the viscosity forces in comparison with inert forces are negligible. This is equivalent to the shear force at the edges (Equation 2.10). In this case f is independent of R_e value and changes only with, e/d. Von Karman set-up an Equation for this situation:

$$f = \frac{1}{4\left[2\log_{10}\left(\frac{d}{e}\right)+1.14\right]^2}$$

Colebrooke and White, leading from this developed:

$$\frac{1}{\sqrt{4f}} = 1.74 - 2\log_{10}\left(2\frac{e}{d} + \frac{18.7}{R_e\sqrt{4f}}\right)$$

And simply to:

$$\frac{1}{\sqrt{f}} = -4\log_{10}\left(2\frac{e}{d} + \frac{1.255}{R_e\sqrt{f}}\right)$$

The f factor is visible on both sides of the Equation and this is also one of the reasons why Moody developed his graph. The advantage of Colebrooke-White Equation is that it can be applied to both smooth and rough pipes. For smooth pipes, the value of e/d = 0. If the R_e value is very high, the term in which R_e appears will fall away, and for high turbulent flow the Equation 2.13 will simply be:

$$f = \left[4\log_{10}\left(\frac{\frac{e}{d}}{3.7}\right)\right]^{-2} \qquad \text{.. (2.13)}$$

This is therefore the Equation for f for full turbulent flow. Atkinson also derived an Equation for pressure loss for turbulent flow of air in the underground working environment (Equation 2.14). The use of both the Equations from Atkinson and the above mentioned should derive to the approximate same answer.

$$= \left(\frac{\quad}{A^3}\right) \times \frac{\rho}{1.2} \times Q^2$$

$$= R_t Q^2 \times \frac{\rho}{1.2} \qquad \text{.............................. (2.14)}$$

Where:

p	=	frictional pressure loss (Pa)
k	=	Atkinson's friction factor (Ns^2/m^4)
C, L	=	Circumference of tunnel (C) and length of tunnel (L) (m)
A	=	Area of tunnel (m^2)
ρ	=	density and 1.2 is standard density in $kg/m^{3.}$

Animation

If we assume the density as 1.2 kg/m3 the Atkinson's Equation can be simplified as follows:

$$p = R_t Q^2$$

Definitions

2.1.2 Air Power as Used in Airflow Calculations

The term **air power** refers to the amount of power (kW) in the air when it flows at a specific quantity and pressure in an airway or also relates to the minimum power necessary to enable the flow of air in a tunnel.

Animation

According to the elementary laws for airflow through a mine the following applies:

- For air to flow from one point to another there must be a difference in pressure between these two points, called **ventilating pressure**.
- Air will always flow from a **high pressure point to a low pressure point.**
- With a **constant resistance** and a **pressure difference increase** between these two points, the **quantity will also increase and vice versa**.
- With a **constant pressure difference** between these two points and a **resistance increase**, the **quantity will decrease and vice versa**.
- To **increase** the **quantity**, **decrease** the **resistance** or **increase** the **pressure** difference between these two points.

Animation

Animation

The air power related to a specific airway or tunnel can be calculated as follows:

$$W_A = \frac{P \times Q}{1\,000} \quad \text{...} \quad (2.15)$$

Where:

W_a	=	Air power (kW)
P	=	Pressure (Pa)
Q	=	Quantity (m^3/s)
1 000	=	to convert to kW.

But, remember, we said earlier that $p = R_t Q^2$ and by substituting P in the above Equation with RQ^2, we can now say:

$$W_a = \frac{P \times Q}{1000} \text{ and } P = R \times Q^2 \times \frac{\rho}{1.2}$$

$$W_a = \frac{R_t Q^2 \times Q}{1\,000} \quad \text{if density is } 1.2 \text{ kg/m}^3$$

$$= \frac{R_t Q^3}{1000}$$

Thus, another Equation for air power (W_a) is:

$$W_a = \frac{R_t Q^3}{1000}$$

Where:

R = Resistance (Ns^2/m^8)

Through manipulation of the Equation above the following Equations can be derived:

$$W_a = \frac{PQ}{1000} \quad \text{or} \quad W_a = \frac{RQ^3}{1000}$$

Then:

$$P = \frac{1000\,W_a}{Q} \quad \text{or} \quad R = \frac{1000\,W_a}{Q^3}$$

And

$$Q = \frac{1000\,W_a}{P} \quad \text{or} \quad Q = \sqrt[3]{\frac{1000\,W_a}{R}}$$

NB

In using any one of the above Equations to calculate air power, remember that **pressure must always be in Pascal** (Pa).

Rule

Fan output power = Air power = Power imparted into the air.

WORKED EXAMPLE 2

A fan handles 5.5 m^3/s @ 0.9 kPa; calculate the air power involved.

Answer

Given that:

$$Q = 5.5\ m^3/s$$

$$P = 0.9\ kPa = 900\ Pa$$

$$W_a = ?$$

$$W_a = \frac{PQ}{1000}$$

$$= \frac{5.5 \times 900}{1000}$$

$$= 4.95\ kW$$

$\therefore$ **Air power involved say 5.0 kW**

WORKED EXAMPLE 3

The calculated output power of a fan is 12 kW, which handles 7.0 m^3/s; calculate at what pressure this fan operates.

Answer

Given that:

W_a = Fan output power = 12 kW

$$Q = 7.0\ m^3/s$$

$$P = ?$$

Now:

$$W_a = \frac{PQ}{1000}$$

Re-arranging the Equation by manipulation:

$$P = \frac{1\,000\ W_a}{Q}$$

$$= \frac{1000 \times 12}{7}$$

$$= 1\,714\ Pa$$

$\therefore$ **The fan pressure = 1 714 Pa**

WORKED EXAMPLE 4

8.2 m^3/s of air flows in a ventilation pipe with a resistance of 12.31 Ns^2/m^8. Calculate the air power.

Answer

Given that:

$Q = 8.2\ m^3/s$

$R = 12.31\ Ns^2/m^8$

$W_a = ?$

$$W_a = \frac{RQ^3}{1\,000}$$

$$= \frac{12.31 \times (8.2)^3}{1\,000}$$

$= 6.78\ kW$

Air power = 6.8 kW

WORKED EXAMPLE 5

5.0 m^3/s of air flows in 570 mm galvanized duct, 210 m in length. Calculate the air power involved.

Answer

Given that:

$Q = 5.0\ m^3/s$

$L = 210\ m$

$D = 570\ mm$

$W_a = ?$

Answer

$C = \pi D$

$= 3.147 \times 0.570$ (Remember D is always in metres)

$= 1.789\ m$

And

$$A = \frac{\pi D^2}{4}$$

$$= \frac{3.147 \times (0.57)^2}{4}$$

$= 0.255\ m^2$

Now:

$$W_a = \frac{RQ^3}{1000}$$

Where:

$$R = \frac{KCL}{A^3}$$

$$= \frac{0.0037 \text{ x } 1.789 \text{ x } 210}{(0.255)^3}$$

$$R = 83.83 \text{ Ns}^2/\text{m}^8$$

Hence:

$$W_a = \frac{RQ^3}{1000}$$

$$= \frac{83.83 \text{ x } (5.0)^3}{1\,000} = 10.47 \text{ kW}$$

∴ Air power involved = 10.5 kW

WORKED EXAMPLE 6

Air flows in a 1.5 km haulage with a resistance of 0.444 Ns^2/m^8 and the air power involved is 15 kW. Calculate the quantity of air flowing in this haulage.

Answer

Given that:

$$R = 0.444 \text{ Ns}^2/\text{m}^8$$

$$W_a = 15 \text{ kW}$$

$$Q = ?$$

Now:

$$W_a = \frac{RQ^3}{1000}$$

Re-arranging the Equation:

$$Q = \sqrt[3]{\frac{1\,000 \ W_a}{R}}$$

$$= \sqrt[3]{\frac{1\,000 \text{ x } 15}{0.444}}$$

$$= 32.3 \text{m}^3/\text{s}$$

∴ Quantity flowing = 32 m^3/s

Objective

Table 2.1 shows typical K values used for specific tunnels and pipes in the underground mining environment. It is expected that the student must be able to reproduce these K values to be applied in air pressure and or air power calculations.

Table 2.1 Typical K factors used in the mining industry

Airway	K (N s^2/m^4)
Right Angled timber shaft	0.045 - 0.09
Cement lined shaft	
• not equipped	0.0037
• with RSJ ‘buttons’	0.0075 - 0.06
Underground tunnel	0.011 - 0.018
Galvanised pipe	0.0037
Flexible ventilation pipes	0.003
Fiberglass pipes	0.0025

2.1.3 System Resistance Curves

A system’s resistance can be calculated and also represented graphically by plotting the system resistance curve on graph paper.

There are three methods employed in obtaining resistance curves:

- Where the resistance of the system has been given, for example 1.21 Ns^2/m^8
- Where the ventilation requirements for the system have been given, for example as follows: 20 m^3/s @ 150 Pa $\left(R = \frac{P}{Q^2}\right)$
- Where the characteristics of the system have been given e.g. a 3 m x 3 m airway, 1100 m long with a friction factor of 0.031 Ns^2/m^4 $\left(R = \frac{KCL}{A^3}\right)$

The next section will give a brief explanation of how these system resistance curves can be drawn.

A. Where the system resistance curve is given:

$$R = 1.21\ Ns^2/m^8$$

Step 1: Assume quantities (10, 20, 30..., use $P = RQ^2$ and calculate the pressures

Q	10	20	30	40	m^3/s
P	121	484	1 089	1 936	Pa

Step 2: Plot these quantities (x - axis) against their corresponding pressures (y - axis).

B. Where the system's ventilation requirements are given ($20\ m^3/s$ @ 150 Pa):

Step 1: Calculate the resistance of the system

$$P = RQ^2$$

$$\therefore R = \frac{P}{Q^2}$$

$$= \frac{150}{20^2}$$

$$= 0.375\ Ns^2/m^8$$

Step 2: Assume quantities (10, 20, 30,...) use $P = RQ^2$ and calculate the pressures.

Q	10	20	30	40	50	60	m^3/s
P	38	150	338	600	938	1 350	Pa

Step 3: Plot these quantities against their corresponding pressures.

C. Where the system's characteristics are given:

Let's assume haulage dimensions of 3 m x 3 m, 1 000 m long and $k = 0.031\ Ns^2/m^8$.

Step 1: Calculate the resistance of the system using

$$R = \frac{KCL}{A^3}$$

$$= \frac{0.031 \times 12 \times 1\,100}{9^3}$$

$$= 0.56\ Ns^2/m^8$$

Step 2: Assume quantities (10, 20, 30,...), use $P = RQ^2$ and calculate the pressures

Q	10	20	30	40	50	60	m^3/s
P	56	224	504	896	1 400	2 016	Pa

Step 3: Plot these quantities against their corresponding pressures.

2.1.4 Airways in Series and Parallel and Combined

Airways underground can be represented as in series or parallel or in a combination of both. It is therefore important that the student understands:

- Under which conditions these will be employed
- What effect the placement strategy will have on the total pressure losses within the system
- The resistances associated with each combination. The next sections will deal with this in detail.

2.1.4.1 Series

Airways are in **series** when one airway is followed by another airway, with each airway having its own physical characteristics.

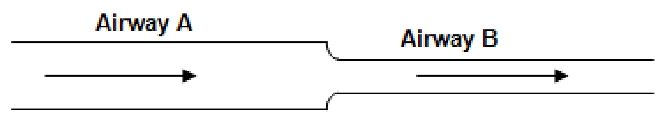


Figure 2.6 Graphical representation of airways in series

In Figure 2.6, airway 'A' is in series with airway 'B'. It is obvious that the total quantity of air flowing through the system is equal to the quantity of air in 'A', which is equal to the quantity of air flowing in 'B' (this only applicable if the density of the air stays constant.)

If the density of the air changes then we have to refer to the mass flow of the air and this is expressed in kg/s.

We normally refer to the mass flow as per kg of dry air as the mass flow of dry air stays constant within a specific airway.

$$m = m_1 \text{ (kg/s)} = m_2 = \rho Q$$

For the density assumed as constant we can say that $Q_T = Q_A = Q_B$. The total pressure loss in the system is the sum of the pressure loss in airway 'A' and airway 'B'.

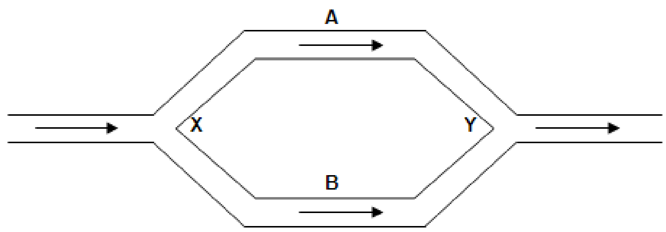
$$P_T = P_A + P_B$$

The total resistance of the system is the sum of the resistance of airway 'A' and airway 'B'.

$$R_T = R_A + R_B \qquad (2.16)$$

2.1.4.2 Parallel

Airways are in parallel when they are 'side by side' (A and B) and have a common inlet and return (X and Y respectively) as shown in Figure 2.7.



Animation

Figure 2.7 Graphical representation of airways in parallel

In Figure 2.7 the points X and Y are common to both airways (A and B). Therefore the pressure loss of airway A equals the pressure loss of airway B, and equals the pressure loss over the whole system.

$$P_T = P_A = P_B$$

The total quantity flowing in the system is equal to the sum of air flowing in airway A and airway B.

$$Q_T = Q_A + Q_B$$

The overall resistance is given by the Equation:

$$\frac{1}{\sqrt{R_T}} = \frac{1}{\sqrt{R_A}} + \frac{1}{\sqrt{R_B}} \quad \text{...} \quad (2.17)$$

If it is necessary to increase the quantity of air flowing through the above system, then it will be possible to increase the volume flow through these airways if they are put in parallel (resistance then lower) and also provided that the pressure available to ensure the airflow remains the same.

WORKED EXAMPLE 7

Two airways in series, one having a resistance of 0.10 Ns^2/m^8 and the other a resistance of 0.07 Ns^2/m^8, are developed to deliver 30 m^3/s of air to the working section. Calculate the pressure required.

Answer

$$R_T = R_A + R_B$$
$$= 0.10 + 0.07$$
$$= 0.17 \text{ Ns}^2/\text{m}^8$$

And

$$P_T = R_T Q_T^2$$
$$= 0.17 \text{ X } (30)^2$$
$$= 153 \text{ Pa}$$

WORKED EXAMPLE 8

Calculate the pressure required to pass 80 m^3/s of air through two parallel airways, each airway having a resistance of 0.09 Ns^2/m^8.

Answer

$$\frac{1}{\sqrt{R_T}} = \frac{1}{\sqrt{R_A}} + \frac{1}{\sqrt{R_B}}$$

$$\frac{1}{\sqrt{R_T}} = \frac{1}{\sqrt{0.09}} + \frac{1}{\sqrt{0.09}}$$
$$= \frac{1}{0.3} + \frac{1}{0.3}$$
$$= 3.33 + 3.33$$

$$\frac{1}{\sqrt{R_T}} = 6.66$$

$$R_T = \left(\frac{1}{6.66}\right)^2$$
$$= 0.023$$

$$P_T = R_T \; Q_T^2$$
$$= 0.023 \text{ x } (80)^2$$
$$= 147 \text{ Pa}$$

$\therefore$ **Pressure required = 147 Pa**

Rule

It should be noted from this example that the overall resistance of airways in parallel is always less than the resistance of any single airway in the parallel system.

2.1.4.3 Combined Series and Parallel

Figure 2.8 shows diagrammatically the layout of a section of a mine where there is a combination of airways in series and parallel (in this example the sketch is shown as a line diagram (Figure 2.8) not to complicate the drawing (previous examples were simpler). The following example will explain clearly how the calculation will be done.

WORKED EXAMPLE 9

The line diagram below represents a combination of airways in series and parallel.

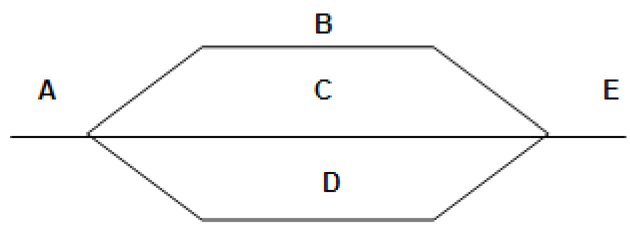


Figure 2.8 Line diagram representing combination of airways in series and parallel

The resistances of the various airways shown in Figure 2.8 are as follows:

A	=	0.009 Ns^2/m^8
B	=	0.20 Ns^2/m^8
C	=	0.18 Ns^2/m^8
D	=	0.15 Ns^2/m^8
E	=	0.01 Ns^2/m^8

If the total quantity flowing through this section is 90 m^3/s calculate the following (remember that in this example we assume that the density remains constant):

- The overall resistance of the section
- The overall pressure required
- The volumes in each airway.

Answer

$$\frac{1}{\sqrt{R_P}} = \frac{1}{\sqrt{R_B}} + \frac{1}{\sqrt{R_C}} + \frac{1}{\sqrt{R_D}}$$

$$= \frac{1}{\sqrt{0.20}} + \frac{1}{\sqrt{0.18}} + \frac{1}{\sqrt{0.15}}$$

$$\frac{1}{\sqrt{R_P}} = \frac{1}{0.447} + \frac{1}{0.424} + \frac{1}{0.387}$$

$$= 7.18$$

$$\therefore R_{parallel} = \left(\frac{1}{7.18}\right)^2$$

$$= 0.0194 \text{ Ns}^2/\text{m}^8$$

And

$$R_T = R_A + R_P + R_E$$

$$= 0.009 + 0.0194 + 0.010$$

$$= 0.0384 \text{ Ns2/m8}$$

$$\therefore \text{Overall resistance} = 0.0384 \text{ Ns}^2/\text{m}^8$$

$$P_T = R_T\, Q_T^2$$

$$= 0.0384 \times (90)^2$$

$$= 311 \text{ Pa}$$

$$\therefore \text{Overall pressure required} = 311 \text{ Pa}$$

$$P_{parallel} = R_P Q_P^2$$

$$= 0.0194 \times (90)^2$$

$$= 157 \text{ Pa}$$

But

$$P_p = P_B = P_C = P_D$$

Also:

$$P_B = R_B Q_B^2$$

$$\therefore Q_B = \sqrt{\frac{P_B}{R_B}}$$

$$= \sqrt{\frac{157}{0.20}}$$

$$= 28.0 m^3/s$$

$$Q_C = \sqrt{\frac{P_C}{R_C}}$$

$$= \sqrt{\frac{157}{0.18}}$$

$$= 29.6 m^3/s$$

$$Q_D = \sqrt{\frac{P_D}{R_D}}$$

$$= \sqrt{\frac{157}{0.15}}$$

$$= 32.4 m^3/s$$

Check $Q_B + Q_C + Q_D = 90.0 m^3/s$

$\therefore$Quantities through airways 'A', 'B', 'C', 'D' and 'E' are 90; 28; 29.6; 32.4 and 90 m^3/s respectively.

2.1.5 Graphical Representation of Airways

In the previous section, it was shown how the actual resistance totals can be calculated for airways in series, airways in parallel and airways combined in series and parallel. It is sometimes necessary to show and draw these airways in graph format so as to plot the actual resistance pertaining to each airway combination.

The section to follow will deal with how to plot the resistance curves applicable to various airway combinations.

2.1.5.1 Graphical Representation of Airways in Series

Laws

Airways in series can be solved graphically by plotting the system resistance curve for each airway and then following the series laws referring to quantity and pressure.

$$Q_T = Q_1 = Q_2$$

$$P_T = P_1 + P_2$$

When plotting airways in series, on a constant quantity line, you add the pressure loss of the one airway to the pressure loss of the other airway.

Because the quantity in each airway is equal, any quantity line can be selected and the pressure required by airway 2 can be added to the pressure required by airway 1 (both on the same quantity line) to obtain the total pressure required by both airways.

See the graphical example that follows. A point can then be plotted and other points can be obtained in the same way until sufficient points are obtained to plot a system curve representing both airways together. See Figure 2.9.

NB

The points on the system curves where the pressures and quantities are read off are sometimes referred to as 'System operating points' or S.O.P. They are marked on the graph as SOP_1 and SOP_2.

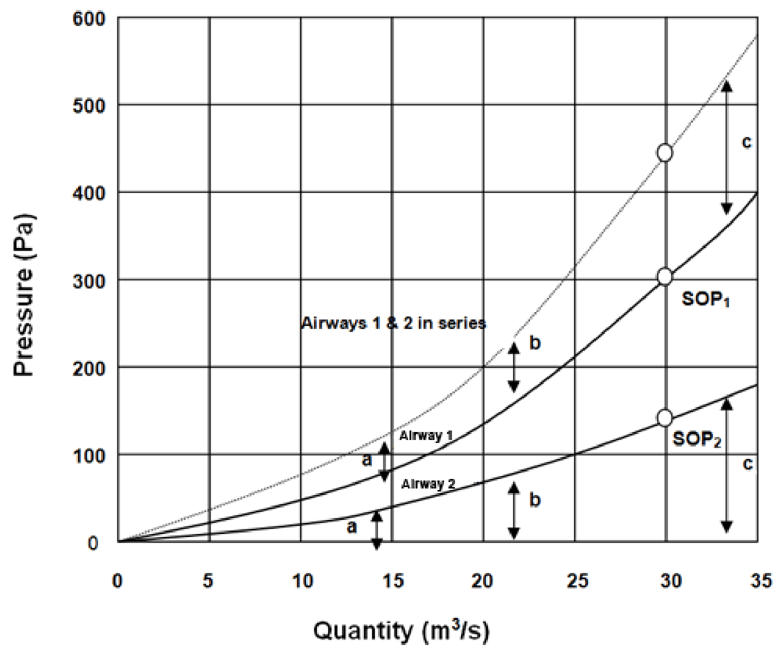


Figure 2.9 Plotting of airway resistances in series

WORKED EXAMPLE 10

The dotted curve in Figure 2.9 is the combined curve for the two airways in series. This curve is an imaginary curve, which represents both airways together. Calculations concerning these two airways in series can now be solved. For example if a pressure drop of 300 Pa exists across airway 1, determine the quantity and pressure drop in each airway and in the whole system.

Answer

Draw a horizontal line from 300 Pa to intersect airway 1. Drop a vertical line to the quantity scale and read off the quantity in airway 1 = 30 m^3/s.

$P_1 = 300$ Pa and $Q_1 = 30\ m^3/s$

But for airways in series:

$Q_T \quad = \quad Q_1 \quad = \quad Q_2$ (only true if the air densities in airway 1 and 2 are the same)

$\therefore\ Q_T \quad = \quad 30\ m^3/s$ and $Q_2 \quad = \quad 30\ m^3/s$

Where the vertical line through 30 m^3/s cuts airway 2, read off the pressure drop in airway 2 = 150 Pa. Where the 30 m^3/s line cuts the combined curve read off the total pressure drop in the system = 450 Pa.

Therefore:

$Q_1 \quad = \quad 30\ m^3/s, \qquad P_1 \quad = \quad 300$ Pa

$Q_2 \quad = \quad 30\ m^3/s, \qquad P_2 \quad = \quad 150$ Pa

$Q_T \quad = \quad 30\ m^3/s, \qquad P_T \quad = \quad 450$ Pa

Check:

$P_1 + P_2 \quad = \quad P_T$

$150 + 300 \quad = \quad 450$

If there are more than two airways in series, the same principles apply: On a constant quantity line the pressures required by each airway must be added together to give the total pressure drop.

2.1.5.2 Graphical Representation of Airways in Parallel

Airways in parallel can be solved graphically by plotting the system resistance curve for each airway and then following the parallel laws referring to quantity and pressure.

Laws

$Q_T \quad = \quad Q_1 \quad + \quad Q_2$

$P_T \quad = \quad P_1 \quad = \quad P_2$

That is, when plotting airways in parallel, on constant pressure lines, you add the quantity of the one airway to the quantity of the other airway. Because the pressures are all equal, any pressure line can be selected and the quantity in airway 1 can be added to the quantity in airway 2 (both on the same pressure line) to obtain the total quantity in both airways.

A point can now be plotted and other points can be obtained in the same way until sufficient points have been obtained to plot a system curve representing both airways together. This is shown in Figure 2.10.

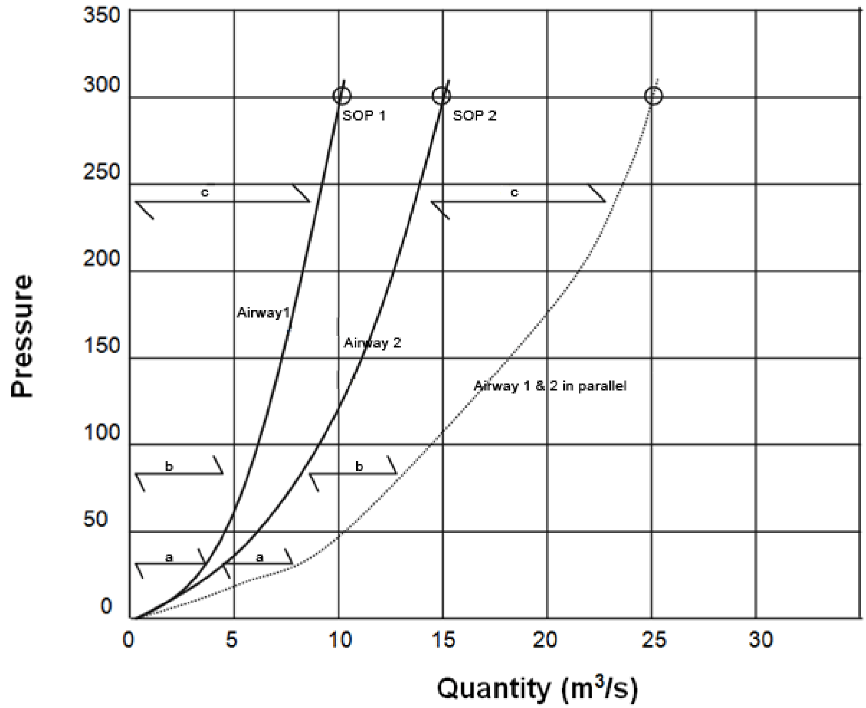


Figure 2.10 Plotting of airway resistances in parallel

The dotted curve is the combined curve for the two airways in parallel. This is an imaginary curve, which represents both airways together. Problems concerning these two airways in parallel can now be solved.

WORKED EXAMPLE 11

If the total quantity flowing through the system is 25 m^3/s, determine the quantity and pressure drop in each airway and the whole system.

Answer

Draw a vertical line from 25 m^3/s to intersect the ‘total’ curve. Draw a horizontal line across to the pressure scale and read off the total pressure drop over the whole system = 300 Pa.

$$P_T = 300 \text{ Pa}$$

And

$$Q_T = 25 \text{ m}^3/\text{s}$$

But for parallel airways:

$$P_T = P_1 = P_2$$

$$\therefore P_1 = 300 \text{ Pa}$$

And

$$P_2 = 300 \text{ Pa}$$

Where the horizontal line through 300 Pa cuts airway 1 and airway 2, read off the quantity in each airway.

Q_1	=	10 m^3/s	Q_2	=	15 m^3/s
Q_1	=	10 m^3/s	P_1	=	300 Pa
Q_2	=	15 m^3/s	P_2	=	300 Pa
Q_T	=	25 m^3/s	P_T	=	300 Pa

Check:

$$Q_1 + Q_2 = Q_T = 10 + 15 = 25 \text{ m}^3/\text{s}$$

Rule

Once again, the ‘system operating point’ for each airway is marked as SOP_1, SOP_2 etc. If there are more than two airways in parallel, the same principles apply: on a constant pressure line the quantity in each airway must be added together to give the total quantity.

2.1.5.3 Graphical Representation of Combined Series and Parallel

NB

The principles for airways in series and airways in parallel can be applied to a network of airways combined in series and parallel by following the same basic laws. The only other point to remember is that the parallel airways should be solved first beginning with the smallest parallel circuit.

WORKED EXAMPLE 12

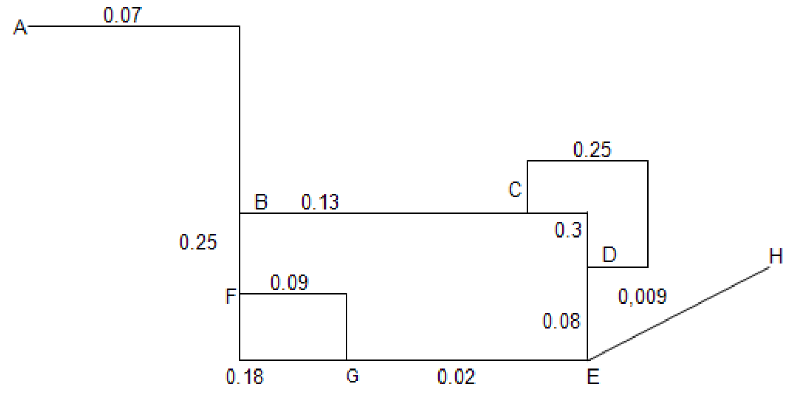


Figure 2.11 Line diagram of combined airways

Figure 2.11 represents a network of airways with resistances given in Ns^2/m^8. Determine the total resistance of the whole network. Figure 2.11 can be re-drawn in a simplified form (Figure 2.12).

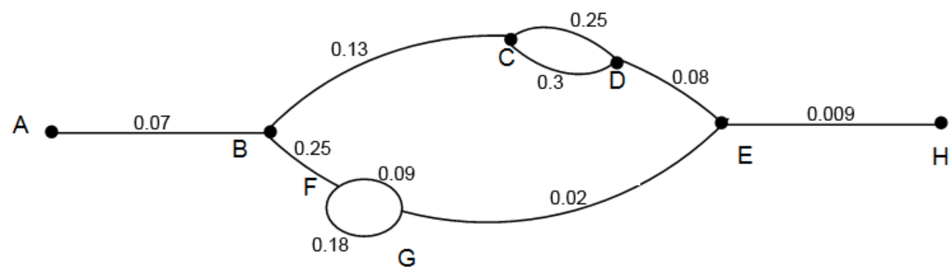


Figure 2.12

Solve parallel systems first, starting with the smaller parallel, within the bigger parallel systems.

$$\text{FG:} \quad \frac{1}{\sqrt{R_{FG}}} = \frac{1}{\sqrt{R_{FGTop}}} + \frac{1}{\sqrt{R_{FGBot}}}$$

$$\frac{1}{\sqrt{R_{FG}}} = \frac{1}{\sqrt{0.09}} + \frac{1}{\sqrt{0.18}}$$

$$R_{FG} = 0.0309 \text{ Ns}^2/\text{m}^8$$

The system can now be shown as:

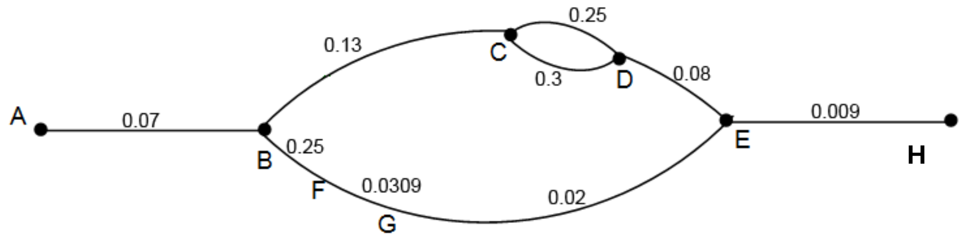


Figure 2.13

Now BF is in series with FG, which is in series with GE

$$\therefore R_{BFGE} = R_{BF} + RFG + R_{GE}$$

$$= 0.25 + 0.0309 + 0.02$$

$$= 0.3009 \text{ Ns}^2/\text{m}^8$$

The system can now be shown as:

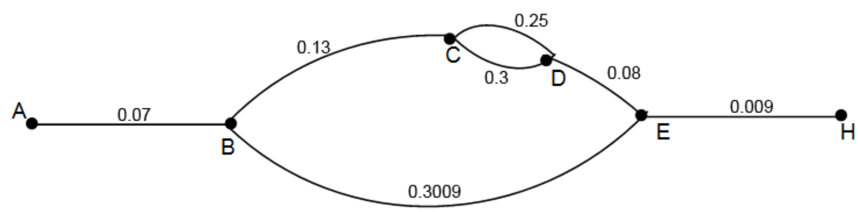


Figure 2.14

$$\text{CD:} \quad \frac{1}{\sqrt{R_{CD}}} = \frac{1}{\sqrt{R_{CDTop}}} + \frac{1}{\sqrt{R_{CDBot}}}$$

$$\frac{1}{\sqrt{R_{CD}}} = \frac{1}{\sqrt{0.25}} + \frac{1}{\sqrt{0.3}}$$

$$R_{CD} = 0.0683 \text{ Ns}^2/\text{m}^8$$

The system can now be shown as:

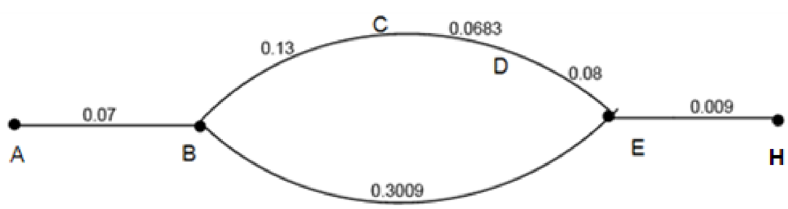


Figure 2.15

Now BC is in series with CD, which is in series with DE

$$\therefore R_{BCDE} = R_{BC} + R_{CD} + R_{DE}$$
$$= 0.13 + 0.0683 + 0.08$$
$$= 0.2783 \text{ Ns}^2/\text{m}^8$$

The system can now be shown as:

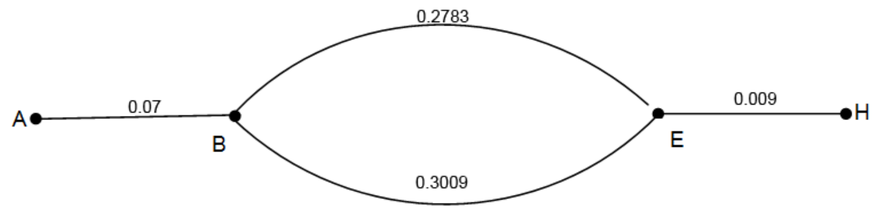


Figure 2.16

$$\text{BE:} \quad \frac{1}{\sqrt{R_{BE}}} = \frac{1}{\sqrt{R_{BETop}}} + \frac{1}{\sqrt{R_{BEBot}}}$$

$$\frac{1}{\sqrt{R_{BE}}} = \frac{1}{\sqrt{0.2783}} + \frac{1}{\sqrt{0.3009}}$$

$$R_{BE} = 0.0723 \text{ Ns}^2/\text{m}^8$$

The system now becomes:

	0.07		0.0723		0.009	
A	____________	B	___________	E	____________	H

Now AB is in series with BE, which is in series with EH

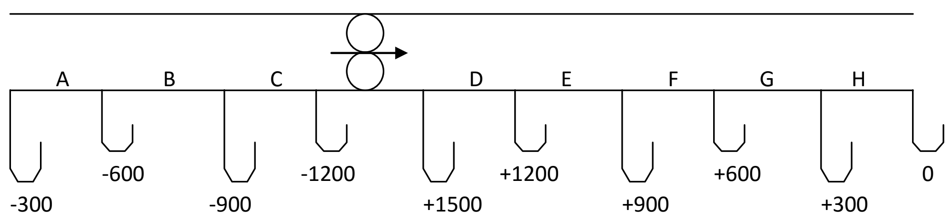
$$\therefore \quad R_T = R_{AB} + R_{BE} + R_{EH}$$
$$= 0.07 + 0.0723 + 0.009$$
$$= 0.1513 \text{ Ns}^2/\text{m}^8$$

2.1.6 Interpretation of Static Pressure Readings Taken in a Ventilation Column

Objective

The next section is a very important section in terms of explaining the pressure losses within a ventilation column/pipe underground. The principles and concepts that were required in Chapter 1 with the knowledge gained in Chapter 2 thus far will help the student to apply the knowledge gained. Figure 2.17 represents a ventilation pipe underground from which pressure measurements were done.

Static pressure readings taken at equal distances apart in a ventilation column of uniform size can be analysed and areas of high resistance or leakage can be pinpointed, but an understanding of the laws of air flow and the fundamental Equation, $P = RQ^2$ is essential for this analysis.



Animation

Figure 2.17 Static pressure measurements in ventilation pipe

Figure 2.17 also represents a duct of uniform size, and static pressure readings taken are taken at equal distances apart (why is this important?).

Definition

Fan pressure: The fan pressure is obtained by the **difference** between readings on each side of the fan.

Fan pressure	=	(+ 1 500) - (- 1 200) Pa
	=	+ 1500 + 1 200 Pa
	=	2 700 Pa

Pressure loss in each section: The pressure loss in each section is obtained by the **difference** between the readings on each side of the section.

For example Section C pressure loss	=	(-900) - (-1 200) Pa
	=	-900 + 1 200 Pa
	=	300 Pa

NB

Leakages or restrictions: In this particular example, which is ideal, the pressure loss in each section from A to H is 300 Pa. This means there are no abnormalities (leakages or restrictions) in the column. Remembering the fundamental Equation, $P = RQ^2$, this is to be expected; each section has the same R ($R = \frac{KCL}{A^3}$) and the same Q.

Rule

However, if there is a leakage in any section, the value of the term 'Q' in the Equation changes and the pressure loss for that section changes. If there is a restriction in any section, the value of R in the Equation changes and the pressure loss for that section changes.

To determine whether a change in pressure loss is caused by a leakage or a restriction, you must know the elementary laws of airflow; these are illustrated in the example that follows.

WORKED EXAMPLE 13

Figure 2.18 represents a column of uniform size with static pressure readings taken equal distances apart. Determine the fan pressure, the pressure loss in each section, and, whether there are any leakages or restriction in the column.

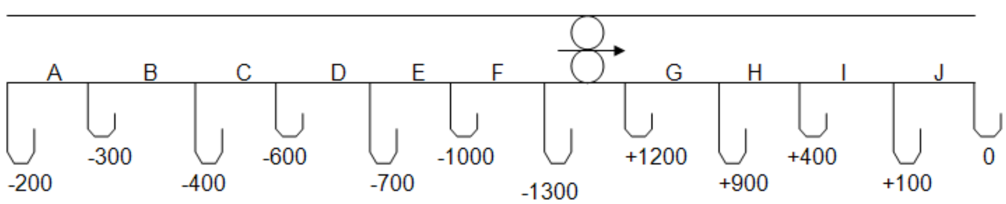


Figure 2.18 Fan Column with static pressure readings

Answer

(i) Fan pressure: Fan pressure = (+1 200) - (-1 300) Pa
= (+1 200 + 1 300) Pa
= 2 500 Pa

(ii) Pressure losses: By difference, the pressure losses in each section are as follows (All pressure losses in Pascal)

A	B	C	D	E	F	G	H	I	J
100	100	200	100	300	300	300	500	300	100

(iii) Leakages or restrictions:

- 100 Pa is used in sections A
- 100 Pa is used up in section B. Here it can be deducted that 100 Pa appears to be the ‘normal’ amount of pressure needed for each section of pipe.
- 200 Pa is used up in section C. This is more than ‘normal’ and from the laws of air flow, this can mean one of two things, either (1) there is a restriction in this section, or (2) there is more air flowing in this section. (Air leaking into section C).
- To decide which is correct, it is necessary to examine the following section D. Here the pressure loss has reverted back to the ‘normal’ loss of 100 Pa.

  This rules out possibility (2), because there would have been more air flowing in section D as well. Hence there is a restriction in Section C.
- 100 Pa is used up in section D, which is ‘normal’.
- 300 Pa is used up in section E. This is more than ‘normal’ and again, can either be a restriction or leakage in. Looking at section F that is also 300 Pa, it appears that in this case it is leakage in at E. This means that the air quantity in the column has now increased, and the new ‘normal’ pressure loss for each section should be 300 Pa.
- 300 Pa is used up in section F, which is ‘normal’.
- 300 Pa is used up in section G, which is ‘normal’.

NB

- 500 Pa is used up in section H. This is more than normal, and because it is on the delivery side of the fan, it can mean only that there is a restriction in the section (i.e. there cannot be more air flowing in the column as air cannot leak in on the delivery side of the fan).
- 300 Pa is used up in section I, which is back to ‘normal’.
- 100 Pa is used up in section J. This is less than normal and can mean only that air has leaked out of the column, i.e. there is less air flowing in section J, hence less pressure loss.

The conclusions are summarized:

A	B	C	D	E
100	**100**	**200**	**100**	**300**
Normal	Normal	Restriction	Normal	Leakage in

F	*8	G	H	I	J
300		**300**	**500**	**300**	**100**
Normal		NormaL	Restriction	Normal	Leakage out

*8: Fan between F and G

2.1.7 Introduction to Ventilation Door, Ventilation Brattices and Regulators

In the underground mining environment several types of appliances are used to control or redirect the flow of air. This is done through the use of door, regulators, brattices and venturis. The following section will explain the use of each in detail.

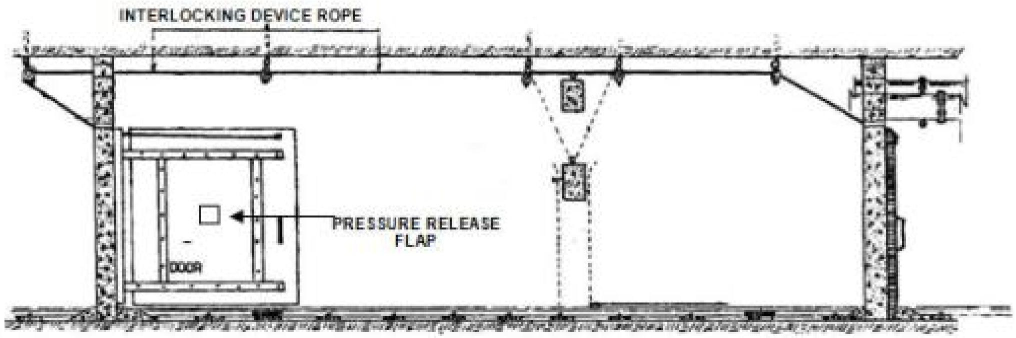
2.1.7.1 Ventilation Doors (Airlocks)

Image

Ventilation doors (2 or 3) are installed in series to form an air lock at various places underground. The objective of installing such air locks is to prevent the flow of air to workings, while at the same time allowing travelling and transport to take place. The requirements of an ideal airlock are:

- It should not leak excessively.
- It must be self-closing and kept closed at all times.
- The installation must be as such that only one door can be opened at any time (‘interlocking’).
- It must be economical, efficient and require minimum maintenance.
- It should be robust, strong and easy to open.
- Each door must be equipped with a proper, handle on both sides pressure release flap, and an effective water trap.
- It should be painted with black and yellow chevron lines (or any other colour as per your mine’s standard) to make it visible.

Figure 2.19 illustrates an air lock (two doors) with interlocking (not always used).



Animation

Figure 2.19 Side view of ventilation doors used underground

References

For more information on ventilation doors, read Environmental Engineering in SA Mines, pages 299-303.

2.1.7.2 Pressure Release Flap

A pressure release flap should be installed on all ventilation doors and should be large enough to equalize the pressure across the door quickly, to facilitate the easy opening of such a door. It is worthwhile to realize that a pressure release flap cannot fulfill its function if there is excessive leakage through the door.

2.1.7.3 Water Trap

Definition

Rule

All air locks, through which drains pass, must be equipped with effective water traps. A **water trap** is a device for allowing water to flow through an air lock without allowing air to leak through. This device is designed on the principle of a vertical manometer. In every case there must be a difference in water levels between the high and low pressure sides of the door, equal to the pressure across the door.

The following features are common to all types of water traps (refer to Figure 2.20):

- The sump must be large enough to enable mud settlement to be cleared from both sides of the partition.
- The gap between the bottom of the partition and the bottom of the sump should be large enough to allow an accumulation of mud.
- The sump should be deep enough to allow the trap to function when no water is flowing in the drain.
- The plan area of the high pressure side should be larger than that on the low pressure side.

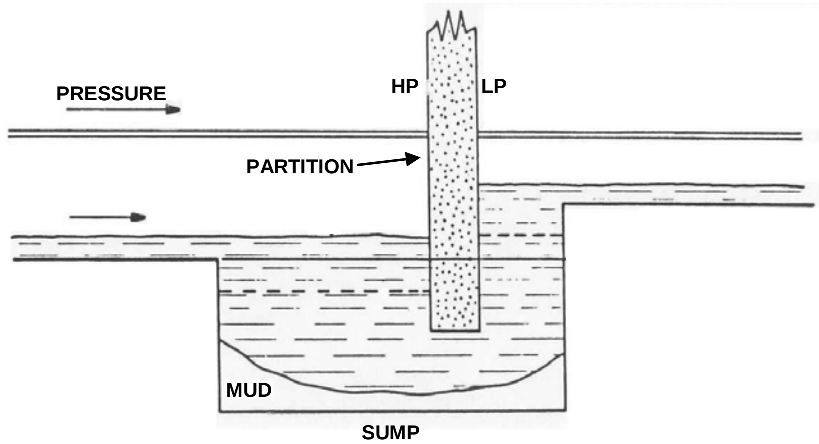


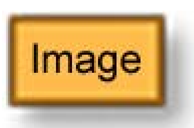
Figure 2.20 A typical water trap for use with ventilation doors (HP = High pressure side, LP=Low Pressure side)

2.1.7.4 Stoppings / Walls

They are installed in workings to stop / block the flow of air completely and can be divided into:

- **Temporary Stoppings:** They are constructed with timber and or plastic sheeting, conveyor belting, ventilation curtains etc. this type of stopping is mainly used when a temporary modification to the ventilation system is required or for test purposes underground during air flow tests.

  They are also frequently installed until such time as they can be replaced by permanent stoppings.



- **Permanent Stoppings**: These are solid cast concrete walls or constructed of vermiculite bricks or concrete bricks. A water trap (100 mm diameter pipe on footwall) and gas drain (25 mm diameter pipe against hanging wall) should be fitted through all permanent stoppings to cater for any possible accumulation of water and / or flammable gas, respectively, behind these stoppings. Figure 2.21 shows a sketch of a typical cast concrete wall.

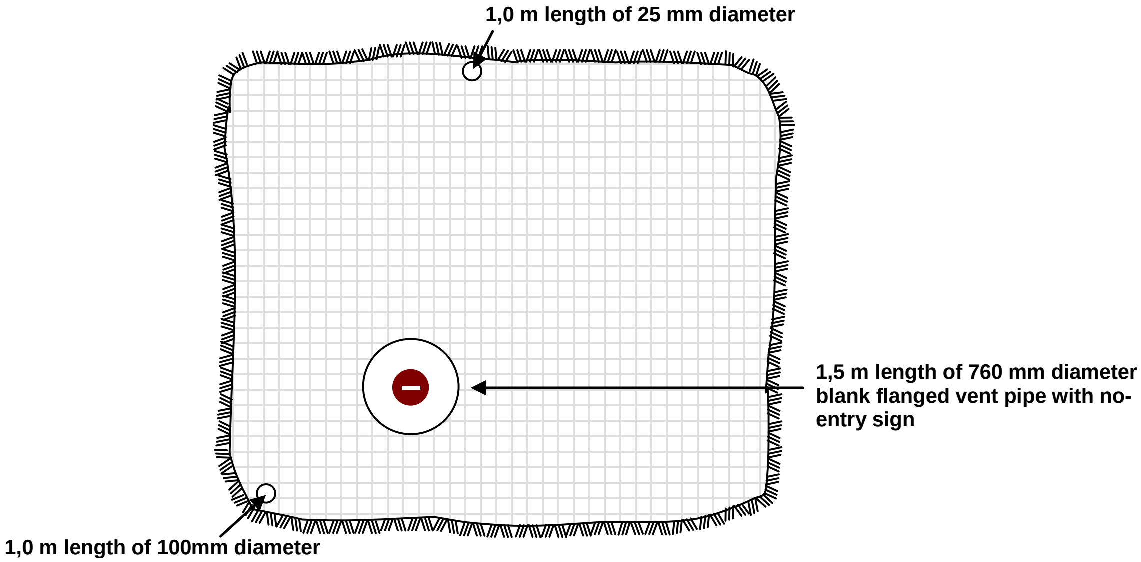
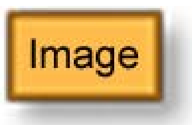


Figure 2.21 Typical cast concrete wall used underground



- **Explosion Proof Stoppings**: These stoppings are built when a section of a mine needs to be sealed off due to a fire or when flammable gas (CH_4) is known to be present and the possibility of a flammable gas explosion exists. There is a specific Code of Practice applicable to different mines in respect of this type of stopping, as there are various ways of constructing explosion proof stoppings in coal and gold mines.

- **Barricades**: A barricade is an appliance erected to prohibit people from unauthorized entry into abandoned or worked out areas, while at the same time allowing air to enter such workings, where necessary (e.g. return airways, intake airways no longer required for travelling and transporting etc). Barricades can be divided into:
  - Emergency temporary barricades
  - Permanent through ventilation barricades.

  Figure 2.22 shows examples of emergency temporary and permanent through ventilation barricades/doors (through ventilation refers to when there is a natural flow of air from one working environment to another induced by fans).

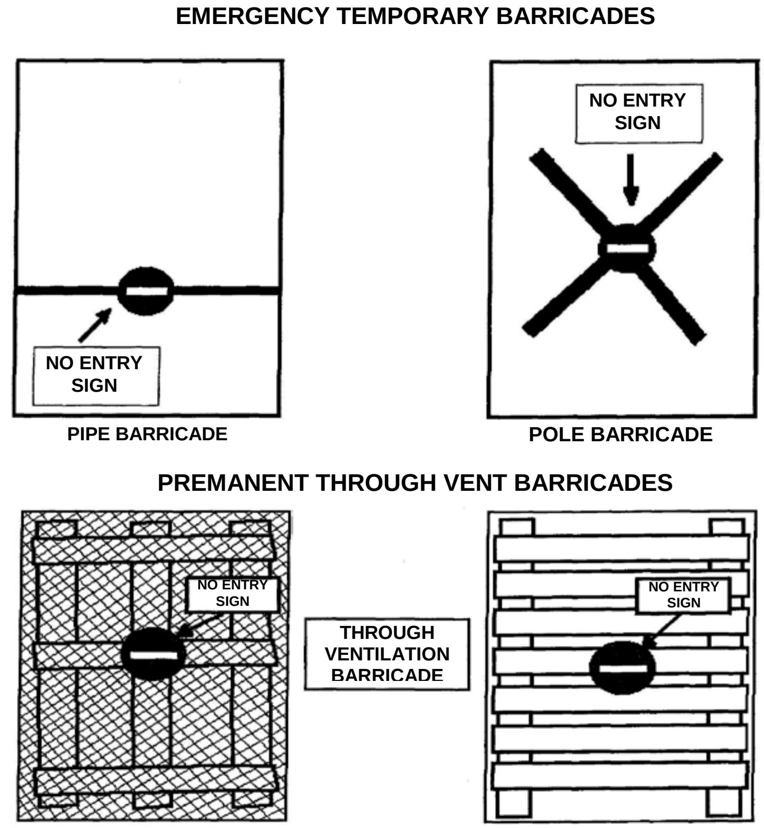
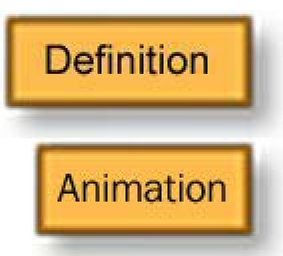


Figure 2.22 Examples of temporary and permanent through ventilation barricades

2.1.7.5 Regulators



A **Regulator** is an opening in a stopping that will allow a pre-determined volume of air or specific quantity of air to pass through the regulator.

A regulator increases the resistance of the system in which it is installed and hence ‘uses up’ some of the available ventilating pressure which results in a decrease of the air volume / quantity (i.e. one of the laws for airflow - at a constant pressure, the air quantity will decrease when the resistance increases.

There are various means to regulate the air quantity and Figure 2.23 illustrates some different types of regulators.

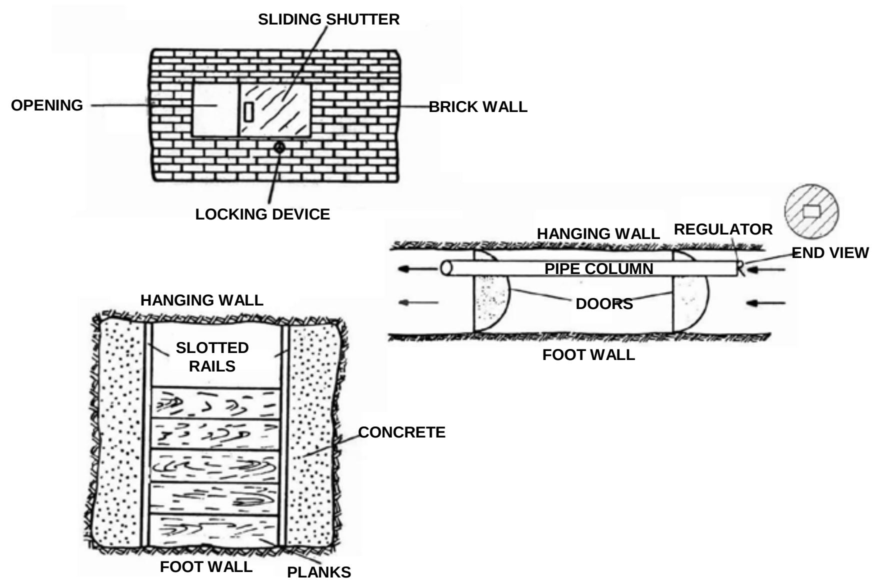


Figure 2.23 Examples of Regulators as used in the underground working environment

These ventilation appliances discussed above are utilized to distribute and control the available air from the shaft to the intake of all workings.

2.1.7.6 Stope and Development Ventilation Appliances

The following ventilation appliances are utilized to distribute and control the air from the intake of the working to the working faces:

- **Brattices:** Brattices are installed in centre or travelling gullies (tunnels underground) to prevent air from by-passing the faces and flowing directly to the return of the workings.

  They are constructed of a heavy material such as conveyor belting and suspended from the hanging wall on a steel pipe or lagging.

  It is important that the conveyor belt strips overlap with each other to form a proper seal. Figure 2.24 illustrates a brattice installation (V100 is a type of plastic used underground).

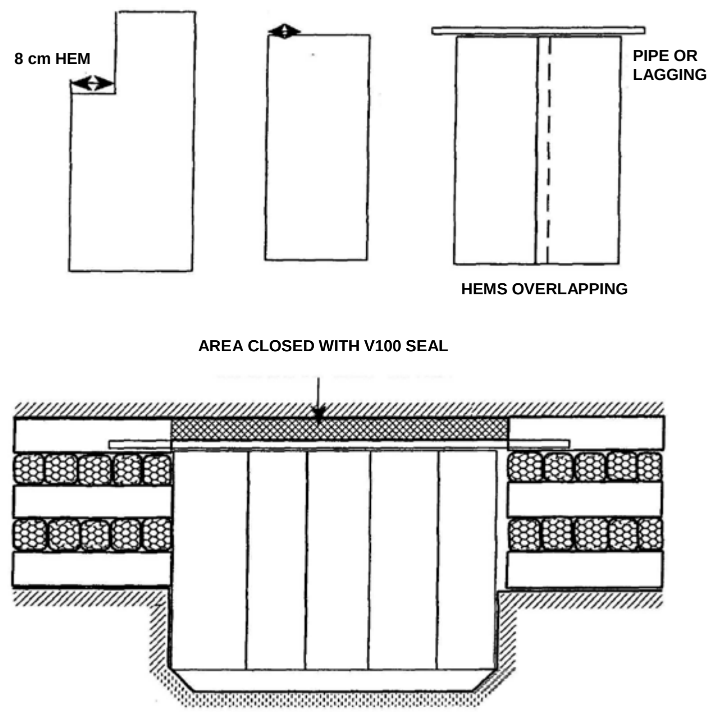


Figure 2.24 Example of brattices installed in gulley underground to direct air

- **Ventilation Curtains**: A variety of plastic based materials are available with little variation in effectiveness. The top of the curtain has to be glued (or using prattley-putty and wire) to the hanging wall, ensuring that all uneven gaps in the hanging are closed off properly to prevent short-circuiting of the air through the worked out areas.

  The bottom of the curtain must overlap the footwall and lashed with waste rock to seal the base of the curtain.

  These ventilation curtains are installed in conjunction with the brattices on strike / dip to direct (maximum) air to the working faces.

  Ventilation curtains are easily damaged and continuous repair and maintenance are required.

- **Stone Walls**: Stonewalls are commonly used as permanent seals and also fulfilling an integral part of the support system. They are constructed from waste rock, which may be over-sprayed with a sealant, such as vermiculite or any fire retardant plaster. They can be build as strike (perpendicular to the dip of the reef) and dip seals (parallel to the dip of the reef) in stopes.

- **Ventilation Piping (Ducting)**: Ventilation piping are used in conjunction with auxiliary fans to convey air to and from working places / faces not in through ventilation, such as development ends, up-dips in stopes, workshops not in through ventilation, back-stopes (stopes with no second outlet for the through air and then supplied with auxiliary fans). Various types of piping are available such as galvanised iron, fibre-glass and plastic flexible ducting. It is important to note that a danger of sparking is possible, if ducting is not made of anti-static material. Types of ducting and material used should meet the required safety standards on your mine.
- **Air-movers (Venturi)**: Air-movers are sometimes used to ventilate stope face upper limits / corners, prospects, winch chambers and places where auxiliary fans with ventilation ducting cannot at some occasions be installed. It can also be used to dissipate or remove any gas accumulations in workings.

The air-mover works on the principle that a jet of air discharges into a throat at a high velocity and entrains the air from the surrounding atmosphere. It is important to note, that an air-mover should at all times be installed in the last point of through ventilation to be effective - if not, it will re-circulates the air, resulting in adverse thermal conditions.

Tests conducted on a Free State mine, where pillar mining took place, yielded that an air-mover installed in the last point of through ventilation and place at the intake of a 25 m x 150 mm diameter PVC pipe, gave almost the same results as 4 kW, 380 mm diameter fan with ducting.

Unfortunately, air-movers are driven by compressed air, which is very expensive, and should be used only in area where they are absolutely necessary. Figure 2.25 illustrates an air-mover.

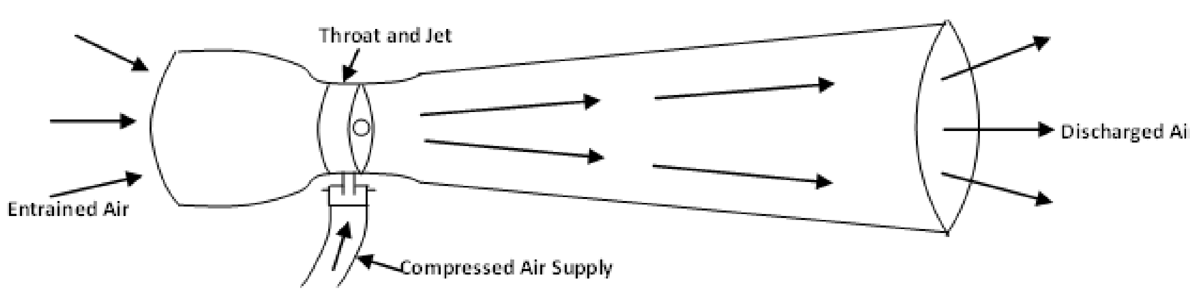


Figure 2.25 A typical Venturi used underground

2.1.7.7 Hoods

A hood is an appliance fitted to an exhaust extraction system at the point where air enters the system in order to control the contaminants entering such system. Hoods can be used on many different extraction systems, such as lab fumes, soldering fumes, paint dips, decreasing vapours, grinding, dust, etc. There are also various designs in hoods and Figure 2.26 illustrates a few designs.

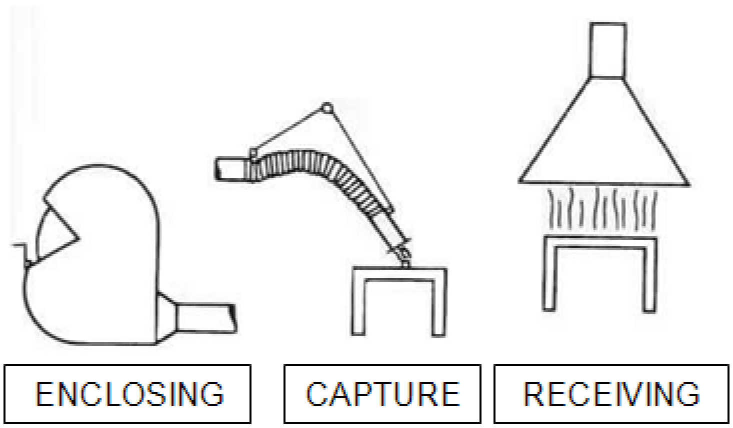


Figure 2.26 Typical hoods use for extraction

2.1.8 Regulator Calculations

Definition

Regulators are openings of specific size, which are used to control the flow of air. A regulator reduces the flow of air by limiting the area through which the air can flow. This increases the overall resistance of the system and less air will flow through the system.

A regulator 'uses up' pressure in a system, which would otherwise have been used overcoming the friction in the system. Consider Figure 2.27.

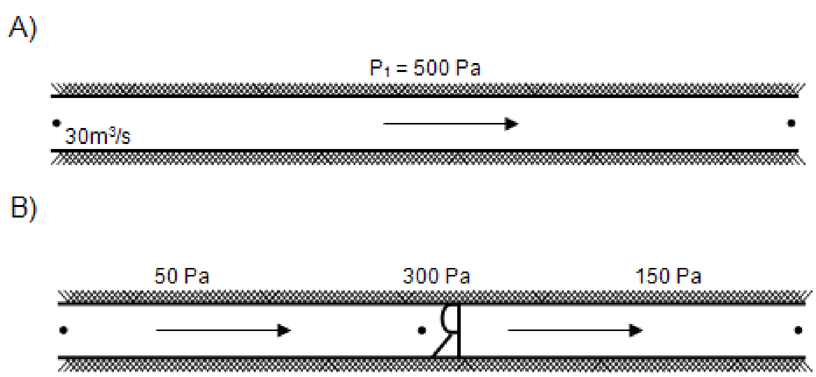


Figure 2.27 Tunnel A (no regulator) and Tunnel B (with regulator)

Sketch (A) shows an airway where 30 m^3/s of air is flowing with a pressure drop of 500 Pa, that is 500 Pa is used up in overcoming the friction in the airway. Sketch (B) shows the same airway with a regulator installed.

Here only 200 Pa is used to overcome the friction of the airway and 300 Pa is used to overcome the resistance of the regulator. (The overall pressure drop over the system remains 500 Pa). To determine the quantity flowing in (B), $P \alpha Q^2$:

$$\frac{P_1}{Q_1^2} = \frac{P_2}{Q_1^2}$$

$$Q_2^2 = Q_1^2 \text{ x } \frac{P_2}{P_1}$$

$$Q_2^2 = \sqrt{Q_1^2 \ \frac{P_2}{P_1}}$$

$$= \sqrt{30^2 \text{x} \frac{200}{500}}$$

$$= 18.97 \text{ say } 19m^3/s$$

Hence, the regulator has reduced the quantity of air flowing in the airway by using up 300 Pa pressure which would otherwise have been used on the actual airway.

Despite the fact that the overall pressure drop, across the system remains 500 Pa (200 Pa on the airway and 300 Pa on the regulator), the overall resistance of the system has increased because of the reduced quantity.

$$P = RQ^2$$

$$\therefore R = \frac{P}{Q^2}$$

Resistance of (A) $\quad R_A = \frac{P}{Q^2}$

$$= \frac{500}{30^2}$$

$$= 0.555 \ Ns^2/m^8$$

Resistance of (B) $\quad R_B = \frac{P}{Q^2}$

$$= \frac{500}{19^2}$$

$$= 1.385 \ Ns^2/m^8$$

The size of the regulator required can be calculated using the Equation 2.19

$$A_r = 1.2\,Q\sqrt{\frac{\rho}{p}} \quad \text{...} \quad (2.18)$$

Where:

A_r = area of regulator (m^2)
Q = air quantity through regulator (m^3/s)
P = pressure used up by regulator
ρ = air density (kg/m^3)

WORKED EXAMPLE 14

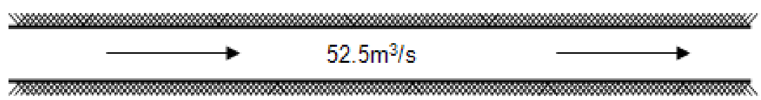


Figure 2.28 Airflow quantity in a tunnel

The pressure loss in the above airways is 720 Pa and the air density is 1.058 kg/m^3. What size regulator would be required to reduce the air quantity to 30 m^3/s?

Answer

To pass 52.5 m^2/s of air requires a pressure of 720 Pa. To pass 30 m^3/s of air requires a pressure of:

$$P_2 = P_1 \times \frac{Q_2^2}{Q_1^2}$$

$$= 720 \times \frac{30^2}{52.5^2}$$

$$= 235 \text{ Pa}$$

This will be the pressure used up on the actual airway, and the regulator must use up the rest.

$\therefore$ Pressure used up by regulator = (720 - 235) Pa
= 485 Pa

Area of regulator:

$$A_r = 1.2 \times Q \times \sqrt{\frac{\rho}{p}}$$

$$= 1.2 \times 30 \times \sqrt{\frac{1.058}{485}}$$

$$= 1.68\text{, say } 1.7 \text{ m}^2$$

Another problem, which occurs sometimes, is the calculation of the regulator size in a column with a fan.

WORKED EXAMPLE 15

A fan in a leak-less, 200 m long, 760 mm diameter galvanized iron column delivers a quantity of 8.0 m^3/s of air at a pressure of 1 210 Pa in a development end at a prevailing density of 1.1 kg/m^3. It is required to reduce this quantity of air to 5.0 m^3/s by means of a, regulator. Determine the size of the regulator. (The curves for the fan and the existing system are in Figure 2.29).

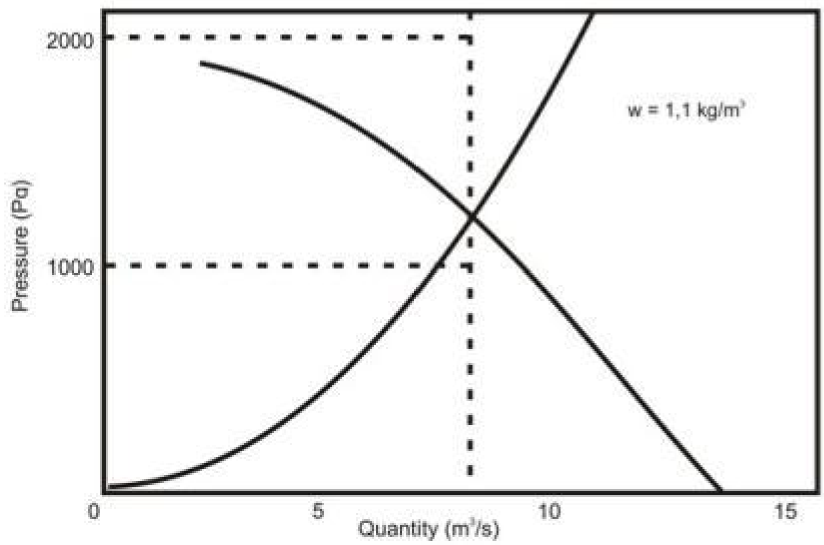


Figure 2.29 Fan pressure and airway resistance graph

Answer

When 8.0 m^3/s of air flows through the fan it operates at a pressure of 1 210 Pa. When 5.0 m^3/s of air flows through the fan it operates at a pressure of 1 660 Pa.

However, at 5.0 m^3/s the system requires a pressure of only:

$$P_2 = P_1 \times \frac{Q_2^2}{Q_1^2}$$

$$= 1\,210 \times \frac{25}{64}$$

$$= 473 \text{ Pa}$$

Therefore the regulator will have to destroy (use up) a pressure of

$$(1\ 660 - 473) \quad = \quad 1\ 187 \text{ Pa}$$

$$\text{Area of regulator} \quad = \quad A_r \quad = \quad 1.2Q\sqrt{\frac{\rho}{p}}$$

$$= \quad 1.2 \times 5 \times \sqrt{\frac{1.1}{1187}}$$

$$= \quad 0.183 \text{ m}^2$$

If a new system resistance curve is drawn (as shown in Figure 2.30), the pressure used up by the regulator is clearly seen to be 1 187 Pa.

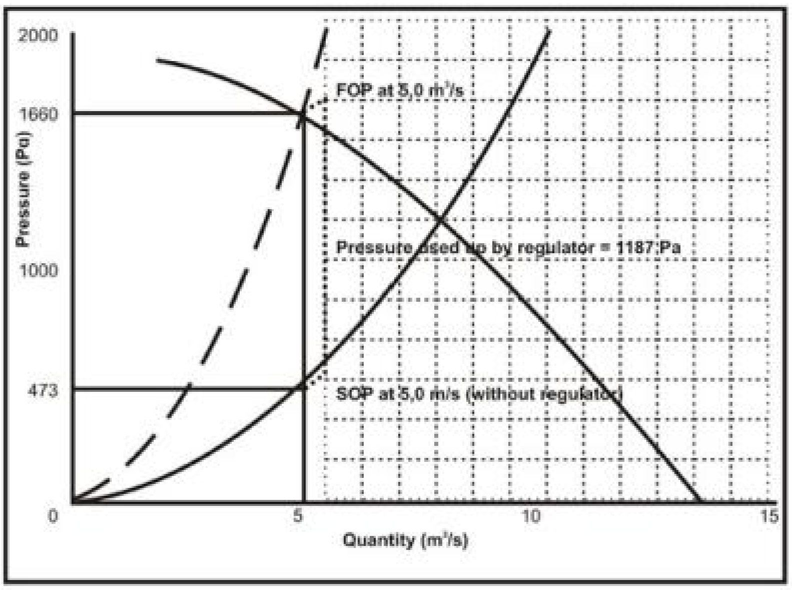


Figure 2.30 Graphs showing SOP and FOP

Common mistakes in the use of the regulator Equation and the principles involved are:

NB

- Use of the wrong 'p' in the regulator Equation. The general mistake is to calculate the pressure required for the reduced quantity (235 Pa - Worked Example 14) and to use this pressure in the regulator Equation instead of the difference between the original and the reduced pressure.
- Use of the wrong 'Q' in the regulator Equation. The original air quantity is mistakenly used in the Equation instead of the reduced quantity, which will flow through the regulator.

- When airways are in parallel and a regulator is installed in one of these airways it is mistakenly believed that air quantity in the other airways will increase by an amount equal to the reduction in air quantity in the regulator airway. This is not correct and is best explained by means of Worked Example 16.

WORKED EXAMPLE 16

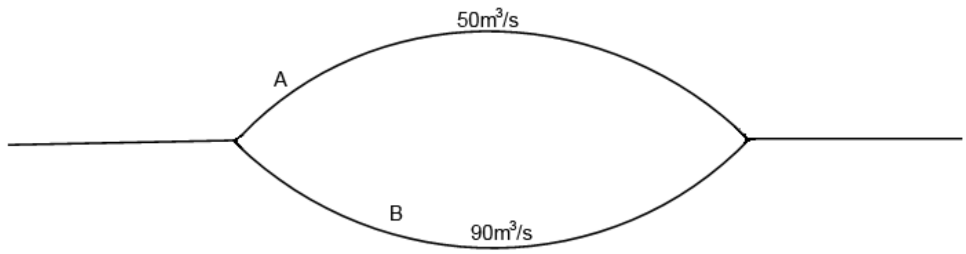


Figure 2.31 Simplified sketch of airway

The pressure drop over the parallel airways A and B shown in Figure 2.31 is 900 Pa. The air density is 1.15 kg/m^3. Determine the size of regulator required to reduce the quantity of air flowing in airway B to 60m^3/s.

Answer

To pass 90 m^3/s through B requires a pressure of 900 Pa.

$\therefore$ To pass 60 m^3/s through B requires a pressure of:

$$P_2 = P_1 \times \frac{Q_2^2}{Q_1^2}$$

$$= 900 \times \frac{60^2}{90^2}$$

$$= 400 \text{ Pa}$$

$\therefore$ The regulator uses up a pressure of: (900 - 400) Pa.

$$= 500 \text{ Pa}$$

Size of regulator required, $A_r = 1.2Q \sqrt{\frac{\rho}{p}}$

$$= 1.2 \times 60 \times \sqrt{\frac{1.15}{500}}$$

$$= 3.45 \text{ say } 3.5 \text{ m}^2$$

Hence, with the regulator installed, the pressure drop over airway B remains 900 Pa. The pressure drop over airway A also remains 900 Pa (parallel airways) and hence the quantity in airway A remains 50 m^3/s. so there appears to be 30 m^3/s of air ‘missing’ from airway B. This 30 m^3/s of air is ‘lost’ because the overall resistance of the system has increased, resulting in an increase in fan pressure and a decrease in fan quantity. In a complex system of airways, the installation of a regulator in one of the airways affects the overall system resistance. The fan operating point calculations involved are therefore extremely complex and usually a computer is used.

NB

2.1.9 Booster Fans

Image

Whereas regulators are used to decrease the quantity of air flowing in an airway, booster fans are used to increase the quantity of air flowing in an airway.

WORKED EXAMPLE 17

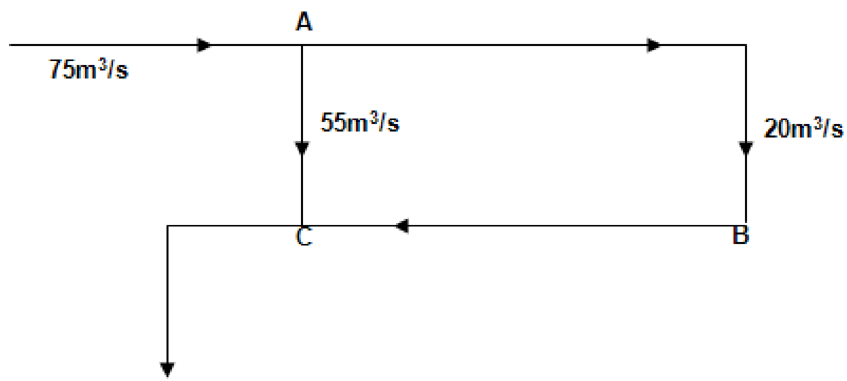


Figure 2.32 Air distribution in a section of a mine

Figure 2.32 shows how air is distributed in a section of a mine. The pressure loss A - C is 800 Pa. It is desired to increase the quantity flowing in ABC to 35 m^3/s by installing a booster fan. Determine the duty of this booster fan.

Answer

Pressure drop AC = pressure drop ABC = 800 Pa (parallel airways), this pressure is supplied by other fans in another part of the section. To pass 20 m^3/s of air requires a pressure of 800 Pa, to pass 35 m^3/s of air requires a pressure of:

$$P_2 = P_1 \times \frac{Q_2^2}{Q_1^2}$$

$$= 800 \times \frac{35^2}{20^2}$$

$$= 2\,450 \text{ Pa}$$

But here is already a pressure difference of 800 Pa across airway ABC that is supplied by other fans.

$\therefore$ Booster fan pressure = (2 450 - 800) Pa
= 1 650 Pa

The air quantity to be handled by the booster fan must obviously be the total quantity of 35m^3/s.

$\therefore$ Duty of required booster fan = 35 m^3/s @ 1 650 Pa

The obvious question, which arises from these results, is, "*How can AC still have a pressure drop of 800 Pa and carry 55 m^3/s when the pressure drop in ABC is 2 450 Pa and the two airways are in parallel*"?

The answer to this is because the fan adds pressure to the air, i.e. the fan is a pressure 'gain' in ABC.

Pressure loss due to friction in ABC	=	2 450 Pa
Pressure gain (due to booster fan in ABC)	=	1 650 Pa
Difference	=	800 Pa

This anomaly is perhaps best explained by the use of absolute pressures at the various points in Figure 2.33. Assume that an absolute pressure of 95.0 kPa existed at A; assume that the booster fan is situated at B (between B_1 and B_2) and that half the pressure loss in ABC is used up in AB_1 and the other half in B_2C.

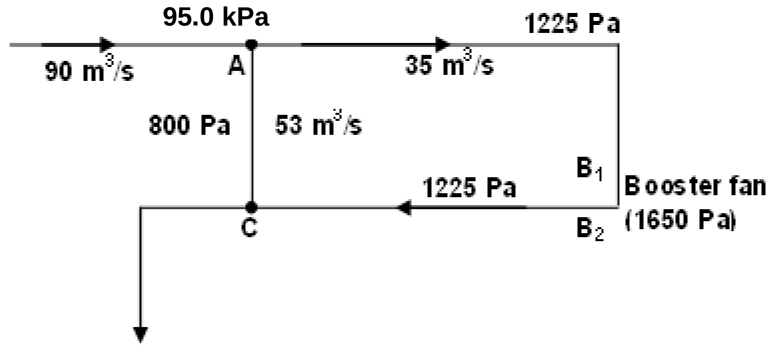


Figure 2.33 Absolute pressures

Absolute pressure at A	=	95.0 kPa
Pressure loss AC	=	800 Pa
∴ Absolute pressure at C	=	94.2 kPa
Absolute pressure at A	=	95.0 kPa
Pressure loss AB_1	=	1 225 Pa (assumed half-way)
∴ Absolute pressure at B_1	=	93.775 kPa (95.0 – 1.225)

Now booster fan ads 1 650 Pa pressure

∴ Absolute pressure at B_2	=	95.425 kPa (93.775 + 1.650)
Pressure loss B_2C	=	1 225 Pa (assumed half-way)
∴ Absolute pressure at C	=	94.2 kPa (95.425 – 1. 225)

(Which is the same pressure obtained at C following route A-C)

Hence, airways ABC and AC still have the same pressure 'difference' which, satisfies the laws of airways in parallel. These pressures also satisfy Equation 1.6 derived in the section on Bernoulli, which, in simple terms, is:

$$TP_1 = FTP = TP_2 + P_L$$

Applying this to ABC gives:

$$95.0 + 1.65 = 94.2 + 2.45$$
$$96.65\ \text{kPa} = 96.65\ \text{kPa}$$

And applying it to AC gives:

$$95 + 0 = 94.2 + 0.8$$
$$95.0 = 95.0$$

NB

Probably the most important point to remember about the calculation of booster fan duties is that a booster fan handles all the air quantity in the airway in which it is situated, but only supplies the extra pressure needed for the airway.

However, as in the case with the regulator in a complex system, a booster fan, which is used to increase the quantity in a specific airway, will also affect the rest of the system, and once again the calculations involved are complex and therefore a computer program is usually used.

2.1.10 Pressure across Doors and Stoppings

Most environmental Control Officials take pressures across ventilation doors (air locks) and stoppings as a matter of routine, but in general little use is made of these readings. Such readings can be very valuable. Some circuits containing ventilation doors and stoppings are a lot more complicated than others, but the principle remains the same.

WORKED EXAMPLE 18

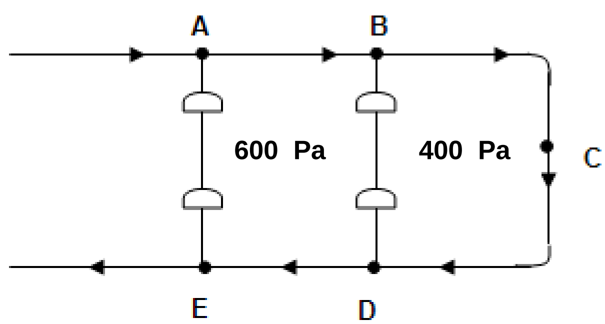


Figure 2.34 Airway with ventilation doors

NB

Refer to Figure 2.34. ABC is an intake airway and CDE a return airway. Connections between the airways are closed off with ventilation doors and the pressures across these doors are shown. (Connections AE and BD could have been closed off with stoppings and the analysis would be the same).

Answer

Despite the fact that it is closed off, connection AE is in parallel with ABCDE. Hence, the pressure of 600 Pa across the doors in AE must be equal to the pressure loss of airway ABCDE.

The same with BD, which is in parallel with BCD. Hence, the pressure of 400 Pa across BD must be equal to the pressure loss of BCD.

(i) 600 Pa is required for ABCDE.

Of this:

(ii) 400 Pa is required for BCD

$\therefore$ 200 Pa is required for AB and DE together

(Because BCD is common to (i) and (ii)).

Now if AB and DE were identical airways (with the same resistance), it could be deduced that 100 Pa is used in AB and 100 Pa is used up in DE (because AB and DE are in series). The example is simplified in Figure 2.35.

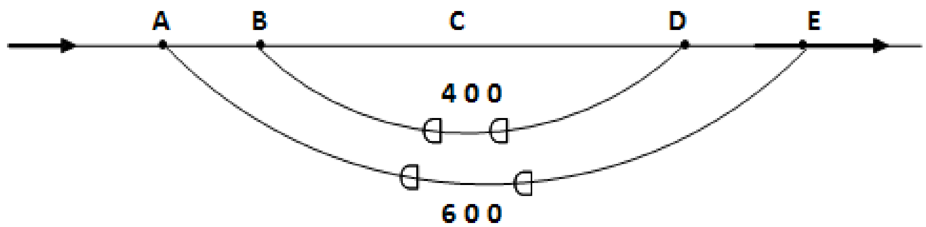


Figure 2.35 Simplified airway with ventilation doors

AE is in parallel with ABCDE, hence the pressure on the doors in AE equals the pressure drop in ABCDE, i.e. 600 Pa. BD is in parallel with BCD hence the pressure on the doors in BD equals the pressure drop in BCD, i.e 400 Pa. But BCD (400 Pa) is part of ABCDE (600 Pa), hence the pressure drop in AB and DE equals 200 Pa.

Finally, if AB and DE had identical resistances, the pressure loss in each airway would be 100 Pa (AB and DE are in series).

WORKED EXAMPLE 19

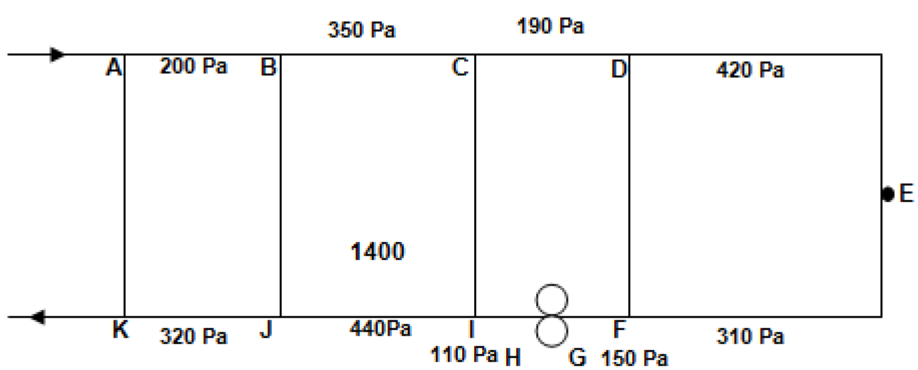


Figure 2.36 Proposed layout of section of a mine

Figure 2.36 represents the proposed layout of a section of a mine where ABCD will serve as an intake airway and FGHIJK as the return airway. A booster fan is to be installed between G and H and its pressure will be 1 400 Pa. Doors are to be installed in the connections between intake and return, but it is not yet known which way the doors should open.

Determine:

- Which way the doors should open
- The expected pressure across each airlock.

Answer

Assume any nominal pressure at A and work your way around the circuit determining the pressure at each point and use these pressures to determine which way the doors should open.

Now let us assume the pressure at A = 5 000 Pa

∴ Pressure at B	=	5 000 - 200	=	4 800 Pa
Pressure at C	=	4 800 - 350	=	4 450 Pa
Pressure at D	=	4 450 - 190	=	4 260 Pa
Pressure at E	=	4 260 - 420	=	3 840 Pa
Pressure at F	=	3 840 - 310	=	3 530 Pa
Pressure at G	=	3 530 - 150	=	3 380 Pa
Pressure at H	=	3 380 **+** 1 400	=	4 780 Pa
		(fan add additional pressure to the system)		
Pressure at I	=	4 780 - 110	=	4 670 Pa
Pressure at J	=	4 670 - 440	=	4 230 Pa
Pressure at K	=	4 230 - 320	=	3 910 Pa

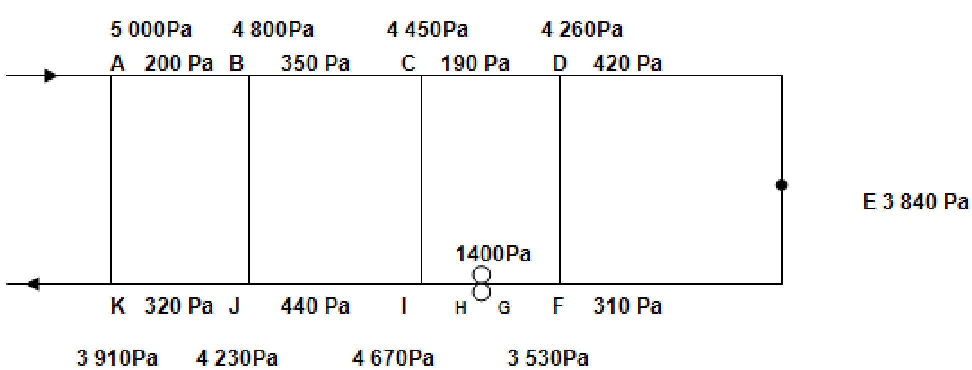


Figure 2.37 Resultant air pressures for airway

Now remember the Elementary Laws for air to flow:

- For air to flow there must be a pressure difference.
- Air always flows from a high pressure to a low pressure.

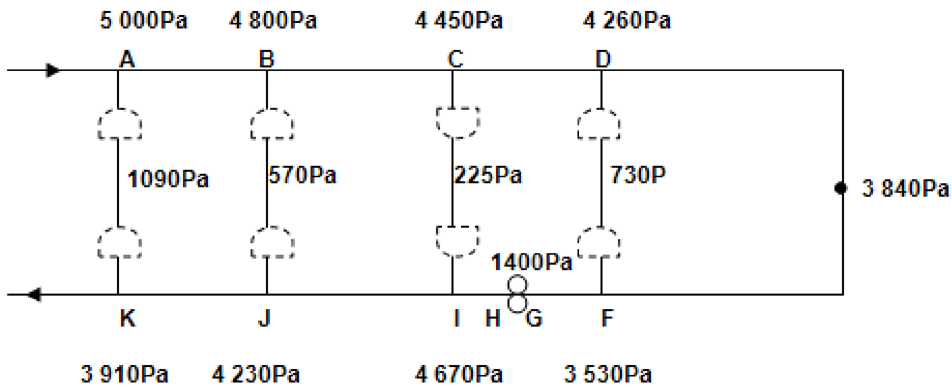


Figure 2.38 Ventilation doors- opening directions

The doors will open towards A, B, and D (high pressure points at A, B and D, doors will be pushed closed against the frames and pressure release flap will ensure that the pressure between doors will be equalised and opening of door made possible, without pressure release flap, opening of door will be very hard to do). For doors between C and I the pressure will be higher at I and therefore the doors will open to the other side (as shown in Figure 2.38).

WORKED EXAMPLE 20

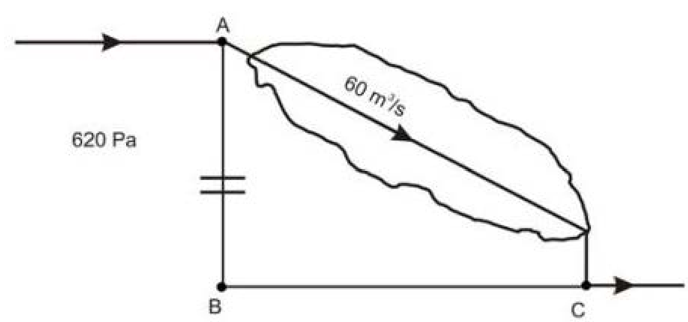


Figure 2.39 Typical mine ventilation layout

Figure 2.39 represents a section of a mine. 60 m^3/s of air flows through the stopes AC. The pressure across the stopping in AB is 620 Pa. AB is a 250 m long 2 m x 2 m raise with a K-factor of 0.013 Ns^2/m^4. BC is a 4 x 3.5 m crosscut, 200 m long, with a K-factor of 0.011 Ns^2/m^4.

The mining department intends to start mining in section AB and the stopping is to be knocked out, assuming total quantity increases to 80 m^3/s and at density of 1.2 kg/m^3, how much air will flow in each section?

Answer

Pressure across stopping in AB = 620 Pa

$\therefore$ Pressure drop in AC = 620 Pa

Quantity in AC = 60 m^3/s

$$\therefore \text{Resistance of AC } R_{AC} = \frac{P}{Q^2} = \frac{620}{60^2} = 0.1722 \text{ Ns}^2/\text{m}^8$$

$$\text{Resistance of AB, } R_{AB} = \frac{KCL}{A^3} = \frac{0.013 \times 8 \times 250}{4^3} = 0.4063 \text{ Ns}^2/\text{m}^8$$

$$\text{Resistance of BC, } R_{BC} = \frac{KCL}{A^3} = \frac{0.011 \times 15 \times 200}{14^3} = 0.012 \text{ Ns}^2/\text{m}^8$$

$\therefore$ Resistance of ABC, (Series) $R_{ABC} = R_{AB} + R_{BC}$

$$= 0.4063 + 0.012$$

$$= 0.4183 \text{ Ns}^2/\text{m}^8$$

But AC and ABC are in parallel.

$$\therefore \frac{1}{\sqrt{R_T}} = \frac{1}{\sqrt{R_{AC}}} + \frac{1}{\sqrt{R_{ABC}}}$$

$$\therefore \frac{1}{\sqrt{R_T}} = \frac{1}{\sqrt{0.1722}} + \frac{1}{\sqrt{0.4183}}$$

$$R_T = 0.0639 \text{ Ns}^2/\text{m}^8$$

And the new total quantity, Q_T = $80\ m^3/s$

$\therefore$ New total pressure drop, P_T = $R_T Q_T^2$

= $0.0639\ 0 \times 80 \times 80$

= 409 Pa (less pressure because of reduced resistance)

And because AC and ABC are in parallel:

$$P_T = P_{AC} = P_{ABC} = 409 \text{ Pa}$$

So:

$$Q_{AC} = \sqrt{\frac{P_{AC}}{R_{AC}}}$$

$$= \sqrt{\frac{409}{0.1722}}$$

$$= 48.7 m^3/s$$

And $$Q_{ABC} = \sqrt{\frac{P_{ABC}}{R_{ABC}}}$$

$$= \sqrt{\frac{409}{0.4183}}$$

$$= 31.3 m^3/s$$

Check:

$$Q_T = Q_1 + Q_2; \quad 80.0 = (48.7 + 31.3) = 80\ m^3/s$$

Hence, 48.7 m^3/s of air will flow through the old stopes in AC and 31.3 m^3/s of air will flow through the new stopes in AB.

NB

The resistance of an airway is not affected by the air density. It is, however, convenient to correct resistance for density as this negates having to remember to correct the required pressure for density at a later stage.

2.2 The Measurement of Airflow

The effectiveness and economic aspects of a ventilation system can only be determined, if the mass flow rate is measured accurately.

Air velocity, volume flow and density are the primary factors required in calculations of fan efficiency and heat transfer calculations. All these lessons are done with the aim of making certain decisions with regards to certain conditions.

Till now we have seen aspects such as volume flows, velocity and area of airways in calculations. Previously it was also indicated that the mass flow rate of air could be determined as follows:

$$\text{Mass flow rate (m)} = \text{Volume flow rate (Q)} \times \text{density } (\rho)$$
$$= m^3/s \times kg/m^3 = kg/s$$

It was also mentioned that the air quantity in an airway could be determined as the product of the area of the airway and the velocity of the air in that airway (in m^3/s).

2.2.1 Determining Air Velocity

Airflow volume cannot directly be determined and therefore it is necessary to measure certain physical aspects of the air or gas, or to use tracer gas techniques to determine through chemical analysis the various compositions of gases in the air. The main physical aspects of how air velocity is measured are as follows:

- The mechanical aspect of light gasses or blades, which come into contact with the air stream
- The dynamic pressure effect of air in motion
- The effect of velocity on the cooling of a warm body.

2.2.2 Mechanical Effect of Air in Motion

Instruments which work with the mechanical effect of moving air are:

- Aluminum dust
- Rotameter
- Velometer
- Anemometer
- Orifice plate.

2.2.2.1 Anemometer

Photo

The Anemometer is primarily used for the measurement of velocities greater than 0.5 m/s. There are multipurpose anemometers, which measure air velocity from 0.5 – 30 m/s, and then high-speed anemometers, which measure air velocities in the order of 5-50 m/s.

Certain aspects to be considered when measuring with an anemometer are as follows:

- **Density**: The anemometer is usually calibrated at 1 kg/m^3, where underground densities are typically 30% higher, and a correction factor will therefore have to be applied.

- **Effects of non-uniform distribution**: The instrument must not be used in areas which are smaller than 6x the diameter of the anemometer.
- **Effects of inertia**: It is recommended that readings be taken over a period of more than 50 seconds.
- **Pulsating flow**: Due to the sensitivity of the instrument to sudden decrease in velocity.

Refer to the section on how instruments must be used underground. The traversing and centre spot methods must be described in detail. Procedures with regards to the correct use of the instrument must also be known.

2.2.2.2 Velometer

References

Know that the instrument exists and the basic working of it (VOEE chapter 3).

2.2.2.3 Rotameter

See VOEE chapter 3 for a description of this instrument.

2.2.2.4 Orifice Plate

The orifice plate works on the principle of pressure difference between two points. The venturi works on a similar principle. Refer to Figure 2. 40.

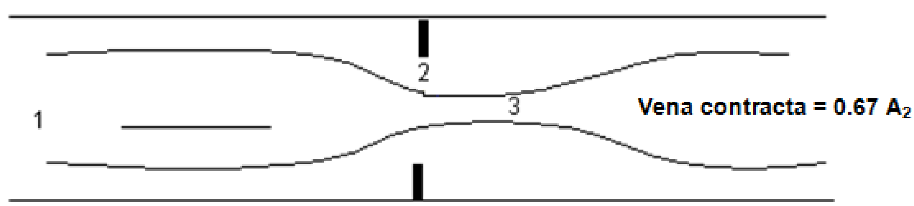


Figure 2.40 Orifice plate

The Equation to calculate airflow volume is:

$$Q = 1.11 \times d^2 \times C_d \times E \times \sqrt{\frac{\Delta P}{\rho}} \quad \text{in } m^3/s \text{ ...... (2.19)}$$

Where:

Q	=	Volume flow (m^3/s)		
d	=	diameter of orifice (m)		
C_d	=	Discharge coefficient	=	$C_c \times C_v$

Where:

C_c	=	A_3 / A_2 (A_2 = area of orifice)

But:

A_3 (vena contracta) = $0.67A_2$

C_v = velocity coefficient (normally the value = 0.97)

Thus:

C_d = 0.67 x 0.97 = 0.65 (a good average value to use)

ΔP = Static pressure difference over orifice / venturi (Pa)

And:

$$E = \sqrt{\frac{1}{(1-m^2)}}$$

Where:

E = Velocity of approach factor

m = A_2 / A_1 (A_1 = area of the pipe or airway)

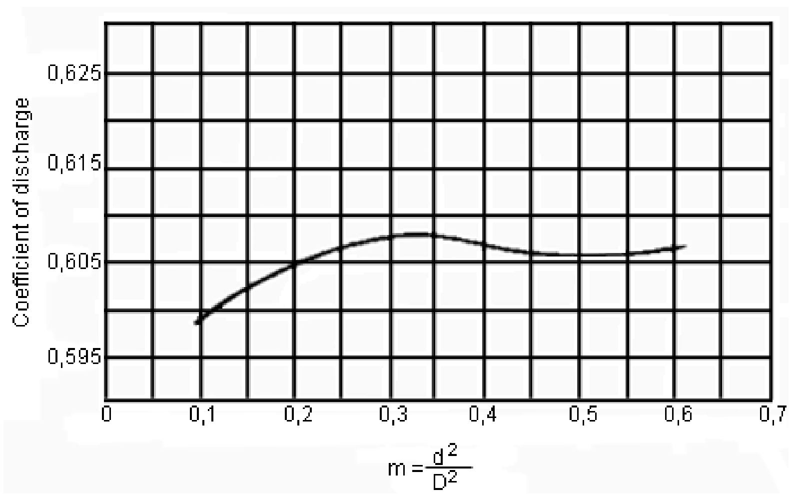


Figure 2.41 Relationship of Co-efficient of discharge and m

From Figure 2.41 it is obvious that the coefficient of discharge is relatively constant for m values between 0.3 and 0.6. The use of an orifice plate or venturi induces a pressure difference over the narrowing, which are the differential height and the area ratio value of m as shown in Figure 2.42. From this Figure it is clear that a venturi has a smaller pressure loss to that of an orifice plate.

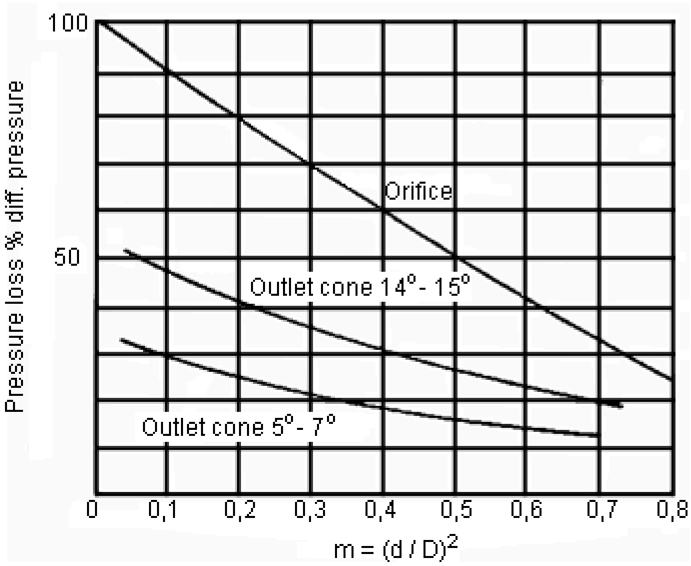


Figure 2.42 Pressure Loss versus m for venture and orifice plate

Although obvious advantages of the venturi, high costs and high manufacturing tolerances prevents it from being used extensively; Nowadays electronic flow meters are very much more commonly used.

2.2.3 Other Airflow Measurement Techniques

In the mining industry, there are a number of measurement techniques to measure airflow. We will now discuss some of these.

2.2.3.1 Tracer Gas Technique

The Tracer gas technique has the advantages over anemometer systems when dealing with complex systems. With this technique the volume flow is measured directly and area readings are thus unnecessary. Areas where turbulent flow and pulsating flow are found are now easily measured. Leakages, re-circulation and the time air spends in an area can easily be identified and analysed.

Quality of gas used:

- Must be cheap
- Non-poisonous
- Odourless
- Mix easily with mine air
- Non-reactive to the surrounding
- Easily traceable.

Method of use

The gas must be released as a fine spray to ensure that it mixes with air. The mass of gas released is determined by the loss in mass of the gas container. The gas concentrations are then strategically measured in places downstream for as long as the instruments measure gas concentrations.

From these concentrations, a graph can be drawn of gas concentration versus time. The graph in Figure 2.43 shows the concentration expressed in parts per million and the time in minutes.

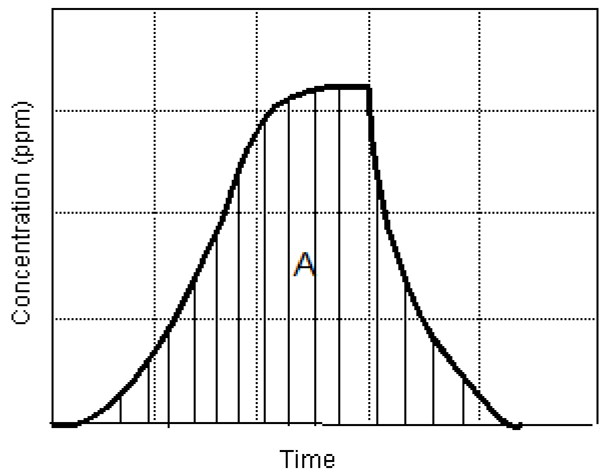


Figure 2.43 Trace gas plot

The volume of gas which flows through the test area is a function of the air volume flow and the time and is given in the following Equation:

$$V_g = \frac{m}{\rho_g} = Q\int Cdt = QA \quad \text{……………………. (2.20)}$$

Therefore:

$$Q = \frac{m}{\rho_g A} \quad \text{………………………………………………………. (2.21)}$$

Where:

V_g	=	Volume of the tracer gas
m	=	mass of tracer gas in kg
Q	=	airflow volume m^3/s
C	=	Trace gas concentration in volume, parts per million
A	=	Area under the graph (Cdt)

This method is mainly used to measure the air volume in shafts where, as a result of practical problems, convention methods are not possible.

2.2.3.2 Kata Thermometer

Photo

The Kata thermometer is used to determine the cooling power of air underground. The use will be demonstrated in the practical class. We also refer to the reject temperature for a specific airway and this relates to the final wet-bulb temperature that the Equation for determining the Kata reading is as follows (the issue of dry-bulb and wet-bulb temperature and the difference related to each will be dealt with in detail in Chapter 5). For the present, it is only important to know that the wet bulb temperature is used in the calculation. The wet- and dry-bulb temperatures are determined with the use of a whirling hygrometer. The Equation to determine the wet kata is the following:

$$K = \frac{F}{T} \quad \text{...} \quad (2.22)$$

Where:

K	=	Kata reading (mille calorie / cm^2 / s), (note not k the friction factor)
F	=	Factor read from wet kata thermometer
T	=	Time in seconds.

The Kata reading can also be given in terms of air velocity and wet-bulb temperature:

$$K = 0{,}7\theta + \theta\sqrt{U} \quad \text{...} \quad (2.23)$$

Where:

Excel

K	=	Kata reading
θ	=	36.5°C – unventilated wet-bulb temperature (Not a big difference if ventilated wet-bulb temperature is used)
U	=	velocity in m/s

NB

What is the difference between ventilated and unventilated wet-bulb temperature? From the above we can get the Equation for velocity U:

$$U = \left(\frac{K - 0.7\theta}{\theta}\right)^2 \quad \text{.............................} \quad (2.24)$$

Rule

The above method is not very accurate for velocities less than 0.1 m/s. The values that are obtained for the kata thermometer can be described as follows.

The higher the kata reading the better the conditions. A value of below 5 is deemed to be dangerous a value between 5 and 10 needs remedial action and a value more than 10 is seen as acceptable. The kata reading is therefore an indicator of the cooling power of the air which relates directly to the productivity of the workers. The higher the Kata reading the more productive the workers will be and the more safer the workplace in terms of exposure to heat related illnesses such as heat exhaustion and heat stroke which can be fatal.

2.2.3.3 Hot Wire Anemometer

Principle: If a heated wire is kept at a constant temperature, the current will vary with the speed of the air passing over the wire, or alternatively if the current is constant the temperature of the wire will change with the speed of the air. The hot wire anemometer is a very sensitive instrument and is designed to measure airflow rates at 20°C and a barometric pressure of 101.3 kPa. The Equation to give the actual velocity is:

$$U_{actual} = U_{observed} \times \frac{\rho_{std}}{\rho_{actual}} \quad \text{………..........…….....} \quad (2.25)$$

Where:

U_{actual}	=	Actual velocity measured in m/s
$U_{observed}$	=	observed velocity
ρ_s	=	standard air density, 1.2 kg/m^3
ρ_a	=	actual air density, kg/m^3

The anemometer works on the principle of the Wheatstone bridge, where one leg of the bridge is the filament. Increasing the heat current to keep the filament at a constant temperature is accomplished by altering the resistance. The current change is directly proportional to velocity. The hot wire anemometer can only be used in areas where there is no methane present.

SELF STUDY PROBLEM

It is expected of you to investigate the temperature and working conditions in a specific working place. During your investigation you received the following values from a kata thermometer and anemometer. The air velocity through the working place is 0.3 m/s and the kata reading is 5.2. Determine the wet-bulb temperature in the working place and make a comment with regards to the working place if the design reject wet-bulb temperature is 28.5°C.

Answer

The workplace is not safe to work in (32.3 ºC). Temperature higher than the design, reject temperature.

SELF STUDY

References

Read the article ‘Back to basics- why we ventilate mines, or what is adequate ventilation?’ by John Marks.

Chapter

3

VENTILATION PRACTICES AND PRINCIPLES

LEARNING OUTCOMES

- Secondary and Primary Ventilation in Gold Mines
- Production section ventilation in gold mines
- Ventilation design guidelines
- Effects of mechanisation of ventilation
- Ventilation of sinking shafts
- Multi-blast operations
- Differentiate different types of ventilation methods
- Calculation of re-entry intervals for development ends
- Distribution and control of underground air.

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Intermediate Workbook Paper 1, MVS of SA.

Mine Ventilation Practitioner's Data Book, Mine Ventilation of South Africa, p UV-DE 2 - UV-DE 7.

Mine Environmental Control (MEC) Certificate, Chamber of Mines, South Africa, Workbook 1.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

3.1 Introduction

Definition

Mine ventilation is the continuous supply of adequate and qualitative air to all parts of a mine underground, where people are required to travel or work. This continuous supply of air is required to:

SHE

- Supply oxygen for breathing purposes and must be above 19% by volume
- Remove heat and provide comfortable working conditions and hence improve production
- Dilute and remove noxious and flammable gases that may be encountered during mining operations
- Dilute and remove hazardous airborne pollutants created by various mining operations underground. (i.e: dust, fumes, aerosols, vapours etc.)

Objective

All these reasons above are to create and maintain an underground working environment that is conducive to the productivity, health and safety of people.

3.2 Principles of Mine Ventilation

Animation

The fresh air ventilating a mine enters at the downcast shaft. It drawn through the working places where it becomes contaminated and is removed from the mine via the up-cast shafts. As was mentioned before, a typical deep-level mine has:

- One or more downcast shafts where the fresh air from surface enters the mine
- Intake airways through which the air flows to the workings
- Various connections between the workings
- Return airways through which the air passes from the workings to the up-cast shaft (or shafts) and out to surface.

Fans are used to exhaust air through the mine since natural ventilation is normally inadequate and unreliable.

Image

To distribute and control the air to the different levels and workings, ventilation appliances such as ventilation doors, walls, regulators, booster fans and auxiliary fans are used. These were discussed in detail in Chapter 2.

Animation

Ascentional and descentional distribution of air will now be discussed in detail.

3.2.1 Ascentional Ventilation

Definition

Ascentional ventilation is the most common method of ventilation. Fresh air is taken through (down) the downcast shaft, directly to the bottom levels of the mine and then allowed to ascent (up-cast) through the workings.

Image

Click on the image icon to take you to an underground mine plan

The return air from the workings is then transported through return airways on the top levels to the up-cast and out to surface via the main up-cast fans. This is diagrammatically illustrated in Figure 3.1.

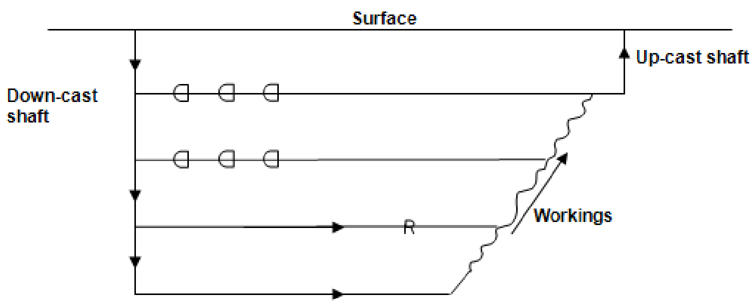

Figure 3.1 Example of ascential ventilation flow through mine

3.2.2 Descentional Ventilation

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This method is not recommended, especially where the presence of flammable gas is known. It is a known fact that flammable gas roof layering can occur against the ventilation flow (counter flow), as a result of the gas specific gravity (lighter than air) characteristics. Thus, before down-casting air through workings, this phenomenon should be borne in mind.

Descentional ventilation is the opposite of ascentional ventilation, fresh air is taken through the down-cast shaft, directly to the top levels of the mine, and then allowed to descend (downcast) through the workings. The return air from the workings is then transported through the return airways on the bottom levels to the up-cast shaft and out to surface via the main up-cast shaft fans. This is diagrammatically illustrated in Figure 3.2.

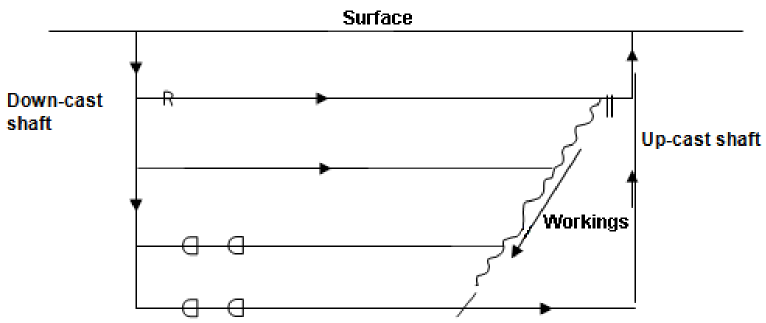

Figure 3.2 Example of descential ventilation flow through mine

3.3 Distribution and Control of Air

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A mine is usually divided into sections or ventilation districts and the total volume of air down-casting must be distributed and controlled between these various ventilation districts or sections. As air will always take the shortest route or path of least resistance, different ventilation appliances are used to ensure a sufficient quantity of air reaches each ventilation district. Effective installation and maintenance of such ventilation appliances are also important to prevent air wastage and hence maintain adequate air volumes to all districts.

Adequate air volumes may enter a stoping - or development end section, but yet not all this air will reach the working faces where it is really required. In-stope and development end ventilation appliances must be used to meet this objective. The ventilation appliances that must be used are discussed in the notes that follow.

NB

Briefly, to control and maintain satisfactory ventilation and thermal conditions underground, the following are essential:

- Adequate shaft systems (Up-cast and down-cast shafts)
- Large intake and return airways to ensure air velocities do not exceed 3.0 m/s and 5.0 m/s respectively
- Adequate fan power
- Proper splitting or distribution of available air to all ventilation districts
- Effective control of available air (reduce leakage of air and air wastage)
- Refrigeration (applicable to deep level mining).

3.3.1 Stope and Development Ventilation Appliances

Image

These appliances (ventilation brattices, regulators, stonewalls etc were discussed in detail in Chapter 2) are utilized to distribute and control the air from the intake of the working to the working faces. Fans, mainly used for development ends (sometimes also in stopes to improve airflow conditions) will be discussed in detail later in this document.

3.4 Ventilation of Working Places

3.4.1 Stope Ventilation

Animation

Stopes are the most important working places in a gold mine as this is where the reef is mined. More workers are employed in a stope than in any other type of working and it is essential that adequate quantities of air are provided and controlled to maintain safe and healthy conditions.

SHE

The major problem in stope ventilation is to ensure that as much available air as possible is directed and kept in the areas where men work and as little air as possible should be allowed to leak into worked-out areas.

Through research it has been shown that productivity performance is directly affected by the environment, for example in an environment where the wet-bulb temperature is 32°C and the air velocity is 0.5 m/s, the productivity performance of workers would be 83%. Whereas, if the wet-bulb temperature were to be kept constant, the air velocity increased to 4 m/s, there would be an increase in performance to 95%. This is a substantial saving in labour requirements to meet the necessary production requirements. Efforts should be made to prevent air flowing where it is not required.

Animation

The air must be controlled by brattices and effective strike and dip walls or curtains in such a way that the workers derive the maximum benefit from it.

In some cases, fans or venturi blowers are also used to increase ventilation flow to maintain a velocity above the set limit.

To direct air onto the faces:

- Strike walls or curtains should be kept as close as possible to the face (without restricting the airflow) with a maximum distance of 9.0 m from the face
- Ventilation brattices or doors should be installed at selected positions in dip-gullies and travelling ways
- Accumulation of rock and rubble restricting the airflow should be prevented at the stope intake and return, as well as all faces
- The stope ore-passes should never be completely emptied. Leaving some broken rock in the box forms a ventilation seal
- Dip gullies and faces that are no longer required should be sealed off.

Figure 3.3 shows a typical layout for a semi longwall layout.

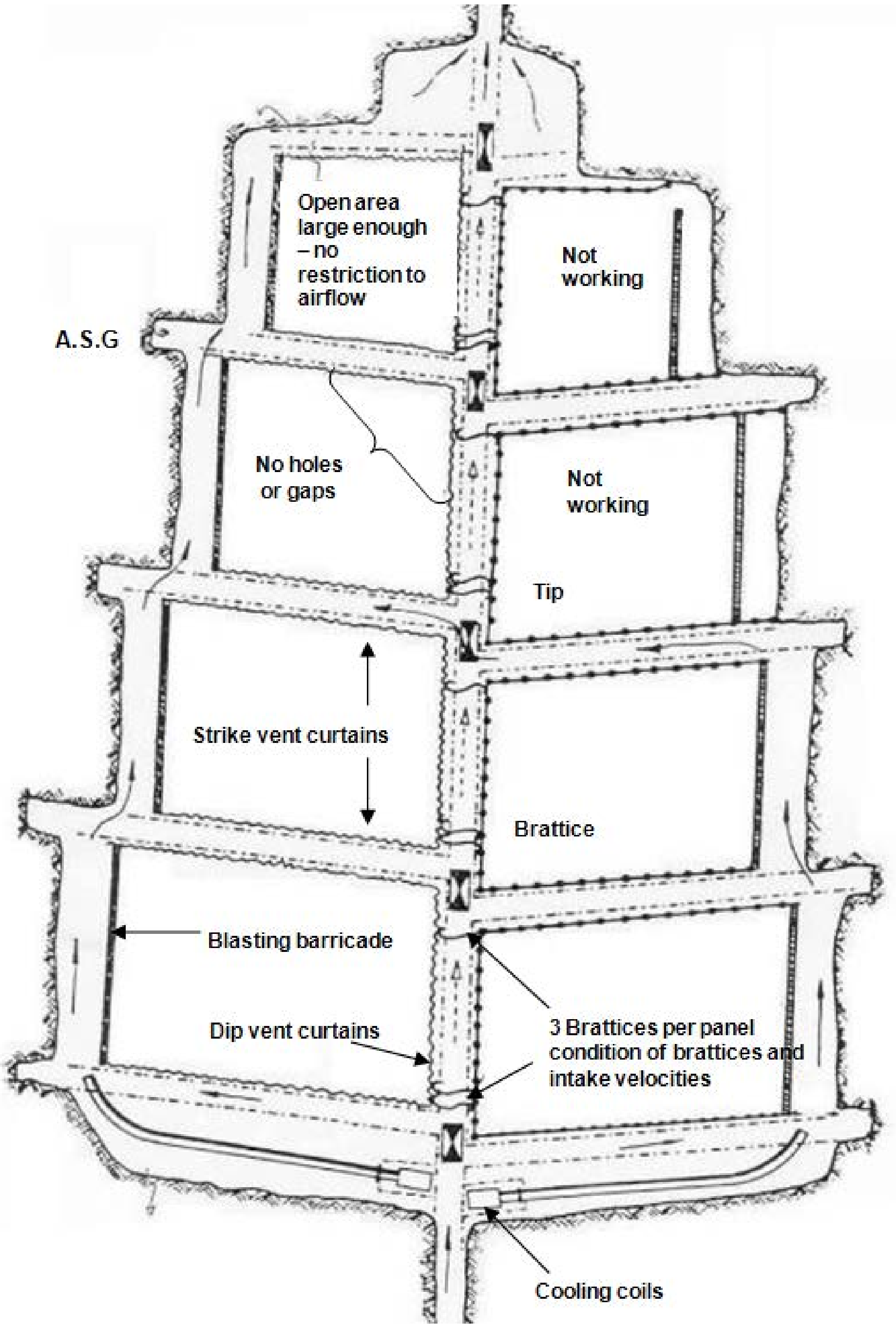

Figure 3.3 A typical ventilation layout for a conventional semi-longwall stope.

3.4.2 Development End Ventilation

Definition

A **development end** is a tunnel shaped excavation driven into virgin ground, with no natural through ventilation and without a second outlet or escape way. They can be horizontal (e.g. crosscuts, haulages, drives etc.), inclined (e.g. raises, boxholes, shafts, etc.), declined (e.g. winzes, shafts) and / or vertical (for example shafts).

NB

To open up the ore body in order to mine or retrieve the gold bearing ore through stoping, a large amount of tunneling has to be done. These tunnels are called '**development ends**' and have no through ventilation. They have to be ventilated by mechanical means, i.e. with the aid of auxiliary fans and ventilation ducts or pipes. The air can be forced to, or exhausted from the face. Air should be used economically and leakage kept to a minimum. Heat and airborne pollutant control should be carefully exercised.

Animation

3.4.2.1 Forcing System

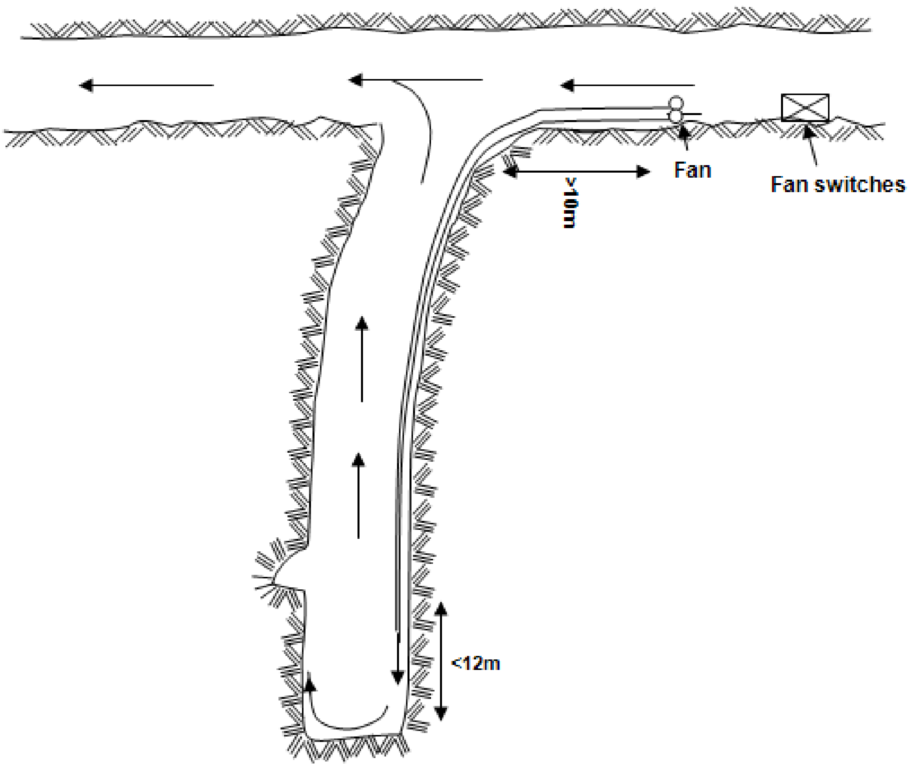

Figure 3.4 Development layout X-cut and haulage force ventilated

Air is acquired from a point at least 10 metres upstream in the nearest through ventilation and with the aid of a fan forced through the ventilation duct and discharged to within 12 metres from the face.

Advantages

The advantages of this system are as follows:

- Air flows to the face inside the pipe at a high velocity, thus good quality air is delivered at the face
- The fresh air is discharged right at the face, where the workers derive the maximum benefit from it
- A single fan and a single column are required
- The fan and motor are always in fresh air, causing less wear and tear on the fan
- Leakage in the column is outward and hence easily detectable.

Disadvantages

The disadvantages of this system are as follows:

- Persons travelling or working in the return do so in contaminated return air from the face
- A long re-entry period is required; hence it is unsuitable for multi-blast development ends
- The return air usually flows back into the general air-stream and causes contamination.

Animation

3.4.2.2 Exhaust System

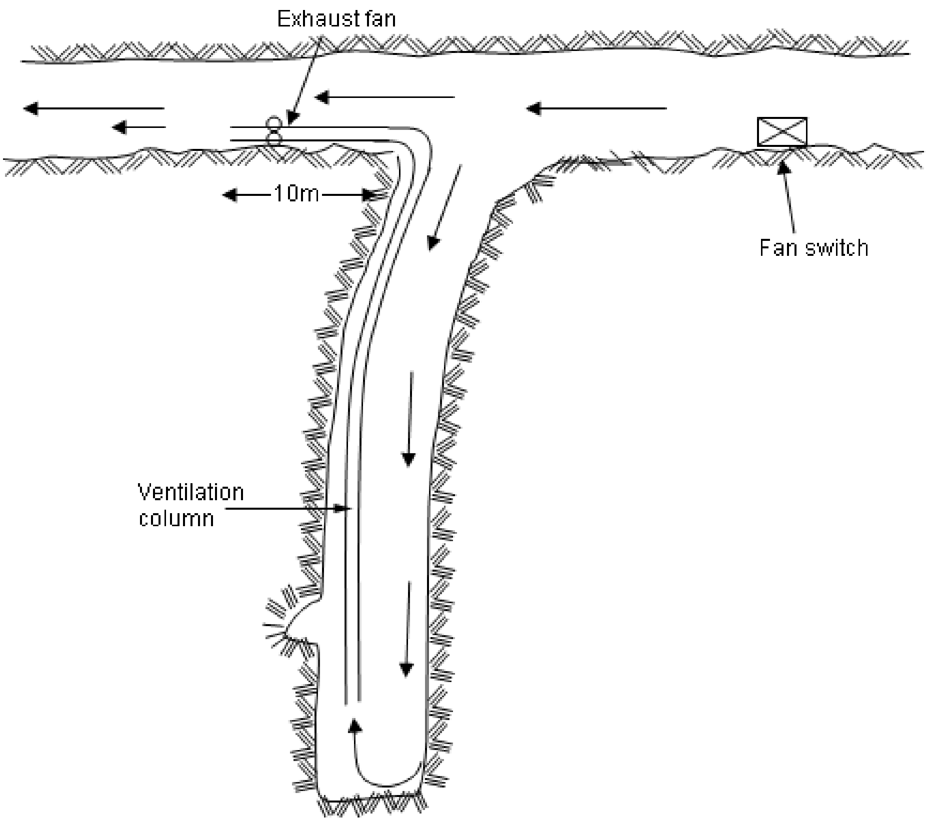

Figure 3.5 Development layout X-cut and haulage exhaust ventilated

With this system a fan is installed in through ventilation, 10 metres downstream from the development end break away. A ventilation column is attached to this fan and extended into the development end, up to as close as possible to the face to exhaust the air from the face. As this system does not effectively ventilate the face, it is not commonly used.

The advantages of this system are as follows:

Advantages

- Persons working in or travelling in the end away from the face derive maximum benefit from the fresh air
- Return air can be controlled
- Single fan and column is required.

The disadvantages are as follows:

Disadvantages

- Face is not effectively ventilated; therefore a gas build up at the face can easily occur
- Fan is situated in return air, increasing chances of methane ignition and result in more wear and tear on the fan
- Quality and quantity of the air supplied to the face is poor - worker on the face derives the minimum benefit
- Leakage on the column is inward and not easily detected.

Animation

3.4.2.3 Exhaust Overlap System

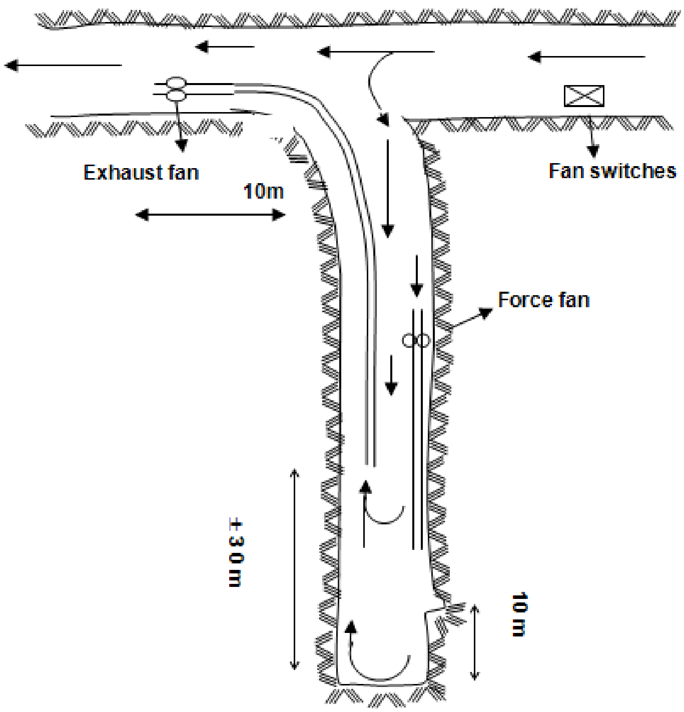

Figure 3.6 Development layout X-cut and haulage Force Exhaust Overlap

An exhaust fan, at least 10 m in through ventilation is installed and connected to the ventilation column, which is extended into the development end up to ±30 m from the face. A second, smaller size force fan is installed, 10 m upstream from the exhaust column intake. A ventilation column is fitted to this fan and extended to within 12.0 m from the face, to force air onto the face.

NB

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The force fan must not handle more than two-thirds of the exhaust fan intake volume, to ensure an adequate volume of air, and hence air velocity is maintained in the overlap section. This is necessary to prevent the possibility of any noxious and/or flammable gas accumulation in the overlap. It is also important that the force fan must be electrically interconnected with the exhaust fan. This is to ensure that should the exhaust fan fail or stop, the force fan will then also stop, to prevent any re-circulation of the force fan.

The advantages of this system are as follows:

Advantages

- Persons travelling and working 'in' the development end do so in fresh air, as the return air is inside the exhaust pipe
- Short re-entry periods are possible when used in multi-blast or high speed development ends
- Return air is under control.

The disadvantages of this system are as follows:

Disadvantages

- Intake air moves slowly along the drive and picks up heat dust and loco fumes on the way. Hence the air supplied to the face is inferior in quality compared to that supplied by the force system
- Two columns and two fans are required
- Poor conditions can exist in the overlap section
- There is a danger of gas accumulation here
- Hence overlap distance in excess of 10 metres and air velocities above 0.3 m/s
- If the exhaust column is not kept under negative pressure (i.e. when fans are installed in series at intervals in the column) contaminated air from the face will leak into the air flowing to the face. This is particularly undesirable in multi-blast development ends when smoke leaking out of the column will cause problems
- Leakage in the exhaust column is inward, hence not easily detectable.

3.4.2.4 Layout for Fans in Twin End Layout

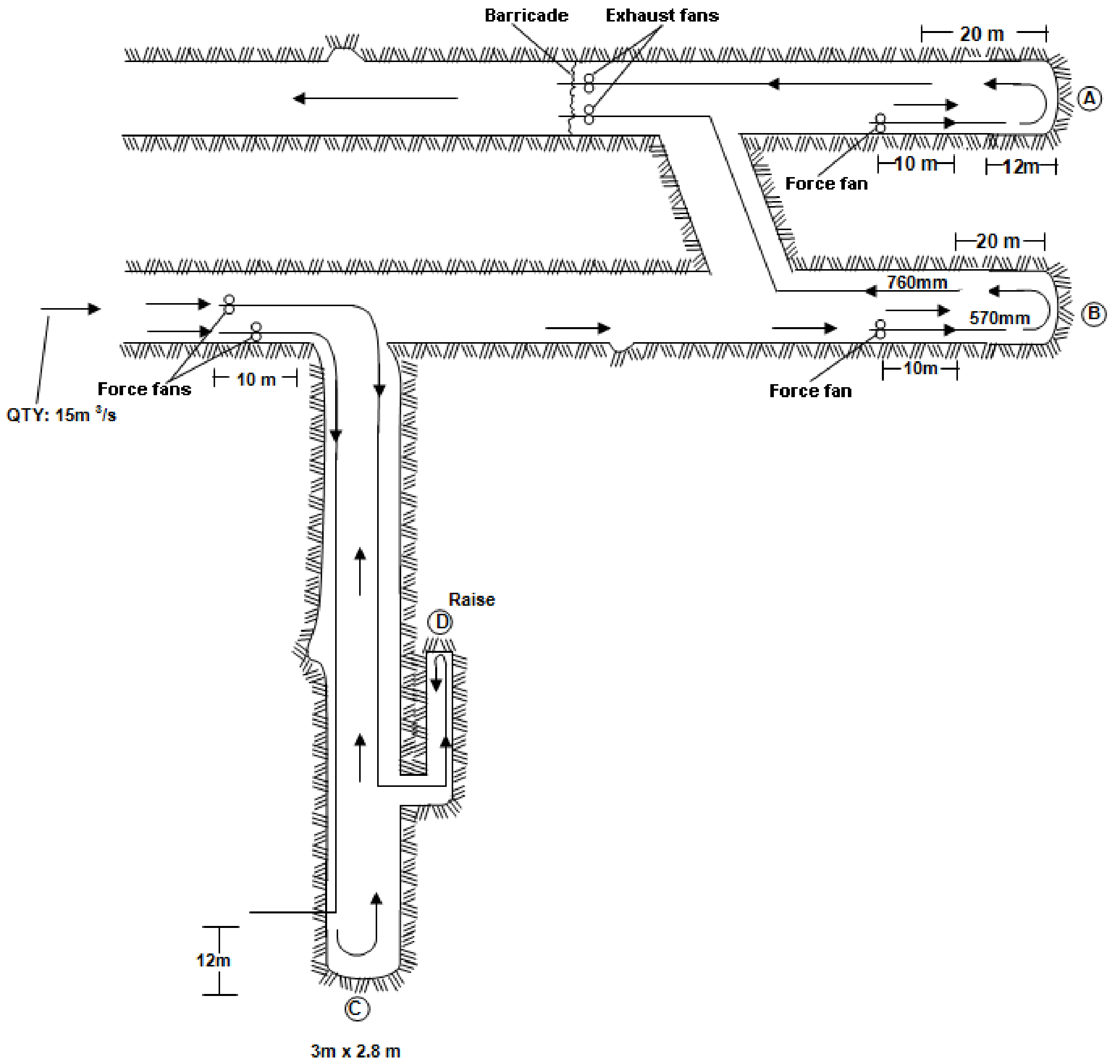

Figure 3.7 Twin end fan ventilation layout

(Note: 570 and 760 mm refers to pipe diameters)

Exhaust Overlap System

Rules

- Force fans and exhaust fans must be electrically interconnected
- Minimum velocity in overlap 0.5 m/s.

3.5 Ventilation Design Guidelines

3.5.1 Stopes

Wet-bulb temperature	27.5°C
Air velocity	1.0 m/s
Wet kata	Above 12.0
Specific Cooling Power (SCP)	300 W/m^2
Dust	Below 0.1mg/m^3
Work Performance	100%

3.5.2 Development

Wet-bulb temperature	27.5°C
Air quantity delivered	Above 0.3 m^3/s/m^2
Wet kata	Above 12.0
SCP	Above 300 W/m^2
Dust	Below 0.1 mg/m^3
Work Performance	100%
Force column delivery from face	less than 12 m

3.5.3 Guidelines When All Work Must Stop If:

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- The wet-bulb temperature exceeds 32.5°C and dry-bulb exceeds 37.0°C
- The wet kata is below 6.0 and SCP below 125 W/m^2
- If flammable gas tests indicate the presence of such gas in the general atmosphere, no matter how low the concentration
- If any scraper rope friction against timber is present or any other fire hazard.

NB

Normal work may only re-commence when ventilation conditions have improved to such an extent that they fall outside the cut-off limits, confirmed by a re-survey.

3.6 The Effects of Mechanized Mining on Ventilation

This method of mining in coal mines and hard rock mines has increased over the past few years, as it permits greater extraction of the coal and hence requires less labour.

SHE

However, an increase in the degree of mechanized mining creates additional ventilation, health, safety and other related problems that result in the following effects:

- Increases the face advance rates
- Increases the amount of dust produced into the air
- Increases the heat content of the air
- Increases the possibility that more flammable gas can be released into the air
- Increases the amount of gases, fumes and other contaminants in the air, where diesel driven equipment is used
- Increases the risk of flammable gas ignition and coal dust explosions where diesel and / or electrical driven machines are used
- Increases the risk of underground fires
- Increases the risk of exceeding the recommended equivalent noise exposure limit
- Makes the installation, extension and maintenance of ventilation equipment difficult.

Some of the above effects necessitate the implementation and control of the following:

- Equip machines with dust collection and suppression equipment to control dust
- Where diesel machines are used, provisions to dilute exhaust gases and other contaminants must be introduced
- Air requirements must be increased to cater for the increased heat loads (i.e. the amount of air required per kilowatt of diesel power - m^3/s per kW)
- Additional auxiliary ventilation is often required in order to concentrate the air, resulting in exceptionally high air velocities, causing uncomfortable working conditions (dusty conditions can cause health and safety hazards, e.g. foreign bodies in the eyes, etc.)
- Where the rate of emission of flammable gas is high, methane drainage needs to be implemented, as the amount of air required to provide safe conditions would be excessive.

- Implement an ongoing program of risk assessment and action plans for reducing potential fire hazards and flammable gas / coal dust explosions
- Improve the hearing conservation program (e.g. attenuation - properly designed silencers on vehicles, noise zoning - identify areas with high risks exposure to noise, make use of ear protection compulsory, etc.)
- Comply with legal requirements concerning diesel and electrical equipment.

3.7 Ventilation of Sinking Shafts

Generally speaking, the same principles and practices, advantages and disadvantages apply to the ventilation of sinking shafts as for the ventilation of development ends.

The issue is complicated slightly by certain other factors, which necessitate the circulation of larger air velocities.

These factors include:

- **Re-entry Interval:** Short re-entry periods are nearly always required in sinking shafts as practically all shaft sinking is done on multi-blast systems. The sinking crew is keen to return to the stage as soon as possible after the blast to recommence work.
- **Labour force:** Many men are employed at the face and on the sinking stage.
- **Face area:** The face area of a sinking shaft is large - as much as 80 m^2 - and therefore requires a large air volume in order to comply with the regulations.
- **Blowing-over:** The face of the shaft is blown-over with compressed air to check for misfires. Large amounts of dust are created by this process and must be diluted with air.
- **Rock Drills:** Sometimes as many as 30 to 40 rock drills may be operating simultaneously.
- **Rock Temperatures:** These are high as shafts are usually the first tunnels driven into an area, and the surrounding rock has never been cooled before.

3.7.1 Methods of Ventilating Sinking Shafts

3.7.1.1 Forcing System

A plug of smoke and dust will travel up the shaft after the blast, and if the re-entry period is short, the workers travelling down the shaft will pass through this plug. Although blasting dust concentrations are high and fumes are believed to make this dust even more dangerous, the time spent in passing through this plug is very short.

3.7.1.2 Exhaust System

The ventilation columns should, for obvious reasons, extend to within 2 to 6 metres from the shaft bottom. This increases the danger of the ventilation pipes being damaged during the blast.

3.7.1.3 Exhaust-Overlap System

This system is difficult to use in vertical shafts as the forcing column is difficult to install and maintain.

3.7.1.4 Force-Exhaust System

NB

Two separate ventilation pipes are used; one for forcing the air down the shaft and one for returning the air directly to surface from shaft bottom.

Care should be taken to let the quantity of exhaust air be greater than the quantity of force air at the shaft bottom, to allow fresh air to travel down the shaft. This benefits men re-entering after the blast.

This method is expensive as it requires two full-length ventilation columns.

3.7.1.5 Reversing the Airflow Direction

NB

This is one of the best ways of ventilating a shaft. One pipe column is used as a forcing system while men are working at the face, but when they leave the bottom before blasting, which is done electrically from the shaft collar, the direction of air flow in the ventilation column is reversed until the next shift enters. Dust and fumes are removed through the pipe, the face is cleaned faster and the workers will not pass through a plug of smoke and dust as they enter the shaft.

Reversing the airflow in the ventilation column is achieved by reversing the direction of rotation of the fan, if it is of the axial flow type, but this result in decreased fan efficiency and takes time to operate.

Animation

3.7.1.6 The Four-Gate System

This is a system for reversing the air flow in a single pipe column. (Figure 3.8).

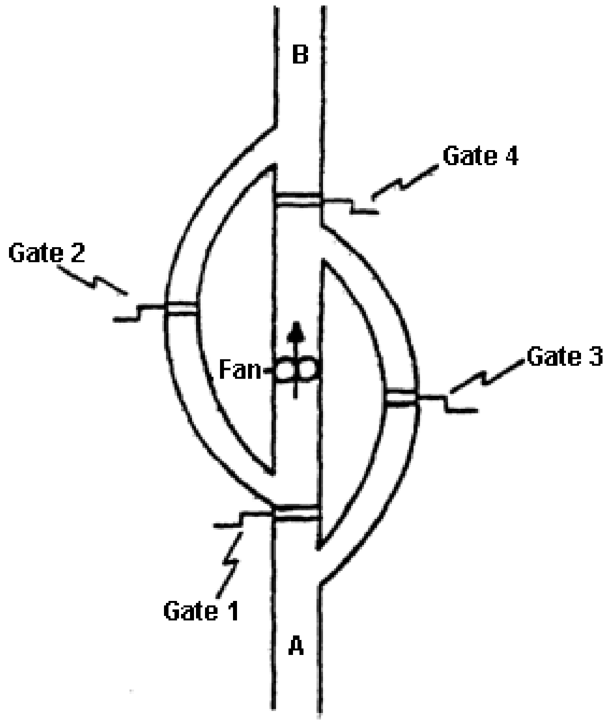

Figure 3.8 The four gate shaft sinking ventilation system

If gates 1 and 4 are open and gates 2 and 3 are shut, air flows from A to B, and with this arrangement the resistance is at a minimum, i.e. the fan is used for a condition in which the maximum airflow is required.

To reverse the direction of the airflow, open gates 2 and 3 and close gates 1 and 4. Air now flows from B to A, although the fan continues to rotate in the same direction.

Because of the two bends, the resistance of the system is increased and slightly smaller air quantity will flow. The gates can consist of rigid metal plates sliding in a slot cut into the pipe and suitably packed to prevent a loss of air through the slots.

The changeover can be made manually or hydraulically in a few minutes. Another way of explaining the four-gate system is shown in Figure 3.9.

When gates 2 and 4 are open and gates 1 and 3 are shut or closed, the system forces air down the shaft. To reverse the airflow direction, gates 1 and 3 are opened and gates 2 and 4 are closed and it changes to an exhaust system.

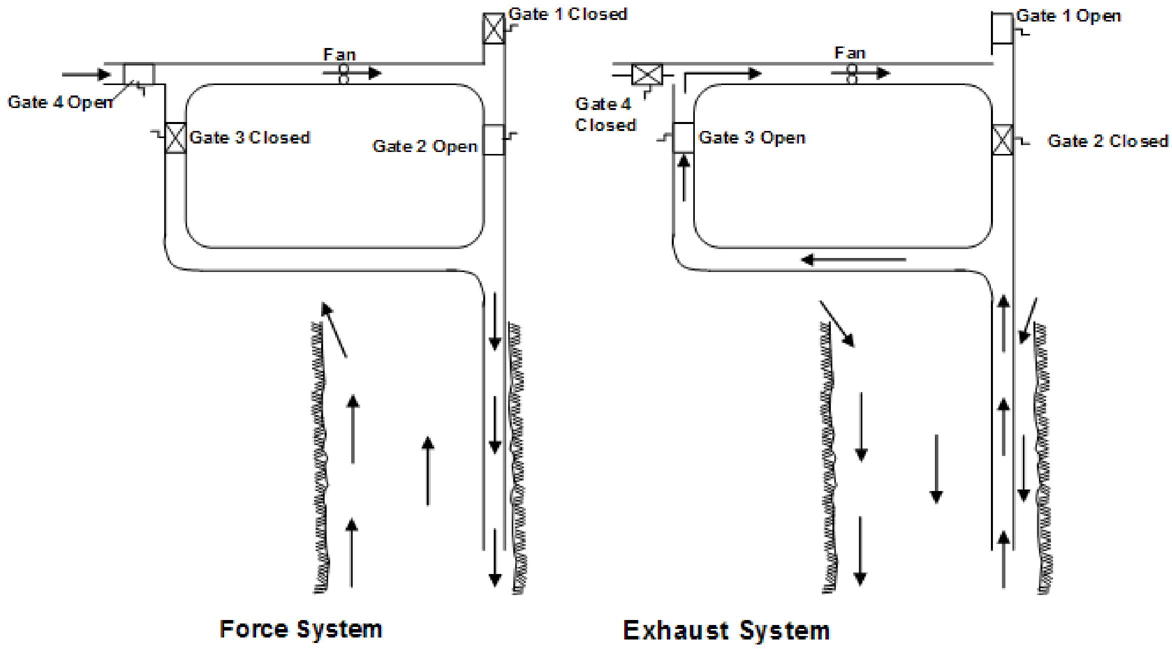

Figure 3.9 Explanation of four gate ventilation system for sinking shafts

Photo

A typical example of a four gate system used at the Impala 11C shaft.

3.7.1.7 Downcast and Upcast Compartments

Definition

These are formed by dividing the shaft by an internal vertical wall, using one side as the downcast side and the other compartment as the upcast side. If the shaft is of the rectangular vertical type and the brattice wall is installed during sinking, it can be used to provide the partition. With this method, a considerable amount of air will flow due to the **natural ventilation pressure** (NVP) that is hot air that rises and assists the fan in getting the contaminated air out of the mine.

With a sub-vertical or sub-incline shaft, the return air must not be released into the intake air going down the mine, for obvious reasons.

The return air should either be taken to the upcast system of the mine, or piped directly to surface, or filtered to remove dust and gases and cooled before being put into the intake air for further use. Removing dust and fumes is done by a fume filter and dust filters. Cooling can be done by a spot-cooler or heat exchanger coils connected to a plant already installed and in use underground.

Self study

Read the article by L Mackay: “Sinking Shafts- Ventilation and Hygiene Challenges”.

3.8 Multi-blasting Operations: Re-entry Interval after Blasting and Permission to Blast More than Once in 24 Hours

3.8.1 Relevant Provisions of the MHSA

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The employer must assess the hazards and respond to the risks to health and safety, in terms of specifications as highlighted in the MHSA, to which employees may be exposed while they are at work. According to this act, the employer must establish and maintain a system of occupational hygiene measurements and engage the part-time or full-time services of a person qualified in occupational hygiene techniques to measure levels of exposure to hazards at the mine. It also states that Every system of occupational hygiene measurements must be appropriate in terms of the hazards to which employees are, or may be, exposed and must provide information, which the employer can use to eliminate, control and minimise such health risks and hazards.

Records must also be kept of all such measurements in order that it can be linked, as far as practicable, to an employee’s records of medical surveillance.

3.8.2 Definitions

In order to differentiate between mining operations using multi-blasting and time blasting, these definitions will apply.

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It should be noted that in all three definitions cognisance must be taken of the occupational hygiene regulations, which requires that no persons must be exposed to airborne contaminants.

3.8.2.1 Time-Blasting

Definitions

Blasting operations taking place not more than once in any 24 hour cycle.

3.8.2.2 Multi-Blasting

Multiple blasting including shaft sinking operations, which could take place during any working shift. Such blasting may only take place where efficacious means of separating intake and return air e.g. a dedicated return airway is provided.

3.8.2.3 Fixed-Time Multi-Blasting

Blasting more than once per 24 hours but, not more than once per shift for both, stoping and development, taking cognisance of a re-entry period sufficient to clear all airways where persons are expected to work or travel.

The re-entry period must be determined by a risk assessment and validated whenever key factors (such as air quantity, contaminants etc.) that can have a significant effect on the re-entry conditions, change.

3.8.3 Re-Entry Intervals

SHE

In terms of a Regulation specified in the MHSA the employer must ensure that the occupational exposure to health hazards of employees is maintained below the limits set out in relevant schedules as specified in the act.

The intervals which must expire before persons are allowed to re-enter the working of your mine in which blasting has taken place, should be fixed as follows:

3.8.3.1 Nil Re-Entry Interval

A re-entry interval need not be observed where persons are expected to work or travel if uncontaminated through ventilation has been established and is effective / operational.

3.8.3.2 General Re-Entry Interval

Should blasting fumes however contaminate the air in any of the workings the general re-entry interval, as set out in the paragraph below, must be observed in those workings.

A general re-entry interval after the blast in all ventilation districts must be observed as per MHSA (Mine Health and Safety Act) and *“The employer must ensure that the occupational exposure to health hazards of employees is maintained below the limits set out in the schedules mentioned before*”. This re-entry interval must be determined after a detailed and recorded risk assessment.

3.8.3.3 Multi-Blast Re-Entry Interval

In terms of regulations mentioned in the MHSA the employer must ensure that the occupational exposure to health hazards of employees is maintained below the limits set out in the schedules mentioned before.

A minimum 30-minute re-entry interval must be observed and the following provisions must be made applicable to all multi-blast development ends or shafts being sunk.

Minimum Air Quantities required (relative to the air density at the working face):

- The quantity of air forced shall be established through a risk assessment process to ensure that the air supplied is of a quality as set out in the schedules mentioned before. This however, must not be less than 0.25 m^3/s for every square meter of face area, for all multi-blast development ends.
- The quantity of air exhausted from the development end should be not less than twice more than the quantity of air supplied by the force column. A minimum force exhaust ratio of 1:2 should be maintained at all times to ensure that no uncontrolled re-circulation takes place in the overlap section.

Rule

- The number of air changes required, calculated on the volume of air between the face and the intake of the force column, shall be determined through a risk assessment process to ensure that on re-entry after the blast, the air in the development end is of a quality as set out in the Schedules of the act related to it, and should **not be less than 8.**

3.8.3.4 Fixed-Time Multi-Blast Re-Entry Interval

For blasting more than once in 24 hours, but not more than once per shift, a minimum re-entry interval will be determined after a detailed and recorded risk assessment with the following provisions made applicable to all fixed-time multi-blast development ends or stopes.

Minimum Air Quantities required (relative to the air density at the working face):

Rule

- The quantity of air forced shall be established through a risk assessment process to ensure that the air supplied is of a quality as set out in the Schedules related to it in the Act and should not be less than 0.15 m^3/s for every square metre of face area, for all multi-blast development ends.

Rule

- The minimum stope face velocity averaged across the height of the stope should be determined through a risk assessment process to ensure that the quality of air is such as meets the requirements set out in the Schedules related to it in the Act and should not be less than 0.25 m/s. This should vary when determining the desired re-entry interval.

Rule

- The number of air changes required, calculated on the volume of air between the face and the intake of the force column, shall be determined through a risk assessment process to ensure that on re-entry after the blast the air in the development end is of a quality as meets the requirements set out in the Schedules related to it in the Act and should **not be less than 8.** The risk assessment must take into account all areas that may be contaminated.

3.9 Calculation of the Re-entry Interval for a Development End

$$\frac{\text{Volume of end (L x W x H) x number of air changes}}{\text{Force fan air quantity } (Q_F) \text{ x } 60 \text{ (minutes)}}$$

$\therefore$ Re-entry period (minutes):

$$\frac{\text{Volumetric capacity x number of air changes (8)}}{Q_F \text{ x } 60}$$

$\therefore$ For an end 100 m long, 4 m high x 4 m wide, with a force quantity of 10 m^3/s and 8 air changes:

$$= \frac{100 \text{ x } 4 \text{ x } 4 \text{ x } 8}{10 \text{ x } 60} \text{ minutes}$$

= 21 minutes

For multiple ends, ventilated **sequentially**, the re-entry period will be the sum of the individual end re-entries, **plus** 8 x the volumetric capacity of the haulage connecting the ends to the RAW.

NB

If at any stage blasting fumes from any end being multi-blasted contaminates any working places in the vicinity, then multi-blasting must cease and conventional time blasting, or fixed-time blasting, must be done until conditions have been rectified for multi-blasting.

3.9.1 Calculation of the Re-entry Interval for Sequentially Ventilated Development Ends

WORKED EXAMPLE 1

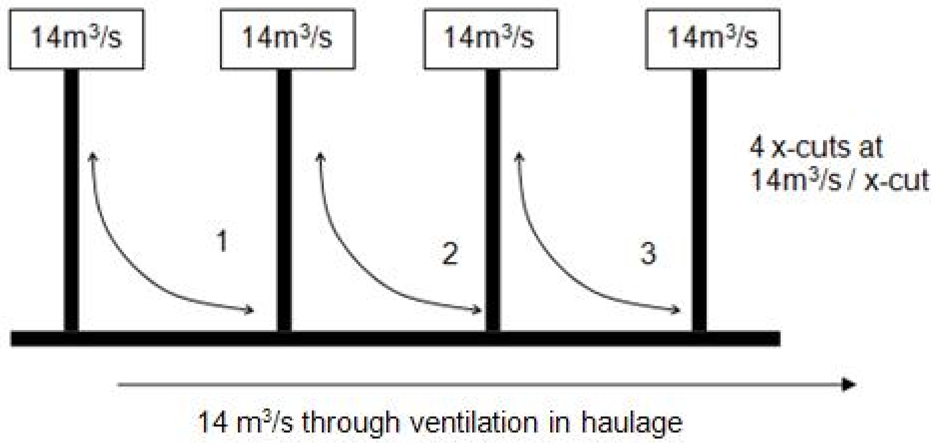

Figure 3.10 Haulage and 4 X-cuts ventilation layout

From Figure 3.10:

- Through ventilation passes each of the x-cuts and gets drawn into the development end by means of a force fan. The contaminated air from the first x-cut then flows to the second x-cut and so forth until it has gone through all the x-cuts. The same amount of air (assume no leakages) that enters the haulage and subsequent 4 x-cuts will be returned contaminated to the **return airway** (RAW).
- The numbers 1, 2 and 3 represents the combined airways (haulage and end) that need to be cleared after blasting.
- The above therefore represents a series air flow network.

Assumptions:

1. All ends are at maximum length (180 m)
2. Spacing of ends: 120 m apart
3. No ventilation column leakage
4. 14 m^3/s force ventilation per end (Q_F).

Re-entry per end, including the time taken to clear the haulage to the next end, per 8 air changes:

$$\begin{array}{cccc} & \underline{\textbf{End}} & & \underline{\textbf{Haulage}} \\ = & \dfrac{L \times W \times H \times 8}{Q_F \times 60} & + & \dfrac{L \times W \times H \times 8}{Q_F \times 60} \\ = & \dfrac{180 \times 4 \times 4 \times 8}{14 \times 60} & + & \dfrac{120 \times 4 \times 4 \times 8}{14 \times 60} \\ = & 27.4 & + & 18.3 \text{ minutes} \\ = & \multicolumn{3}{l}{45.7 \text{ minutes, say } 46 \text{ minutes}} \end{array}$$

Re-entry to the last end (no haulage to clear) = 27.4 minutes

Total re-entry, based on 8 air changes

$$\begin{array}{ll} = & (46 \times 3 \text{ combinations}) + 27.4 \\ = & 165.4 \text{ minutes, say } 165 \text{ minutes} \end{array}$$

Re-entry interval to this section after the general blast is 2 ¾ hours.

WORKED EXAMPLE 2

The amount of through ventilation is now 100 m^3/s.

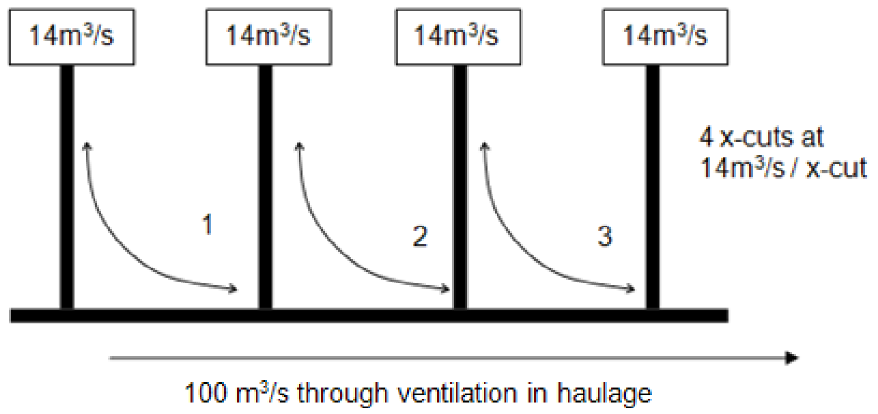

Figure 3.11 Haulage and 4 X-cuts ventilation layout (increased haulage airflow quantity)

From Figure 3.11:

- Through ventilation passes each of the x-cuts and gets drawn into the development end by means of a force fan. The contaminated air from the first x-cut then flows to the second x-cut and so forth until it has gone through all the x-cuts.

  The same amount of air (assume no leakages) that enters the haulage and subsequent 4 x-cuts will be returned contaminated to the return airway (RAW).
- The numbers 1, 2 and 3 represents the combined airways (haulage and end) that need to be cleared after blasting.
- The above therefore represents a series airflow network.

Assumptions:

1. All ends are at maximum length (180 m)
2. Spacing of ends: 120 m apart
3. No ventilation column leakage
4. 14 m^3/s force ventilation per end (Q_F)

Re-entry per individual end, including the time taken to clear the haulage to the next end, per 8 air changes:

$$\underline{\textbf{End}} \qquad\qquad \underline{\textbf{Haulage}}$$

$$= \frac{L \times W \times H \times 8}{Q_F \times 60} + \frac{L \times W \times H \times 8}{Q_F \times 60}$$

$$= \frac{180 \times 4 \times 4 \times 8}{14 \times 60} + \frac{120 \times 4 \times 4 \times 8}{100 \times 60}$$

$$= 27.4 + 2.6 \text{ minutes}$$

$$= 30 \text{ minutes}$$

Re-entry to the last end (no haulage to clear) = 27.4 minutes

Total re-entry, based on 8 air changes

$$= (30 \times 3 \text{ combinations}) + 27.4$$

$$= 117.4 \text{ minutes, say 120 minutes}$$

Re-entry interval to this section after the general blast is 2 hours.

3.10 Air Quantity Design Guidelines for Development Ends

When the quantity of air for a specific development has been determined, the pipe column sizes and the fans needed for the supply of the air have to be determined. There are certain guidelines and these are given in Table 3.1.

Table 3.1

Pipe column sizes	**Design quantity**	**Fan motor size (normally axial flow fans)**
305 mm	0.6 - 1.0 m^3/s	0.75 kW
410 mm	2.0 - 3.0 m^3/s	4 kW
570 mm	3.5 - 7.0 m^3/s	7.5 - 22 kW
760 mm	8.0 - 14.0 m^3/s	45 kW

Chapter

4

FANS: SPEED, PERFORMANCE AND FAN TYPES

LEARNING OUTCOMES

Knowledge and understanding of

- The difference between a fan and a blower
- Specific speed and performance curves
- The difference between centrifugal fans, axial flow fans, mixed flow fans
- The various air control methods
- The application of fan performance curves, namely: electrical power and effect of velocity, density and pressure
- The concept of semi-series and semi-parallel fan placements.

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Brake, DJ, 'Fan total pressure or fan static pressure: which is correct by solving ventilation problems?' *Journal of the Mine Ventilation Society of South Africa*, January/March, 2002, p6-11.

Saving power and reducing noise in auxiliary mine ventilation', *Journal of the Mine Ventilation Society of South Africa,* July/September 2005, p96-97.

Mine Environmental Control (MEC) Certificate, Chamber of Mines, South Africa, Workbook 1.

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4.1 Introduction

Fans are necessary to distribute air in the mines of today and in these modules you will learn about their application, efficiencies and methods of altering their performance by changing speed or by installing them in air densities different to their characteristic curves. In addition, the effects of natural ventilation pressure on fan performance will be discussed. As was mentioned before, an evaseè is fitted to a fan or to a system to regain some of the velocity pressure at the discharge and the effects of fitting an evaseè are also described.

In many situations underground and on surface fans are installed either in series and / or in parallel to provide the necessary duty. Examples of fans installed in this manner will be shown in this chapter. In practice many fans are not installed truly in parallel or truly in series but are in 'semi-series' or 'semi-parallel'. The class notes and other resources give a detailed explanation of solving problems when this is the case.

NB

It is most important to remember that in practice and in examination questions, the problem may entail some or all of these effects simultaneously. For example, the problem may involve changing the characteristic curves of the fan for density, applying natural ventilation pressure (NVP) and then placing the fans in parallel. Problems of this nature are very complex and should be done step-by-step.

NB

A common error made by students is that of failing to ensure that the **system resistance curve and the fan characteristic curves are at the same density**. It should be remembered that the fan, in practice, will operate at the density of the air in the system and hence density corrections should be made accordingly.

4.2 Fan Description

Definition

In addition to the definition of a fan, a **fan** is an appliance that converts mechanical energy delivered to the fan, into potential energy (pressure) and kinetic energy (velocity). Pressure of course, is necessary to overcome the resistance (or pressure loss associated with system) of a particular duct or system in which the fan is operating. Fans may vary in diameter, power rating, air volume handled and pressure created.

Animation

The main parts of a fan consist of the following:

- The principal component of a fan is the impeller, consisting of the hub and blades
- The impeller is mounted to a shaft that is directly or indirectly coupled to a motor and mounted inside the casing or housing of the fan.

In addition, a fan is generally equipped with an inlet cone, an outlet evaseè and guide vanes to improve flow characteristics.

Fans can be divided into:

- Main Fans
    - o Axial Flow Fans
    - o Centrifugal or Radial – Flow Fans
- Auxiliary Fans (axial and centrifugal or mixed flow air fans)
- Booster Fans (axial and centrifugal)

4.2.1 Main Types of Fans

These fans are normally situated on surface at the top of the up-cast shaft that exhaust air through the mine, and is commonly known as the 'lungs' of a mine. These fans can either be electric driven axial flow or centrifugal fans, that can handle volumes of air ranging from 25-450 m^3/s at pressures of 400 Pa to 9.0 kPa and power ratings from 50 kW to 4 100 kW (4.1 MW).

4.2.1.1 Axial Flow Fans

Definition

Fans in which the direction of airflow entering and leaving the fan is parallel to the axis or shaft of the fan are referred to as **axial flow fans.** When an axial flow fan operates, the blades scoop up the air on the one side (negative pressure) of the impeller and push it to the other side (positive pressure), thereby causing a flow of air parallel to shaft or axis of the fan. Simultaneously, the air is given a twist when leaving the impeller, causing it to leave the impeller with a spiral / swirl motion. To counteract this motion, a set of stationary blades is usually installed on either the inlet or outlet side of the impeller, which also improves the efficiency of the fan. These fixed blades are the inlet or outlet guide vanes. An example of an axial flow fan is showed in Figure 4.1.

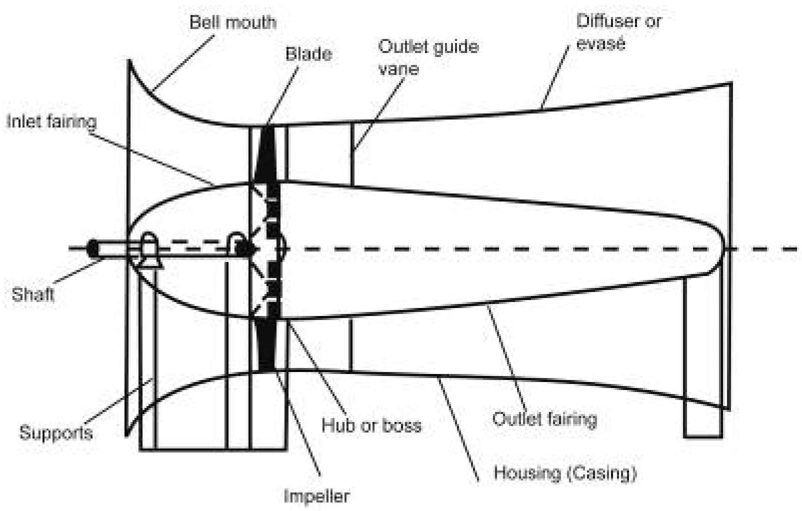


Figure 4.1 Construction of axial flow fans.

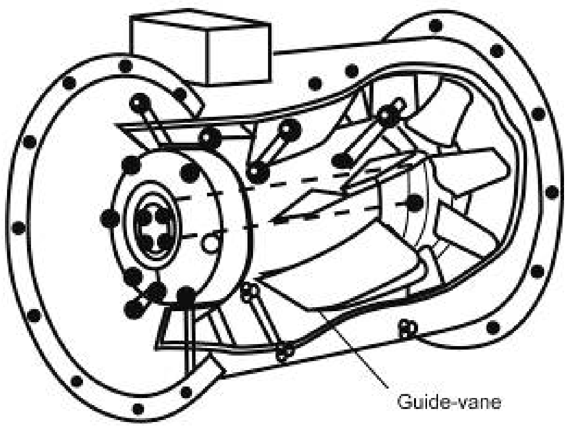


Figure 4.2 Down stream guide-vane fan

Fans with two or three impellers, one behind the other, each with its own set of guide vanes are called two-stage or three stage fans.

Definition

Contra-rotating fans are fans that have two impellers, driven by separate motors and revolving in opposite direction. Axial flow fans also have casings and in large efficient fans consist of three parts - a large cylindrical section round the impeller, an inlet section with a bell mouth to reduce entrance losses and a diverging outlet section or evaseè, to reduce shock losses by allowing the air to slow down gradually.

To further improve the efficiency of the fan by ensuring smooth flow of the air, a short snub noise inlet fairing on the upstream side (fan inlet) and a long, pointed outlet fairing on the down-stream side (fan outlet) are sometimes installed.

Fan blades can be made of flat steel plate, called laminar blades or otherwise formed into a special aerodynamic shape, called aerofoil blades.

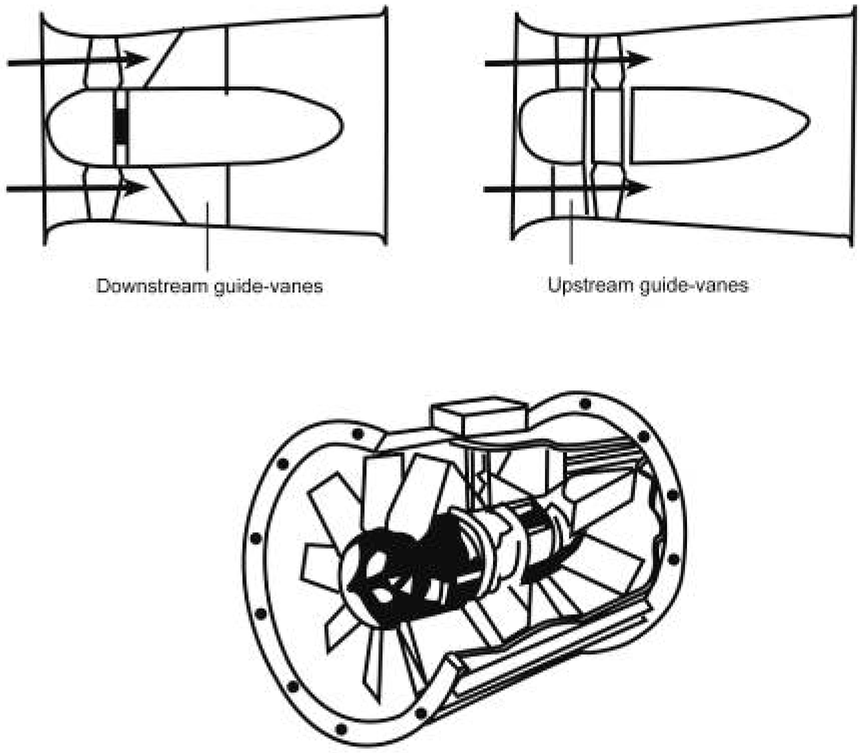


Figure 4.3 An axial-flow fan with contra-rotating impellers

4.2.1.2 Centrifugal or Radial- Flow Fans

Definition

Centrifugal or radial- flow fans are fans in which the direction of airflow entering the fan is parallel to the axis or shaft of the fan and is discharged radially (i.e. at right angles to the fan axis or shaft).

A centrifugal fan works on a different principle to axial flow fans. ‘Centrifugal’ means ‘fleeing from the centre’. Anything that is revolved tends to leave the centre and will do so if allowed. An object attached to a string and swing around, will move in circles and the moment the string is released, it will fly away.

Definition

The impeller consists of two rings with blades fitted between them, similar to the paddle wheel of an old steam ship or a water mill. When the fan operates, air is drawn parallel to the shaft into the open ends of the impeller (the eye) and discharged radially through the blades. When air enters on both sides of the impeller, it is called a **double inlet** fan. If air enters one side only, it is called a **single inlet fan**.

The blades can either be laminar or aero-foil shape and can be arranged in three different ways:

- Radially or straight
- Forward curved / inclined
- Backward curved.

NB

Figure 4.4 illustrate these blade arrangements and it must be noted that these blade arrangements each has a real impact on the power consumption related to that fan and this will be dealt with in detail later.

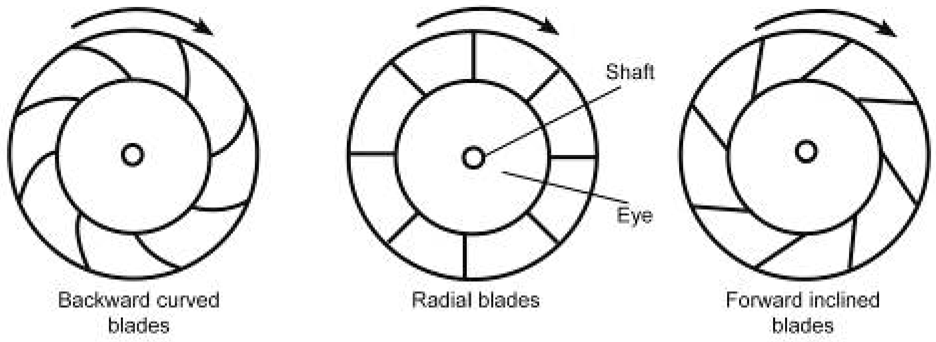


Figure 4.4 Various types of fan blade arrangements

Definition

Centrifugal fans normally do not have inlet guide vanes, but do have an evaseè (on outlet side) and usually a short inlet cone. The casing around the impeller is shaped like a spiral curve and is called the **scroll** or **volute**.

The point where the scroll is nearest to the impeller is called the **cut-off** or **clearance.**

The discharge of centrifugal fans can be **top or bottom horizontal** discharge and **top or bottom vertical** discharge. These fans are also classed as either **left or right-hand drive** and hence this expression indicates on which side of the fan the motor is situated. When one stands behind the fan and faces the direction in which the air is discharged, it is a left-hand drive if the motor is at one's left and vice versa.

These fans can be direct driven, when the motor shaft is in line with and directly connect to the fan impeller, or it can be indirectly driven through gears, belts or friction clutch. See Figure 4.5 and Figure 4.6 for centrifugal fans.

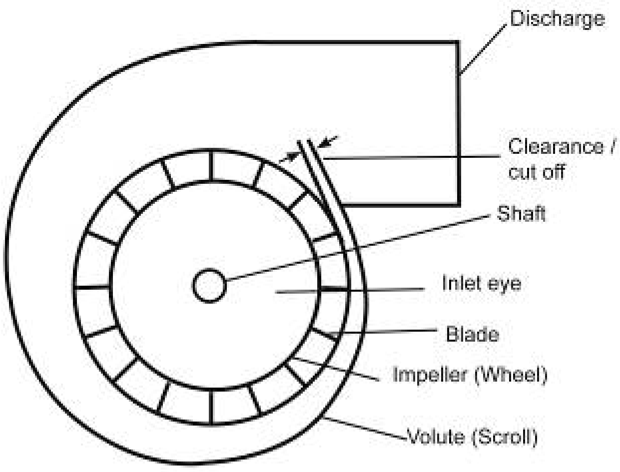


Figure 4.5 Construction of a centrifugal fan

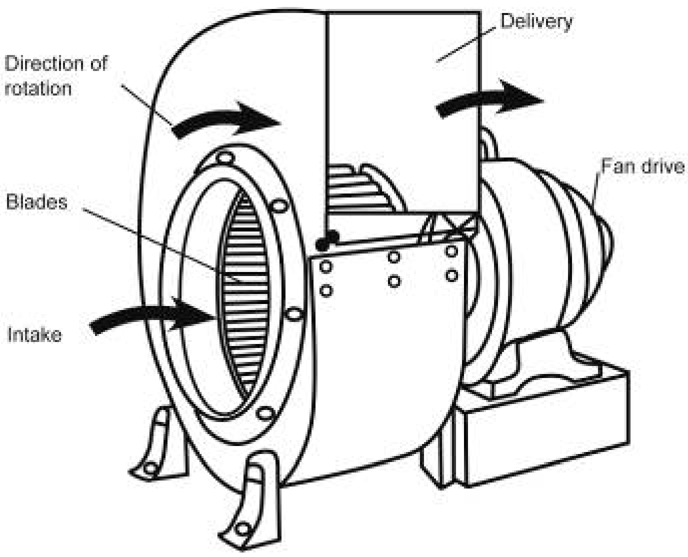


Figure 4.6 A typical centrifugal fan (left hand drive, top horizontal discharge)

Table 4.1 shows a detailed breakdown of the advantages and disadvantages of axial flow fans compared to centrifugal fans.

Table 4.1 Comparison of Fans for Use in Auxiliary Ventilation

Characteristics	Axial Flow Fans	Centrifugal Fans
Advantages	• Easy to install • More flexible • Cheaper and smaller.	• Operates at a lower speed • Less noisy • Less wear and tear.
Disadvantages	• Bad stall zone • Noisy.	• Difficult to install-large excavations required • Require complicated pipe layouts • Bulky.
Size	• Less bulky than centrifugal fans, especially for fan volumes of below 165 m^3/s and pressures of 3 000 Pa or less.	
Ease of installation	• Easier to install because there is no change of direction of the air flowing through the fan. This advantage is particularly marked when fans have to be installed in ventilation columns.	
Flexibility	• Axial-flow fans with adjustable pitch blades are more flexible and cover a wider range of duties than centrifugal fans, even when these are equipped with adjustable inlet guide vanes.	
Efficiency	• At one time regarded as having a greater efficiency than centrifugal fans.	• Improvement in the design of the latter type has resulted in these fans now equaling, and sometimes exceeding, the performance of axial-flow fans.
Large volumes and high pressure		• Cheaper for large volume flow (over 235 m^3/s) or high pressures (over 3 000 Pa).
Reliability		• Because centrifugal fans usually run at lower speeds than axial-flow fans of the same performance, there is less wear and consequently they are more reliable.
Stall characteristics	• Violent.	• Very mild.
Noise	• Noisier	• Marked advantage
Preference	• In the past axial - flow fans were preferred as main auxiliary underground fans.	• Today centrifugal fans are preferred as main fans whether sited underground or on surface.

Research is being carried out continuously to improve the design and performance of large and small fans. Table 4.2 shows the effect on volume flow, pressure and efficiency with reversal in rotation.

Table 4.2 Effects of reversal of rotation

	Pressure	Quantity	Efficiency	Direction of Airflow
Axial	Reduced	Reduced	Reduced	Opposite
Centrifugal	Reduced	Reduced	Reduced	Same

4.2.2 Auxiliary Fans

These fans are used underground to ventilate working places not in through ventilation (i.e. where air would not naturally enter or also referred to as 'dead' ends), such as:

- Development ends
- Back- stopes
- Up-dip stoping panels
- Workshops etc.

Definition

The definition for a '**dead end'** is an end which has advanced more than twice its width and then needs to be mechanically ventilated with fans. Usually, these fans are electric driven axial flow fans, that can handle air volumes of up to 15 m^3/s @ 2.1 kPa, power ratings from 4 kW to 45 kW and varies in diameter from 380 mm to 760 mm, but other sizes may exist.

4.2.3 Booster Fans

The installation of booster fans underground sometimes becomes unavoidable. Longer airways have to be serviced when workings approach the extremities, resulting in an increased resistance of a mine / shaft system. In order to overcome the increased resistance of such system, booster fans need to be installed to assist the main fans. Booster fans generally handle between 60 and 90 m^3/s at pressures ranging up to 3.0 kPa.

NB

These installations can either be axial flow or centrifugal fans, but axial flow fans are favoured, as normally little or no additional excavations are required for installation purposes. One important factor to be considered when installing these units are that they should be installed as close as possible to the up-cast shaft to prevent recirculation, which can contaminate intake air with heat, blasting fumes and noxious gases from fires.

4.3 Fan Testing

Usage

A fan test column is used to obtain the characteristic curves of a small mine fan. This test column consists of a suitable length of ventilation piping, with an adjustable regulator at one end. Various reducers are used to accommodate most of the common sizes of fans in the column and facilities are available to measure accurately the pressure and quantity of the fan.

Objectives

Objectives of testing fans:

- To check whether the fan performance complies with the manufacturer's specifications (this is normally done as soon as possible after installation).
- To check the fan performance after overhaul.
- To obtain performance figures which may not be readily available, e.g. at different pitch settings, or ranges not covered by existing curves.
- To determine whether the fan is still performing satisfactorily. Prolonged use may have resulted in inefficiency due to wear or accumulations of dirt on various parts.

A typical test column is illustrated in Figure 4.7.

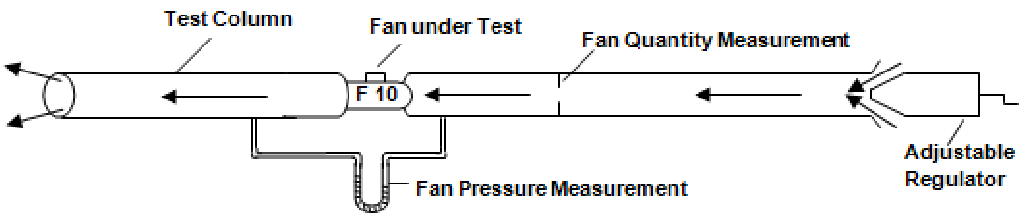


Figure 4.7 Fan testing bench arrangement

When the fan under test has been fitted in the column, the regulator is opened fully and the fan is started. The fan works against a low resistance (because of the short column) and will handle a large quantity of air at a low pressure. The quantity and pressure are measured and recorded as a 'test point'.

The regulator is then closed slightly increasing the resistance against which the fan operates, and the quantity and pressure are measured and recorded as another 'test point'. The quantity is obviously lower and the pressure higher because of the increased resistance.

This procedure is repeated until sufficient 'test points' have been obtained to draw the characteristic curve of the fan over its full range of performance. The air density and fan speed are also determined while the fan is under test, because the characteristic curve drawn is only valid for that particular speed and density.

For each particular regulator setting, the input power to the fan is also measured. The efficiency of the fan at various duties can then be calculated and curves can be drawn for fan power and fan efficiency.

4.4 Fan Performance

Just like any machine or person, a fan has certain characteristics or distinguished qualities with respect to work that it is able to perform under different circumstances. These fan characteristics can be told by means of graphs, which could then be described as fan characteristic curves.

Definition

Fan characteristics curves indicate how much air a fan can deliver at any particular pressure and how much power is required to drive the fan and the fan efficiency, in each case.

NB

As emphasised above, these curves are applicable to a particular fan when it is driven at **a specific speed (r/s) and given air density (kg/m³)**. These parameters should always be specified on the graph. Figure 4.8 shows the characteristic curve related to a fan as well as the power and efficiency curve which is obtained from the fan test being done.

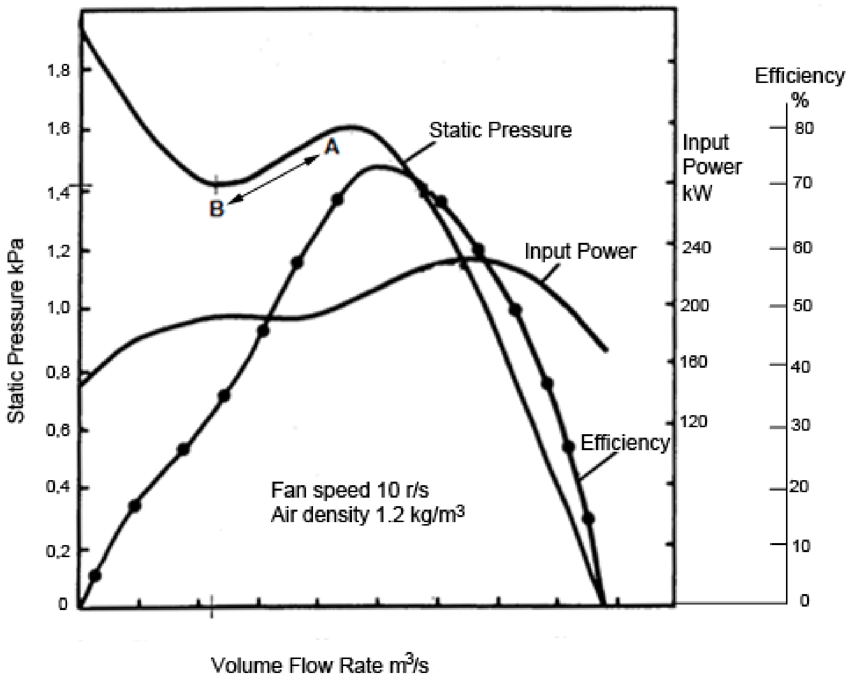


Figure 4.8 Characteristic curves for fans (fan, power and efficiency)

4.4.1 Fan (Characteristic) Curve

In Figure 4.8, the lower right-hand portion of the pressure - volume curve marked 'static pressure', indicates that this fan will handle 175 m^3/s, at virtually no pressure, i.e. running with no ducts attached to it in open air. However, when this fan is installed in a system or mine, the quantity will decrease but at a higher pressure.

The higher the resistance of the system in which this fan operates, the lesser the quantity and the greater the pressure put into the system, until it reaches a point A (Figure 4.8), at which the fan will deliver ± 90 m^3/s at a pressure of 1.6 kPa. If the system resistance increases further, the fan will still deliver less air, but at a lower pressure up to point B (45 m^3/s @ 1.42 kPa). This portion A-B (Figure 4.8) is called the 'Stall Zone' of the curve / fan.

Definition

Thus, the **stall zone** is that portion of the fan characteristic curve in which both the fan pressure and quantity increase simultaneously, as opposed to the normal case where the quantity decreases as the fan pressure increases and vice versa.

NB

When the fan is operating in the stall zone, the air tends to separate from the fan blade surfaces and is not a good thing or recommended.

The signs of a fan operating in the stall zone are:

- The fan vibrates excessively, which may cause mechanical failure
- The fan makes unusual noises, it whines
- An ammeter measure the input amps can be seen to fluctuate markedly
- The quantity and pressure of the fan alter considerably.

The immediate action to be taken when a fan is operating in the stall zone is to:

ML

- Alter the characteristics of the system in which the fan operates, so that the fan operates on a different part of the curve, by short-circuiting some of the air through the system.

4.4.2 Fan Input Power Curves

This curve (shown in Figure 4.8 as well) shows the power required at the fan shaft with different air quantities being delivered. Obviously, the motor driving the fan will have to deliver greater power than the power delivered to the fan shaft, if there are any losses in the drive (that is fan inefficiency). The shape of the input power curve is important and there are two different types of power curves.

4.4.2.1 Overloading Power Curves

Definition

An **overloading power curve** is a power curve where the power increases continuously as the quantity increases until eventually the power become too high and the fan will trip or burn out. This is graphically shown in Figure 4.9.

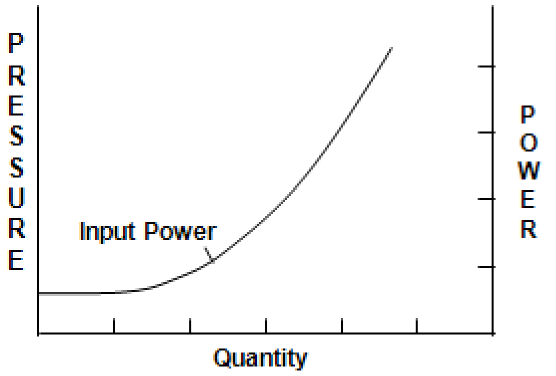


Figure 4.9 Example of overloading power curve

4.4.2.2 Non Overloading Power Curve

Definition

A **non-overloading power curve** is a power curve where the power increases as the quantity increases up to a certain point, when the power will then slowly decrease while the quantity continues to increase and is shown in Figure 4.10.

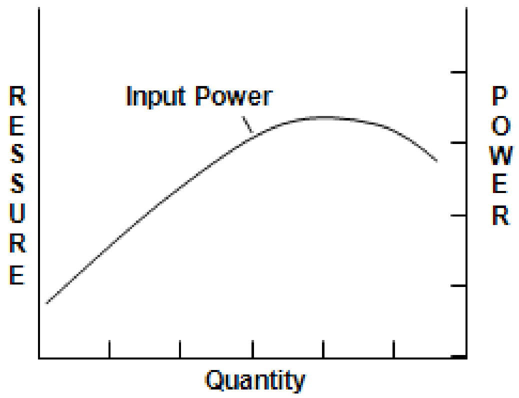
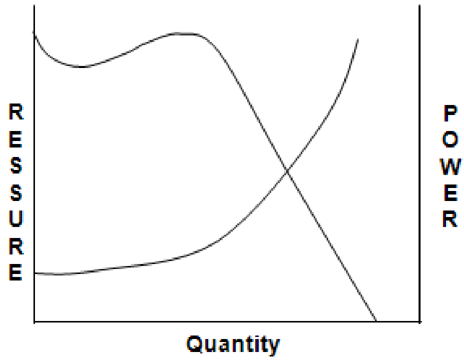


Figure 4.10 Example of a non overloading power curve

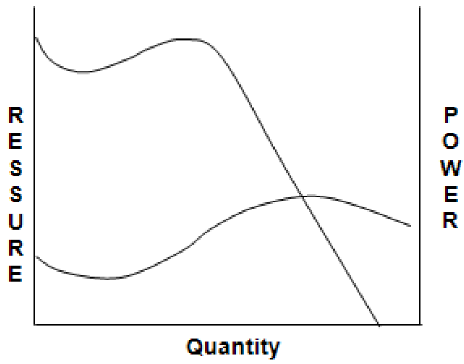
4.4.3 Typical Power Curves for Various Types of Fans

As was mentioned before the type of blade has a major influence on the input power related to a fan. These are graphically shown for each type of blade curve. (Figure 4.11)

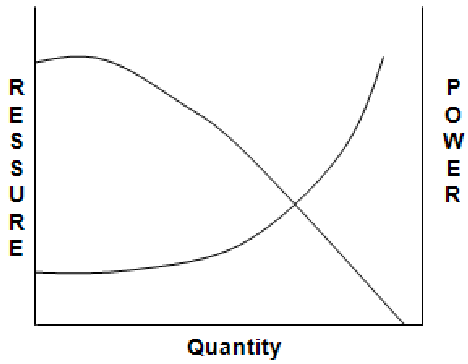


Fan with overloading power curve
Centrifugal fan with forward curved blades

Fan with non-overloading power curve
Centrifugal fan with backward curved blades



Fan with non-overloading
Power curve for axial flow fan



Fan with overloading power curve
Centrifugal fan with straight blades

Figure 4.11 Examples of power curves for different types of blades

4.5 Fan Input Power, Efficiencies, Pulley Sizes

NB

The following section will deal with the various calculations pertaining to fans. It is important to note that solutions pertaining to fans can either, be done graphically or through calculations, please read the questions accordingly (questions answered using graphically when calculations were required will be marked as incorrect).

Animation

4.5.1 Fan Input Power and Efficiency Calculations

Most mine fans are electrically driven by 3-phase alternating current (A.C.) motors. The electrical power consumed by such a motor can be calculated from:

$$W = E\ I\ p_f \sqrt{3} \quad \text{(4.1)}$$

Where:

W = power (kW) if E is in kV
E = voltage (kV)
I = current (A)
P_f = power factor

Definitions

This power W is called the **motor input power**. The power delivered to the shaft of the motor is called the **motor output power**. This can be obtained by multiplying the motor input power (obtained from the above formula) by the efficiency of the motor. Motors are generally related on their output power, for example, a 10 kW motor is a motor, which has an output power of 10 kW. With no energy losses (this is normally assumed as such), the motor output power is equivalent to the **drive input power**.

The power delivered to the fan is called the **fan input power** and is equivalent to the **drive output power**. This can be obtained by multiplying the motor output power (or drive input power) by the efficiency of the drive.

The power added to the air by the fan (fan output power) is known as the **air power**. This can be obtained by using the well-known air power formula shown previously.

$$W_a = \frac{p \times Q}{1\ 000}$$ (from chapter 2)

The above explanation started at the motor input power and worked towards the fan output power, the power becoming less in each case because of the inefficiency of the motor, drive and fan.

If one had to start at the fan output power and work towards the motor input power (against the direction of the transmission of power) the power would become greater in each case. In this, the power would be multiplied by $\frac{100}{\text{efficiency}}$, where the efficiency is expressed as an exact number such as 70% (which can be assumed as the **overall** efficiency of the fan if this number is not given to the student).

Figure 4.12 shows the various definitions, which have been discussed much more clearly.

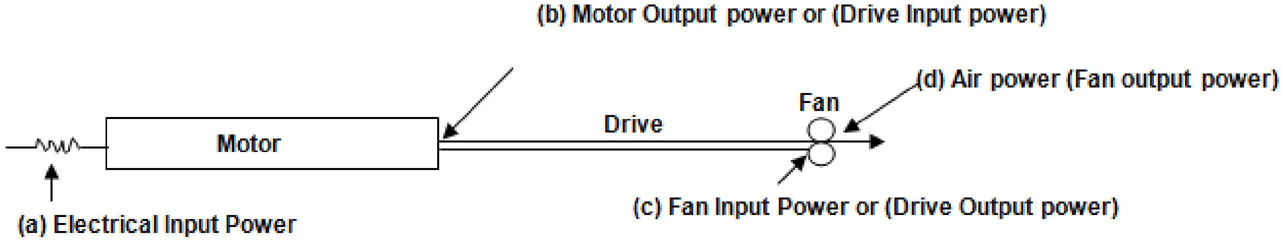


Figure 4.12 An example of fan related power descriptions

From Figure 4.12 it can be seen that (refer to a, b, c and d accordingly from the figure above):

$$\text{Motor Efficiency} = \frac{\text{motor output power}}{\text{motor input power}} \times 100 = \frac{b}{a} \times 100\%$$

$$\text{Drive Efficiency} = \frac{\text{fan input power}}{\text{motor output power}} \times 100 = \frac{c}{b} \times 100\%$$

$$\text{Fan Efficiency} = \frac{\text{air power}}{\text{fan input power}} \times 100 = \frac{d}{c} \times 100\%$$

$$\text{Overall fan Efficiency} = \frac{\text{air power}}{\text{motor input power}} \times 100 = \frac{d}{a} \times 100\%$$

Definition

There are various types of fan drives available. **Direct drive fans** are fans in which the motor shaft is connected directly onto the fan shaft and the motor drives the fan directly. Fans can also be driven indirectly by belts and pulleys, gearboxes or a friction clutch.

The following worked examples show how the various powers and efficiencies are calculated (when doing these examples a sketch of the fan, drive and motor should be made as was shown before).

WORKED EXAMPLE 1

A surface cooling tower fan handles 200 m^3/s at a pressure of 780 Pa. The fan is driven by a 3-phase A.C. Motor, which draws 84.6 amps from a 1 500 V supply. The power factor is 0.91. The motor is 94% efficient and the drive is 97% efficient.

Determine:

i. Motor input power
ii. Motor output power
iii. Fan input power
iv. Fan efficiency
v. Overall fan efficiency.

Answer

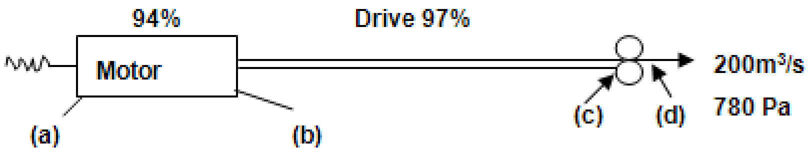


Figure 4.13 Graphical representation fan and motor

i. Motor input power (a) $= W = E\,I\,pf\,\sqrt{3}$

$= 1.5 \times 84.6 \times 0.91 \times \sqrt{3}$

$=$ 200 kW

ii. Motor output power (b) $= (200 \times \frac{94}{100})$ kW

$=$ 188 kW

iii. Fan input power (c) = drive output power

= drive input power x drive efficiency

= motor output x drive eff

$= (188 \times \frac{97}{100})$ kW

= 182.36 say 182 kW

iv. Fan efficiency $= \dfrac{\text{air power}}{\text{fan input power}} \times 100\%$

$= \dfrac{d}{c} \times 100\%$

Air power W_a $= \dfrac{p \times Q}{1000}$

$= \dfrac{200 \times 780}{1000}$

$= 156$ kW

$= \dfrac{156}{182} \times 100\%$

$= 85.7\ \%$

v. Overall fan efficiency $= \dfrac{\text{air power}}{\text{motor input power}} \times 100\%$

$= \dfrac{d}{a} \times 100\%$

$= \dfrac{156}{200} \times 100\%$

$= 78\%$

WORKED EXAMPLE 2

A fan with an 82% efficient belt drive handles 82.2 m^3/s at a pressure of 2 340 Pa. The fan is 76% efficient and the motor is 95% efficient.

Determine:

i. The input to the motor

ii. The overall efficiency

iii. The current drawn by the motor, given that the voltage is 6 600 V, the current has a time lag of 24° behind the voltage and the number of phases is 3.

Answer

i.

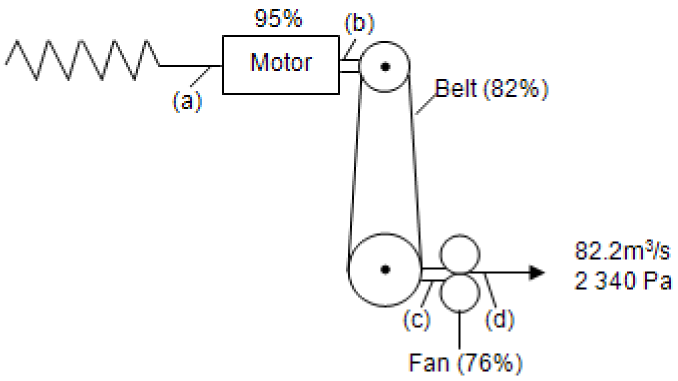


Figure 4.14 Fan and fan belt arrangement

$$\text{Air power } W_a = \frac{p \times Q}{1000}$$

$$= \frac{2340 \times 82.2}{1000}$$

$$= 192.35 \text{ say } 192 \text{ kW}$$

$$= \text{output power of fan at (d)}$$

$$\therefore \text{Input power to fan} = \text{air power} \times \frac{100}{\text{fan eff}}$$

$$= 192 \times \frac{100}{76}$$

$$= 252.6 \text{ kW}$$

$$= \text{output power of drive at (c)}$$

$$\therefore \text{Input power to drive} = \text{output power of drive} \times \frac{100}{\text{drive eff}}$$

$$= 252.6 \times \frac{100}{82}$$

$$= 308.049 \text{ say } 308 \text{ kW}$$

$$= \text{output power of motor at (b)}$$

$$\therefore \text{Input power to motor} = \text{output power of motor} \times \frac{100}{\text{motor eff}}$$

$$308.049 \times \frac{100}{95}$$

$$= 324.21 \text{ say } 325 \text{ kW}$$

ii. Overall efficiency $= \dfrac{\text{air power}}{\text{motor input power}} \times 100\%$

$$= \frac{d}{a} \times 100\%$$

$$= \frac{192}{325} \times 100\%$$

$$= 59.07 \text{ say } 59.1\%$$

The overall efficiency in this case could also have been obtained by:

Overall eff. = (motor eff. x drive eff. x fan eff. x 100%)

$$= \frac{95}{100} \times \frac{82}{100} \times \frac{76}{100} \times 100\%$$

$$= 59.07 \text{ say } 59.1\%$$

iii. Input power to motor:

$$W = E\,I\,pf\,\sqrt{3}$$

$\therefore$ Current consumed by motor:

$$= \frac{w}{Epf\sqrt{3}}$$

$$= \frac{325}{6.6 \times \cos 24^{o} \times \sqrt{3}}$$

$$= 31\text{amp}$$

4.5.2 Fan Speeds with Belt Drives

NB

The speed of a belt-driven fan can be altered relatively simply by changing the diameter of either the motor pulley or the fan pulley. The fan speed varies inversely with the diameter of the fan pulley.

Rule

This means that a smaller fan pulley results in an increased fan speed and a larger fan pulley results in a decreased fan speed.

Rule

The fan speed varies **directly** with the diameter of the **motor pulley**. This means that a smaller motor pulley results in a lower fan speed and a larger motor pulley results in an increased fan speed. This explanation results in the Equation:

$$\frac{\text{fan}_{\text{speed}}}{\text{motor}_{\text{speed}}} = \frac{\text{motor}_{\text{pulley diameter}}}{\text{fan}_{\text{pulley diameter}}} \quad \text{....................................... (4.2)}$$

Tip

This Equation can be manipulated to obtain any other term which may be required. However, when faced with a problem of this nature, it is usually easier to draw a sketch of the installation and use logic to derive the Equation.

WORKED EXAMPLE 3

A fan runs at a speed of 12.3 r/s. The fan pulley diameter is 195 mm and the motor pulley diameter is 150 mm, determine two ways of increasing the fan speed to 18 r/s.

Answer

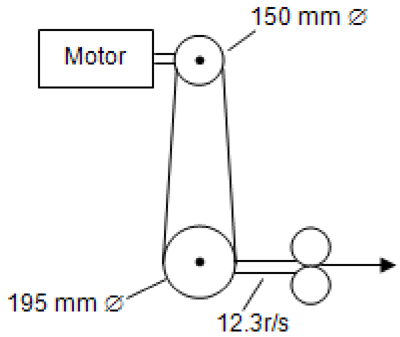


Figure 4.15 Fan and fan belt arrangement

From Figure 4.15 it can be seen that in order to increase the fan speed either:

i. The fan pulley diameter should be decreased

ii. The motor pulley diameter should be increased

i. Decrease fan pulley diameter.

New fan pulley diameter as follows:

$$= \text{old fan pulley diameter x } \frac{\text{old fan speed}}{\text{new fan speed}}$$

$$= 195 \text{ x } \frac{12.3}{18.0}$$

$$= 133.25 \text{ say } 133.3 \text{ mm}$$

ii. Increase motor pulley diameter

New motor pulley diameter as follows:

$$= \text{old motor pulley diameter x } \frac{\text{new fan speed}}{\text{old fan speed}}$$

$$= 150 \text{ x } \frac{18}{12.3}$$

$$= 219.5 \text{ say } 220 \text{ mm}$$

It must be stressed that **either** of the above two changes will result in the fan speed increasing to 18 r/s. If both the fan pulley diameter was decreased **and** the motor pulley diameter increased the resultant fan speed would be (check if you agree with the answer and method below):

Also include:

$$= 12.3 \times \left(\frac{195}{133.3}\right)^2 \qquad = 12.3 \times \left(\frac{220}{150}\right)^2$$

$$= 26.32 \text{ r/s} \qquad = 26.45 \text{ r/s}$$

approximately the same.

4.5.3 Fan Efficiency Curve

The efficiency curve is derived from the two curves mentioned above, ie; the fan characteristic curve and input power curve.

NB

Please take note that different types of power curves can be plotted and represented in different questions. Make sure which power curve is referred given. It can be the fan input power curve or the fan output power curve and sometimes the fan motor input power curve or fan motor output power curve can be given.

Definition

The **efficiency of any machine** is defined as the ratio of the useful work output to the energy input, expressed in percentage. The useful work done by the fan is to move air in terms of the quantity of air it moves and the pressure it puts into the air. These two terms are combined into Air power (fan output power).

The air power at various quantities and their corresponding pressures are calculated and divided by the corresponding input power to give efficiencies. The efficiency curve is then drawn by plotting these efficiencies against the corresponding quantities.

4.5.4 System Resistance Curves Used with Fan Curves

It was previously shown in Chapter 2 that the air flowing through a system obeys the law:

$P = RQ^2$ (if air density is assumed as standard 1.2 kg/m^3)

Where:

P	=	Pressure required (Pa), small prefers to pressure loss
R	=	Resistance (Ns^2/m^8)
Q	=	Quantity (m^3/s)

NB

Providing R does not change, P is proportional to Q^2. If the pressure required to pass a certain quantity of air through a system is known, the pressure required to pass other quantities of air through the system can be determined easily. (Note that the resistance of the system does not change). It can also be expressed as follows and a very important relationship to be kept in mind if new pressures pertaining to new quantities needs to be determined.

$$\frac{P_2}{P_1} = \left(\frac{Q_2}{Q_1}\right)^2 \text{ or } P_2 = P_1 \times \left(\frac{Q_2}{Q_1}\right)^2 \quad \text{……………………(4.3)}$$

if the resistance remain the same.

WORKED EXAMPLE 4

The ventilating pressure required to pass 4 m^3/s of air through a system is 150 Pa. What pressure will be required to pass 6 m^3/s of air through this system? Let suffix 1 indicate the first set of conditions and suffix 2 the second.

In the first condition:

$$P_1 = R_1 Q_1^2 \quad \therefore \quad R_1 = \frac{P_1}{Q_1^2} \qquad (\mathbf{a})$$

In the second condition:

$$P_2 = R_2 Q_2^2 \quad \therefore \quad R_2 = \frac{P_2}{Q_2^2} \qquad (\mathbf{b})$$

But the resistance in the system does not change.

$$\therefore R_1 = R_2$$

Substituting the Equations (a) and (b) in Equation (**c**)

Then:

$$R_1 = R_2 = \frac{P_1}{Q_1^2} = \frac{P_2}{Q_2^2} \qquad (\mathbf{c})$$

And

$$P_2 = \frac{P_1 Q_2^2}{Q_1^2} \quad \text{(as shown before for Equation 4.3)}$$

Given:

P_1 = 150 Pa, Q_1 = 4 m^3/s and Q_2 = 6 m^3/s

Then:

$$P_2 = 150 \times \left(\frac{6}{4}\right)^2 = 338 \text{ Pa}$$

By substituting the value of any other quantity for Q_2 in Equation (d), it is possible to calculate the pressure requirements for various quantities in this system.

WORKED EXAMPLE 5

What will be the pressure required to pass 6.67 m^3/s of air through the system given in the first example?

Equation 4.3 stated:

$$P_2 = \frac{p_1 Q_2^2}{Q_1^2}$$

$$P_1 = 150 \text{ Pa}, \qquad Q_1 = 4 \text{ m}^3/\text{s}$$

And

$$Q_2 = 6.67 \text{ m}^3/\text{s}$$

$$\therefore P_2 = \frac{150 \times (6.67)^2}{(4)^2} = 417 \text{ Pa}$$

System resistance curves have been discussed in detail in Chapter 2 before.

4.5.5 Fan Operating Point (F.O.P)

Definition

If the mine, or system resistance curve is plotted on the same graph as the fan characteristic curve (both curves must be plotted for the same density), the two curves will intersect at a particular point on the graph. This is the point at which the fan would operate if placed in the mine or system at the stated density and is known as the '**Fan Operating Point**' of the fan.

From the graph, the operating point of the fan, that is the pressure and volume at which the fan would operate can be obtained. If the original graph shows also the power and efficiency curves, these values can also be obtained. This is illustrated in Figure 4.16.

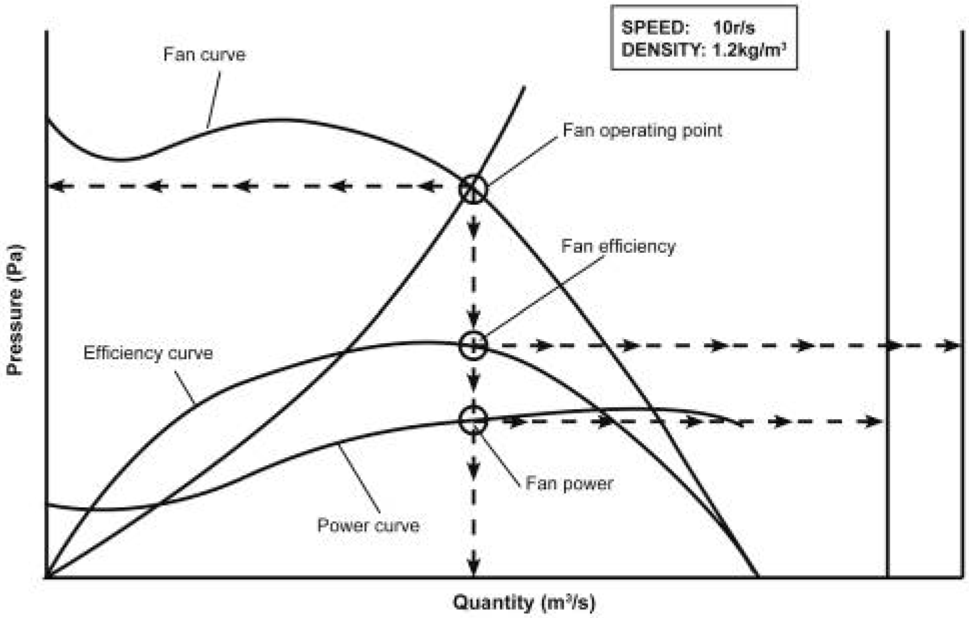


Figure 4.16 Example of resistance curve and fan curve to show FOP

WORKED EXAMPLE 6

The following figures were obtained from a fan test conducted on a direct driven axial flow fan (results in Table 4.3):

- Fan speed: 13.5 r/s
- Air density: 1.012 kg/m^3

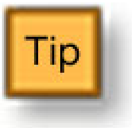
Table 4.3 Figures were obtained from a fan test conducted on a direct driven axial flow fan

Quantity (m^3/s)	Pressure (Pa)	Fan Input Power (kW)
0	930	3.6
2	840	4.6
4	870	5.9
6	910	7.8
8	860	10.0
10	730	12.4
12	490	14.7
13	310	15.0
14	0	14.6

From these figures draw a complete fan performance curve, including power and efficiency curves. Determine the fan operating point, input power and efficiency of this fan if it is to be installed in 760 mm galvanised duct, 54 m long, having a friction factor of 0.0035 Ns^2/m^8 at a prevailing density of 1.012 kg/m^3.

Answer

From the information given in the table above, without any additional calculations, draw the following curves, after selecting a suitable scale to be used on the graph paper. (The information given in the table above can be used as a guide to the scale to be used).



- Fan characteristic or performance curve by plotting quantity (horizontal or x-axis) against pressure (vertical or y-axis).
- Fan input power curve by plotting quantity (x-axis) against input power (Y-axis). (Remember: always plot **fan input power** curve, not air power curve. Candidates sometimes tend to plot the air power curve instead).

These plotted fan and power curves are shown in Figure 4.17.

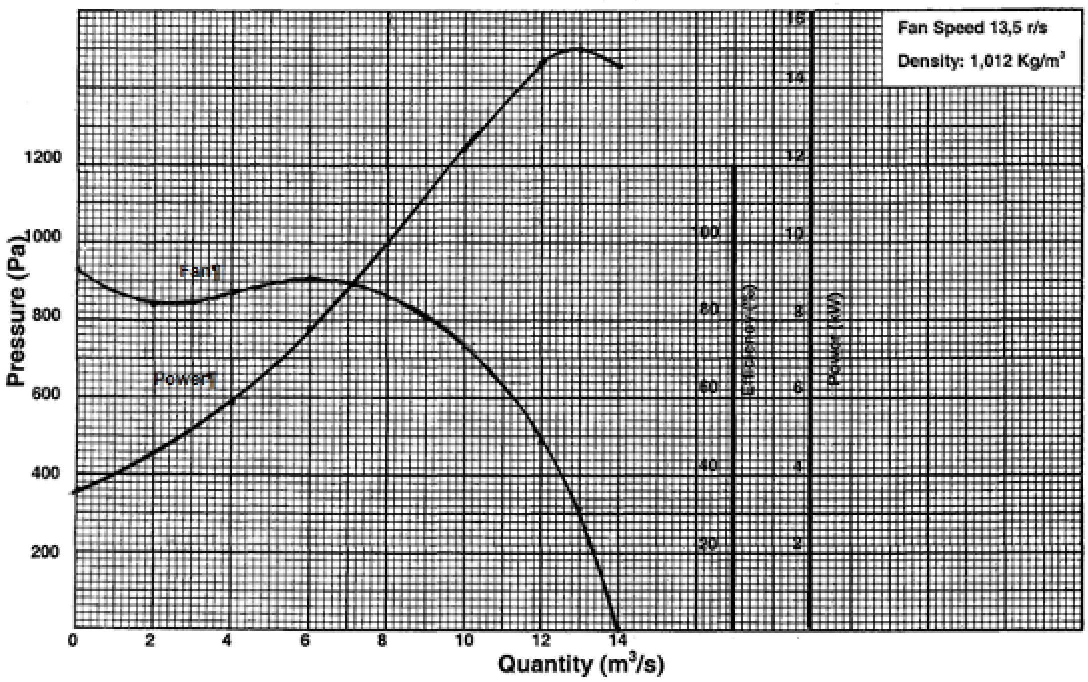


Figure 4.17 Fan and fan input power curve

Now that the fan performance and input power curves are plotted, some calculations need to be done, before the fan efficiency curve can be drawn.

$$\text{Fan Efficiency} = \frac{\text{Airpower}}{\text{Fan input power}} \times 100\%$$

$$\text{And Air power } (W_a) = \frac{P \times Q}{1000}$$

Now using the same table as given above, add two additional columns to this table with headings 'Air power' and 'Efficiency'.

Table 4.4 illustrates as follows:

Table 4.4 Summary of calculated results

Quantity (m^3/s)	Pressure (Pa)	Fan input Power (kW)	Air power (kW) $W_a = \frac{P \times Q}{1\,000}$	Fan Efficiency (%)
0	930	3.6	0	$\frac{0}{3.6} \times 100 = 0$
2	840	4.6	1.7	37.0
4	870	5.9	3.5	59.0
6	910	7.8	5.5	70.0
8	860	10.0	6.9	69.0
10	730	12.4	7.3	59.0
12	490	14.7	5.9	40.0
13	310	15.0	4.0	27.0
14	0	14.6	0	0

From the Table 4.4, draw the efficiency curve by plotting quantity (x-axis) against efficiency (y-axis) on the same graph paper. This plotted curve is shown in Figure 4.18 together with the fan and fan input power curves.

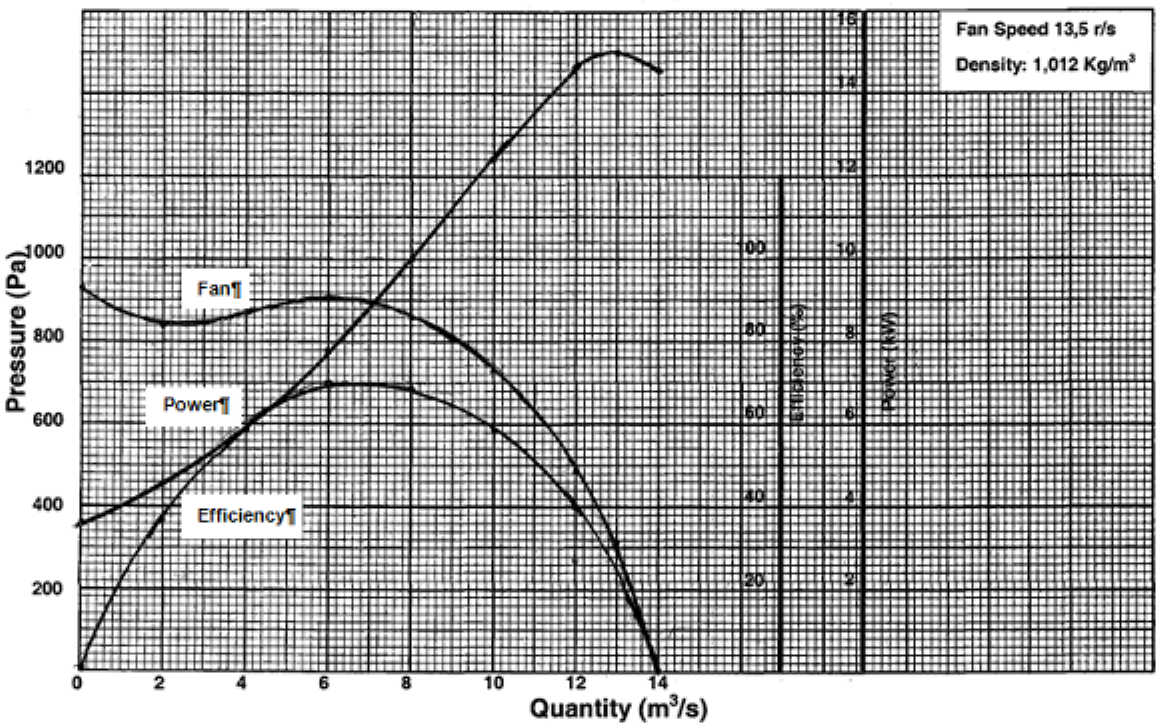


Figure 4.18 Plotted fan, fan input power and efficiency curves

Before the fan operating point, fan input power and fan efficiency can be determined, the system resistance must be calculated and then plotted on the same graph.

Given that:

This fan is to be installed in a 760 mm galvanised duct, 54 m long with a friction factor of 0.0035 Ns^2/m^4, the system resistance for the pipe can now be calculated:

NB

(Very important note is that the term $\frac{\rho}{1.2}$ was not included AS both fan and system are at a density of 1.012 kg/m^3). Also, remember that a change in density only relates to a pressure difference change and that the resistance (physical property of the pipe or tunnel) does not change because of density changes in the air.

$$R = \frac{KCL}{A^3}$$

$$= \frac{0.0035 \times 2.386 \times 54}{(0.453)^3}$$

$$= 4.85 \ Ns^2/m^8$$

Now to calculate the points to be plotted on the graph for the system resistance curve, use $P = RQ^2$ (as was shown in Chapter 2). Assume quantities and calculate their corresponding pressures using the resistance calculated above for the system.

Q	2	4	6	8	10	12	14
P	19	78	175	310	485	698	951

Now plot the above pressures (y-axis) against quantities (x-axis). This system characteristic curve with all the other characteristic curves is shown in Figure 4.19.

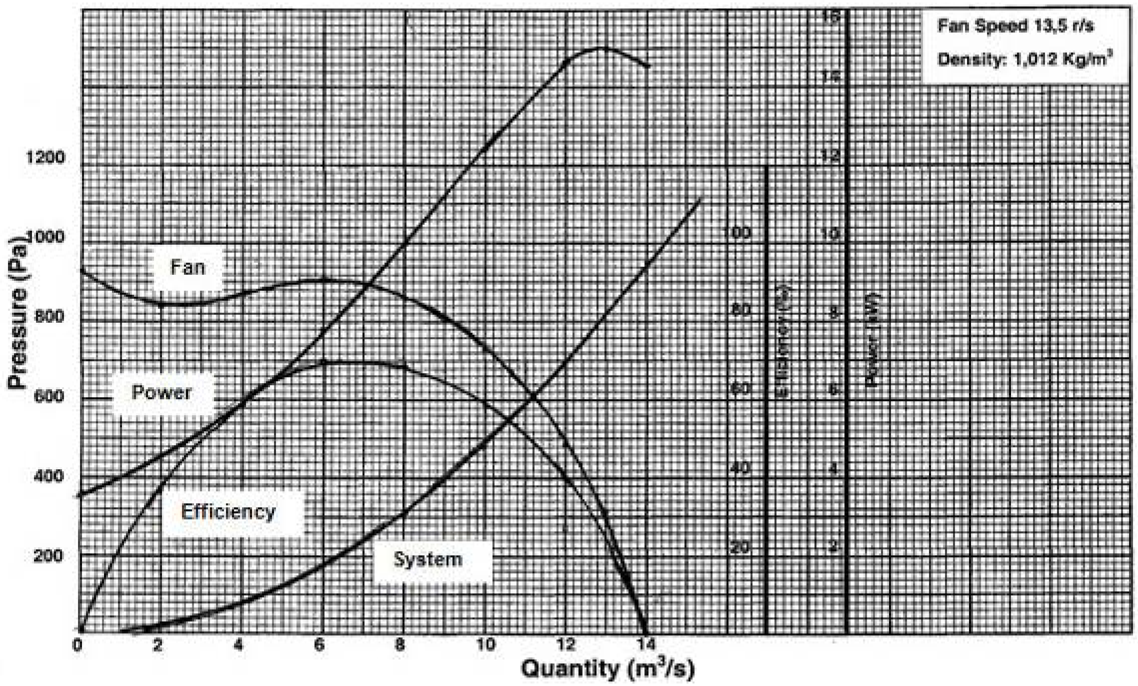


Figure 4.19 System resistance curve incorporated with fan curves

Read the graph as follows:

- Where the system resistance curve intersects the fan characteristic curve, is the Fan Operating Point (FOP)
- From this 'point', go horizontally on constant pressure line to the left and read off the pressure - 610 Pa. This is the pressure at which the fan must operate to overcome the resistance of the system
- Again from FOP, go vertically down on the constant quantity line and read off the quantity-11.2 m^3/s. This is the quantity the fan will handle at 610 Pa
- Where this vertical quantity line intersects the efficiency curve, go horizontally to the right and read off the efficiency of the fan on the efficiency scale-49%

- Read off the input power of the fan. Again, start at FOP and go vertically up to intersect the input power curve. Then go horizontally to the right and read off the input power of the fan on the input power scale-13.8 kW.

Thus FOP	=	11.2 m^3/s @ 610 Pa
Fan input power	=	13.8 kW
Fan efficiency	=	49%

Figure 4.20 illustrates how to read off the necessary information.

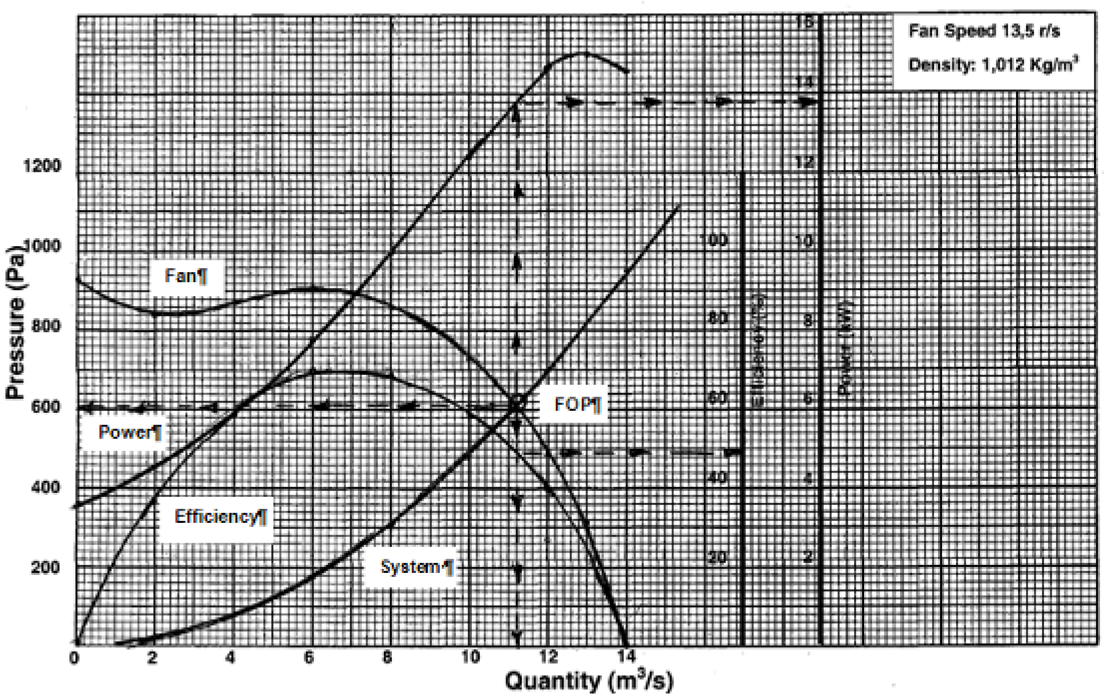


Figure 4.20 Final Graph to read off respective values

Remember, alternatively, the system resistance can be given, or the ventilation requirements can be given (that is the fan is to be installed in a system that must handle 11.2 m^3/s @ 610 Pa).

4.6 Fan Pressures and Fan Specifications

This aspect has been dealt with extensively in Chapter 1 and the student can refer back to it for revision purposes. Also refer to the Chapter on Fan Engineering in the Handbook, Mine Environmental Engineering for South African Mines.

4.7 Methods of Controlling Fan Performance

A fan is usually designed for one particular duty. It can however, happen that the fan will be required to operate efficiently at a number of different duties or different conditions.

These different requirements can be caused by:

- Increased resistance encountered
- Required increase in volume
- Required increase in pressure
- Change in air density.

Different duties of the same fan can be achieved by changing different parameters associated with it. This is clearly shown in Table 4.5.

Table 4.5 Summary of methods of changing duties for fans

Axial-flow fans	Centrifugal fans
Changed the speed of the fan (relatively simple with indirect drive)	Change the speed of the fan (relatively simple with indirect drive)
Change the pitch of the blades (if possible)	Alter damper control (if possible)
Remove some blades (balance must be maintained)	Baffle the fan (inefficient and expensive)
Remove one or more stages (if a multi-stage fan)	Alter length of blades (variable tip)
Change pitch of guide vanes (if possible)	Fit evaseè
Baffle the fan (inefficient and expensive)	
Fit evaseè	

4.7.1 Control Methods for Centrifugal Axial-Flow Fans

There are several ways of controlling the output of a centrifugal fan, these are:

- System dampers
- Inlet damper control
- Inlet guide vane control
- Speed control.

References

These are discussed in detail in Ventilation and Occupational Environment Engineering in Mines, Chapter 7. Fan Engineering and needs to be studied as well.

NB

The impacts of these various methods on the power consumed by the fans are also showed in this chapter. The most efficient of all these methods of controlling the output of a centrifugal fan is speed control as the efficiency of the fan is not affected when the speed of the fan is controlled.

4.8 Selection of Fans

NB

As the characteristic curves of different fans vary considerably, the selection of a fan for a particular duty depends on the requirements specified. If the duty of the fan is likely to be changed in the future, then an adjustable pitch fan would probably be most suitable. If the resistance of the mine is likely to increase, for example due to pillar extraction in main downcast shafts with the possible loss of the shaft as an effective airway, then a fan with a steep characteristic curve, that is a fan curve which shows relatively small volume changes for relatively large changes in pressure would be most suitable.

On the other hand, if the resistance of the mine is likely to be decreased, that is due to the development of additional return airways and up-cast shafts (in parallel mode), then a fan with a relatively flat characteristic curve, that is a fan curve showing comparatively large volume changes with relatively small pressure changes, would be most suitable.

All other things being equal, the fan, which performs at maximum efficiency at the required duty, would be selected. Other points to be borne in mind when selecting a fan are:

- Ease of installation
- Noise level, especially if the fan is to be situated close to hoists or on surface near residential areas
- Reliability
- Ease of maintenance.

4.9 Siting of Fans (Where to put Fans)

The following are some of the points, which must be considered when determining the position of main fans:

- **Heat**: A fan is a machine and the input energy is converted into work and heat (inefficient portion). Fans **should not be** placed in the downcast system of hot deep mines unless this is unavoidable.
- **Accessibility**: Fans should be accessible in the event of a breakdown. This applies particularly to breakdowns, which may occur during the blasting or re-entry periods. Fans should, to ensure accessibility, be situated on surface or in the downcast air.

NB

- **Noise**: Fans should be situated where the noise level causes the least disturbance. Noisy fans situated near a residential area are a guarantee of complaints. Centrifugal fans are quieter than axial-flow fans and this is an important factor when considering the type of surface fan to install.
- **Leakage**: High pressure should be avoided to minimize air leakage. Airlocks should be planned before finalizing the fan site. Fans placed in series and some distance apart should be sited so that unavoidable air leakages will be from downcast to the up-cast system.

Rule

The practice in modern deep South African gold mines is to site the main fans at the top of the up-cast shaft with small booster fans used underground if required.

4.10 Safety Devices

Many safety devices can be fitted at fan installations. These include:

SHE

- Warning devices to indicate hot bearing temperatures
- Trip contacts
- Excess vibration cut-outs
- Self-closing louvres (doors) which operate if a power failure occurs
- Sensing devices for automatic pitch control
- A standby motor.

Rule

A telephone should be situated at all main fan sites.

4.11 Effects of NVP, Density, Speed and Evasee's on Fan Performance

There are several factors which influences the fan's performances. These are:

- The NVP
- Density of the air
- Speed of the fan
- Evaseès.

NB

These will now be discussed in detail. It is very important for the student to also understand the impact of multiple combinations of the above on the performance of the fan. Various aspects of the fan's performance will be influenced and each should be noted and understood accordingly.

4.11.1 Air Density Change

NB

When a fan is tested, whether by the manufacturer or by personnel on the mine, it is important to obtain the density of the air during the fan test. The air density is usually obtained at the **inlet** of the fan.

Rule

The performance curve (fan curve) of the fan is then plotted at the prevailing air density (density during the test). The air density at which fans are tested can obviously vary a great deal. It is because of this reason that manufacturers generally supply fan curves at a standard air density of 1.2 kg/m^3. When fan curves are plotted at a standard density of 1.2 kg/m^3, it then becomes relatively simple to compare the performance of various types of fans.

However, the performance curve of the fan must be altered to suit the air density at the position where the fan is to be installed before a correct selection can be made. The fan performance curves can be altered for air density changes by applying the following **laws** mentioned. (subscript '1' represents 'old' conditions and subscript '2' represents 'new' conditions):

Laws

1. Quantity remains constant, $Q_1 = Q_2$
2. Pressure varies directly as air density, $p \alpha \rho$

$$\therefore p_2 = p_1 \times \frac{\rho_2}{\rho_1}$$

3. Power varies directly as air density, power $p \alpha \rho$

$$\therefore \text{power}_2 = \text{power}_1 \times \frac{\rho_2}{\rho_1}$$

4. Efficiency (eff) remains constant, $\text{eff}_1 = \text{eff}_2$

NB

When plotting fan curves, all parameters **are plotted against quantity**. Quantity remains constant during an air density change (because the density affects the pressure in the airway as much as it affects the pressure of the fan). Hence the above parameters need only be moved up or down on a line of constant quantity when there is an air density change.

WORKED EXAMPLE 7

Figure 4.21 shows the performance curves of an axial flow fan. These curves were plotted at an air density of 1.2 kg/m^3. Re-plot these curves at a density of 1.3 kg/m^3.

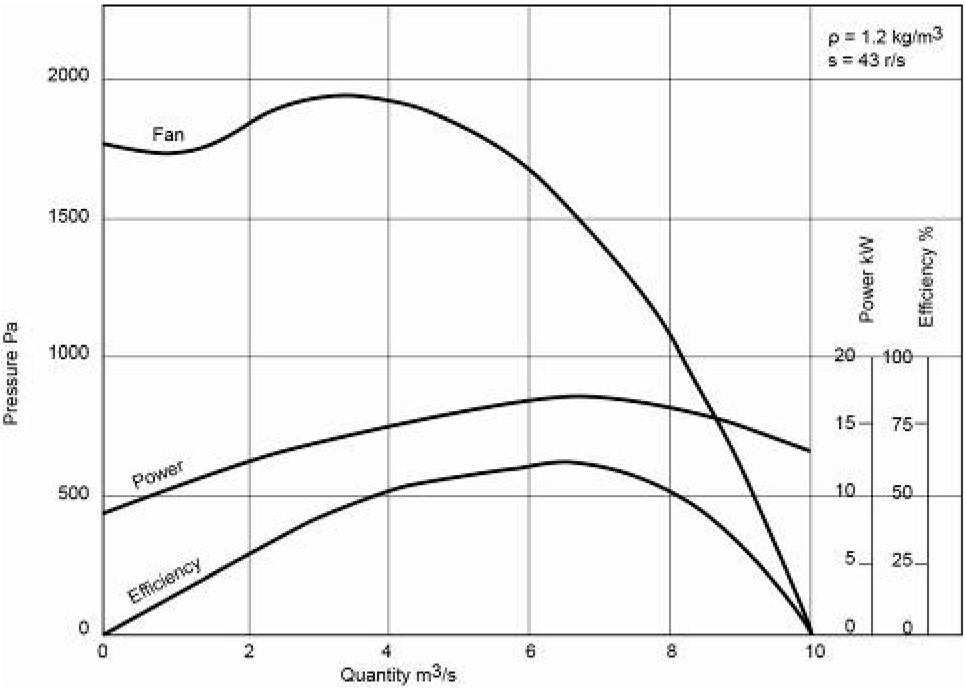


Figure 4.21 Fan curve at density 1.2 kg/m^3

Answer

a. Fan curve

i. Select any point on the original fan curve, for example 0 m^3/s at 1 770 Pa.

Quantity remains constant, so new quantity = 0 m^3/s

New pressure:

$$p_2 = p_1 \times \frac{\rho_2}{\rho_1}$$

$$= 1\,770 \times \frac{1.3}{1.2}$$

$$= 1\,918 \text{ Pa}$$

A ‘new’ point can now be plotted for 0 m^3/s @ 1 918 Pa.

(It shows that the ‘old’ point has moved vertically upwards, pressure increasing because density increased).

ii. Select another point on the original fan curve for example 2 m^3/s at 1 820 Pa. Quantity remains constant, so new quantity is 2 m^3/s.

New pressure:

$$p_2 = p_1 \times \frac{\rho_2}{\rho_1}$$

$$= 1\,820 \times \frac{1.3}{1.2}$$

$$= 1\,820 \times 1.083$$

$$= 1\,971 \text{ Pa}$$

Another ‘new’ point can be plotted for 2 m^3/s @ 1 971 Pa

(Again the ‘old’ point has moved vertically upwards).

iii. Other points on the original power curve can be corrected for density in a similar fashion until sufficient values are obtained to draw the complete power curve at the new density.

Note:

NB

(The ratio of $\frac{\text{new air density}}{\text{old air density}} = \frac{1.3}{1.2} = 1.083$, which can be used as a constant multiplier to establish all the new pressure points).

b. Power curve

i. Select any point on the original power curve, for example 0 m^3/s at 8.7 kW

Quantity remains constant, so new quantity is 0 m^3/s

New power:

$$power_2 = power_1 \times \frac{\rho_2}{\rho_1}$$

$$= 9.43 \text{ kW}$$

A ‘new’ point can now be plotted for 0 m^3/s @ 9.43 kW.

ii. Select another point on the original power curve e.g 2 m^3/s at 12.3 kW. Quantity remains constant, so new quantity 2 m^3/s.

New power:

$$power_2 = power_1 \times \frac{\rho_2}{\rho_1}$$
$$= 12.3 \times 1.083$$
$$= 13.32 \text{ kW}$$

A 'new' point can now be plotted for 2m^3/s @ 13.32 kW.

iii. Other points on the original power curve can be corrected for density in a similar fashion until sufficient values are obtained to draw the complete power curve at the new density. Again the ratio:

$$\frac{\rho_2}{\rho_1} = 1.083 \text{ can be used as a constant figure.}$$

c. Efficiency curve

The efficiency curve is plotted in the graph below; it must be remembered that the efficiency remains constant when the air density changes. The corrected fan curves are in Figure 4.22. The broken lines represent the new fan curve and new fan power curve at a new density of 1.3 kg/m^3. The efficiency line remains the same (speed of the fan at 43 r/s).

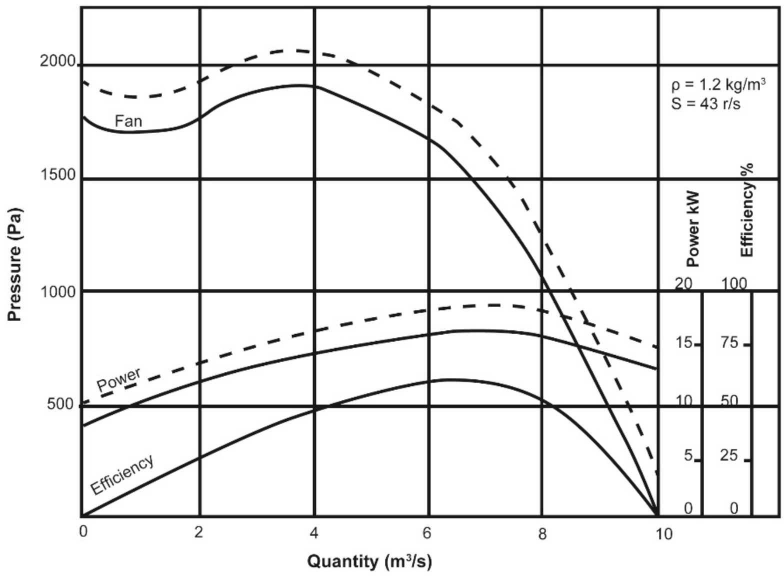


Figure 4.22 Corrected fan, power and efficiency curves for air density change

NB

Notice that the corrected fan curve intersects the original fan curve when the pressure equals zero. This will always happen and the reasons are that; quantity remains constant (at 10.0 m^3/s), and the old pressure equals zero, so no matter what ratio the old pressure is multiplied by, it will still remain zero.

Rule

It is also worth remembering that an increased air density will push the fan curve and power curve 'up', while a decreased air density will push the fan curve and power curve 'down'.

4.11.2 Speed Changes

NB

An important point to note when conducting a fan test is the speed at which the fan runs. Any change in the fan speed will obviously result in very different performance characteristics.

Hence when the curves of the fan are plotted from the fan test results, they are valid **only at the speed at which the fan was running during the test.**

If the fan is to be used at a speed other than that which is specified on the fan curve, the curves have to be corrected to the relevant speed.

Laws

The fan performance curves can be altered for fan speed applying the following **laws** (subscript '1' represents 'old' conditions and subscript '2' represents 'new' conditions):

1. Quantity varies directly as fan speed, $Q \alpha s$ (s the speed)

$$\therefore Q_2 = Q_1 \times \frac{s_2}{s_1}$$

2. Pressure varies directly as fan speed squared, $p \alpha s^2$

$$\therefore P_2 = P_1 \times \left(\frac{s_2}{s_1}\right)^2$$

3. Power varies directly as fan speed cubed, power αs^3

$$\therefore Power_2 = Power_1 \times \left(\frac{s_2}{s_1}\right)^3$$

4. Efficiency remains constant, $eff_1 = eff_2$

Unlike the air density change, both quantity and pressure will change with speed change.

WORKED EXAMPLE 8

Figure 4.23 shows the performance curves of an axial flow fan. These curves were plotted at a fan speed of 11.5 r/s. Re-plot these curves at a new fan speed of 13.8 r/s.

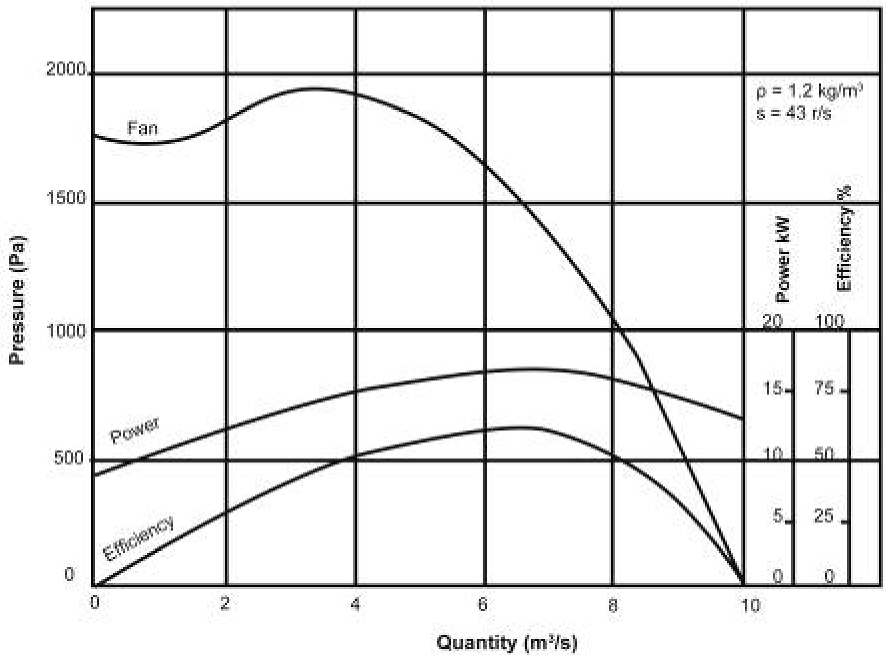


Figure 4.23 Fan, power and efficiency curves at 11.5r/s

Answer

Select a complete range of points from the original fan curve and original power curve, and tabulate as shown (Table 4.6) (read off values for power and fan pressure for each Q value selected):

Table 4.6 Fan performance data

Quantity (m^3/s)	Pressure (Pa)	Power (kW)
0.0	1 770	8.7
2.0	1 820	12.3
4.0	1 910	15
6.0	1 670	16.5
8.0	1 090	16
9.0	610	15
10.0	0	13

Remember when tabulating these figures that power is plotted against quantity, so each power read off should be read according to the quantity selected. Any number of points on the original curves could have been selected. Table 4.6 represents all the 'old' conditions denoted by subscript '1'.

a. Quantity: Quantity varies directly as fan speed and new quantity.

$$Q_2 = Q_1 \times \frac{s_2}{s_1}$$

Hence each quantity in the above table must be multiplied by:

$$\frac{13.8}{11.5} = 1.2$$

b. Pressure: Pressure varies directly as fan speed squared,

$$P_2 = P_1 \times \left(\frac{s_2}{s_1}\right)^2$$

Hence each pressure in the above table must be multiplied by the new factor and new pressure factor then:

$$\left(\frac{13.8}{11.5}\right)^2 = 1.44$$

c. Power: Power varies directly as fan speed cubed, and new power,

$$Power_2 = Power_1 \times \left(\frac{s_2}{s_1}\right)^3$$

Hence each power in the above table must be multiplied by the new power factor which is:

$$\left(\frac{13.8}{11.5}\right)^3 = 1.728 \text{ say, } 1.73$$

The new updated results are shown in the Table 4.7.

Table 4.7 Updated fan performance data

Q_1 (m^3/s)	$Q_2 = Q_1 \times 1.2$ (m^3/s)	P_1 (Pa)	$P_2 = P_1 \times 1.44$ (Pa)	$Power_1$ (kW)	$Power_2 = power_1 \times 1.73$ (kW)
0.0	0	1 770	2 549	8.7	15.0
2.0	2.4	1 820	2 621	12.3	21.3
4.0	4.8	1 910	2 750	15	26.0
6.0	7.2	1 670	2 405	16.5	28.6
8.0	9.6	1 090	1 570	16	27.7
9.0	10.8	610	878	15	26.0
10.0	12.0	0	0	13	22.5

NB

(Note: Two tabulations were made for the purpose of explanation. It is necessary to do both. Table 4.7 would be sufficient). The new curves can now be drawn using the results shown in the table above. Figure 4.24 show the new curves in broken lines.

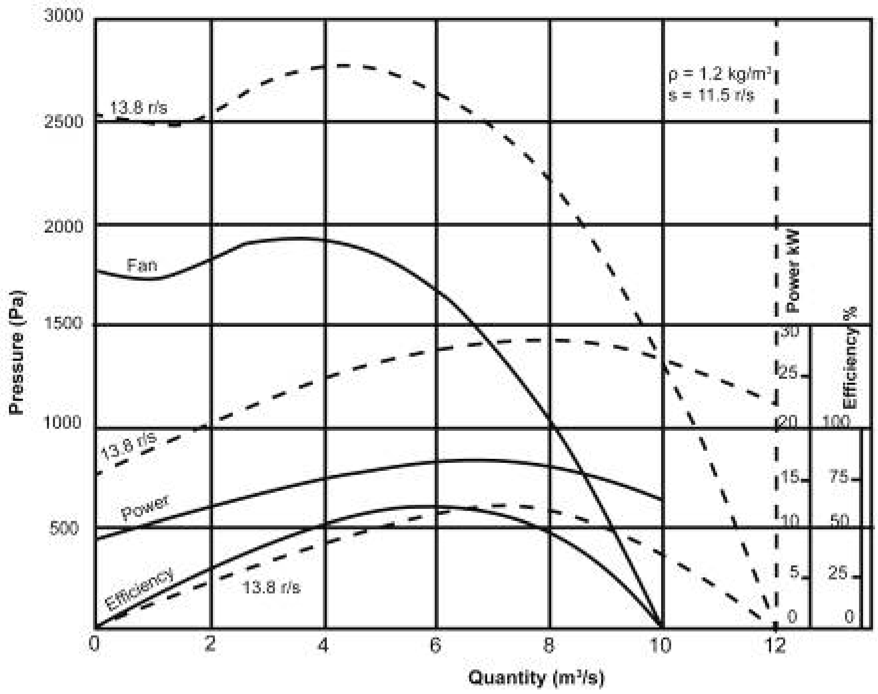


Figure 4.24 New fan, power and efficiency curves for increased speed change

- The new fan curve runs more or less parallel to the old one. The fan curve moves up for increased speed and down for decreased speed.
- The new power curve is plotted against **new quantity**. It is a common mistake to plot new power against old quantity.
- Speeding up a fan results in a great increase in power because of power's cubic variation with speed, for example if the fan speed was doubled the fan would require $(2)^3 = 8$ times as much power. The reverse is also true for a decrease in fan speed.

NB

NB

- The new efficiency curve is plotted on the graph. It should be remembered that when there is a speed change, the efficiency remains constant. However, efficiency is plotted against quantity and the quantity does change with speed. Hence the new efficiency curve moves to the right when the fan speed increases and to the left when it decreases.
- Any point on the original fan curve moves to its new position as if it was moving along a square law curve (Note: A system resistance curve is a square law curve).

This can be proved by determining the relationship between pressure and quantity for any point on the original curve and comparing it with the same relationship of the corresponding point on the new fan curve. If the relationship is the same, the point has moved to its new position by following a square law curve. For example in the previous graph, the point marked ‘a’ is any point on the original fan curve. With the increased speed, ‘a’ moves to ‘a1’ in the new can curve.

$$\text{Relationship 'a'} = \frac{P}{Q^2}$$

$$= \frac{P}{Q^2}$$

$$= 17.03\ Ns^2/m^8$$

$$\text{Relationship '}a_1\text{'} = \frac{P}{Q^2}$$

$$= \frac{1570}{9.6^2}$$

$$= 17.04\ Ns^2/m^8$$

Hence point ‘a’ moved to ‘a_1’ as if it were moving along a square law curve. This is a very useful point to remember concerning fan speed changes.

4.11.3 Density and Speed Change

It is sometimes necessary to change fan curves for both speed and density. This can either be done separately, that is first to change the curve for speed and then change that curve for density, or vice versa), or the changes can be made simultaneously.

It can be represented as follows:

$$P_2 = P_1 \times \frac{\rho_2}{\rho_1} \times \left(\frac{s_2}{s_1}\right)^2 \text{ Pa}$$

$$\text{Power}_2 = \text{Power}_1 \times \frac{\rho_2}{\rho_1} \times \left(\frac{s_2}{s_1}\right)^3 \text{ kW}$$

4.11.4 Natural Ventilation Pressure

NB

Natural ventilation pressure can have quite a substantial effect on the operating point of a fan and there is more than one method of determining the fan operating point when NVP is involved. The value of NVP is **constant for any quantity**.

Rule

Hence it can be represented graphically as a horizontal straight line through a single pressure value.

Attribute	NVP Assisting	NVP Opposing
Air direction	Move air in the same direction as the fan	Move air in the opposite direction to the fan.
Resistance curve	**'Lower' the system resistance curve** against which the fan operates. This means the fan will deliver more air at a lower pressure (refer to graph).	'Raise' the system resistance curve against which the fan operates. This means that the fan will handle less air at a higher pressure.
Curve	The value of NVP can be subtracted from the system resistance curve at several different quantities until sufficient points are obtained to plot a (system-NVP) curve.	The amount of NVP can be added onto the system resistance curve at several different quantities until sufficient points are obtain to plot a (system + NVP) curve.
FOP	Given by the intersection of the (system-NVP) curve and the fan curve.	Found at the intersection of the (system+ NVP) curve and the fan curve.

WORKED EXAMPLE 9

Figure 4.25 shows the performance curve of a mine fan. The fan is to be installed in a system, which has a resistance of 0.329 Ns^2/m^8. NVP of 300 Pa will **assist** the fan after installation. Determine the operating point of the fan and the system after installation.

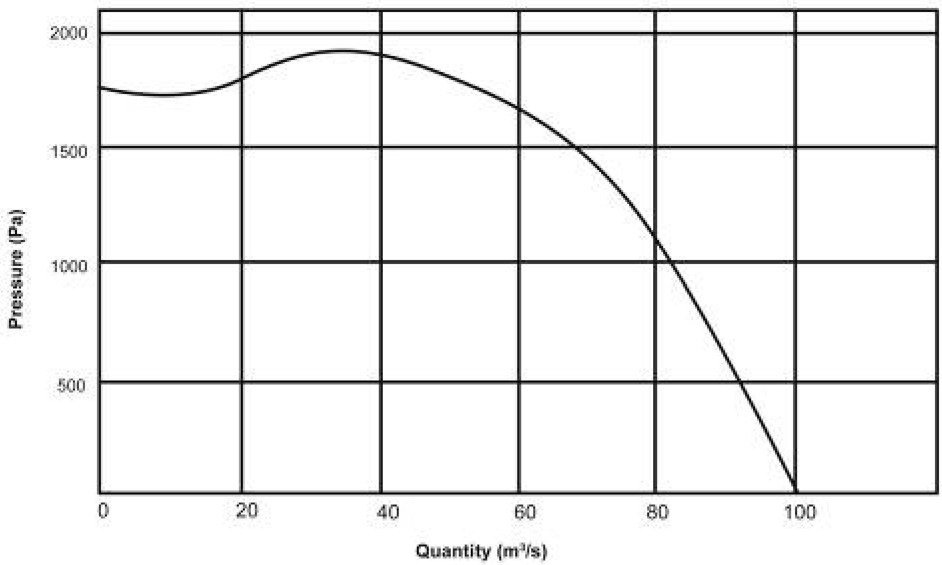


Figure 4.25 Original fan curve for system

Answer

A check should be made that the fan curve, system resistance curve and NVP are all at the same air density, which should be the air density where the fan is to be installed. No mention is made of air density in this case so it can be assumed that the fan, system and NVP are all at the same density.

i. Plot the system resistance curve for R = 0.329 Ns^2/m^8 by using the equation $P = RQ^2$.

ii. Plot a curve representing (system - NVP) by subtracting 300 Pa from the system curve at various quantities for example:

1. At 0 m^3/s, system pressure = 0 Pa
   $\therefore$(system - NVP) point = (0 - 300) Pa
   = -300 Pa (at 0 m^3/s)

2. At 20 m^3/s, system pressure = 132 Pa
   $\therefore$(system - NVP) point = 132 - 300) Pa
   = -168 Pa (at 20 m^3/s)

3. At 40 m^3/s, system pressure = 526 Pa
   $\therefore$(system - NVP) point = (526 - 300) Pa
   = 226 Pa (at 40 m^3/s)

4. At 60 m^3/s, system pressure = 1 184 Pa
   $\therefore$(system - NVP) point = (1 184 - 300) Pa
   = 884 Pa (at 60 m^3/s)

Continue using this method until sufficient points have been obtained to plot a (system - NVP) curve.

iii. Where the (system - NVP) curve cuts the fan curve, read off the **new** operating **point of the fan** as 71.5 m^3/s @ 1 380 Pa.

iv. The (system - NVP) curve is an **'imaginary'** curve which helps in finding the fan operating point, **but is not an actual system resistance curve.**

The system operating point (SOP) can **only be found on the actual system resistance curve**. Clearly the system must handle the same volume of air as the fan, so where the 71.5 m^3/s quantity line cuts the system resistance curve, read off the operating point of the system as:

71.5 m^3/s @ 1 680 Pa

v. Check:

In order to pass 71.5 m^3/s of air through the system, a pressure of 1 680 Pa is required (read from the system curve). Under normal circumstances the fan would then have to produce a pressure of 1 680 Pa. However, because NVP assists the fan, the fan only has to produce 1 380 Pa, the other 300 Pa being produced by the (assisting) NVP.

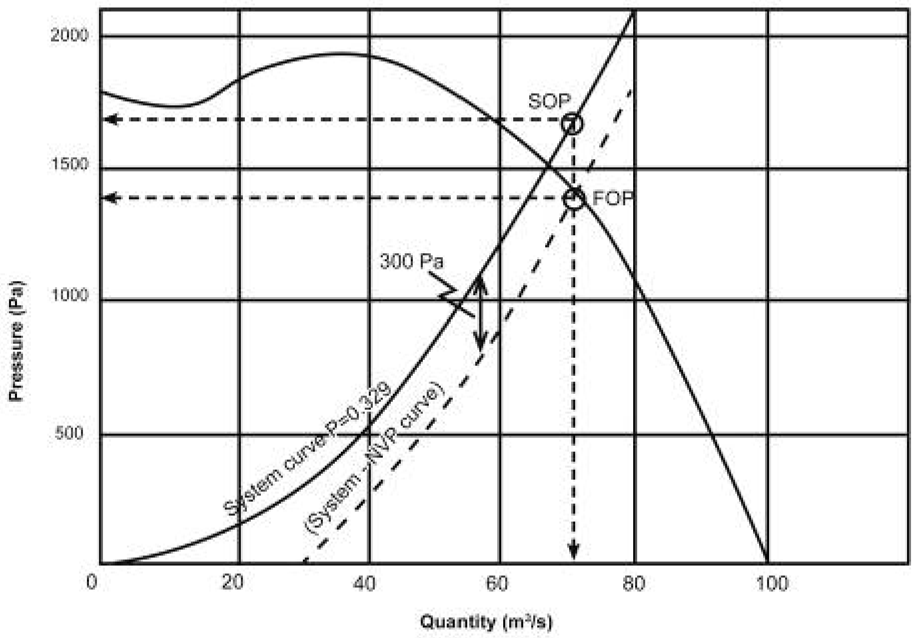


Figure 4.26 New system resistance curve with NVP assisting

WORKED EXAMPLE 10

Figure 4.27 shows the performance curve of a mine fan, which is to be installed in a system, which has a resistance of 0.216 Ns^2/m^8. The fan will be opposed by NVP of 400 Pa after installation. Determine the operating point of the fan and the system after installation.

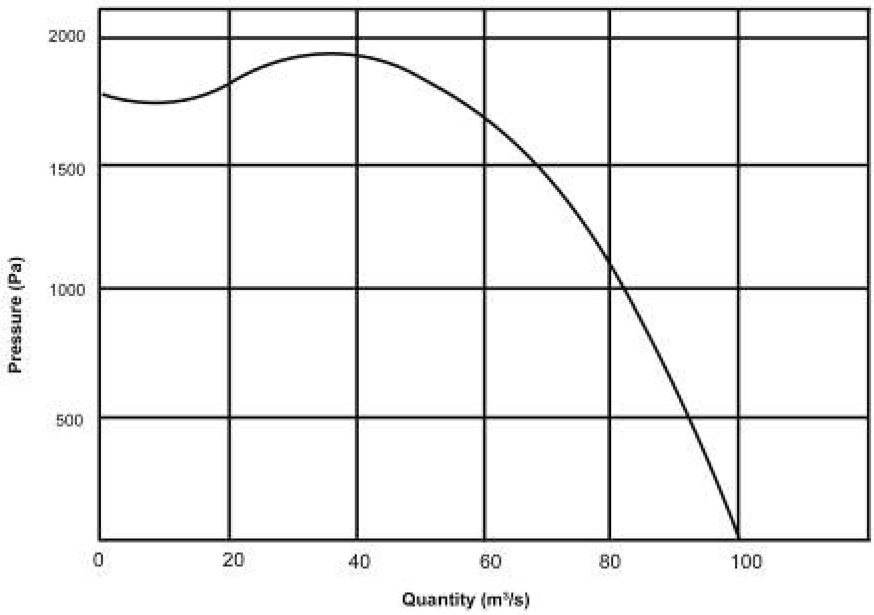


Figure 4.27 Original fan curve for system

Answer

Once again it is assumed that the fan, system and NVP are already all at the air density where the fan is to be installed.

i. Plot the system resistance curve for R = 0.216 Ns^2/m^8 by using the Equation $P = RQ^2$.

ii. Plot a curve representing (system + NVP) by adding 400 Pa to the system curve at various quantities e.g.

1. At 0 m^3/s, system pressure = 0 Pa
   $\therefore$(system + NVP) point = (0 + 400) Pa
   = 400 Pa (at 0 m^3/s)

2. At 20 m^3/s, system pressure = 86 Pa
   $\therefore$(system + NVP) point = (86 + 400) Pa
   = 486 Pa (at 20 m^3/s)

3. At 40 m^3/s, system pressure = 346 Pa
   $\therefore$(system + NVP) point = (346 + 400) Pa
   = 746 Pa (at 40 m^3/s)

Continue using this method until sufficient points are obtained to plot a (system + NVP) curve.

iii. Where the (system + NVP) curve cuts the fan curve, read off the operating point of the fan as:

69.7 m^3/s @ 1 450 Pa

iv. Once again the (system + NVP) curve is an 'imaginary' curve which helps to find the fan operating point, but is not an actual system resistance curve. The system operating point can only be found on the actual system resistance curve. The system must handle the same volume of air as the fan, so where the 69.7 m^3/s line cuts the system resistance curve, read off the system operating points as:

69.7 m^3/s @ 1 050 Pa

v. To pass 69.7 m^3/s of air through the system requires a pressure of 1 050 Pa (read from the system curve). Under normal circumstances the fan would then have to produce a pressure of 1 050 Pa. However, because NVP opposes the fan, the fan has to produce 1 450 Pa that is enough additional pressure to overcome the system resistance and the opposing NVP.

The final pressure of the fan is therefore shown as follows:

400 Pa $_{\text{NVP}}$ + 1050 Pa $_{\text{system}}$ = 1 450 Pa $_{\text{fan pressure}}$.

The resultant curves are shown in Figure 4.28.

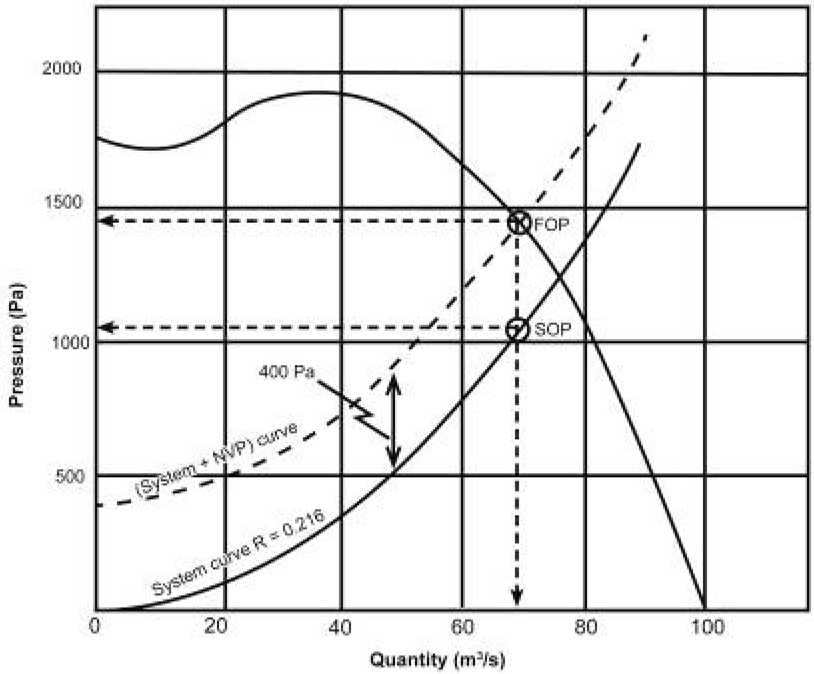


Figure 4.28 Final system resistance curve NVP opposing

4.11.5 Effects of Evaseès

NB

If a fan is operating in a system and an evaseè is fitted at the discharge end of the fan, both the fan and the system operating points will change. This change is usually beneficial as it results in a larger quantity of air through the system. The evaseè can either be fitted to the fan discharge (in exhausting systems) or to the system discharge (in forcing systems).

Rule

Figure 4.29 shows an evaseè fitted to the discharge side of an exhaust fan (Rule of thumb: The evaseè is 'closest' to the fan and will therefore influence the fan curve)

Fitting an Evaseè to a Fan Discharge

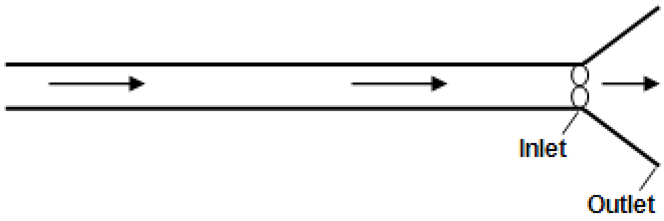


Figure 4.29 Fan fitted to discharge side of exhaust fan

NB

The fitting an evaseè to the fan discharge results in an improved fan performance curve with no change in the system resistance. The magnitude of the improvement in fan performance depends on the characteristics of the actual evaseè (including efficiency).

WORKED EXAMPLE 11

An 85% efficient evaseè with an inlet area of 0.5 m^2 and an outlet area of 1.2 m^2 is to be fitted to the discharge of an axial flow fan. Plot the new fan characteristics curve. The prevailing air density is 1.2 kg/m^3.

Answer

(The completed fan curve is shown at the end of this answer).

i. Assume the fan handles a quantity of 4 m^3/s

Then 4 m^3/s flows through the evaseè (also remember that Q = UA).

Velocity pressure at the evaseè inlet

$$= \frac{U_{in}^2 \rho}{2}$$

$$= \frac{\left(\frac{4}{0.5}\right)^2 \times 1.2}{2}$$

$$= 38\text{Pa}$$

Velocity pressure at evaseè outlet, VP

$$= \frac{U_{out}^2 \rho}{2}$$

$$= \frac{\left(\frac{4}{1.2}\right)^2 \times 1.2}{2}$$

$$= 7 \text{ Pa}$$

Loss in $VP_{in} - VP_{out}$ = theoretical gain in static pressure.

$$= (38 - 7) \text{ Pa}$$

$$= 31 \text{ Pa}$$

$$\text{Actual gain in SP} = \frac{85}{100} \times 31$$

$$= 26 \text{ Pa}$$

The actual gain in SP or 26 Pa is now added to the fan curve at a quantity of 4 m^3/s, to obtain one point on the new fan curve (marked 'a' on graph):

ii. Assume the fan handles a quantity of 6 m^3/s

Actual gain in SP

$$= (VP_{in} - VP_{out}) \times \frac{\text{evaseè efficiency}}{100}$$

$$= \left(\frac{\left(\frac{6}{0.5}\right)^2 \times 1.2}{2} \right) - \left(\frac{\left(\frac{6}{1.2}\right)^2 \times 1.2}{2} \right) \times \frac{85}{100}$$

$$= (86 - 15) \times \frac{85}{100}$$

$$= 60 \text{ Pa}$$

The actual gain in SP of 60 Pa is added to the fan curve at a quantity of 6 m^3/s to get another point on the new fan curve (marked 'b' on the graph, Figure 4.30). This procedure is repeated until sufficient points are obtained to plot the complete new fan curve as shown in Figure 4.30.

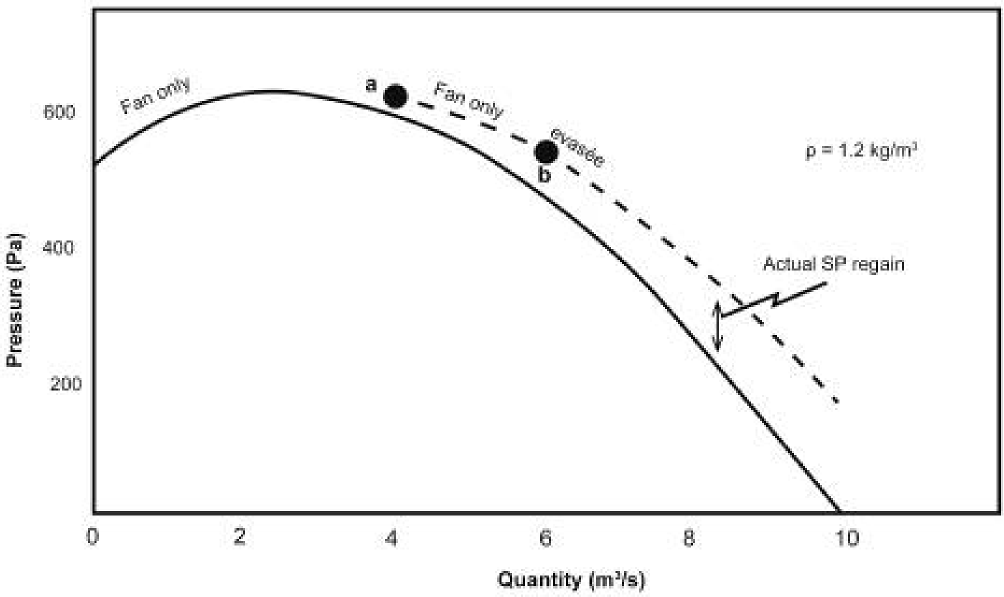


Figure 4.30 New plotted fan curve for evaseè (imaginary curve)

NB

iii. It must be remembered that when an evaseè is fitted to the fan outlet, the fan + evaseè are considered as a single unit, represented by the new fan curve which has been plotted.

Fitting an Evaseè to a System Discharge

Figure 4.31 shows an evaseè fitted at the discharge end of a column attached to the fan. Fitting an evaseè to a system discharge results in a reduced system resistance curve, with no change in the fan curve.

Rule

(Rule of thumb: The evasee is 'closest' to the column and will therefore influence the system resistance curve associated with the column). Again, the magnitude of the reduction in system resistance depends on the characteristics of the evaseè.

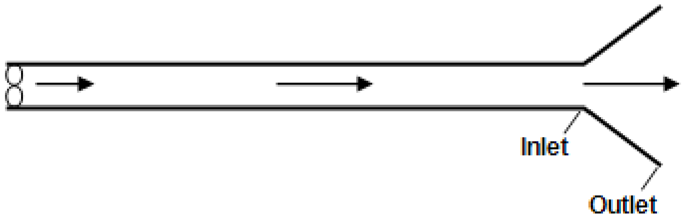


Figure 4.31 Evaseè fitted at discharge end of column

WORKED EXAMPLE 12

An 80 % efficient evaseè with an inlet area of 0.5 m^2 and an outlet area of 1.2 m^2 is fitted to the discharge of a ventilation column. The air density is 1.0 kg/m^3. Plot the new system resistance curve.

Answer

(The completed system curve is shown at the end of this answer).

i. Assume a quantity of 8 m^3/s flows through the system (also remember Q = UA).

Then 8 m^3/s flows through the evaseè.

$$VP_{in} \text{ at evaseè inlet} = \frac{U_{in}^2 \rho}{2}$$

$$= \frac{\left(\frac{8}{0.5}\right)^2 \times 1.0}{2}$$

$$= 128 \text{ Pa}$$

$$VP_{out} \text{ at evaseè outlet} = \frac{U_{out}^2 \rho}{2}$$

$$= \frac{\left(\frac{8}{1.2}\right)^2 \times 1.0}{2}$$

$$= 22 \text{ Pa}$$

Theoretical gain in SP $= VP_{in} - VP_{out}$

$= (128 - 22)$ Pa

$= 106$ Pa

Actual gain in SP $= (106 \times \frac{80}{100})$

$= 85$ Pa

This actual gain in SP of 85 Pa is now **subtracted from the system resistance curve** at a quantity of 8 m^3/s to obtain one point on the new system resistance curve (marked 'a' on the graph).

At 8 m^3/s old system pressure = 600 Pa

$\therefore$ At 8 m^3/s new system pressure = 515 Pa

New system resistance, $R = \frac{P}{Q^2}$

$= \frac{515}{8^2}$

$= 8.05 \text{ Ns}^2/\text{m}^8$

Use this resistance and the formula $P = RQ^2$ to plot the new system curve as shown in Figure 4.32.

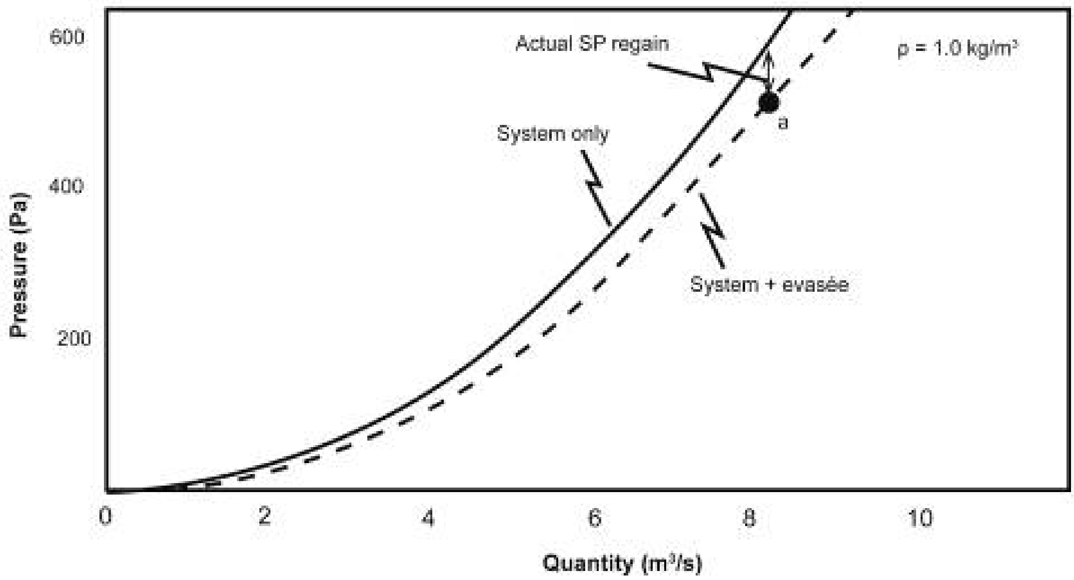


Figure 4.32 Effect of evaseè on system resistance

ii. It must be remembered that when an evaseè is fitted to the system discharge, the system plus evaseè are considered as a single unit represented by the new system resistance curve, which has been plotted.

4.12 Fans in Series and Parallel

4.12.1 Fans in Series

Definition

When fans are installed one behind the other so that each fan handles the same quantity of air, the fans are said to be in **series**. The fans could either be installed one directly behind the other or they could be installed some distance apart. Provided they handle the same quantity of air they are still in series.

With two fans installed in series the same quantity of air will pass through each of the fans, but each fan will add its own pressure to the air (according to its capabilities).

NB

Only with identical fans (i.e. ones which have the same performance curve) **will each fan add the same amount of pressure to the air.** The total amount of pressure added to the air by fans in series is the sum of the individual pressures added by each fan.

Laws

Hence two laws for fans in series can be written as:

1. $Q_T = Q_1 = Q_2$
2. $P_T = P_1 + P_2$

If the performance curves of each of the fans, and the resistance of the system in which they will operate is known, a combined performance curve of both fans can be drawn and the fan and system operating points can be obtained. This is done by applying the above laws, as shown in the Worked Example 13.

WORKED EXAMPLE 13

Figure 4.33 shows the performance curves of two different fans, which are to be installed in a system, which requires a pressure of 210 Pa to pass a quantity of 2 m^3/s. Determine the operating point of each fan and the system after installation.

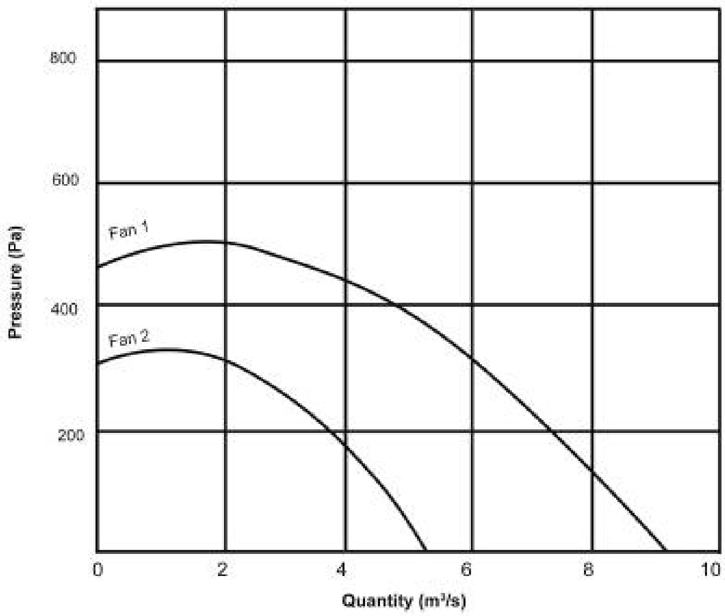


Figure 4.33 Two fans to be installed in series

i. This is done by the following method, which is very similar to plotting airways in series where the pressures are added at constant quantity.

1. If the airflow was 0 m^3/s, 0 m^3/s would flow through fan 1, 0 m^3/s would flow through fan 2 and 0 m^3/s would flow through both fans combined.

   Therefore:

   $Q_T = Q_1 = Q_2$

   But at 0 m^3/s, pressure added by fan 1 = 470 Pa

   And at 0 m^3/s pressure added by fan 2 = 315 Pa

   $\therefore$ Pressure added by both fans together,

   $P_T = P_1 + P_2$

   $= 470 + 315$

   $\therefore P_T = 785$ Pa

   Now a point on the combined curve of both fans can be plotted at 0 m^3/s @ 785 Pa.

2. At 2 m^3/s, $Q_T = Q_1 = Q_2 = 2\ m^3/s$

   Also at 2 m^3/s, $P_1 = 500$ Pa

   And

   $P_2 = 330$ Pa

   $\therefore P_T = 830$ Pa

   Plot a point for 2 m^3/s @ 830 Pa

3. Continue this method until sufficient points are obtained to plot the combined curve of both fans.

NB

(**Note**: At a quantity of 5.4 m^3/s the pressure added by fan 2 is 0, so that theoretically, the combined pressure of both fans equals the pressure added by the fan 1 above. For any higher quantities the combined curve follows the curve of the larger fan. However, in practice the smaller fan acts as a resistance and hence the available fan pressure is reduced).

Rule

ii. The combined curve which has now been plotted is an 'imaginary' curve representing both fans together in series i.e. as if there was only one fan. It should be remembered that fans only operate on their own curves and systems only operate on their own curves.

iii. Resistance of system, $R = \dfrac{P}{Q^2}$

$$= \frac{210}{2 \times 2}$$

$$= 52.5 \text{ Ns}^2/\text{m}^8$$

Plot the system resistance curve for R = 52.5 Ns^2/m^8 using the Equation, $P = RQ^2$.

iv. Where the system resistance curve cuts the combined curve, read off the total quantity through the system and the total pressure drop across the system: 3.6 m^3/s @ 680 Pa.

This is in fact the system operating point because the system handles the total quantity and requires the total pressure drop. (Also the point is on the system curve).

Now $Q_T = Q_1 = Q_2 = 3.6\ m^3/s$

$\therefore$ Move vertically down from the system operating point to each individual fan curve to obtain the operating point of each fan. This is shown in Figure 4.34.

Fan 1 operating point = 3.6 m^3/s @ 460 Pa

Fan 2 operating point = 3.6 m^3/s @ 220 Pa

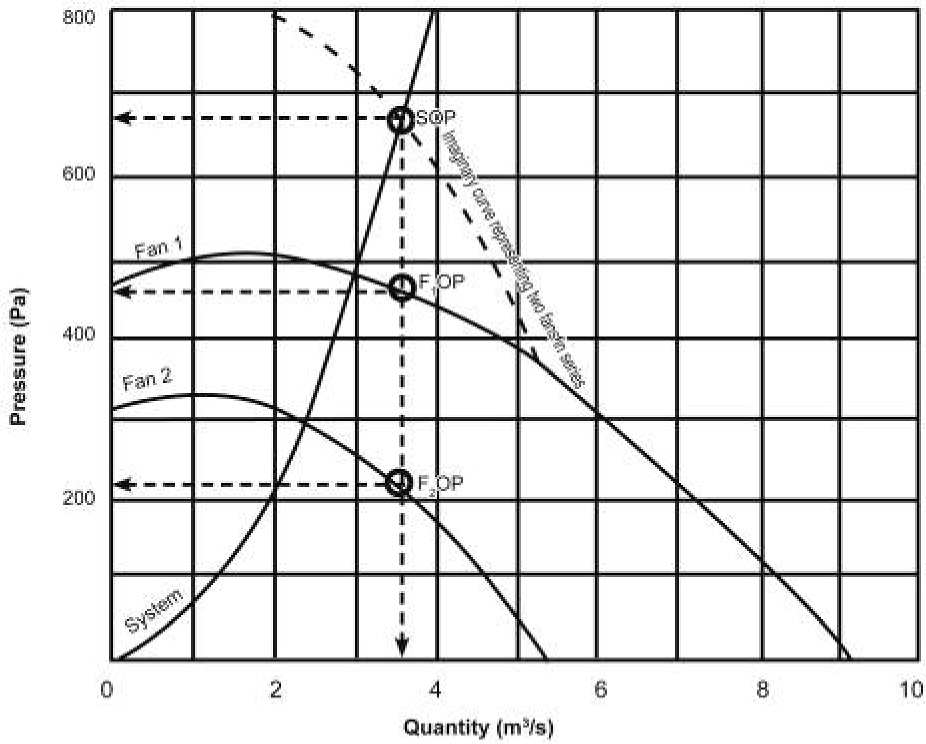


Figure 4.34 Resultant fan curve for two different fans in series

v. Check:

Q_T	=	Q_1	=	Q_2
3.6	=	3.6	=	3.6 m^3/s
P_T	=	P_1	+	P_2
680	=	460	+	220 Pa

!

vi. A common mistake made when analysing the results of plotting fans in series is to read off the individual fan operating points at the point where the system curve cuts each fan curve. This is **completely wrong** as it would mean that each fan and the system handle different quantities of air, so the law $Q_T = Q_1 = Q_2$ is not satisfied.

Fans are normally installed in series when there is a high system resistance**.** For example it would have been wrong to install the two fans mentioned in the last worked example in series if the system resistance curve cut fan 1 at say, 6 m^3/s @ 315 Pa, as fan 1 would have been able to pass this quantity without the assistance of fan 2.

NB

It should be remembered that the main idea behind plotting fan and system curves, and obtaining operating points, is to be able to predict in advance what the fan and system operating points will be.

SHE

This helps to avoid making mistakes such as putting fans in series when they should have been put in parallel, or replacing a burnt-out fan with one which immediately goes into a stall, etc.

The same principles as described above can be used to plot more than two fans in series. It is always useful to draw a sketch of the installation to make the situation a little clearer.

4.12.2 Fans in (True) Parallel

Definition

When fans are installed side by side in such a way that they draw air from the same source and deliver it to the same place, the fans are to be in **parallel** that is they have a **common intake and a common return**. Figure 4.35 (A- D) show a few examples of fans in **true** parallel.

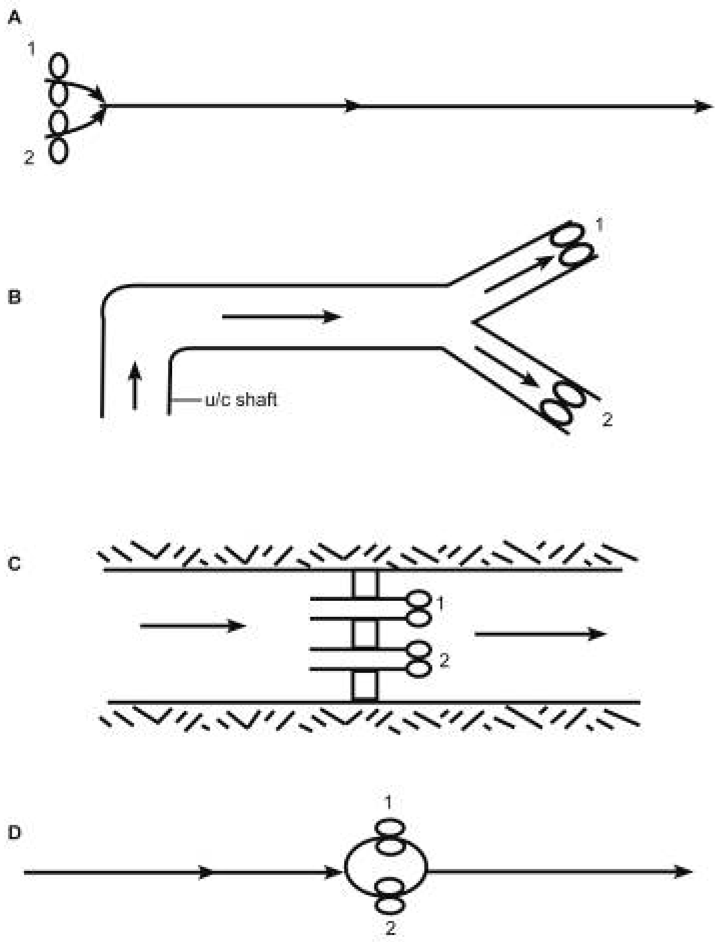


Figure 4:35 (A-D) Fans in true parallel

In case (A) and case (D) the y-pieces in which the fans are fitted must be reasonably short in length so that the fans still have a common intake and return.

NB

When fans are installed in parallel each fan must produce the same amount of pressure because they have the same intake point and the same discharge point (as with parallel airways). For the same reasons the total amount of pressure added by both fans equals the amount of pressure added by each individual fan. (Note: Not the sum of the pressures added by each fan).

However, each fan will handle a different quantity of air according to its capabilities. Only identical fans (those ones which have the same fan curve) will handle the same quantity of air if they are installed in parallel. The total quantity of air handled by fans in parallel is the sum of the individual quantities of the fans.

Hence two laws for fans in parallel can be written as:

Laws

1. $Q_T = Q_1 + Q_2$
2. $P_T = P_1 = P_2$

A combined performance curve representing the two fans in parallel can be drawn from the individual performance curves of each fan. The fan and system operating points can then be obtained by plotting the system resistance curve of the system in which the fans are to be installed. This can be done by application of the above laws as shown in the Worked Example 14.

WORKED EXAMPLE 14

Figure 4.36 shows the performance curves of two different fans which are to be installed in 105.4 m of 760 mm diameter column which has a K-factor of 0.0035 Ns^2/m^4. The prevailing density is 1.2 kg/m^3. Determine the operating point of each fan and the system after installation.

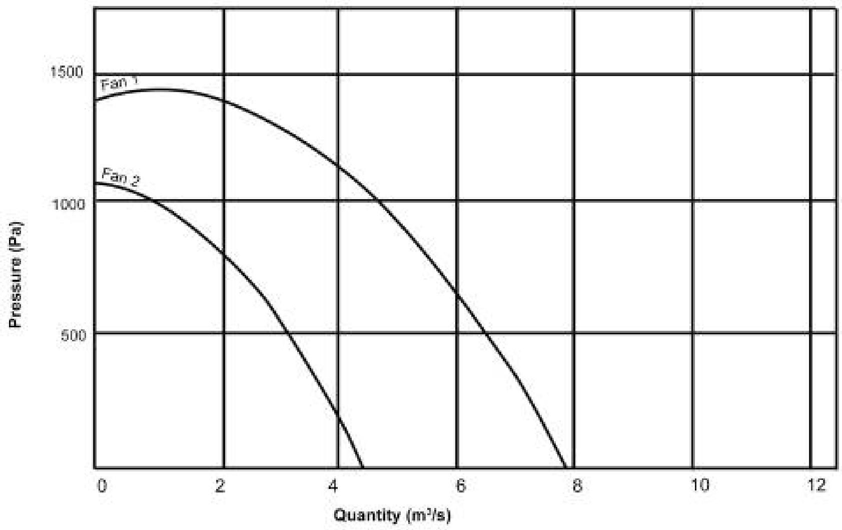


Figure 4.36 Two different fans to be installed in parallel

Answer

i. Plot a combined curve representing both fans in parallel. This can be done using the following method: (the method is similar to plotting airways in parallel where the quantities are added at constant pressure).

If the pressure was 0 Pa, and the pressure added by both fans would be 0 Pa.

i.e: $P_T = P_1 = P_2$

But at 0 Pa, quantity of fan 1 = 7.8 m^3/s

And at 0 Pa, quantity of fan 2 = 4.4 m^3/s

$\therefore$ Quantity handled by both fans together.

$Q_T = Q_1 + Q_2$

$= 7.8 + 4.4$

$\therefore Q_T = 12.2$ m^3/s

A point on the combined curve of both fans can now be plotted at 12.2 m^3/s @ 0 Pa

ii. At 250 Pa:

$P_T = P_1 = P_2 = 250$ Pa

Also at 250 Pa:

$Q_1 = 7.2$ m^3/s

And

$Q_2 = 3.9$ m^3/s

$\therefore Q_T = 11.1$ m^3/s

A second point can be plotted for 11.1 m^3/s @ 250 Pa

iii. At 500 Pa:

$P_T = P_1 = P_2 = 500$ Pa

Also at 500 Pa:

$Q_1 = 6.5$ m^3/s

And

$Q_2 = 3.3$ m^3/s

$\therefore Q_T = 9.8$ m^3/s

A third point can be plotted for 9.8 m^3/s @ 500 Pa.

iv. Continue this method until sufficient points are obtained to plot the combined curve of both fans.

NB

(Note: At pressure of 1 100 Pa the quantity handled by fan 2 is 0 m^3/s, so the combined quantity of both fans equals the quantity of fan 1 alone. For any higher pressures, the theoretical combined curve follows the curve of the larger fan, fan 1)

- Again, the combined curve which has now been plotted is an 'imaginary' curve representing both fans together in parallel, i.e. as if there was only one fan. It must be remembered that fans only operate on their own curves and systems only operate on their own curves.
- Resistance of system, R $= \frac{KCL}{A^3} \times \frac{\rho}{1.2}$

$$= \left(\frac{0.0035 \times 2.39 \times 105.4 \times 1.2}{0.4543^3 \times 1.2} \right)$$

$$= 9.43 \ Ns^2/m^8$$

Plot the system resistance curve for R = 9.69 Ns^2/m^8 using the Equation $P = RQ^2$.

- Where the system resistance curve cuts the combined curve, read off the total quantity through the system and the total pressure drop across the system read off as (system operating point):

8.5 m^3/s @ 700 Pa

Now:

$P_T = P_1 = P_2 =$ 700 Pa

$\therefore$ Move horizontally across from the system operating point to each individual fan curve to obtain the operating point of each fan as shown in Figure 4.37:

Fan 1 operating point = 5.9 m^3/s @ 700 Pa

Fan 2 operating point = 2.6 m^3/s @ 700 Pa

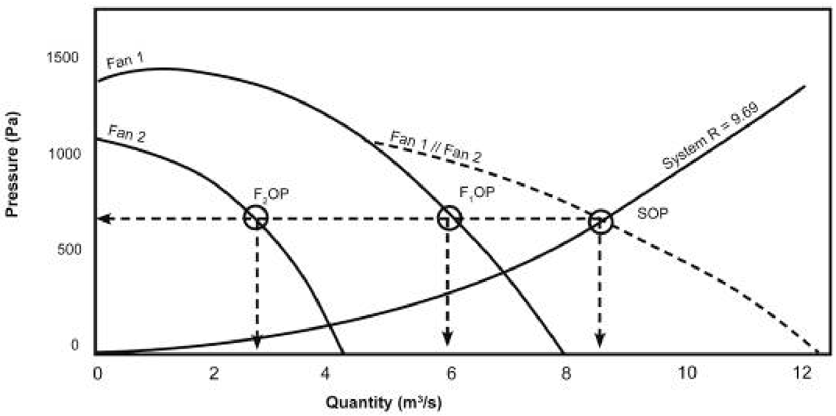


Figure 4.37 Fan operating and system operating points for two different fans in parallel

Check:

Q_T	=	Q_1	+	Q_2
8.5	=	5.9	+	2.6
P_T	=	P_1	=	P_2
700	=	700	=	700

- **Again a common mistake is to read off individual fan operating points at the point where the system curve cuts each fan curve**. This is completely wrong as it would mean that each fan and the system would operate at difference pressures and the law $P_T = P_1 = P_2$ would not be satisfied.
- When one or more of the fans in parallel has a marked stall zone, this region of the combined curve becomes difficult to plot. This is because for certain pressures, there is more than one quantity, which has to be added to the other fan curve. The result of this is that two and sometimes even three points are plotted at the same pressure for the combined curve Figure 4.38 clarifies this.

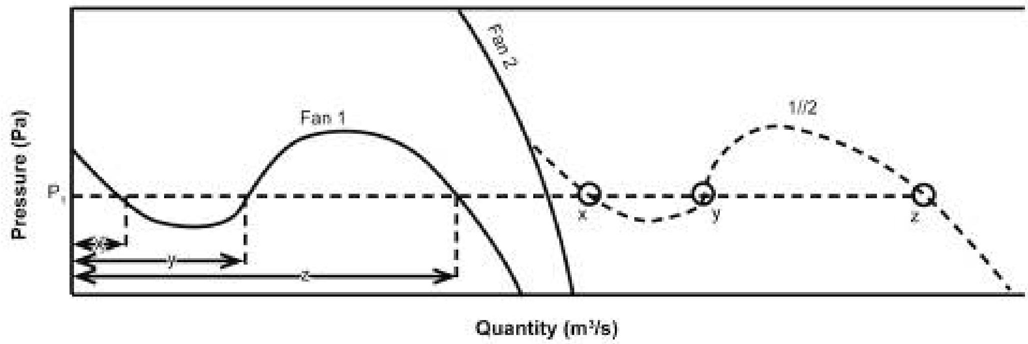


Figure 4.38 Fans in parallel with marked stall zone

From Figure 4.38, at the single pressure, P1, three different air quantities can be read from the curve for fan 1 (x, y and z). These three different quantities are added to the quantity of fan 2 at the same pressure and the points x, y and z are obtained. This makes the combined curve's 'stall zone' rather steeply inclined towards the left of the graph and sometimes resembles a figure of eight as shown in Figure 4.39.

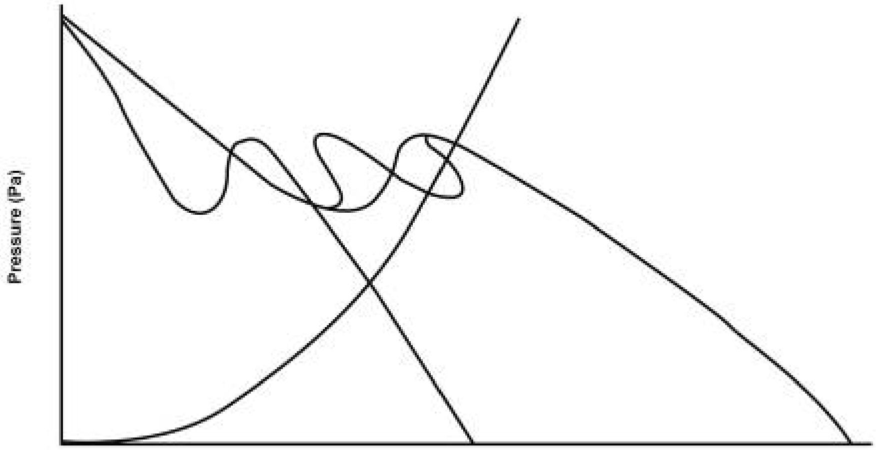


Figure 4.39 Fans in parallel showing distinct stall zone characteristics when combined

Rule

Fans are normally installed in parallel when there is a low system resistance, e.g. If the system curve cut fan 1 curve in the last example at 4 m^3/s @ 1 175 Pa, it would be wrong to place the fans in parallel as fan 1 would be able to cope on its own. The principles described above can also be used to plot more than two fans in parallel. Once again a sketch of the installation should be drawn for clarity.

4.12.3 Fans Combined in Series and Parallel

By following the principles and methods described for airways in series and for airways in parallel, it becomes relatively simple to plot combined curves for combinations of fans in series and parallel.

In the airflow section, when it was necessary to plot out system curves for a network of airways in series and parallel, the method used was to combine the various systems together (in series or in parallel) until one system resistance curve remained. This final system curve then represented the whole network.

The method is much the same for fans in series or parallel or combined in series and parallel.

Objective

The idea is to obtain one fan curve which represents all the fans together that is as if there was only one fan (hence the term ‘imaginary’ fan curve). The analysis can then be started at the intersection of this one fan curve and the system resistance curve.

It is not necessary to elaborate on this with a number of detailed worked examples, as the number of possible combinations is virtually unlimited. However, for each of the sketches below, a brief explanation of the method and a rough graph is given. Let us consider Figure 4.40 first.

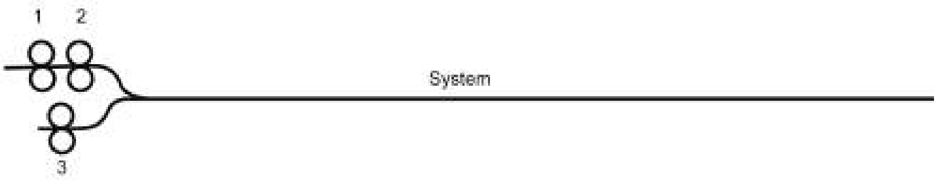


Figure 4.40 Combined system of fans series and parallel

i. Plot the combined curve for fans 1 and 2 in series. Call this curve x.

ii. Plot the combined curve of fan 3 and curve x in parallel. Call this curve y.

iii. Read off the system operating point at the intersection of y with the system curve.

iv. The last combination was placing fan 3 and x in parallel. Hence use parallel laws to go back on constant pressure to the two individual curves plotted in parallel (that is Fan 3 and curve x). This gives the operating point of fan 3 because it is on fan '3' curve but not of fan 1 and 2 (Remember curve x is 'imaginary')

v. This point on curve x gives the total quantity through fans 1 and 2 which equals the quantity through fan 1 and equals the quantity through fan 2.

$\therefore$Go back on constant quantity to the two individual fans to give the operating point of fan 1 and 2.

vi. Checks:

$Q_1 = Q_2$

$Q_{1+2}\ Q_3 = Q_{system}$

$P_1 + P_2 = P_3 = P_{system}$

vii. A rough graph of the above procedure is shown in Figure 4.41.

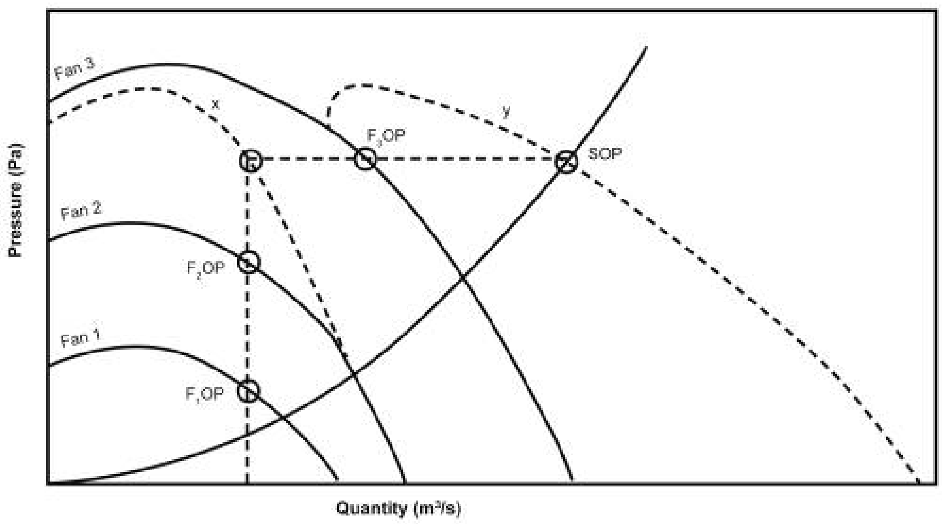


Figure 4.41 Solution for example 1 done above

1. Let's consider example 2 shown in Figure 4.42.

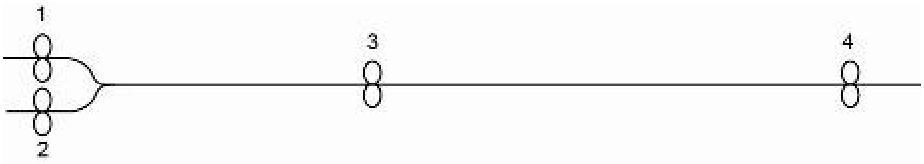


Figure 4.42 Fans 1 and 2 in parallel, with fans 3 and 4 in series

i. Plot the combined curve for fans 1 and 2 in parallel. Call this curve x.

ii. Plot the combined curve of fan 3, fan 4 and curve x in series. Call this curve y.

iii. Where curve y cuts the system curve, read off the system operating point.

NB

iv. The last combination was placing fan 3, fan 4 and curve x in series. Hence use series laws to go back on constant quantity to fan 3, fan 4 and curve x. This gives the operating points of fan 3 and fan 4 but not of fans 1 and 2. (Remember curve x is 'imaginary').

v. This point on curve x gives the total pressure added by fans 1 and 2.

But $P_T = P_1 = P_2$

$\therefore$ Go back on constant pressure to the individual fans curves of fans 1 and 2 and obtain fans 1's operating point and fan 2's operating point.

vi. Checks:

$$\begin{aligned} Q_1 + Q_2 &= Q_3 = Q_4 = Q_{system} \\ P_1 &= P_2 \\ P_1 &= P_2 + P_3 + P_4 = P_{system} \end{aligned}$$

vii. A rough graph of the above procedure is shown in Figure 4.43.

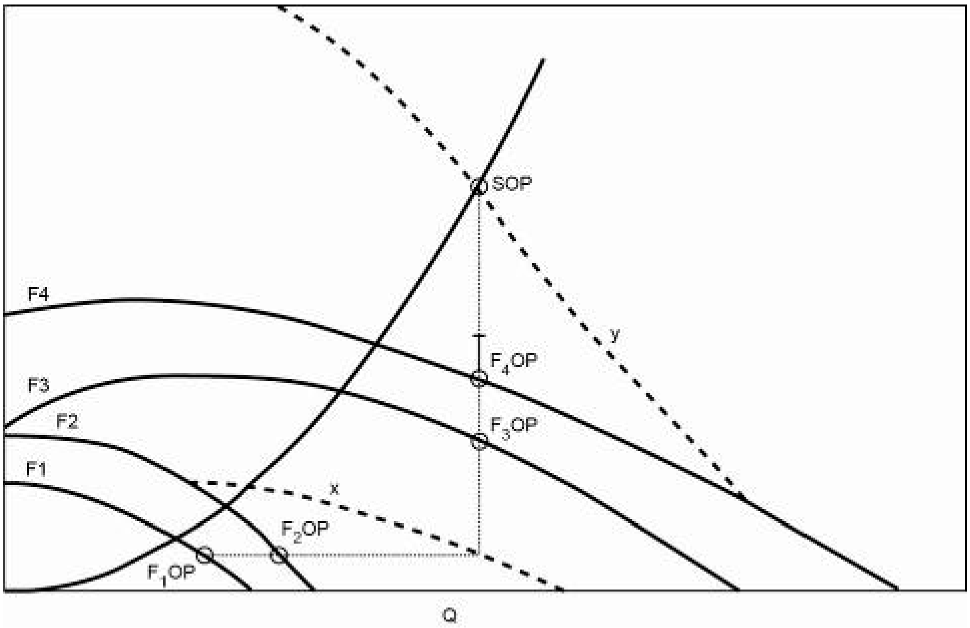


Figure 4.43 Final configuration for example 2 fan combinations

4.12.4 Fan in Series and Parallel with NVP

By applying the principles learned in the examples mentioned above, analysis of fans in series and parallel with NVP becomes relatively simple. As shown in the previous section, the various fan curves are combined together to eventually form one 'imaginary' fan curve representing all the fans.

The system resistance curve is then drawn as well as the (system '+' or '-'NVP) curve. Analysis then starts at the point where the combined fan curve cuts the (system + NVP) curve.

NB

This intersection point x does not give any actual operating points as it is not on an actual fan curve, nor an actual system curve. However, as shown in the NVP section, the system operating point can be found vertically above or below point x, on the actual system curve (above if NVP is assisting and below if NVP is opposing).

Analysis then restarts at point x. By moving back from point x on constant pressure (if the fans were plotted in parallel), or on constant volume (if the fans were plotted in series), the various fan operating points can then be found.

Once all fan and system operating points have been found, the appropriate checks on pressures and quantities should be made.

4.12.5 Fans in Semi Series or Semi Parallel

In the sections before we dealt with examples of true parallel and true series and combinations of both. Fan placements are sometimes not true parallel or true serie placements and then we refer to these combinations as either semi series or semi parallel and both are very distinct in their application. With the solution of these combinations we make use of a system of residual curves.

4.12.5.1 Residual Curves

Definitions

The word ‘residual’ simply means ‘remaining’ or ‘leftover’. When applied to fan curves a **residual fan curve** is the curve representing what usefully remains after the fan has overcome an airway or system. When applied to system curves, a **residual system curve** is the curve representing that portion of the system which remains after the fan has done all it can to overcome the resistance of the system.

NB

Residual curves can be very useful when solving problems concerning complex fan installations. As mentioned, in practice, fans are not always in ‘true’ series or in ‘true’ parallel, but sometimes in ‘semi-series’ and ‘semi-parallel’. Residual curves can also be used to solve airways in ‘semi-parallel’. Figure 4.44 shows a fan installation in a system of three airways. Airways ‘b’ and ‘c’ have a common point at the inlet ‘x’, but **far** removed points at their discharge. Hence the airways b and c are said to be ‘semi-parallel airways’.

The fan supplies the pressure necessary to overcome the resistance in all three airways. Whatever pressure the fan supplies at point ‘x’ will operate equally in both airways b and c. This means that ‘b’ and ‘c’ will receive an equal amount of pressure, but will handle different quantities if their resistances are different. Hence ‘b’ and ‘c’ can be treated as if they were in true parallel.

Worked Example 15 uses the residual curve methods of solving airways in semi-parallel.

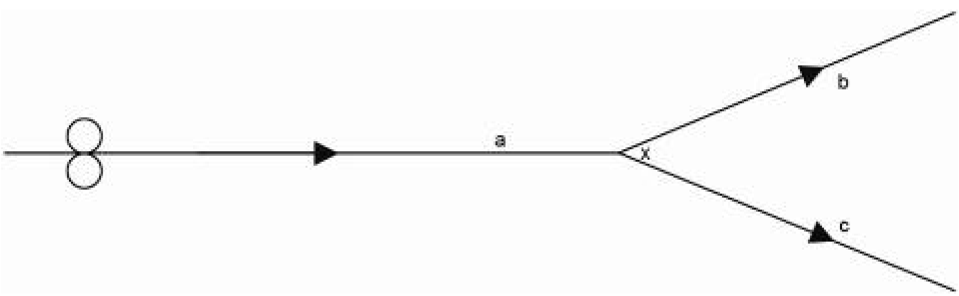


Figure 4.44 Semi-parallel airways

WORKED EXAMPLE 15

Figure 4.45 shows a layout of 3 air ways and an exhaust fan. The performance curve of a fan, which is to be installed in a system of airways as shown in the following sketch:

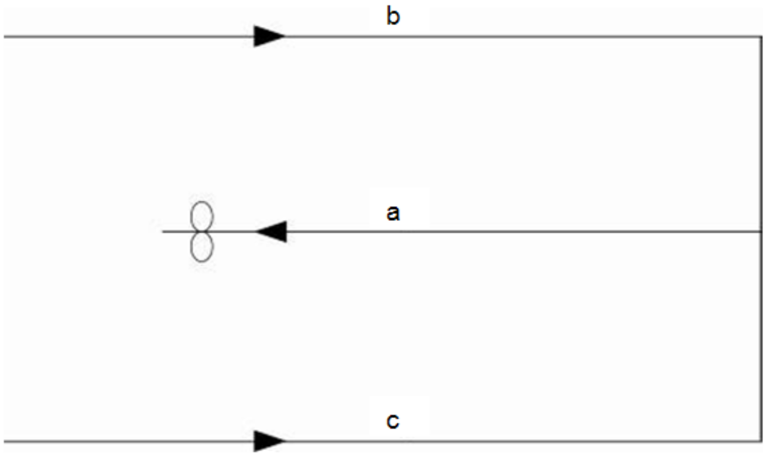


Figure 4.45 Layout of fan and three airways

The system resistance curve of each airway and fan curve are also given and shown in Figure 4.46. Determine the operating point of the fan and of each system after installation.

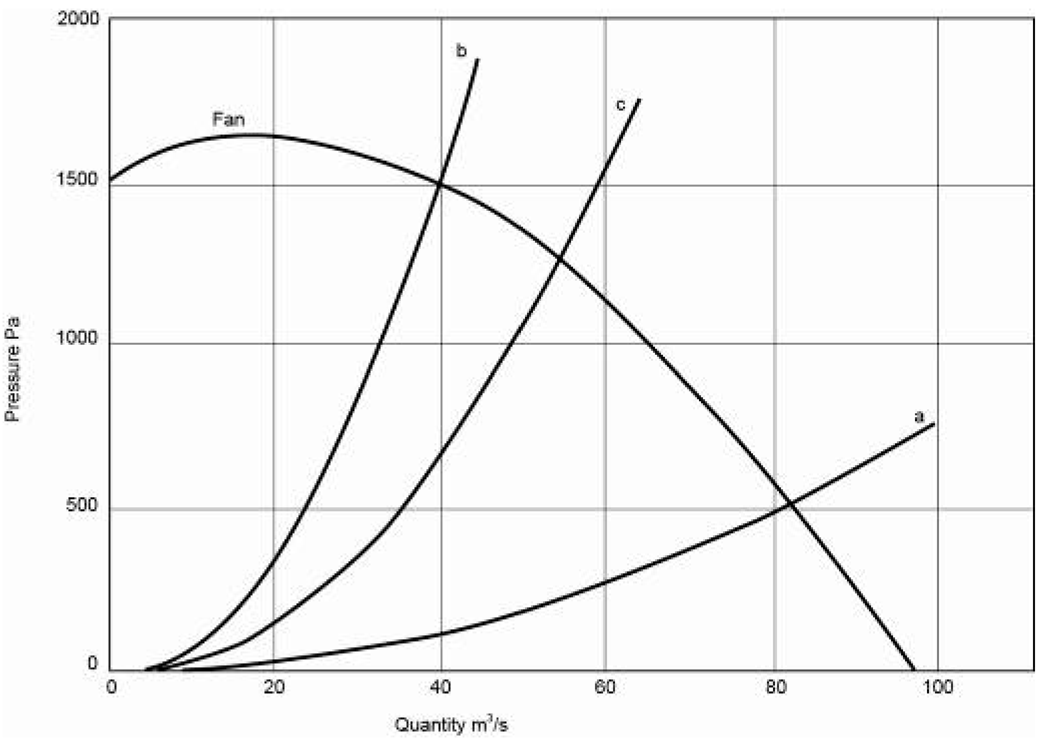


Figure 4.46 Fan curve and resistance curves for system

Answer

Airways ‘b’ and ‘c’ are semi-parallel airways. The fan must first overcome the resistance of airway ‘a’ before it can work on airways ‘b’ and ‘c’. Hence,

i. Draw the residual fan curve of the fan minus system ‘a’ resistance. This is done by subtracting system a’s pressure from the fan pressure at constant quantity, for example:

1.	At 0 m^3/s P of system ‘a’	=	0
	At 0 m^3/s P of fan	=	1 525
	At 0 m^3/s residual fan pressure	=	(1 525 - 0) Pa
		=	1 525 Pa
2.	At 60 m^3/s P of system a	=	280
	At 60 m^3/s P of fan	=	1 140
	At 60m^3/s residual fan pressure	=	(1 140 - 280) Pa
		=	860 Pa
3.	At 82 m^3/s P of system a	=	515
	At 82 m^3/s P of fan	=	515
	At 82 m^3/s residual fan pressure	=	(515 - 515) Pa
		=	0 Pa

The above three points are merely examples of the method. Obviously more points are needed to plot the complete fan residual curve.

(The complete fan residual curve is represented by the broken-line curve marked (fan - a) in Figure 4.47. The fan residual curve is an **‘imaginary’** curve which represents the remainder of the fan performance after the resistance of system a is completely overcome, that is as if an ‘imaginary’ fan was now situated at the return of b and c, and airway ‘a’ no longer existed. The sketch could be simplified to:

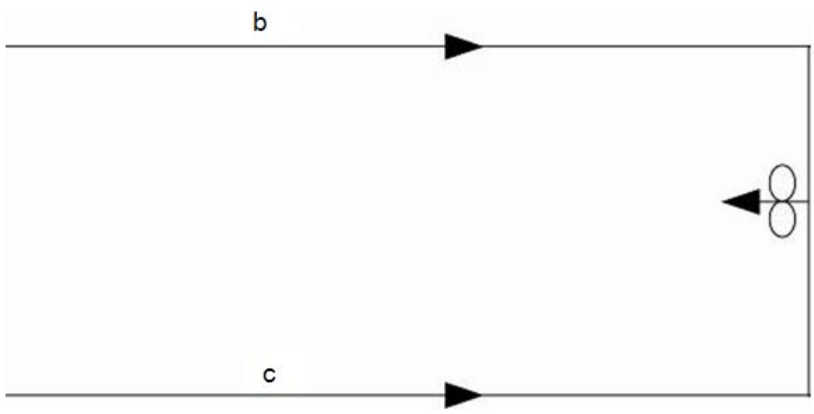


Figure 4.47 Fan and parallel airways

ii. Plot the combined curve for airways b and c in parallel by adding the quantities at constant pressure. Call this curve x. The sketch could now be simplified to:

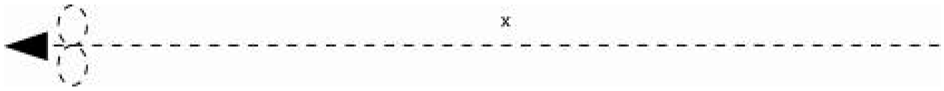


Figure 4.48 Fan and system imaginary curves

iii. The problem has now been simplified to a **single fan curve** and **a single system curve**, both of which are **'imaginary'** curves. Analysis starts at the intersection of x with the fan residual curve. Call this point the 'start point'. With all complex fan and system networks the 'start point' gives the total quantity through the whole system and can be read off as 66 m^3/s.

Referring to the original sketch, (Figure 4.46) only the fan and system 'a' handle the total quantity of 66 m^3/s the fan operating point can be read of from the graph as:

66 m^3/s @ 1 000 Pa

Where the 66 m^3/s quantity line cuts system 'a' read off system a's operating point.

$S.O.P._a$ = 66 m^3/s @ 330 Pa

Return to the start point. The start point was obtained by plotting 'b' and 'c' in parallel. Therefore move back from the operating points of 'b' and 'c'. The system operating points for 'b' and 'c' can now be read off as:

$S.O.P_b$ = 26.5 m^3/s @ 670 Pa

$S.O.P_c$ = 39.5 m^3/s @ 670 Pa

iv. Checks:

Q_{fan}	=	Q_a	=	Q_b	+	Q_c				
66	=	66	=	26.5	+	39.5				
P_{fan}	-	P_a	=	P_{b+c}	=	P_b	=	P_c		
1000	-	330	=	670	=	670	=	670		

Figure 4.49 shows the graphical representation of the solution.

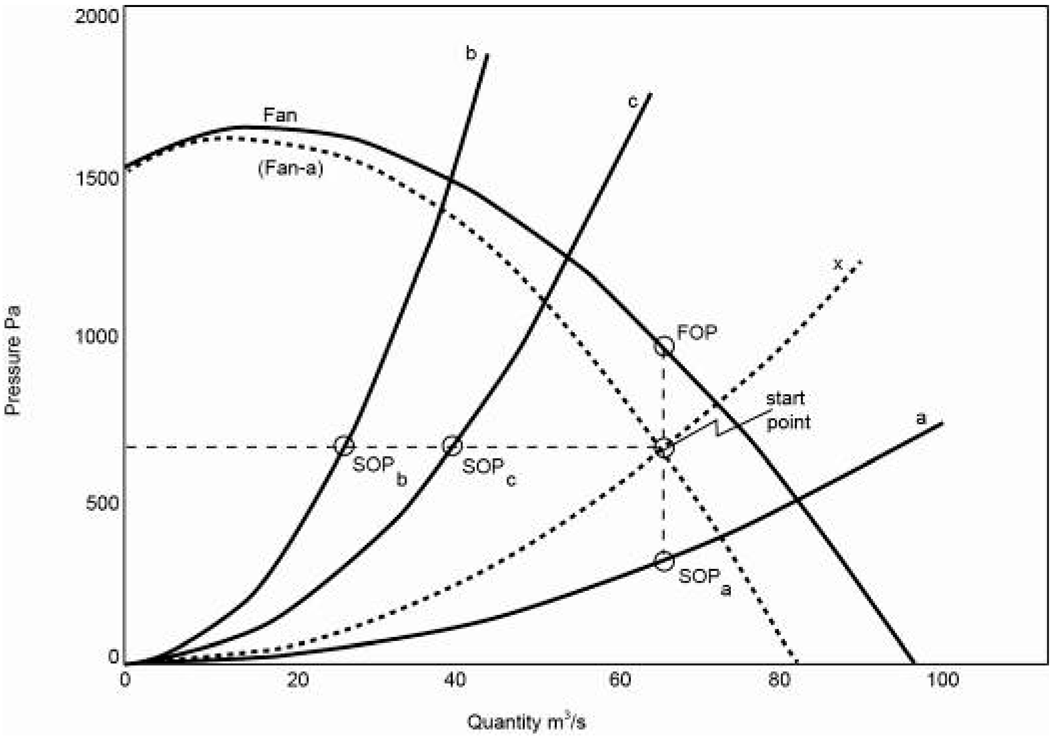


Figure 4.49 Graphical solution for Worked Example 15

Notes

i. The above example could also have been solved without using the residual method by combining the three system curves into one single curve using the laws of airways in series and parallel. The fan operating point would then be at the intersection of this single curve with the fan curve. System operating points can then be traced back using the methods explained in the section on airways in series and parallel.

ii. As was explained in the series and parallel section the basic method is to combine fan curves together and system curves together form 'imaginary' curves until the problem is reduced to a single fan and a single system. This same method applies to all residual problems.

iii. Residual curves are 'imaginary' curves and it must be remembered that **fans and system only operate on their actual own curves**.

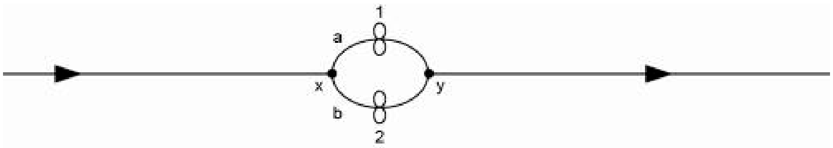
iv. The method used is exactly the same whether the fan is forcing or exhausting air.

4.12.5.2 General Notes on Residual Curves

i. It should be remembered that any residual curve is an 'imaginary' curve which helps in solving the problems. Fans only operate on their own curves and systems only operate on their own curves.

ii. One of the differences experienced with fans in semi-series and semi-parallel is that the problem is often mistaken for fans in true series or true parallel. The following sketches (Figure 4.50: A-D and Figure 4.50: E,F) will attempt to clarify some of these misunderstandings:

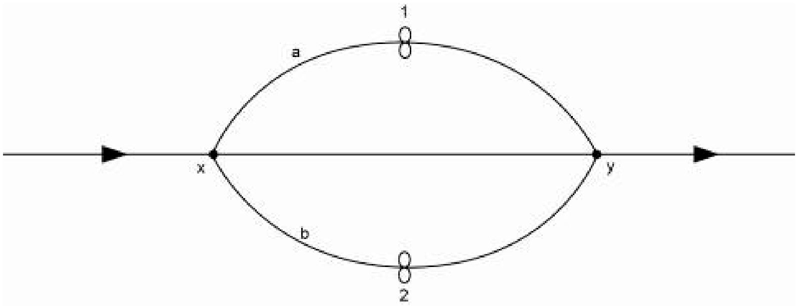
A.



B.

Systems 'a' and 'b' in sketches A and B are short in length. Hence sketches (A) and (B) show fans in true parallel. This is because systems 'a' and 'b' are short enough for the fans to draw air from a common point 'x' and to deliver it to a common point 'y'.

C.



D.

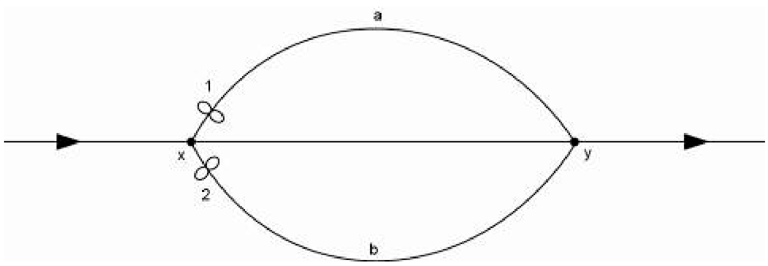
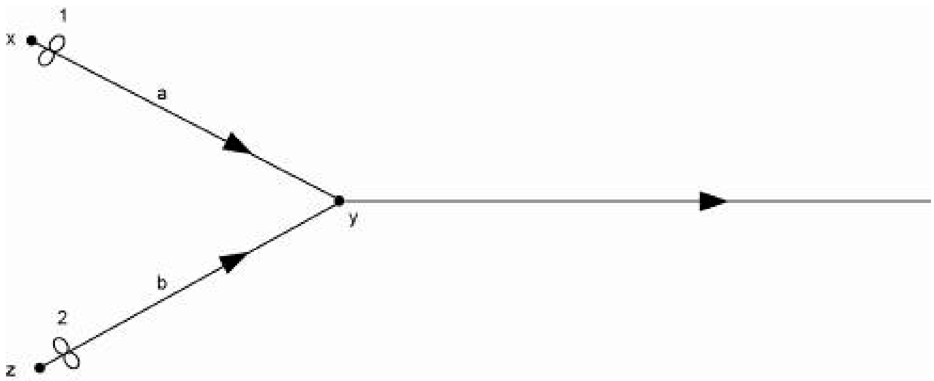


Figure 4.50 (A-D) Fan and duct combinations examples

E.



F.

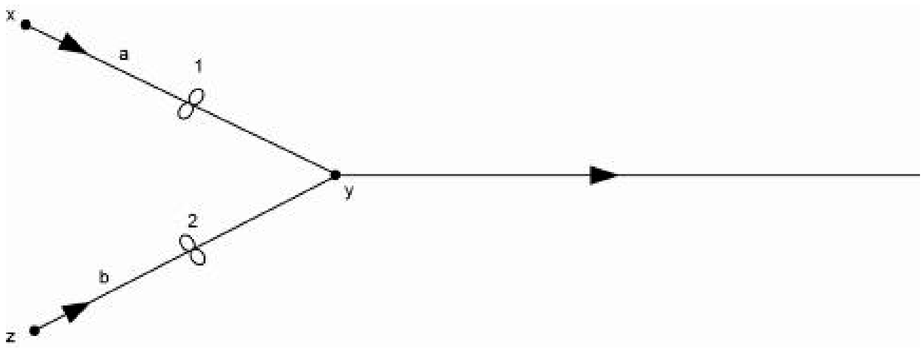
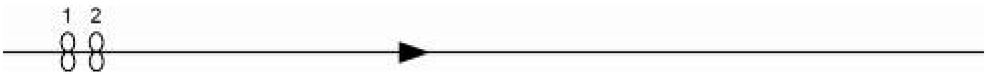


Figure 4.50 (E, F) Fan and duct combinations examples

NB

In sketch (C) systems ‘a’ and ‘b’ are lengthy systems and the fans are in semi-parallel. The fans could have been installed close to ‘x’ where they would have the same intake (as shown in sketch (D)), but they still do not have the same discharge. The same reasoning applies to sketches (E) and (F), which are also examples of fans in semi-parallel. In other words **when fans are in semi-parallel the actual position of the fan in the system is unimportant**.

G.



H.

I.

J.

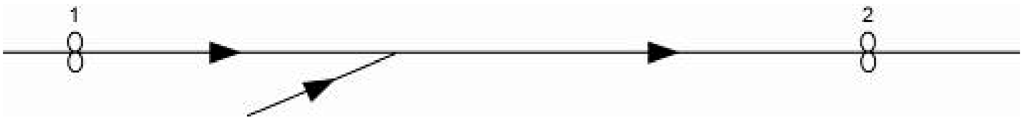


Figure 4.50 (G-J) Fan and duct combinations examples

Sketches (G) and (H) show fans in true series, as the fans are installed one behind the other and each fan handles the same quantity of air. However, as soon as there is leakage between the two fans as shown in sketch (I), the fans are no longer in true series **as they do not handle the same quantity of air**. The fans are then said to be in semi series.

iii. The layout of the installation of the fans and systems can often look complicated in residual problems. It is always advisable to draw simplified sketches of such layouts. Figure 4.51 shows a complicated system with fans and air ways and can be simplified to the example shown in Figure 4.52.

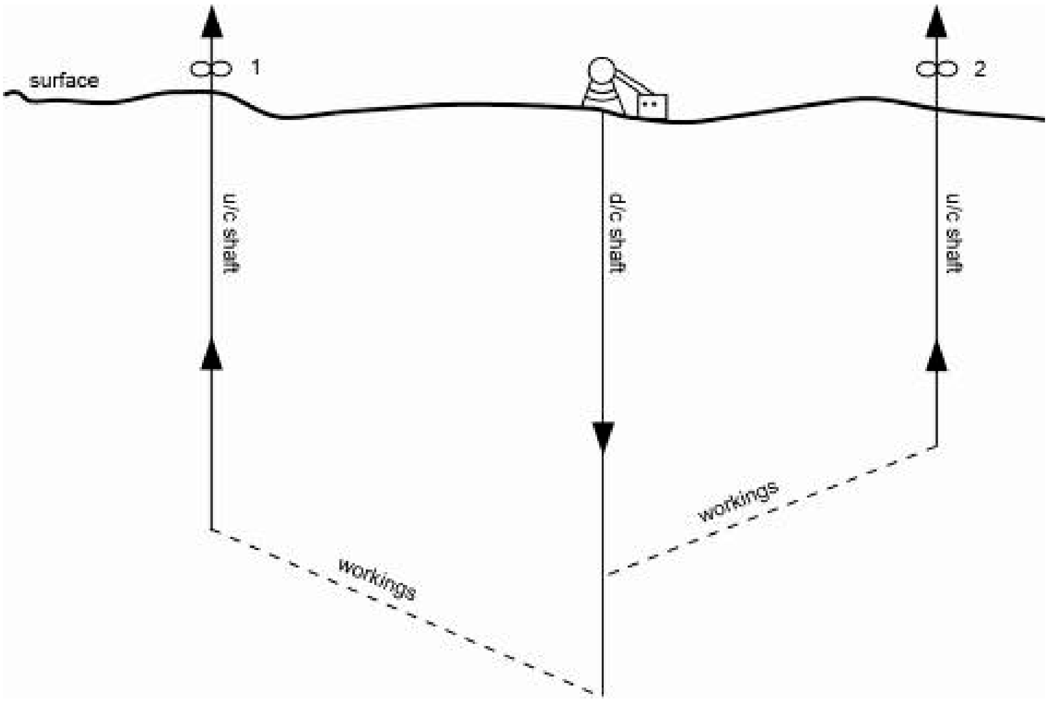


Figure 4.51 Complicated layout of air ways

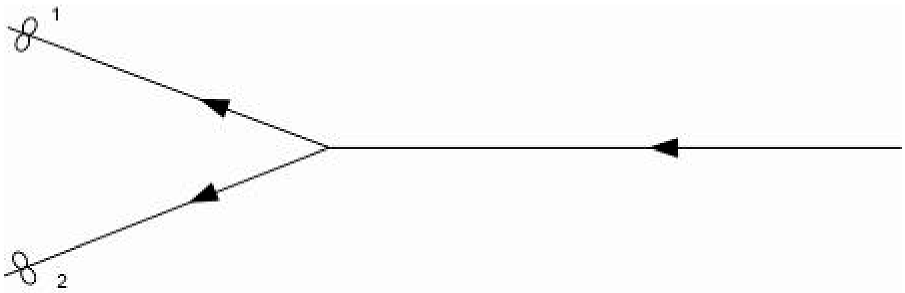


Figure 4.52 Simplified version of Figure 4.51

This then makes the analysis a little easier.

4.12.5.3 Procedure for Solving Fans in Semi-Series (Excluding Air Duct z)

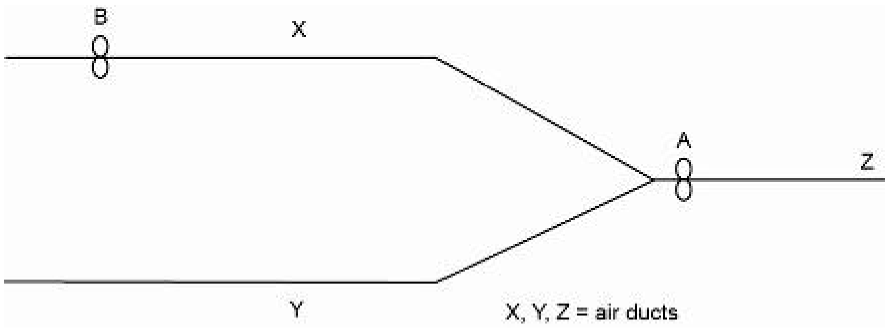


Figure 4.53 Example of fans in semi series network

The following procedure can be followed to solve the layout:

- Simplify the parallel branches. Branch ‘Y’ does not have a fan and therefore branch ‘X’ needs to be ‘readjusted’ to represent a branch ‘without’ a fan, therefore including the fact that the fan in branch ‘X’ reduces the resistance thereof. Subtract the fan pressure curve from the resistance curve to establish a new **imaginary** resistance curve ‘C’.
- Now add the curve ‘C’ to the resistance curve of ‘Y’ in parallel, to get one curve that represents the two resistance curves ‘C’ and ‘Y’ as one curve.
- The network was therefore simplified by the ‘breaking up’ of the parallel network.
- This combined resistance curve can now be added to the network with the fan and solved (this is now a simplified serie network).
- With the inclusion of air duct ‘Z’ in the serie network, it becomes a little more complicated, but as long as the standard fan laws are adhered to, it should not create any problems.
- Always remember the relationship between the quantity Q and the pressure P.

4.12.5.4 Procedure for Solving Semi-parallel Networks (without Fan in Series Branch, only Air Duct Z)

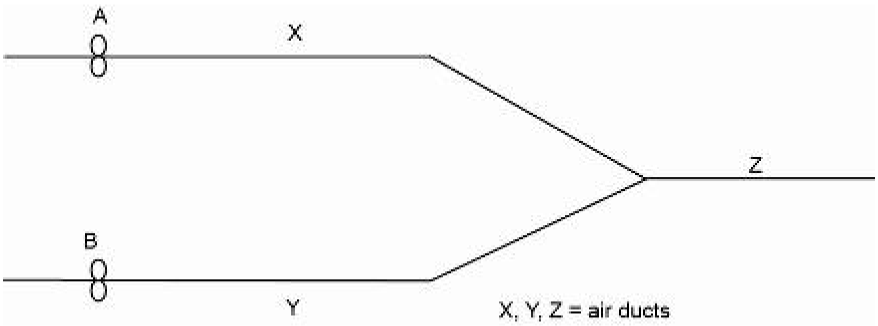


Figure 4.54 Example of fans in semi parallel network

The same principles form the previous example applies:

- Simplify the parallel network by ‘subtracting’ the resistance curves ‘X’ and ‘Y’ from the two fan curves ‘A’ and ‘B’ respectively. This will give two new imaginary fan curves, which includes the effect of the two respective resistances.
- These two imaginary fan curves can now be added together in parallel to give a new imaginary fan curve that represents the two fans in parallel.

- Having this one fan curve and the resistance curve for Z the network can be solved.
- The point where the combined fan curve is intersected by the resistance curve for ‘Z’ is called the **start** point for solving the network.
- Draw a line horizontally back to the imaginary fan curves and ‘up’ to the original fan curves, the various fan duty points can be determined. The airflow and pressure of the fans can be determined in this way. The quantity and pressure in the various ducts can also be determined in the same manner.
- If a fan ‘C’ is added in the serie portion of the network, it also becomes more complicated. The new fan can be added to the combined parallel fans and where this new combined fan curve in serie intersects the resistance curve ‘Z’ is the new duty point for fan ‘C’.
- From this point onwards the various fan laws pertaining to the pressure and the quantity needs to be applied to solve the rest of the network.

WORKED EXAMPLE 16

Figure 4.55 represents a downcast shaft, an up-cast shaft and two ventilation districts of a shallow mine.

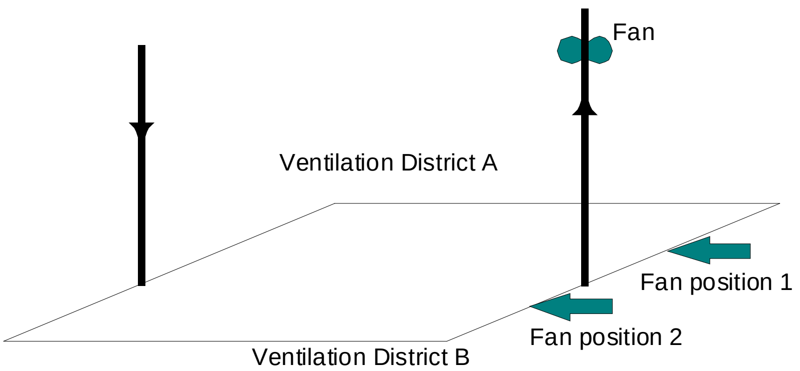


Figure 4.55 Ventilation district layout

Table 4.8 indicates the pressure loss, which would be required to pass 100 m^3/s of air through various parts of the mine.

Table 4.8 Pressure versus air quantity

Location	Pressure drop (Pa)	Air Quantity (m^3/s)
Downcast shaft	120	100
Upcast shaft	80	100
Ventilation district A	300	100
Ventilation district B	500	100

The attached graph shows the characteristic of one of two identical fans. Determine graphically the quantity of air flowing through each ventilation district:

a) If the fans are installed in parallel on top of the up-cast shaft
b) If one fan is installed at position 1 in District A, and the other at position 2 in District B, ignore changes in air density.

A common mistake made with the answer to this question is that the student assumes that 100 m^3/s of air flows through the mine, which it is obviously not.

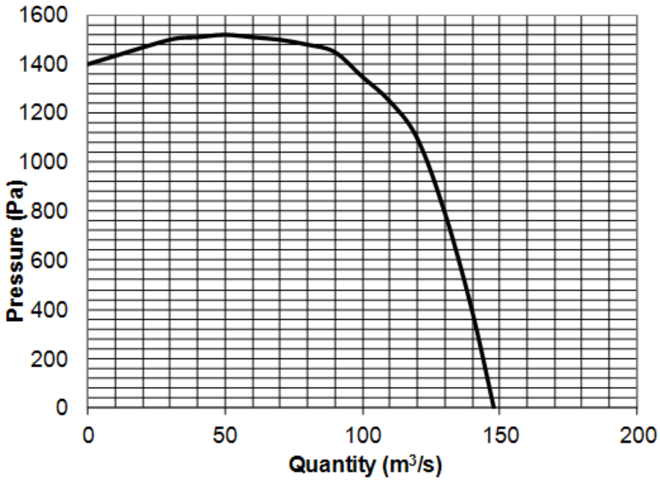


Figure 4.56 Fan curve for example

Answer

a)	Quantity flowing through district A	118 m^3/s
	Quantity flowing through district B	91 m^3/s
	Total quantity	209 m^3/s
b)	Quantity flowing through district A	109.0 m^3/s
	Quantity flowing through district B	98.5 m^3/s
	Total quantity	207.5 m^3/s

Can you determine the pressures related to the system and fans as well?

SELF STUDY

Read the following articles:

References

- 'Fan total pressure or fan static pressure: which is correct by solving ventilation problems'? D.J Brake
- 'Saving power and reducing noise in auxiliary mine ventilation'. *Journal of the Mine Society of South Africa*, July/September 2005.

Chapter

5

PSYCHROMETRY AND THERMODYNAMIC PRINCIPLES OF MINE AIRFLOW

LEARNING OUTCOMES

Study Objectives 5.1

- Know various psychrometric properties of air and with the aid of a psychrometric chart, determine the values
- Be able to solve various psychrometric calculations with the help of charts
- Gas laws, ideal gas, universal gas constant and gas mixtures
- Psychrometric properties.

Study Objectives 5.2

- Develop the Equations
- Evaluate the Integrals
- Comparison of Bernoulli and Pseudo's Equations
- Application of Equations
- Application of thermodynamic aspects to a mine ventilation network.

REFERENCES

Brake, DJ, 'Calculations of the natural (unventilated) wet-bulb temperature, psychrometric dry-bulb temperature and wet-bulb globe temperature from standard psychrometric measurements', *Journal of the Mine Ventilation Society of South Africa* Oct/Dec 2001, p108-112.

Mine Environmental Control (MEC) Certificate, Chamber of Mines, South Africa, Workbook 2.

Kislig, RE, 'A theoretical and critical analysis of the various methods of determining energy and pressure losses in shafts', *Journal of the Mine Ventilation Society of South Africa,* Oct/Dec 1999, p134-143.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

5.1 Psychrometric Characteristics

This section (5.1) is going to concentrate on the thermodynamic aspects of air and water vapour. It is very important in the whole mine ventilation process, because the properties of air change underground. Attention will be given to the determination of psychrometric properties of air with the help of psychrometric charts as well as formulae. With psychrometric calculations we assume air and water vapour to be an ideal gas and that Dalton's law partial pressures holds true. We will set up psychrometric calculations for various situations.

Air by itself is a mixture of various gasses, of which the largest component is Nitrogen (78.09%), Oxygen (20.95%), Argon (0.93%) and Carbon Dioxide (0.03%). If we talk about **dry air**, it is a combination of the previous mentioned gasses. When we talk about **air**, it includes a combination of dry air as well as water vapour.

Psychrometry is about the physical properties of air and water vapour mixtures. Examples of these properties are:

- Barometric pressure
- Temperature (wet-bulb and dry-bulb temperatures, measured with a whirling hygrometer).
- Humidity
- Heat capacity and so forth.

Air is a mixture of gasses, but for our purposes, we treat it as a single gas. This can be done, because there is minimal variation in the relationship of the gasses in air. The atmosphere has a certain amount of water vapour. This amount of water vapour varies; this variation determines the degree of saturation of the air.

The less the moisture contents in the air, the dryer the air (and the further the gap between the wet-bulb and dry-bulb temperature readings which are measured simultaneously).

Rule

When air is 100% water saturated, we can say it is at full humidity (in this case the dry-bulb and wet-bulb temperature are exactly the same).

NB

It is also very important to note that the wet-bulb temperature is always less than the dry-bulb temperature and at best equal to the dry-bulb temperature (at 100% humidity). If air is not fully saturated, evaporation will continue until such a state. Air, which does not take up water vapour, is known as dry air, although this is not really possible. The concept of dry air is still important, and we have already encountered the term.

5.1.1 Gas Laws

Air is defined as a mixture of various gasses and these gasses agree with certain gas laws. Of these is:

Laws

- Boyle's Law
- Charles' Law
- Universal Gas Law.

Animation

5.1.1.1 Boyle's Law

This law says that if a constant mass of gas is kept at a constant temperature, the pressure will be inversely proportional to the volume.

$$P_1V_1 = P_2V_2 = P_3V_3 = P_nV_n \quad \text{.......................................} \quad (5.1)$$

Where:

P	=	absolute pressure (kPa, Pa) and
V	=	Volume (m^3), Volume rate (m^3/s).

Animation

5.1.1.2 Charles' Law

This law stipulates that if a gas with a specific mass is kept at a constant pressure, the volume of gas is directly proportional to the absolute temperature.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} = \frac{V_3}{T_3} = \frac{V_n}{T_n} \quad \text{.......................................} \quad (5.2)$$

Where:

V	=	volume (m^3), Volume rate (m^3/s)
T	=	Absolute temperature in Kelvin (Temperature in ^oC + 273.15) K

5.1.1.3 The Universal Gas Law (Ideal)

This law combines Charles's ($\frac{V_1}{T_1} = \frac{V_2}{T_2}$) and Boyles ($P_1V_1 = P_2V_2$) law into one law that is known as the universal gas law. The law includes pressure, volume and temperature and is expressed as:

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = \frac{P_3V_3}{T_3} = \frac{P_nV_n}{T_n} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \; (5.3)$$

Another method of expressing the universal gas law is as follows:

$$PV = \frac{m}{M}RT \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \; (5.4)$$

Where:

R	=	Universal Gas constant (kJ/ (kg mol K))
	=	8.31436 kJ/kg mol K
V	=	volume (m^3)
T	=	absolute temperature (K)
P	=	absolute pressure (kPa)
m	=	mass of gas (kg)
M	=	molecular mass

To get the psychrometric results for instances of dry air and water vapour, we shall rewrite the ideal gas law as:

$$P_aV = \frac{m_a}{M}RT$$

Where 'a' refers to dry air and the gas constant is given as 0.287035 kJ/(kgK).

For water vapour the Equation is written as:

$$P_wV = \frac{m_w}{M_w}RT$$

Where:

'w' refers to water vapour.

5.1.2 Dry Air

As already mentioned; many calculations in ventilation use 'per kilogram dry air'. This does not necessarily mean that the air is dry, but that a reading is taken with some relationship to the mass of dry air and by so doing ignored the mass of water vapour.

The reason for the use of the term dry air in ventilation calculations is that the mass of dry air stays constant as it flows from one section of the mine to the next and we assume that there are no leakages. Therefore the mass of the water vapour in the air changes gradually. Calculations are made relatively simple when using properties that stay constant.

5.1.2.1 Properties of Air-Water Vapour Mixtures

An air-water vapour mixture has certain physical properties that are identified below: (see relationship between heat and moisture content changes in air attached).

- **Dry-bulb Temperature (°C):** Is the temperature that is measured by a normal dry-bulb thermometer and is an indication of the sensible heat.
- **Wet-bulb Temperature (°C):** Is the temperature measured by a thermometer where the bulb is covered with a wet cotton cover and is exposed to an air velocity of 3 m/s and is shielded from radiation which is greater than the dry-bulb temperature and is an indication of the energy content of the air in kJ/kg.

- **Adiabatic Process:** Is when no heat is added or lost in the system, with air, which changes in temperature. (A bicycle pump, which is isolated such that heat cannot be added or removed, is an example of such heat transfer).
- **Isothermal Process:** This is when the temperature of the air is constant, while the heat content changes.
- **Dew-Point Temperature (°C):** This is the temperature where condensation takes place. At this temperature, the saturation vapour pressure is equal to the actual vapour pressure.
- **Relative Specific Humidity (g/kg):** Is the mass of water present per unit mass of air.
- **Vapour Pressure (kPa):** Is the pressure of water vapour in the mixture of air and water vapour. This pressure is a maximum when the air is saturated and is then known as saturated vapour pressure.
- **Individual (Partial) Pressure:** Is the pressure made up by each gas component from a mixture of gases adding up to the total pressure.

- **Relative Humidity (%):** Expressed as a percentage and is:

$$\frac{\text{Actual vapour pressure at current temperature}}{\text{Saturated vapour pressure at dry bulb temperature}} \times 100\%$$

- **Enthalpy or Total Heat (kJ/kg):** Is the energy per unit mass for a fluid, as a result of the random movement of the fluid particles. Enthalpy includes the thermal energy stored as ‘internal’ energy and energy as a result of the pressure of the fluid. It is thus the sum of the sensible heat, latent heat (heat associated with phase changes solid-liquid-gas) and super heat of air and water vapour mixture, calculated at a specific temperature (wet- and dry-bulb temperature) and pressure.
- **Sigma Heat (kJ/kg):** This is the enthalpy of air-water vapour mixture, minus the total heat of the liquid at the atmospheric wet bulb temperature.
- **Absolute Humidity (g/m^3):** Is the actual mass of water vapour per cubic metre of air.
- **Super heat (kJ/kg):** Is the heat, which increases the temperature of a gas, or vapour above its saturated temperature.

All of the above properties can be mathematically calculated, and this is what the rest of the chapter will cover.

5.1.2.2 Psychrometric Property Formulas

NB

By manipulation of the ideal gas law for dry air and water vapour, we can mathematically model the following psychrometric properties. It is important to note that in order to calculate these properties that the wet bulb, dry-bulb and Barometric Pressure (BP) is needed.

VAPOUR CONTENT (r, kg/kg)

$$r = \frac{0.622\ P_w}{P - P_w} \quad \text{(kg/kg)} \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \ (5.5)$$

Where:

P = Absolute pressure (Barometric pressure (BP) = $P_a + P_w$)

The following Equation is for P_w

$$P_w = P'_{ws} - AP \times (t_{db} - t_{wb})$$

Where:

P_w	=	Vapour pressure (kPa)
P'_{ws}	=	Saturated vapour pressure (kPa)
A	=	Psychrometric constant = 0.000644 $^oC^{-1}$

References

The psychrometric constant A is a constant, which is dependant on the pressure and wet-bulb temperature. In Chapter 18 of VOEE, Figure 18.3 shows this aspect. The correct psychrometric constant must be used for specific conditions.

A handy and trustworthy Equation to relate saturated vapour

$$P'_{ws} = 0.6105 \exp\left(\frac{17.27\ t_{wb}}{237.3 + t_{wb}}\right) \quad \text{(kPa)} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(5.6)}$$

Where:

P'_{ws} = Saturated vapour pressure

SPECIFIC VOLUME (ν, m^3/kg)

$$\nu = \frac{0.287035\ T}{P - P_w} \quad (m^3/kg) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(5.7)}$$

With:

T = $273.15 + t_{db}$

DENSITY (ρ, kg/m^3)

$$\rho = \frac{1 + r}{\nu} \quad (kg/m^3) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(5.8)}$$

ENTHALPY (H, kJ/kg)

The total enthalpy:

H = $H_a + H_w$

With the enthalpy of dry air:

H_a = $C_{pa} \times t_{db}$ = $1.005\ t_{db}$

The enthalpy of the water vapour component:

H_w = $r\ H'_w$

Where:

H'_w = $1.8\ t_{db} + 2501$

H = $1.005 t_{db} + r\ (1.8 t_{db} + 2501)$ …………………. (5.9)

SIGMA HEAT (S, kJ/kg)

Definition

Sigma heat is important in mine ventilation requirements. It is defined as the enthalpy of the air minus a small water enthalpy term. This term is the enthalpy of the water vapour content, which is associated with a unit mass of dry air, and is calculated as if the water vapour is liquid water at a wet-bulb temperature.

Sigma heat is expressed as per unit mass dry air.

$$S = H - r\,H'_{wl}$$

With:

$$H'_{wl} = 4.183\,t_{wb} + 0.2$$

$$S = H - r\,(4.183t_{wb} + 0.2) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (5.10)$$

5.1.2.3 Psychrometric Properties of Air at Saturated Conditions

Up to this stage no remark has been made to the limit of water vapour that can be absorbed in a mixture. This is a limit to the amount of water vapour pressure (P_w) that can exist in any space.

The limit is dependant primarily on temperature and is given by the vapour pressure temperature relationship for vapour pressure.

Although generally speaking when air is saturated, we must understand that it is the space around the air that is saturated, **and not the air itself**. The calculations for these circumstances mentioned above will be done in exactly the same manner.

5.1.2.4 Partially Saturated Conditions

If the air is not saturated the vapour pressure P_w will have a lower value than the saturated vapour pressure. The actual vapour pressure will depend on the amount of water vapour present in the air.

There are many methods to determine the **humidity of air**, but the most widely used on mines is to measure the wet-bulb temperature.

Definition

A **wet-bulb thermometer** is a normal thermometer whereby the bulb of the thermometer is covered by cotton and is kept damp. Evaporation of the water from the wet area results in the bulbs temperature being lowered to a point below the dry-bulb air temperature. The air temperature is normally called the dry-bulb temperature to differentiate it from the wet-bulb temperature. The amount of temperature decrease is an indication of vapour pressure. We have referred to the wet and dry-bulb temperatures in Chapter 2 of this course and also mentioned that it measured with the use of a whirling hygrometer.

5.1.2.5 Psychrometric Properties Dependant on Saturation

The two aspects here are relative humidity and the dew point temperature.

RELATIVE HUMIDITY (ϕ, %)

This is expressed as a percentage and is:

$$\phi = \frac{P_w}{P''_{ws}} \times 100\ \% \qquad (5.11)$$

Where:

P_w	=	Actual vapour pressure
P''_{ws}	=	Saturated vapour pressure at dry-bulb temperature

And

$$P''_{ws} = 0.6105 \exp\left(\frac{17.27\ t_{db}}{237.3 + t_{db}}\right) \text{ (kPa)} \qquad (5.12)$$

DEW POINT TEMPERATURE (t_{dp}, °C)

Definition

Rule

The **dew point temperature** is defined as the temperature to which air must be cooled for condensation to begin. This is the point where the air is saturated. During the cooling process there is no change in the vapour content and therefore the vapour pressure also remains constant. The dew point temperature is the point where the saturated vapour pressure is equal to the actual vapour pressure.

The Equation for the calculation of the dew point temperature is:

$$t_{dp} = \frac{237.3\ x}{(17.27 - x)}\ ^{\circ}\mathrm{C} \qquad (5.13)$$

Where:

$$x = \log_e \frac{P_w}{0.6105} \qquad (5.14)$$

WORKED EXAMPLE 1

(Heat Source Calculation)

The following readings are taken at an underground booster fan:

- Q = 186 m^3/s
- Temperature 21.5/28.9°C at 102.3 kPa
- Static pressure increase over fan 5.5 kPa
- Electric Power input to motor 1 296 kW.

Calculate the expected temperature of the air leaving the fan.

Answer

From Psychrometric calculations (Homework):

- r_{in} = 0.01288 kg/kg
- Enthalpy = 61.93 kJ/kg
- ν = 0.8649 m^3/kg.

The fan is horizontal (fans are normally installed horizontal unless otherwise stated), therefore there is no potential energy and we assume that we can ignore the kinetic energy component.

The energy Equation looks as follows:

$q + W = M_a (H_2 - H_1)$, no water present, therefore no vapour.

Some of the electrical power is converted to heat (motor losses) and the rest to mechanical work, W.

Therefore:

NB

$q + W = 1\,296$ kW

(This is a very important aspect to remember)

$$M_a = \frac{Q}{\nu} = \frac{186.0}{0.8649} = 215.1 \text{ kg/s}$$

$$H_2 - H_1 = \frac{(q+W)}{M_a} = \frac{1296}{215.1} = 6.03 \text{ kJ/kg}$$

$$H_2 = 6.03 + 61.93 = 67.95 \text{ kJ/kg}$$

What does this give us?

r_2 = 0.01288 kg/kg (Why?)

H_2 = 67.95 kJ/kg

Q

The temperature can be read off from the Psychrometric charts or calculated with the formulae mentioned before. At what pressure will we look up the temperature using the Psychrometric charts (what value for P_2 will we use)?

A

Psychrometric chart

For the purpose of this course the value of the wet-bulb and dry-bulb temperatures will never be calculated, but always read off from the applicable psychrometric chart. The temperature that was read off from the 107.5 kPa chart (which is the closest to the pressure of 107.8 kPa) is as follows:

The **answer** is +- 23.8 / 34.7°C

Q

How much heat is given off (the following Equation can be used)?

$$q = \left[\frac{M_a}{(H_2 - H_1)}\right] - W$$

(check answer at home)

A

From the above, what effect of heat transfer do we have if we blow the same amount of air over a heater (barometric pressure remains constant or unchanged)?

The t_{wb} temperature increase of the air as a result of the heater will be less, as there is no increase in pressure (prove this statement with your own calculation).

Psychrometric chart

The Psychrometric charts are only shown graphically in increments of 2.5 kPa, which means that if we have any barometric pressure (BP) which is not a whole number, for example 80, 82.5, 85, 87.5, 90 etc. Then the Psychrometric properties cannot be read off the chart directly and has to be interpolated or calculated as per the Equations mentioned before.

Using the charts simplifies the procedure very much and will in most cases be used for solving Psychrometric properties (charts will be made available if it is required to use them).

Psychrometric chart

⌧ WORKED EXAMPLE 2

Calculate the psychometric properties of air at a temperature of 13.9/19.6ºC and at a barometric pressure of 94.983 kPa.

Excel

Answer - See Psychometric calculations (variables temperature and barometric pressures)

5.1.2.6 Psychrometric Charts (Barenbrug)

Each of the physical properties defined above, can be calculated by means of equations, but to simplify this, Psychrometric charts can be used to read off these properties**.** Remember, **Psychrometric charts are only applicable to air, and not water.**

NB

In order to determine any of these physical properties from a psychrometric chart, it is necessary to measure or know three properties. The **most common and usually measured properties** are:

- **Barometric Pressure (BP) (kPa)**
- **Wet-Bulb Temperature (ºC)**
- **Dry-Bulb Temperature (ºC)**

Psychrometric chart

Animation

As mentioned before there are Psychrometric charts available, ranging from 80 kPa to 130 kPa, at intervals of 2.5 kPa, this means that these charts start at 80 kPa followed by 82.5 kPa, 85.0 kPa, 87.5 kPa and so forth up to 130 kPa.

PROCEDURE TO USE A PSYCHROMETRIC CHART

The notes that follow, describe the application of a psychrometric chart, step by step. The same temperature as for the calculation example is used so as to demonstrate that the answers obtained from each is very similar (however the calculated values always more correct).

WORKED EXAMPLE 3

Psychrometric chart

The barometric pressure in an underground working place is measured as 97.5 kPa and the air temperature as 30.0 / 34.0°C. Use a Psychrometric chart to determine the following physical properties of the air:

(i) Moisture content (Apparent specific humidity)

(ii) Apparent specific volume

(iii) Apparent density

(iv) Vapour pressure

(v) Relative humidity

(vi) Sigma heat

(vii) Dew point temperature

(viii) Enthalpy (Total heat).

Psychrometric chart

Answer

First select the correct chart according to the barometric pressure that is the 97.5 kPa chart (always select the chart closest to the given barometric pressure, for the example, if the given BP is 98.5 kPa, then the 97.5 kPa chart is closer to 98.5 kPa than for instance the 100 kPa and hence the 97.5 kPa chart must be used). If however the required pressure lies between the two applicable charts and a more correct answer is required, then a process of interpolation between the two charts will have to be followed given barometric pressure, for the example, if the given BP is 98.5 kPa, then the 97.5 kPa chart is closer to 98.5 kPa than for instance the 100 kPa and hence the 97.5 kPa chart must be used). If however the required pressure lies between the two applicable charts and a more correct answer is required, then a process of interpolation between the two charts will have to be followed.

Mark the required dry-bulb temperature line (34.0°C) at the bottom and the top of the chart. Use a ruler and join these two points. Identify the wet-bulb line (30.0°C) and, on this constant wet-bulb line go diagonally down to the right, to intersect the dry-bulb (34.0°C) line. Where these two temperature lines (wet-bulb and dry-bulb) intersect, make a 'point'.

This is illustrated as follows:

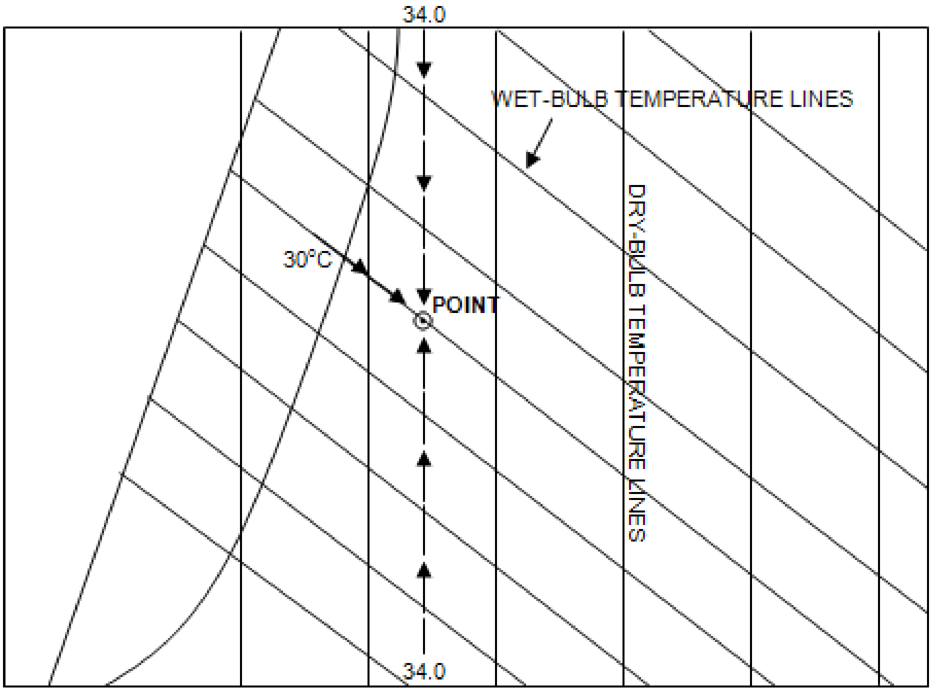


Figure 5.1 Diagrammatic procedure to plot temperature on a psychrometric chart

(i) Moisture Content (r) or Apparent Specific Humidity (g/kg)

From the 'point' marked above and shown in Figure 5.1 and also showed in Figure 5.2, move horizontally to the right to the scale marked 'apparent specific humidity'. Read off the moisture content as 26.3 g/kg (note that every small 'block' is equal to 0.5 g/kg on the actual chart). Figure 5.2 illustrates how to locate the point.

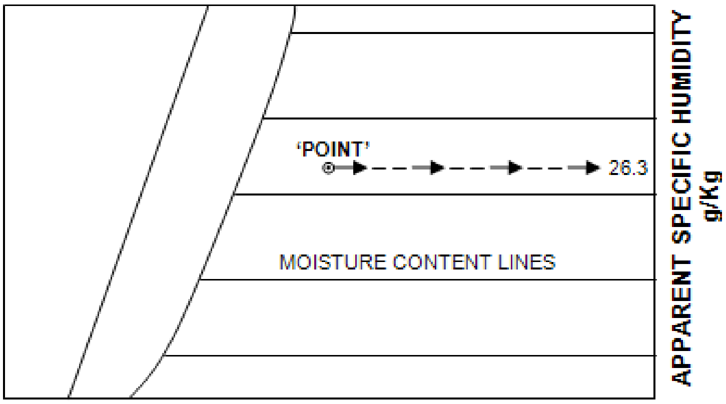


Figure 5.2 Plotting of apparent specific humidity

(ii) Apparent Specific Volume (m^3/kg)

The apparent specific volume lines are those straight lines running slightly to the left of vertical. The 'point' is between the 0.94 m^3/kg and the 0.95 m^3/kg lines. By accurate estimation between these two lines, the apparent specific volume is 0.944 m^3/kg.

This is illustrated as follows:

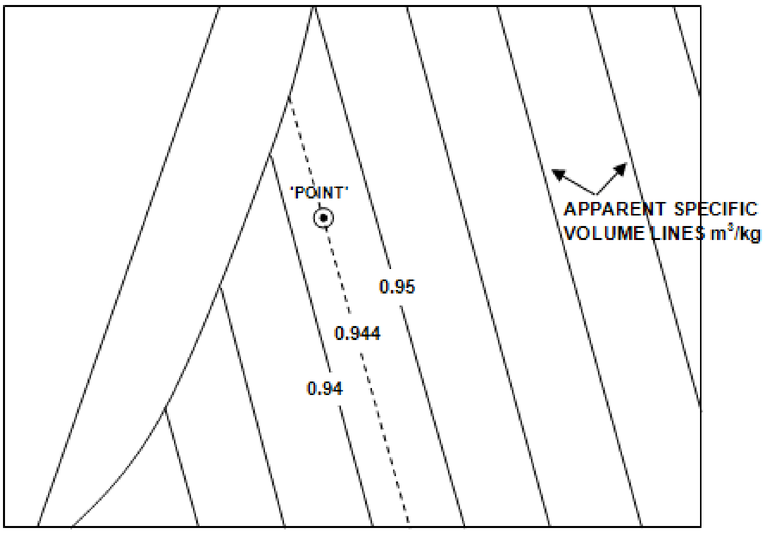


Figure 5.3 Finding apparent specific volume on psychrometric chart

(iii) Apparent Density (kg/m^3)

Apparent density cannot be read off the psychrometric chart, but can be calculated from the apparent specific volume. The Apparent density is the reciprocal of apparent specific volume and can be determined as follows:

$$= \frac{1}{\text{apparent specific volume}} \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(5.15)}$$

$$= \frac{1}{0.944}$$

$$= 1.06 \text{ kg/m}^3 \text{ (note, specific volume unit m}^3\text{/kg)}$$

(iv) Vapour Pressure (kPa)

From the 'point', move horizontally to the left to the vapour pressure scale. Read off the vapour pressure as 3.98 kPa (note that the vapour pressure scale is different from the apparent specific humidity or moisture content scale).

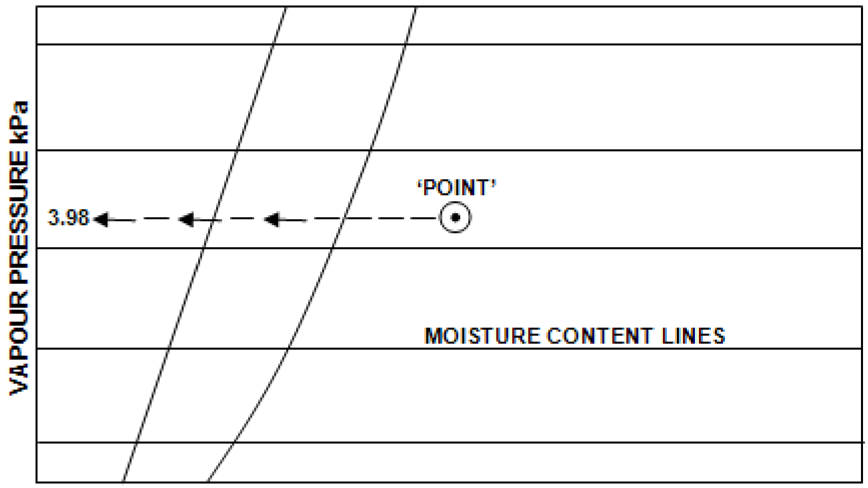


Figure 5.4 Finding vapour pressure on psychrometric chart

(v) Relative Humidity (%)

These lines are the only curved lines on the Psychrometric chart, running from the bottom left corner to the top right corner. The 'point' is approximately midway between the 70% and 80% lines. Read of the relative humidity as 76%. Figure 5.5 illustrates the point as follows:

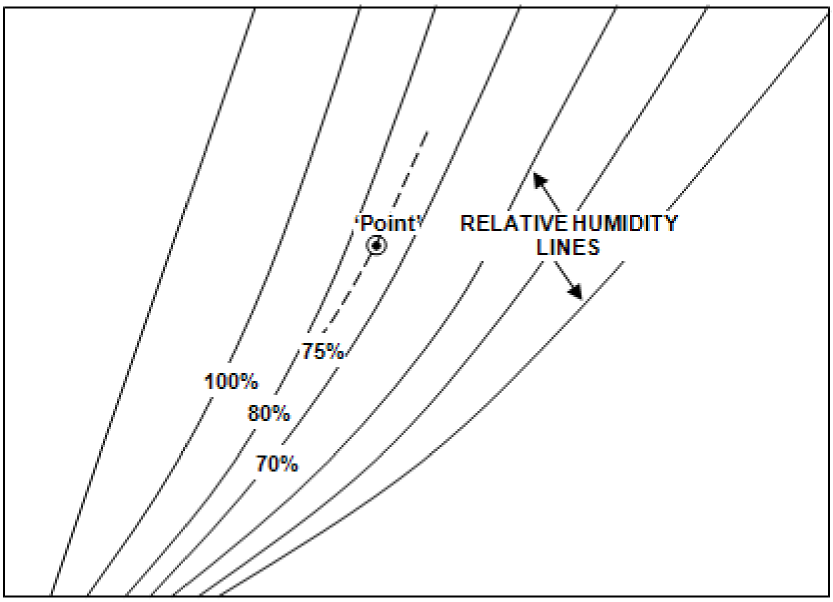


Figure 5.5 Finding relative humidity on a psychrometric chart

NB

Note that at any point on the 100% relative humidity line, that the wet-bulb and dry-bulb temperatures are the same, which means the air is then 'fully saturated'.

(vi) Sigma Heat (kJ/kg)

These sigma heat lines run parallel to the constant wet-bulb lines and the scale is marked on the straight line above the 100% relative humidity line.

From the 'point', move parallel to the left on the constant wet-bulb line to intersect the sigma heat (ΣH) scale. Read off the sigma heat as 98.9 kJ/kg. This is illustrated in Figure 5.6.

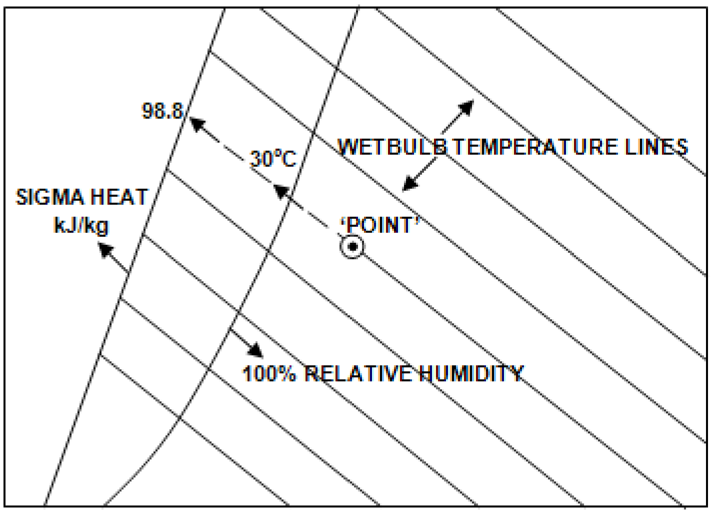


Figure 5.6 Finding Sigma heat on a psychrometric chart

NB

Note, that in determining the sigma heat, only the wet-bulb temperature can be used, ignoring the dry-bulb temperature, because the constant wet-bulb temperature lines run parallel to the sigma heat scale.

(vii) Dew Point Temperature (ºC)

To determine dew point temperature, move from the 'point' horizontally to the left to intersect the 100% relative humidity line. Read of the wet-bulb temperature as 28.9ºC. Hence, the dry-bulb temperature will also be 28.9ºC as the relative humidity is a 100%, which means the air is fully saturated.

$\therefore$ Dew point temperature 28.9 / 28.9ºC

Figure 5.7 illustrates.

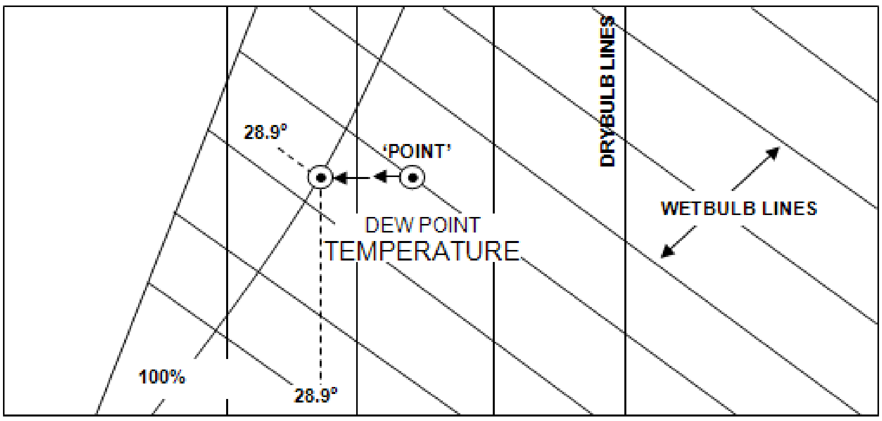


Figure 5.7 Finding the dew point temperature on a psychrometric chart

(viii) Enthalpy or Total Heat (kJ/kg)

The enthalpy scale is marked around the outside border of the Psychrometric chart, similar to a picture frame. The first scale starts at the bottom left corner at zero and moves vertically up, around the top left corner to the right top corner to 200. Similar, the second scale starts at the bottom right corner at zero, and moves horizontally to the right, around the bottom right corner and vertically up to the top right corner to 200.

To read off enthalpy, place the point of a pencil on the 'point' and place a ruler against the pencil to slope from the top left to the bottom right corners.

With the ruler still against the pencil, swivel the ruler, until the same enthalpy values can be read in the top left and bottom right sides of the two scales. Now read off the value of the enthalpy as; 102 kJ/kg.

This is illustrated in Figure 5.8:

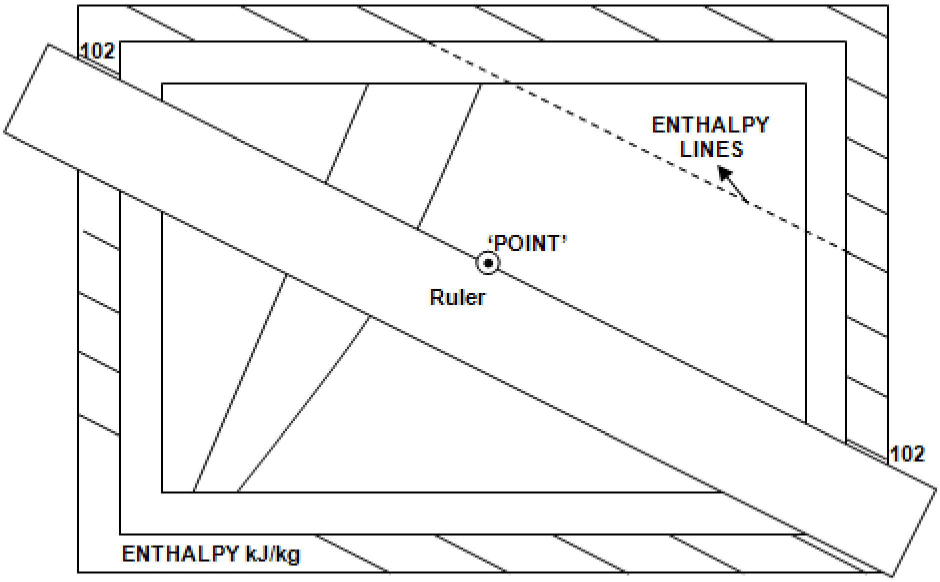


Figure 5.8 Finding Enthalpy on a psychrometric chart

NB

Note, that the enthalpy or total heat content (102 kJ/kg) is slightly higher than the sigma heat content of 98.8 kJ/kg, because total heat also includes latent heat and super heat, this however amounts to a very small portion (102 - 98.9 = 3.2 kJ/kg).

INTERPOLATION BETWEEN DIFFERENT PSYCHROMETRIC CHARTS

Rule

It sometimes happens that no chart is available for a particular measured barometric pressure. Common practice is to use the closest chart to the measured barometric pressure. However, to determine the required physical properties, more accurately, interpolation between the two nearest charts to the measured barometric pressure is necessary.

WORKED EXAMPLE 4

The barometric pressure in an underground stope is 111.0 kPa and the air temperature 28.0 / 32.0°C. Using Psychrometric charts, determine the sigma heat content of the air (all the answers read off can be checked by using the excel spreadsheet in Worked Example 2.)

Psychrometric chart

Answer

Using the temperature 28.0/32.0ºC and the 110 kPa Psychrometric chart the sigma heat is read off as 82.0 kJ/kg.

From the 112.5 kPa chart the value for the sigma heat is read off as, 80.8 kJ/kg. A pressure difference of 2.5 kPa therefore gives a difference of 1.2 kJ/kg in sigma heat.

Now, from 110.0 kPa to 111.0 kPa, there is a difference of 1 kPa

$\therefore$ Difference in sigma heat = $\frac{1}{2.5} \times 1.2$

= 0.48 kJ/kg

$\therefore$ Sigma heat at 111.0 kPa = (82.0 – 0.48) kJ/kg

= 81.52 kJ/kg

NB

(Note: Subtract the 0.48, because the Sigma heat values **decrease** from 110.0 to 112.5 kPa and therefore the answer of 81.52 kJ/kg).

Alternatively:

From 111.0 kPa to 112.5 kPa, there is a difference of 1.5 kPa and now the difference in sigma heat = $\frac{1.5}{2.5} \times 1.2$

= 0.72 kJ/kg

$\therefore$ Sigma heat at 111.0 kPa = 80.8 + 0.72

= 81.52 kJ/kg

NB

(Note: Now add the 0.72 kJ/kg to 80.8 kJ/kg because the sigma heat values **increase** from 112.5 kPa to 110.0 kPa and therefore this alternative method can be used to check the answer in the first part of this answer).

This same method of interpolation can be used to calculate any of the physical properties, more accurately.

THE LOMBARD'S AIR DENSITY CHART

To determine the true density of an air-water vapour mixture the 'Lombard's Air Density Chart' can normally be used in routine daily work. There are, however, various types of air density charts available, but in these notes, we will concentrate on the application of the abovementioned chart. Where extreme accuracy is required, true density should be calculated using the appropriate formulae, which will not be discussed in these notes.

The Lombard's Air Density Chart is shown in Figure 5.9 and we will only consider some part of it.

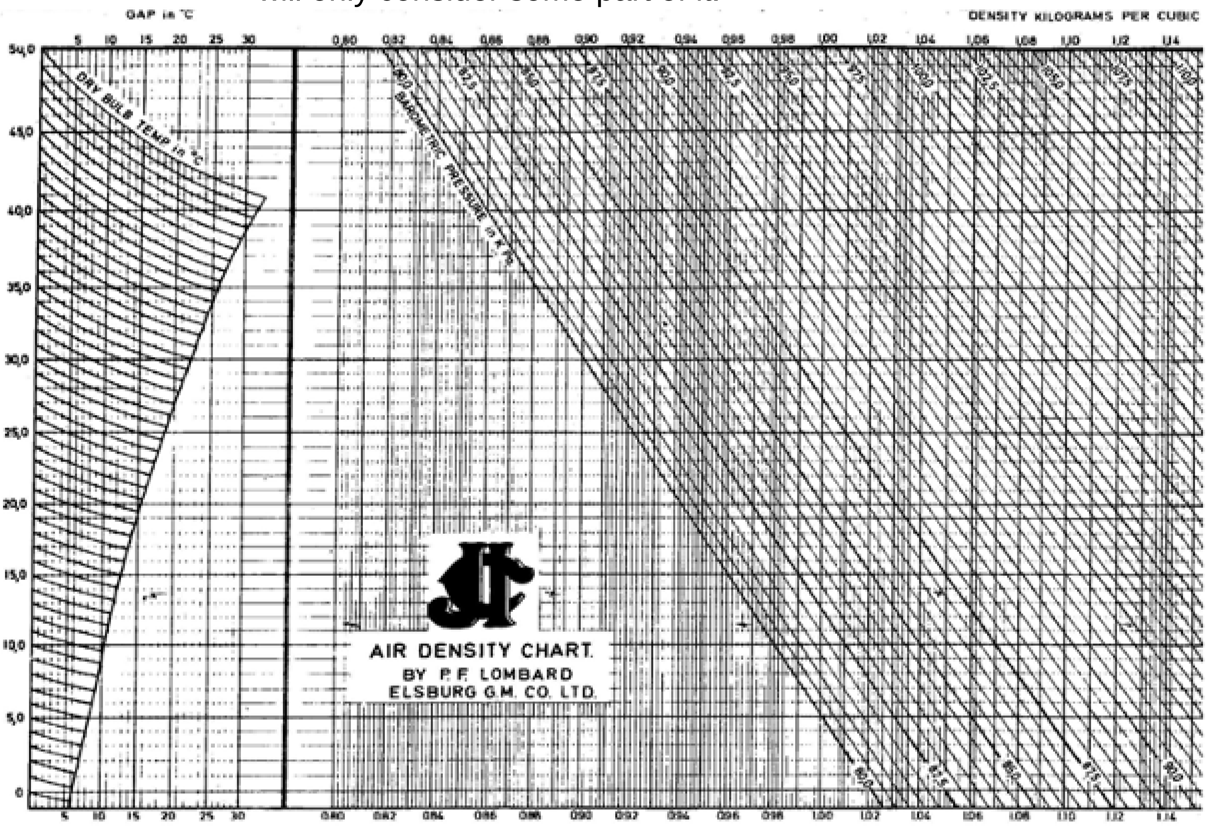


Figure 5.9 Lombard's density chart

- On the vertical axis of this chart is the scale for dry-bulb temperature, ranging from 0ºC to 50ºC (note that these dry-bulb temperature lines are curved and sloping from the top left corner towards the bottom right).
- On the horizontal axis, bottom and top, there is a portion on this scale indicating 'Gap in ºC', ranging from 0ºC to 30ºC.
- On the balance of the horizontal axis (bottom and top) is the density scale starting at 0.80 kg/m^3 and continue to beyond 1.14 kg/m^3 (remember the above chart is only part of the total Lombard's Density chart).
- The barometric pressure lines are the straight inclined lines to the left of vertical ranging from 80.0 kPa and continue to 130.0 kPa.

Application of Lombard's Air Density Chart

A step-by-step description of how to use the chart to determine the true density of the air will follow.

WORKED EXAMPLE 5

Determine the true density from the following measured properties of an air-water vapour mixture. The Barometric Pressure is 95.0 kPa and the air temperature is 20.0 / 25.0^{o}C.

Answer

- First determine the gap between the given wet-bulb and dry-bulb temperatures

  Gap = Dry-bulb wet-bulb temperature

  = 25.0 - 20.0

  = 5.0^{o}C

- Get the 'point' where the vertical gap line of 5.0^{o}C intersects the curved dry-bulb temperature line of 25.0^{o}C, mark the 'point'
- From this 'point' move horizontally to the right, using a ruler, to intersect the 95.0 kPa line (inclined straight lines running from top left to bottom right)
- At the intersection of this 95.0 kPa line, move vertically up or down and read off the air density at the top or bottom scale. The true air density is 1.10 kg/m^3.

Figure 5.10 illustrates these steps.

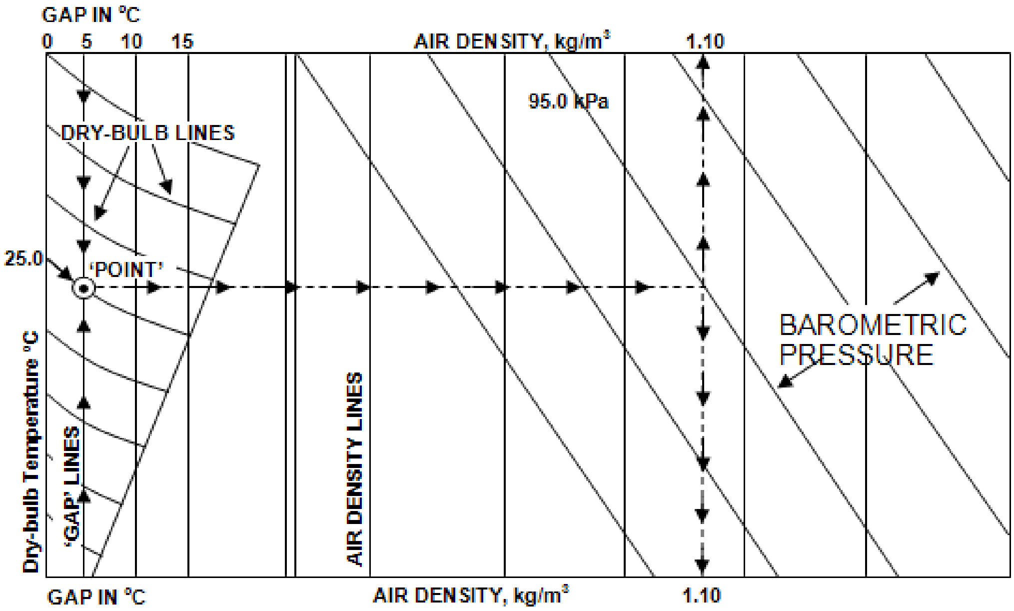


Figure 5.10 Finding true density on Lombard's air density chart

THE MASS FLOW OF DRY AIR

As discussed before, the concept of dry air is useful in ventilation calculations, because the mass of dry air remains constant in any circuit, provided that there is no air leakage.

As was shown in Chapter 2 of this course, there are two Equations that can be used to calculate the mass flow of air:

(i) Mass flow (m) = Quantity (Q) x Density (ρ) or

(ii) Mass flow = $\frac{\text{Quantity}}{\text{Apparent Specific Volume (ASV)}}$

$$m = \frac{Q}{ASV}$$

Where:

m	=	Mass flow of dry air (kg/s)
Q	=	Quantity of air water vapour mixture (m^3/s)
ρ	=	Density (kg/m^3)
ASV	=	Apparent Specific Volume (m^3/kg)

WORKED EXAMPLE 6

The air quantity measured is 60 m^3/s. Calculate the mass flow of the air for the following cases (these are 4 different scenarios):

a)	Apparent Specific Volume:	0.87 m^3/kg
b)	Density:	1.24 kg/m^3
c)	Barometric pressure:	100 kPa
	Air Temperature:	25.0 / 31.7°C
d)	Barometric pressure:	95 kPa
	Dry-Bulb temperature:	30°C
	Gas constant dry air:	0.2871 kJ/kgK

Answer

a) Apparent Specific Volume is given

$$\therefore \text{Mass flow} = \frac{\text{Air Quantity}}{\text{ASV}} = \frac{60}{0.87} = 68.9 \text{ kg/s}$$

b) Density is given

$$\therefore \text{Mass flow} = Q \times \rho = 60 \times 1.24 = 74.4 \text{ kg/s}$$

Psychrometric chart

c) Barometric pressure and temperature are given, therefore we use the 100 kPa Psychrometric chart and with the temperature given, read off the apparent specific volume.

$\therefore$ Apparent specific volume

$$\therefore \text{Mass flow} = \frac{\text{Air Quantity}}{\text{ASV}}$$

$$= \frac{60}{0.90}$$

$$= 66.7 \text{ kg/s}$$

d) Barometric pressure, dry-bulb temperature and gas constant are given. Thus, the 'ideal gas law' must be used:

$$\frac{PV}{T} = R$$

Now if $\frac{PV}{T} = R$

$$\text{Then } V = \frac{RT}{P}$$

$$= \frac{0.2871 \times (273 + 30)}{95}$$

$$= 0.916 \text{ m}^3\text{/kg}$$

NB

Note: The temperature is expressed in Kelvin and the **dry-bulb temperature** is used, if the dry-bulb and wet-bulb temperature were given, a common mistake made is to use the wrong temperature)

$$\therefore \text{Mass flow} = \frac{\text{Air Quantity}}{\text{ASV}}$$

$$= \frac{60}{0.916}$$

$$= 65.5 \text{ kg/s}$$

WORKED EXAMPLE 7

320.0 m^3/s of air enters a downcast shaft at a density of 0.97 kg/m^3 and at the bottom of the shaft the apparent specific volume is 0.871 m^3/kg. Calculate the air quantity at the shaft bottom if it is assumed that there is no air leakage.

Answer

Mass flow of dry air at top of shaft	=	$Q \times \rho$
	=	320.0×0.97
	=	310.4 kg/s

But the mass flow of dry air remains constant.

$\therefore$ Mass flow at shaft bottom = 310.4 kg/s

Given that apparent specific volume at shaft bottom as 0.871 m^3/kg and knowing that $m = \frac{Q}{ASV}$, then re-arranging this formula:

$$
\begin{aligned}
Q &= m \times ASV \\
&= 310.4 \times 0.871 \\
&= 270.4 m^3/s
\end{aligned}
$$

Therefore the quantity of the air at the shaft bottom is 270.4m^3/s.

NB

The quantity of air changes as it moves through the mine because of the change in air density.

Rule

If it is therefore assumed that the density remains constant, then the quantity through the mine will remain constant as well (provided that no air losses occur).

WORKED EXAMPLE 8

Refer to Figure 5.11:

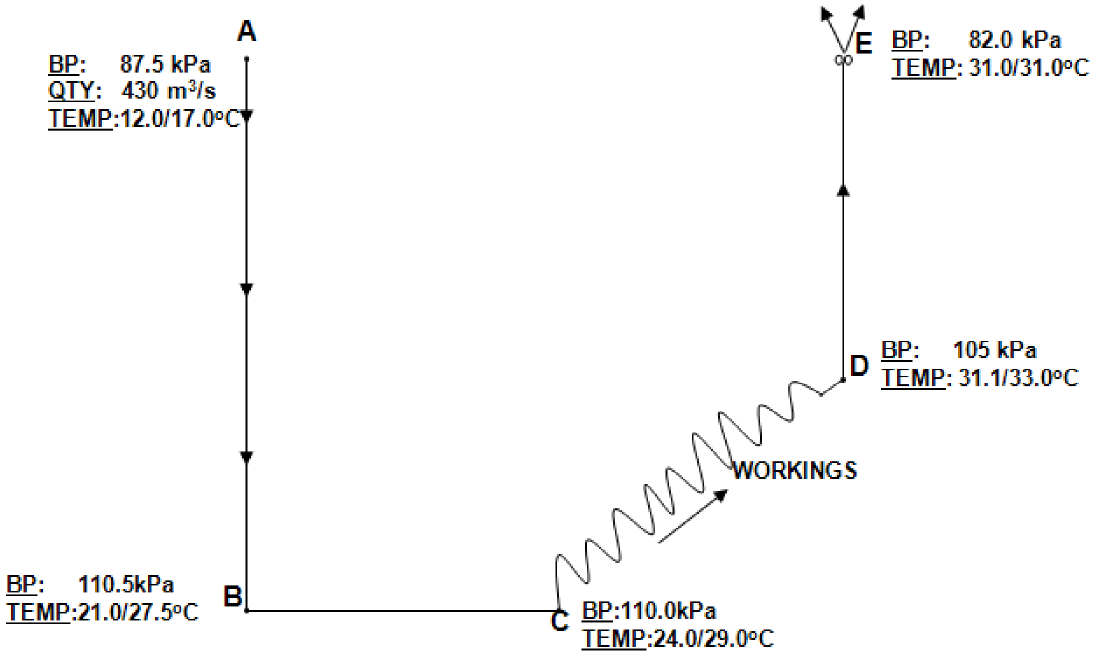


Figure 5.11 Schematic drawing of air flow through mine

From Figure 5.11, calculate the quantities at B, C, D and E assuming there are no air leakages.

Psychrometric chart

Answer

Mass flow at A can be determined as follows (where the ASV 0.965 m^3/kg read off from the 87.5 kPa Psychrometric chart):

$$= \frac{Q}{ASV}$$

$$= \frac{430}{0.965}$$

$$= \mathbf{445.6\ kg/s}$$

But mass flow remains constant throughout the system. Therefore mass flow A (m_A) = 445.6 kg/s = m_B = m_C = m_D = m_E

$$\text{Now if} \quad m = \frac{Q}{ASV}$$

$$\text{Then} \quad Q = m \times ASV$$

$$\therefore \mathbf{Q_B} = M \times ASV$$

Psychrometric chart

Where the ASV is 0.799 m^3/kg read off from 110.0 kPa Psychrometric chart.

$$= 445.6 \times 0.799$$

$$\mathbf{= 356.0\ m^3/s}$$

$$\mathbf{Q_C} = m \times ASV$$

Where the ASV is 0.808 m^3/kg read off from the 110.0 kPa chart.

$$= \quad 445.6 \times 0.808$$

$$\mathbf{= \quad 360.0\ m^3/s}$$

$$Q_D \quad = \quad m \times ASV$$

Where the ASV is 0.873 m^3/kg read off from the 105 kPa chart.

$$= \quad 445.6 \times 0.873$$

$$\mathbf{= \quad 389.0\ m^3/s}$$

$$Q_E \quad = \quad m \times ASV$$

Psychrometric chart

Where the ASV is 1.12 m^3/kg read off from the 82.5 kPa chart.

$$= \quad 445.6 \times 1.12$$

$$\mathbf{= \quad 499.0\ m^3/s}$$

NB

Note: B (105.5 kPa) and E (82.0 kPa) the closest Psychrometric chart to the given barometric pressures have been used, when interpolation needs to be done it will be stated as such.

SELF STUDY PROBLEM

Psychrometric chart

By using the relevant Psychrometric charts, complete the following Table (use the Psychrometric calculations to check your read off answers).

Table 5.1 Psychrometric properties (three conditions)

Question number	(a)	(b)	(c)	Units
Barometric pressure	95	112.5	87.5	kPa
Temperature	28.0/31.0	31.5/32.0	25.5/31.5	°C
Sigma heat				
Total heat content				
Moisture content				
Vapour pressure				
Relative humidity				
Dew point temperature				
Apparent specific volume				
Apparent density				

SELF STUDY PROBLEM

Consider Figure 5.12. By making use of the psychrometric charts, calculate the airflow volume between points 1 and 3. (Assume no leakages).

Answer

44.4 m^3/s

41.4 m^3/s

40.6 m^3/s

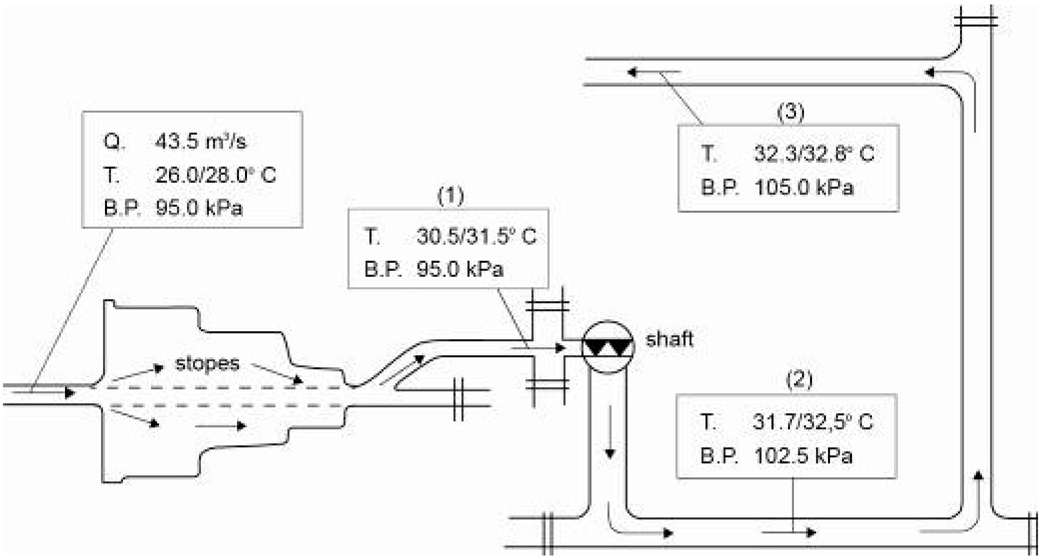


Figure 5.12 Diagrammatic representation of air flow through mine

5.2 Heat and Moisture Content Changes in Air

When an air-water vapour mixture flows along an underground system, the barometric pressure and temperature of the air changes continuously resulting, in a change of all the physical properties of the air.

NB

The most important physical property changes that will be discussed are the sigma heat content (kJ/kg) and moisture content (g/kg) of the air. The Psychrometric chart is made up of a combination of sensible and latent heating or cooling. A line or curve connecting two points on a Psychrometric chart as a result of a combination of sensible heat (SH), sensible cooling (SC) and latent heating (LH) or latent cooling (LC) at constant pressure can have any shape and any direction depending upon the predominance of the one process over the other. Figure 5.13 shows the subdivision of a Psychrometric chart according to the types and magnitude of heating or cooling.

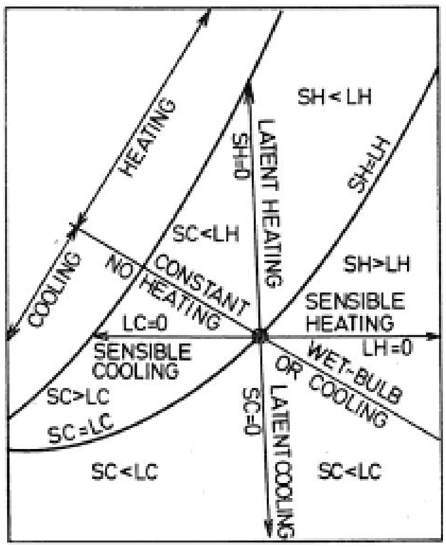


Figure 5.13 Combination of sensible and latent heating and cooling

When heat is added to the air, the sigma heat content of the air will increase, and vice versa (sensible heat process). The same with water vapour, if water vapour evaporates into the air as a result of heating the air, the moisture content of the air will increase. When water vapour condenses out of the air, as a result of cooling the air, the moisture content of the air will decrease (Latent heat process).

Refer to Figure 5.14 to explain the cut off position for evaporation and condensation of water vapour.

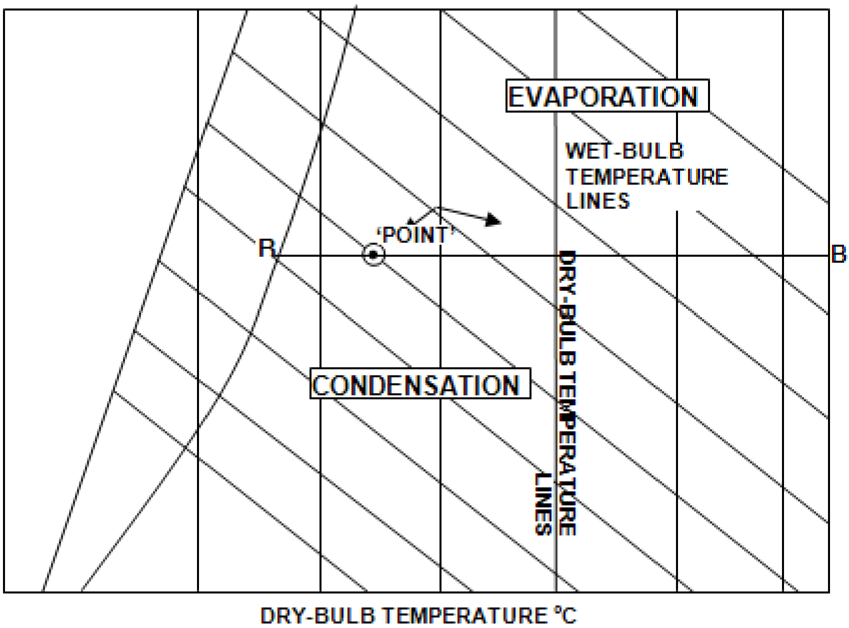


Figure 5.14 Relationship of evaporation and condensation with regards temperature point

Draw a horizontal line ‘BB’ through the ‘point’.

- **Evaporation** of water vapour **into the air** takes place, if the final condition lies anywhere above line ‘B’ and to the right of the 100% relative humidity line (i.e. the moisture content of the air increases). Thus, **with an increase in air temperature water vapour will evaporate into the air.**

- **Condensation** of water vapour **out of the air** takes place, if the final condition lies anywhere below line B to the right of the 100% relative humidity line (i.e. the moisture content of the air decreases). Thus, **with a decrease in air temperature, water vapour will condense out of the air.**

The following needs to be considered when sensible and latent heat calculations are done:

- **Sensible Heat Requirements (change in temperature without change in state)**

  Use the static heat Equation $W = M \times C_p \times \Delta t$

- **Latent Heat Requirements (change in state without change in temperature**):
  - To change ice at 0ºC to water at 0ºC and vice versa, the amount of energy needed is 335 kJ per kg of ice.
  - To change water at 100ºC to vapour at 100ºC and vice versa, 2 440 kJ/kg of ice is necessary.

WORKED EXAMPLE 9

(Combined Sensible Heat and Latent Heat calculation)

Calculate the amount of heat required to change 2 kg of ice at -2ºC to superheated water vapour at 105ºC. If the process took 0.0833 hours, calculate the power involved.

Answer

The process can be best described by Figure 5.15 (it shows the various energy processes involved, that is latent heat and sensible heat to heat 2 kg of ice from -2ºC to 105ºC). Legends used in sketch are as follows:

- **SH - Sensible Heat**
- **LH - Latent Heat**

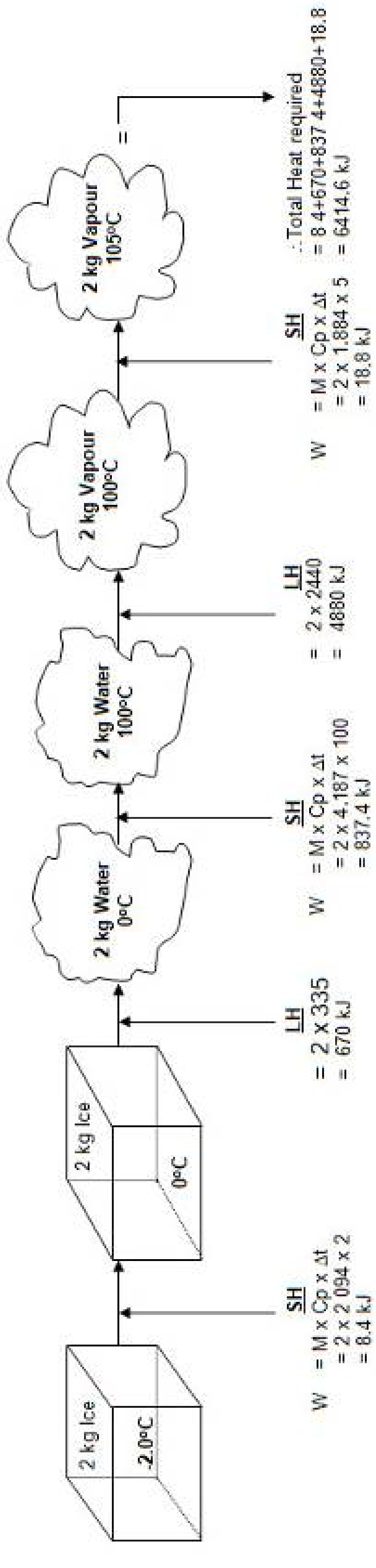


Figure 5.15 Phases of heat change from ice to vapour for water

The summarised results are shown in Table 5.2.

Table 5.2 Tabulation of results

Process Change		**SH/ LH**	**Calculation**	**Answer**	
To change 2 kg ice at $-2^{o}C$	to ice at $0^{o}C$	SH	= 2 x 2.094 x 2	= 6.188	kJ
To change 2 kg Ice at $0^{o}C$	to water at $0^{o}C$	LH	= 2 x 335	= 670	kJ
To change 2 kg water at $0^{o}C$	to water at $100^{o}C$	SH	= 2 x 4.187 x 100	= 837.4	kJ
To change 2 kg water at $100^{o}C$	to vapour at $100^{o}C$	LH	= 2 x 2440	= 4 880	kJ
To change 2 kg vapour at $100^{o}C$	to vapour at $105^{o}C$	SH	= 2 x 1.884 x 5	= 18.8	kJ
∴Total heat/energy required				**= 6 415**	**kJ**

Now the process took 0.0833 hours and 0.0833 x 60 x 60 = 300 seconds

$$\therefore \quad \text{Power involved} = \frac{\text{Total Heat/energy Required}}{\text{Seconds}}$$

$$= \frac{6\ 415}{300}$$

$$= 21.4 \text{ kW}$$

5.2.1 Change in Heat Content of Air

(Ignoring auto-decompression – auto decompression and auto compression will be covered in detail later).

Let's start this section with a simple example that will show how the increase in heat content of the air can be calculated (in kJ/kg or kW).

WORKED EXAMPLE 10

When 40.0 m^3/s of air at a BP of 107.5 kPa flows along an underground section at a temperature of 26.0 / 29.5 ^{o}C, it already contains a certain amount of heat present in the air, which is called the 'heat content' of the air. See Figure 5.16.

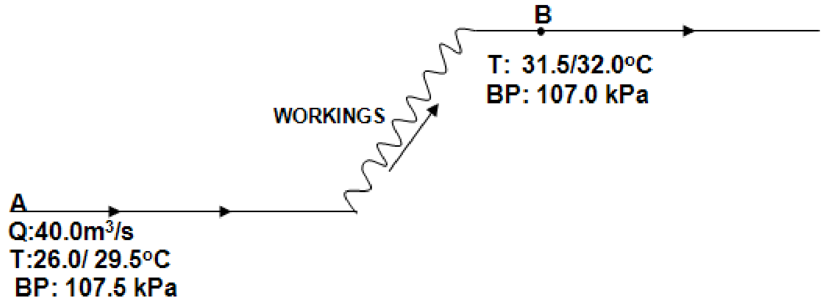


Figure 5.16 Schematic representation of air flow through mine

Psychrometric chart

Referring to the 107.5 kPa Psychrometric chart and using the temperature of 26.0 / 29.5ºC, one can determine the sigma heat as 75 kJ/kg, which is the heat content of the air at A.

Now, as this air flows from point A along the haulage and workings, heat is added to the air from many external sources, such as heat-flow from rock, machinery and so forth. Therefore at point B the temperature has increased to 31.5/32.0ºC. Referring to the 107.5 kPa Psychrometric chart, using this temperature, the sigma heat is now 99.6 kJ/kg, which is the heat content of the air at B.

By calculating the difference in sigma heat between points A and B, one can now say that the amount of heat added to the air is 24.6 (99.6 - 75.0) kJ/kg. **This means that 24.6 kJ of heat has been added per 1 kg of air.** What is the mass flow of air flowing in this section?

$$\text{Mass flow} = \frac{\text{Quantity}}{\text{ASV}}$$

From the 107.5 kPa Psychrometric chart at a temperature of 26.0/29.5ºC, the apparent specific volume (ASV) is read off as 0.833 m^3/kg.

$$\therefore \text{Mass flow of air} = \frac{40}{0.833}$$

$$= \mathbf{48.02\ kg/s}$$

What is the total amount of heat added to the air in kilowatts?

Now, total heat added = Mass flow$_{air}$ x Δ Sigma heat

$$\text{kW heat} = m \times \Delta S$$

$$= \frac{kg}{s} \times \frac{kJ}{kg}$$

= kJ/s **(and 1 kJ/s = 1 kW)**

∴Total heat **added** = 48.02 x 24.6

= 1 182 kW

Alternatively:

Total heat at A = mass flow A x S_A

= 48.02 x 75

= 3 601 kW or 3.6 MW

Total heat at B = Mass flow B x S_B

= 48.02 x 99.6

= 4 783 kW or 4.8 MW

∴Total Heat **added** = Total heat B - Total heat A

= 4 783 – 3 601

= 1 182 kW

The same methods as explained above will apply when the air is cooled down or reconditioned by means of cooling coils, bulk air coolers and so forth using refrigeration. Worked Example 11 will explain it better.

WORKED EXAMPLE 11

Refer to the Figure 5.17.

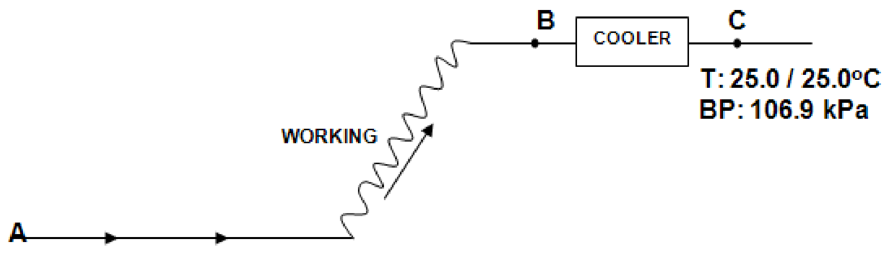


Figure 5.17 Schematic representation of air flow through mine

At B the measured temperatures were 31.5 / 32.0°C. The air now passes through a cooler and the temperature at C is 25.0°C saturated. Calculate the total amount of heat removed from the air (use numbers from previous Worked Example numbers).

Answer

Mass flow at B = 48.02 kg/s

ΔS at B = 99.6 kJ/kg

Psychrometric chart

From 107.5 kPa Psychrometric chart at a temperature of 25.0°C saturated, the sigma heat at C is 71.2 kJ/kg.

∴Difference in Sigma Heat H = S_B - S_C

= 99.6 - 71.2

= 28.4 kJ/kg

That is the heat removed is 28.4 kJ per kg of air.

Now total heat **removed**	=	Mass flow x ΔS
	=	48.02 x 28.4
	=	1 364 kW or 1.4 MW

∴Total heat removed from the air = 1364 kW = Cooler duty

Note:

- **Always assume that the total heat removed from the air, is equal to the total heat added to the water flowing through the cooler that is cooler duty.**
- **Also, always assume that the air leaving a cooler (coils, sprays etc.) is at a saturated temperature,** unless otherwise stated.

5.2.2 Changes in Moisture Content of the Air

The effect of changes in moisture content of the air on the temperature of the air will be explained with the help of Worked Example 12.

WORKED EXAMPLE 12

Using the same numbers as from Worked Example 11:

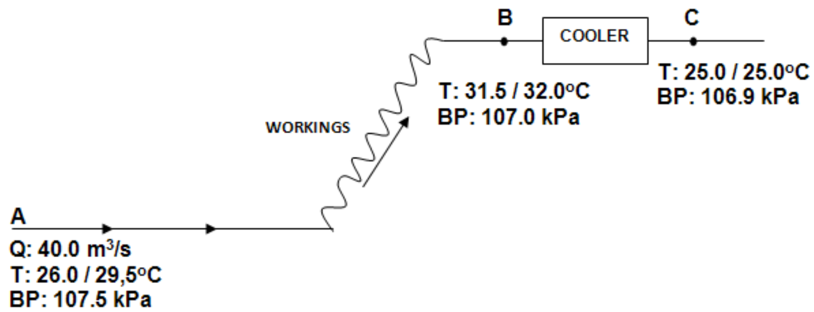


Figure 5.18 Schematic representation of air flow through mine

Answer

A to B:

From A to B the air temperature has increased, therefore water vapour has evaporated into the air. Now, determine the moisture content (apparent specific humidity) at points A and B using the 107.5 kPa Psychrometric chart and the respective air temperatures at A and B.

Moisture content (r) at A	=	18.6 g/kg
Moisture content (r) at B	=	27.8 g/kg

By calculating the difference in moisture content between A and B, one can now say that the amount of water vapour evaporated into the air is:

$\Delta r \quad = \quad 27.8 - 18.6 = 9.2$ g/kg

This means that 9.2 grams of water vapour has evaporated into 1 kg of air, and the mass flow of the air is 48.02 kg/s (calculated previously). What is the total amount of water vapour that evaporated into the air in litres per second (ℓ/s) and litres per 8 hour shift?

Total amount evaporated $= m \times \Delta r$

$= \frac{kg}{s} \times \frac{g}{kg}$

$=$ g/s

To convert g/s to kg/s = $\frac{\textbf{g/s}}{\textbf{1000}}$ = kg/s = ℓ/s (1 kg ^{H_2O} = 1 litre ^{H_2O})

$\therefore$ Total evaporated $= \frac{m \times \Delta r}{1000}$

$= \frac{48.02 \times 9.2}{1000}$

$= 0.44\ \ell/s$

And **litres per 8 hour shift can be determined as follows:**

$= 0.44 \times 60_{\text{minutes}} \times 60_{\text{seconds}} \times 8_{\text{hrs}}$

$=$ 12 723 ℓ per 8 hour shift

B to C:

From B to C the air temperature has decreased, therefore water vapour has **condensed** out of the air.

At B the moisture content = 27.8 g/kg

At C the moisture content = 19.0 g/kg

$\therefore$ Moisture content difference = 8.8 g/kg

Now total water condensed $= \frac{m \times \Delta r}{1000}$

$= \frac{48.02 \times 8.8}{1\,000}$

$= 0.42\ \ell/o$

And litres / 8 hour shift (as before):

$= 0.42 \times 60 \times 60 \times 8$

$=$ 12 170 ℓ per 8 hours

Animation

5.2.3 Heat Absorption and Heat Rejection Systems

Heat absorption takes place at the workings, where heat is removed from the air by cooling coils, sprays, bulk air coolers and so forth. Heat rejection takes place normally in main return airways (RAW's) where the heat absorbed from the air at the workings is rejected by condenser cooling towers or sprays. The concept of condensers, evaporator and the towers associated with it will be dealt with in detail in the chapters on refrigeration and refrigeration components.

NB

The above and the comparison between these two processes will be discussed in detail in these said notes. Important to note, is that both heat absorption and heat rejection systems consist of two main circuits, namely:

- A water circuit
- An air circuit (heat given off from air to water or vice versa).

When the performance (duty) or any other parameters of such appliance to absorb or reject heat, are to be calculated, on the air side, Psychrometric charts are used as mentioned before.

However, looking at the performance of these appliances from a **water-side** point of view, **Psychrometric charts cannot be used, and hence the heat flow equation ($q = m \times C_p \times \Delta t$) must be used and will also be dealt with in the chapter on heat, for now it is important to note the following.**

Under ideal conditions the heat exchange process between water and air can be expressed as follows:

$$(m \times c_p \times \Delta t)_{water} = (m \times \Delta S)_{air}$$

$$kW_{water} = kW_{air}$$

Rule

Where C_p is the specific heat capacity of water which is 4.187 kJ/kg°C. By using the same values for the Worked Example as before Worked Example 13 will explain it better.

WORKED EXAMPLE 13

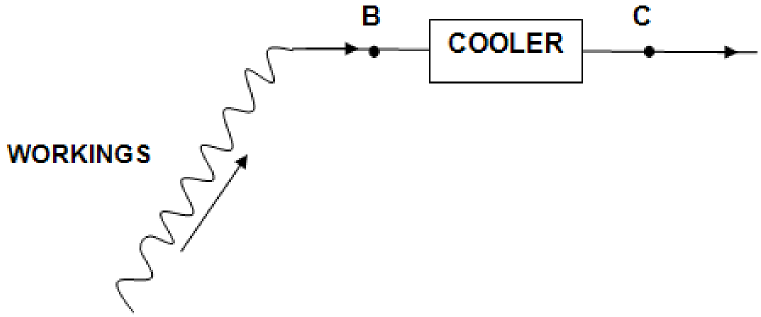


Figure 5.19 Schematic representation of airflow through mine

From Figure 5.19, calculate the water-flow rate through the cooler if the water entering the cooler is 8.0°C and the water leaving the cooler is 19.6°C.

Answer

From the air side, it was calculated that the heat removed from the air by the cooler is 1 364 kW (Using Psychrometric charts).

Assume that the heat removed from the air is equal to the heat added to the water = 1 364 kW.

Thus, in the flow heat Equation, $q = m \times C_p \times \Delta t$, with q as 1364 kW.

Re-arranging this Equation to determine the water-flow rate, then

Where:

q	=	1 364 kW
Cp	=	4.187 kJ/kg°C
Δt	=	19.6- 8.0 = 11.6°C
m_w	=	water-flow rate?
$\therefore m_w$	=	$\frac{1364}{4.187 \times 11.6}$
	=	28.1 ℓ/o

$\therefore$Water-flow rate = 28.1 ℓ/o

5.2.4 Airways that Combine

NB

It does happen that airways underground flow into one another, which means that airways with air with different psychrometric characteristics will influence one another in such a way that the the properties of the air will be different after the airways have joined. This is an important aspect to know, as it is often used to mix cold air from the bulk air coolers with hot air from the workplaces to make it cool again. For example, if airways A, B, C and so forth join each other into one final airway E, and the resultant psychrometric properties for airway E, must be determined, it can be done as follows (for the Sigma heat (S) value and moisture content (r)):

$$S_E = \frac{S_A \text{ x } m_A + S_B \text{ x } m_B + ...+ S_n m_n}{m_E} \text{ .... } (5.16)$$

The above mentioned Equation could also have been represented as follows:

$$S_E = \frac{kW_A + kW_B + ...+ kW_n}{m_E}$$

The combined moisture content for the airways combining can be determined as follows:

$$r_E = \frac{r_A \text{ x } m_A + r_B \text{ x } m_B + ...+ r_n \text{ x } m_n}{m_E} \text{ .... } (5.17)$$

5.2.5 Application of Thermodynamics to Mine Airflow

Definition

Thermodynamics is the science which studies the effects of heat and work on the properties of fluids. Up until now we have considered some very basic aspects and applications applicable to Psychrometry. It is however important to understand and apply aspects of thermodynamics applicable to mine airflow circuits, because as we already know that Bernoulli's Equation can only be used for certain conditions and that is:

- Ideal Fluids
- Slow moving fluids
- Incompressible fluids.

Definition

Incompressible fluids include gasses and other fluids in situations whereby the pressure does not change by more than a few kilopascals. Although all gases fall into the compressible fluid class, the density is negligibly small if the change is only a few kilopascals. Only with large changes in pressure will there be a noticeable difference in density, which must then be taken into account. Air, which flows through fans in mine ventilation networks, is treated as an ideal gas.

It is clear that Bernoulli doesn't stand true for compressible fluids and therefore we must derive thermodynamic equations to satisfy this. In the following notes we are going to first look at the case where r **(moisture content) stays constant and the density changes and secondly where density and moisture content changes**, naturally the most practical situation.

5.2.5.1 Properties of Substances with Relation to Thermodynamics

The properties of gasses with relation to thermodynamics are pressure, temperature and specific volume (inverse of density). If the specific volume and pressure or temperature is specified, then all the other properties of a liquid are bound. Other properties of gasses, which have already been dealt with are:

- Thermal capacity (specific heat capacity)
- Latent heat
- Enthalpy
- Entropy
- Viscosity.

NB

Heat and work are the two fundamental forms of which energy can be added or removed from any substance.

Definitions

Heat is the flow of energy as a result of a temperature difference (from warm to cold environment, **visa versa not possible**). Increased temperature in a gas results in increased energy content (**internal energy**) of the gas. Work is the flow of energy, which takes place in the driving shaft of a fan or pump.

Laws

The increase in energy is primarily as a result of an increase in pressure with an increase in temperature. This leads us to the two laws of thermodynamics, which is:

- First Law: Conservation of Energy
- Second Law: Energy, which is dissipated ('lost') due to friction, cannot be converted back to its original form.

NB

Derivation of the Bernoulli equation will also show how **natural ventilation energy** (NVE) and **natural ventilation pressure** (NVP) helps or hinders the flow of air through a mine. The thermodynamic derivation is also important in pressure and network analysis.

5.2.5.2 The 'Steady State' Energy Equation

The Equation is used as a basis for all flow processes. The term 'Steady' means **that the process does not change with time**.

Consider a situation where a fluid is displaced through a column by means of a fan from point '1' to point '2' (Figure 5.20). The abbreviations in the sketch are as follows:

- Pressure (P)
- Velocity (U)
- Temperature (T, Kelvin)
- Enthalpy (H)
- Height from datum/reference line (Z).

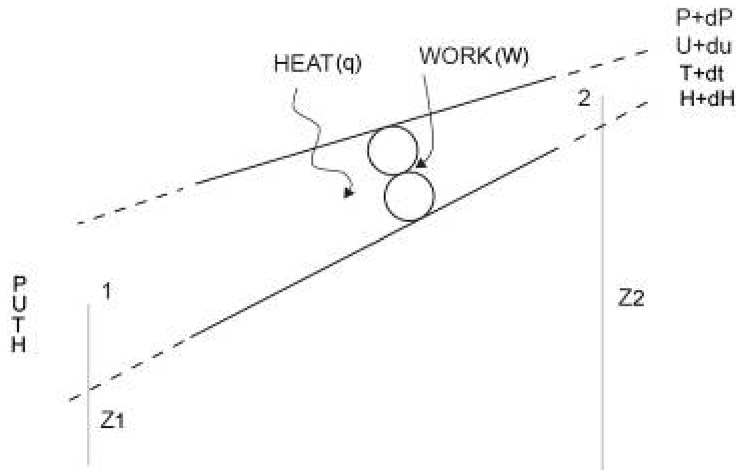


Figure 5.20 Fluid displacement through fan

Three definite types of energy must be considered if we want to establish an energy balance for any system that is:

- Heat from an external source (q)
- Work delivered by a fan (W)
- The energy content of the fluid which enters or leaves the 'control film'.

The energy content of any fluid consists of 3 properties, namely:

- Enthalpy
- Kinetic energy (as a result of the velocity of the fluid)
- The gravitational energy (as a result of the position relative to the gravitational field of the earth).

Definition

The **enthalpy** is the energy per unit mass of the fluid as a result of random movement of the molecules. Enthalpy includes the thermal energy as internal energy and energy as a result of pressure in the liquid.

From the Figure 5.20 the energy balance can be described as follows:

$$\text{(Inflow energy quantities} \quad = \quad \text{outflow energy quantities)}$$

$$dW + dq + M(H + gz + \frac{1}{2}u^2) \quad = \quad M((H + dH) + g(Z + dZ) + \frac{1}{2}(u + du)^2) \qquad \text{……... (5.18)}$$

The term $(u + du)^2 = u^2 + 2udu + du^2$ with du^2 negligibly small.

The steady flow energy Equation can thus be expressed simply as:

$$dW + dq \quad = \quad M[dH + gdZ + udu] \quad = \quad M[dH + gdZ + d(\frac{u^2}{2})]$$

$$dW + dq \quad = \quad M[dH + gdZ + d(\frac{u^2}{2})] \qquad \text{............................ (5.19)}$$

It is also important to note that the symbols q, W and M refer to the flow of heat, work and mass, while the terms in the right hands side represent energy per unit mass. In setting up the energy balance above no reference is made to friction or other losses which may occur as the fluid passes from position 1 to position 2. This is because that friction or any other losses take place within the control system and thus do not influence the Equation and therefore valid for an equally for processes with and without friction. The equation therefore applies to **both compressible and incompressible fluids**.

ENTHALPY

The steady flow energy Equation does not contain a pressure term, but this is included in the enthalpy term, $dH = vdP + Tds$.

Where:

dH	=	change in total energy (Enthalpy) content of the fluid
νdP	=	pressure component, ν(specific volume), dP (pressure difference), **(useful energy)**
Tds	=	entropy component, T (temperature), ds (entropy), **(wasted energy)**

Tds represents the total 'wasted' heat, which is added to the air as it flows through a part of the airway. There are two sources of heat namely:

- External source dq
- Frictional energy dF.

The entropy component is thus, **Tds = dq + dF**

Definition

A perfect pump or fan where all the energy is used to increase the νdP term is known as an **isentropic energy system** (Constant entropy).

Rule

Absolute pressure must always be used for calculations.

THERMAL CAPACITY

Definition

Thermal capacity is defined as the increase in enthalpy per unit temperature.

$$C_p = \frac{dH}{dT} \text{ at constant pressure.. (5.20)}$$

From the steady flow energy equation, with no work, kinetic energy difference negligibly small and with no elevation, the Equation is written as follows:

$$dq = MdH \text{ and with } C_p = \frac{dH}{dT} \;=>\; dH = C_p dT\text{.....................(5.21)}$$

$H_2 - H_1 = C_p\,(T_2 - T_1)$ for ideal gas at any pressure which gives us:

$$dq = M\,C_p dT\text{... (5.22)}$$

NB

The above Equation is not applicable for gas mixtures where there is a change in the moisture content and is applicable for non-gasses where the change in pressure is **less than 1 MPa**. Calculation methods discussed under psychrometry must then be used.

The thermal or heat capacity of water, dry air or water vapour is as follows:

Rules

- C_p water — 4.187 kJ/kg°C or Kelvin
- C_p dry air — 1.005 kJ/kg°C or Kelvin
- C_p water vapour (steam) — 1.884 kJ/kg°C or Kelvin
- C_p Ice — 2.094 kJ/kg°C or Kelvin

5.2.5.3 Thermodynamic Principles of Mine Airflow with Moisture Content Constant

The steady flow energy Equation simplified shown before is as follows:

$$dW + dq \quad = \quad M[dH + gdZ + udu] \quad = \quad M[dH + gdZ + d(\frac{u^2}{2})]$$

From the previous section:

dH = νdP + Tds and with Tds = dq + dF (friction energy losses)

From the previous section, gives us the Equation:

dH = vdP + dq + dF (do not confuse F with force (Newton))

The steady flow energy equation can therefore be re-written as follows:

$$dW \quad = \quad vdP + dF + gdZ + udu$$

This is known as the energy Equation and by multiplying with the density (ρ) throughout the following **pressure Equation** can be derived:

$$\rho dW \quad = \quad dP + \rho dF + \rho gdZ + \rho udu \quad \text{or also as}$$

$$dp_f \quad = \quad dP + dp \quad + \rho gdZ + \rho udu \text{ (pressure Equation)}$$

Where:

ρdW = dp_f (fan pressure)

ρdF = dp (friction pressure loss)

And if integrated over a particular air way or shaft becomes:

$$\int dW \quad = \quad \int \nu dP + \int dF + \int gdZ + \int udu$$

All integrals evaluated between point 1 the beginning of the air way and point 2 the end of the air way gives us:

$$W_f \quad = \quad \int vdP + F + g\,(z_2 - z_1) + \frac{1}{2}(u_2^2 - u_1^2) \qquad \text{................. (5.23)}$$

Where:

W_f = fan energy

5.2.5.4 Application of the Energy Equation

Figure 5.21 shows a graphical representation of air flowing through a mine with moisture constant and density changing.

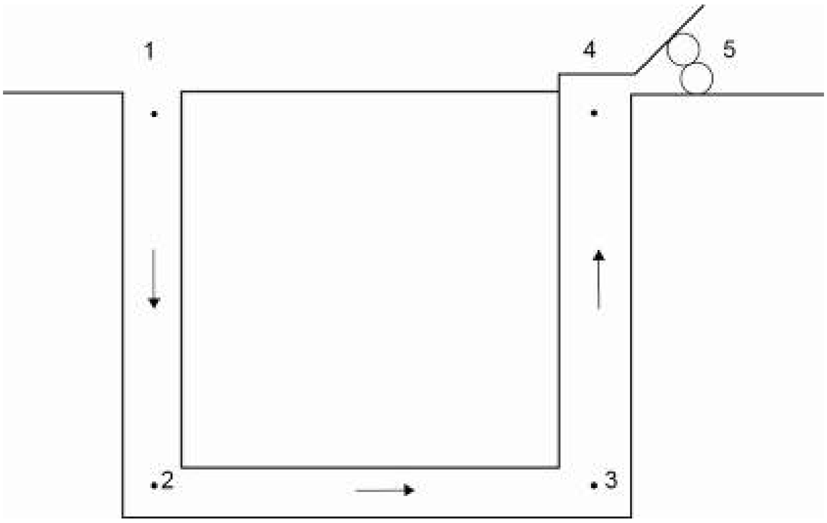


Figure 5.21 Airflow through a mine

Downcast Shaft (position 1 to 1):

$\int \nu dP + F + g\,(Z_2 - Z_1) = 0$ (no fan and velocity term small enough to ignore)

Thus:

$F = g\,(Z_1 - Z_2) - \int \nu dP$ (F = Total friction energy loss)

It can be represented as the pressure versus specific volume graph shown in Figure 5.22.

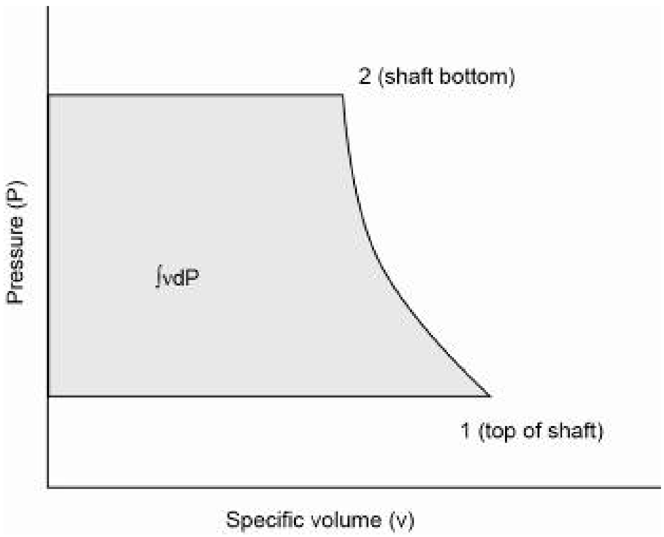


Figure 5.22 Pressure versus specific volume

Integration of the pressure Equation $dp_f = dP + dp + gdZ + udu$ gives:

$$p_f = (P_2 - P_1) + p + g\int \rho dZ + \int \rho\, udu \quad \text{.............................} (5.24)$$

For the same downcast shaft (position 1 to 2) we get the following (assume velocity terms as negligibly small):

$$0 = (P_2 - P_1) + p + g\int \rho dZ$$

$$p = (P_1 - P_2) - g\int \rho dZ$$

This is graphically represented in Figure 5.23.

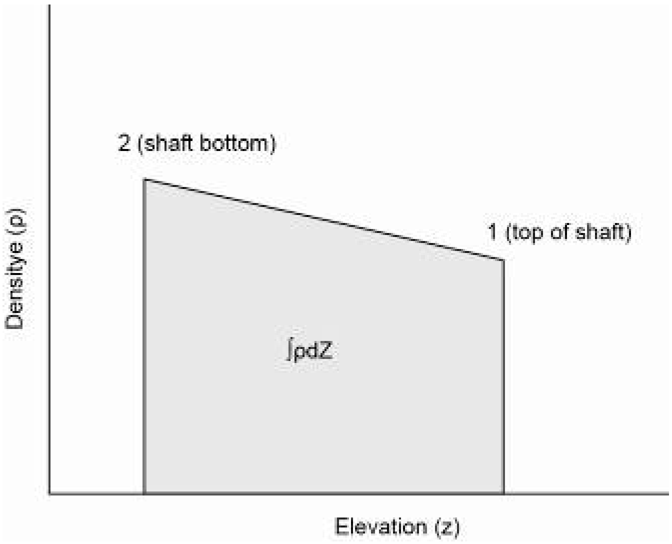


Figure 5.23 Density versus elevation

Using the energy Equation for the rest of the air way, the following can be derived:

For workings (point 2 to 3):

$\int \nu dP + F + 0 =$ 0 (no fan, workplaces horizontal)

For upcast shaft (point 3 to 4):

$\int \nu dP + F + g(Z_4 - Z_3) = 0$ (no fan in shaft (only at the exit to the shaft))

For upcast shaft fan (point 4 to 5):

$\int \nu dP + 0 + 0 = W_f$, (fan horizontal, assume no friction energy losses)

For return airway (point 5 to point 1):

0 + 0 + 0 = 0 no fan, horizontal, no friction losses, no pressure change

The summation of above Equations gives the following for the total air way: $\oint \nu dP + \sum F + 0 \quad = \quad W_{fan}$ with dZ = 0 (shafts with same elevation), rewritten we get the energy balance Equation:

$$\sum F \quad = \quad W_{fan} - \oint \nu dP \text{ with more than 1 fan } \sum W_{fan} \quad \text{...... (5.25)}$$

Definition

The term $\int \nu dP$ is known as the **natural ventilation energy component (NVE)**. This term exists as a result of addition of energy to the mine air as it flows through the mine.

The mine exists as a heat machine and uses a part of the heat energy, which is added to the air to do work, and this is to help with circulation of the air through the mine (not very effective). What happens to air when it gets hot (where does it normally go)?

NB

The situation can be represented as a pressure specific volume graph, and this is shown in Figure 5.24. The specific volume of the air at the top of shaft is greater than that at the bottom of the up cast shaft due to an increase in heat in the mine workings (the density of the air in the up cast shaft is less than in the downcast shaft) and as a result the path traced out encloses a finite area

The area under the graph is thus negative as shown in Figure 5.25 below and the natural ventilation energy (NVE) or $-\int \nu dP$)) is thus positive and assists the fan in overcoming frictional energy 'losses' in the mine. That the enclosed area has a negative value is apparent, if it is remembered that during the original derivation of the energy balance Equation $\int \nu dP$ was defined as:

$$\oint \nu dP = \int \nu dP_{\text{down cast}} + \int \nu dP_{\text{workings}} + \int \nu dP_{\text{up cast}} + \int \nu dP_{\text{fan}}$$

$$= \int \nu dP_{\text{down cast}} + (\text{very small}) + \int \nu dP_{\text{up cast}} + (\text{very} \quad \text{small})$$

$$= \int \nu dP_{\text{down cast}} + \int \nu dP_{\text{up cast}}$$

Rule

$\int \nu dP_{\text{down cast}}$ has a positive value, whereas $\int \nu dP_{\text{up cast}}$ has a negative value, but is numerically larger, and the sum of the two terms is therefore negative, and $\int \nu dP$ has a negative value. This is graphically shown in Figure 5.24.

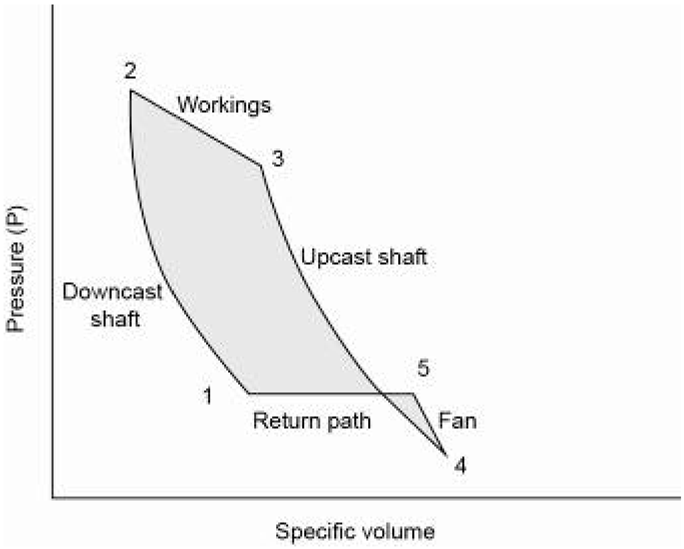


Figure 5.24 Pressure versus specific volume

$\int \nu dP$ is calculated by using the following Equation:

$$\int \nu dP = \frac{k}{k-1}(P_2\nu_2 - P_1\nu_1) \quad \text{.. (5.26)}$$

With:

$$k = \frac{\log\left(\frac{P_2}{P_1}\right)}{\log\left(\frac{\nu_2}{\nu_1}\right)} \quad \text{.. (5.27)}$$

From the formula $\sum F = W_f - \int \nu dP$, we can see that $\int \nu dP$ becomes positive and if we want to determine W_f a smaller fan input energy will be required. Fan will thus be assisted by the natural ventilation energy component.

The total fan input power $W_f = \sum F - \int \nu dP$

Where:

F	=	friction energy loss of the fan
W_f	=	fan air power (or PQ as previously identified) per unit mass

5.2.5.5 Application of the Pressure Equation

The pressure Equation, $p_f = (P_2 - P_1) + p + g\int \rho dZ + \int \rho\, udu$, can also be used for each airway in the mine, and can be expressed for different situations:

Downcast Shaft (point 1 to 2)

$(P_2 - P_1) + p + g \int \rho dZ = 0$, (no fan and velocity terms negligible).

Therefore:

$$p = (P_1 - P_2) - g \int \rho dZ$$

For Workings (point 2 to 3)

$(P_3 – P_2) + p + 0 = 0$ (no fan, workings horizontal)

Upcast shaft (point 3 to 4)

$(P_4 - P_3) + p + g \int \rho dZ = 0$ (no fan in shaft)

Upcast shaft fan (point 4 to 5)

$(P_5 - P_4) + 0 + 0 = p_f$ (fan horizontal, no friction pressure loss)

Return path (point 5 to 1)

0 + 0 + 0 = 0 (no fan, horizontal, no friction losses, no pressure difference, assumed same level of elevation)

Summation of the above Equations gives:

$$0 + \sum p + g \int \rho dZ = p_{fan} \quad \text{.................................. (5.28)}$$

For more than one fan in the system the Equation changes as follows:

$$p_f - g \int \rho dZ = \sum p \quad \text{(pressure balance Equation).......... (5.29)}$$

Where:

p_f = fan pressure

$g\int \rho dZ$ = Natural Ventilation Pressure (NVP)

p = total frictional pressure loss

In simpler terms the following formula can be used:

$$P_{fan} = p_{friction\ losses} \pm \text{Natural ventilation pressure (NVP)}$$

The Equation can be interpreted the same as the energy Equation. In this instance natural ventilation pressure exists due to a difference in density in the upcast and downcast shafts.

The ‘+’ sign is used when the NVP is opposing the fan and the ‘-‘sign is used when the NVP is assisting the fan. This principle can be applied the same for the NVE previously discussed.

This difference is a result of heat flow in the air and pressure differences due to the fan and possibly pressure differences due to elevation differences between the two shafts. A density versus elevation diagram sums up the situation as follows (shown in Figure 5.25):

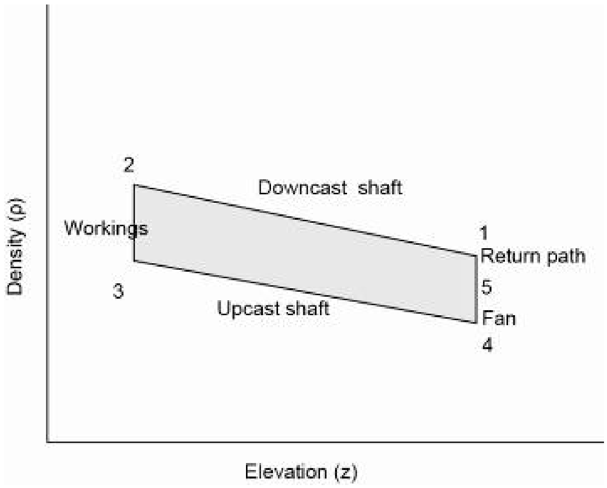


Figure 5.25 Density versus elevation

From the Figure 5.25 we see that the density of the downcast shaft is higher than that of the upcast shaft. The integral from 1 to 2 is negative, and the integral from 3 to 4 is positive and therefore the sum of the two integrals is negative (The area under the graph is thus negative). From Equation 5.29 it is clear that the fan is helped by the natural ventilation pressure. The natural ventilation pressure plays an important role in air flow and fan pressure requirements.

5.2.5.6 Thermodynamic Principles of Mine Airflow with Change in Moisture Content and Air Density

This is the more practical situation, in the sense that the moisture composition of the air gradually changes as a result of condensation and evaporation. Furthermore, the water droplets can flow in the same direction (at different velocities) or opposite that of the airflow. Let’s consider a situation of a ‘control film’ environment shown in Figure 5.26, but take into consideration that the moisture content changes. In this case, water droplets flow in the same direction as airflow.

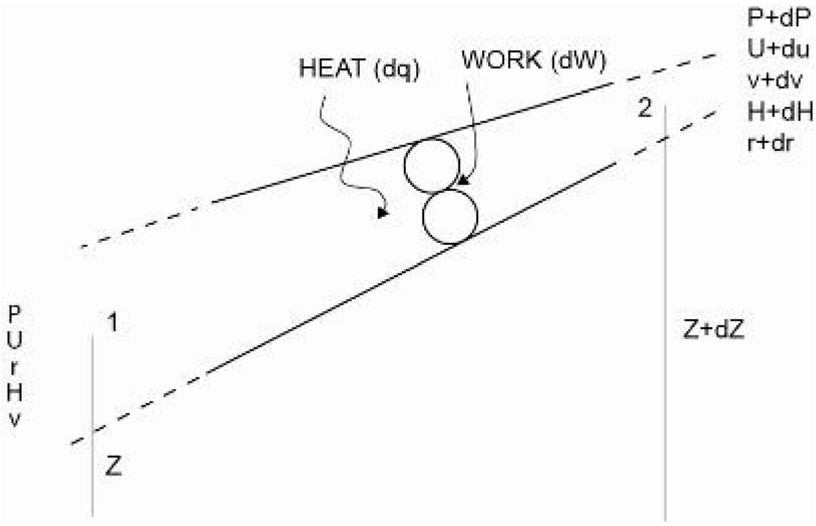


Figure 5.26 Controlled film environment including moisture changes

The symbols can be explained as follows and shown in Table 5.3.

Table 5.3 Summary of symbols relating to controlled film environment

	Inlet conditions			Outlet conditions		
	Dry air	**Moisture**	**Droplets**	**Dry air**	**Moisture**	**Droplets**
Miosture content		r	A		r + dr	a + da
Velocity	U_a	U_a	U_w	$U_a + du_a$	$U_a + du_a$	$U_w + du_w$
Enthalpy	H_a	H_m	H_w	$H_a + dH_a$	$H_m + dH_m$	$H_w + dH_w$

Where:

a = air
m = moisture
w = water

Important to remember, is that the mass flow of dry air stays constant and that all relevant amounts are expressed per unit mass dry air. Application of steady flow energy Equation and from the mass balance assuming no water and no droplets, then:

dr + da = 0, we get the following result:

$$dW + dq = g(1+r)dZ + gadZ + (1+r)u_a du_a + \frac{1}{2}u_a^2 dr + au_w du_w + \frac{1}{2}u_w^2 da + dH_a + rdH_m + H_m dr + adH_w + H_w d_a$$

The next term, which we now look at, is enthalpy (H) of the fluid and here we must not confuse H with h, which is the symbol for height in Bernoulli's Equation.

Definition

Enthalpy has been previously been defined as the energy per unit mass of a fluid as a result of random movement of the fluid's molecules. The enthalpy of the fluid is given as:

$$H \quad = \quad H_{air} + H_{water} \quad \text{...} \quad (5.30)$$

Or

$$H \quad = \quad (H_a + rH_m) + aH_w$$

And from this we get:

$$dH \quad = \quad dH_a + rdH_m + H_m dr + adH_w + H_w da$$

The steady flow energy Equation for changes in moisture content can be expressed as:

$$dW + dq = g(1+r)dZ + gadZ + (1+r)u_a du_a + \frac{1}{2}u_a^2 dr + au_w du_w + \frac{1}{2}u_w^2 da + dH$$

Again, we can say, dH = vdP+dq+dF (ν is now volume per unit mass dry air) and from this we get the final energy Equation for change in moisture content of air:

$$dW = \nu dP + dF + g(1+r)dZ + gadZ + (1+r)u_a du_a + \frac{1}{2}u_a^2 dr + au_w du_w + \frac{1}{2}u_w^2 da$$

This Equation can be expressed in pressure terms, by multiplying $1/\upsilon$ (ρ / (1+r)) to the Equation:

$$dp_f = dP + dp + g\rho dZ + \frac{ga}{\nu}dZ + \rho u_a du_a + \frac{1}{2}\frac{u_a^2}{\nu}dr + \frac{au_w}{\nu}du_w + \frac{1}{2}\frac{u_w^2}{\nu}da$$

If we set a = da = dr = 0 (no moisture and water droplets applicable), we will get the Equation which we previously used. Although these Equations must be integrated in order to account for a full shaft or airway situation. The only difference is when the droplets move in the opposite direction to airflow, the terms gadz and ga/ν dz will become negative. The integrated Equations for the case **where water droplets move in the same direction** as the airflow is:

$$W_f = \int \nu dP + F + g\int(1+r)dZ + g\int adZ + \int(1+r)u_a du_a + \frac{1}{2}\int u_a^2 dr + \int au_w du_w + \frac{1}{2}\int u_w^2 da$$

…. (5.31)

And the pressure Equation:

$$p_f = (P_2 - P_1) + p + g\int \rho dZ + g\int \frac{a}{\upsilon}dZ + \int \rho u_a du_a + \frac{1}{2}\int \frac{u_a^2}{\upsilon}dr + \int \frac{au_w}{\upsilon}du_w + \frac{1}{2}\int \frac{u_w^2}{\upsilon}da$$

…. (5.32)

5.2.5.7 Application to a Complete Mine Airway

If we now leave out the velocity terms in the previous two Equations, we will get:

$$W_f = \int \nu dP + F + g\int (1+r)dZ + g\int a dZ \qquad \text{.................................... (5.33)}$$

$$p_f = (P_2 - P_1) + p + g\int \rho dZ + g\int \frac{a}{\nu} dZ \qquad \text{.................................... (5.34)}$$

These two Equations can be expressed for a complete airway as such:

$$W_f = \oint \nu dP + \sum F + g\oint (1+r)dZ + g\oint a dZ \qquad \text{................................ (5.35)}$$

$$p_f = \sum p + g\oint p dZ + g\oint \frac{a}{v} dZ \qquad \text{......................................(5.36)}$$

With these Equations one must remember that the terms W_f, ν, r and a are expressed in terms of unit mass flow of dry air. The energy Equation can be rewritten as:

$$W_f = \oint \nu dP + \sum F + g\oint (1+r)dZ + g\oint a dZ$$

$$W_f - \oint \nu dP \quad = \quad \sum F + g\oint (1+r)dZ + g\oint a dZ \qquad \text{...................... (5.37)}$$

NB

It is the same as the previous energy balance Equation, except that fan energy plus natural ventilation must not only overcome friction energy losses **but must also remove water droplets from the mine.**

The pressure Equation above can be manipulated to show as follows:

$$p_f - g\oint \rho dZ \quad = \quad \Sigma p + g\oint \frac{a}{\nu} dZ \qquad \text{.. (5.38)}$$

As in this case, some of the fan energy must be used to add additional pressure as well as to remove water droplets from the mine. In this pressure equation there is not a similar term $g\oint r dZ$ as in the energy Equation. This term is found in the energy Equation as a direct result of the fact that all energy terms in the energy Equation are expressed as per unit mass of dry air. A similar term can be substituted into the pressure Equation, which is then rewritten as follows:

$$g\oint \rho\, dZ + g\oint \frac{a}{v} dZ \quad = \quad g\oint \frac{1+r}{v} dZ \; + \; g\oint \frac{a}{v} dZ$$

$$g\oint \rho dZ + g\oint \frac{a}{\upsilon} dZ = g\oint \frac{1+r}{\upsilon} dZ + g\oint \frac{a}{\upsilon} dZ \qquad \text{.................................... (5.39)}$$

Because:

$$\rho = \frac{1+r}{\nu}$$

$$g\oint \frac{1}{v}\, dZ + g\oint \frac{r}{v}\, dZ + g\oint \frac{a}{v}\, dZ \qquad \text{........................ (5.40)}$$

$$P_f - g\oint \frac{1}{v}\, dZ = \sum p + g\oint \frac{r}{v}\, dZ + g\oint \frac{a}{v}\, dZ \qquad \text{....................... (5.41)}$$

The natural ventilation pressure is now:

$$-g\oint \frac{1}{v}\, dZ$$

Self study

The table and calculation on page 29 of VOEE will explain the application in more detail. The student must take note of how the different table values are determined and final energy and pressure Equations are set up.

5.2.6 Comparison of Bernoulli's Equation and Thermo-Equations

Comparison of the pressure Equations:

$$p_f = (P_2 - P_1) + p + g\ (Z_2 - Z_1) + \frac{1}{2}(u_2^2 - u_1^2) \quad \text{(Bernoulli)}$$

$$p_f = (P_2 - P_1) + p + g\ dZ + udu \quad \text{(Thermo)}$$

If the difference in density is very small, the equation changes to:

$$g\,\rho \int dZ = g\,\rho\,(Z_2 - Z_1) \text{ and}$$

$$\rho \int udu = \frac{1}{2}\rho\,(u_2^2 - u_1^2)$$

This is the same for incompressible fluids.

5.2.7 Pseudo Pressure Equation

This is just standardising of the pressure Equation. From the original energy Equation:

$$W_f = \int vdP + F + g(Z_2 - Z_1) + \frac{1}{2}(u_2^2 - u_1^2)$$

If we multiply right through with density (ρ) we get:

$$\rho W_f = \rho\int vdP + \rho F + \rho g(Z_2 - Z_1) + \frac{1}{2}\rho(u_2^2 - u_1^2) \qquad \text{................. (5.42)}$$

With the density of air being 1 or 1.2 kg/m^3, all the terms in the Equation is N/m^2 or Pa. This is not a true pressure Equation and is therefore named a pseudo Equation.

Where:

ρW_f = fan pressure (to standard density)

ρF = pressure loss (to standard density)

SELF STUDY

Read the following articles:

References

- 'A theoretical and critical analysis of the various methods of determining energy and pressure losses in shafts' by R.E Kislig.
- Calculations of the natural (unventilated) wet-bulb temperature, psychrometric dry-bulb temperature and wet-bulb globe temperature from standard psychrometric measurements by D.J Brake.

Chapter

6

HEAT

LEARNING OUTCOMES

Knowledge and understanding of

- Heat sources
- Estimation of mine heat load
- The nature of heat transfer
- Heat transfer by conduction, convection and radiation
- Coefficient for overall heat transfer
- Fouling factor, arithmetic average and logarithmic average
- Heat transfer on wet surfaces
- Heat flow from rock to air in tunnels
- Heat flow on flat surfaces
- Graphs of Virgin Rock Temperature
- Influence of heat transfer coefficient
- Graph of heat flow versus surface temperature.

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Jones, MQW and Rawlins, CA, 'Thermal properties of backfill from a deep South African gold mine', *Journal of the Mine Ventilation Society of South Africa,* Oct/Dec 2001, p100-105.

Kebonte, SA, and Biffi, M, 'The feasibility of using heat sinks at depth', *Journal of the Mine Ventilation Society of South Africa,* Jan/March 2001, p6-12.

Rawlins, CA 'Underground mine heat loads and associated reduction methodologies', *Journal of the Mine Ventilation Society of South Africa* Jan/March 2004, p25-30.

Ventilation and Occupational Environment Engineering in Mines. Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

Animation

References

6.1 Heat Sources

In this section we are going to concentrate on the sources of heat underground and how it influences the environment. Heat sources and its identification have a big part in the determination of the heat load for a deep underground mine.

Also refer to Heat Sources (VOEE pages 427 to 435).

The underground environment is exposed to many sources of heat. It can be divided into two groups, namely:

- Primary sources
- Secondary sources.

6.1.1 Primary Sources of Heat Underground

The main source of heat in a metal mine is heat flow from the rock surrounding the workings. This is because in general such mines are deeper and the rock is at a high Virgin Rock Temperature (VRT).

NB

With the increasing use of diesel equipment underground, however, this has presented the ventilation practitioner with a new problem. Large diesel engines are intense sources of heat, and because diesel vehicles often work in excavations that are not all that well supplied with ventilation air, there can be very high local temperatures. An additional problem is the exhaust gas given off by diesel engines, which has to be diluted by ventilation air. In collieries, which are generally shallow, the main source of heat is usually that given off by machinery. Even so, high temperatures are seldom encountered in coal mines. In base metal mines, where the ore body is massive and the VRT is not high, the principal source of heat is again machinery.

6.1.1.1 Heat Flow from Rocks

Heat flows into the ventilation air from the surface of the rock exposed by an excavation by:

- Convection of heat from the rock surface into the air
- Radiation from the surface of the rock to objects in the excavation.

NB

It follows that the factors given below are important in determining the rate of heat flow to the air, and the temperature rise that it experiences as a result:

- The total area of rock that is exposed to the ventilation air.
- Thermal factors such as the conductivity of the rock and the emissivity of the rock surface (to be explained in detail later in this chapter).

- The difference in temperature between the rock surface and the **dry-bulb temperature** of the air. This depends on the age of the excavation. Rock surface temperatures are higher in new excavations than ones that have been exposed for several years.
- The time of contact between the rock surface and the air.
- The degree of wetness of the rock surface.

6.1.1.2 Heat from Machinery

The amount of heat generated by machinery depends on the sources of power used, and the type of work done by the machine. The usual sources of energy underground are diesel fuel and electricity.

The type of work done by a machine falls into two classes:

Example

References

- One is work that includes the hoisting of some load through a vertical distance. This may be broken rock in a skip, water in a pipeline, or men and materials in a hoist. In these case most of the energy supplied to hoist or pump involved **appears as an increase in the potential energy of the load hoisted.**(example 8 on page 430 of VOEE The only heat dissipated into the underground air is that arising out of the inefficiency of the machine. For instance, if a motor-pump combination has an overall efficiency of 85%, and is taking 100 kW of electrical power, 85 kW goes into increasing the potential energy of the water, and 15 kW appears as heat.
- If, however, no hoisting of any material is taking place, as with a scraper winch or a front-end loader, for instance, all of the power input to this equipment is used to overcome friction, and so all of it ultimately is dissipated into the air as heat.

WORKED EXAMPLE 1

A mine ventilated with 750 m^3/s of air at density of 1.2 kg/m^3 produces 225 000 tons of rock per 30 day month at a depth where the average virgin rock temperature is 40ºC. By the time this rock reaches surface in a skip, it has been cooled to 20ºC.

a) What is the average rate at which heat is released by the broken rock?

b) By how much will this heat energy increase the temperature of the air?

Answer

a) Use the steady sensible heat Equation to solve the problem.

$$\text{heat in rock (kJ)} = \text{mass of rock} \times C_{p\,rock} \times \Delta t$$

$$= 2\,25\,000 \times 1000 \times (0.873) \times (40 - 20)$$

$$= 3\;928\;500\;000 \text{ kJ/month}$$

$$\text{heat in rock (kJ/s)} = \frac{3\,928\,500\,000}{30 \text{ days} \times 24_{hrs} \times 60_{min} \times 60_{sec}}$$

$$= 1\,515.6 \text{ say } 1516 \text{ kW}$$

b) The increase in temperature of the air now needs to be determined, once again the sensible heat Equation is used (m is now the mass flow rate in kg/s):

$$\text{Heating of } 750 \text{ m}^3\text{/s of air by } 1^{\circ}\text{C requires} = m_{(\rho \times Q)kg/s} \times c_{pa} \times \Delta t$$

$$= (750 \times 1.2) \times 1.005 \times 1$$

$$= 904.5 \text{ say } 905 \text{ kJ/s or kW}$$

$$\text{In ratio the increase in air temperature} = \frac{1515}{905} = 1.67 \text{ say } 1.7^{\circ}\text{C}$$

HEAT FROM DIESEL EQUIPMENT

References

Significant quantities of heat are generated underground by plant such as front-end loaders, bulldozers and dump trucks, with drill-rigs, multi-purpose vehicles, utility vehicles and the traditional track-bound diesel locomotives adding their (smaller) share of heat. Locomotives are set to work on horizontal levels and therefore there is no work against gravity. The heat that is produced is equivalent to the calorific value (example 10, page 433 of VOEE) of the diesel used (read the section on diesel equipment on page 432 of VOEE).

NB

In all of this equipment, the input power ultimately appears as heat, and the average rate at which they generate heat can be calculated from knowledge of the monthly (say) consumption of fuel and its calorific value. Alternatively, it can be found by knowing the rated power of all the engines, their utilisation factors (fractions of time at which the engines are running), and those fractions of time at which the engines are being run at full power.

Rule

A rule of thumb to determine the heat load as a result of diesel machinery working underground is that it is normally three times the rated kW capacity of the machine.

!

This is the reason why diesel machinery are very inefficient in terms of the amount of heat produced versus the amount of work done by these types of machinery.

HEAT FROM ELECTRICAL EQUIPMENT

Underground equipment drawing large quantities of electrical power include hoists, pumps and fans. The latter include not just main and booster fans, but also the sum total of all the auxiliary fans being used in the mine. In the case of hoists and pumps, the heat dissipated by them to the surrounding air is equivalent to their inefficiency as explained above. An electric motor loses energy by so-called copper losses, and by mechanical friction. Mechanical friction is also the reason for energy loss in a hoist or pump.

Mechanical ventilation such as with fans increases the airflow through the mine and provides additional cooling for the workers, because it increases the air velocity and decreases the temperature. The energy losses in the motor and the aerodynamic losses of the fan are immediately carried over to the air as heat. The compression of air as it passes through a fan also increases the heat content of the air. Therefore, **all** of the energy input to a fan appears as heat. The heat, which is therefore carried over to the air by the booster fans is equal to the thermal **equivalent of the electrical energy which enters the fan motor** *(*see example 7 on page 429 of VOEE).

The same is essentially true for electric locomotives. This is because they operate either horizontally or at low gradients underground, so the heat they dissipate is the thermal equivalent of the power they consume. In the case of battery locomotives, the heat generated at the batter charging stations should also be taken into account.

WORKED EXAMPLE 2

The following measurements were taken at an underground booster fan inlet:

Intake air volume	236 m^3/s
Pressure across fan	5 kPa
Temperature	24/27ºC
Barometric pressure	115 kPa
Power input to fan motor	1 775 kW
Motor efficiency	95%

i. Determine the conditions at the fan discharge

ii. What are the efficiency and overall efficiency of the fan?

Psychrometric chart

Answer

i. Using the 115 kPa Psychrometric chart, the characteristics of the air can be determined.

The specific volume of the air in the main airway = 0.762 m^3/kg

The quantity of air in the main airway (given) = 236 m^3/s

Sigma heat of the air entering the fan = 64.5 kJ/kg

Apparent specific humidity of air entering fan = 15.3 g/kg

$$\text{mass flow of the air} = \frac{Q}{\nu} = \frac{236}{0.762} = 310 \text{ kg/s}$$

The increase in heat of the air, per kg of mass flow of air, can be determined as follows:

$$\text{unit increase in heat} = \frac{kW}{m} = \frac{1\,775}{310} = 5.73 \text{ kJ/kg}$$

Sigma heat of air leaving fan (S_{out}) = $S_{in} + \Delta S$

= (64.5 + 5.73)

= 70.2kJ/kg

The barometric pressure fan outlet = 115 + 5

= 120 kPa

Psychrometric chart

The temperature of the air can be read off the 120 kPa psychrometric chart, Sigma heat of 70.2 kJ/kg and assuming the specific humidity remains constant (note, **120 kPa not the 115 kPa, a common mistake made by students**).

Temperature of the air leaving the fan is 26.0/32.2ºC

(ii) Fan air power = pQ

= 5 x 236

= 1 180 kW

Fan motor input power = 1 775 kW (given)

Fan input power = fan motor input power x $motor_{efficiency}$

= 1 775 x 0.95

= 1 686 kW

The fan efficiency can now be determined as follows:

$$\text{fan efficiency} = \frac{\text{air power}}{\text{fan input power}} \times 100\%$$

$$= \frac{1\,180}{1\,686} \times 100\%$$

$$= 70\ \%$$

The fan overall efficiency can now be determined as follows:

$$\text{fan overall efficiency} = \frac{\text{air power}}{\text{fan motor input power}} \times 100\%$$

$$= \frac{1\,180}{1\,775} \times 100\%$$

$$= 66\ \%$$

WORKED EXAMPLE 3

The following measurements were taken at the inlet of an underground hoist chamber:

Temperature	26.3/29.5ºC
Barometric pressure	115 kPa
Hoist motor input power	2 170 kW
Motor efficiency	87%

What is the volume of air that needs to be supplied to the hoist chamber to ensure that the wet-bulb temperature in the chamber does not exceed a reject temperature of 28.5ºC? State any assumptions you make in giving your answer.

Answer

Psychrometric chart

Using the 115 kPa Psychrometric chart, the characteristics of the air can be determined and also remembering that the reject temperature is always expressed as the wet-bulb temperature unless otherwise stated (in this case then the wet-bulb temperature is 28.5ºC):

The specific volume of the air entering chamber	= 0.778 m^3/kg
Sigma heat of the air entering chamber (26.3ºC)	= 72.8 kJ/kg
Sigma heat of the air leaving chamber (28.5ºC)	= 83.5 kJ/kg
Specific humidity of air entering chamber	= 17.5 g/kg
The allowable increase in Sigma heat	= 83.5 - 72.8
	= 10.7 kJ/kg

The waste heat given off to the air by the motor can be determined as follows (87% motor efficiency means 13% given off as heat):

$$\text{waste heat to air} = \frac{(100 - 87)}{100} \times 2\,170$$
$$= 0.13 \times 2\,170$$
$$= 282 \text{ kW}$$

The mass flow of air (kg/s) required can be determined as follows:

$$\text{mass flow of air} = \frac{Q}{\nu}$$
$$= \frac{282}{10.7}$$
$$= 26.4 \text{ kg/s}$$

$$\text{Air volume flow, Q} = m \times \nu$$
$$= 26.4 \times 0.778$$
$$= 20.5 \text{ m}^3/\text{s}$$

The volume flow was asked, this is something that a lot of students miss that there is a difference between the mass flow (kg/s) and the volume flow of air (m^3/s).

6.1.1.3 Methods of Reducing Heat from Machinery

Reduction of the quantity of heat being dissipated by machinery is not really possible, but what can be done is to locate plant so that air returning from it passes directly to a return airway and then to surface. For example, a large underground booster fan should be installed near the bottom of the upcast shaft.

Pumps and hoists cannot in practice be located in return airways because they have to be operated and maintained by mine staff. Also, the machinery itself is adversely affected by the dust, gases and moisture in return air.

Hoist and pump chambers can be ventilated by part of the downcast air, which is then either taken directly to the return, or cooled (cold water cooling hot air) by means of a refrigeration plant before being re-mixed with the remaining downcast air.

The arrangement used will depend on:

- Air temperatures reached
- Availability of refrigeration
- Quantity of downcast air flowing.

READING

References

- Section 2.1 on page 571 of EE
- Section headed 'Artificial Sources of Heat' on page 429 to 435 of VOEE
- 'Le Roux's Notes on Mine Environmental Control' Notes 6.1 and 6.3.

6.1.2 Secondary Sources of Heat in Mines

6.1.2.1 Personnel

It has been estimated that the waste heat given off by workers underground can amount to as much as 10% of the total heat of a mine. This is unavoidable and results from the metabolism of the body.

6.1.2.2 Oxidation of Timber and Other Materials

On the Witwatersrand gold mines, the decomposition of iron sulphides, as well as the decay of timber used for support, gives off heat. However, this is not a large source of heat and can be overlooked.

NB

In a shallow metal mine, where the effects of the geothermal temperature gradient and auto-compression are small, the heat from oxidation is more significant. However, in such mines heat is not a problem.

The oxidation of coal can produce as much as 80 to 85% of the total amount of heat put into the air. In certain South African coal mines this could therefore be regarded as a major source of heat.

6.1.2.3 Lamps and Lighting

This is a small source of heat and cannot be avoided. However, with improved illumination in mines, it could become important in future.

6.1.2.4 Compressed Air Pipes

Compressed air leaving a surface compressor is usually hot, and when piped directly down a downcast shaft there is generally a transfer of heat from the pipe to the ventilation air.

However, in many cases the compressed air is deliberately cooled before it enters the mine, in which case heat transfer is reduced or eliminated.

6.1.2.5 Electrical Cables and Plant

As already discussed, electricity is used underground for powering motors fitted to pumps, fans and electric locomotives. But the cables used to convey electrical power and underground transformers are subject to so-called copper losses and iron losses, which show up as heat in the ventilation air. Still, this is not a large source of heat.

6.1.2.6 Heat from Water Piping

Heat from hot pipes, such as those carrying condenser water from a refrigerator, do not constitute a heat load on the mine if such pipes are located in return airways. When this is not possible, they should be covered with suitable thermal insulation.

6.1.2.7 Explosives

NB

Heat from this source is not usually an important factor as it is dissipated by the ventilation air during the re-entry period. It is however important to note that heat generated from explosives are a large component of the total heat generated during mining operations and it is therefore essential that there will be enough air available at the right quality (temperature) to remove all the heat before the next shift are supposed to start work. Worked Example 4 will show the importance thereof.

WORKED EXAMPLE 4

You are the ventilation engineer on a shaft and you have to make an estimate of the heat load potential of a new kind of explosive to be tested on your shaft. The following technical details of the explosive are available:

- ANFO based explosives heat release 3 700 kJ/kg of explosives
- The charge density of ANFO explosives 0.8 kg/ton of rock blasted.

It is known that the heat generated by the explosives will be removed in two stages, namely with the blasting fumes and some will be retained in the rock as heat. With high fragmentation it is assumed that 45% of the heat will be removed with the fumes.

It is also estimated that 20% of the heat will be removed within the first hour after blasting. The amount of rock to be blasted is 2 000 tons; assume the C_p of rock as 950 J/kg ºC.

Determine the following:

a) The total amount of heat produced by the explosives (kJ)
b) How much of the total heat produced remains in the rock and is not removed with the fumes (kJ)
c) The amount of heat removed with the fumes within the first hour after blasting
d) The rise in temperature of the rock blasted within the first hour
e) What would the rise in temperature in the rock be (within the first hour) if the explosives used were a nitro glycerine based explosive with a heat release rate of 5 800 kJ/kg explosive used. Assume the same charge density.

Answer

a) The total amount of heat produced can be calculated as follows:

$$\text{heat produced by explosives} = \text{mass of explosives used} \times \text{heat release rate of explosives}$$
$$= (2\ 000 \times 0.8) \times 3\ 700$$
$$= 5\ 920\ 000 \text{ kJ}$$

b) The energy given off as heat was given as 55% then:

$$\text{heat remaining in the rock} = \text{total amount of heat released} \times \text{percentage remaining}$$
$$= 5\ 920\ 000 \times (100 - 45)\%$$
$$= 3\ 256 \text{ MJ}$$

c) Expressed as a kW unit in other words (kJ/s)

$$\text{amount of heat removed in the first hour} = \frac{\text{total amount of heat released}}{60_{min} \times 60_{sec}} \times \%\text{ removed}$$
$$= \frac{5\ 920\ 000}{3\ 600} \times 0.2$$
$$= 329 \text{ kW}$$

d) Once again the sensible heat Equation is used.

$$\text{heat in rock (kJ)} = \text{mass of rock} \times C_{p\ rock} \times \Delta t$$
$$\Delta t = \frac{\text{heat in rock (kJ)}}{\text{mass of rock} \times C_{p\ rock}}$$
$$= \frac{5\ 920\ 000 \times \frac{80}{100}}{2\ 000 \times 1\ 000 \times 0.950}$$
$$= 2.49 \text{ say } 2.5^{o}\text{C}$$

e) As before the sensible heat Equation to be used:

$$\text{heat in rock (kJ)} = \text{mass of rock} \times C_{p\ rock} \times \Delta t$$
$$\Delta t = \frac{\text{heat in rock (kJ)}}{\text{mass of rock} \times C_{p\ rock}}$$
$$= \frac{(2000 \times 0.8_{kg/kg} \times 5\ 800) \times 0.8_{\%\ remaining}}{2\ 000 \times 1\ 000 \times 0.950}$$
$$= 3.9^{o}\text{C}$$

With all factors remaining the same the temperature could also have been calculated as a ratio. With ANFO used (3 700 kJ/kg) the rise in temperature was 2.49ºC. With nitro-glycerine based explosives used the rise would be:

$$\frac{5\ 800 \times 2.49}{3\ 700}\ ^{o}\text{C} = 3.9^{o}\text{C}$$

6.1.2.8 Movement of Rock Masses

Mining operations cause the rock above the excavations to subside under the action of gravity. This movement causes crushing, fracturing and grinding within the body of rock, and the friction involved is converted to heat.

This is an unavoidable source of heat, but it is distributed throughout the rock mass, and the effect is to increase slightly the temperature of the strata, with a very small proportion of this heat entering the ventilation air.

6.1.2.9 Fissure Water in Underground Excavations

Definition

NB

Fissure water is water that comes for an underground reservoir which is heated up as a result of the surrounding VRT. Fissure water, depending on the flow rate from the rock into the workings as well as the virgin rock temperature with which it was in contact. An important note here is that the heat flow that will take place from the hot fissure water to the air is very much dependent on the temperature difference between the fissure water and the **dry-bulb temperature** of the air. Worked Example 5 will show how the heat load associated with fissure water can be calculated.

WORKED EXAMPLE 5

A mine produces 2 million litres of fissure water per day. The water temperature is 45ºC and the air temperature in the haulage in which the water is release is 25/28ºC. What heat load is imposed on the mine?

Answer

$$\begin{aligned}\text{heat load imposed} &= \text{mass flow of water} \times c_{pw} \times \Delta t \\ &= \frac{2\,000\,000}{24 \times 60 \times 60} \times 4.187 \times (45 - 28)^{\circ}\mathrm{C} \\ &= 1\,648 \text{ kW or } 1.65 \text{ MW}\end{aligned}$$

NB

A common mistake made with these kind of heat flow calculations is that the wrong temperature is used (important to note that the dry-bulb temperature must be used to determine the heat load due to fissure water).

6.1.3 Auto Compression and Auto De-Compression

NB

It is important to note that auto compression is not a source of heat, but because the air temperature increases with depth due to pressure increase, it can be discussed as a heat source. In a mine shaft the increase in enthalpy is expressed as:

$$H_2 - H_1 = g\,(z_1 - z_2)$$

Rule

The enthalpy increase is thus **9.79 kJ/kg for each 1 000 m going down a mine**. Psychrometric properties of air are usually expressed in kg dry air. If this happens, the enthalpy increase, as a result of auto-compression, is dependant on moisture content. The thermal capacity of air changes with moisture content.

Consider the following situations:

Dry Air

Moisture Content	=	0 kg/kg
Enthalpy increase	=	9.79 kJ/kg per 1 000 m
Thermal capacity	=	1.005 kJ/kg^{o}C
Dry-bulb temperature increase	=	9.79 / 1.005
	=	9.741^{o}C / 1 000 m

Moist Air

Moisture content (assumed)	=	0.015 kg/kg
Enthalpy increase	=	9.79 x 1.015
	=	9.937 kJ/kg / 1 000 m
Thermal capacity	=	1.032 kJ/kg^{o}C
Wet-bulb temperature increase	=	9.937 / 1.032
	=	9.629^{o}C / 1 000 m

Shafts are not normally dry and therefore the moisture content plays a large part from the calculations above. In most practical situations the increase in wet-bulb temperature is independent of the wetness of the shaft, due to the non-linear properties of the Psychrometric properties of air.

Rule

When water flows down a pipe in a shaft, the increase in enthalpy is 9.79 kJ/kg per 1 000 m. As long as the water is in the pipe, the pressure will increase. The increase in pressure when friction is negligible is 9.79 MPa per 1 000 m.

At the bottom, the conditions increase the water temperature at a rate which is dependent on the temperature of the water at the beginning. The temperature increase is small, around 0.2°C per 1 000 m, if the water enters at 30°C

Definition

If the water moves at **terminal velocity** in the pipe (terminal velocity being the maximum velocity that the water will reach in an open ended pipe where the down ward acceleration force will equal the upward frictional force within the pipe), and is not kept at high pressure and is not passed through an expansion valve, the situation will be as follows:

$$\Delta t = 9.79 / 4.187 = 2.338^{o}C / 1\,000 \text{ m}$$

The increase in temperature is better described as an increase due to friction losses and not **as a result of auto compression** as is the case with air.

Auto compression cannot be discarded, but in a deep mine the heat effect of auto-compression can be countered by cooling. This is done by cooling of the air above ground or below ground or both. Local cooling units can also be installed at the work place. The increase in temperature of water can be countered by passing the water through a cooling system. The effect of auto de-compression, which is the exact opposite of auto compression, must also sometimes be considered. The following example will explain the situation with regards auto de-compression of air much better.

WORKED EXAMPLE 6

Figure 6.1 shows a section of a mine. Determine the Entalphy of the air leaving the stope at B. The distance between A and B is 600 m.

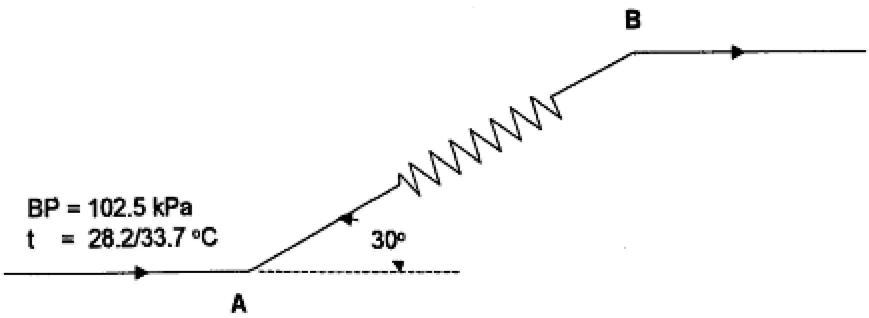

Figure 6.1 Upcast flow of air in a stope raise

Answer

First, calculate the vertical distance through which the air travels. Using the sine formula:

$$\Delta h = 600 \sin 30^{\circ}$$

$$= 300 \text{ m}$$

The decrease in enthalpy is therefore:

$$\Delta H = 9.79 \times 300/1\,000$$

$$= 2.94 \text{ kJ/kg.}$$

Psychrometric chart

Using a Psychrometric chart for a barometric pressure of 102.5 kPa, we find that the enthalpy of the air entering the stope is 89.7 kJ/kg (this value could also have been calculated). Thus, the Enthalpy of the air leaving the stope is determined as follows:

$$89.7 - 2.94 = 86.76 \text{ say } 86.8 \text{ kJ/kg}$$

Psychrometric chart

The wet-bulb temperature of the air leaving the stope at point B can then be determined by reading off the value of the temperature by using the 102.5 kPa Psychrometric chart and using the Enthalpy value of 86.8kJ/kg. Under normal conditions underground the dry-bulb temperature is approximately 2ºC higher than the wet-bulb temperature which is read off from the Psychrometric chart.

NB

At this point it is very important to note that auto compression and auto de-compression of air plays a significant role in the actual temperature of the air leaving a work place underground, and should be included in calculations in this regard, unless otherwise stated (ignore the effects of auto compression or auto de-compression). The following Equation will help you understand how to calculate the heat load/generated in a working place if the entry temperature and exit temperature has been given and it is required to calculate the heat generated in the stope and considering the effect of either auto-compression or auto-decompression.

$$\text{Enthalpy}_{\text{out}} = \text{Enthalpy}_{\text{in}} + \frac{\text{kW}_{\text{heat generated}}}{\text{m}_{\text{air}}} \pm \text{auto-(de)compression effect} \quad \ldots (6.1)$$

The term $\frac{\text{kW}_{\text{heat generated}}}{\text{m}_{\text{air}}}$ is the increase in Enthalpy (kJ/kg) of the air due to the influence of external sources such as machines, people and so forth.

In ventilation calculation underground we normally use the Sigma heat as it is the difference between Sigma heat and Enthalpy is negligibly small as we are working with relatively small pressures where the compressibility effect of the air is not that much. Enthalpy as was shown with the definition thereof earlier, includes the effect of moisture within the air, but as mentioned is a relatively small effect. The Equation above can therefore also be expressed as follows and the one normally used.

$$\text{Sigma heat}_{\text{out}} = \text{Sigma heat}_{\text{in}} + \frac{\text{kW}_{\text{heat generated}}}{\text{m}_{\text{air}}} \pm 9.79 \text{ kJ/kg/1 000 m}$$

The plus minus sign is an i ndication of whether auto-compression ('+' sign) is to be considered or whether auto de-compression (the '-'sign) needs to be considered.

The term $\frac{\text{kW}_{\text{heat generated}}}{\text{m}_{\text{air}}}$ is now the increase in Sigma heat of the air due to the influence of external sources such as machines, people and so forth.

A short summary of heat sources is shown in Table 6.1 and the estimated relative effect of heat in the mining context. The values may differ drastically from mine to mine and are here only for illustration purposes.

Table 6.1 Summary of the distribution of heat sources

Source	Relative Heat Effect
Heat from Rock	35%
Auto-compression	20%
Machinery	15%
Workers	10%
Oxidation	10%
Explosives	8%
Lamps	Negligibly small
Warm pipes/ electrical cables	2%
Rock movement	Unknown
Fires	Usually zero
Warm water (fissure water)	Normally low contribution, but can be very high

6.1.4 Virgin Rock Temperature (VRT)

It is a well-known fact that the temperature of the rock increases with depth. The VRT of the rock in the Wits is lower than that of the Free State and the reason for this is the difference in the thermal conductivity of the strata. The rock in the Wits is a better-layered rock and the rock in the Free State is a better-insulated rock (holds heat back). This is the reason that the deeper you go in the Free State mine the hotter it is. Figure 6.2 shows how the VRT differs for rock types and depths.

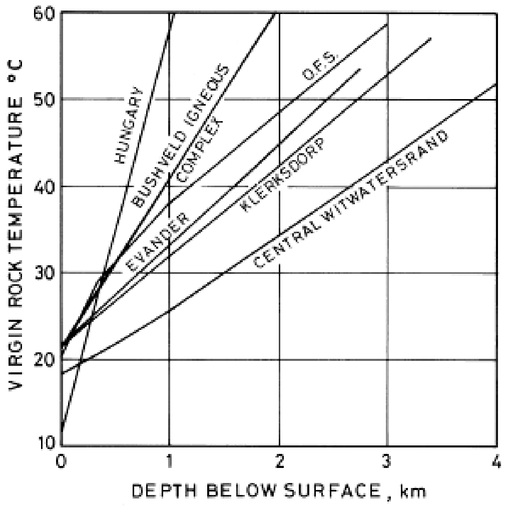

Figure 6.2 Different Virgin Rock temperatures for different areas

The determination of the VRT for the Witwatersrand is as follows:

$$(T_s + 6D + 1.1\ D^2)\ ^oC \quad (6.2)$$

Where:

D	=	Depth of layer above (km)
T_s	=	Surface rock temperature

And the value for T_s (Witwatersrand) is 18.3ºC if the surface rock temperature is unknown. For the Free State area the VRT can be determined as follows:

$$(T_s + 25.5\ D_1 + 14.2\ D_2 + 8.2\ D_3)\ ^oC \quad (6.3)$$

Where:

D_1	=	Thickness of Karoo (km)
D_2	=	Thickness of Lava (km)
D_3	=	Thickness of Quartzite (km)

NB

And the value for the T_s (Free State) is 20.0ºC if surface temperature is unknown. It is important to note that heat flow from the rock can be transferred through convection or conduction. These two terms will be explained in detail later.

6.2 Estimation of Mine Heat Load

In this section we look at the estimation of the heat load of a mine. The previous section discussed the calculation and effect of different heat sources.

The methods used were:

- Empirically developed
- Methods based on the application of methods for calculations of individual heat sources.

In the life of a mine, during different project stages, various heat sources are encountered. In the early stages there is not a lot of information available and it is unnecessary to do detailed calculations of the heat load and therefore empirical methods are used. As the mine planning progresses, it is more important to use more detailed methods.

6.2.1 Empirical Methods

Advantage

These, are based on the reading of actual mines and thus has a great advantage.

Disadvantages

This has two disadvantages in the sense that the observations are scattered and the problem exists in extrapolating.

6.2.2 Barenbrug Graph

The graph shows two important aspects:

- How the heat produced per ton mined per month varies with the average VRT, the mining method and the area in which the mine is located.
- Secondly, the graph shows how much of this heat can be removed by ventilation.

The difference in the two values shows the amount of cooling required. Figure 6.3 shows a graph for a Free State mine.

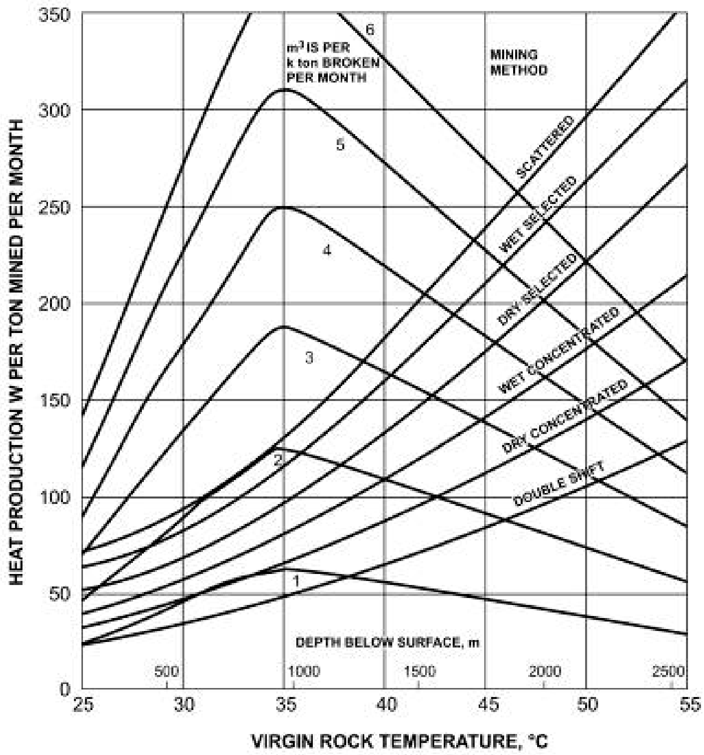

Figure 6.3 Barenbrug graph for a Free State mine heat load prediction

Method to use graph:

- Get a mining method line on the graph (e.g. 'wet cons')
- VRT temperature given (or calculated) 50ºC
- Depth underground 2 200 m
- m^3/s/kton broken 3 m^3/s/kton mined

The vertical difference in the mining method line and the air required line gives us the cooling required. If we mine 250 000 ton in the above situation, what is the cooling requirement? The answer is approximately 17.5 MW [(180-110) x 250 000]).

6.2.3 Stroh Graph

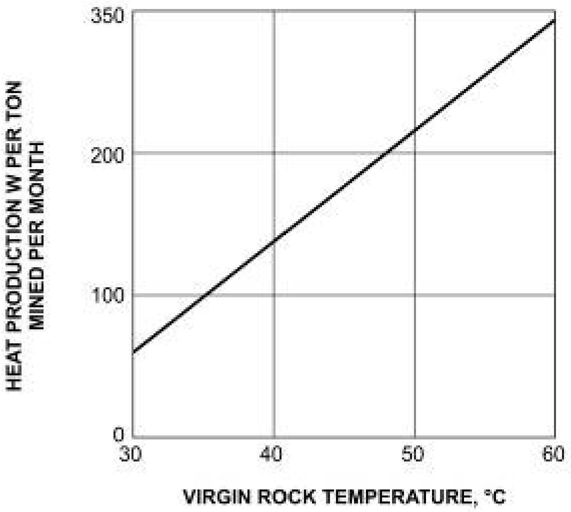

Figure 6.4 Stroh graph for heat load prediction

This is a simple substitute, but works on the same principle as Barenbrug. Figure 6.4 shows the values for a specific mining method and the temperature found in the Free State.

6.2.4 Whillier Graph (Cooling required, empirical and theoretical, any hot mine)

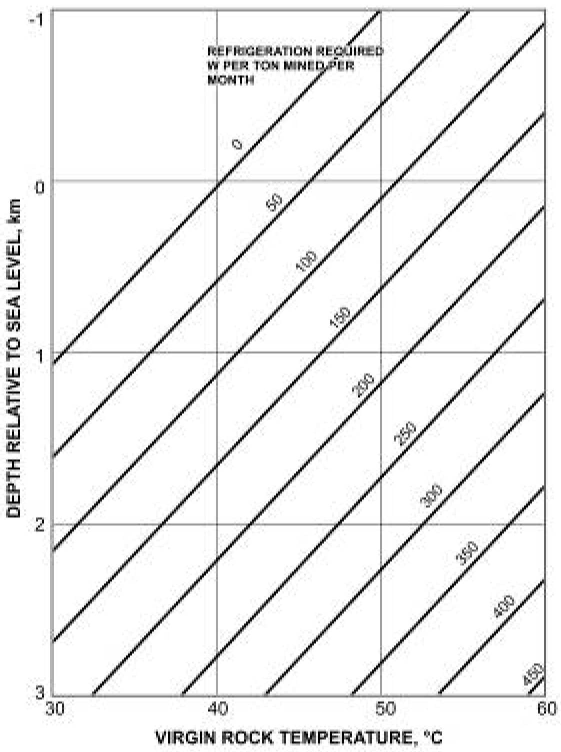

Figure 6.5 Whillier graph for heat load prediction

[Figure 6.5](#) includes the effect of auto-compression. From this graph the cooling in W/ton can be read directly. Here we use the **depth beneath sea level** and VRT.

6.3 Methods to Reduce the Effects of Heat

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With the ever-increasing depths of mining, it is essential to take all possible precautions to reduce the effects of heat on the workers.

The main methods of reducing the effects of heat are:

- Reducing heat-flow from rock
- Removal of machinery
- Dilution by ventilation
- Acclimatisation
- Refrigeration.

6.3.1 Reducing Heat-Flow from Rock

Q

What can be done to reduce the amount of heat transfer from rock to air?

A

- With regards the area of the rock. It is better to keep the size of ventilated excavations as small as possible, consistent with the demands of the mining operations. Nothing can be done about the conductivity and emissivity of the rock, but trials have been carried out in which layers of insulating plastic foam have been sprayed onto the rock surfaces of intake airways, in order to reduce the rate of temperature rise of the air.
- Time of contact is related to the speed at which air is flowing. Slow moving air is in contact with a rock surface for a relatively longer time and so heats up faster than air flowing at a higher speed. In general, air quantities and hence air speeds could be increased by reducing the resistance of the mine excavations and installing larger fans. Ventilation staff should also be alert to leakage, and make every effort to reduce it. Higher air speeds not only reduce heat pickup in intake airways, but also increase the cooling power of the air in which production staff is working.

NB

- The wetness of the rock plays an important role in the heat transfer potential. If the rock surface is wet, and the ventilation air not saturated, water evaporates and the rock surface is cooled.

  This means the difference between the temperature of the rock well behind the rock surface (the VRT) and the surface temperature increases. This brings about an increase in the heat flow by conduction from the rock behind the surface.

- To reduce heat-flow from rock, the ambient dry-bulb temperature should also be kept as close as possible to the temperature of the rock. To achieve this, it means that the evaporation of water into the air must be reduced to a minimum. The effect of water evaporating into the air is illustrated in Figure 6.6.

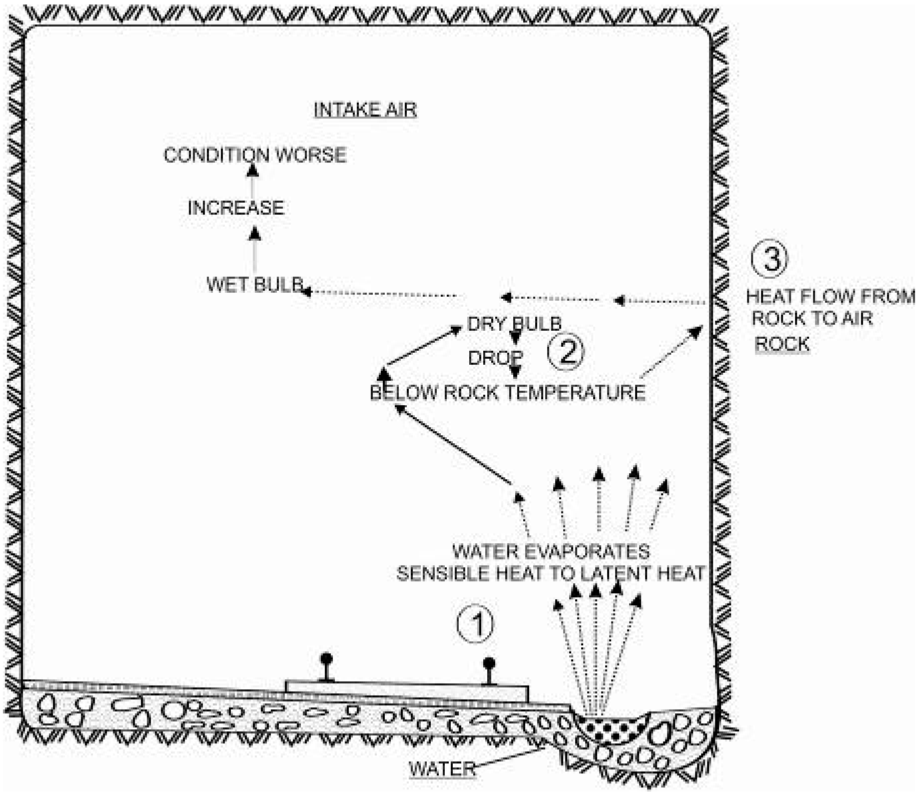


Figure 6.6 Evaporation effect of water on air

By referring to Figure 6.6 the following explanation is important:

1. When water evaporates, sensible heat is taken from intake air and changed to latent heat, evaporating the water
2. When sensible heat is removed from intake air, the dry-bulb temperature drops to below the temperature of the rock
3. Heat starts to flow from the rock to the air
4. Heat flow from the rock increases the wet-bulb temperature of the air making conditions worse.

It is also important to note that heat can be transferred to the air at a higher rate from a wet surface than from a dry surface.

In order to reduce or minimise the heat-flow from the rock to the air, the following control measures can be implemented:

- Keep downcast shafts dry by sealing off water fissures, maintain shaft water columns, providing water tight skips, minimise water usage on shaft stations (washing) and providing drainage rings at regular intervals in the shaft.
- Keep shaft stations, intake airways and cross-cuts as dry as possible.
- Ensuring open drains in haulages and cross-cuts are covered to prevent evaporation of water and regular maintenance of drains to prevent overflowing. Alternatively, drain water can be pumped in columns back to the shaft and then cascaded down annex holes to the settlers at shaft bottom where the water is treated and cleaned for re-use.
- Adhere to watering down standards and procedures for dust suppression in all workings and exercise strict water controls, thus minimising water wastage.
- Maintain all water pipes and minimise leakage of water from pipes.
- When the service water is chilled, it is advantageous, from a heat control-point of view to use more water under controlled conditions.

Some other advantages of using chilled water in terms of heat control measures underground, the following can be said:

Advantages

- Cool water flowing in the drains of intake airways has the effect of reducing the rise in temperature that would otherwise occur
- Broken rock is cooled more quickly by the chilled water
- Rock stays wet for a longer period of time and therefore less water is needed to combat dust generation
- Wet-bulb temperature is reduced and stays lower for a longer period of time
- Chilled service water is actually capable of condensing water out of hot humid air.

6.3.2 Removal of Machinery

In almost all mines, electricity is the major source of energy and all this energy consumed is converted into either heat energy or useful work.

In addition to the abovementioned energy source, other energy sources are diesel driven and compressed air operated equipment, which also adds to the total heat load in a mine.

SHE

It is therefore of utmost importance that regular inspections and maintenance be done on all these equipment to ensure they operate at maximum efficiency and hence that less heat energy is produced and added to the air.

Alternatively, where possible; machinery and equipment should be installed on the return air-side of workings. This is not always practicable, as some appliances/equipment must be operated and maintained continuously and need to be situated on the intake air side of workings (fresh air), for example:

- Hoists
- Pumps
- Refrigeration plants
- Booster fan
- Motors and so forth.

If machinery are placed in return air, bear in mind that the dust, gases and moisture in such air will have an a diverse effect on machinery, resulting in more wear and tear than normal.

If fresh air is used to ventilate machinery (e.g. hoists, pumps and refrigeration plants), the return air from such machinery can be mixed with the balance of the fresh air, it can be piped directly to a return airway, or it can be reconditioned (cooled) before re-introduced into the fresh air.

6.3.3 Dilution by Ventilation

By increasing the air volume it means that the air velocity will also increase. With increased air velocities, the contact time of the air with the heat source (rock, machinery and so forth.) is reduced, resulting in less heat pick-up by the air and improved thermal conditions.

Increased air velocities also improve the cooling power of the air. Air volumes / velocities can be increased by:

- Reduce air leakage to a minimum and minimise air wastage
- Exercise good air control and distribution
- Reduce resistance to airflow
- Increase pressures by installing larger fans, reduce resistance and so forth.

6.3.4 Acclimatization

Definition

Acclimatization is the physiological adaptation of the body to prolonged daily heat stress exposures with environments above 27.5°C wet-bulb. The ability to work increased and the risk to heat disorders decreased with acclimatization.

In principle, heat acclimatization is based on exposing men to the heat stress levels that they are ultimately expected to experience in a controlled and progressive manner. This can be achieved by slowly increasing the work rate in a fixed environment, or by keeping the work rate fixed and increasing the environmental thermal conditions or by gradually increasing both.

As the physiological response between men varies, some men can develop dangerously high levels of body temperatures during acclimatization and should be checked and monitored regularly during this process. Several practices of acclimatization are used in the South African gold mining industry, such as Climatic Room Acclimatization on surface or underground, micro-climate acclimatization, vitamin C supplemented acclimatization and lately Natural Acclimatization Underground (NAU).

Natural Acclimatization Underground is a process whereby heat tolerant employees are acclimatized in their normal underground working environment for a specified period of time, provided that the wet-bulb temperature does not exceed 32.5°C, and is not below 27.5°C. This heat stress management process is briefly discussed below.

6.3.4.1 Pre Selection

All employees (novices, returnees and ex leave) must be medically examined and declared fit for underground work.

6.3.4.2 Heat Tolerance Screening (HTS)

Objective

The objective of this process is to identify employees that are heat tolerant and heat intolerant.

Heat tolerance screening is conducted in a climatic chamber on surface under qualified supervision, with the following controlled environmental conditions:

- Wet-bulb: 28.0ºC
- Dry-bulb: 29.5ºC
- Air velocity: 0.3 - 0.5 m/s in all areas of the chamber

With the above environmental conditions, employees conduct bench stepping (30.5 cm fixed height) for a period of 30 minutes at 24 steps per minute.

At the end of the 30-minute bench stepping, employees are classified as:

- **Heat tolerant:** those individuals whose oral temperature does not exceed 37.6°C and can commence with NAU
- **Heat intolerant:** those individuals whose oral temperature is in excess of 37.6°C and should not be allowed to work in areas conducive to heat illness (27.5°C and above).

6.3.4.3 Natural Acclimatization Underground (NAU)

Heat tolerant employees spend twelve shifts underground in their normal working environment (27.5°C – 32.5°C wet-bulb). They must work on day shift only, not working over time and on a 60/15 minute work / rest cycle. Drinking water must be freely available and employees must consume approximately 500 ml during the rest cycle. The above process must be performed under supervision of an appointed competent person, who must also check for signs of heat disorders during the rest cycles. Employees must be withdrawn if any signs of heat illnesses are noticed.

Figure 6.7 displays the heat tolerance screening process through a diagram.

SHE

Note: The Mines Health and Safety Act, requires that all the above data of the individual be incorporated into the medical surveillance programme.

6.3.5 Refrigeration

When the above methods failed to reduce hot conditions, refrigeration can be introduced as a last resort, because refrigeration constitutes great expenses in purchasing, operating and maintaining them. For more information on refrigeration, refer to the notes on refrigeration.

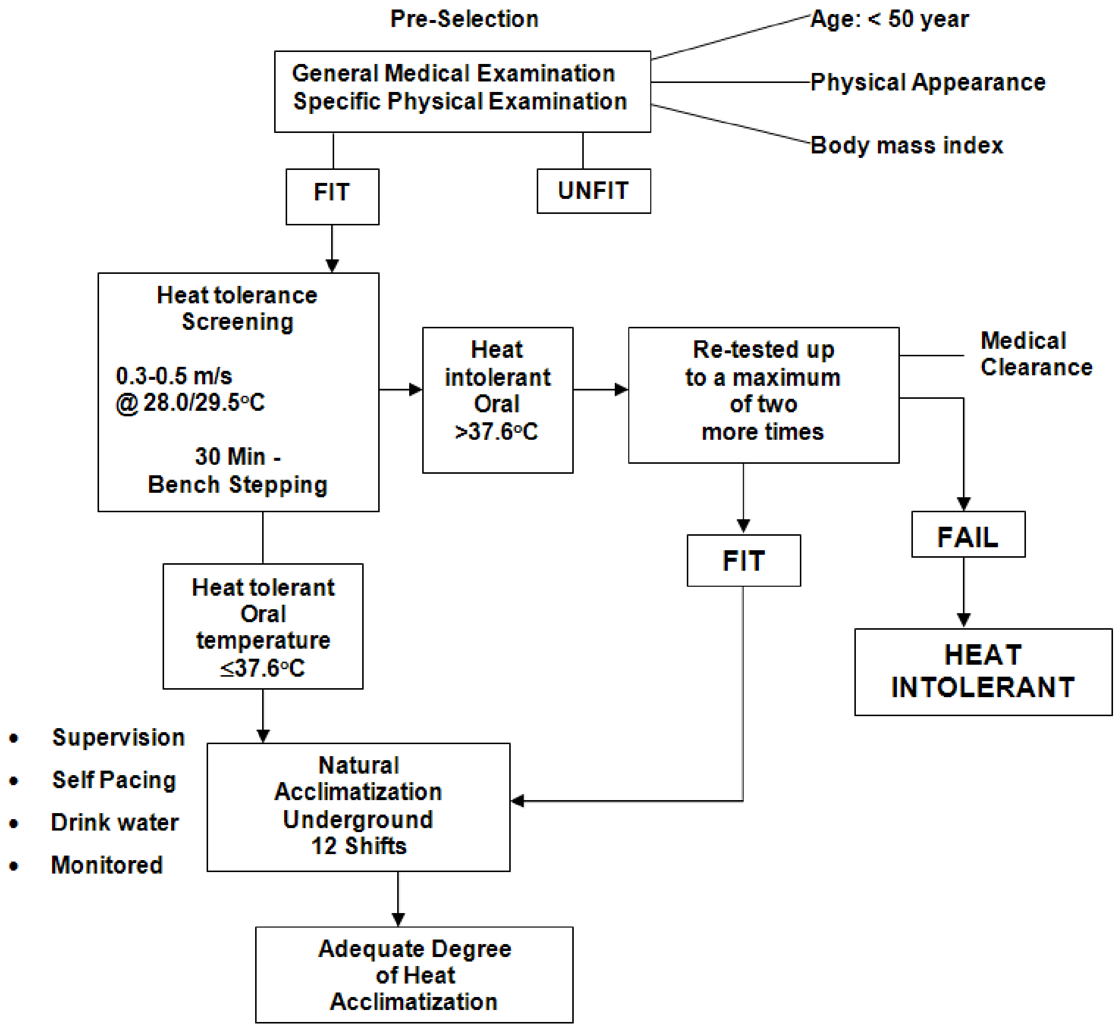


Figure 6.7 Heat tolerance screening diagram

6.4 Conduction, Convection and Radiation

In this section we are going to look at the different aspects of heat transfer. Heat transfer takes place in three ways, namely:

- Conduction
- Convection
- Radiation.

6.4.1 Nature of Heat Transfer

Definition

Heat transfer can be seen as the flow of thermal energy from one body to another as a result of temperature difference between two bodies. Air that flows over a heater will get warmer as a result of heat which is transferred from the heater to the air. The rate at which heat transfers (watt) from the heater to the air is equal to I^2R, where I is the current that flows through the element and R is the resistance of the element.

In other words, if there are no losses we can make the following deduction. If the airflow rate m (kg/s), the rate of energy transfer is:

$$
\begin{aligned}
I^2R &= \text{rate of heat transfer (Watts)} \\
&= m \times \text{increase in enthalpy of air} \\
&= m \times C_{pa} \times (t_2 - t_1)
\end{aligned}
$$

Definition

In this case the increase in enthalpy is equal to the product of the thermal capacity of the air C_{pa} and the increase in temperature $(t_2 - t_1)$, because there is no change in moisture content in the air. The term $\mathbf{C_{pa}}$ is known as the specific heat of the air at a constant pressure.

WORKED EXAMPLE 7

In a fan type heater, air is blown through at a rate of 0.2 m^3/s with a temperature of 15°C. The current flowing through the element is 8 amps and the resistance is 28 ohms. Determine the temperature of the air when it leaves the heater. Assume that the air density is 1.1 kg/m^3 and that the thermal capacity (C_{pa}) of the air is 1.015 kJ/kg°C.

Answer

Self study

From a simple calculation we find the outlet temperature as 23.03°C (this to be done for home work). We must however also remember that:

$$m = \rho \times Q = kg/m^3 \times m^3/s = kg/s.$$

6.4.1.1 Conduction

Definition

Conduction is the heat transfer process that occurs in solid bodies, such as metals, rock, the walls of pipes, or in stationary fluids. The factors, which affect the rate of heat transfer by conduction, are:

- Difference in temperature
- Cross-sectional area of the body
- Conductivity of the material of which the body is made
- Thickness of the solid body.

These are shown in Figure 6.8.

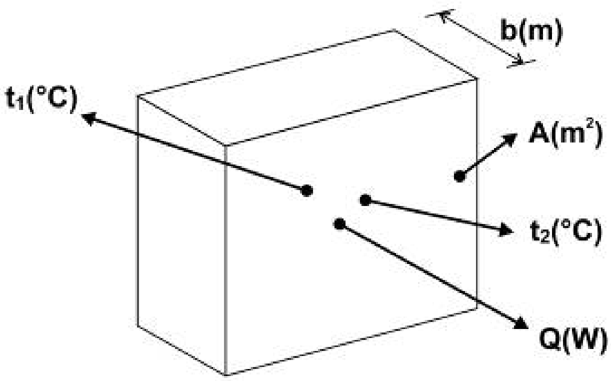


Figure 6.8 Description of factors influencing heat flow through conduction

NB

The rate at which heat flows by conduction increases if the first three factors mentioned increase, the last factor mentioned decreases. The first of these, temperature difference, can be thought of as the **driving force that causes the heat to flow**. The rate of heat flow is greater when the cross-sectional area of flow is greater and when the thickness of the material is less.

The size of the body therefore plays an important role in the amount of heat that can be transferred. The size of the body is therefore expressed in terms of the cross sectional area and thickness as shown above.

Another shape to consider is to look at a pipe where it is necessary to determine the conduction from the inside to the outside of a pipe. We have an inside and outside point, over which we must get an average area where conduction will take place. Figure 6.9 illustrates this point better.

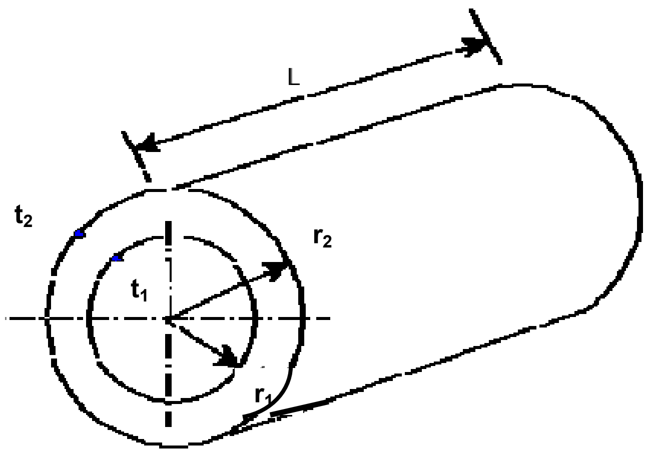


Figure 6.9 Illustration of conduction through pipe

NB

To calculate the area A_1 and A_2 (related to radius 1 and radius 2), it is **important to note** that the **surface area** over which the heat transfer will take place needs to be considered (where L is the length of the pipe column):

$$A_2 = 2\pi r_2 L$$

$$A_1 = 2\pi r_1 L$$

There are two methods to determine the average area, namely:

- Summing the inside and outside areas and then dividing by two
- Calculating a logarithmic average as follows:

$$A_m = \frac{(A_2 - A_1)}{\log_e\left(\frac{A_2}{A_1}\right)}$$

$$A_m = \frac{(A_2 - A_1)}{2.303 \log_{10}\left(\frac{A_2}{A_1}\right)} \quad \text{.......................... (6.4)}$$

NB

It is important to note that the percentage error between the first and second methods will get bigger as the ratio A_2/A_1 increases. The logarithmic average for the area **is the most correct** and the heat transfer calculated by this will give a more correct value than by using the summation average.

The relationship between these different factors mentioned before is given by the following Equation, which is used to calculate the amount of heat flow (from the hot to the cold body):

$$q = \frac{kA(t_1 - t_2)}{b} \quad \text{.. (6.5)}$$

Where:

Q	=	the rate of heat flow by conduction (W)
A	=	the cross-sectional area of flow (m^2)
t_1 and t_2	=	temperatures of opposite faces of body(ºC)
b	=	thickness (m)
k	=	conductivity of the material (W/m^{o}C)

NB

Shown in Table 6.2 is some average values of the conductivity's of several types of material commonly found in mining practice. You should memorise them.

Table 6.2 Conductivity values k for different materials

Copper	380 W/m°C
Quartzite	6 W/m°C
Brick	0.7 W/m°C
Wood	0.17 W/m°C
Steel	45 W/m°C
Insulation	0.034 W/m°C (plastic foam)
Concrete	1.7 W/m°C
Glass	1.0 W/m°C

6.4.1.2 Convection

Heat transfer by convection takes place when a moving fluid, such as water or air, goes over a solid body with a different temperature. The factors that influence the amount of heat transfer are again the temperature difference, the area and a new term called the heat transfer coefficient (h_c) for convection. This coefficient is dependent on the boundary layer thickness, which is dependent on the Reynolds number value discussed before (R_e).

The boundary value is very small if the fluid flows at a high velocity. In practice this means that the heat transfer as a result of convection increases if the velocity of the fluid increases.

This is so, because the heat transfer which must take place must do so across the boundary layer, which becomes **smaller as the velocity increases**.

Definition

A simple definition of **boundary layer** is that it is the minimum distance from a heat source where there is no immediate increase in temperature and velocity of the fluid, adjacent to the heat source.

The R_e value for the fluid is a dimensionless parameter, which characterises velocity, turbulence and size of a fluid. The R_e value is calculated by (dealt with in Chapter 2):

$$R_e = \frac{\rho D u}{\mu}$$

Where:

u	=	velocity (m/s)
D	=	diameter (m)
ρ	=	density (kg/m^3)
μ	=	viscosity (Ns/m^2)

For air:

$$R_e = \frac{67000\ D\ u\ \rho}{1.2}$$

For water:

$$R_e = 10^6 \text{ x u x D (at } 20^oC)$$

It must be noted that the density is dependent on the air pressure and that the density of water decrease with the increase in temperature.

The viscosity (µ) of water decreases with increase in temperature, whilst the viscosity of air increases with the increase in temperature.

THE CONVECTION HEAT TRANSFER CO-EFFICIENT h_c

Because the thickness of the boundary layer is in general not known, the previous conduction Equation cannot be used when we talk about convection. The following Equation is considered:

$$q_c = h_c A (t_1 - t_2) \qquad (6.6)$$

Where:

q_c	=	rate of heat transfer between the fluid and surface (W)
A	=	area over which transfer takes place (m^2)
$(t_1 - t_2)$	=	temperature (^oC)
h_c	=	convection coefficient ($W/m^{2o}C$)

The calculation of convection is done by the prediction of the coefficient for a specific fluid.

PREDICTION OF h_c

As mentioned before, the resistance against convection is mainly dependent on the thickness of the boundary layer, which in turn is dependent on the R_e value. The relationship between the R_e value and the heat transfer coefficient (k is the heat transfer co-efficient of the solid material) is as follows:

$$\frac{h_c D}{k} = \text{constant x } Re^n$$

For **water that is inside** a pipe:

$$h_c = \frac{5\,680(1+0.015t)u^{0.8}}{d^{\ 0.2}}$$

Where:

d = diameter of the pipe in mm and

t = temperature of the water.

For **air inside** a pipe:

$$\frac{h_c D}{k} = 0.02 \times R_e^{0.8}$$

For **air outside** a pipe:

$$\frac{h_c D}{k} = 0.2 \times R_e^{0.6}$$

From the above Equations, a set of graphs have been drawn. (Figure 6.10, Figure 6.11 and Figure 6.12).

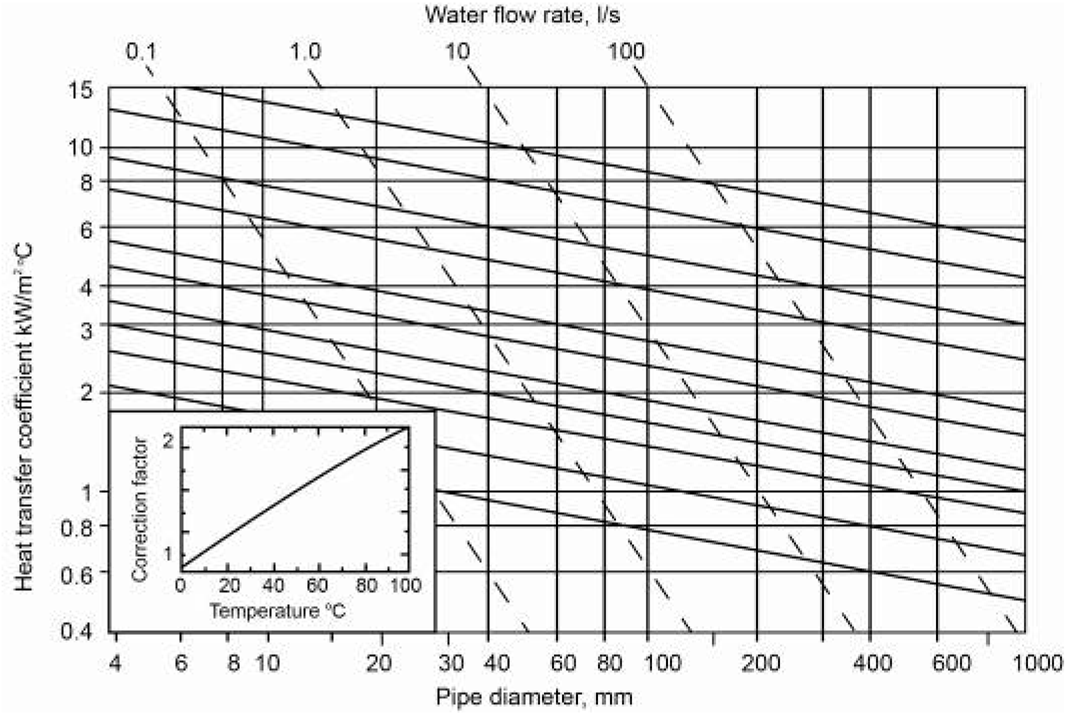


Figure 6.10 Water flow in a pipe

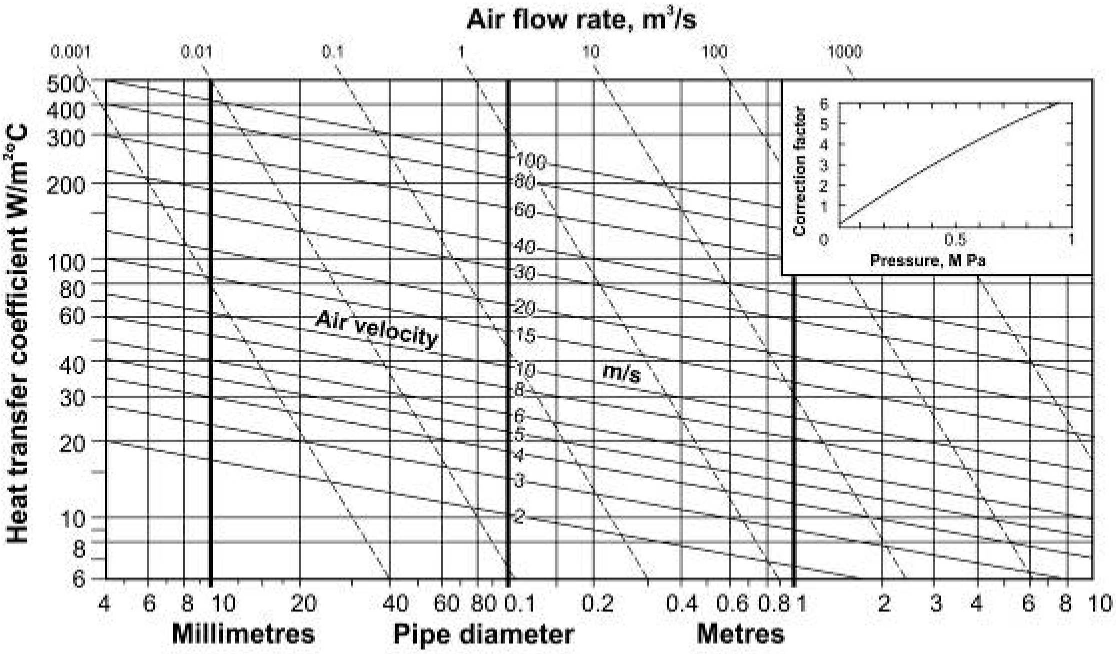


Figure 6.11 Airflow in a pipe

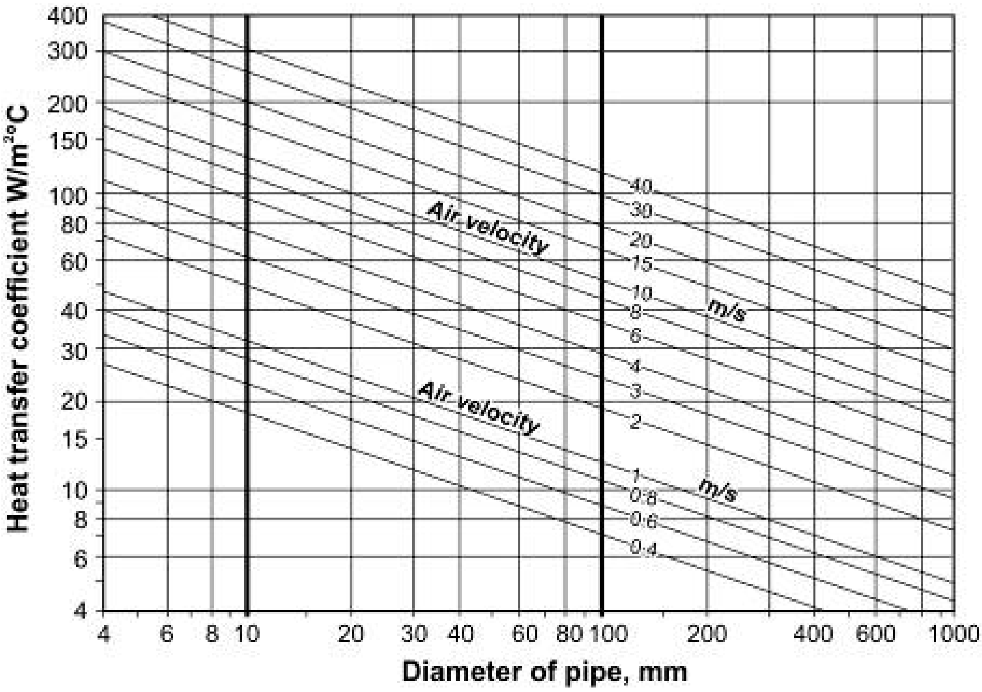


Figure 6.12 Airflow outside a pipe

WORKED EXAMPLE 8

Water flows in a warm pipe in an airway. Assume the length of the airway to be 100 m. This may be water from the cooling plant of an underground system. In this case the temperature of the water will be cooled and the temperature of the cooler air that flows over the pipe that carries the hot water will increase. Figure 6.13 shows the situation where water and air flows in the same direction.

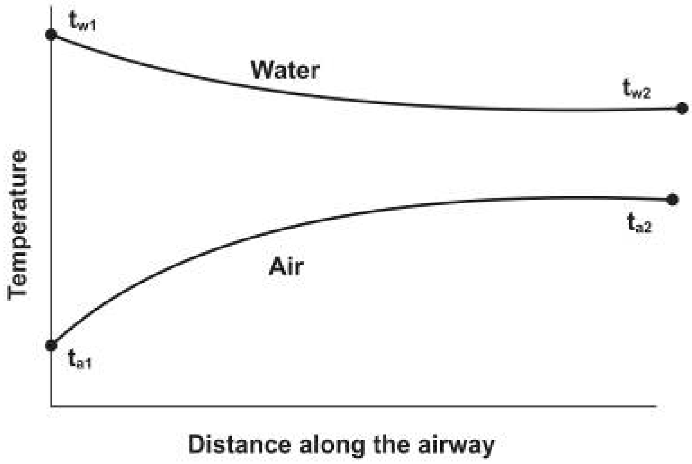


Figure 6.13 Heat exchange that takes place between water and air

If the water temperature changes from t_{w1} to t_{w2}, we know that from the law of conservation of energy the air temperature from t_{a1} to t_{a2} from the equation will change accordingly:

$$\text{heat flow rate} = m_w \times c_{pw} \times (t_{w1} - t_{w2}) = m_a \times c_{pa} \times (t_{a2} - t_{a1})$$

Where:

C_{pa} = 1.015 kJ/kg^{o}C for atmospheric air
C_{pw} = 4.187 kJ/kg^{o}C for water

Q

The question is therefore that if the temperature of the air at 1 and the temperature of the water at 1 are known, what will the final temperature of the air and water be?

Answer

Definitions

To determine this we must look at the heat transfer analysis of the system. The heat transfer through solids is **conduction**, from a solid surface to an adjacent liquid is **convection** and from a solid to another solid through air is **radiation**. In the case where there is evaporation or condensation on a surface, which is in contact with air, there is additional latent heat transfer, which is proportional to the rate of evaporation or condensation. Some of these processes can take place without influencing the other.

NB

The various heat transfer processes and calculations will be dealt with in detail later in this chapter, for now it is important to understand that the heat given off by one fluid will be absorbed by the other and vice versa.

WORKED EXAMPLE 9

The velocity of air in a front is 1 m/s. Calculate the convection from the body of a worker. Assume that the average diameter of a worker is 200 mm and further that the average skin surface area is 1.5 m^2 and the average skin temperature is 35°C. The conditions in the front are as follows: 29/31°C at 100 kPa.

Answer

The heat transfer coefficient can be read from Figure 6.12 (airflow outside a pipe)

$$h_c = 9.5 \text{ W/m}^2\,^{o}\text{C}$$

The rate at which a man's body gives off heat:

$$\begin{aligned} q_c &= h_c \times A\,(t_{skin} - t_{dry\ bulb}) \\ &= 9.5 \times 1.5 \times (35 - 31) \\ &= 57 \text{ Watt} \end{aligned}$$

Self study

Example 2 on page 368 of VOEE

6.4.1.3 Heat Transfer through Radiation

Radiation heat transfer takes place between two bodies over a medium, example heat from the rock surface of a tunnel through air to a chilled water pipe. The driving force for this process is the difference in temperature between the two surfaces. Other factors which influence the amount of heat transferred are:

- The measure in which the one surface 'sees' the other and the amount of energy that is radiated. These two properties are combined in a factor called emissivity and view factor (F_{ev}).
- The areas of the two surfaces: In the formula where the heat transfer rate is calculated, the smaller area is used. Thus when the amount of heat transferred by radiation between a pipe and tunnel must be calculated, the area of the pipe will be used.
- A coefficient that is known as the radiation heat transfer co-efficient h_R.

RADIATION HEAT TRANSFER (THE EQUATION)

All surfaces radiate energy at a rate that is dependent on the 4th power of the absolute temperature. The net heat transfer between two surfaces, which are in each other's presence, and can 'see' each other, will be the difference in energy that radiates from the various surfaces.

The Equation is:

$$q_R = 5.67\left[\left(\frac{T_1}{100}\right)^4 - \left(\frac{T_2}{100}\right)^4\right] \text{ x } A_1 \text{ x } F_{ev} \text{ .... } (6.7)$$

Where:

q_R	=	net radiated heat flow (Watt)
T	=	absolute temperature (K)
A_1	=	smaller area (m^2)
F_{ev}	=	emissivity and view factor
5.67	=	Stefan-Boltzman constant

EMISSIVITY AND VIEW FACTOR F_{EV}

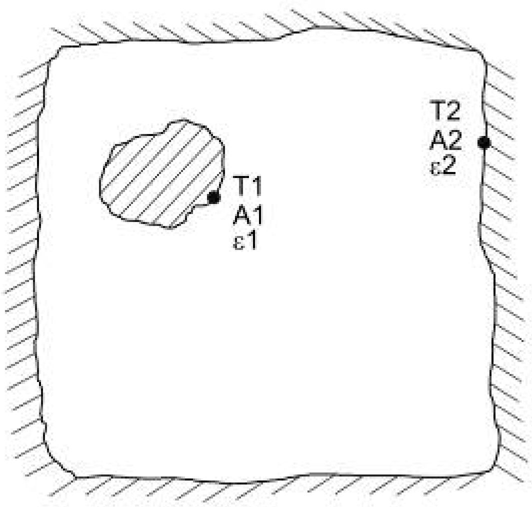


Figure 6.14 Emissivity and view factor graphical explanation

Rule

From Figure 6.14 the factor will be described as follows. A black surface reflects no energy which falls on it and thus has the ability to radiate energy as a result of the maximum absorption of the energy. Reflective surfaces reflect most of the energy which falls on it and thus radiate less energy out. When we refer to the rate at which energy is radiated through another surface, the emissivity energy E will always be less for a black surface.

Definition

The ratio E/E_{black} of emissivity energy of a surface to that of a black surface at a set temperature is called the **emissivity** (ε).

Definition

Non-black surfaces do not absorb all the energy that falls on the surface. The amount that is absorbed is the **absorbability** (α), which just like emissivity has a maximum value of 1 for a black surface. For a given surface there is a close relationship between α and emissivity and for certain circumstances, α = emissivity and can without a large error be equated to each other. It therefore implies that shining surfaces/metals have a low emissivity and that mat/dull surfaces have a high emissivity.

Figure 6.14 represents the two bodies with an inter-medium air. View factor is used to specify the amount of energy that will radiate from a specific surface to the other. Most materials in engineering work have a high emissivity of 95% or more.

In general, the determination of the emissivity and view factor is a very complex process.

For mining conditions we can combine emissivity and the view factor, F_{ev} and with reference to Figure 6.14 calculate it as follows:

$$F_{ev} = \frac{1}{\left[\frac{1}{\varepsilon_1} + \frac{A_1}{A_2}\left(\frac{1}{\varepsilon_2} - 1\right)\right]} \quad \text{.............................. (6.8)}$$

WORKED EXAMPLE 10

A galvanised steel pipe with a diameter of 70 mm takes water at a temperature of 70°C through an office with dimensions 5 (l) x 4 (w) x 3 (h) meters. The pipe goes through the length side of the office. Determine the emissivity and view factor, F_{ev} and the rate of heat transfer as a result of radiation. Assume that all the surfaces in the office have an em issivity of 0.95 and that the emissivity of the pipe is 0.8. The office has an average temperature of 24°C.

A_1 (smaller area, pipe) = 1.1 m^2 → [π x 70/1 000 x 5]

A_2 (large area) = 94 m^2 → 2[(3x4) + (4x5) + (5x3)]

$$F_{ev} = \frac{1}{\left[\frac{1}{\varepsilon_1} + \frac{A_1}{A_2}\left(\frac{1}{\varepsilon_2} - 1\right)\right]}$$

$$= \frac{1}{\left[\frac{1}{0.8} + \frac{1.1}{94}\left(\frac{1}{0.95} - 1\right)\right]}$$

$$= 0.8$$

T_1 = 343.15 K

T_2 = 297.15 K

$(T_1/100)^4$ = 138.66

$(T_2/100)^4$ = 77.97

From the Equation for q_R:

$$q_R = 5.67\left[\left(\frac{T_1}{100}\right)^4 - \left(\frac{T_2}{100}\right)^4\right] \text{ x } A_1 \text{ x } F_{ev}$$

$$= 5.67\left[\left(\frac{343.15}{100}\right)^4 - \left(\frac{297.15}{100}\right)^4\right] \text{ x } 1.1 \text{ x } 0.8$$

$$= 303 \text{ W over the 5 m length}$$

RADIATIVE HEAT TRANSFER CO-EFFICIENT h_R ($W/m^{2o}C$)

For conditions in the mining industry, it is acceptable to use the following simple Equation for the determination of radiation heat transfer. The Equation is:

$$q_R = h_R A_1 (t_1 - t_2) F_{ev} \quad \text{.............................} \quad (6.9)$$

Where:

h_R = the radiation heat transfer coefficient ($W/m^{2o}C$).

The numerical value for h_R is dependent on the average value of the two temperatures and is thus independent of t_1, or t_2, and:

$$t_{av} = \frac{(t_1 + t_2)}{2}$$

Numerical values for h_R can be read from Table 6.3.

Table 6.3 Numerical values for Radiation heat transfer co-efficient

t_{av} ^oC	10	20	30	40	50	60
h_R ($W/m^{2o}C$)	5.15	5.71	6.32	6.97	7.65	8.39

The numerical value of h_R can also be calculated with:

$$h_R = 0.2268\left(\frac{T_{avg}}{100}\right)^3 = 4.62 \text{ x } \left(1 + \frac{t_1 + t_2}{546.3}\right)^3$$

Where:

$$T_{av} = 273.15 + \frac{(t_1 + t_2)}{2}$$

EXAMPLE QUESTION 11

A clerical worker at a metallurgical plant located in the Lowveld is complaining that her office is too hot, and is especially aware of the heat coming off one wall, which separates her office from the furnace area. This wall measures 4 m by 2 m and consists of brick 100 mm wide with a plaster layer 20 mm thick on either side. The surface temperatures of the wall are 110^oC on the furnace side and 65^oC on the inside. The air in the office is held at 27^oC by an air conditioner. You may take the plaster to have the thermal conductivity of concrete. The worker has a surface temperature (skin and clothing) that averages 28^oC, a surface area facing the wall of 0.8 m^2, and an emissivity and view factor with the wall of 0.95.

At what rate is heat passing through the wall by conduction? Hence, what is the convective heat transfer co-efficient for the wall? How much heat is the worker herself receiving from radiation?

(**Answer**: 4 371W; 14.4 W/m^{2o}C; 212 W)

6.5 Overall Heat Transfer Coefficient

In this section we are going to look at the joint effect of all the heat transfer processes, as well as the heat transfer on wet surfaces.

6.5.1 Overall Coefficient of Heat Transfer

In most practical situations the rate of heat transfer is governed by all three of the processes, depending on the particular configuration of the equipment involved.

NB

To illustrate the method of calculation to be used a case where **hot water flowing inside a steel pipe,** will be considered. The calculations will be done for a pipe length L. It should be noted that the total rate of heat transfer is proportional to the length of the pipe.

The method of calculation follows directly from the fact that the same amount of heat that is convected from the water to the pipe wall must in turn be conducted radially outwards though the steel pipe and through the insulation until it reaches the outer surface of the insulation.

From this point it passes outwards by the joint processes of convection to the air and radiation to the surroundings. The Equation for each portion of heat flow path must be written.

Usually the only known temperatures are that of the water, t_1 and air, t_5. The other temperatures at the interfaces must be assumed, and then subsequently eliminated from the equations by algebraic manipulation.

For calculation of pipe areas, the use of logarithmic mean is unnecessary and the error is minimal. It is thus acceptable to use the arithmetic mean.

By referring to Figure 6.15, the terminology will be bet ter understood.

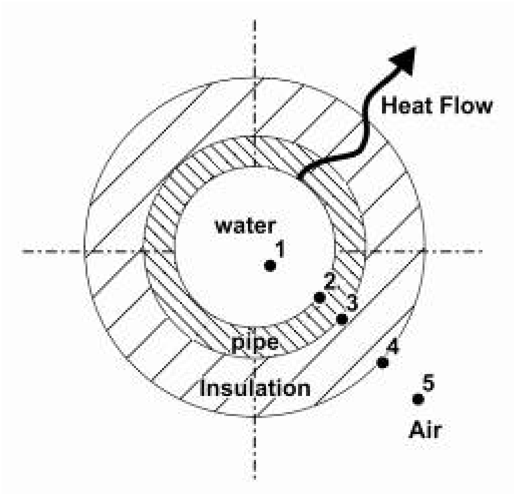


Figure 6.15 Insulated pipe

From Figure 6.15 the following Equations can be derived:

Convection from water to pipe $q = h_2 A_2 (t_1 - t_2)$

Conduction of heat through pipe $q = \dfrac{k_{pipe} A_{23} \times (t_2 - t_3)}{(r_3 - r_2)}$

Conduction heat through insulation $q = \dfrac{k_{insulation} A_{34} \times (t_3 - t_4)}{(r_4 - r_3)}$

Conv. and radiation on outer surface $q = (h_4 + 0.95\ h_R)\ A_4\ (t_4 - t_5)$

Where:

$A_{34} = \dfrac{(A_3 + A_4)}{2}$

$A_{23} = \dfrac{(A_2 + A_3)}{2}$

$A_2 = 2\ \pi\ r_2\ L$

$A_3 = 2\ \pi\ r_3\ L$

$A_4 = 2\ \pi\ r_4\ L$

k_{ins} = Thermal conductivity of insulation

k_{pipe} = thermal conductivity of steel pipe

h_2	=	convection coefficient between water and inside of pipe.
L	=	pipe length
h_4	=	convection coefficient between atmospheric air and insulation
h_R	=	radiation coefficient - the value 0.95 is an assumption for emissivity and view factor, F_{ev}

The unknown temperatures t_2, t_3, and t_4 can be eliminated from the above Equations, because the heat quantity transferred through the pipe (q) remains the same. The general Equation is therefore as follows:

$$q = UA\,(t_1 - t_5) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(6.10)}$$

Where:

$$UA = \frac{1}{\dfrac{1}{h_2A_2} + \dfrac{r_3 - r_2}{k_{pipe}A_{23}} + \dfrac{r_4 - r_3}{k_{ins}A_{34}} + \dfrac{1}{(h_4 + 0.95h_R)A_4}} \quad \ldots \text{(6.11)}$$

Where:

U	=	Overall heat transfer coefficient $W/m^{2o}C$ and UA in $W/^{o}C$.

In the case of heat exchangers, the UA factor has an additional role. When it is desired to transfer heat from one fluid to another under specified temperature conditions, suitable equipment must be purchased. In effect, it is the UA factor which is being bought. Therefore a cost comparison of the different designs of heat exchangers must be done on the basis of the UA factor.

NB

Where an insulating quality is needed the product with the lowest UA will be preferred and for heat exchangers the product with the highest UA value will be chosen.

In calculations involving heat transfer coefficients, it is always easier to calculate the UA product and not the U value itself. This greatly simplifies the mean areas for the cylindrical portions.

WORKED EXAMPLE 12

A steel pipe 150 mm inside diameter and 160 mm outside diameter is insulated, and carries **cold water** (10°C) at a velocity of 0.5 m/s in an underground airway. The air in the airway has a temperature of 24/27°C at a barometric pressure of 100 kPa and has a fluid velocity of 3 m/s.

The pipe is covered with insulation material 30 mm thick. Assume a thermal conductivity of 0.035 W/m°C and determine the following:

i. The UA factor for a 1m length of insulated pipe
ii. The rate of heat gain by the water, per meter length of pipe
iii. Temperature of the inner and outer surfaces of the pipe and of the outer surface of the insulation.

Answer

i. In order to determine the UA factor from the Equation it is first necessary to evaluate the various r, A and h terms (and important piece of advice now is not to change the numbering around as is shown in Figure 6.15 (here the heat flow was from the inside of the pipe outwards, keep the direction of numbering the same as it is applicable to the equations mentioned before):

r_2 = 0.075 m
r_3 = 0.08 m
r_4 = 0.11 m
$r_3 - r_2$ = 0.005 m
$r_4 - r_3$ = 0.03 m
A_2 = 0.471 m^2
A_3 = 0.503 m^2
A_4 = 0.691 m^2
A_{23} = 0.487 m^2
A_{34} = 0.597 m^2
$k_{pipe\ (Steel)}$ = 45 W/m°C
k_{ins} = 0.035 W/m°C

h_2 = 1400 W/m^2°C from Figure 6.10 (water flow in pipe)
h_4 = 18 W/m^2°C from Figure 6.12 (airflow outside pipe)
h_R = 5.6 W/m^2°C from Table 6.3

Self study

Calculate UA = 0.67 W/°C (for homework exercise)

ii) The rate of heat gain by the water per metre of pipe length is

$$q = UA \times (t_1 - t_5)$$
$$= 0.667 \times (10 - 27)\ °C$$
$$= -11.34 \text{ Watt}$$

(The negative number indicates that the heat flow is in the opposite direction compared to our numbers convention originally chosen, heat flows from hot to cold).

iii) $t_2 - t_1 \quad = \quad \dfrac{q}{h_2 A_2}$

NB

Note that t_2 and t_1 change around as a result of opposite flow of heat (t_2 is greater than t_1 because of change in heat flow direction).

$t_2 \quad = \quad 10.017\ ^{o}C$ (remember that q = 11.34 Watt)

$$t_3 - t_2 = \frac{q\ (r_3 - r_2)}{k_{pipe}\ A_{23}}$$

$t_3 \quad = \quad 10.02^{o}C$ (for home work check the answer)

Self study

$$t_5 - t_4 = \frac{q}{A_4\ (h_4 + 0.95\ h_R)}$$

$t_4 \quad = \quad 26.31^{o}C$ (for home work check the answer)

A few important aspects to note from the above Equations:

- The UA factor is for 1 m of pipe
- The dominant term in the UA factor is the resistance against heat flow due to insulating material. The greatest temperature drop is over the insulation material
- The difference in temperature between the inside and outside of the pipe is negligible
- On the airside of the pipe; resistance is not small enough to ignore
- The outside temperature (26.31°C) of the insulation is relatively high compared to the dew point temperature (check answer as 23°C), which means that there will be no condensation on the pipes, which means that latent heat transfer will not play a role in this case.

6.5.1.1 The Fouling Factor 1/hf

The Equation for the calculation of the UA factor is general for any heat transfer process. For all instances where scaling take place inside tubes, the effect of scaling material on the heat transfer process must be taken into account.

Another term $\dfrac{1}{h_f A_2}$ must be added to the bottom of the UA Equation to include the effect of scaling that takes place in the pipe (especially old pipes). The term $\dfrac{1}{h_f}$ is called the fouling factor:

$$UA = \frac{1}{\dfrac{1}{h_f A_2} + \dfrac{1}{h_2 A_2} + \dfrac{r_3 - r_2}{k_{pipe} A_{23}} + \dfrac{r_4 - r_3}{k_{ins} A_{34}} + \dfrac{1}{(h_4 + 0.95 h_R) A_4}} \quad .\ (6.12)$$

A typical value for the fouling factor that is used in cooling systems is 0.0003 m^2 °C / W, which means that:

$$\frac{1}{h_f} = 0.0003 \text{ and therefore } h_f = \frac{1}{0.0003}$$

NB

In heat exchangers (water/air and water/refrigerant) there is no insulating material around the pipe and the insulation term will fall away.

If the outside fluid was a liquid and not air, the radiation term h_R will also fall way.

6.5.2 Heat Transfer and Wet Surfaces

6.5.2.1 Overall Conduction, Convection and Radiation Transfer

When a surface which is in contact with air, is colder than the surrounding, there will be a transfer of heat in the surface as a result of convection from the air and the radiation as a result of the surrounding surfaces.

If the temperature of the surface is lower than the dew point temperature of the air, condensation will take place.

There will thus be additional convection and radiation heat transfer plus a latent heat component, because in the process of condensation the water vapour releases its latent heat.

The total rate of heat transfer is thus the summation of the above processes:

$$q_{total} = q_c + q_R + q_L$$

When a wet surface is warmer than the dew point temperature of the air, evaporation will take place, which means that the **latent heat transfer will be away** from the surface. (Refer to Figure 6.6).

Example

Situations can exist where the latent heat transfer process are away from the surface, whilst the radiation and convection is towards the surface. An example is in the use of a wet-bulb thermometer.

6.5.2.2 Condensation

When condensation takes place, the latent heat is into the surface. The latent heat of condensation of water vapour is 2 430 kJ/kg and it varies slightly with temperature. The value for latent heat can thus be calculated by:

$$\lambda = 2501 - 2.383\, t \quad \text{.......................................} \quad (6.13)$$

Where:

t = the temperature **of the surface**, which is colder than the dew point temperature of the air.

If the dew point temperature of the air is 20°C and the surface temperature for two instances are 10 and 18°C respectively, then this means that the amount of latent heat transfer for the 10ºC surface is much higher.

The rate of condensation is also dictated by the boundary layer, which is found in convection. If the reading of moisture content are taken from the air at various distances from the surface of condensation or evaporation and the vapour pressure is calculated from these readings, then the variation in vapour pressure with respect to the distance from the surface will be as shown in the Figure 6.16.

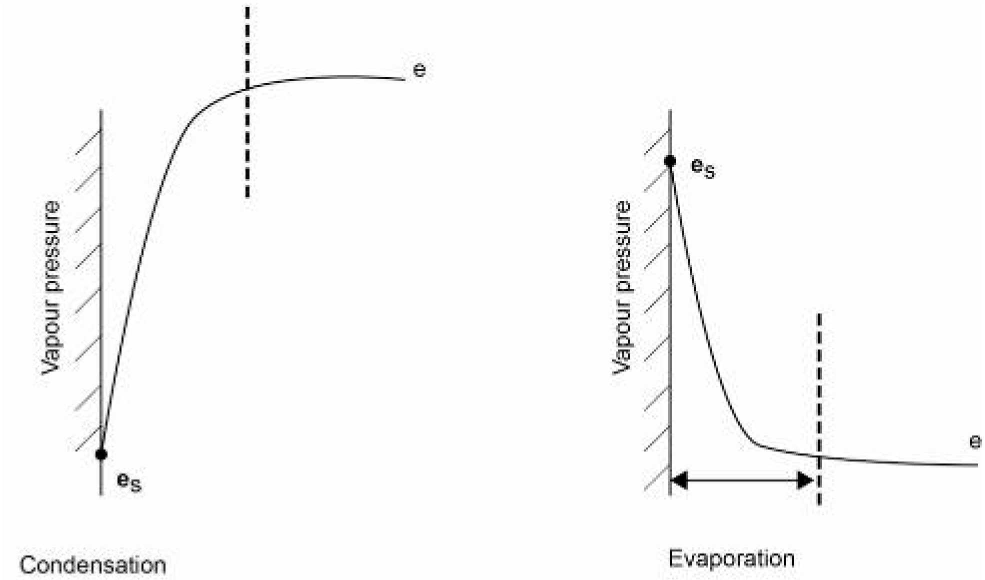


Figure 6.16 Variation in vapour pressure

It was determined experimentally that the rate of condensation is directly proportional to the convection heat transfer coefficient, h_c. The driving force which results in condensation is vapour pressure, just as temperature is the driving force for heat transfer.

It is advantageous to define the equivalent temperature difference for latent heat transfer as the temperature difference which h_cA gives as the same heat transfer as the actual latent heat transfer. Thus:

$$q_L = h_c A \, \Delta t_L \left(\frac{\lambda}{2430}\right) \left(\frac{100}{P}\right)$$

Where:

q_L	=	latent heat transfer in Watt
h_c	=	convection coefficient in W/m^{2o}C
Δ_{tL}	=	equivalent temperature difference for latent heat transfer
A	=	Area on which condensation will take place m^2
λ	=	Latent heat of condensation in kJ/kg
P	=	Barometric Pressure in kPa

By using Figure 6.17, the value of Δt_L can be determined. Other missing values in the above equation can be determined mathematically or by using graphs as indicated before. It is expected of you to be able to determine the value of q_L.

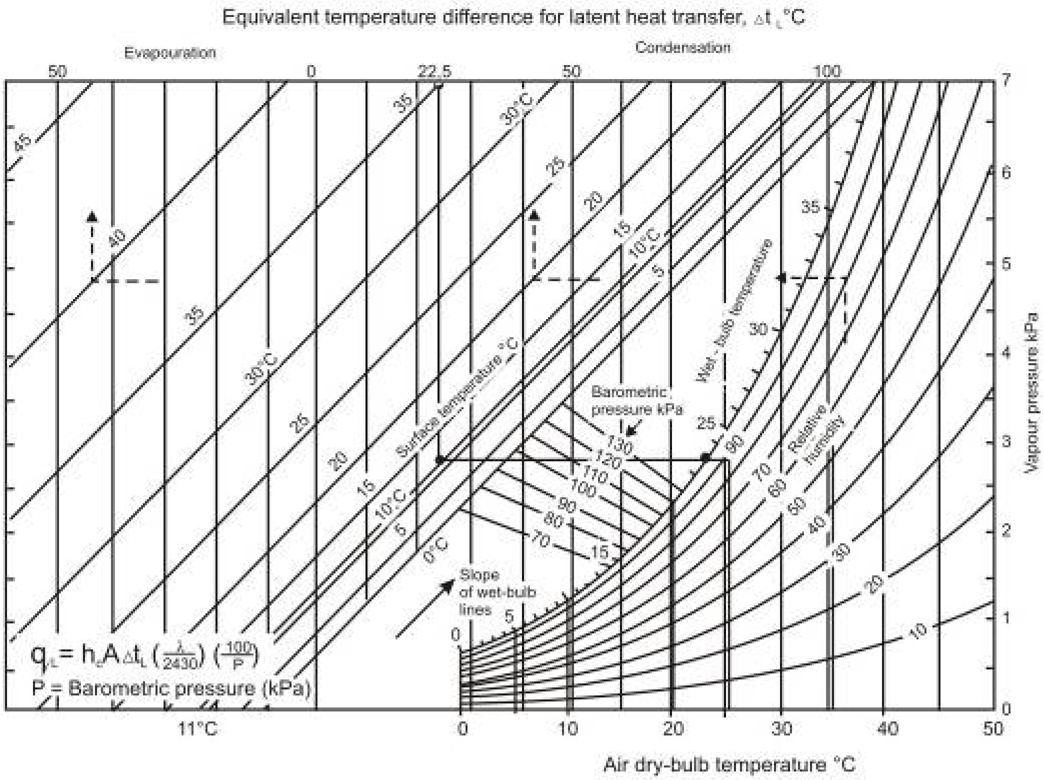


Figure 6.17 Equivalent temperature difference for latent heat transfer

6.6 Unsteady-State Heat Transfer

NB

In this section we are going to look at the various heat transfer processes between rock and air. This aspect is important as it has a large effect on the cooling requirements of a deep underground mine.

6.6.1 Unsteady-State Heat Flow

In mines, the transfer of heat from the rock into the ventilation air never reaches a steady state, although after six months, after a tunnel has been developed; the changes are relatively slow (the greatest amount of heat transfer takes place within the first 6 months after being developed).

After the rock has been exposed for a few weeks, the main resistance to heat flow is the rock itself and has the effect that the difference between the rock surface and the air temperature (dry-bulb) is virtually the same after a few weeks.

Example

The biggest temperature difference is in the rock itself. A typical example is that after 12 hours 75% of the temperature decrease is in the rock itself whilst after 4 days 90 % of the temperature decrease has taken place.

It is because of the fact that the greatest resistance to heat flow from the rock to the air occurs in the rock that predictions of heat flow in the rock must receive careful attention. It is therefore clear that the rate of heat transfer (heat flux) at any stage is proportional to the difference in the virgin rock temperature (VRT) and the rock surface temperature (t_s).

Other factors that influence the rate of heat transfer is:

- Thermal conductivity
- Density and heat capacity of the rock
- Shape of the tunnel
- ‘Age’ of the development.

In mining situations there are two dominant shapes, namely:

- Circular symmetrical (typical of tunnels)
- Plane exposed surface of a semi-infinite solid (narrow reef stoping).

Self study

Section 7 on page 382 of VOEE should be done for self study purposes.

References

Read the following articles:

- 'Underground mine heat loads and associated reduction methodologies' by C.A.Rawlins.
- 'The feasibility of using heat sinks at depth' by S.A Kebonte and M Biffi.
- 'Thermal properties of backfill from a deep South African gold mine' by M.Q.W Jones and C.A Rawlins.

Chapter

7

WATER RETICULATION

LEARNING OUTCOMES

- Use of water and distribution thereof
- Determine pressure loss for water systems
- Use of the Darcy-Weisbach Equation
- Solve a simple chilled water and distribution system by using calculations and charts
- Conservation of energy concepts
- Describe the following measuring techniques:
    - Orifice plate
    - Pitot tube
    - Rotameter
    - Electromagnetic
    - Ultrasonic flow meters.
- Use the orifice plate to determine the flow rate
- Describe the following measurement techniques:
    - Turbine and open channel flow meters
- Turbine and open channel flow meters.

REFERENCES

Mine Environmental Certificate (MEC), Chamber of Mines, South Africa, Workbook 1.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

7.1 Introduction

In this module we will look at various aspects pertaining to water for use on mines. The use of water, (whether for cooling, domestic use or production) and the reticulation of water with regards pumping and or refrigeration systems will receive special attention.

Water consumption in hard rock mines is ± 2 to 4 tons of water per ton of rock milled. On most mines there exists water - carrying veins of such magnitude that Rand Water Board water does not have to be used extensively and in this way saving the mine a lot of money. Some mines have excess water and additional pumping capacity needs to be created. Regulations require a water pressure of at least **150 kPa** per type of machinery in every workplace. The water for drilling and watering down (dust control) is recycled and is purified. It contains:

- Solutions of mineral salts, CO_2 and O_2
- Acids
- Mineral salts in suspension if the solution is over saturated, undissolveable material like mud, fine sand or silica particles and organic material like wood, carbon particles, oil etc.
- Bacteria.

The material in suspension fouls the inside of pipes and forms a thin layer of coarse material that can block the pipes. The impurities in water used in gold mines consist of:

- Approximately silica (the majority)
- Organic material
- Calcium carbonate.

In shallow mines only the excess water is pumped to surface, this can be millions of litres in some cases. The rest is pumped to dams on the different levels and is reticulated. In deep mines where chilled water is used both excess and water needed for the chilling plant is pumped to surface.

7.1.1 The Role of Water in the Mining Industry

Water is one of the essential commodities used in a mine. It also creates some handling and safety concerns. Water is used for the following:

- **Cooling of machinery**: Rock drills, drill steel and diesel powered machines.
- **Dust control**: All rock drills must be supplied with a water tube through which water flows at a minimum of 150 kPa to wet chippings to allay dust. Blasted rock is watered down before it is removed.

- **Cooling of the environment**: Service water or chilled water used in dust control or chilled water used in cooling cars or bulk air coolers create a more favourable working environment.
- **Transport medium**: Fines in plants, coarse material like coal, backfilling and water-jets.
- **Power source**: Hydro powered rock drills and other machinery. Hydropower is still in the development stage and holds a lot of potential.

7.1.2 Methods to Remove Water from Mines

There are various methods of removing water from a mine and these will be discussed in detail in the sections to follow.

7.1.2.1 Drainage Tunnels through Adits

Adits have sometimes been tunnelled for drainage purposes, only in some cases with the objective of dewatering old workings situated fairly close to the surface. It is always desirable that advantage should be taken of natural conditions to expedite or simplify the drainage of mines.

High-lying mines in mountainous areas can be dewatered through developing day lighting tunnels in the bottom of the mine instead of pumping the water out. The length of these tunnels can be quite long, sometimes even kilometres in length, to be used in dewatering a mine.

7.1.2.2 Sealing off Water Containing Veins and Fissures

Definition

In shafts, the concrete lining on the sidewalls of a shaft stops groundwater from seeping into the shaft. Areas containing large amounts of water in the proximity of excavations are sealed off. This process is called **cementation** whereby cement is pumped into a borehole that has intersected a water carrying vein or fissure.

Floodwalls and cement plugs with a walkway and a door on the one side are built in areas where there is a high risk of sudden flooding. Surface pillars are left on the outcrop to prevent rainwater from running into the mine. Surface storm water trenches near the outcrop serve the same purpose.

7.1.2.3 Hoisting Water in Kibbles

This is a method used during shaft sinking.

Disadvantages

The disadvantage of this method is that it disrupts the shaft sinking cycle and is more expensive than high-pressure pump installations. Sometimes the pump stations flood and this method is then also applied to dewater the shaft.

It is also adopted when dewatering a mine, which has been abandoned and allowed to flood.

Definition

Sometimes **bailing** (using tanks attached to hoist rope) is used as a permanent method of dealing with water and a separate hoist and shaft compartments are utilised.

Advantages

It is a less economical method of removing water, but it has the advantage of simplicity and low repair costs.

The bailing tanks are attached to the winding rope, which are cylindrical in shape and can vary from 5 to 18 m^3 water holding capacity (how much water is this in litres?). They are provided with 'valvesor flaps' that open automatically when the tank enters the water at the shaft bottom and closes under the weight of the water in the tank. When full; they are tipped automatically at the surface.

7.1.2.4 Boreholes

Long horizontal or inclined boreholes can be put in with a diamond drill and accumulation of water can be drained off to suit the pumping equipment available.

7.1.2.5 Siphons

Definition

A S**iphon** is a water transferring system or device, which finds a wide field of usefulness underground as well as on surface. They are used for transferring water from a '**higher level to a lower level**' over some obstruction, such as a wall or some other intervening obstacle, which is at a higher level than the top of the water, as shown in Figure 7.1

Rule

The idea is that water must be transferred from point 'X' to point 'Y' over the intervening ridge 'R' shown in Figure 7.1. In practice, the highest point of the siphon should not be more than 6 m vertically higher than the water level at 'X'. The grade of the pipe should be uniform (to avoid accumulation of air at the high points) and all joints must be air tight to prevent leakages (any leaks may stop a siphon from working).

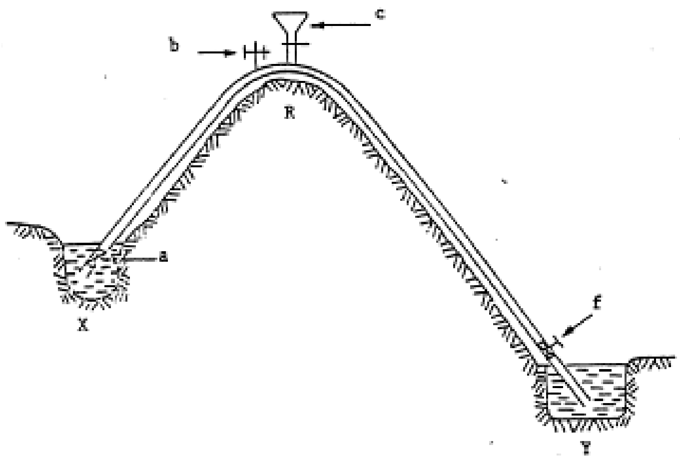


Figure 7.1 The use of the siphon method

A non-return valve is placed at the inlet side of the pipe (point ‘a’), an ordinary through valve above the water level at the discharge end (point ‘f’), a funnel with a valve at the highest point (point ‘c’), and an ordinary valve near the funnel (point ‘b’).

To start the siphon, it should be filled with water by pouring water into the funnel at point ‘c’. Instead of a funnel, the highest point can be connected directly to the main mine water supply if it is in any way possible. The valve at ‘f’ must be closed so that the pipe on the down flow side can be filled with water. The valve at ‘b’ should be open and the non-return valve at ‘a’ will prevent water flowing back into the dam at ‘X’. The valve at ‘b’ is to allow air to escape while the pipe is being filled on the ‘up flow’ side.

When both the pipes are completely filled, it is indicated by only water coming from the valve at point ‘b’. Both the valves (‘b’ and ‘c’) are now closed and when the valve at ‘f’ is opened, the water will begin to flow through the siphon, provided that all air has been removed from the pipes.

ML

Mining Engineers manage mines and it is often expected of them to give advice pertaining to pumping of water and reticulation systems required for refrigeration for a specific working environment. Chilled water systems (refrigerated water) are employed in hot mines and mining engineers who manages these mines will have to have a thorough understanding of chilled water systems. They therefore also need to understand the design aspects of chilled water systems. The purpose of this chapter is to introduce the design principles of these reticulation systems.

Objective

In the simplest sense of the word two systems are used on a mine:

Definitions

- **Open ended pipe system.** Water fall free down a pipe column to a dam underground (called a cascade system) and a **closed circuit arrangement** where through a series of pressure reducing valves the pressure in a pipe column is controlled and the pressure in the pipe due to the height of the column is used in driving machinery underground; for example rock drills
- **Hydro power equipment** (that is equipment that is driven through the use of water under pressure, either made available through the height of the column of water or pressure increase with pumps).

NB

Pressure losses forms a very important component in understanding pressures needed for pumping water in a water reticulation and it is therefore important to understand how these pressure losses are determined and how it eventually influences the effectiveness of a reticulation system. The next section will explain this in detail.

Definition

7.2 The Darcy Weisbach Equation

The above-mentioned Equation is used in general to determine the **pressure head** (friction losses in terms of meters head) when water or chilled water flows in a pipe and is expressed as:

$$H_f = \frac{\lambda L U^2}{2\ g\ D} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (7.1)$$

Where:

H_f	=	Friction pressure head loss in meter
λ	=	Friction coefficient (dimensionless)
L	=	Pipe length (m)
U	=	Average fluid velocity (m/s)
g	=	Gravitational acceleration (m/s^2)
D	=	Diameter for round pipe (m)

In the case where **non-circular ducts** are used, D is replaced by the hydraulic diameter that is determined as follows (this aspect was dealt with in detail in Chapter 2):

$$D_h = \frac{4A}{C} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (7.2)$$

Where:

A	=	Area (m^2)
C	=	Circumference of the pipe (m)

The friction coefficient can be calculated from the following Equation:

$$\lambda = \frac{\Delta p}{\left(\frac{\rho\ U^2}{2}\right) L\ D_h} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (7.3)$$

NB

Note: The term $\frac{\rho U^2}{2}$ is the velocity pressure as shown before.

Where:

ρ = Fluid density.

Definition

Or where there is **turbulent** flow (the Reynolds number between 2 500 and 1 000 000 indicates turbulent flow), the following Equation can be used:

$$\lambda = \frac{0.316}{R_e^{0.25}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (7.4)$$

Where:

R_e = Reynolds number

$$R_e = \frac{\rho DU}{\mu} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(7.5)}$$

Where:

μ = dynamic viscosity coefficient (Ns/m^2 or kg/m/s)

Self study

Alternatively we can use the Stanton and Nikuradse diagrams to determine the value of λ. As a result of the scale and possible correlation errors the answer will thus not be as accurate as that which was calculated above.

The Darcy Weisbach Equation is fully explained in Chapter 1 of VOEE (page 16, 19, and the chart on page 16 of VOEE).

WORKED EXAMPLE 1

68 litres/second of water at a temperature of 18ºC flows along a 150 mm **internal** diameter pipe, which is 1 460 m long. Calculate the pressure head loss along this pipe.

Answer

Normally in a problem like this you are given enough information to determine what the height is. It is irrelevant at this stage whether the pipe is vertical or horizontal. In this example however you will have to determine the R_e number first. Read off the value for the dynamic viscosity (μ) from Figure 1.11, page 12 of EE.

$$R_e = \frac{\rho DU}{\mu}$$

$$= \frac{1000 \times 0.15 \times 3.85}{10^{-3}}$$

$$= 577\,500$$

The water velocity is calculated as 3.85 m/s (the quantity of water flow is 0.068 m^3/s and the area of the inside of the pipe calculated as):

0.01767 m^2 ($U = \frac{Q}{A}$).

NB

The assumption was made that 1 m^3 of water is equal to 1 000 litres of water at a water density of 1 000 kg/m^3 (this always to be assumed if the density of the water is not given).

From the Stanton Nikuradse diagram the value of λ can be read off as approximately 13.5×10^{-3}.

The head loss can now be determined as follows:

$$H_f = \frac{\lambda \, L \, U^2}{2 \, g \, D}$$

$$= \frac{13.5 \times 10^{-3} \times 1\,460 \times (3.85)^2}{2 \times 9.79 \times 0.15}$$

$$= 99.5 \text{ m}$$

Definition

When water is standing still in a pipe that is vertical (as in a shaft), the vertical height is normally referred to as the **static height** and used to calculate the static pressure (H_S), or in simple terms, **'the standing still'** pressure.

NB

At this point it is also important to note that previously the pressure loss was expressed in Pa for air and in some cases converted to kPa.

As the density of water is approximately a 1 000 times more than for air, the answer for pressure losses pertaining to water is normally expressed in kPa or MPa. It can therefore be stated that under frictionless or no flow conditions the pressure increase in a vertical pipe is:

$$\Delta p = \frac{\rho_w g \, \Delta h}{1\,000} \text{ kPa}$$

If the density of the water is assumed as 1 000 kg/m^3, then the Equation above can be simplified re-written as follows:

$$\Delta p = \frac{1\,000 \, g \, \Delta h}{1\,000} \text{kPa}$$

$$= g \, \Delta h \text{ in kPa}$$

The pressure head Δh expressed in metre is therefore calculated from this formula as $\left(\frac{p_L}{\rho g}\right)$.

The increase in pressure corresponding to a depth of 1 m is therefore 9.79 kPa. For normal flow conditions where friction losses occurs the pressure increase is:

$$\Delta p = 9.79 \times (\Delta h - H_f) \text{ kPa}$$

WORKED EXAMPLE 2

A pipe column is planned for a new section of a mine. This column will be installed in a portion of a vertical shaft for a total distance of 150 m and will then be installed in a horizontal haulage to a point 1 600 m away from the shaft.

The vertical pipe will be of 200 mm internal diameter and the horizontal pipe will be 150 mm internal diameter. The required water flow rate is 100 litres/second at a discharge pressure of 20 m head (ignore pressure losses due to bends, valves etc.). Will this column be adequately sized given that the dynamic viscosity of the water is 10^{-3} (Ns/m^2 or kg/m/s). If not, what pressure would a pump have to supply to overcome friction?

Answer

Firstly consider the shaft column. Assume that the water is delivered at the top of the column at 0 m pressure (free flowing). The available pressure for the complete system is therefore 150 m. Using the Darcy Weisbach Equation:

$$H_f = \frac{\lambda L U^2}{2gD}$$

Where:

$$U = \frac{Q}{A} = \frac{0.1}{0.031} = 3.183 \text{ m/s}$$

NB

100 litres/second is 0.1 m^3/s and the area of a 200 mm inside diameter pipe is calculated as 0.031 m^2 $\left(\frac{\pi D^2}{4}\right)$.

$$R_e = \frac{\rho D U}{\mu}$$

$$= \frac{1\,000 \times 0.2 \times 3.183}{10^{-3}}$$

$$= 636\,600$$

From the Stanton Nikuradse diagram the value of λ can be read off as approximately 13 x 10^{-3}.

$$H_f = \frac{\lambda L U^2}{2gD}$$

$$= \frac{13 \times 10^{-3} \times 150 \times (3.183)^2}{2 \times 9.79 \times 0.2}$$

$$= 5.045 \text{ say } 5 \text{ m}$$

Repeat this process for the horizontal pipe (calculate the value of U first (U = 5.659 say 5.7 m/s)

$$R_e = \frac{\rho DU}{\mu}$$

$$= \frac{1\ 000 \times 0.15 \times 5.7}{10^{-3}}$$

$$= 855\ 000$$

From the Stanton Nikuradse diagram the value of λ can be read off as approximately 12.5×10^{-3}.

$$H_f = \frac{\lambda L U^2}{2gD}$$

$$= \frac{12.5 \times 10^{-3} \times 1\ 600 \times 5.7^2}{2 \times 9.79 \times 0.15}$$

$$= 221.2 \text{ say } 221\text{m}$$

The total head loss for the full length of pipe = 221 + 5
= 226 m

But only 150 m is available (the vertical height of column representing the potential energy or the ability of the water to perform work) and a deliver head of 20m is required (this relates to the amount of pressure that is necessary to perform work). The additional head required can therefore be determined as follows:

Additional head = 226 + 20 – 150
= 96 m head at a flow rate of 100 litres / second

NB

This amount can be expressed in kPa to show the duty of the pump.

$$p = \frac{\rho g H}{1\ 000}$$

$$= \frac{1\ 000 \times 9.79 \times 96}{1\ 000}$$

$$= 939.8 \text{ say } 940 \text{ kPa}$$

The pump duty is therefore 100 litres / second at 940 kPa.

7.3 Pipe Friction Chart

Besides calculating head loss due to friction by means of Equations such as the Darcy Weisbach Equation, it is also possible to determine head loss due to friction by means of a friction chart shown in Figure 7.2. By studying the chart you will see that in order for you to use the chart, at least two of the following factors need to be known:

- Water flow rate
- Pipe diameter
- Water velocity in the pipe
- Head loss.

The pipe friction chart gives sufficiently accurate information for most mining situations and caters for normal temperature ranges encountered in chilled water reticulation circuits. Another Equation to calculate the head loss in pipes is:

$$H_f = 2.04 \times 10^{-9} \frac{m^{1.92}}{D^{5.13}} L \times AF \quad (7.6)$$

Where:

H_f = Frictional pressurre loss (m)

D = Inside diameter of pipe (m)

L = Length of the pipe column (m)

AF = Age factor (dimensionless)

m = Mass flow of the water (litres/second)

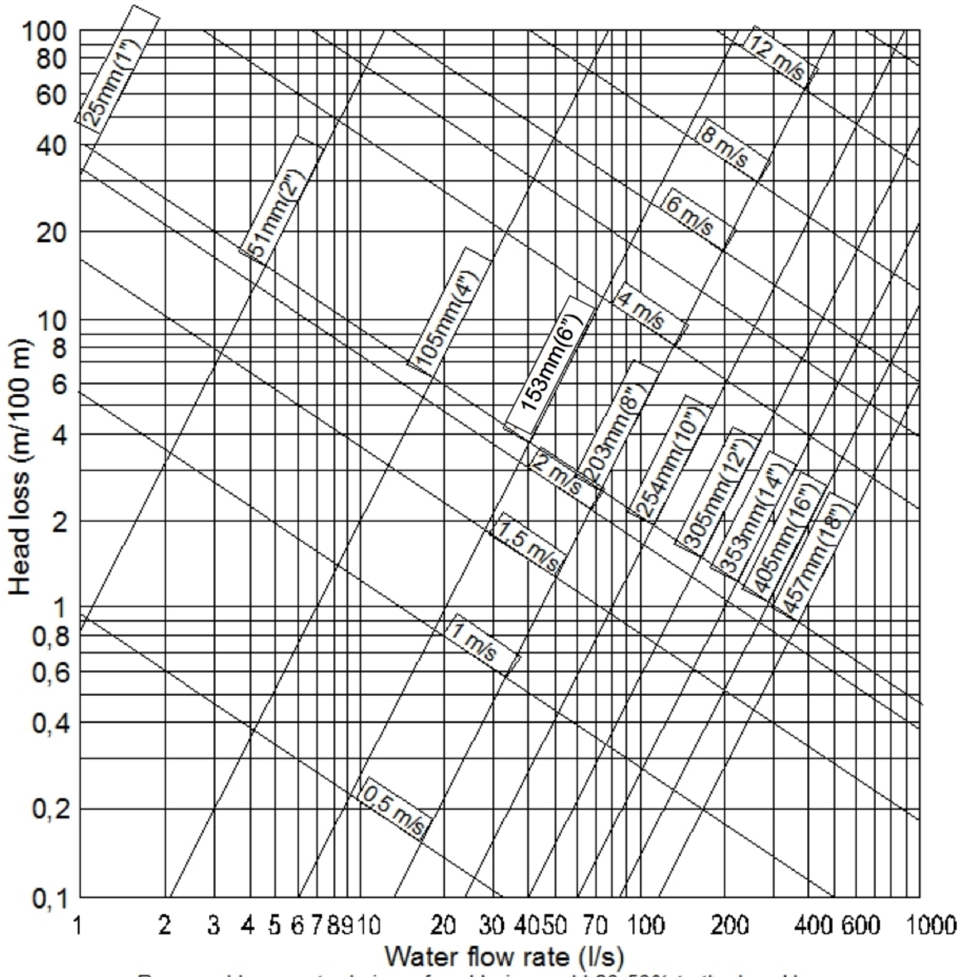


Figure 7.2 Pipe friction chart

NB

It is important to note that this formula, as well as the chart shown in Figure 7.2, relates to reasonably new steel pipes. For new pipes the age factor is 1 and for old pipes the age factor (VOEE, p 504) given in Table 7.1 should be used.

Table 7.1 Age factors for pipes

Age in years	New	10	15	20	30
Factor	1.0	1.3	1.45	1.6	2.0

Bends, T-pieces, valves and other restrictions add to the pressure loss in a system. These losses can be accounted for by increasing the length of the column to simulate the added resistance of the fittings. Equivalent pipe lengths for various fittings are given in Figure 7.3.

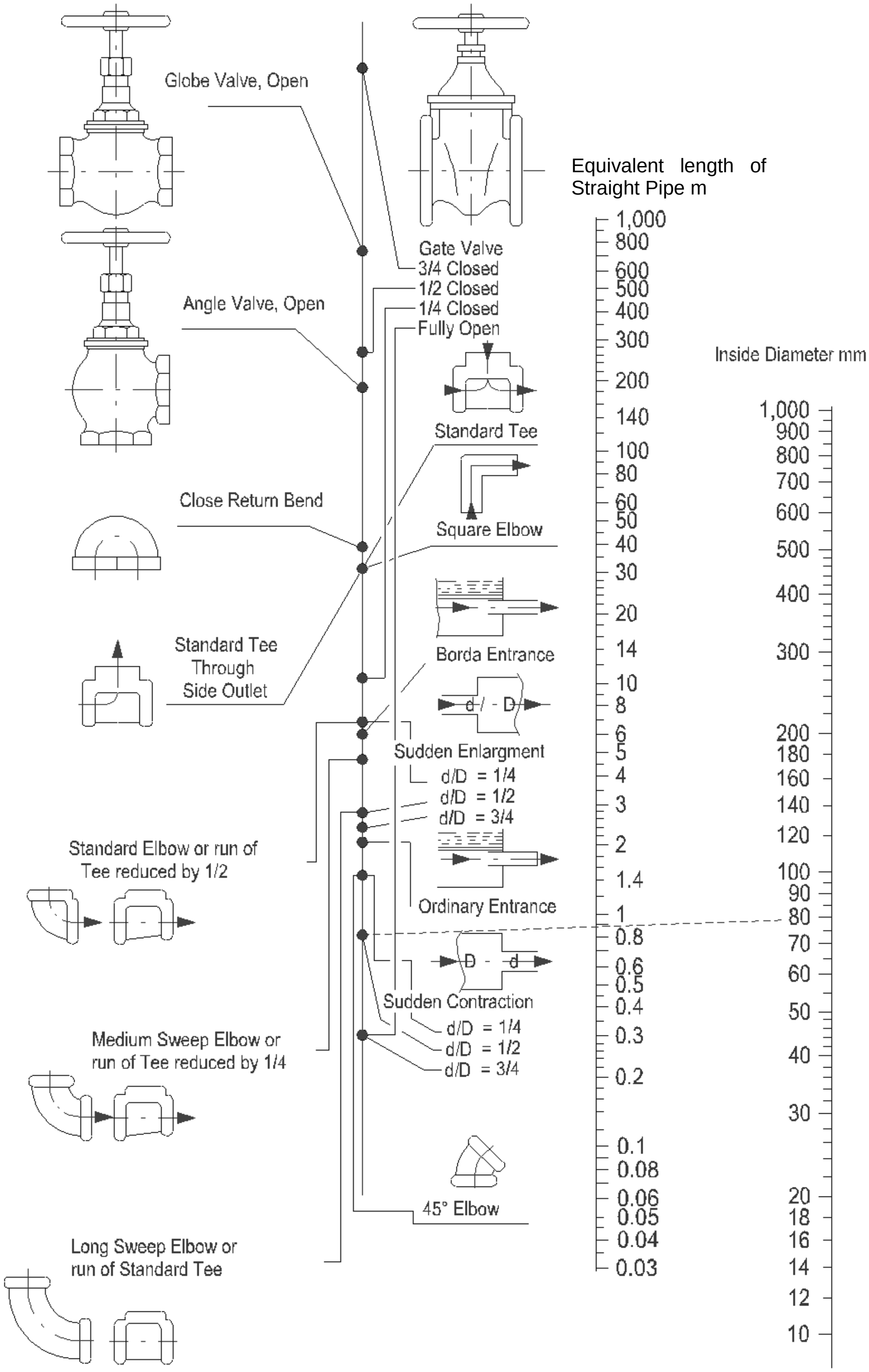


Figure 7.3 Approximate equivalent length of straight pipe for pipe fittings and valves

This chart **does not give the approximate head losses** for fittings, but rather an **equivalent length of pipe** that would represent the same head loss as the pipe fitting in question. The nomogram (Figure 7.3) is used as follows:

- Draw a line from the applicable identification point/position (dot) on the left hand side to the inside diameter line on the right-hand side.
- The 'loss', expressed as an equivalent length, is then read off the centre line.

WORKED EXAMPLE 3

Calculate the gauge pressure of the water at the bottom of a 1 000 m vertical pipe with a diameter of 155 mm in which 50 ℓ/s is flowing. The barometric pressure at the bottom of the pipe is 100 kPa. The barometric pressure on surface needs to be calculated first.

Answer

(i) Theoretical air pressure increase (kPa)

$$= \frac{\rho g \Delta h}{1\ 000}$$

$$= \frac{1_{avg\ air} \times 9.79 \times 1\ 000}{1\ 000}$$

$$= 9.\ 79 \text{ say } 10 \text{ kPa}$$

Let's assume frictionless / ideal conditions then the pressure at the top can be calculated as follows using the:

$$P_{atm\ bottom} = P_{atm\ top} + \text{Pressure increase}$$

$$P_{atm\ top} = P_{atm\ bottom} - 10$$

$$= 100 - 10$$

$$= 90 \text{ kPa}$$

(ii) The head loss due to friction is 5.5 m/100 m as read from Figure 7.2.

$$h_f = 5.5 \times \frac{1\ 000}{100} = 55 \text{ m}$$

The relationship between the absolute, atmospheric and gauge pressure is discussed in detail in Chapter 1.

$$P_{absolute} = P_{atmospheric} + P_{gauge\ pressure}$$

$$\therefore P_{gauge\ pressure} = P_{absolute} - P_{atmospheric}$$

$$= (P_{atm} + P_{gauge\ pressure\ increase}) - P_{atm}$$

$$= 90 + [(1\ 000 - 55) \times 9.79] - 100$$

$$= 90 + 9\ 252 - 100$$

$$= 9\ 242 \text{ kPa}$$

If a valve is installed at the discharge from a vertical pipe then the pressure rating of the pipe must be rated for at least the static pressure corresponding to no flow.

WORKED EXAMPLE 4

A square elbow is installed in a 200 mm pipe column. What will be the pressure loss expressed as an equivalent length for this elbow?

Answer
Connect the identification point for the elbow (6th point from top position/identification point) to the inside diameter (200 mm) on the right-hand line. The answer is then read off as 13 m. Worked Example 5 takes into account all of the above mentioned factors.

WORKED EXAMPLE 5

Figure 7.4 represents a water reticulation circuit installed in a mine.

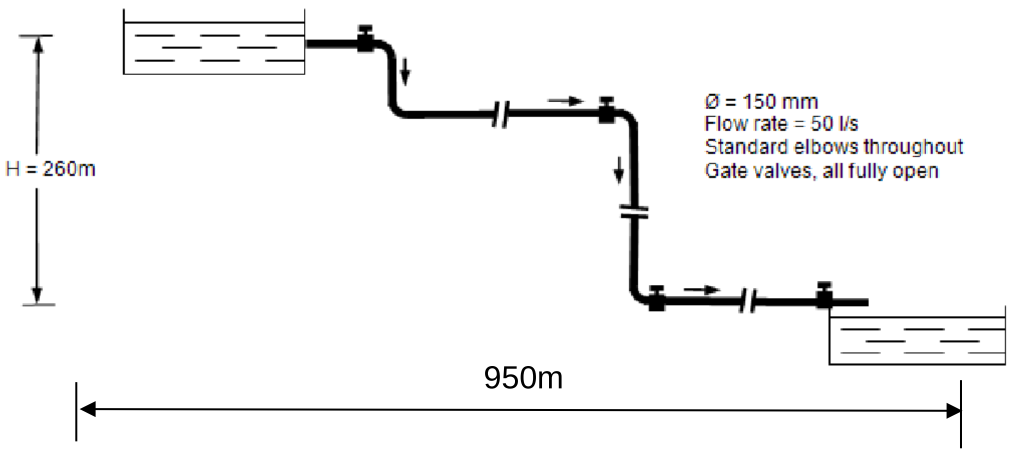


Figure 7.4 Simple water reticulation layout

Determine the following:

(i) The eventual head loss in (m) for this system which will be in use for 10 years expressed in metres and kilopascals. Comment on your answer.

(ii) The velocity of the water in the pipe.

Answer

(i) Total column length	=	(vertical + horizontal) distance
	=	260 + 950 m
	=	1 210 m

Additional pipe length to be added for losses through pipe fittings:

Equivalent pipe length for elbows = 4 x 5 m
= 20 m

Equivalent pipe length for valves = 4 x 1 m
= 4 m

Additional pipe length for square entry = 2.6 m

Therefore total column length = 1 210 + 20 + 4 + 2.6
= 1 236.6 say 1 237 m

Head loss through column read off from Figure 7.2 as 5.3 m/100 m

$$= 5.3 \times \frac{1237}{100}$$

= 65.56 m head loss

Allowance for age (10 years from Table 7.1):

= 65.56 x 1.3

= 85.2 m head loss

(ii) Head loss expressed in kilopascals:

$$= \frac{\rho_w \times g \times H_f}{1\,000}$$

$$= \frac{1\,000 \times 85.2 \times 9.79}{1\,000}$$

= 834.1 kPa

Comments

(i) The available head (260 m) far exceeds the losses (85.2 m) along the column. For the required flow rate of 50 ℓ/s this column is oversized and a smaller column may appear more practical. However, a quick look at the next standard pipe down (105 mm) gives a head loss of 40 m/100 m which is equivalent to 484 m loss to simply overcome friction in the pipe, disregarding all the other losses. It would therefore be recommended that this column, which allows some spare capacity, should be installed.

(ii) The water velocity in the pipe is read off from Figure 7.2 as approximately 3.7 m/s.

WORKED EXAMPLE 6

What would the head loss be, expressed in m/100 m in a reasonably 105 mm new pipe carrying 40 litres/second?

Answer

On the chart, find the point along the bottom horizontal axis indicating 40 litres / second. Plot vertically upwards to where this line intersects the diagonal line indicating the 105 mm pipe.

Now read horizontally across to the left where you will read off the approximate answer of 27.5 m/100 m.

WORKED EXAMPLE 7

From Worked Example 6, determine the head loss of the pipe if it is 1 240 m long.

$$H_f = \frac{27.5}{100} \text{ x length of pipe}$$

$$= \frac{27.5}{100} \text{x} 1240$$

$$= 341 \text{ m}$$

This value can also be expressed in terms of pressure (Pascal or kilo Pascal). As mentioned before the pressure in the water reticulation system is normally very high and expressed in kPa and also as MPa. The Equation to convert m head to kPa is as follows:

$$p = \frac{\rho g h}{1000}$$

The previous example can therefore be converted as follows:

$$p = \frac{\rho g h}{1\ 000}$$

$$= \frac{1\ 000 \text{ x } 9.79 \text{ x } 341}{1\ 000}$$

$$= 3\ 338.4 \text{ kPa}$$

In other words, for the 40 litres/second to flow for a distance of 1 240 m you would use up 3 338.4 kPa of pressure or expressed in metres; 341 m.

7.3.1 Terminal Velocity

The term '**terminal velocity'** can be defined as follows:

Definition

'The velocity of water flowing down an open-ended vertical column, where the friction losses are equivalent to the gravitational acceleration'.

This means simply that the water cannot flow any faster down the open column, because the shear and drag forces within the column fully cancel out the gravitational force of the earth.

An aspect pertaining to terminal velocity is the fact that the terminal velocity applicable to various pipe columns can be read off the pipe friction chart directly, so also the water flow rate (Figure 7.2).

It is done as follows:

- The maximum pressure head loss is applicable where pressure head loss is 100 m / 100 m of pipe, or 1 (the value can never be more than one and is the maximum pressure head loss expressed in m/m).
- On this line various velocities for different size pipes can be read off directly
- The water flow rate can also be read off from this chart. By going from the point where the 100 m / 100 m pressure head loss line intersects with the pipe column diameter line, go down vertically along the water flow rate line and read off the applicable water flow rate in litres/second.

7.3.2 Calculation of Flow Rates and Storage Dam Sizes in Chilled Service Water Systems

The cyclical nature of mining activities presents some difficulties in designing systems, these are:

- Water consumption
- Peak flow rates
- Time cycles and daily variations must always be catered for.

There are no known methods of determining these parameters other than by experience.

Rule

As a general rule, two-thirds of water consumption occurs during one-third of the day, which is during the main shift. Peak flow rates are from two to three times the average. Total water consumption is best expressed as a ratio of water used and tonnage mined. Unfortunately this ratio can vary from 0.5 to 3.0 or even higher. A ratio of 1.0 ton per ton of rock mined can be used as a starting point for unknown conditions.

NB

The most general form of measuring water flow as mentioned before is in m^3/s and litres/second respectively. Dams are usually expressed in terms of their hour capacity at a specific flow rate and the formula is expressed as:

$$q = \frac{Volume}{time} = \frac{m^3}{s} \quad \text{.......................................} \quad (7.7)$$

Furthermore:

$$1 \text{ litre/second} = \frac{\ell / s}{1000} \times 3600 = [\ell/s \times 3.6]\, m^3/\text{hour}$$

$$T = \frac{\text{Volume}}{3.6 \times q_{\ell/s}} \quad \text{.......................................} \quad (7.8)$$

Where:

T	=	hours, water store time
V	=	volumetric capacity of the dam (m^3)
Q	=	water flow rate out of the dam litres/second.

NB

Note: A very important design aspect pertaining to water flow rates in a pipe is that the economic water flow velocity has been found to be 2 m/s. If the velocity of the fluid has not been given, an economic velocity of 2 m/s has to be assumed for calculations pertaining to water reticulation systems.

Rule

A thumb rule pertaining to flow rate of water in new pipes is also expressed as follows:

The water flow rate in ℓ/s $\approx$ (pipe diameter in inch)^2 (1 inch $\pm$ = 25 mm)

For example, a 100 mm inside diameter pipe is '4' and there based on an economic flow rate of 2 m/s a flow rate of 4^2, or 16 litres/second can be expected.

WORKED EXAMPLE 8

Calculate the chilled service water requirements for a mine breaking 150 000 tons per 24-shift month and blasting once per day. Total water consumption is 1.5 tons per ton of rock mined and the 8-hour main shift uses 2.5 times the average. Calculate the storage dam size assuming the refrigeration plant delivers water at the average daily flow rate.

Answer

Total monthly water consumption:

$$= \quad 150\ 000 \times 1.5$$

$$= \quad 225\ 000\ m^3$$

Mean daily flow rate (from basic arithmetic and logic)

$$= \quad \frac{225000}{24} \times \frac{1000}{24 \times 3600}$$

$$= \quad 109.0\ \ell/s$$

Maximum flow rate

$$= \quad 109 \times 2.5$$

$$= \quad 273.0\ \ell/s$$

Rule

If the flow rate for the main 8-hour shift was not known, then the rule of **two-thirds consumption** during this period would lead to a flow rate of 217 ℓ/s.

Storage capacity

$$= \quad \frac{(273 - 109) \times 8 \times 3600}{1000}$$

$$= \quad 4\ 723\ m^3$$

This is the minimum storage capacity required and does not cater for plant failure. If four hours of reserve water is required:

Total storage capacity (from basic arithmetic and logic)

$$= \quad \left(\frac{109 \times 4 \times 3600}{1000} \right) + 4\ 723$$

$$= \quad 6\ 292\ m^3$$

NB

!

Storage dams must be carefully designed to ensure that their total capacity is available for use. It should be further noted that when reserve capacity is used it may only be possible to make this up over the following weekend. The hot water or clear water sumps at shaft bottom must have a capacity equal to or greater than the total water quantity in circulation. This is necessary to prevent flooding in the event of pump failure.

When water for air cooling duties is provided in a common main with service water, no additional chilled water storage capacity is required over and above that provided for the service water. It is evident that air cooling water is continually in circulation and does not require storage other than to provide for breakdown in the system.

Hot water storage dams at shaft bottom and at the refrigeration plant are occasionally designed to have a large capacity to cater for batch pumping.

This procedure provides for control of maximum demand (pumping only at off-peak periods). The difference between hot water dams and cold water dams as used in the underground mining environment will be discussed in detail in Chapter 8.

7.4 Temperature Increase of Water in Vertical Flow Situations

ML

When water flows into a mine it loses potential energy. Unless energy recovery systems are used, all this energy is converted into heat, and results in an increase in water temperature. This is one of the main motivations for installing energy recovery devices that reduce the temperature increase of water flowing underground.

This can be calculated from:

$$g\Delta h = C_p \Delta t \quad \text{or}$$

$$\Delta t = \frac{g\Delta h}{C_p}$$

For a 1 000 m change in elevation:

$$g\Delta h = 9.79 \times 1\,000$$

$$= 9.79 \text{ kJ/kg}$$

NB

A very important point to note is that the term gΔh relates to the increase or decrease of Enthalpy when air moves down and up an air way respectively.

Since $C_{pw} = 4.187$ kJ/kg° C

$$\Delta t = \frac{9.79}{4.187} = 2.34^{\circ}\text{C per 1000 m}$$

If an energy recovery turbine generator set was installed with an overall efficiency of 75%, then 75% of the potential energy would be converted into electrical energy and 25% would be converted into heat energy, which would increase the temperature of the water. Therefore the electrical power generated per ℓ/s per 1 000 m would be:

9.79 x 0.75 = 7.34 kW per ℓ/s per 1 000 m

And the temperature increase would be:

2.34 x 0.25 = 0.59 ºC per 1 000 m

The above applies to the temperature increase and pressure drop across the turbine, which is in addition to the frictional losses in the upstream pipeline.

The overall temperature increase is the result of the dissipation of the potential energy of water in its passage down the mine and occurs irrespective of whether the energy is dissipated due to friction in the pipe or by turbulence as the water discharges into a dam.

It should be noted that the temperature increase only occurs when the pressure in the pipe is reduced hence the temperature increase can be avoided by keeping the water in closed circuit.

7.4.1 The Joule-Thompson Effect

When water flows in a column, a certain rise in the temperature of the water is noted (the temperature of air increases with depth and this is commonly referred to as '**auto-compression**' but his is not the case with water flowing down a pipe in a shaft). It is important to note that auto-compression relates to the vertical distance displaced. However, strictly speaking, the term 'auto-compression' is incorrect when applied to water. The correct term, which will be used here, is the '**Joule-Thomson effect'** and this can be calculated as follows:

$$\Delta T = \mu \Delta P$$

Where:

ΔT	=	**change** in temperature (K or ^{o}C)
μ*	=	Joule-Thomson coefficient (-2.4×10^{-7} K/Pa)
ΔP	=	frictional pressure drop (negative) (Pa)

The use of the μ in this case is the same as the symbol used for dynamic viscosity. This is, however, not the same term; do not confuse it with μ in the Reynolds number.

The Joule-Thomson effect **also applies to horizontal pipes** (because of the pressure loss between two points), in which case the pressure loss or drop can be measured and applied to the Equation.

WORKED EXAMPLE 9

Water at 10ºC flows down a 1 000 m long column in a shaft and discharges into a dam at the bottom of the shaft. What is the temperature of the water entering the dam?

(i) Under ideal conditions

(ii) If there was a head loss of 3.7 m / 100 m of pipe length.

Answer

(i) If the pipe referred to was closed off at the bottom, the water pressure would have been:

$$p = \frac{\rho g H}{1\ 000}\ \text{(kPa)}$$

$$= \frac{1000 \times 9.79 \times 1000}{1000}$$

$$= 9\ 790\ \text{kPa}$$

$$= 9\ 790\ 000\ \text{Pa}$$

As this pressure is dissipated (used up or destroyed) when the water is discharged into the dam, the difference in temperature at the bottom of the shaft will be:

$$\Delta T = -\ 2.4 \times 10^{-7} \times (-9\ 790\ 000)$$

$$= +\ 2.35\ \text{K or } ^{o}\text{C}$$

The water entering the pipe is 10 ºC or 283.15 K. It will therefore be 2.35 ºC higher at the bottom of the shaft, expressed as follows:

$$(10 + 2.35)\ ^{o}\text{C} = 12.35\ ^{o}\text{C, or}$$

$$(283\ 15 + 2.35)\text{K} = 285.5\ \text{K}$$

The formula for the increase in water temperature for water flowing down vertical pipes previously used, namely

$\frac{g\Delta h}{C_p}$ gives the same approximate answer, but the Joule-Thomson

Equation is more accurate and is also used for water flow along horizontal pipes, as it is based on the actual pressure loss along the length of the pipe.

NB

$g\Delta h = C_p$. Here $g\Delta h$ refers to the enthalpy for 1 000m change in elevation

$$g\Delta h = 9.79 \times 1\ 000$$

$$= 9.79\ \text{kJ/kg}$$

$$\Delta t = \frac{g\Delta h}{C_{p\ water}}$$

$$= \frac{9.79}{4.187}$$

$$= 2.34°\text{C per } 1\ 000\ \text{m}$$

(ii) The head loss was given for a specific reason (to be used in the arguments to solve the problem). The head loss has been given as 3.7 m per 100 m and is therefore 3.7/100 x 1 000 = 37 m. The **available head** is therefore:

$$= 1\ 000 - 37$$

$$= 963\ \text{m}$$

By using the formula:

$$p = \rho g h$$

$$= 1\,000 \times 9.79 \times 963$$

$$= 9\,428\,000 \text{ Pa}$$

As this pressure is dissipated (used up) when the water is discharged into the dam, the temperature difference of the water at the bottom of the shaft can be calculated as follows:

$$\Delta T = \mu \Delta P$$

$$= -2.4 \times 10^{-7} \times (-9\,428\,000)$$

$$= 2.26 \text{ say } 2.3 \text{ Kelvin}$$

New temperature $= 10^{o}C + 2.3^{o}C = 12.3^{o}C$

NB

This is a slight difference compared to the answer in (i), slightly lower. This calculation shows that the increase in temperature is dependent on the amount of pressure available to do work. In the second case, with pipe friction included for the example, it shows that because less pressure is available to do work, the lower the increase in temperature. **The initial gut feel was that the temperature would increase because of friction, which is not true.**

!

A common mistake made by the student is that the pressure is used as kPa, which will give a very small temperature rise answer and the '-' sign is also forgotten.

7.5 Cavitation

Definition

Rule

In chilled water circuits (and other water flow circuits) in mines, **cavitation** is normally a result of water boiling in a pipe creating vapour bubbles and the subsequent collapse of these vapour bubbles. When the absolute water pressure in a pipe reduces (for example due to pipe friction), the water **may** boil. At an absolute pressure of 100 kPa water boils at 99.6 ºC, whereas **at an absolute pressure of 5 kPa it boils at 32.9 ºC**.

Definition

The temperature at which water boils depends upon pressure and there are standard tables that give the temperature and corresponding pressure (known as the **saturation vapour pressure)** at which water boils.

For any water temperature, if the pressure reduces to below the saturation vapour pressure, the water flashes to steam. When this occurs the steam blocks the pipe, which reduces the water flow and hence the frictional pressure drop. This reduction in frictional pressure drop increases the pressure in the pipe to above the saturation pressure and the steam bubbles collapse.

When this occurs there will be no blockage in the pipe due to the bubbles and the flow will increase again.

NB

This process of formation and subsequent collapse of steam bubbles can be heard as a characteristic banging in mine water reticulation pipes. In general, cavitation can be avoided by ensuring that the absolute water pressure does not reduce below the saturation vapour pressure corresponding to the temperature of the water.

7.5.1 Cavitation with Gravity Feed from a Dam

Cavitation frequently occurs when water is gravity fed from one dam to a lower dam. Consider the idealised situation when water flows between two dams at different elevations.

The water flow will be determined by the difference in elevation and the length of pipe and pipe friction. To illustrate how cavitation can occur consider the simplified Worked Example 10.

WORKED EXAMPLE 10

The vertical distance between two dams is 50 m and the length of pipe on each horizontal level is 25 m. The barometric pressure at the dam is 120 kPa and the water temperature is 7°C. Assume that there are no valves in the system and ignore the pressure losses in bends and entrances to the pipe. Determine whether cavitation will take place.

Answer

As a first approximation the frictional pressure loss will be an amount of 50 m / 100 m (that is 0.5) per m of pipe. After 24 m of horizontal pipeline the pressure in the pipe will be less than 1 kPa and cavitation will occur.

7.6 Pumps

Pumps for lifting or conveying liquids can be broadly divided into a number of different categories, according to some distinctive feature or with reference to a particular duty. Without taking into consideration special applications, the majority of pumps could be classified under one of the following headings:

Definitions

- **Reciprocating pumps**; in which a plunger or piston is mechanically reciprocated in a liquid cylinder.
- **Rotary pumps**; where the liquid is forced through the pump cylinder or casing by means of rotating drums or pistons, with the aid of centrifugal force.

- **Centrifugal pumps**; in which pressure is generated through a centrifugal force by a rotating vaned wheel.
- **Airlift pumps**; in which the water is raised by the buoyancy of an aerated column of water in a submerged tube.

When confronted with a pumping problem, one obviously wishes to know the most suitable and efficient type and size of pump for that particular service. In some cases it might be possible to say that experience has shown that a particular type has given satisfactory results over a number of years, but more frequently two or more different types might be equally efficient and reliable.

In order to give a satisfactory answer to such a problem, it is necessary to take into consideration all the prevailing circumstances, together with the guaranteed, or probable, efficiencies of suitable types of pumps.

Modern pumps of all types have undoubtedly reached a high degree of efficiency, but probably the greatest development has been in the range of operations performed by centrifugal pumps. For a great number of purposes the direct coupled and the belt driven pumps have replaced the **older** reciprocating pump, in spite of the fact that the efficiency of the centrifugal pump is lower (in comparison). It has however many advantages and this will be discussed in detail later in this section.

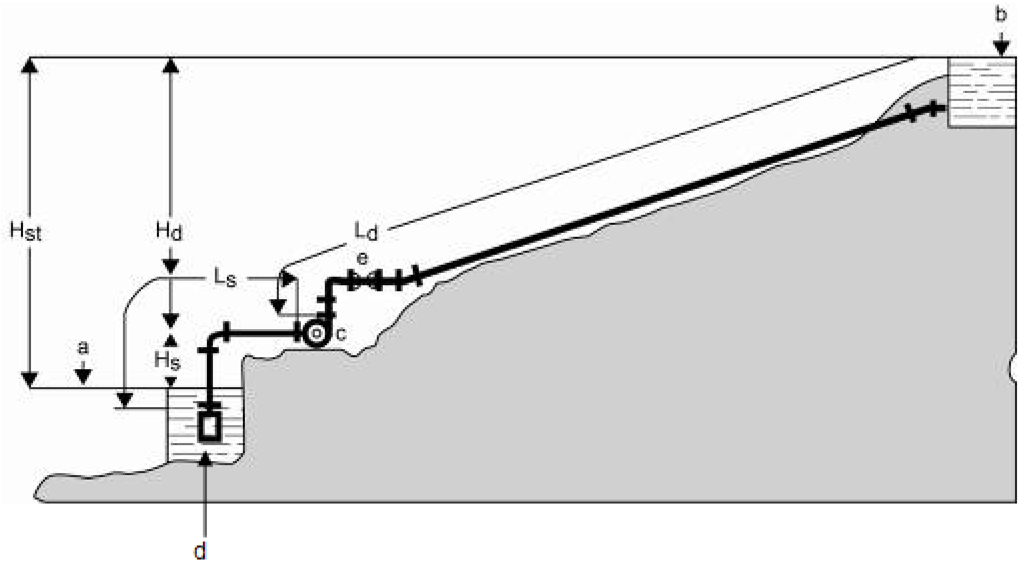
However, the centrifugal pump is unable to compete with the reciprocating and rotary type of pumps applied in areas where high pressure and fluctuating outputs are required, particularly when the heat in the exhaust steam from direct-acting or steam driven pumps can be usefully employed (not extensively used on mines).

NB

The duty of a pump is the amount of work capable by the pump (duty of a pump will be discussed in detail later in this chapter, but similar to the duty discussed for fans). The range of duties performed by pumps varies a lot, and nowadays pumps are designed and developed by manufacturers to meet specified duties. In the process of selecting a pump for a specific type of work environment (apart from considering the duty) or layout, there is very important data that needs to be considered. Figure 7.5 shows some of these aspects graphically and the aspects that needs to be considered are:

- What kind of liquid needs to be pumped?
- The temperature of the liquid (do you know why?)
- Total delivery amount required (litres/second)
- Total static head (H_{st}) in metres between the lowest liquid intake level and highest level in the discharge reservoir

- Static suction head (H_s) between lowest liquid intake level and pump centre line
- Length and inside diameter of the delivery pipe (this is necessary to determine the friction head (H_f) associated with the delivery pipe)
- How is the pump to be driven (direct-coupled to an electric motor, petrol (not allowed underground) or diesel engine, steam turbine or belt driven)
- If the motor is to be tendered also for electric drive, please give details of main supply (alternating current (ac) or direct current (dc, or battery supply), also single or 3-phase supply, voltage and for ac supply, the frequency (Hz).



a- Lowest liquid intake level
b - Highest discharge level
c - Centrifugal pump
d- Strainer with foot valve
e - Regulating valve

H_d - Delivery head
L_d - Length of delivery pipe
L_s- Length of suction pipe
H_s- Suction head
H_{st} - Total static head

Figure 7.5 Factors to be considered in a water reticulation system where pumping is needed

Some of the factors mentioned will be dealt with again in detail in the section where various pump layouts and calculations are considered. For now we can therefore summerise in saying, that in the mining industry, pumps may be grouped in two main classes:

- Those having valves
- Those without valves.

Pump types	With valves	Without valves
Type	Piston, plunger and bucket pumps	Centrifugal, rotary and air lift pumps
Flow	Intermittent flow	Continuous flow of liquid

This section will be divided into 4 parts, namely:

- Types of pumps
- General layouts for pumps
- Settling of impurities in water
- Pump related calculations.

7.6.1 Detail Description Pertaining to Different Types of Pumps

There are several types of pumps applicable for use in mines and they are as follows:

7.6.1.1 Reciprocating Pumps (Plunger or Piston)

Reciprocating pumps have differences in the moving portion inside the cylinder.

This moving portion may consist of a bucket, a plunger or a piston. The main difference between a piston and a plunger is the following:

- A piston is shorter than the stroke, whereas a plunger is longer
- A piston must have a packing (such as rings inlaid in its rim) to provide a tight joint. With a plunger, the packing is carried in a stuffing box at the end of the cylinder. Refer to Figure 7.6.

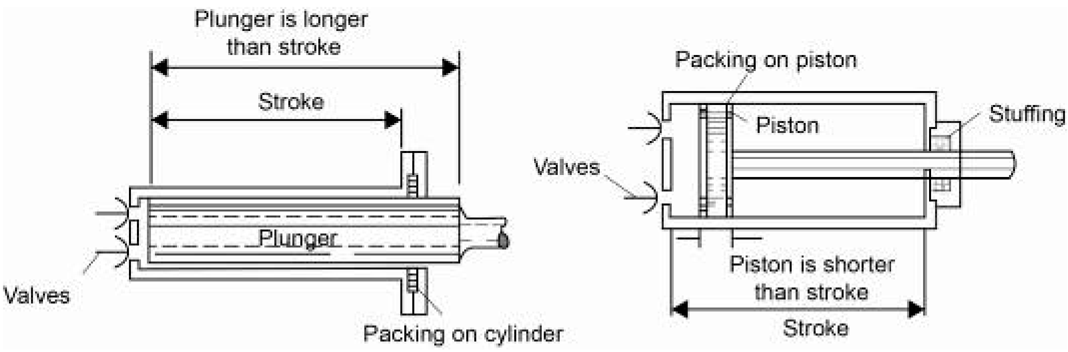


Figure 7.6 Difference between plunger and piston

7.6.1.2 Piston Pump

A piston pump as shown in Figure 7.7 almost always has suction and delivery valves connected to each end of the cylinder and is therefore a double acting pump. This type of pump operates most satisfactorily against medium pressures.

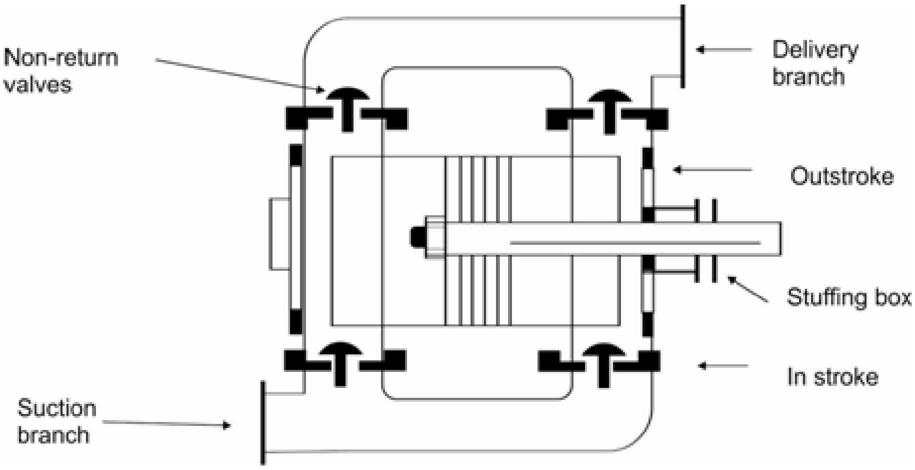


Figure 7.7 Double acting piston pump (diagrammatic)

As the piston moves to the right, water is drawn through the left suction valve and delivered through the right delivery valve. On moving to the left, water is drawn through the right suction valve and delivered through the delivery valve situated on the left side.

7.6.1.3 Plunger Pump

Example

Usage

An externally packed pump (Cameron type) is shown in Figure 7.8 and is an example of a plunger pump. The pump is driven by compressed air and is commonly used in underground mines as the source of power is compressed air and can therefore be used in areas where there is no access to electrical power.

The air cylinder on the right hand side has a piston, which is directly connected to the water plunger. When the air piston operates with a reciprocating movement from left to right or from right to left, the water plunger similarly follows. The plunger operates in two separate water boxes, which are only connected via the suction and delivery pipes. Each water box has two valves, that is, an inlet and an outlet valve.

Referring to Figure 7.8, it will be noticed that when the plunger moves from the left hand side to the right hand side, the water in the box is compressed, therefore the upper valve will open, allowing the water to flow out into the delivery, and the bottom valve will close, preventing water running back into the pump. While this is happening, a vacuum is being created in the left box, which will cause the top valve to close and the bottom valve will open, sucking in more water.

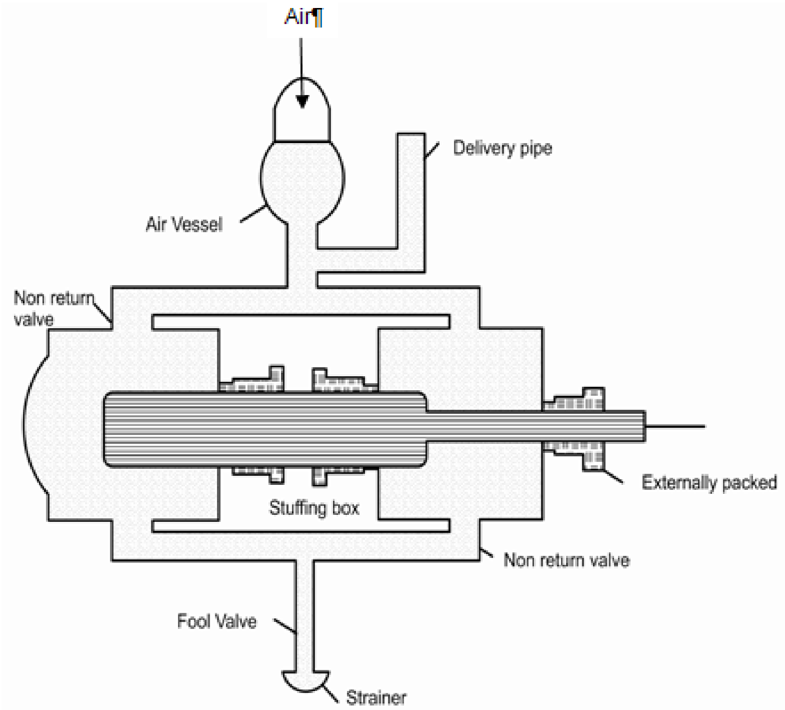


Figure 7.8 Section through a plunger pump (externally packed)

PROBLEMS EXPERIENCED WITH PLUNGER PUMPS

The following common problems are experienced with plunger pumps:

Definition

Disadvantages

- **Slip** is the leakage of water in a pump due to several reasons:
  - If the strainer is not completely submerged under the water, then air can slip in with the water into the suction pipe, leading to a condition where the pump is pumping both water and air.
  - If the foot valve is not efficient, then on the delivery stroke, water slips back into the sump, leading to a loss in efficiency.
  - If any joints or flanges are loose on the suction column, air will slip in again with the water, resulting in inefficient operation.
  - If the suction valves do not open and close promptly and completely (e.g. due to bad seating, poor springs or grit in the valves) water will slip back into the sump on the delivery stroke.
  - If the internal packing is not good, water will slip into the right chamber when the plunger is travelling to the left and vice versa (refer to Figure 7.8), leading to inefficient operation.

- If the internal packing is not good, water will slip out on the delivery stroke and air will slip in on the suction stroke.

Definition

- **Cavitation** in pumping is the formation of a vacuum space in the cylinder or suction pipe (refer to Figure 7.9).

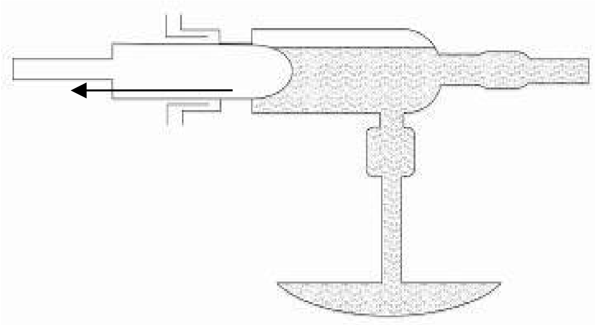


Figure 7.9 Formation of vacuum space (cavitation)

Causes of cavitation and remedies:

- If the suction head is too great, then atmospheric pressure cannot push the column of water up the suction pipe. This leads to the formation of a cavity behind the plunger. This can be eliminated by a flooded suction.
- If the suction pipe diameter is too small, it will not be capable of feeding water into the cylinder at a fast enough rate. This will lead to the formation of a cavity behind the retreating plunger. This is eliminated by making the suction pipe 30% larger than the delivery pipe.
- If the plunger is travelling to fast, on the suction stroke, then it will not give the column of water coming up the suction pipe a chance to 'catch up' with it. This leads to more cavitation. This is prevented by operating the pump at a reasonable speed, which will vary under different conditions.

- **Pump that refuses to deliver water.** The failure of a reciprocating pump to deliver water may be due to the following reasons:
  - Pump is not properly primed
  - Suction head is too high
  - Various mechanical failures
  - Air leakages (past packing etc.)
  - Clogged strainer or strainer does not lie deep enough in the water
  - Delivery valve or foot valve may be damaged

- o Water may be too hot (formation of vapour)
- o Excessive air in water (suction pipe may be too close to a wall – air bubbles).

Definition

- **Water hammer** occurs when there is a sudden change in water flow in a pipeline. In mines this frequently occurs when a valve suddenly closes in a vertical shaft pipeline. Water which is flowing towards the valve will try to continue to flow and this will result in a higher pressure at the upstream side of the valve, which will result in water flowing away from the valve causing a reduction in pressure. Similarly on the downstream side of the valve the water will try to move away from valve leaving a vacuum. This will cause pressure waves to flow through the pipeline.

  As the pressure waves move through the pipe, the fluid compresses and expands, which can result in pressure fluctuations that are significantly greater than the normal working pressure of the pipe and cause damage to the pipe.

QUANTIFYING THE EFFECT OF WATER HAMMER

References

Water hammer is a highly transient and complex phenomenon and is normally analysed using specially developed computer programs which take into account the characteristics of many parameters including the layout of the network, pipes (material, wall thickness and diameter) and opening and closing speeds of valves. A simple analysis (Massey, 1972) allows the maximum pressure rise due to water hammer to be calculated from:

$$\Delta P = \frac{cU}{g} \quad \text{...} \quad (7.9)$$

Where:

ΔP	=	**Maximum** pressure rise due to water hammer (m of water) when the water flow is instantaneously stopped
c	=	Velocity of sound in water taking into account the properties of water and the pipe material
U	=	Water velocity
g	=	Gravitational constant (m/s^2).

Rule

For normal mine steel pipes, c is about 1 400 m/s and the above formulae can be simplified to $\Delta P = 140$ U. The prediction of water hammer pressures is extremely complex and it is recommended that if a mine ventilation engineer is concerned with the possibility of water hammer then he should consult a water hammer specialist.

The common causes of water hammer include the opening or closing of quick-acting valves and pump failures.

WATER HAMMER ON PISTON AND PLUNGER PUMPS

On the delivery stroke of the plunger or piston, water is travelling up the delivery pipe. During the suction stroke some of this water falls back again. On the next delivery stroke therefore, the plunger, when going forward, meets a column of water falling back and hits it with a violent blow, creating the familiar ‘knock’ in the pipe column, called water hammer. Refer to Figure 7.10.

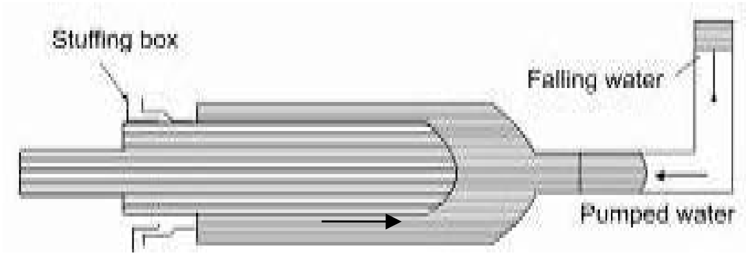


Figure 7.10 Explanation of water hammer

NEUTRALISING WATER HAMMER

Referring to Figure 7.11, it will be noticed that during the delivery stroke of the pump, the water in the air vessel is pushed up, compressing the air, which is trapped in it. On the suction stroke the non-return valve closes and the air under compression continues to push the water up the delivery column, with consequently little interruption in the flow of the water. The air vessel therefore:

- Reduces the effects of water hammer
- Smoothes out the flow of water in the delivery pipes.

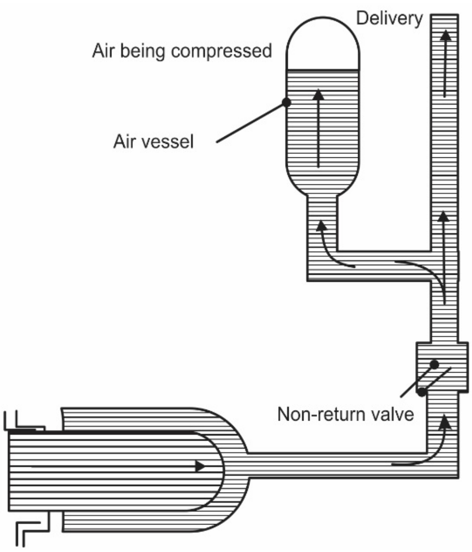


Figure 7.11 Neutralising water hammer

7.6.1.4 Centrifugal Pumps

CONSTRUCTION AND OPERATION

An open or closed wheel with curved blades or vanes, termed the impeller, rotates at a high speed in a pump casing as illustrated in Figure 7.12. The casing is initially filled with the liquid to be pumped, and on setting the pump in motion, the liquid is caught up by the blades. Due to the centrifugal force the liquid is continuously kept moving from the centre to the outer circumference.

The liquid is therefore delivered form the suction pipe into the delivery pipe, while further liquid is drawn into the suction so that the pump delivers continuously. When large quantities of water have to be lifted against low heads, the pump is fitted with impellers of relatively small diameter, but greater lift to provide larger passages for the liquid. Small quantities pumped against high heads require large diameter impellers.

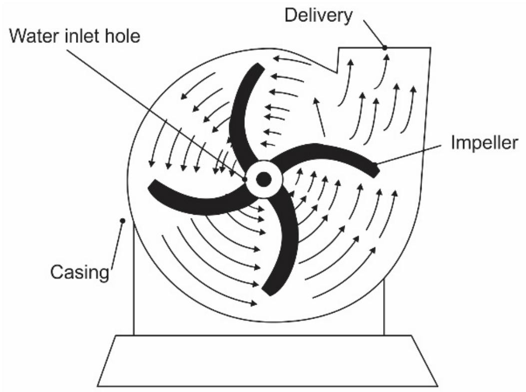


Figure 7.12 Section of a centrifugal pump

PRIMING DEVICES APPLICABLE ON CENTRIFUGAL PUMPS

Before any centrifugal pump (and so also piston and plunger pumps), is started, it must be fully primed, this means that the whole casing and suction pipe must be completely filled with the liquid, which is to be pumped. If this is not done, the pump may be ruined and, of course no liquid will be delivered. Figure 7.13, Figure 7.14 and Figure 7.15 adequately explain the different priming practices applied.

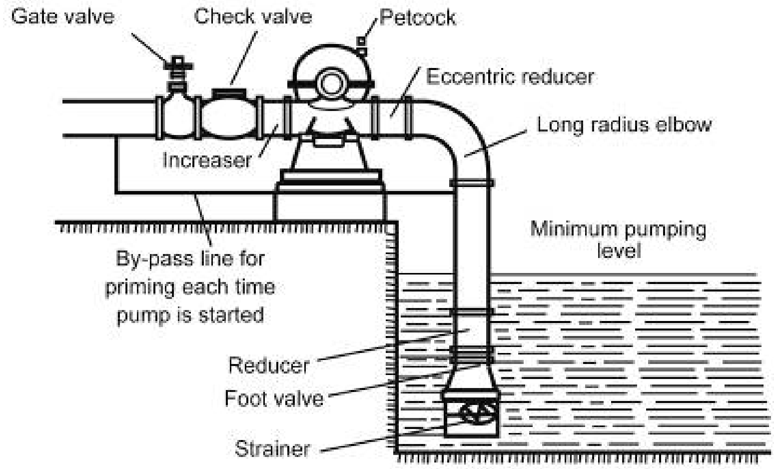


Figure 7.13 Arrangement for foot-valve priming

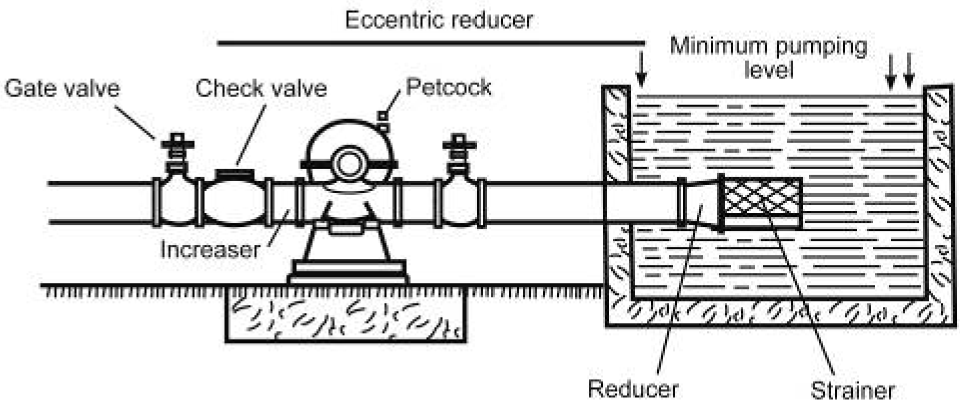


Figure 7.14 Piping arrangement with positive head on suction

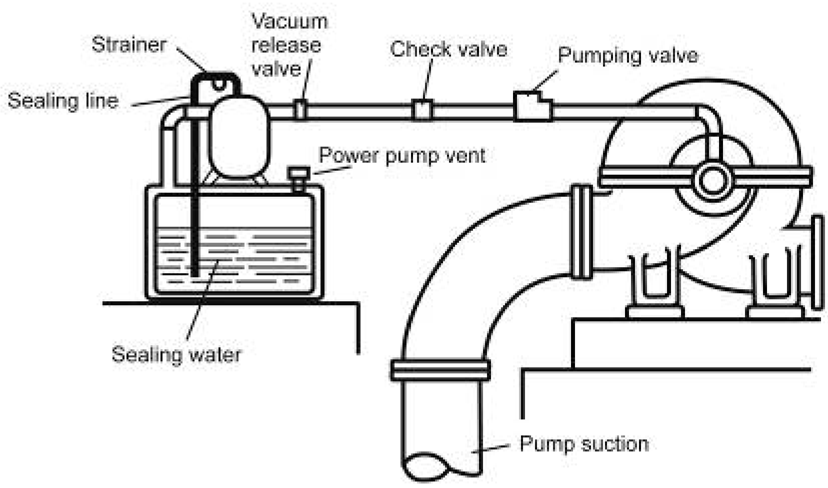


Figure 7.15 Automatic priming arrangement with special primer pump

CAUSES OF CENTRIFUGAL PUMPS FAILING TO DELIVER LIQUID

- Suction head is too large
- Strainer is blocked and liquid cannot therefore flow to pump casing
- Pump is not properly primed
- Air leakages at inlet pipes and packed stuffing boxes
- Pump may be badly worn or some or other mechanical fault
- Strainer not properly submerged under water, thus permitting air to be drawn in with the water
- Speed of pump too low to build up pressure.

Although water flows down a mine shaft by means of gravitation, the water must be pumped out of the mine (or sometimes pumps needs to be introduced in columns to overcome the pressure losses in the system, especially where specific pressures for doing work is required (drilling, cleaning and so forth). Water is also pumped from various levels to various sections, in refrigeration water circuits and to boost water pressure where required.

NB

It is important to know the pump requirements for underground situations. Centrifugal pumps are the most commonly used underground. The various characteristics of centrifugal pumps, that is pressure curves, flow rates and other pump related issues will now be dealt with in detail. Pumps and fans are very similar and the details pertaining to fans were discussed in detail in Chapter 4.

7.6.1.5 Comparison of Plunger and Piston Pumps versus Centrifugal Pumps

Plunger and Piston Pumps	
Advantages	**Disadvantages**
More economical for capacities of 1 million litres per day.	Solid particles in the valve seats create a possibility of blockage.
Wear and tear of the components is lower than centrifugal pumps. Thus application in pumping dirty water and mud is possible.	Water lash damages pipes - joints that leak and broken columns.
Applied in areas where high pressure and fluctuating outputs are required.	Plunger and piston pumps need constant supervision.

Centrifugal Pumps	
Advantages	**Disadvantages**
Cheaper for high capacities and large height difference applications.	Lower efficiencies. Small pumps 50% and large pumps 72%.
Occupies less space – important in bad ground conditions.	Centrifugal pumps are designed for specific capacities and height difference. The efficiency of these pumps are reduced with changes in working conditions.
Simple design – no valves and reciprocating parts. If the water is purified and neutralised the maintenance is low.	Clean water is essential – mud or gravel, even in small quantities are extremely corrosive to the impeller and casing due to the high turning speeds, unless the pump was designed to be highly corrosion resistant.
Constant discharge at constant speed – no air vessel is necessary due to the absence of the water lash effect.	For maximum efficiency the edges of the guide vanes of a turbine pump must be sharp. These edges are damaged at a high rate in acidic conditions.
Water speed can be up to 2.4 m/s in contrast to the 1.5 m/s maximum of Plunger and piston pumps.	
The high rotation speeds of centrifugal pumps make direct connection between the pump and the motor possible and gearboxes are not essential.	
If discharge pipes get blocked or if the main discharge valve is closed there is no pressure build up.	
A blockage in the intake pipes is not dangerous. The impeller keeps on turning with less power consumption.	
Low initial costs.	
Low maintenance cost, as all motion is rotary.	
No sensitive parts such as the moving valves of reciprocating pumps (centrifugal pumps have no valves).	
Small and light foundations.	
Motor for centrifugal pump is smaller than motor for reciprocating pumps due to higher speed of centrifugal pump.	
Little if any vibration.	
Discharge of liquid is continuous and without shock.	
The flow of water can be controlled automatically from full flow to no flow, without shutting down the pump or damage to the pump or pipes.	

7.7 Various Pump Layouts and Configurations

With the placement of pumps there are various parameters and descriptions that need to be considered. Various definitions also applies and for the purpose of calculations a few terminologies have to be considered (some have been mentioned before).

7.7.1 Suction Head

Definition

By **suction head** or lift is meant the vacuum, which the pump creates in order to draw water into the suction side of the pump. The water is raised by means of the atmospheric pressure, which varies from altitude and weather conditions (as shown by a barometer). When the plunger or piston has completed a suction stroke, a vacuum is formed as shown in Figure 7.16. This normally referred to as depressed suction (negative). This will tend to 'suck' the water up into the suction pipe.

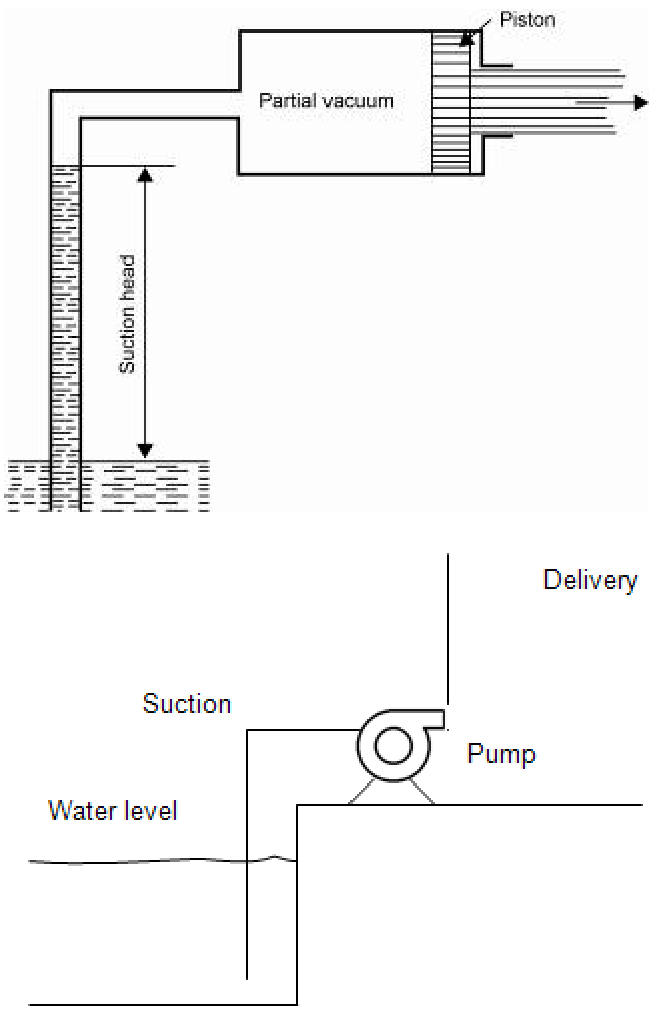


Figure 7.16 Depressed or negative suction

7.7.2 Flooded Suction

Definition

Trouble in the operation of a plunger or piston pump is more likely to occur from conditions arising on the suction side than from any other cause. This can be eliminated by arranging the water to flow down into the pump suction that is by providing a **flooded suction**. This is shown in Figure 7.17.

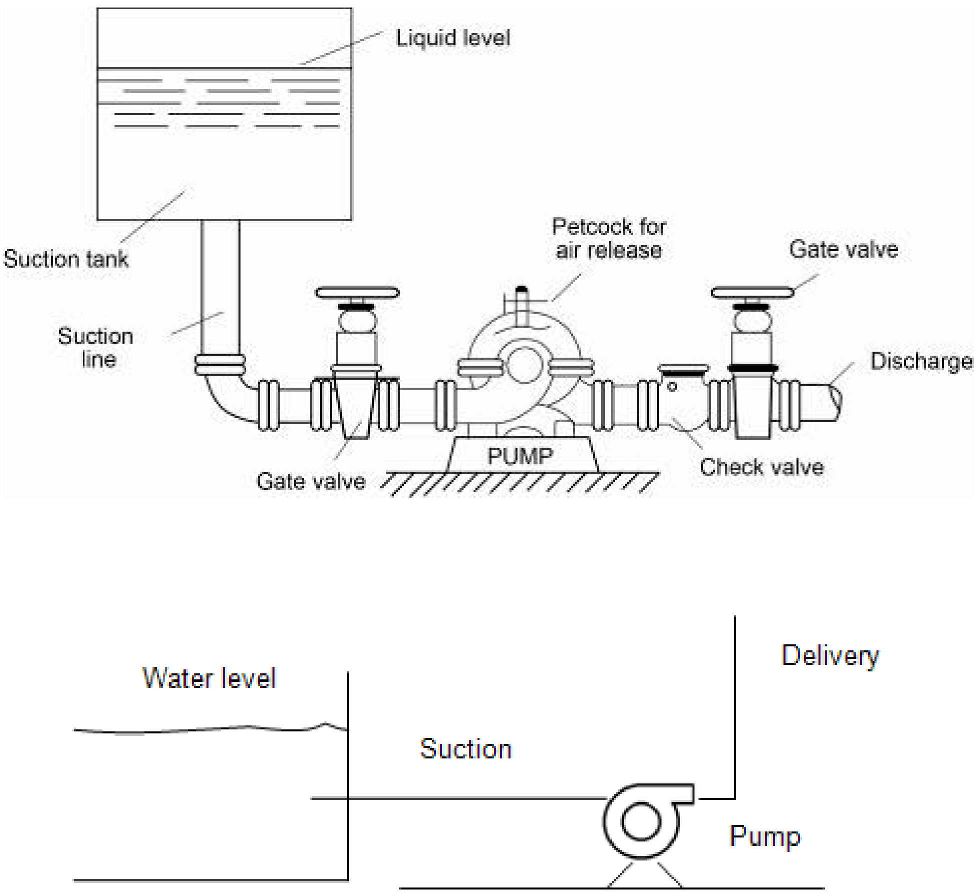


Figure 7.17 Flooded or positive suction

7.7.3 Static Head of Pump

Definition

This is the vertical height measured from the water surface to the point where the water is flowing out of the delivery pipe (Figure 7.18).

$$H_{(st)atic} = H_{(s)uction} + H_{(d)elivery}$$

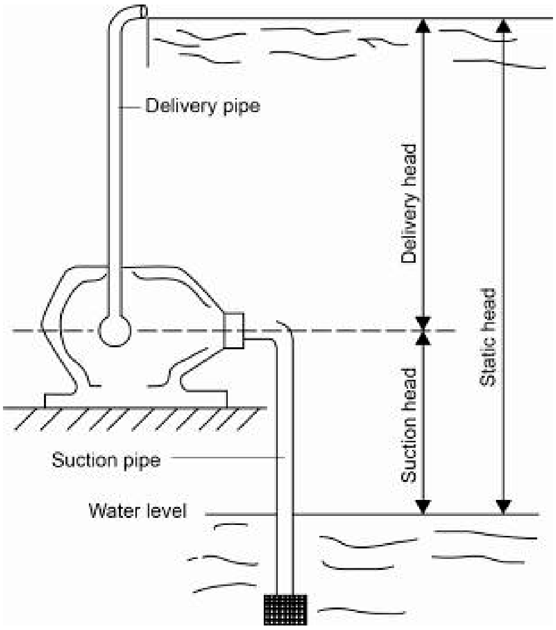


Figure 7.18 Static head(s) applicable to pumps

7.7.4 Pump Placements in Serie

When pumps are placed in serie it increases the head that the pump can handle. In other words the pumping pressure is increased. Figure 7.19 show such a configuration schematically.

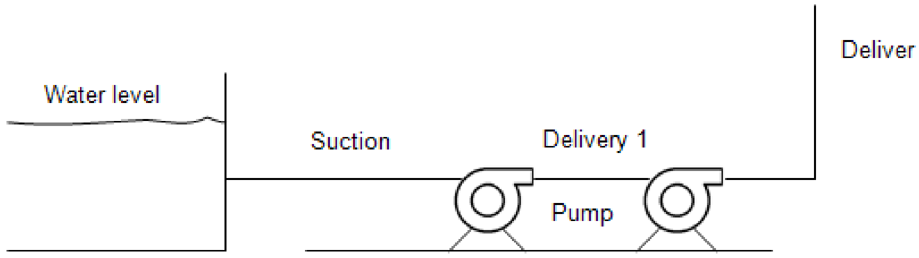


Figure 7.19 Pumps in serie

7.7.5 Pump Placements in Parallel

When pumps are placed in parallel it increases the volume flow delivered by the two pumps combined. Figure 7.20 show such a configuration schematically.

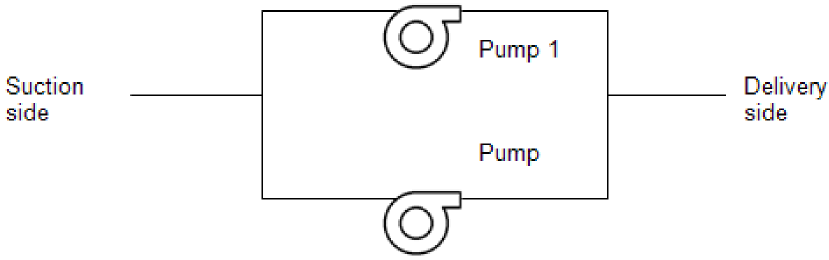


Figure 7.20 Pumps in parallel

7.7.6 Pump Characteristic Curves

7.7.6.1 Total Manometric Head (H_T)

Definition

The head shown in the characteristic curves of pumps is the total or **manometric head** and is defined as the increase in the pressure that takes place between the inlet (suction) side of the pump, and the discharge side, and is expressed in metres (liquid) column. This head includes the frictional losses in the column. Figure 7.21 indicates this head.

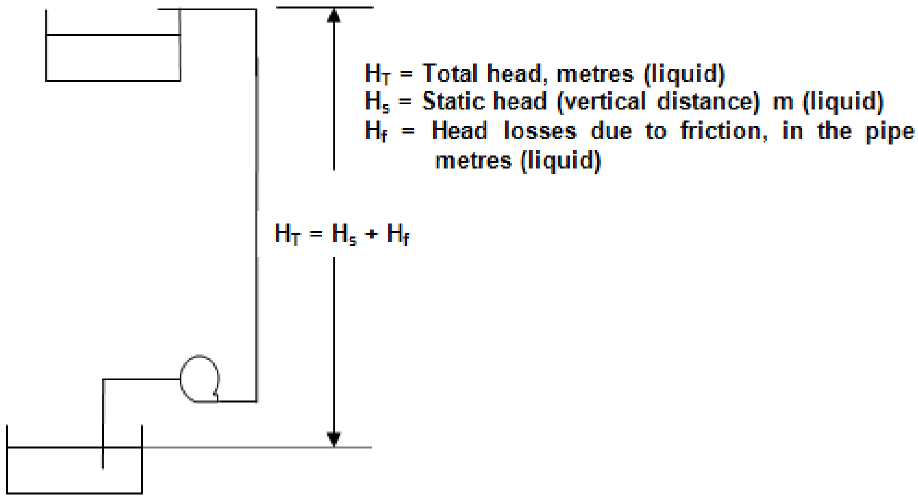


Figure 7.21 Diagrammatic layout of an underground pump layout

NB

What is important to realise is that the **total head** (**manometric head H_T)** for a pump running at constant speed **does not change with the specific gravity (SG) of the liquid being pumped**. In other words, a pump lifting water for say ‘x’ metres will also lift mercury for ‘x’ metres.

Definition

It brings us to the concept of specific gravity (SG). The **specific gravity** of a liquid is the ratio in density compared the density of water and can be expressed as follows:

$$SG_{liquid} = \frac{\rho_{liquid}}{\rho_{water}} \quad \text{.......} \quad (7.10)$$

Rule

However, the **pressure reading** over the pump in kPa will not remain constant, but will vary according to the **specific gravity** of the liquid being pumped. Pump characteristic curves are therefore drawn in metres head versus volumetric flow, q in m^3/s or litres/second

7.7.6.2 Static Head (H_s)

Definition

The **static head** (as mentioned before) on the other hand is defined as the difference, in metres (liquid), between two levels of the liquid being pumped (that is **without taking frictional** losses into account).

7.7.6.3 System Curves

The system curve looks similar to that of a system resistance curve for airflow, except that once again it is plotted as metres head (loss) versus volumetric flow (m^3/s or litres/second).

The drawing of the system resistance curve for a specific pipe size column is done in a similar way as was done for air ways. It is done as follows:

- From the pipe friction chart (Figure 7.2) the pressure head loss in m / 100 m must be read off for different flow rates for a specific pipe diameter
- The length of the pipe column then needs to be multiplied for each pressure head loss read off from the chart
- Plot the values obtained against the flow rate that it relates to
- For horizontal pipes the graph is plotted from origin (as for air ways) as shown in Figure 7.23.
- To plot the final system curve, the static head is added to the system resistance (H_f) incurred in the system. This is shown in Figure 7.22.

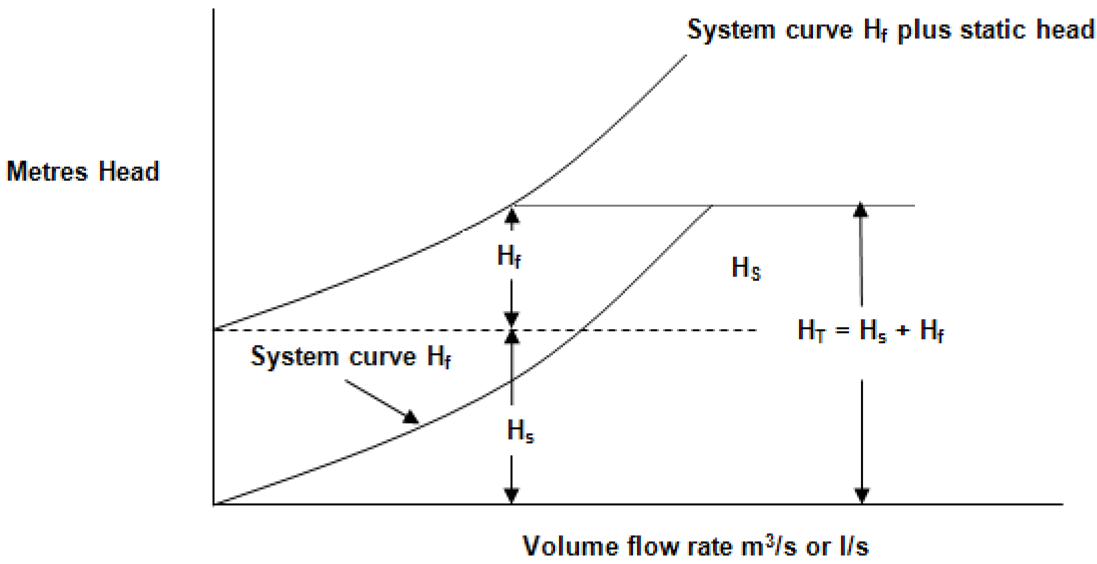


Figure 7.22 Graph showing static and friction head

Where the final system curve (also manometric or total curve) intersects the pump operating curve, the pump operating point can be read off as shown in Figure 7.23.

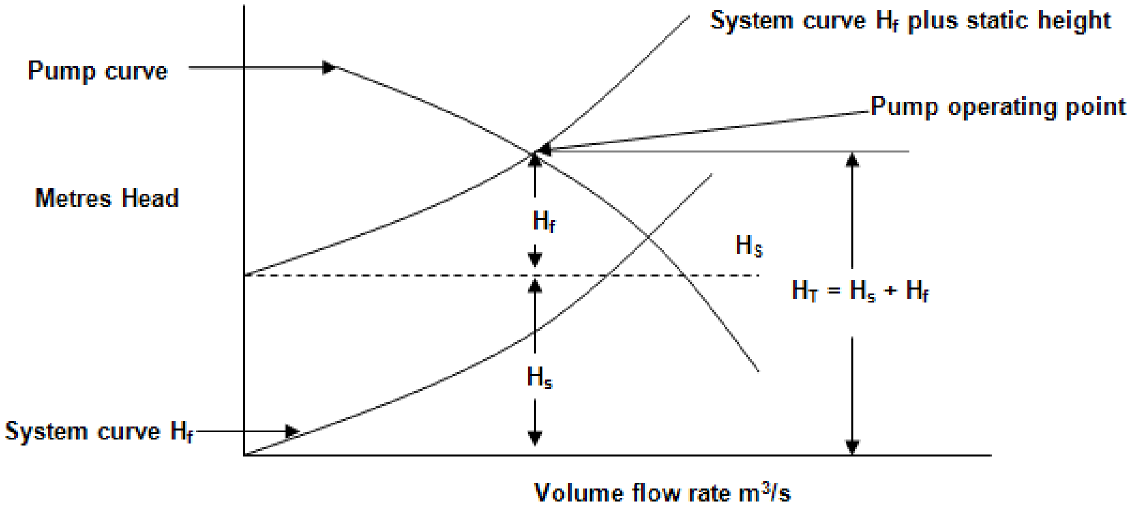


Figure 7.23 Graphic representation of pump operating point

The pump head can now also be converted to kPa as was done before.

7.7.6.4 Pump Characteristic Curves Summary

Pumps are an integral part of any chilled water distribution system. The performance of any pump operating at a given speed is represented by four characteristic curves.

They are as follows (and graphical detail shown in Figure 7.24):

- **Curve 1: Head versus flow curve:** For any pump a family of head versus flow curves can be drawn which show the relationship between head and flow for a given size impeller. Generally the head decreases as the flow increases. The operating point is determined as the intersection between the system resistance curve and the head flow curve. For a given size of impeller the highest head occurs when there is zero flow and is known as the **shut-off head**. The size of the impeller can be checked by briefly operating the pump against a closed head and measuring the head at zero flow. The size of the impeller can be determined from the shut-off head.
- **Curve 2: Efficiency versus flow curve:** This is shown on the head-flow curve. In general the efficiency increases as flow increases to peak efficiency after which, as the flow increases further, the efficiency reduces. The efficiency (%) is defined as the output or work performed divided by the input work.

  The output work is the flow rate (ℓ/s) multiplied by the head (normally expressed in MPa, where for ventilation air pressure it is expressed as Pa and/or kPa). This output work is similar to that of the air power expression as is used in airflow power calculations.

The input work is absorbed power of the pump motor and can be expressed in kW and/or MW. The pump should be selected at close to its maximum efficiency. Ideally the design operating point should be selected so that if the friction head (due to changes in circuit resistance) and subsequent total head increases, then the efficiency increases.

- **Curve 3: Required Nett Positive Suction Head (NPSH) versus flow rate curve** Pump manufacturers provide curves which give the required NPSH to avoid cavitation in the pump versus flow rate. In general the NPSH flow rate curve indicates an increase in NPSH as the flow increases. The available NPSH of a system is calculated as follows:

Available NPSH = BP- SH - Friction – VP_s (kPa)

Where:

BP = Barometric pressure (kPa)

SH = Static height difference between the surface of the water and the centre of the pump (kPa, 1 m of water is approximately 10 kPa)

Friction = Pipe friction in the suction piping (kPa)

VP_s = Saturated vapour pressure corresponding to the temperature of the water (kPa).

NB

An important point to note is that the available NPSH must be greater than the required NPSH given by the pump manufacturer.

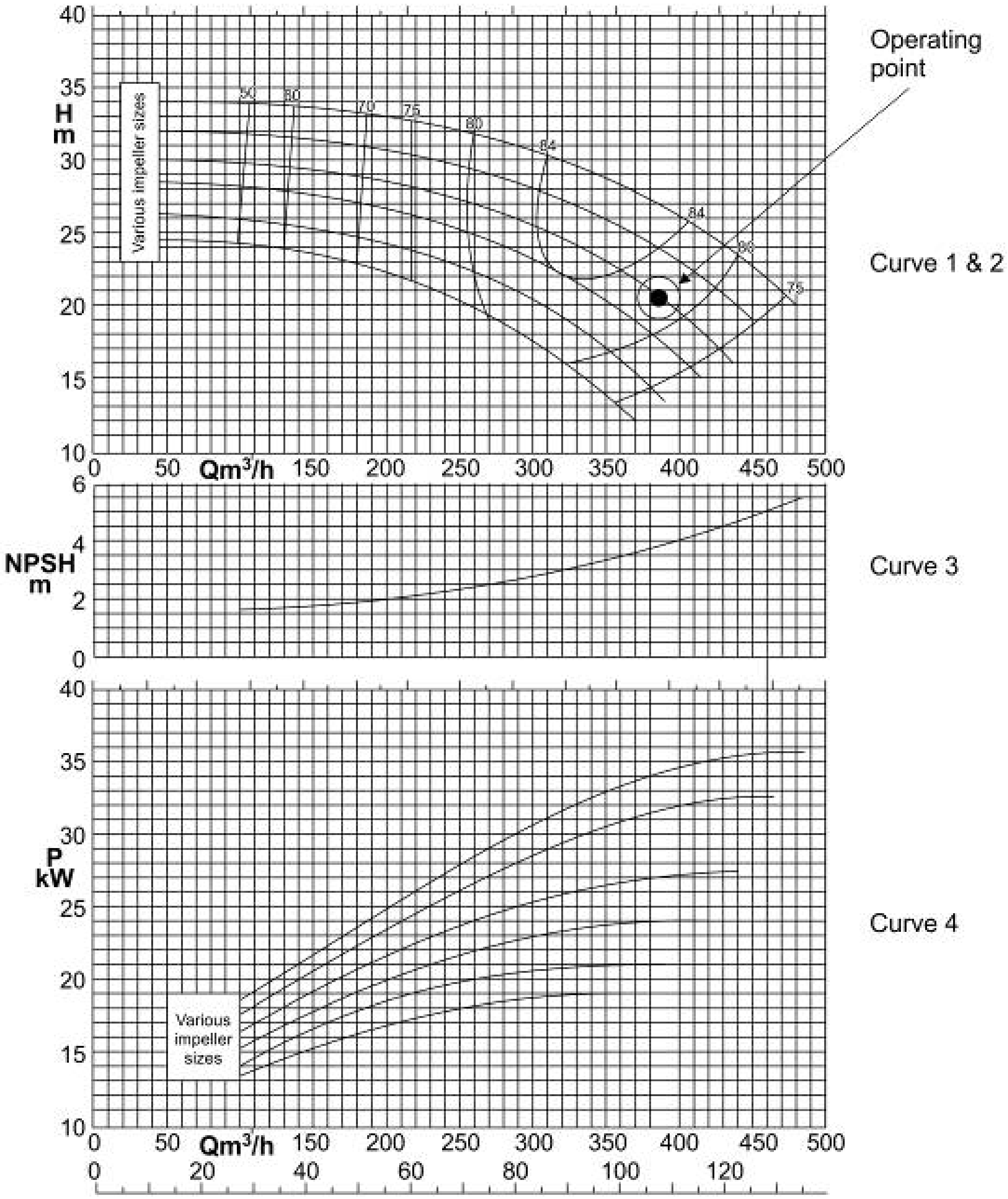


Figure 7.24 Typical pump characteristics curves

- **Curve 4: Power versus flow curve:** Normally, the power absorbed by the pump motor increases as the flow rate increases. Obviously if a pipe breaks then the flow rate from a pump will increase and it is normal practice to size the electric motor to cater for the maximum flow. It is also important to note that the exact shape of each curve varies with pump design.

NB

7.7.6.5 Pump NPSH (Net Positive Suction Head) Calculations

The net positive suction head is calculated as the difference between the fluid total pressure[1] on the suction side of a pump (evaluated at the pump impeller centre line) and the liquid vapour pressure head[2]. Manufacturers test their pumps directly by detecting the onset of cavitation to determine a value for the NPSH. If the pressure at the suction to the pump reduces to below this value then cavitation will occur.

7.8 Determination of Pump Power Requirements

The following applies to pumping calculations:

- Hydrostatic Pressure

$$P_{st} = \frac{\rho g H_{st}}{1000}$$

Where:

P_{st} = Hydrostatic pressure in kPa
g = gravitational acceleration as 9.79 m/s^2
H_{st} = static liquid head in metres (m)
ρ = density of fluid being pump in kg/m^3
(water = 1 000 kg/m^3)

- Total Head (Manometric Head)

H_T = $H_{st} + H_f$

Where:

H_T = Total head in metres (m)
H_{st} = static liquid head in metres (m)
H = friction head in metres (m)

- Water Power (Similar to Air Power used earlier)

$$P_w(\text{water power}) = \frac{q \times \rho \times g \times H_T}{1000_{(\text{convert l/s to m}^3\text{/s})} \times 1000_{(\text{convert Pa to kPa})}} \text{ in kW}$$

Where:

q = quantity of water in litres/second
g = gravitational acceleration as 9.79 m/s^2
H_T = Total head in metres (m).

NB

The water power Equation applies when clear water is pumped, when mud is pumped the equation for the water power must be multiplied by the specific gravity (SG) of the water with solids.

- Pump Shaft Input Power

The power requirement at the shaft of a pump (i.e. pump input power, take note, not pump motor input power), is calculated by the following formula:

$$\text{pump shaft input power (water power)}_{\text{in}} = \frac{q \times \rho \times g \times H_T}{\eta_{\text{pump}} \times 1000_{(\text{convert l/s to m}^3\text{/s})} \times 1000_{(\text{convert Pa to kPa})}}$$

Where:

q	=	volume flow rate in litres/second
ρ	=	density of the fluid being pumped in kg/m^3
g	=	gravitational acceleration m/s^2
H_T	=	Total head in metres
η_{pump}	=	efficiency of the pump.

NB

Note: The pump efficiency referred to above is the total or manometric efficiency of the pump and can be determined by the following formula:

$$\eta_{\text{pump}} = \frac{\text{pump output power}}{\text{** pump shaft input power}}$$

**Note: This is not the pump motor input power (equal to the pump motor output power); the pump motor input power is also dependent on the motor efficiency.

- Pump Motor Input Power

The pump motor input power must also take the efficiency of the pump motor into consideration and therefore can be written as follows:

$$\text{pump motor input power} = \frac{q \times \rho \times g \times H_T}{\eta_{\text{pump}} \times \eta_{\text{pump motor}} \times 1000_{(\text{convert l/s to m}^3\text{/s})} \times 1000_{(\text{convert Pa to kPa})}}$$

Where:

$\eta_{\text{pump motor}}$ = efficiency of the pump motor

It is quite often required to estimate the time required to remove an amount of water as a result of flooding underground.

For this to be done the following needs to be known:

- The amount of water to be removed (pumped) in litres or kilolitres
- The capacity of the pump or pumps to be used in litres/second or kilolitres/second
- The capacity of any other method of removing the water (that is hoist etc).

WORKED EXAMPLE 11

The following information is available regarding a pump installed in an underground water circuit:

Total friction losses in the circuit	15 m
Pump static head	42.5 m
Pump volumetric flow rate	172 litres/second
Pump efficiency	68%

Determine:

(i) The pump total head
(ii) Pump shaft power
(iii) Pump total head expressed in kPa.

Answer

(i) $H_T = H_S + H_f$

$= 42.5 + 15$

$= 57.5$ m

(ii)

$$\text{Pump shaft input power} = \frac{q \times H_T \times g \times \rho_w \times SG}{1000 \times \eta_w}$$

(SG for water is 1)

$$= \frac{172 \times 57.5 \times 9.79 \times 1000 \times 1}{1000_{(1)} \times 1000_{(2)} \times 0.68}$$

(NOTE: divide by $1000_{(1)}$ to convert Pa to kPa)

$= 142.4$ kW

(NOTE: divide by $1000_{(2)}$ to convert ℓ/s to m^3/s)

(iii) To convert metres head to kPa:

$$p = \frac{\rho g h}{1000}$$

$$= \frac{1000 \times 9.79 \times 57.5}{1000}$$

$= 562.9$ kPa

NB

Note: The pump efficiency referred to above is the total or manometric efficiency of the pump and can be determined by the following formula (the equation very similar to that used for fans):

$$\eta_{pump} = \frac{\text{pump output power}}{^{**}\text{pump shaft input power}} \times 100\% \qquad \text{.................. (7.11)}$$

**This is not the pump motor input power; the pump motor input power is also dependent on the motor efficiency.

WORKED EXAMPLE 12

Determine the operating point for a pump with pump curve shown in Figure 7.25 and to be installed in a system which will consist of a 150 mm diameter, reasonably new steel pipe (ignore losses for bends). The total length of the column is 100 m and the total vertical distance is 75 m. Figure 7.26 shows a graphical representation of the pump column (it must be noted that in this case the static head is taken as from the level of the dam underground, as we assume that the dam will always be full and the level of water in the dam will assist the pump in terms of the height of the water in the dam).

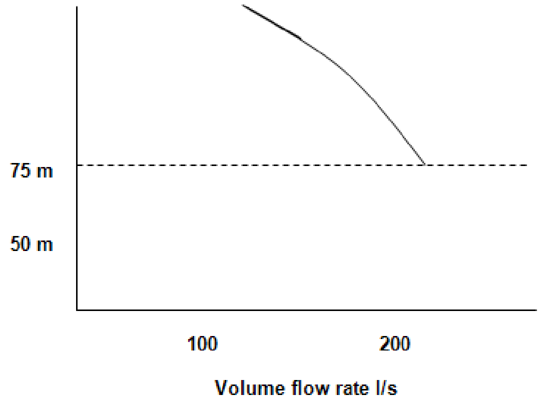


Figure 7.25 Pump curve shown with 75 m static height

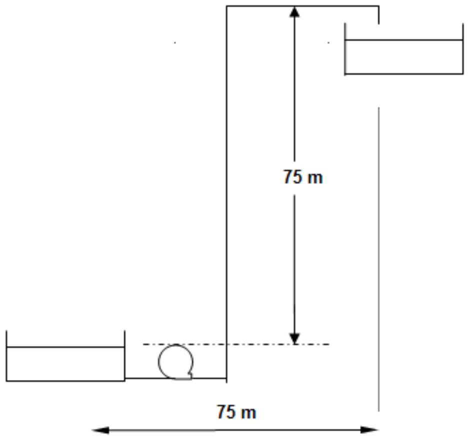


Figure 7.26 Graphical representation of pump column layout

Answer

To determine the head loss for the column, you must use the friction chart given previously (Figure 7.2).

From this graph for a 150 mm pipe you can read off the head losses for various water flow rates:

10 litres/second = 0.26 m / 100 m

0 litres/second = 0.8 m / 100 m

and so forth the other pressure loss head for other flow rates as mentioned before (taken over the full length of the pipe column).

Once the friction losses have been determined, you can plot the friction curve H_f below (curve A), and add it to the pump static head required, H_S (curve B). Where curve B intersects the pump characteristic curve (H_T), the system operating point can be read off. This is shown in Figure 7.27.

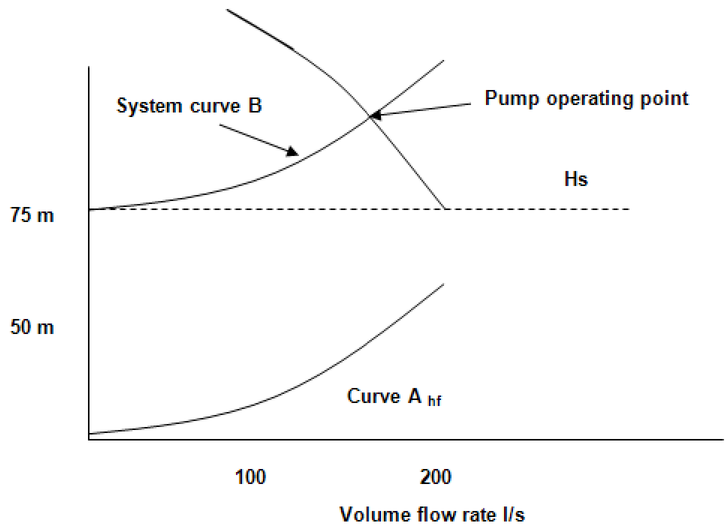


Figure 7.27 Determining pump operating point

Read off the answer from the graph, the pump operating point is 115 litres/second at 100 m head.

It was previously mentioned that the various shock losses, which occur, could be read off from the nomogram.

Another type of shock loss can now also be mentioned and that is the inlet and outlet shock losses, and can be determined as follows:

shock losses = in Pa

With the symbols as previously mentioned (U = velocity of water through the pipe).

NB

Note: The outlet shock losses are only applicable where the discharge pipe discharges into a container or dam. In practice these values must always be added to the H_f term when plotting the system curve.

7.9 Other Water Pumping Related Issues

The intake pipe column must be equipped with a sieve or a grid to keep coarse objects and particles from damaging pumps. This insert must be airtight. A water lash valve is essential in piston plunger pumps to absorb the water lash that can cause pipe columns to burst.

Another protection device used with piston plunger pumps is a relieve valve. If the discharge valve stays closed when the pump is started, the relieve valve is opened on the discharge side of the pump and the water is circulated to the intake side of the pump. The relieve valve is designed only to open when the pressure are above normal (if this does not happen the pipes will burst).

Rule

The following figures are approximate maximum water speeds applicable for different pumps, but as a general rule or assumption the economic water speed associated with pumping can be taken as 2 m/s (unless specifically given or requested as mentioned below):

a) Piston and plunger pumps:	1.2 m/s - intake column
	1.5 m/s - discharge column
b) Centrifugal pumps:	1.8 m/s - intake column
	2.4 m/s - discharge column

7.9.1 Corrosion and Erosion in Pumps and Water Pipes

This is attributed to acid in the water, air, CO_2 and muddy water that contains fine gravel. CaO is used to neutralise mine service water, $CaO + H_2O = Ca\,(OH)_2$,this then reacts with the acid, $Ca(OH)_2 + H_2SO_4 = CaSO_4 + 2H_2O$. The $CaSO_4$ (gypsum) precipitates in the pipes together with other impurities and forms a hard core inside the pipes. $NaCO_3$ prevents this core forming, but is quite expensive. Special chemicals like Aquoil and Iron sulphate that have the same function are currently used when CaO is used for neutralising water.

If the water pressure suddenly falls the O_2 and CO_2 in solution forms bubbles that are bad for pipes and pumps. These low-pressure zones form at bends, leakages or where water free falls in parts of the column. A general cause or bubble formation is a negative pressure (sucking) pipe column that is too long and thin.

Pumps are effective protected against bubble formation by placing pump chambers in high-pressure zones like below water dams.

NB

Ordinary fine silt mud is not detrimental at low water speeds, but this statement is only valid when it contains no coarse gravel. Gravel causes a freshly eroded surface that is corroded by the acid in the water.

7.9.2 Clearing and Purifying Impurities of Mine Water

NB

Due to the impurities contained in underground water, the handling and direction thereof is of high importance. The following groups of impurities are contained in water:

a) **Undissolved solid substances**:

- o Particles, which sink – fine ore and waste and crystallized salts
- o Floating particles – very fine ore and waste (-100 µm), wood splinters, organic material and finely crystallized salts.

b) **Dissolved substances**:

- o Chlorine, copper, iron, manganese and zinc.
- o Sulphates – whole series
- o Carbonates – whole series
- o Poisonous substances – Ar, Pb, Na and cyanide.

c) **pH – Defined as**:

7 Acidic

+ 7 Alkaline

7 Neutral

d) **Bacteria**: Whole series with the most prominent being ecolli (stomach diseases).

Disadvantages

The disadvantages of impurities are:

- o Erosion
- o Fouling
- o Chemical corrosion.

CLARIFYING

!

Before water is pumped out of the mine, mud and silt is removed from it. If this is not done it will cause major damage to pumps as the muddy water is highly abrasive. Mud pumping is done with mud pumps, specially designed for this purpose. The drains in the tunnels underground serve as the primary catchments for mud and silt, which is loaded out from the drains, left to dry and loaded into hoppers.

The runoff water is still murky and not suitable for re-use before it is clarified. Clarifying of mine water is done with the help of flocculants in underground settling dams excavated in the rock mass.

NB

The most important designs are of the:

- Square vertical type
- Cylindrical
- Conical type
- V-formed elongated type.

Figure 7.28 shows a functional diagram of a cylindrical conical type of settling dam.

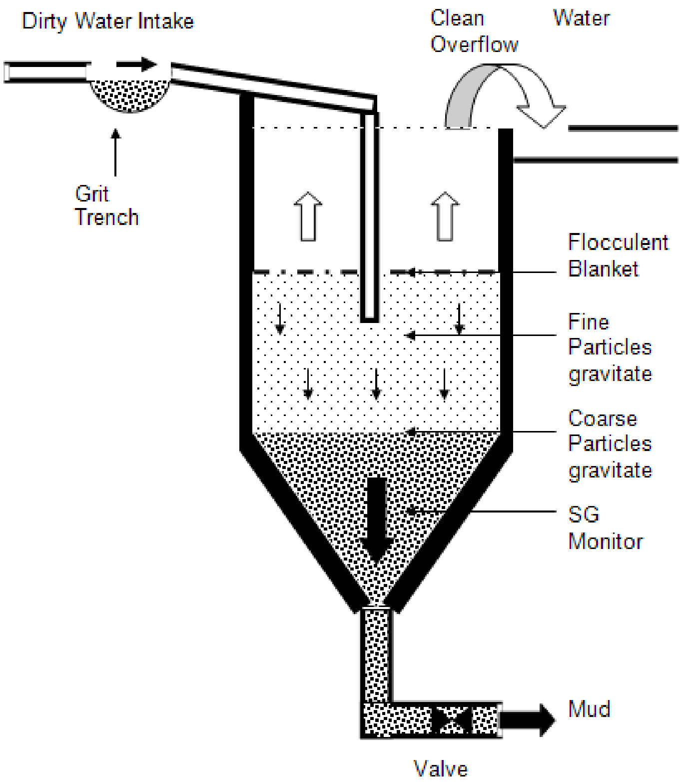


Figure 7.28 Functional diagram of a conical settling dam.

Water purification takes place through the following steps:

- Adding chemicals for pH control, flocculants for accelerated settling and chlorine for further purification.
- Water passes over a grit trench where the coarse particles settle out and is removed with a scraper to raise the capacity and efficiency of the conical settling dam.
- Most of the dust particles settle out and some fine particles rise with the water due to turbulence.
- At a certain height the upward flow and gravitation is in equilibrium. The dust particles collide and form clods that are heavy enough to settle.
- Clean water overflows to the clean water dam.
- The SG of the settled mud is monitored and the underflow is controlled at a SG of ± 1.2 (12% solids).

- For effective settling the upward flow velocity of the water is critical. This factor determines the volume of water of water that can be handled, the number and the size of the dams needed. The cost per dam is quite expense and forms an important budget expense for any new mine.

The following are typical dimensions of settling dams:

- Diameter 6 - 10 m
- Total height 30 - 60 m
- Settling height 20 - 40 m
- Flow velocity 1 - 2 mm/s

Not all the water goes through the shaft bottom settling dams; sometimes a part of the fissure or dolomite water is sent through a quick purifying plant and then handled. The mud usually has a high gold content and it is pumped to the concentrator plant or sent though a filter press and the dry cakes are tipped in the shaft ore-passes.

7.10 Water Flow down a Pipe and Energy Recovery

The refrigeration system of many mines is on surface and the chilled water is transported down the mine by insulated pipes. As the water flows down the shaft, we find a similar process to that of auto compression. This is the conversion of potential energy to enthalpy, such that we get the following Equation:

$$H_2 - H_1 = g\,(Z_1 - Z_2) \qquad \text{(7.12)}$$

If all the potential energy is released as friction, such as the case of water being distributed in a cascading system or if the water flows at terminal velocity in the pipe to dam where it is to be stored, the final pressure in the water will be the same as that at the top of the shaft, namely atmospheric pressure.

The temperature increase of the water was also shown as being 2.34ºC/km if g is 9.79 m/s^2 (test the difference in temperature increase by use of Joule Thompson concept).

Disadvantage

A disadvantage of the increase in temperature of the water down the shaft is the fact that the ability to absorb heat is decreased. This becomes a problem if we want to transport water down shafts of more than 2 km. To eliminate this effect, energy recovery systems are used. Not only is 75% of the potential energy regained, but also a decrease in the increase of temperature of water. The smaller increase in temperature Δt is calculated as follows:

$$\Delta t = \frac{(1-\eta) \times g \times (Z_2 - Z_1)}{c_{p\ water}} \qquad \text{(7.13)}$$

Here the symbol Z is used once for the height difference to avoid confusion with H which in this case is referring to the Entalphy and where η is the overall affectivity of the energy recovery turbine and pipes. Pelton turbines are often used for this purpose and have a hydraulic efficiency above 80% for a wide range of flow rates.

If one considers the Equation that relates to potential energy with regards height then it shows that:

Potential energy = mgh in Joule

Where:

m	=	Mass of the body/water/rock (kg)
g	=	Gravitational acceleration (m/s^2)
h or Z	=	Height of falling

If however the mass (m) is expressed as a flow rate, such as kg/s or litres/second (and we also assume that 1 litre = 1 kg), then the unit for the Equation changes from Joule to Joule/second which is Watt.

NB

This is a very important aspect to remember in calculations relating to turbines.

WORKED EXAMPLE 13

A shaft hoists 200 000 tons of ore per month and the average service water use is 100 ℓ/s. This corresponds with a water usage of approximately 1 ton water per ton rock mined. It is assumed that development will take place between 2 000 and 3 000 m. The main dam that will supply water to the mine is at 2 000 m. The used water must be pumped 3 000 m to surface at an average efficiency of 60%. It is practical to use two energy recovery turbines that together have an overall efficiency of 76% for 2 000 m. Assume that the water is chilled to 4°C on surface and after use in the workings is pumped out of the mine at 28°C.

Determine the following:

(i) Average pump power

(ii) Average power recovery

(iii) Net pump power required

(iv) Increase in temperature

(v) Increase in water temperature with energy recovery

(vi) Heat removed from the mine without energy recovery

(vii) Heat removed from the mine with energy recovery.

Answer

It makes sense to first make a diagrammatical sketch of the pipe configuration as shown in Figure 7.29.

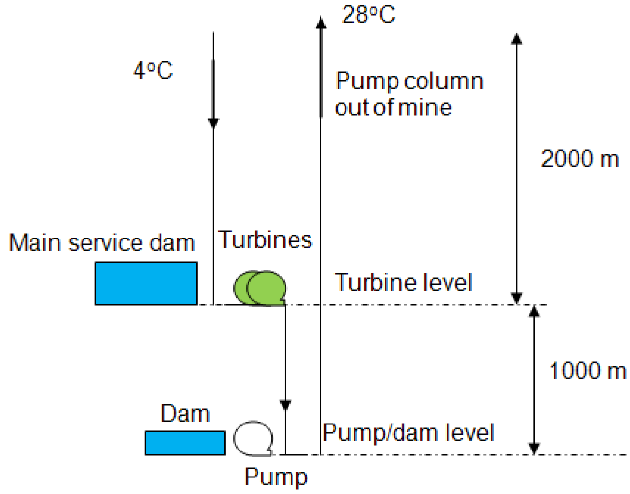


Figure 7.29 Pipe, dam and turbine configuration in shaft

(i) Average pump power:

$$= \frac{m_{\ell/s} \times g \times \Delta Z}{1\,000 \times \eta}$$

$$= \frac{100 \times 9.79 \times 3\,000}{1\,000 \times 0.6}$$

$$= 4\,900 \text{ kW or } 4.9 \text{ MW}$$

NB

Note that 1litre/second = 1kg/s

(ii) Average energy recovery:

$$= \frac{m_{\ell/s}}{1\,000} \times g \times \Delta Z \times \eta_{turbine}$$

$$= \frac{100}{1000} \times 9.79 \times 2000 \times 0.76$$

$$= 1\,490 \text{ kW or } 1.49 \text{ MW}$$

(iii) Net pump power required:

= 'i' – 'ii' = 4900 – 1490

= 3 410 kW

(iv) Increase in temperature (Joule-Thompson):

$$= \Delta Z \times \frac{^{o}C}{km}$$

$$= 3 \times 2.34$$

$$= 7.02^{o}C$$

(v) Increase water temperature with energy recovery:

$$= \eta \times \Delta Z \times \frac{^{o}C}{km}$$

$= 0.76 \times 2 \times 2.34$

$= 3.56\ ^{o}C$

The nett increase in water temperature due to the Joules Thompson effect can be calculated as follows:

The nett increase in temperature = iv' – 'v'

= $7.02\ ^{o}C - 3.56\ ^{o}C$

= $3.46^{o}C$

The assumption must be made that the rate at which water is pumped out of the mine is equal to that of the supply. We are interested in the effective heat that is removed from the mine.

The result from the energy recovery system is that the amount of heat that can be removed from the mine is greater, thus meaning that there is more cool water available for cooling. We can ask ourselves what the temperature increase would be in the system if no energy recovery system were used or what it would be if one were used.

(vi) Without Energy Recovery

Temperature increase can be determined as follows:

(Temp water pumped out of the mine **minus** chilled water temperature **minus** temperature increase as a result of Joule-Thompson effect))

= 28 - 4 - 7.02

= $16.98^{o}C$ say $17^{o}C$

Total heat removed = $m \times C_{pw} \times t$

= $100 \times 4.187 \times 17$

= 7 118 kW or 7.118 MW

(vii) With energy recovery

The temperature increase can be determined as follows:

(Temperature of water pumped out of the mine minus chilled water temperature minus temperature increase as a result of Joule-Thompson effect (**energy recovery reduces temperature increase**))

= 28 - 4 – 3.46

= $20.54^{o}C$

Total heat removed = $m \times C_{pw} \times t$

= $100_{litres/sec} \times 4.187 \times 20.54$

= 8 600 kW or 8.6 MW

By incorporating the energy recovery system, a larger amount of heat can be removed. Further; the increase in heat removal is equal to the energy recovery of the water that flows in the pipe.

WORKED EXAMPLE 14

Water flows down a shaft column for a distance of 1 390 m. The temperature of the water entering the column is 8.3ºC. A turbine, which is 72% efficient, is installed at the bottom of this column. Ignoring all other heat transfer taking place in the column, what will the temperature of the water be at the bottom of this column.

Answer

The expected temperature increase in the water due to the Joule Thompson effect can be calculated as follows:

If the pipe referred to was closed off at the bottom, the water pressure would have been:

$$
\begin{aligned}
p &= \rho g h \\
&= 1\,000 \times 9.79 \times 1\,390 \\
&= 1\,3\,60\,8\,100 \text{ Pa}
\end{aligned}
$$

As this pressure is dissipated (used up) when the water is discharged into the turbine, the temperature difference of the water at the turbine can be calculated as follows:

$$
\begin{aligned}
\Delta T &= \mu \Delta P \\
&= -2.4 \times 10^{-7} \times (-13\,608\,100) \\
&= 3.27 \text{ Kelvin or } ^{\circ}\text{C}
\end{aligned}
$$

As the turbine is only 72% efficient the actual increase in water temperature will be as follows:

$$
\begin{aligned}
\text{nett } \Delta t &= \text{actual } \Delta t - \Delta t \times (\eta_{turbine}) \\
&= 3.27 - (3.27 \times \frac{72}{100}) \\
&= 0.92^{\circ}\text{C}
\end{aligned}
$$

The final temperature of the water leaving the turbine is:

$$
\begin{aligned}
&= 8.3^{\circ}\text{C} + 0.92^{\circ}\text{C} \\
&= 9.22^{\circ}\text{C}
\end{aligned}
$$

Self study

Stroh gives examples on pages 517 to 521 in VOEE and must be done for self study purposes.

SELF STUDY

Read the following article by FJ Nadaraju: “Recovery of energy from water flowing down a mine shaft using a standard pump as a turbine”.

7.10.1 Change in Water Temperature as it is Pumped out of the Mine

Rule

As in the case with air (due to auto de-compression), there is no decrease in water temperature when water is pumped out of the mine. If there is minimal heat transfer from the shaft and other surrounding pipes, the temperature of the water when it reaches surface will be the same as that at the shaft bottom from where it is pumped.

7.11 Closed Circuit Water Reticulation Systems

NB

References

This forms a very important part of the design of chilled water systems in a hot mine. It is expected of the student to go through the design selection systems for selection pipes as described in Chapter 25 of VOEE. These are also applicable for examination purposes. Example 2 on page 506 must be done for self-study purposes. These closed circuit systems as mentioned before, works through a system of pressure reducing valves and therefore differs significantly compared to the open ended pipes as used in a cascade system (level dam to next level dam as described earlier).

7.12 Measuring of Water Flow Rates

South African mines use a great quantity of water in mining operations. Some mines use as much as 5 tons of water for each ton of rock mined. In deep mines this water can be a source of heat or cooling. Chilled water that is pumped to the working place is used to cool the surroundings.

The effectiveness of the cooling system is dependent on the measurement techniques that are used, because accurate flow rates need to be determined.

7.12.1 Measuring Water Flow

7.12.1.1 Classification of Flow Meters

Flow meters are classified in categories in terms of various principles that are used to measure flow rate.

PHYSICAL PRINCIPLES OF FLOW METERS

- The transformation of kinetic energy into pressure energy which is then measured
- Generation of an electrical impulse that is proportional to the water velocity
- The inference of the water velocity from its effects on the speed or intensity of sound of the flowing water
- Movement of vanes suspended in the water stream.

Examples

TRANSFORMATION OF KINETIC ENERGY INTO PRESSURE ENERGY

Flow meters that use this principle are the:

- Pitot tube
- Annubars
- Impact flow meter
- Venturi tube
- Nozzle
- Orifice plate.

The pressure differential for all these instruments is proportional to the square of the water flow rate. In SA mines the most general method to measure water flow rate in a pipe, is the square edged orifice plate. Figure 7.30 shows this method clearly.

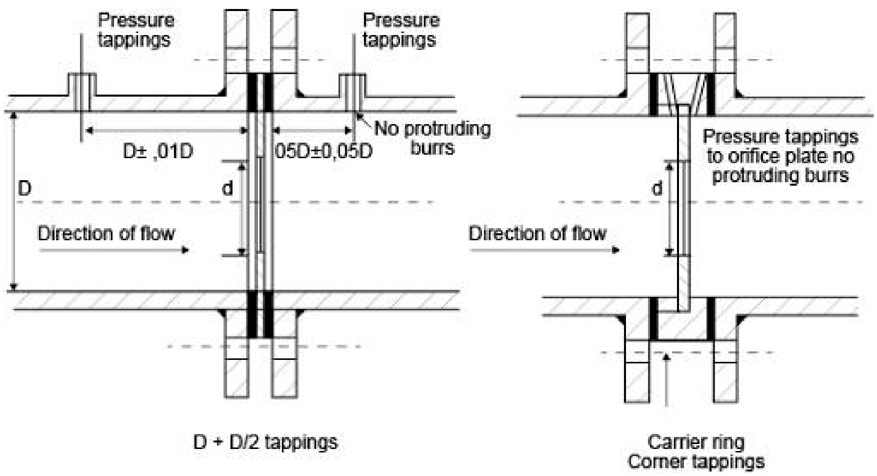


Figure 7.30 Orifice plate flow meters

Here the principle of D and D/2 tappings and corner tappings are shown. With corner tappings the pressure tappings are located immediately adjacent to the upstream and downstream faces of the orifice plate. The pressure tappings in the D and D/2 arrangements are at the distance D upstream of the orifice plate and D/2 downstream of the orifice plate and are welded into the pipe.

Advantages

The differences in advantages between using the two types of orifice plates are as follows:

Table 7.2 Different types of tappings

Corner Tappings	D and D/2 Tappings
Tappings are incorporated in a carrier ring, thus there is less chance of it being wrongly positioned.	Less expensive as no carrier ring is required.
Installation is easier as no pressure tappings have to be welded to the pipe.	Holes are larger so they are less susceptible to blockage.
The device can be installed either way round.	Area ratios greater than 0.5 the pressure differential is larger with D and D/2 tappings for the same flow rate.

Water flow is related to the pressure drop across the orifice plate by the equation:

$$q = C\sqrt{\Delta p} \qquad \text{.......... (7.14)}$$

Where:

C = constant that is a function of the orifice plate diameter d, and the pipe diameter (used as K or K')

D = the position of the pressure tappings and also the units in which the pressure differential is measured.

= water flow (l/s)

NB

= kPa (**take note**).

The value for C (K, K') for specific situations is read from Table 7.3 and Table 7.4.

Table 7.3 Recommend sizes of orifice plates for use with differential pressure meters which measure in inches of water or kPa

Pipe diameter			**Online plate dimensions**			**Factors**		
Nominal mm	Nominal mm	I.D.mm	I.D.mm	O.D.mm	Thickness mm	K	K'	q*
0	(4)	104.8	78.4	160	5	2.5	5	25
150	(6)	155.6	112.5	217	5	5.0	10	50
200	(8)	206.4	156.1	271	8	10	20	100
250	(10)	257.2	192.3	332	8	15	30	150
300	(12)	308.8	224.5	379	10	20	40	200
350	(14)	360.0	253.9	442	10	25	50	250
400	(16)	410.0	280.6	493	12	30	60	300

K is the constant in equation (1) below when the pressure differential is measured in inches of water.

K' is the constant in equation 1 when the pressure differential is measured in kPa.

q* is the water flow rate (ℓ/s) when the pressure differential across the orifice plate is 100 inches of water or 25kPa.

q = constant x $\sqrt{\Delta p}$

Table 7.4 Recommended sizes of orifice plates with differential pressure meters which measure in centimeters of mercury under water

Pipe diameter			**Online plate dimensions**			**Factors**	
Nominal mm	Nominal mm	I.D.mm	I.D.mm	O.D.mm	Thickness mm	K''	q*
100	(4)	104.8	75.6	160	5	5	25
150	(6)	155.6	108.2	217	5	10	50
200	(8)	206.4	150.5	271	8	20	100
250	(10)	257.2	185.2	332	8	30	150
300	(12)	308.8	216.0	379	10	40	200
350	(4)	360.0	243.8	442	10	50	250
400	(16)	410.0	269.1	493	12	60	300

K'' is the constant in equation (1) when the pressure differential is measured in centimeters of mercury under water.

q* is the water flow rate (ℓ/s) when the pressure differential across the orifice plate is 25 centimeters of mercury under water.

q = constant x $\sqrt{\Delta p}$

From the Tables it is apparent that different values of C (K, K') exist for different pipes. When the flow velocity of the water is low, it is difficult to obtain an accurate measurement without equipment designed for low flow rates (to be very sensitive). The reason for this is that the pressure difference over the orifice plate is directly proportional to the square of the flow rate.

When the flow rate is very low, consider a fifth from the original, the pressure difference will be $(0.2)^2$ or 4% of the value, which will mean that the pressure difference readings will not be very accurate. This is one of the limitations of the orifice plate. The orifice is therefore not very accurate for low flow rates.

Possible solutions for the problem is to replace 2 lengths of pipe with a smaller diameter pipe upstream and downstream of the orifice plate.

Or:

To install a smaller diameter pipe in parallel with the main stream pipe.

Or:

Install a different suitable orifice plate.

7.12.1.2 Derivation of the Orifice Equation

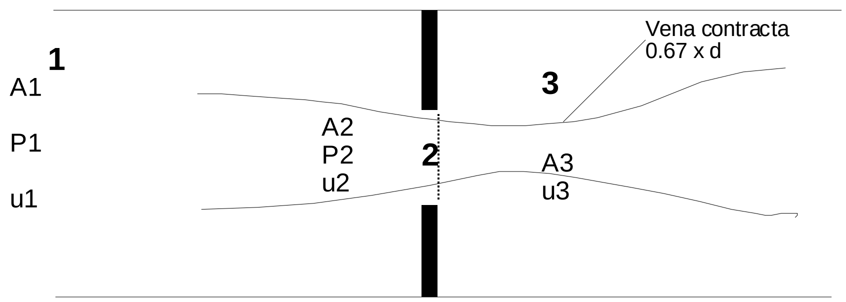


Figure 7.31 Orifice plate description

From Bernoulli's Equation:

$$\frac{P_1}{\rho g} + \frac{u_1^2}{2g} = \frac{P_1}{\rho g} + \frac{u_2^2}{2g}$$

With:

$$Q = A_1U_1 = A_2U_2$$

$$u_1 = \frac{A_2U_2}{A_1}$$

$$u_1^2 = \left(\frac{A_2}{A_1}\right)^2 \times U_2^2$$

With substitution into Bernoulli:

$$u_2 = \sqrt{\frac{2\ \Delta p}{\rho\ (1 - \left(\frac{A_2}{A_1}\right)^2}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (7.15)$$

With $\frac{A_2}{A_1}$ = m the Equation changes to:

$$u_2 = \sqrt{\frac{2\ \Delta p}{\rho\ (1 - m^2)}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (7.16)$$

At position 3, two coefficients are defined as a result of the reduction in the Vena contracta:

1. The velocity coefficient C_v and is expressed as:

$$c_v = \frac{\text{actual average velocity in the vena contracta}}{\text{**velocity with no friction losses}}$$

Where:

** = velocity without orifice plate

2. The compression coefficient C_c expressed as follows:

$$\frac{A_3}{A_2} \text{ is the } \left(\frac{\text{area of jet}}{\text{area of orifice plate}}\right)$$

The average value of C_v is 0.97

A combination of the above-mentioned gives an Equation named the discharge velocity coefficient (C_d) and the Equation looks as follows:

$$C_d = C_c \times C_v \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (7.17)$$

Where:

$$u_2 = C_d \sqrt{\frac{2\ \Delta p}{\rho\ (1 - m^2)}}$$

And

$$q = u_2\ A_2$$

For water

$$q\ (\ell/s) = 1.11 \times 10^3\ d^2\ C_d \times E \times \sqrt{\Delta p} \ldots\ldots\ldots\ldots (7.18)$$

$$\text{with } E = \sqrt{\frac{1}{(1 - m^2)}} \quad \text{the approach velocity factor}$$

When we want to determine the size of an orifice plate, there is a specific procedure that should be followed and this will now be discussed:

- The first phase in the estimation of the size is to determine a relative pressure loss for the expected flow rates.

Rule

- Because a number of mines use a number of different pressure gauges between 1 and 25 kPa, it is acceptable to design for 6 kPa. This allows flow rates, which are between half and double the design, to be measured with fair accuracy.

The value N (number of transfer units) is calculated as follows:

$$N = \frac{9.01 \times 10^{-4} \times q}{D^2\sqrt{\Delta p}} \qquad (7.19)$$

By using Figure 7.32 we can read the value of m and the diameter of the orifice plate can be determined provided the pipe diameter is known.

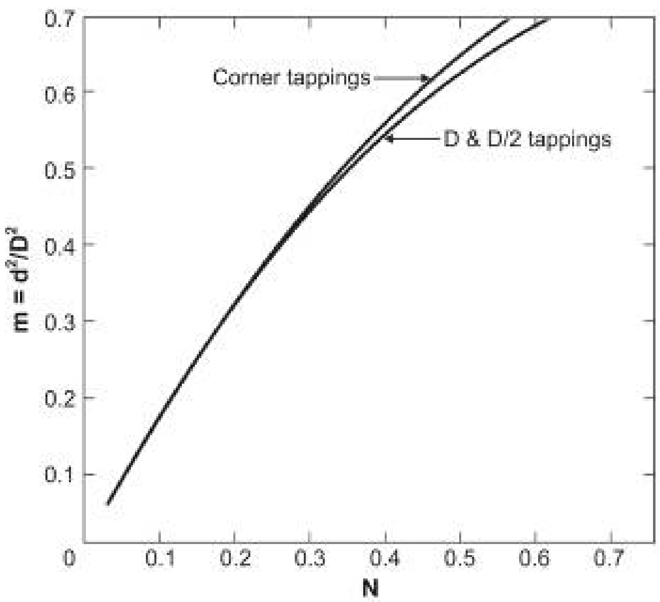


Figure 7.32 m value versus N

Figure 7.33 and Figure 7.34 give the values for the discharge coefficient for corner and D & D/2 tappings as a function of the area m. The flow rate (l/s) can be determined by Equation 7.18.

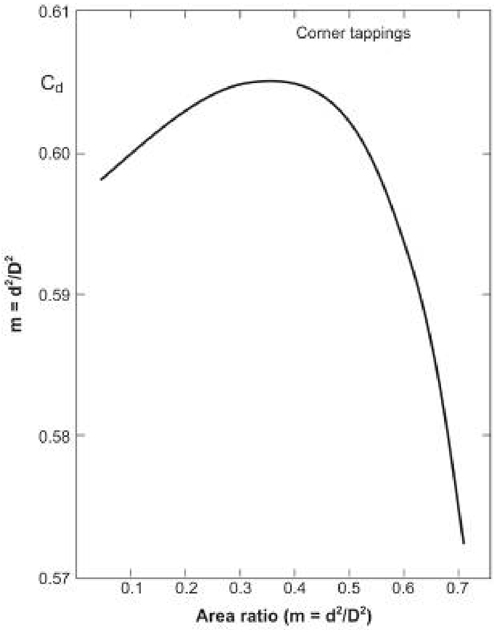


Figure 7.33 C_d value versus m (corner tappings)

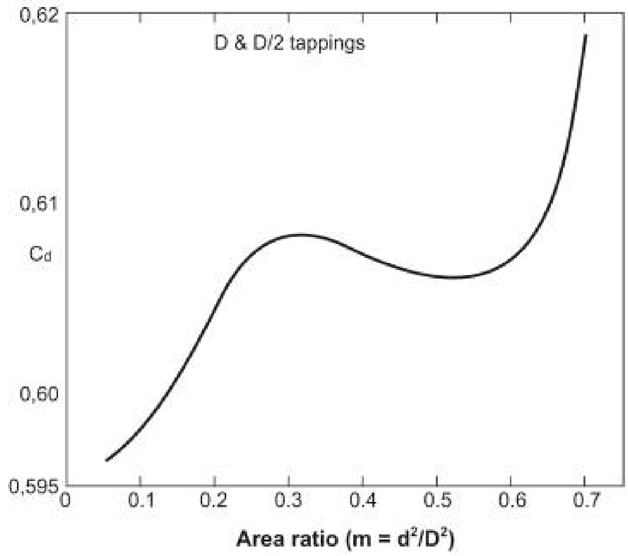


Figure 7.34 C_d value cersus m (D and D/2 tappings)

Turbine Flow Meter (Not extensively used in SA)

Principle of working:

- It is a free running motor that is installed axially in the pipe
- It rotates with an angular velocity proportional to the flow velocity
- By measuring the rotor speed, the flow rate can be obtained
- Magnetic pick up emits electrical pulses each time the turbine blade passes
- Measurements are given digitally.

OPEN CHANNEL FLOW METERS ('WEIR')

The principle on which the meter works is from Bernoulli's Energy Equation and ignoring the approaching water velocity.

Where:

h = $\frac{U^2}{2g}$ and U = $\sqrt{2gh}$

We differentiate between two types of open channel flow meters:

- **The Rectangular Weir**

B = width
h = water height

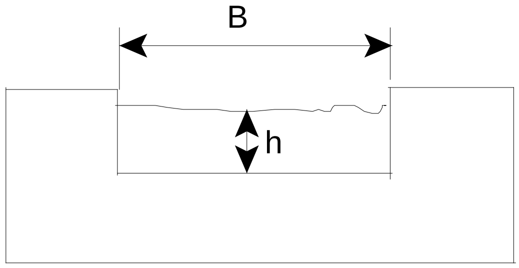


Figure 7.35 The rectangular weir

The derivation of the formula is not important, but the application is. The Equation looks as follows:

$$q \quad = \quad 1000\ B\ C_d \sqrt{\frac{8\ g\ h^3}{9}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ (7.20)$$

Where:

q	=	Litre / second
B	=	width (m)
C_d	=	0.62 (discharge coefficient)
h	=	water height (m)
g	=	gravitational acceleration (9.79 m/s^2)

The simple Equation for the above mentioned is as follows:

$$q \quad = \quad 1\ 830\ B\ \sqrt{h^3} \qquad \text{assume } C_d = 0.62$$

- **The V-weir Flow Meter**

θ – angle of V-cut
h = water height

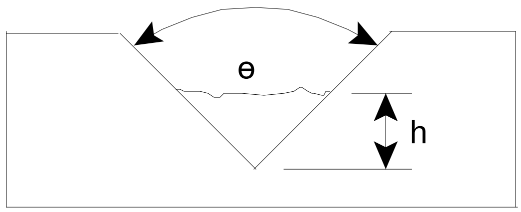


Figure 7.36 The V-weir flow meter

The Equation to measure the volume flow is:

$$q = 1\,000\, C_d \frac{8}{15} \sqrt{2gh^5} \tan\left(\frac{\theta}{2}\right)$$

Which simplifies to:

$$q = 1370\sqrt{h^5} \tan\left(\frac{\theta}{2}\right) \quad \text{if} \quad C_d = 0.58 \qquad \text{............... (7.21)}$$

With θ = degrees

Rule

C_d can vary between 0.58 and 0.69 and is dependent on θ and h.

7.12.1.3 Methods to Estimate the Flow Rate

FREE FLOWING WATER OUT OF PIPE

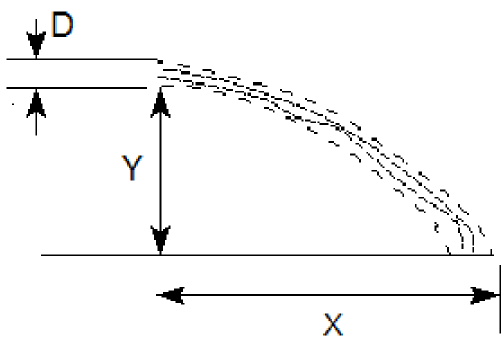


Figure 7.37 Free flowing water from pipe

Consider an open pipe as shown in Figure 7.37. Water flows out of the pipe with a velocity u and with gravitational effects:

Horizontal distance x = ut

Vertical distance = ½ gt^2

By combining the above Equation, we get the following Equation for velocity u:

$$u = \sqrt{\frac{g\,x^2}{2\,y}} \quad \text{... (7.22)}$$

The flow rate (q) in l/s = 1 000 UA (1000 litres = $1m^3$)

And from this the simple Equation:

$$q = 1740\,D^2 \sqrt{\frac{x^2}{y}} \quad \text{in litres / second.................. (7.23)}$$

To obtain a reasonably accurate figure:

- The pipe must be horizontal
- The pipe is flowing full.

MEASUREMENT OF WATER VELOCITY IN AN OPEN CHANNEL

Consider the following Figure 7.38.

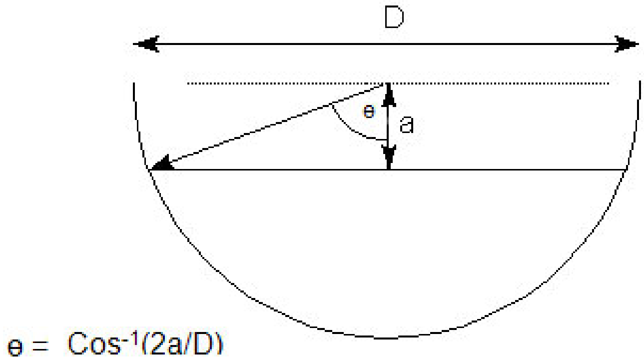


Figure 7.38 Diagrammatic representation of water in mud drain

The velocity can be measured by spraying a cloud of aluminum dust onto the surface of the water stream and measuring the time over a pre-measured distance between two points. Remember that the velocity in the middle is greater than that at the edges, with the average velocity:

$$U_{ave} = 0.8\,U_{measured.}$$

The flow rate can thus be measured and used with the following Equation:

$$q = 200\,D^2\,(\theta - \sin\theta\,\cos\theta)U_{measured} \quad \text{........................... (7.24)}$$

Where area factor = $200\,D^2\,(\theta - \sin\theta\,\cos\theta)$ and θ expressed in radians (2π radians = 360ºC). The area factor can for example include the effect of mud in the drain, and can be obtained from Figure 7.39 as well.

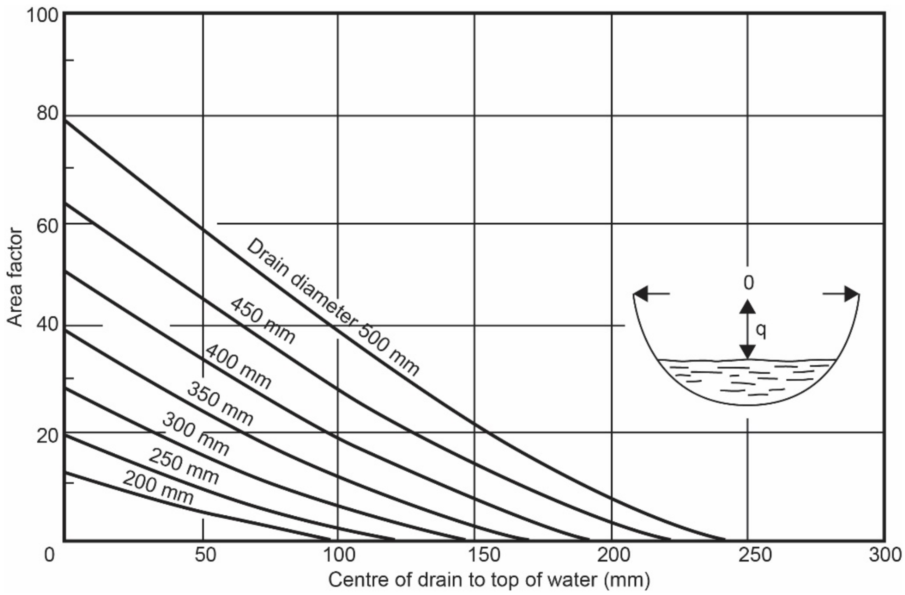


Figure 7.39 Determining Area factor graphically

NB

It is important to remember that the correction factor can be obtained if the mud depth in the pipe and the water depth from the middle of the pipe are given (in other words 2 correction factors are now applicable). The resultant area factor will thus be the difference in the area factor for the 2 given depths.

EXAMPLE PROBLEM FOR SELF STUDY

The average velocity of water in a drain is 2.5 m/s. The diameter of the drain is 500 mm and the height at the middle of the pipe to the water surface is 100 mm. The average depth of the mud, measured at the middle of the pipe is 200 mm. Calculate the flow rate, by:

(i) Determining the area factors
(ii) Using Figure 7.39.

Chapter

8

REFRIGERATION

LEARNING OUTCOMES

8.1 Basic Refrigeration Principles

- Basic refrigeration system
- Concept of vapour compression refrigeration units
- Comparison between fridge plants on surface and underground
- Properties of refrigerants.

8.2 An Underground Refrigeration Plant

- Diagrammatic representation of single stage plant
- Pressure enthalpy diagram, single stage plant
- Performance and duty of a single stage plant.

8.3 A Surface Cooling Plant

- Performance of a single stage plant
- Definitions and calculations for refrigeration plants.

8.4 Siting of a Refrigeration Plant

- Diagrammatic representation of two stage systems
- Performance and ability of multi stage systems
- Cycle variations.

8.5 Pressure Enthalpy Diagrams

- Cooling towers and spray chambers
- Design guidelines
- Performance
- Cooling coil calculations.

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Mine Environmental Control Certificate (MEC), Chamber of Mines, South Africa, Workbook 2.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

8.1 Basic Refrigeration Principles

In deep hot mines refrigeration plays a very important role. The basic principle that is employed to cool air that is going down a mine is that the air is drawn through chilled water (water that is cooled) sprayers, which then in turn absorbs the heat from the hot air. The air that comes from the spray chamber is now cold and can be used to remove heat from the working environment underground. The chilled water absorbs heat from the air and then returns back to the refrigeration plant to be cooled again. Details pertaining to the different heat exchange processes will be dealt with in detail in this chapter.

Figure 8.1 Photo of a modern fridge plant arrangement

8.1.1 The Basic Refrigeration Cycle

Objective

A refrigerator operates on the principle of moving heat from one space to another thereby cooling one substance and heating another. A simple example is a kitchen fridge, where heat is removed from the content inside the fridge and put into the ambient air in the kitchen. This is achieved by using a refrigerant liquid that evaporates at very low temperatures (low boiling point). It boils at a low temperature thereby extracting heat from the refrigerator contents. The gas resulting from the boiling refrigerant liquid is compressed via a compressor increasing its temperature further to allow it to condensate back to a liquid by rejecting heat to the ambient air in the kitchen. The refrigerant liquid can now boil again by extracting heat from the refrigerator content.

NB

In a mine the principle is the same (as for a kitchen fridge) where heat is extracted from intake air and transferred to the return air. On the mine intake air side the air is typically cooled by a heat exchanger which is transferring heat to the 'evaporator' side of the refrigeration machine to boil the refrigerant liquid. The refrigerant gas (result of boiling) is compressed via a compressor to increase its temperature further before it is condensed back to a liquid in a heat exchanger in the return air circuit of the mine (condenser side). The amount of heat rejected to the return air circuit is equal to the **amount of heat removed** from the intake air circuit **plus the electrical energy absorbed by the compressor motor**.

The four main parts (shown in Figure 8.2) of a vapour compression refrigeration machine as used for mining purposes are the:

Animation

Click on the button to open

- Compressor
- Condenser
- Evaporator
- Expansion or float valve

Flowing through the system is a fluid referred to as the refrigerant, which in parts of the cycle is a liquid and in others a vapour and for clarity sake will be discussed first (it is the heat absorbing and discarding medium in the refrigeration machine and responsible for all the heat exchange processes that takes place).

NB

Definition

The refrigerant is contained within the refrigeration machine in a closed circuit. No leaks are allowed and the process is a continuous cycle of heat exchange process, and we will discuss the process as it starts in the evaporator. The direction of flow for the refrigerant is always in the same direction; evaporator>compressor>condenser>expansion valve and back to the evaporator. This also referred to as a **single stage vapour compression cycle** (note the pressure conditions in each cycle).

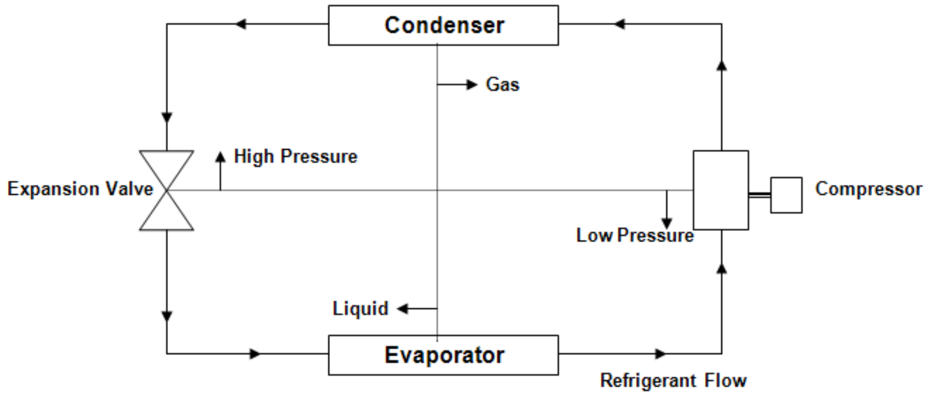


Figure 8.2 Diagrammatic representation of the refrigeration process

8.1.1.1 Refrigerant

Definition

The medium that is used in the refrigeration machine is the **refrigerant.** Different refrigerants do exist and are applied depending on the characteristics required. Each refrigerant has a characteristic boiling point-pressure relationship. The actual temperatures in the plant (that is the atmospheric temperature and the water temperatures expected form the working environment) will have a bearing on the most suitable refrigerant to be used so that the pressures will be kept within acceptable limits. This means that the pressure temperature relationship of a refrigerant must be such as to avoid excessively low pressures in the evaporator and excessively high pressures in the condenser.

NB

!

High pressures require costly construction of the condenser, while low pressures in the evaporator means that the compressor must handle large volume flow rates, and if the pressures are below the barometric pressure, there is the danger of ingress of air or moisture to the machine. Refrigerants must therefore have the following qualities:

- Must be cheap and not affect the lubricant used in the compressor.
- Should also be non-corrosive, non-toxic, non-flammable and non-explosible.

NB

- Must be easily detectable so that leakages from the machine can be located (under no circumstance must ammonia be used underground, why? Ammonia is one of the best refrigerants, but its pungent smell and toxic effects precludes its use underground). It is used as the refrigerant in surface refrigeration plants. The Kyoto protocol also stipulates that the use of it has to be terminated in the very near future (do an internet search to get more detail on the specifications applicable to the Kyoto protocol).

- A refrigerant must have a high latent heat of evaporation so that the refrigerant flow rates can be kept to a minimum (can you explain this from previous chapters? Think of the definition of latent heat).
- Has its own cooling range and must therefore be selected accordingly.
- Has its own critical temperature which is the temperature above which the vapour/gas cannot be made to liquefy no matter how great the pressure.
- The suitability for application is therefore dependent on a number of factors and no particular refrigerant is universally applicable or preferable.

Typical types of refrigerants that are commonly used for refrigeration machines are as follows:

- **R-22** (or HCFC-22), widely used in chillers, out of use as from 2010
- **R-134a** (or HFC-134a), widely used in many screw chillers, replaces R-12
- **R-123** (or HCFC-123), replaced R-11 in low-pressure centrifugal chillers, must be phased out after 2020
- **R-717** (or NH_3, ammonia), used in open drive screw chillers. Not suited for centrifugal chillers due to low molecular weight
- **R-410A**, Ideal for flooded evaporators (large coolers)
- **R-407C**, R-22 replacement, vapour pressure close to that of R-22, not suited for flooded evaporator. Used in reciprocating and other positive displacement compressors
- **R-245a**, replaces R-141b and R-123.

8.1.1.2 Evaporator

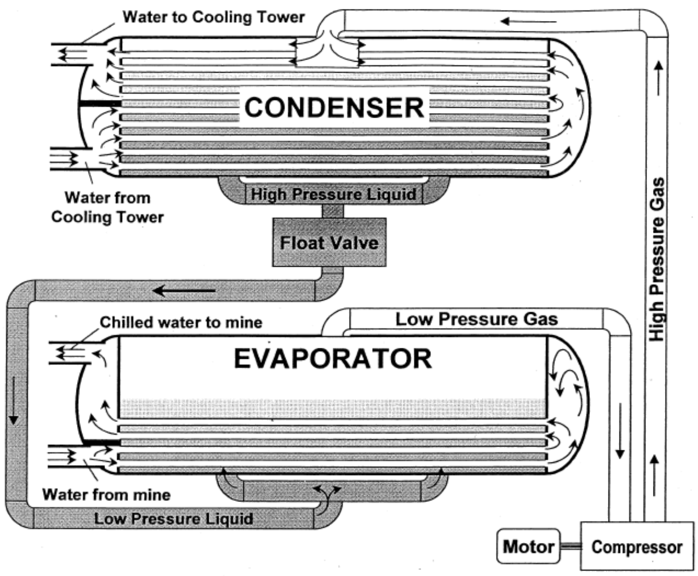
The refrigerant enters the evaporator as a saturated **liquid** under **low** pressure. It takes **heat** from the relatively hot chilled water entering the evaporator, and it is evaporated by the heat (through a **latent** heat process) into a saturated vapour. This process takes place at a **constant** temperature. The refrigerant normally leaves the evaporator as a saturated vapour, but sometimes the refrigerant can pick up so much heat from the chilled water that it can leave the evaporator as a slightly super-heated vapour.

Because heat always travels from the body with the high temperature to the body with the low temperature, two conditions are obvious:

- The water temperature is higher than the **refrigerant evaporation temperature**
- The chilled water temperature leaving the evaporator is still slightly **higher** than the refrigerant evaporation temperature – the chilled water temperature leaving the evaporator can never be the **same** or **lower** than the refrigerant evaporation temperature, for obvious reasons.

Figure 8.2 showed a diagrammatic representation of the process that takes place within the refrigeration machine and Figure 8.3 more detail description of the refrigeration process.

As shown in Figure 8.3, the evaporator is only partly filled with water tubes, so as to ensure that only vapour and no liquid leaves the evaporator. Any liquid entering the compressor could damage it.



Image

Figure 8.3 Typical circuit of a centrifugal water chilling machine

NB

Note that the evaporation process takes place essentially at **constant temperature**. This is because the refrigerant is boiling, and the heat it extracts from the cool water is its latent heat of vaporisation (phase change process from water to vapour). Refrigeration machines makes use of latent heat in this way and have higher cooling capacity and smaller size than an equivalent unit that does not involve a change in phase of the refrigerant.

Shell-and-tube evaporators, similar in construction to shell-and-tube condensers, are normally used in mine refrigeration plants (as shown in Figure 8.4).

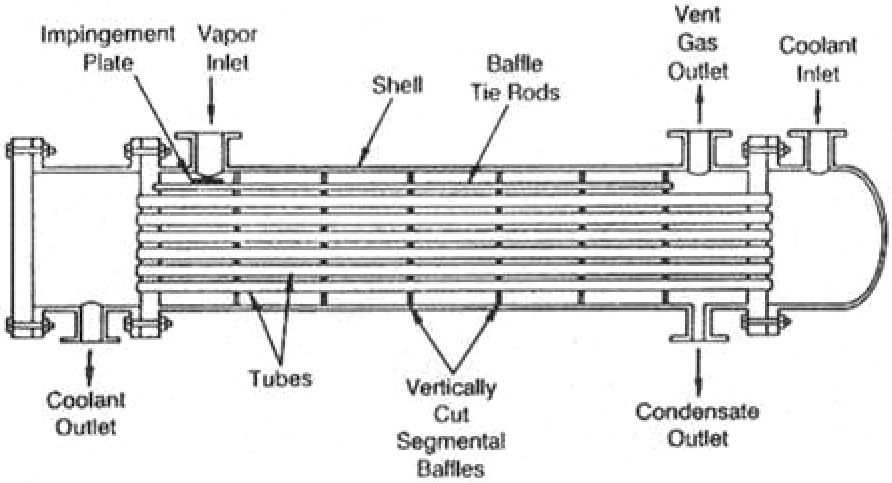


Figure 8.4 Diagrammatical representation of a shell-and-tube heat exchanger

Technology has changed considerably over the last few years and nowadays more modern heat exchangers are used for evaporators and condensers. It is called a gasket welded plate heat exchanger.

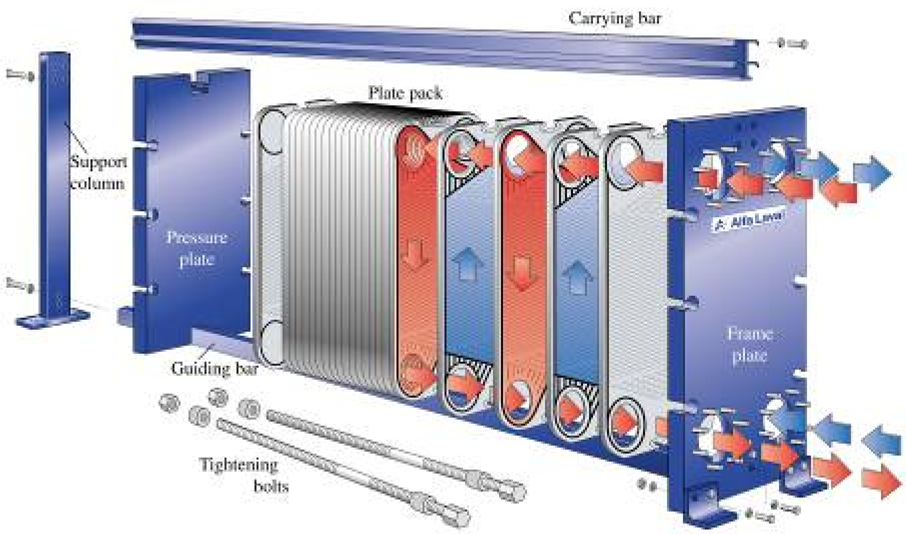


Figure 8.5 A gasket welded plate heat exchanger

Advantages for the use of the welded plate heat exchangers are as follows:

Advantages

- High surface to area ratio
- High heat transfer, turbulence on both sides
- Low Δt down to 1Kelvin (or ºC)
- Compact, compared with shell-and-tube
- Low cost, because of thin plates
- Can easily be opened for inspection and cleaning
- Extra plates can be added for increased capacity
- Plates pressed from stainless steel or higher grade material (titanium, incoloy, hastalloy)
- Gaskets are the weak point (nitrile rubber, hypalon, viton, neoprene)
- Plates can however be welded to overcome above problem, but then cannot be opened.

Fouling in evaporators is generally much less than in condensers since the introduction of chilled service water the possibilities of fouling and corrosion occurring in the evaporator has increased. To avoid this many new evaporators have titanium tubes because this metal is highly resistant to corrosion and fouling.

8.1.1.3 Compressor

The function of the compressor is to compress the refrigerant. Compressors are normally of the single stage multi cylinder type and can be driven by internal combustion engines, steam engines or electrical motors.

Electrical motors are normally used in underground applications. When there is a large difference in temperature and therefore pressure between the evaporator and condenser, multi stage compressors are used. Reciprocating compressors are limited in size to approximately 3.5 MW of refrigeration.

Centrifugal Compressors are normally used in underground systems and have the ability to handle large volumes of gas, and are used in systems where refrigeration is greater than 3.5 MW. When two or more compressors are used, they are connected in parallel. Compressors can be of the open or closed type.

The refrigerant leaves the evaporator as a saturated vapour (or slightly super-heated) under a relatively low pressure and temperature condition, and then enters the compressor inlet in the same state.

The compressor compresses the refrigerant and adds pressure and heat to the refrigerant, thereby increasing the refrigerant temperature drastically into a super-heated state.

If it is an open type compressor, that is if the compressor's electrical motor is situated outside the compressor, and the motor is exposed to the plant room atmosphere, then the heat **lost** by the motor due to its inefficiency is directly put into the plant room atmosphere.

The rest of the power goes into the refrigerant as heat. If the compressor is a hermetically sealed unit (that is if the motor is enclosed inside the compressor shell and is in direct contact with the refrigerant), then **all** the input electrical power into the motor is added to the refrigerant as heat.

The compressor compresses the vapour and discharges it at a higher pressure and higher temperature. Compressors may be a:

- Piston-cylinder type compressor (very rare in mining)
- A centrifugal unit
- Or a turbine compressor.

Photo

Screw type compressors are also used, especially for refrigerators using ammonia as refrigerant.

Examples

In conclusion refrigeration machine are commonly referred to by the type of compressor that it has, the four examples are:

- Centrifugal, primarily large tonnage [above l 000 kW]
- Screw [200 - 1.500kW]
- Scroll compressor [up to 200 kW]
- Reciprocating compressor [up to 500 kW].

8.1.1.4 Condenser

The function of the condenser is to remove heat from the refrigerant as a result of compression and the heat from the evaporator.

This warm gas (super-heated vapour), at a high pressure flows to the condenser, whereby it is circulated through a cooling medium (usually water). The water first removes the sensible heat from the compression **and then the latent heat from the evaporator such as to condense the fluid into a liquid**. When the liquid leaves the condenser it is at a high temperature and pressure. Condensers may be water or air-cooled. In tube type condensers the water flows through the tubes, whilst the refrigerant flows over the outside of the tube where it condenses.

The refrigerant is therefore cooled by the water, first from a super-heated vapour into a saturated vapour (with an accompanying drop in refrigerant temperature), and then, at a **constant** temperature, the refrigerant is condensed from a saturated vapour into a saturated liquid. This condensing temperature is much higher than the condensing water entering or leaving temperature (the cooling medium in the condenser (usually water) therefore removes heat from the refrigerant).

8.1.1.5 Expansion Valve

The expansion valve is the controlling device regulating the refrigerant flow from the condenser into the evaporator and many different types are employed. In some machines an orifice plate is used instead of a float controlled expansion valve, however, in all cases the action on the refrigerant is the same. The refrigerant leaves the condenser as a saturated liquid under a high pressure/temperature condition and enters the expansion valve in this state.

Definition

The expansion valve allows the refrigerant pressure to **decrease drastically** (see Figure 8.2) from the compressor outlet pressure (condensing pressure) to the compressor inlet pressure (evaporating pressure). The compressor outlet pressure is the same as the condenser pressure (inlet or outlet, as it is on the same pressure line of the P-H diagram). This aspect will be dealt with in detail when the P-H diagram is discussed later in this chapter. This decrease-in-pressure process takes place at a **constant** refrigerant **heat content**, therefore the expansion process takes place **isothermally** (iso, means the **same**, thermal means **heat**).

Definition

During this process of sudden pressure reduction some of the liquid refrigerant is released as a vapour (or gas) on the outlet side of the expansion valve. This vapour is called '**Flash gas**'. The reduction in pressure is of course accompanied by a related decrease in refrigerant temperature therefore the refrigerant now enters the evaporator again as a saturated liquid containing a small percentage of flash gas.

SHE

This circle of activity is continuous and takes place under sealed (leakless) conditions. If leakage does take place and the refrigerant leaks out of the circuit:

- The plant performance deteriorates and can stop completely
- The refrigerant can release poisonous gas upon entering the fresh air conditions in the plant room (depending what type of refrigerant is used).

Image

Essentially, the expansion valve is therefore an orifice through which the liquid refrigerant flows. In so doing, it drops from the higher to the lower pressure of the cycle. This low-pressure liquid then enters the evaporator, and the whole cycle repeats itself continuously.

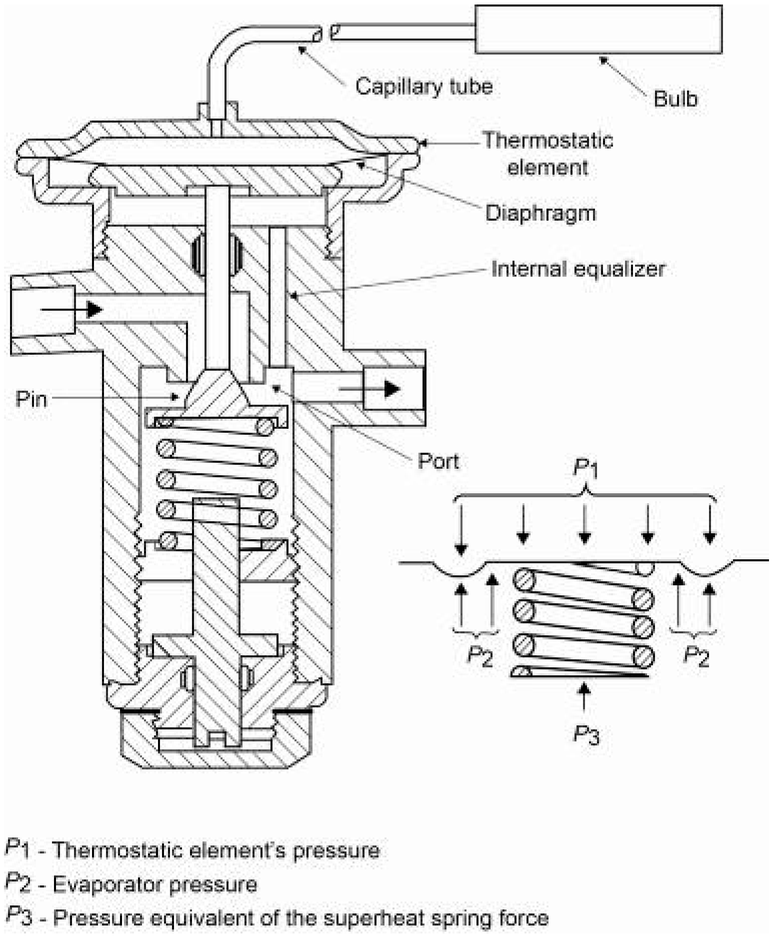


Figure 8.6 Example of a thermostatic expansion valve

8.1.1.6 Typical Operating Parameters

Table 8.1 gives figures for some of the operating parameters of three typical refrigerators actually in use in the mining industry.

- The first column applies to a plant with capacity 3 500 kW or 3.5 MW of refrigeration, located underground, discharging heat to an underground (condenser) cooling tower, and making use of Refrigerant 134a.
- The second is a similar unit, also using Refrigerant 134a, but this time with a capacity of 5 000 kW or 5 MW, and located on surface. Notice the difference in the temperatures of the condenser cooling water.
- The third unit is an ammonia refrigerator, located on surface, driven by a screw compressor, and with plate-type evaporator and condenser, rather than shell-and-tube (make sure you know the difference).

Table 8.1 Comparing typical operating parameters of refrigeration machines

Parameter		Unit 1	Unit 2	Unit 3
Compressor input power	kW	1094	**1206**	1465
Condenser water flow rate	ℓ/s	126	270	407
Condenser inlet water temperature	ºC	40.5	22.0	21.5
Condenser exit water temperature	ºC	49.2	27.5	27.5
Condensing pressure	kPa (abs)	1468	750	1167
Condensing temperature	ºC	54.4	29.0	30.0
Evaporator water flow rate	ℓ/s	60.5	114.0	320
Evaporator inlet water temperature	ºC	18.8	14.5	7.0
Evaporator exit water temperature	ºC	5.0	4.0	0.5
Evaporating pressure	kPa (abs)	342	287	406
Evaporating temperature	ºC	4.3	-0.6	-1.5

Unit 1: Capacity 3 500 kW / 3.5 MW
Located underground
Discharging heat to underground cooling tower
Using refrigerant 134a

Unit 2: Capacity 5 000 kW / 5 MW
Located on surface
Using refrigerant 134a

Unit 3: Ammonia refrigerator 5 700 kW / 5.7 MW
Located on surface
Driven by screw compressor
Plate-type evaporator and condenser

8.2 An Underground Refrigeration Plant

8.2.1 Layout

Figure 8.7 shows the typical layout of a refrigeration plant installed underground.

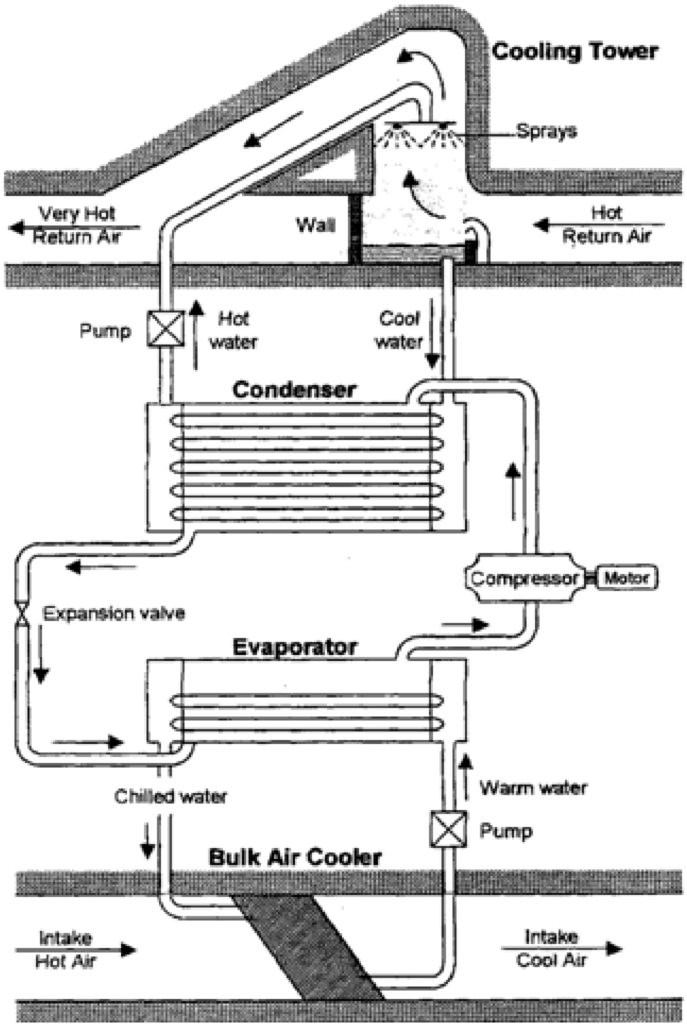


Figure 8.7 Typical layout of a refrigeration plant underground

8.2.1.1 The Hot Water Circuit

Hot water circulates between the condenser and the cooling tower. In the condenser itself, as previously explained, the hot high-pressure refrigerant vapour condenses. This occurs irrespective of whether the condenser is the shell-and-tube type, or the plate type. Now a liquid at high pressure, the refrigerant flows out of the bottom of the condenser and from there to the expansion valve.

During the condensing process, the latent heat of vaporisation of the refrigerant is passed on to the cooling tower, which becomes heated as the result. This water is pumped to a cooling tower, where heat is removed from it, and it then returns to the condenser at a lower temperature. A detailed description of the Hot water circuit (where the process of heat rejection takes place) is shown in Figure 8.8.

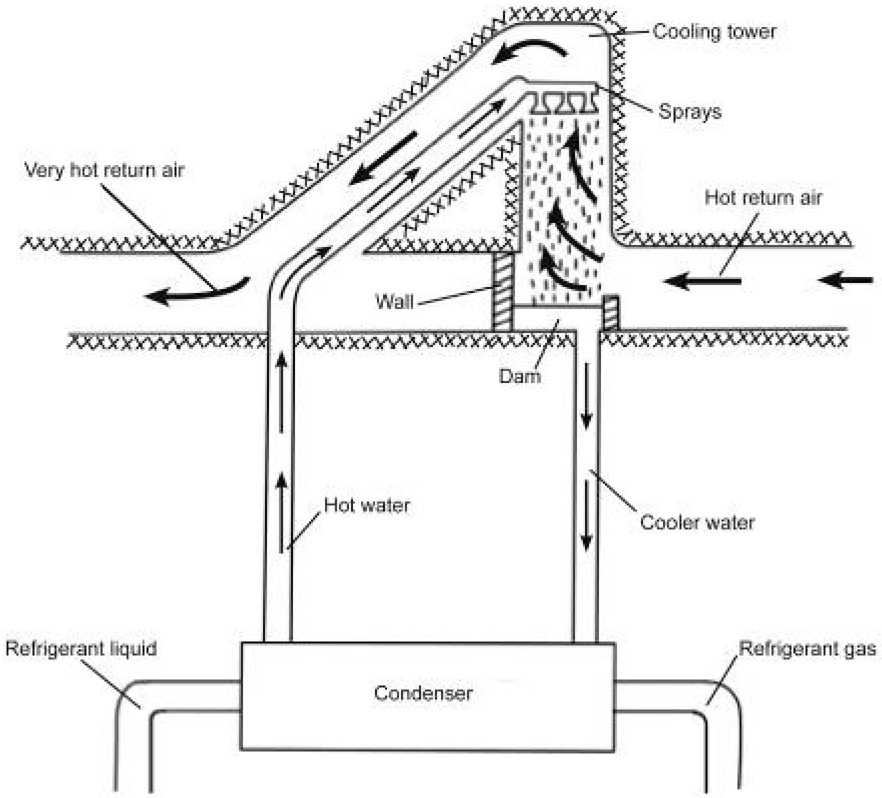


Figure 8.8 A detail layout of the hot water circuit

Typical figures for flow rates of water and refrigerant through a condenser, as well as inlet and exit temperatures, are given in the Table 8.1.

UNDERGROUND (CONDENSER) COOLING TOWERS

The construction of a typical underground cooling tower is shown in Figure 8.9. Also given are typical temperatures of the water entering and leaving the tower, typical temperatures of the air at inlet and exit, and the flow rates of the water and air.

Animation

Quite often, condenser cooling towers are located one or two levels above the so-called refrigeration plant, where the compressors, condensers and evaporators are installed.

This means that pumps must be provided to raise the water from the plant to the condenser cooling towers. The reason for this is that the RAW is often used as the cooling tower and is normally situated at a higher position from the plant where the hot air is on its way out of the mine. The air however is still cooler than the very hot water that comes from the plant and can therefore be used to cool it before it goes back to the fridge plant.

Notice that in the layout shown in Figure 8.9, that the hot water is being sprayed into the air by a set of nozzles pointing upwards at about, 45°, and mounted in a ring-shaped pipe mounted on a suitable ledge.

In other arrangements, there is a rectangular grid near the top of the tower on which a set of sprays is mounted, directed vertically upward. The ring installation has the advantages that:

Advantages

- Access by engineering staff for repair and maintenance is easier
- The resistance to air flow by the grid in the other arrangement is avoided
- These grids, too, are subject to corrosion by the warm, moist return air.

Horizontal spray chambers are also sometimes used as underground condenser cooling towers.

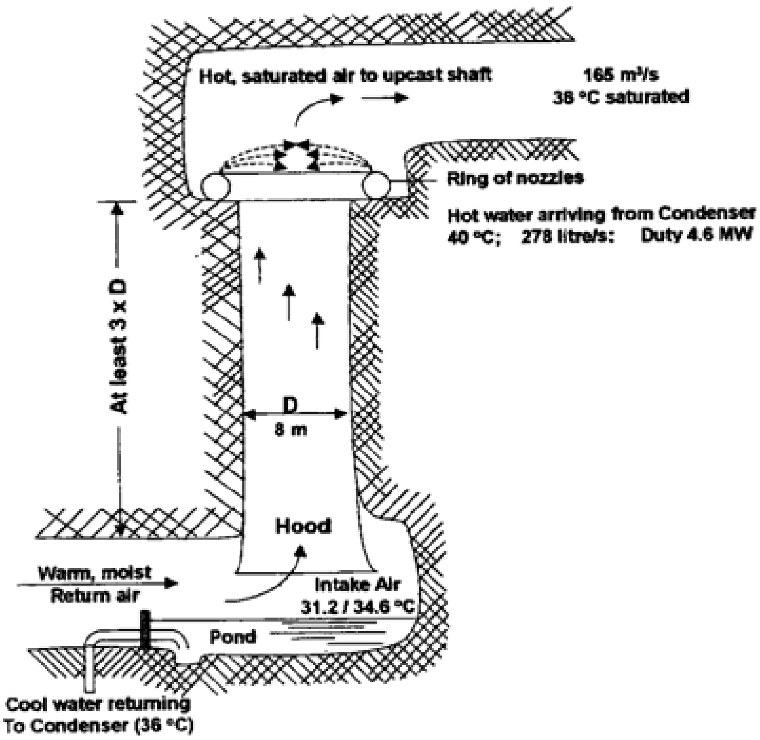


Figure 8.9 A typical underground condenser cooling tower layout

8.2.1.2 The Cold Water Circuit

The cold water circuit of a refrigeration plant begins with the evaporator, where cold water is generated. From here, the water is piped to usually several cooling units in the workings of the mine, where they cool the ventilation air. This is clearly shown in Figure 8.10.

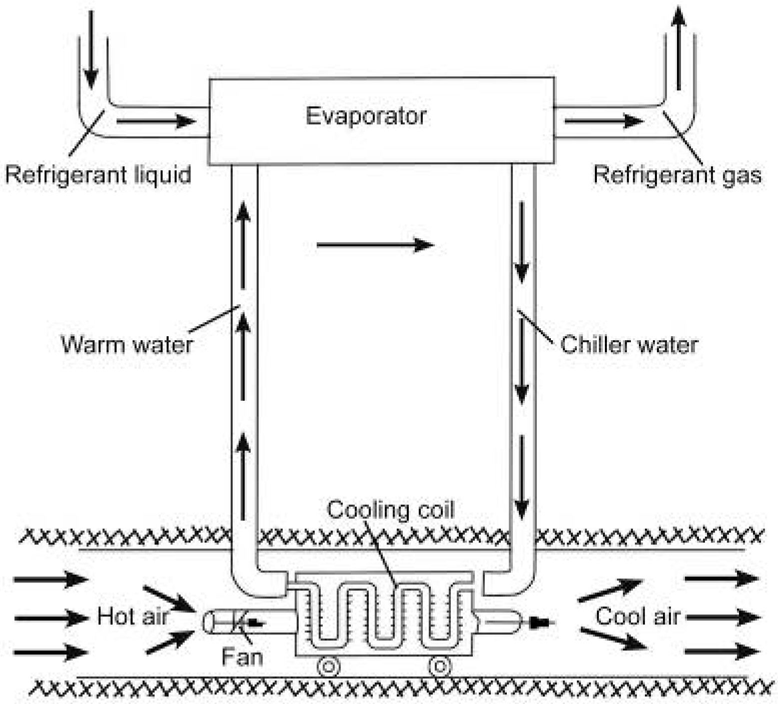


Figure 8.10 Detail of the cold water circuit

NB

Cooling units can be several different types, depending on the quantity of air that is to be cooled, and whether or not the water is to be sprayed into a stream of air, or passed through coils such that the water does not come directly in contact with the air. Closed circuit units will now be discussed in detail. It must however be noted that large volumes of air can be cooled through the spraying of chilled water into the air. Spraying water into ventilation air however contaminates the water with dust and gases, so that it requires treatment before being returned to the condenser.

When relatively large quantities of air are to be cooled, bulk installations consisting of banks of cooling coils are often employed. They are located in intake airways rather than close to the workings, so that they can be left in place for years at a time, and are easily accessible for maintenance. On the other hand, there are cooling cars, which are fitted with wheels for running on tracks, have a small number of cooling coils on board, and are ventilated by means of an auxiliary fan. These will be discussed and shown in the sections to follow.

In summary, a refrigeration plant therefore does in overall terms removes heat from the ventilation air in a working place, exchanges this with the refrigerant in the refrigerator (water-refrigerant heat exchange), discharges it to the cooling water in the condenser (together with the energy input of the compressor), and finally passes it on to up cast ventilation air in the cooling towers.

COOLING COILS

NB

Figure 8.11 shows a cooling coil of the usual design employed in mines to cool ventilation air. They may be assembled in fixed banks, so as to provide air cooling in bulk, or mounted two or three at a time on a car so as to be easily movable underground. Typical figures for air and water flow rates, and inlet and exit temperatures for the two fluids (air and water) are also given. It must be noted here that the refrigerant – water heat exchange is only applicable at the refrigeration machine.

Chilled water from the evaporator of a refrigerator is passed through sets of tubes in an individual coil. Usually, the water is passed back and forth through a coil several times, through separate sets of tubes, and there is a header tank at each end of the coil. The tubes are manufactured of a metal capable of conducting (as was discussed in Chapter 6) heat well and resistant to corrosion by the water for example cupro-nickel (copper and nickel alloy).

NB

Most important of all the tubes are fitted with a large number of circular fins, also of a good heat conducting metal (high UA value).

These have the effect of increasing the area separating the ventilation air flowing across the coil from the water flowing through it, and so improving the rate of heat transfer from air to water.

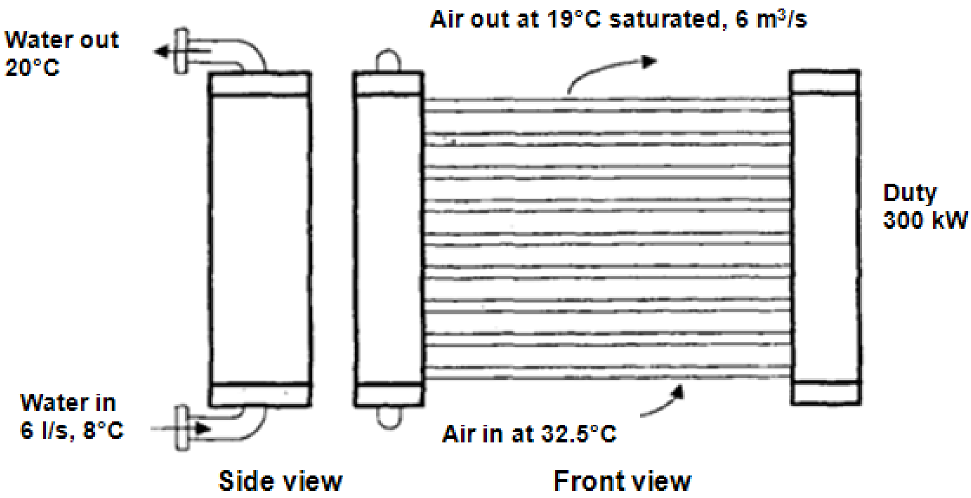


Figure 8.11 Typical example of a cooling coil

BULK AIR COOLERS USING COILS

The layout of a typical underground bulk air cooler consisting of a bank of individual cooling coils is shown in Figure 8.12. In this case there is only one stage, that is, the ventilation air being cooled passes through only one bank of coils.

Self study

Performance figures are also given for a typical bulk cooler of this kind and shown in Figure 8.13 (Can you calculate the duty of the cooling as given in Figure 8.12?)

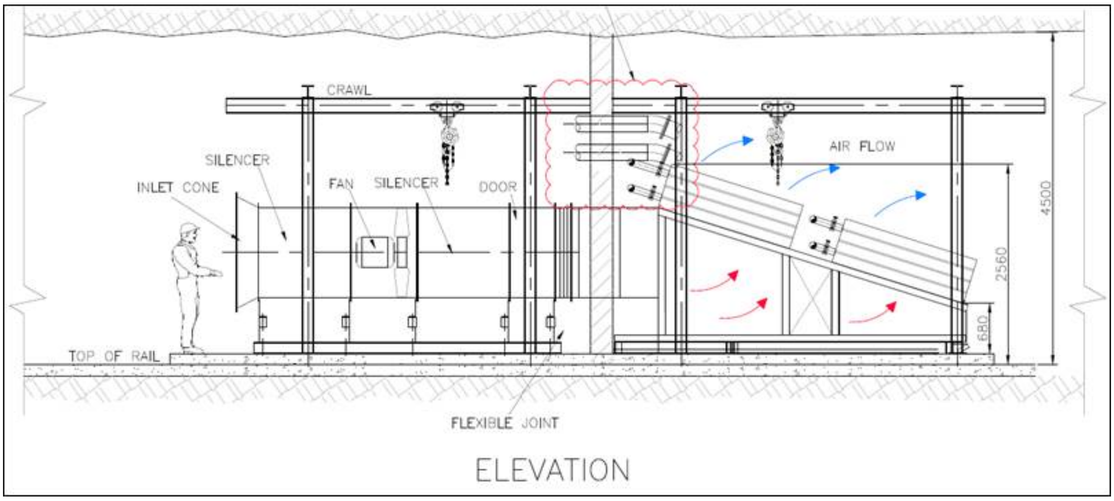


Figure 8.12 Bulk air coolers Indirect-Contact Exhanger Banks (Willemse et al 2012)

COOLING CARS USING COILS

A cooling car fitted with coils works on the same principle as a bulk air cooler, but generally has only two or three coils at the most, and having a rolling stock chassis, can be moved from place to place underground as required. A side view of a coil-type cooling car is shown in Figure 8.13, including data for flow rates and temperatures.

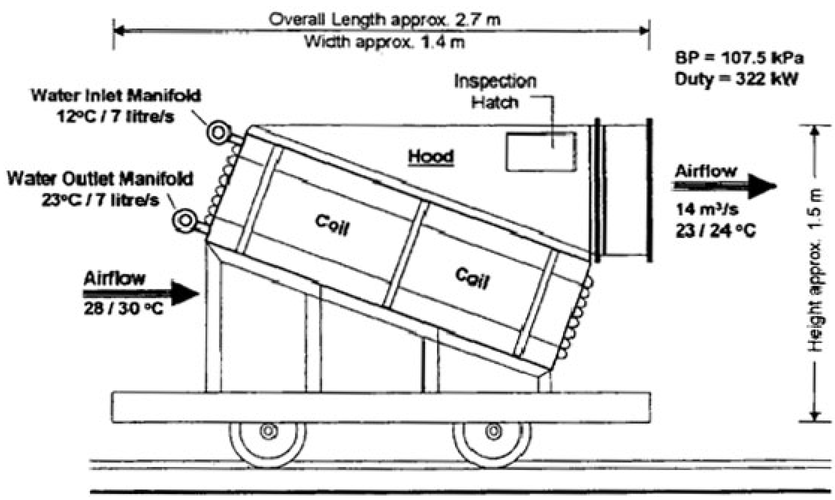


Figure 8.13 Cooling car using coils

COMPARISON BETWEEN CONDENSER COOLING TOWERS AND EVAPORATOR COOLING COILS

Table 8.2 Gives a clear understanding of the difference between these two types of units

Table 8.2 Comparison between condenser and evaporator towers

Condenser cooling tower	Cooling coil (chilled water comes from the evaporator)
Air side: 1. **Air leaving** tower is at a higher temperature **(hotter) than the air entering** tower.	**Air Side:** 1. **Air leaving** cooling coil is at a lower temperature **(cooler) than the air entering** coil.
2. **Air volume (m^3/s) leaving** the tower is **bigger than the air volume entering** the tower.	**2.** **Air volume (m^3/s) leaving** cooling coil is **smaller than the air volume entering** the coil.
3. Water **evaporated** into the air.	**3.** Water **condensed** out of the air.
Water side: 1. **Water entering** the cooling tower is at a higher temperature **(hotter) than the water leaving** the tower.	**Water side:** 1. **Water entering** cooling coil is at a lower temperature **(cooler) than the water leaving** the coil.

Figure 8.14 shows an overview layout of an underground refrigeration machine or fridge plant and the various heat exchange as well as chilled water reticulation associated with it. It also shows the reticulation associated with the water being pump out of the mine (temperature given in this figure are for illustrative use and will change from plant to plant).

As mentioned before and repeated here in a different way (including additional pumping), there are basically 4 main processes and components applicable to a chilled water and clear water pumping reticulation system underground, these are:

- A condenser cooling tower, where heat exchange between the condenser water and hot air (reject air) takes place.
- A refrigeration machine (where the heat exchange takes place between the refrigerant and water (closed circuit) takes place.
- The bulk air coolers (evaporator cooling) , or sometimes closed circuit coolers where heat exchange takes place between the hot air underground and the chilled water from the refrigeration unit/plant.
- And lastly the water that gets collected in the underground settlers (large dams that are used to collect water underground, mud settled out) and clear hot water (because of heat pick up in the mine, pumped back to surface.

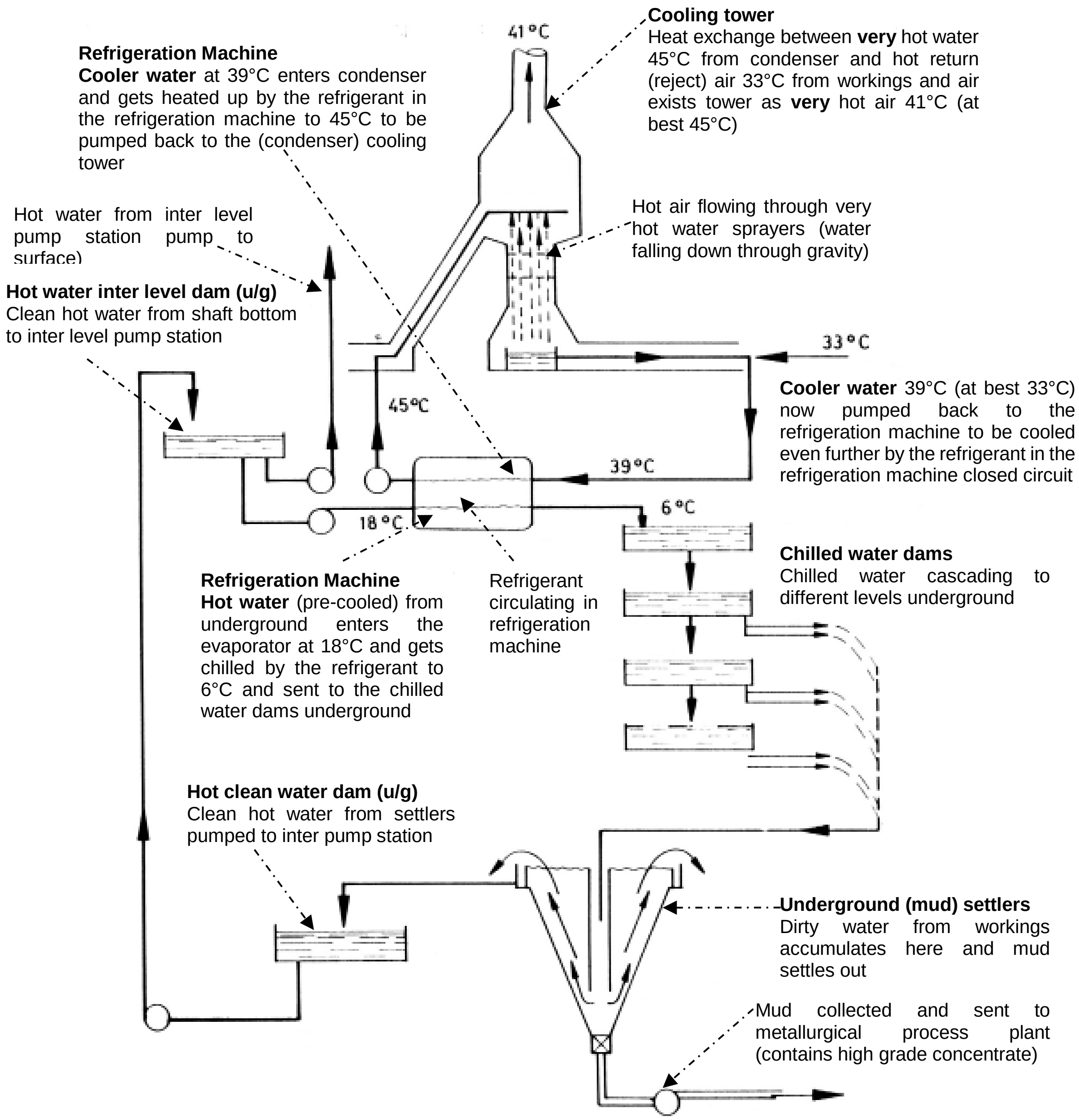


Figure 8.14 Typical chilled water reticulation system underground

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Refrigeration is therefore required in deep hot South African mines to ensure that acceptable working conditions (that is temperatures), are kept underground. This is done to ensure that production is kept to a maximum.

NB

Refrigeration systems are a very expensive component of the work process of a deep hot mine, and therefore it is important that the planning and operating of the system is optimal.

Definition

Refrigeration in summary is therefore a process whereby the purpose is to remove heat from one medium at a specific temperature and to release it elsewhere at a higher temperature.

A refrigeration machine therefore uses mechanical energy to absorb heat at a certain temperature and release heat at a higher temperature. Work is done in a refrigeration system to produce cooling.

8.3 A Surface Cooling Plant

8.3.1 The Basic Layout

Figure 8.15 shows the layout of a typical surface refrigeration plant. Like an underground plant, it produces a supply of cold water, which in this case is sent underground partly as service water and partly to cool ventilation air by means of bulk coolers and cooling cars.

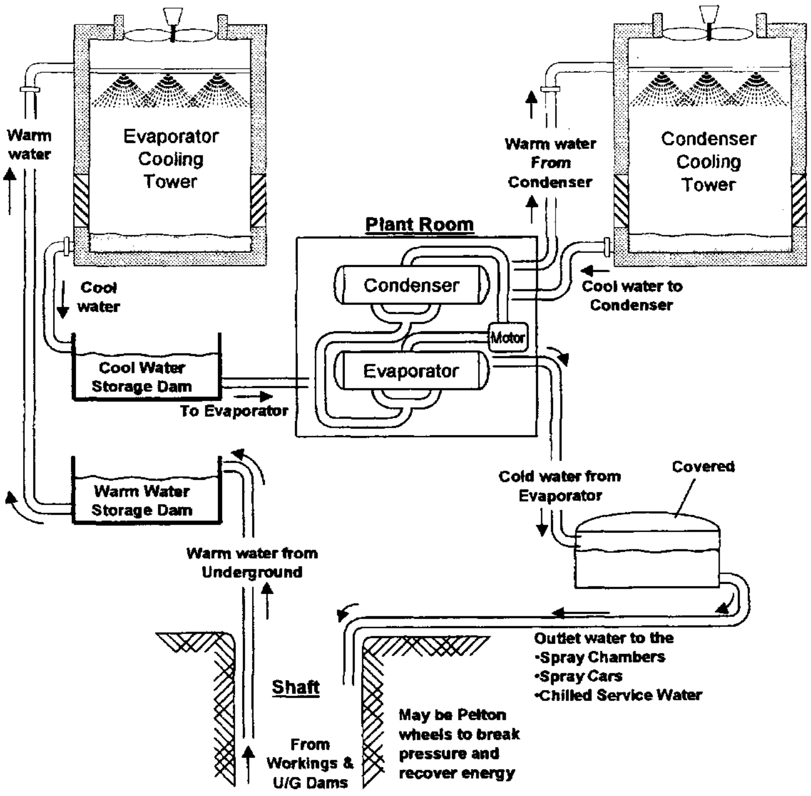


Figure 8.15 Typical working of a surface refrigeration plant

In an installation of this kind the water undergoes the following sequence of events:

First, warm water returning from the mine arrives on surface to which it has been pumped from the settlers and dams underground as was shown in Figure 8.15. After being collected in one or more storage dams, this warm water is pumped to a pre-cooling tower, where it is sprayed downwards into a rising stream of air. Part of the water is evaporated, and as the latent heat required to do this comes from the water itself, it cools down and then passes to another storage dam.

The cool water in this dam now flows through the evaporators of the refrigerator, and then makes its way to one or more covered cold water dams, from which it is sent underground as service water, or to supply underground coolers or surface bulk air coolers (the working of the bulk air cooler will be explained in detail later in this chapter. Quite often, the water going down the shaft is passed at some level in the shaft through water turbines such as Pelton Wheels (as discussed in Chapter 7). This arrangement breaks the pressure of the column of water and makes it possible to recover some of the potential energy of the water.

8.3.1.1 Pre-Cooling Towers

Definition

The process of pre-cooling warm underground water in a cooling tower provides a simple and inexpensive way of cooling it. It is often called '**free' cooling** because it does not entail the expense of a refrigerator to provide the same amount of cooling which would otherwise be needed.

The extent to which a pre-cooling tower can cool water is dictated by the wet-bulb temperature of the air. Generally, it is possible to cool the water to about 2º (above the wet- bulb temperature of the air (if the wet-bulb temperature of the air was say 24ºC, then the temperature of the water cooled would be 26ºC).

Pre-cooling towers are more useful and effective in the Highveld regions than they are in the Bushveld (due to the high ambient temperatures in the Bushveld) and therefore no real 'free' cooling always possible.

Also, ironically, they are more effective in winter, when there is less need for cooling in the mines (most mines switches their plants off in winter to do major services, cooling not needed, except for very deep mines, where the effect of auto compression still plays a major role). Pre-cooling towers need very little power input. Some pumping may be required, and a small motor will be needed to drive the main fan at the top of the tower (to get the air to circulate vertically upwards through the tower and in this way cool the air).

The structure of a typical pre-cooling tower is shown in Figure 8.16 and includes typical flow rates and temperatures. It is also common that modern towers are packed filled, with either splash fill, trickle grid or film type fill material. Thus is done to increase water/air contact time.

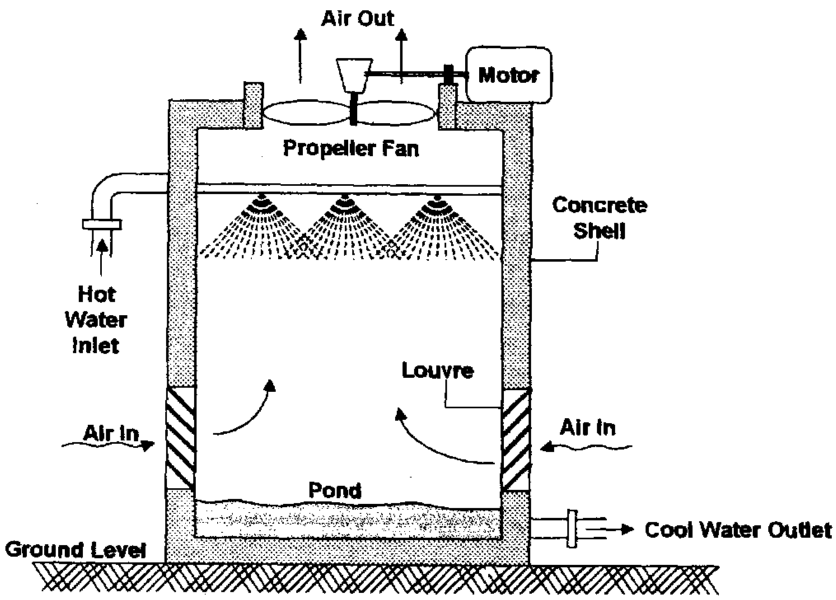


Figure 8.16 A typical layout for a pre-cooling tower

8.3.1.2 Condenser Cooling Towers

These devices are essentially the same as pre-cooling towers in design and construction, the main difference being that their purpose is to cool the condenser water circulating through the refrigeration machine, rather than underground water. For their construction, refer to Figure 8.16 of the diagram of a pre-cooling tower. Typical operational data for a R-134a refrigerator with a duty of 26 MW located at a Witwatersrand mine, in summer is as follows:

Table 8.3 Typical operational data for a condenser cooling tower

Water flow rate	700 litre/s
Airflow rate	250 m^3/s
Water temperature at inlet	48^{o}C
Water temperature returning to condenser	39^{o}C
Inlet air temperature	18 / 28^{o}C
Outlet air temperature	38^{o}C saturated

8.3.1.3 Bulk Air Coolers

There are two different types of bulk air coolers using chilled service water commonly found, and both are illustrated in the figures that follow:

- The first is a cascade cooler shown in Figure 8.17, in which the cold water is introduced into a perforated tray at the top of the unit, from which it trickles down over what is called fill or packing. This often consists of a honeycomb of PVC, or small wooden slats mounted at angles to the vertical. Whatever the precise arrangement, the idea is to spread the cold water out into a film with a large surface area, so that it makes good thermal contact with the air flowing through the fill.
- The second type of common bulk air cooler shown in Figure 8.17 is using cold service water. This is a horizontal (evaporator) spray chamber with two stages of cooling in this case (more stages can be included which will make the cooling of the air much more effective but does have a financial implication. The operating procedure is clearly shown in Figure 8.18. In each case typical operating parameters are given.

Many hot mines have surface bulk air coolers where the chilled water from the evaporator is being sprayed vertically in a bulk air cooler spray chamber.

Towers can be horizontal or vertical. The air is forced through the water sprays with large fans installed in parallel and then channeled down the shaft and mixed with the ambient air.

Chilled water can therefore go down the mine as service water and also used to cool bulk volumes of air to go down the mine, so as to alleviate the effect of auto compression.

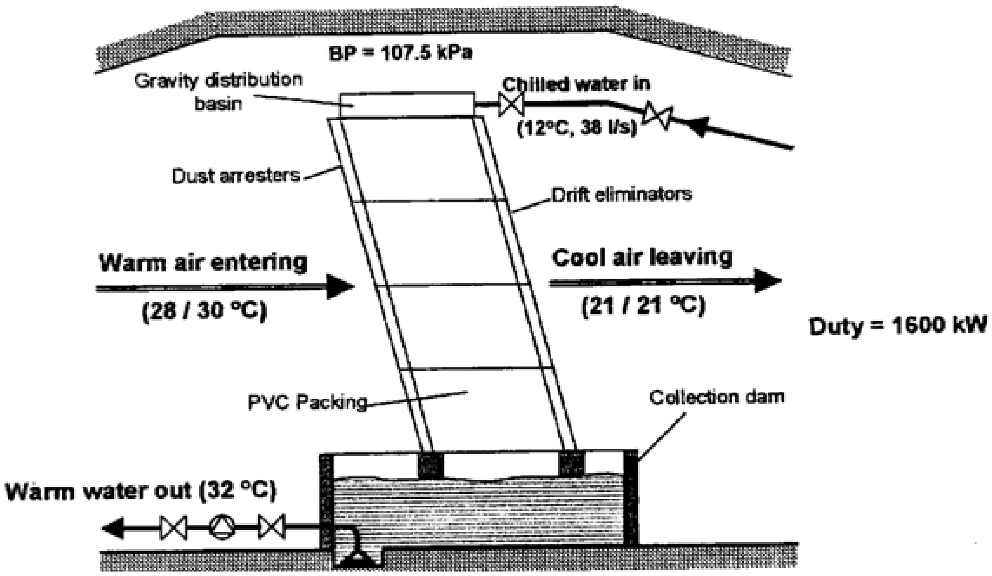
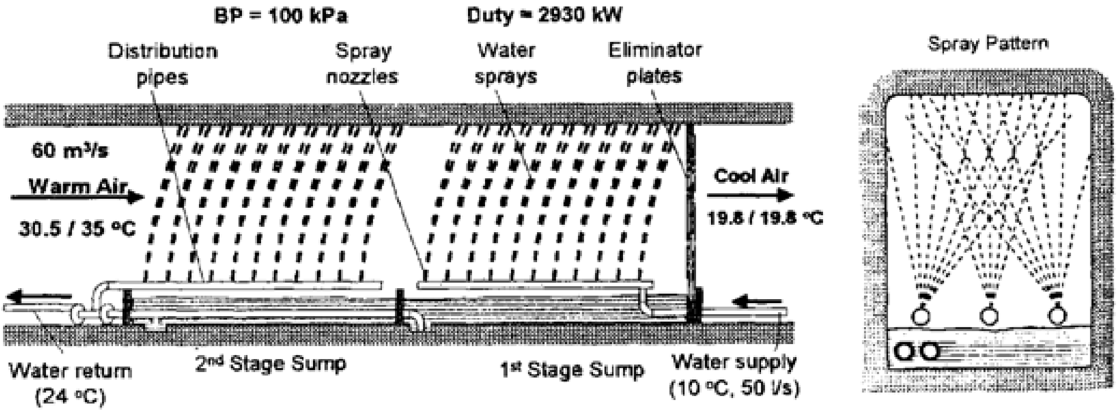


Figure 8.17 A typical cascade bulk air cooler



Animation

Usage

Figure 8.18 Horizontal Evaporator spray chamber bulk air cooling (can also be vertical)

The cascade principle is also commonly used in individual cooling cars that are designed to make use of cold service water. A typical unit is shown in Figure 8.19, together with typical flow rates and temperatures.

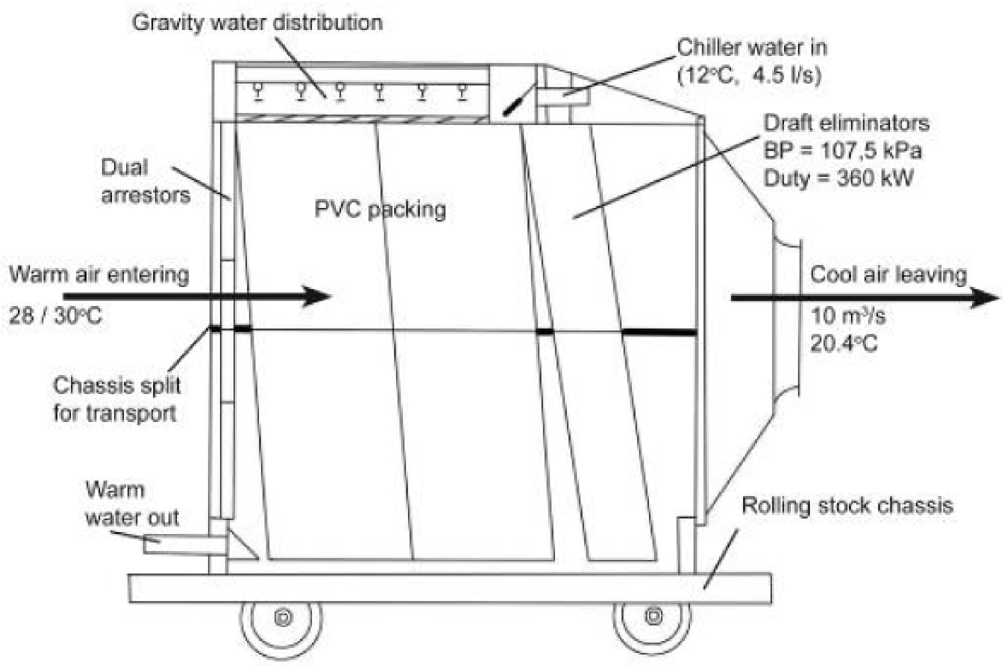


Figure 8.19 Cooling car using cascade principle

WORKED EXAMPLE 1

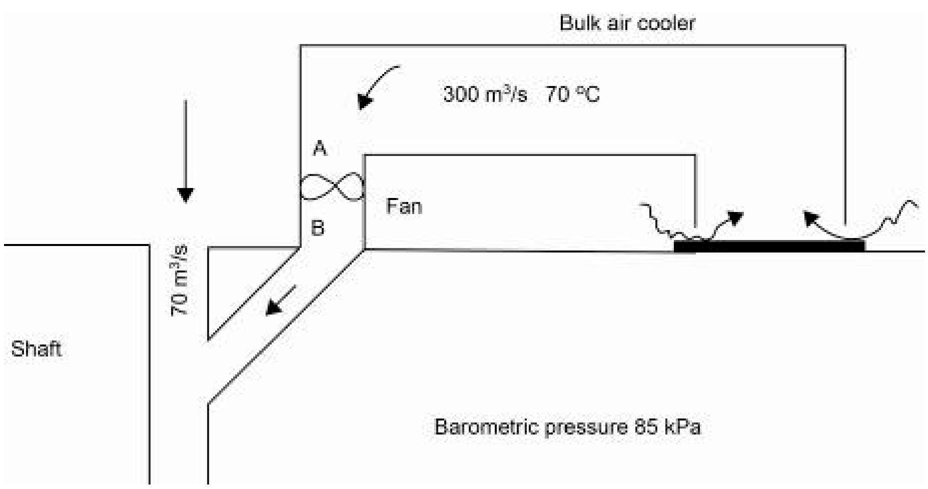


Figure 8.20 Example of surface bulk air cooler

The surface bulk air cooling system as shown in Figure 8.20 is incorporated into a shaft system. An amount of 300 m^3/s of air is cooled to a temperature of 7ºC (this temperature will obviously be a saturated temperature because the air is drawn through the sprayers). A direct driven axial flow fan delivers the cooled air to the shaft. An additional 70 m^3/s of ambient air (air that did not go through the spray chamber) mixes with the cooled air in the shaft. The fan handles 300 m^3/s at a density of 1 kg/m^3 and the pressure across the fan is of 700 Pa.

The three phase electrical input to the fan motor is as follows:

- 550 Volt
- 350 Amps
- Power factor of 0.9.

The mine engineers are concerned about free moisture being caused by condensation at the mixing point of the two air streams in the shaft. There is no additional free water that comes from the bulk air cooler. The following questions needs to be answered (your application of psychrometry understanding and knowledge, which included heat exchange principles, plus the current knowledge of refrigeration practices will now be needed to answer this question):

Psychrometric chart

(i) Will any condensation take place at the mixing point if the following atmospheric conditions prevail. Use a graphical method using the 85 kPa psychrometric chart attached (show the method of deriving these answers)?

- 17.0/25.0ºC
- 17.0/17.0ºC

(ii) During a 1 hour period 5 400 litre of water condenses from the air in the cooling tower. If the relative humidity of the ambient air during this period is 55%, what was the wet-bulb and dry- bulb temperature of the air entering the bulk air?

Answer

(i) The motor power must first be determined:

Motor input power = $E \times I \times p_f \times \sqrt{3}$

= $0.55 \quad 35 \quad 0.9 \quad \sqrt{3}$

= 299.7 say 300 kW

Heat content of air entering the fan using the 85 kPa psychrometric chart:

Sigma heat of air at point A = 25.5 kJ/kg

Heat content (kW) at A = $m_A \times S_A$

= 300 1.0 25.5

= 7 6 0 W

Heat content of air at B = 7 6 0 300

= 7 9 0 W

This is one of the first catches to the question. As the fan draws the air all the energy that is going into the fan motor is transferred to the air either in terms of moving and heating up the air, 300 kW must therefore be added to get the total heat added to the air.
The Sigma heat increase can now be determined as follows:

$$\text{Heat of air at B} = \frac{kW}{m} = \frac{7\,950}{300} = 26.5 \text{ kJ/kg}$$

Apparent specific humidity (A.S.H) of air at A (assume that the moisture content remains constant across the fan):

$r_A = 7.4$ kJ/kg

Psychrometric chart

From the 85 kPa psychrometric chart the temperature of the air leaving the fan can be read off.

Note: This is not strictly correct as in this case we have ignored the pressure increase across the fan as it is very small, 0.7 kPa. If however the pressure increase across the fan is large, say 2.5 kPa, the barometric pressure at the delivery side of the fan will be 87.5 kPa and then you have to use the 87.5 kPa psychrometric to read off the delivery temperature.

Temperature of air at B = 7.5/8.0ºC

Now plot on the 85 kPa psychrometric chart the temperature 17.0/25.0ºC and 7.5/8.0ºC (85 kPa and ignore the effect of the pressure increase of 700 Pa across the fan) and draw a straight line (XY) between the two points.

Measure the distance between the two points as 37 mm.

$$\text{mixing point air ratio (Q)} = \frac{70}{300 + 70} = 0.1892$$

With this ration and the length of the line the mixing point position can be plotted:

The mixing point is therefore = 0.1892 x 37 mm

= 7 mm

This is 7 mm from the 'lower' point as the amount of cool air is much more than the ambient air and the mixed air temperature will tend to be closer to the colder temperature. The temperature of the mixture can be read off as 9.6/10.6ºC and no condensation will take place as the straight line (XY) does not intersect the dew point vertical temperature line at any point (see Figure 8.21.).

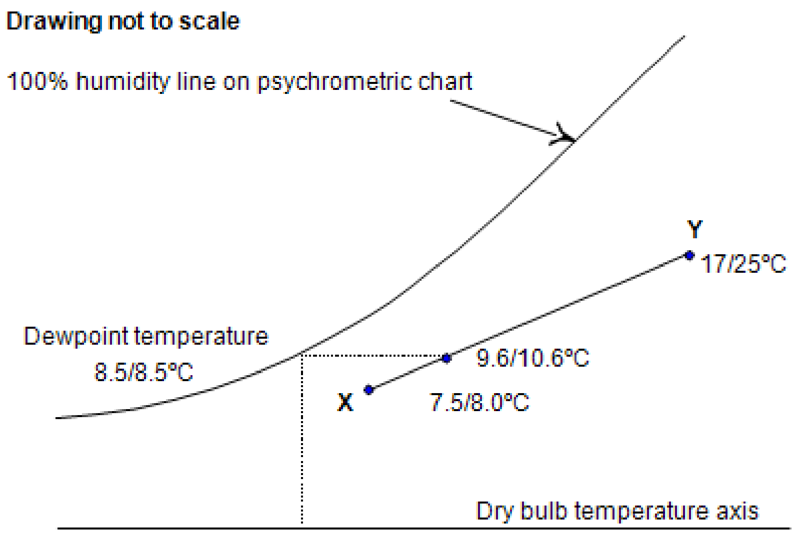


Figure 8.21 Determining dew point temperature

Repeat the procedure for the other conditions:

17.0/17.0ºC: This close and either way (condensation or no condensation) could be correct. The probability exists that no condensation will take place.

13.5ºC/15.5ºC: No condensation will take place.

(ii) Condition of air **out** of the tower:

Sigma heat at 7ºC	=	25.5 kJ/kg
A.S.H (r value)	=	7.4 g/kg
Amount of water in air (m_2)	=	m x r
	=	300 x 7.4
	=	2 220 g/s
	=	2.220 kg/s

Amount of water condensed from the air (taken out of the air) that entered the tower say (m_1) can be calculated as follows (assume that 1 litre of water = 1 kg):

$$\text{amount of water } (m_1) = \frac{m_{water}}{60 \times 60} = \frac{5\ 40}{3\ 600} = 1.5 \text{ kg/s}$$

The condition of the air before entering the tower (r = g/kg) can be determined as follows. Remember that ($m_{out} = m_{in} - m_{condensed}$) and therefore ($m_{in} = m_{out} + m_{condensed}$).

$$\text{condition of air} = \frac{m_{total}}{\text{mass flow of air}} = \frac{1\,500 + 2\,220}{300} = \frac{3\,720}{300} = 12.5 \text{ g/kg}$$

Psychrometric chart

From A.S.H equal to 12.5 g/kg on the 85 kPa psychrometric chart go across and intersect the 55% relative humidity line and read off the temperature:

17.8/24.0ºC

8.4 Siting of a Refrigeration Plant

8.4.1 Locating Plant Underground

Disadvantages

There are several disadvantages to installing refrigeration plants underground. One of these is that the condenser must reject its heat to underground air, which is considerably hotter than air on surface. This reduces the co-efficient of performance (COP) of the plant. The COP rises because the temperatures of the water flowing through the condenser and evaporator get closer to one another. Furthermore, **pre-cooling of water is not possible.** Calculations pertaining to plant performance will be dealt with in detail later in this chapter. The underground location brings several other difficulties:

- Access for maintenance and repair is less easy, as men and materials have to travel in shaft conveyances
- The opportunity is lost to use cheaper refrigerants such as ammonia, which is considered too dangerous for use underground
- Also, the capacity of the plant that can be installed is limited by the amount of upcast air that is available
- The condensing temperature for the same plant will be higher underground than on surface.

Advantages

On the other hand, there are two main advantages to locating a cooling plant underground:

- One of these is that the plant has a higher positional efficiency. This factor compares the amount of cooling actually achieved at the face to that produced by an evaporator (as distances increases from the evaporator to where the cooling is needed more losses can be expected).

Rule

As cold water flows from the refrigerator to a spot or bulk air cooler, some of the cooling is lost from it to the surroundings (that is, heat is gained in the water), because no piping is perfectly insulated. Hence for every kW of heat taken out of the water in the evaporator, less than a kW of heat is taken out of the ventilation air at the face. In general, the farther a plant is from the working area it is intended to cool, the lower will be its positional efficiency.

- The other main advantage of underground plant is that it implies less power having to put into pumping. As already mentioned, since the condenser water and cold water circuits are closed, all that the circulating pumps have to do is overcome the frictional resistance to flow: they do not have to lift the water through vertical distances.

8.4.2 Locating Plant on Surface

Advantages

Surface refrigeration installations have several major advantages:

- They are not limited in capacity by the amount of air available to flow through the condenser cooling towers
- Their COP's are higher, because the condenser can reject heat at a lower temperature
- Access for supervision, maintenance and repair is much easier, and if desired, cheaper refrigerants can be used
- Also, there is the opportunity for pre-cooling.

Disadvantages

The major disadvantages of these plants are as follows:

- The cost of pumping return water from settlers and dams at the bottom of the mine to surface. These costs have been dealt with in several ways:
  - One, already mentioned, is to install energy recovery turbines. These can be used to drive pumps directly (which idea has considerable practical difficulties), or to drive electrical generators, which return power directly to the mine's grid.
  - Another idea has been to send ice down the mine instead of chilled water.

    Because of the latent heat of fusion of the ice, less ice has to be sent down than the equivalent amount of water, this reduces pumping cost, by a factor of about 4.

    On the other hand, the transport of ice imposes a lot of problems.

- Surface plants also have poorer positional efficiency than an underground plant. Furthermore, like downcast air, chilled water going underground loses its cooling properties because of adiabatic compression.

8.4.3 Cooling Air Remote from the Face

It is invariably bulk cooling installations that are found remote from the face. Often, they are placed in major intake airways underground, but they can also be found at the shaft collar. In the latter case, the intention is to arrange that even in summer, the downcast air have the same temperatures as it does in winter. This arrangement has the advantage that return water does not have to be pumped through a large vertical distance.

Advantage

Sometimes, bulk air coolers are provided at some intermediate position in the shaft, where the air has become sufficiently hot as a result of auto-compression to be conducive to heat stroke conditions that affects the workers. In these cases, generally, only a part of the total downcast air is extracted and cooled, before being returned to the main downcast stream (the recirculation of air). Bulk coolers are most often placed in intake airways, again usually at a point where the air has become too hot for environmental safety. Bulk coolers have the following advantages:

SHE

Advantages

- Bulk coolers have the advantage that they are readily accessible for maintenance and repair. Also, they generally remain in one place for several years at least, so that the trouble and expense of moving them periodically is avoided.
- Another advantage is that they do not require as much piping to supply them with water as coolers close to the face do. Pipe work is expensive, especially when it has to be thermally insulated as well.

Disadvantage

Bulk air coolers usually also have poor positional efficiency. This is partly because of the loss of cooling that occurs along the piping leading to the cooler and also because of leakage of ventilation air. Only part of the air that has been cooled by the bulk installation reaches the working face, and the rest is lost.

8.4.4 Cooling Air Near to the Face

Disadvantage

The advantages and disadvantages of cooling units close to the face follow from what has just been said about bulk coolers. A great deal of cost is tied up in long runs of insulated water piping. Next, small individual units have to be moved quite frequently as the production face positions changes. Also, remote coolers are often not very accessible, which makes their inspection maintenance and repair difficult.

Advantage

On the other hand, cooling close to the face has a good positional efficiency, at least in principle. This assumes that the piping carrying cold water to the cooling units is well insulated, and that ventilation controls between the cooler and the face are effective.

8.5 Pressure Enthalpy Diagrams

8.5.1 Introduction

In this section we are going to primarily look at pressure-enthalpy diagrams and the use thereof. These diagrams will clearly show various aspects of the different phases that the refrigerant goes through.

8.5.2 The Pressure-Enthalpy Diagram

Dealing with a well known substance like water first can assist in understanding the refrigeration process.

Q

How much sensible heat is required to raise the temperature of water from 0°C to 100°C (boiling or evaporation point) at sea level?

A

(Use 1kg of water)

Water (as a liquid) temperature increase from 0°C to 100°C (Use the sensible heat Equation)

Heat = $m \times c_p \times \Delta t$ Mw x Cpw

= $1 \times 4.187 \times (100 – 0)^oC$

= 418.7 kJ

Q

How much heat is required to evaporate this 1kg of water at the **evaporation temperature of 100°C**?

A

The **latent** heat of evaporation of water 2440 kJ/kg

= 1kg x 2440 kJ/kg

= 2440 kJ

In other words, if 2440 kJ of latent heat is added to water as a saturated liquid at 100°C, **at sea level** this liquid water will completely **evaporate** into a saturated vapour at 100°C.

Q

How much super-heat (sensible heat) is required to raise the temperature of water vapour from 100°C to 110°C?

A

The saturated vapour temperature is 100°C. To heat the vapour into the super-heated region of 110°C would required:

Heat = $m \times c_p \times \Delta t$ Mw x Cpw x Δt

= $1 \times 1.884 \times (110 – 100)^oC$

= 18.84 kJ

The total amount of heat input into the water was:

= 418.7 + 2440 + 18.84 kJ

= 2877.5 kJ

The percentages if heat additions are:

Sensible heat (0°C – 100°C)	=	14.55%
Latent heat (liquid to vapour at 100°C)	=	84.80%
Super-heat (100°C – 110°C)	=	0.65%

From the above figures it can be seen that the latent heat exchange was by far the greatest component, with the added advantage that it took place at a **constant** temperature. At the Witwatersrand area, with its **lower** barometric pressure, water will boil (evaporate) at ± 95°C. Therefore, **the lower the pressure on the water, the lower is its evaporating temperature (or condensing temperature).**

These aspects, have been dealt with earlier in the course and is repeated to show its applicability in P-H diagrams. If a pressure/temperature scale is plotted against a heat scale for the above calculation, it will look as follows: (Figure 8.22).

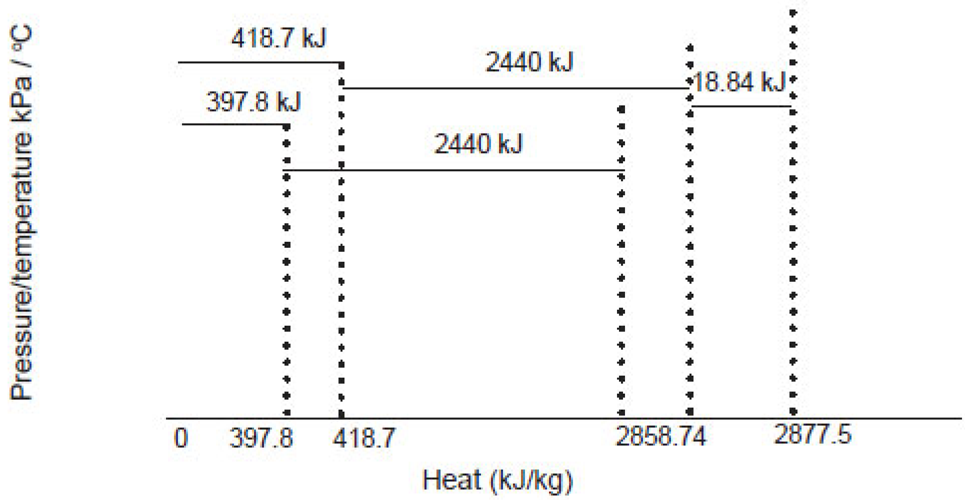


Figure 8.22 Pressure/temperature versus Heat scale

Note, that the lower the pressure/temperature value becomes, the more to the **left** (lower heat value) of the 100°C point will evaporation start.

Any point at a pressure/temperature value higher than 100°C will fall to the **right** of the original point (values were started at sea level pressures and temperatures for different phase changes). If many points are now calculated the graph will look as shown in Figure 8.23.

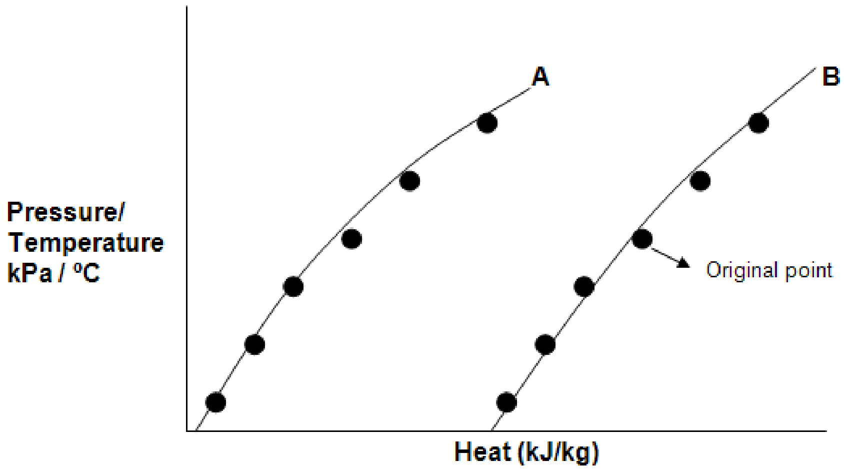


Figure 8.23 Pressure versus Enthalpy diagram plot for water

- All the points under A above show **where evaporation starts** (or where condensation stops)
- All the points under B above show where **evaporation stops** (or condensation starts). The jointed points forms the saturated liquid and saturated vapour line as shown in Figure 8.23.

The graph has now taken the form of a well-known pressure enthalpy diagram used to analyse a refrigeration plant's performance. Each refrigerant obviously has its own graph because each refrigerant has different pressure/temperature/heat characteristics than the others.

When a refrigerant is selected for a particular duty, one of the factors to consider is the air temperatures in the Return Airway (RAW) where heat will be rejected by the cooling tower into the upcast (return) air.

NB

Whatever the RAW air temperature is, **the refrigerant condensing pressure/temperature characteristic must be so much higher than the RAW air temperature that the refrigerant should reject heat into the condenser water circuit.**

Then the condenser water temperature should still be higher than the RAW air temperature so that the water can reject heat into the air.

Definition

The **pressure enthalpy diagram** (P-H diagram) is therefore a chart that shows the various properties of a cooling agent (in our example it was the water) in various situations or conditions.

The pressure enthalpy or P-H diagram shown in Figure 8.24 is a most useful diagram for showing the properties of a refrigerant. The vertical axis of the chart is the absolute pressure of the refrigerant, and the horizontal axis is its specific enthalpy (kJ/kg). A skeleton of a typical P-H chart is shown in Figure 8.24. Notice the two (dashed) lines of constant temperature, and the two solid lines labeled ‘saturated liquid line’ and ‘saturated vapour line’.

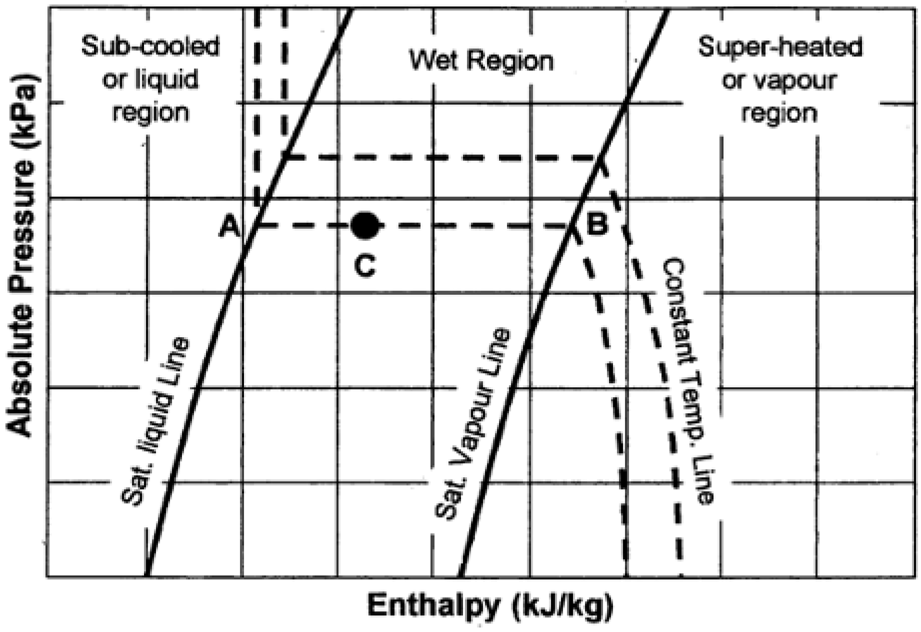


Figure 8.24 A typical P-H diagram

These two solid lines show the boundaries of regions on the chart in which the refrigerant exists in different phases, that is, the liquid phase, vapour phase, and a mixture of the two.

Definition

The liquid region consists of all the points lying to the left of the saturated liquid line. This is also sometimes called the **sub-cooled region**.

NB

Understand that the word ‘saturated’ means that the liquid cannot absorb any more energy without turning into a vapour. It does not mean saturated with moisture, as it would in the case of Psychrometry.

The vapour region consists of all the points to the right of the saturated vapour line.

It is also sometimes called the super heated region. Here again the word ‘saturated’ does not mean saturated with liquid.

The wet region comprises all the points lying between the saturated liquid and vapour lines. In this region, the refrigerant is neither a liquid nor a vapour, but a mixture of the two. You can picture it as a gas containing a mist of liquid droplets. At a point C, for instance, the mixture consists of vapour at the state B with liquid mist at the state A. the nearer C is to A, the greater quantity of liquid in the mixture, and the nearer C is to B, the more vapour.

Notice that in the wet region (from Figure 8.24), constant temperature lines are horizontal. This reflects the fact that during evaporation at constant pressure, the temperature of any substance also remains constant. So if quantity of refrigerant evaporates at the pressure of point A, it stays at the constant temperature of point A until point B is reached. Similarly for condensation: vapour at point B condenses at constant temperature until it reaches A.

Another set of lines that are shown on the P-H diagram is lines of constant density (which is not shown in Figure 8.24), which in its turn is used to calculate the volume flow rate of the refrigerant during that part of the cycle (remember that the mass flow stays constant through the cycle, provided that there are no leaks). We also know that the volume flow rates changes as the density changes.

Chart

An example and results of a complete professional P-H chart, in this case for R-134a, appears below. This is taken from 'The Mine Ventilation Practitioner's Data Book', which contains extensive information on the properties of all the most important refrigerants.

WORKED EXAMPLE 2

As before, the best way of illustrating the use of a P-H chart may be by means of a worked example. Therefore, we take the case of a quantity of R-134a at an **absolute pressure** of 572 kPa and a temperature of 5^{o}C. Answer the following questions, using the P-H diagram for this refrigerant.

Q

(**Heating of the liquid**) How much heat must be supplied to the refrigerant for it to start evaporating? Assume the pressure remains constant.

A

Plot the point representing the initial state of the 5^{o}C refrigerant and mark this point A. This point lies at the intersection of the 572 kPa pressure line and the 5^{o}C temperature line. A is in the sub-cooled region of the chart, which means that initially the refrigerant is in the liquid condition.

Now draw a horizontal line through point A to intersect the saturated liquid line and call this point B. It is at this point that the refrigerant will start evaporating.

Heat added = enthalpy at B - enthalpy at A = 228 - 202 = 26 kJ/kg

Q

(**Pressure reduction**) Is there any other way of getting the refrigerant to start evaporating, that is, without adding heat?

A

Draw a vertical line downwards through A to intersect the saturated liquid line at point C. This represents a process in which the pressure of the refrigerant is lowered. At C, it will begin to evaporate. Note that the phrase ‘without adding heat’ means that the enthalpy remains constant. The pressure at point C is 350 kPa.

Q

(**Latent heat and heating of the liquid**) From the initial state of the refrigerant, how much heat must be added to evaporate it completely? What will its temperature then be? Again assume the pressure remains constant.

A

Draw a horizontal line through A and B and extend it to intersect the saturated vapour line at point D. At D the refrigerant will have evaporated completely and will be saturated vapour. The amount of heat needed to take the refrigerant from state A to state D is:

Enthalpy at D - enthalpy at A = 410 - 202 = 208 kJ/kg

NB

The temperature of the refrigerant will be 20°C at D. This is the same as at B, because both pressure and temperature are constant in the evaporation process.

Q

(**Latent heat**) How much heat is needed to evaporate the refrigerant, that is, take it from state B to state D?

A

Heat added = enthalpy at D - enthalpy at B = 410 – 228 = 182 kJ/kg

This is of course the latent heat of vaporisation of the refrigerant at this pressure.

Q

(**Superheating**) After the refrigerant is evaporated completely, a further 10 kJ/kg of heat is added to it. What sort of heat is this, and what happens to the refrigerant? Again assume the pressure remains constant.

A

Enthalpy of the refrigerant at D is; 410 kJ/kg. Adding 10 kJ/kg to it at constant pressure, will give a point E, say, to the right of D. The enthalpy at E will be 410 + 10 = 420 kJ/kg. Plotting the point E on the chart gives its temperature as 30ºC. At D the refrigerant is superheated vapour, and being 10ºC hotter than its temperature at saturation, we would say it has 10 degrees of superheat. The heat added is sensible heat.

Q

(**De-superheating or cooling**) Assume that the refrigerant is at a pressure of 1318 kPa and a temperature of 70ºC. How much heat must be removed from it for it to start condensing? How much further heat must then be removed from it to cause it to condense completely? At what temperature will it condense? Assume the pressure remains constant.

A

Let point F be the point at which the lines for 70ºC and 1318 kPa intersect. Draw a horizontal line for F to the saturated vapour line at point G. At this point the refrigerant will start condensing. In taking the refrigerant from F to G we are removing superheat. That amount of superheat removed is equal to:

Enthalpy at F - enthalpy at G = 446 - 425 = 21 kJ/kg

Now draw a horizontal line from G to intersect the saturated liquid line at point H. At H the refrigerant will have condensed completely into a liquid. The amount of heat removed will be:

Enthalpy at G - enthalpy at H = 426 - 274 = 152 kJ/kg

This is again the latent heat of condensation at this pressure. The temperature at which the refrigerant condenses can be read off the chart as 50ºC.

Q

(**Sub-cooling**) If a further 15 kJ/kg were removed from the refrigerant, what would happen? Again assume that the pressure remains constant.

A

Draw a horizontal line to the left from point H. At H the enthalpy is 274 kJ/kg, but the removal of 15 kJ/kg gives the new point I, where the enthalpy is 259 kJ/kg. At I the temperature is 43ºC. The refrigerant has been sub-cooled and at point I, is a liquid.

Q

(**Pressure reduction**) If now, the pressure of the refrigerant were reduced from 1318 kPa to 572 kPa, without addition or removal of heat, what would happen to its temperature and state?

A

The phrase ‘without addition or removal of heat’ indicates that during the pressure reduction process, the enthalpy remains constant. Therefore, drop a vertical line through point I until it meets the line of 572 kPa pressure.

This point lies in the wet region: call it point J.

From the chart the temperature of point J is 20ºC. At this point the refrigerant is a mixture of liquid and vapour. However, since J is nearer the saturated liquid line than the saturated vapour line, it is more liquid (by mass) than vapour. The percentage of vapour that is present can be calculated from the enthalpies as follows:

$$\text{Percentage vapour at J} = \frac{\text{Enthalpy at J - Enthalpy at B}}{\text{Enthalpy at D - Enthalpy at B}} \times 100\%$$

$$= \frac{259 - 228}{410 - 228} \times 100$$

$$= 17\%$$

This means that the proportion of liquid present at point J is 83%.

The Processes on a P-H Diagram

Chart

To end this Worked Example, Figure 8.25 as part of Worked Example 3, is a skeleton P-H diagram showing all the processes that have been undergone by the refrigerant (R-134a), together with enthalpy figures.

WORKED EXAMPLE 3

Q

(**Density lines**) Assume that a quantity of R-134a at a pressure of 57 kPa (absolute) and temperature 15ºC is flowing at the rate of 10 kg/s, what is the quantity, that is, the volumetric flow rate?

A

On the sample P-H chart for R-134a given earlier, notice that the region to the right of the saturated vapour line, a number of lines labelled ‘Density’ are shown. These connect points on constant density in kg/m^3, and relate volume to mass. At the point given, the density may be read off the chart as 2.2 kg/m^3.

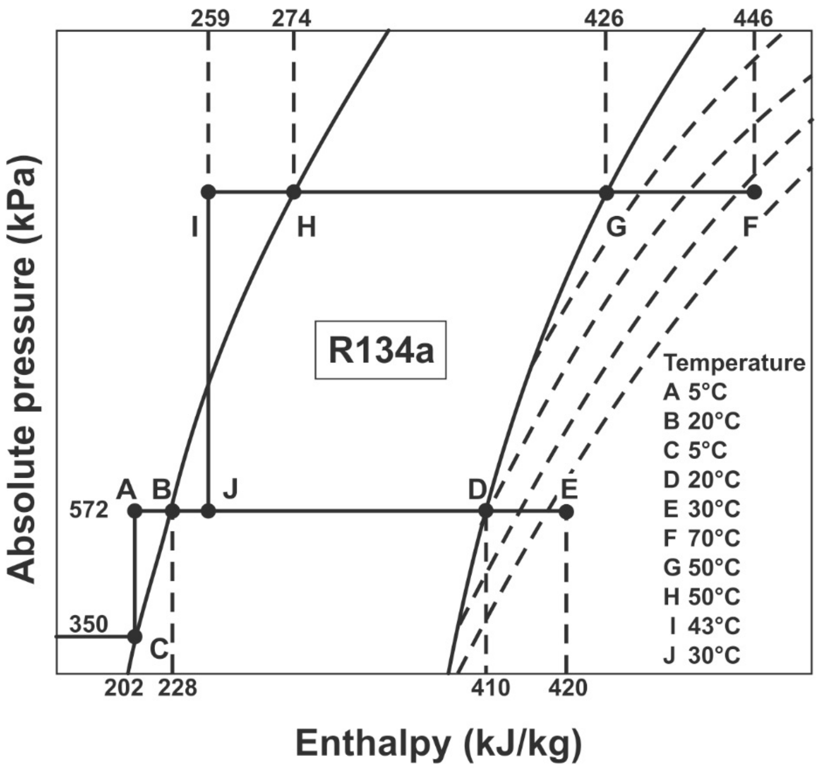


Figure 8.25 Graphical representation of R134a P-H diagram

And therefore:

$$Q = \frac{m}{\rho}$$

$$= \frac{10.0}{2.2}$$

$$= 4.6 \text{ m}^3/\text{s}$$

NB

Notice from the chart that **density increases as pressure** increases.

Q

(**Entropy**) Assume that a mass of Refrigerant 134a is at a pressure of 572 kPa (abs) and a temperature of 30ºC. If this is now compressed to a pressure of 1318 kPa (abs) and a temperature of 150ºC, what is the increase in entropy? What is the increase in enthalpy? How efficient is the compression process.

A

In a P-H diagram, lines of constant entropy slope upwards and to the right in the vapour region to the right of the saturated vapour line. See the sample chart given earlier to confirm this. The symbol S is used to denote entropy.

Definition

Entropy is a property whose physical meaning is difficult to grasp in common-sense terms. It is a relationship between heat transfer and absolute pressure, and we may think of it as a property that remains constant during a frictionless adiabatic process. Compare it with temperature, which is the property that remains constant during an isothermal process. The only reason we have some idea of what temperature means physically is that we can sense it with our skin. Otherwise it would be just as obscure as entropy.

NB

During a frictionless adiabatic process, as we have said, entropy remains constant. But if any friction is present, entropy rises during the process, and the extent of the entropy rise is an indication of how inefficient the process has been. In thermodynamic theory, sources of friction are known as irreversibility's, for a technical reason. Also, it is one of the fundamental principles of thermodynamics (the Second Law) that during an adiabatic process, entropy can only increase or remain constant: it can never decrease. In practice, friction is never absent, so that entropy always rises.

In the present case, the end states of the compression process are plotted on the skeleton P-H chart given in Figure 8.26. Using the actual chart, we find that:

Entropy at A = 1.92 kJ/kg°C Entropy at B = 2.00 kJ/kg°C

Increase in entropy = 2.00 – 1.92 = 0.08 kJ/kg°C

Reading enthalpies off the chart, we also have

Enthalpy at A = 420 kJ/kg Enthalpy at B = 534 kJ/kg

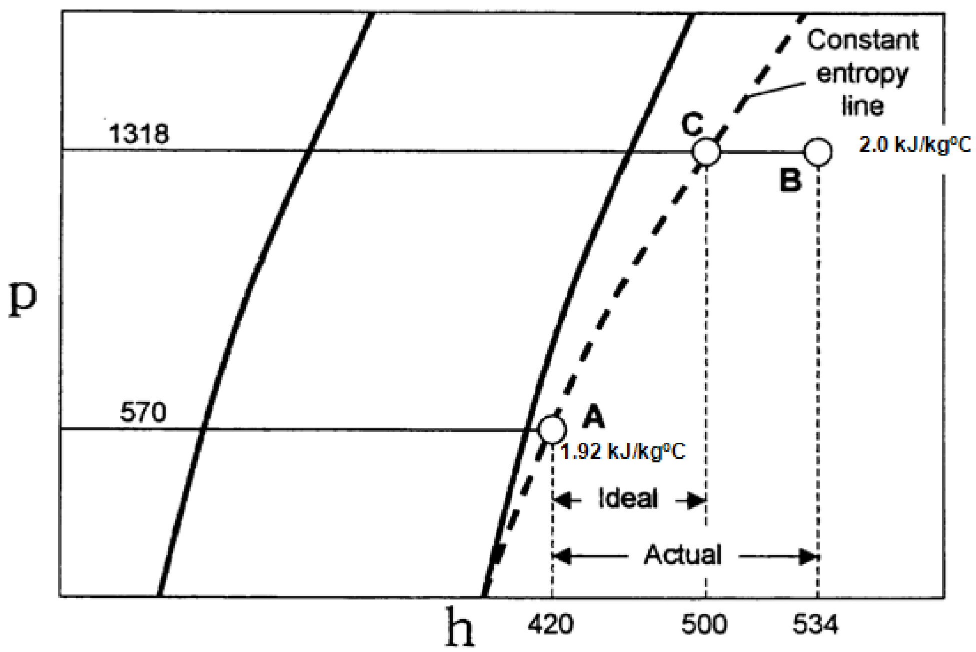


Figure 8.26 Skeleton P-H chart for Worked Example 2

Turning now to the compression process, we have seen that it results in an increase in entropy of 0.08 kJ/kg^{o}C. If it had been frictionless, the entropy would have remained constant at a value of 1.92 kJ/kg^{o}C. Plot a point C, where the line for 1.92 kJ/kg^{o}C entropy crosses the pressure line for 1318 kPa. This would have been the end point of the compression process had it been ideal, that is adiabatic and frictionless. The important point to note is that during the actual process, the enthalpy rise is greater than during the ideal process. The difference in these enthalpy rises is a measure of the efficiency of the process. In fact:

$$\text{Compression efficiency} = \frac{\text{ideal rise in Enthalpy}}{\text{actual rise in Enthalpy}} \times 100\%$$

$$= \frac{\text{Enthalpy at C - Enthalpy at A}}{\text{Enthalpy at B - Enthalpy at A}}$$

$$= \frac{500 - 420}{534 - 420}$$

$$= 0.701 \text{ or } 70.1\%$$

Notice that whether or not the compression process is ideal, the temperature of the **refrigerant increases**. The temperature lines show increase in temperature to the right and vertically upwards from any given point in the vapour phase.

8.5.3 Refrigeration Cycle on a P-H Diagram

The refrigeration cycle for a single-stage vapour compression refrigerator can be plotted on a P-H diagram when only a very few measurements have been made. It is then a simple matter to follow the various processes undergone by the refrigerant as it flows through the four main components of the plant. When used in this way the P-H diagram is very valuable aid in checking refrigerator performance. The measurements required to plot a single-stage cycle are:

- The evaporating pressure (absolute) or temperature
- The condensing pressure (absolute) or temperature
- The temperature of the vapour leaving the compressor.

Recall that in the wet region of the chart, pressure and temperature are not independent of one another. It is for this reason that at the upper and lower pressures of the cycle, either the pressure or the temperature need be known. The refrigeration cycle can now be plotted in the following way.

Draw a horizontal line between the saturated liquid and vapour lines at the condensing pressure or temperature. Mark the extremes of this line as A and X as shown in Figure 8.27.

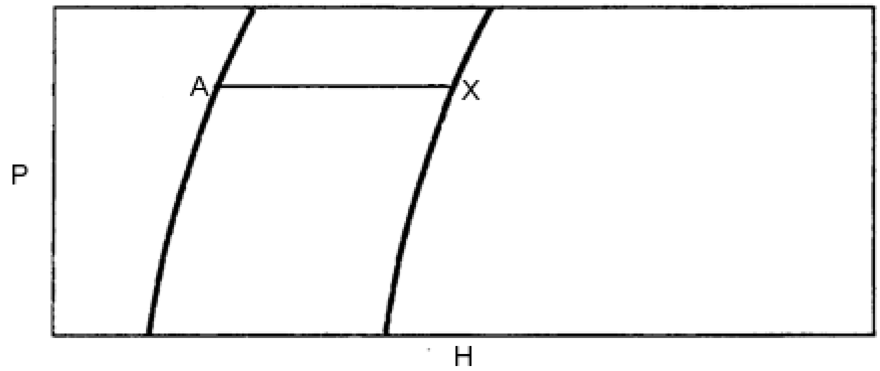


Figure 8.27 P-H diagram indicating region between saturated liquid and vapour lines

Now extend this line to the right of the saturated vapour line to intersect the compressor outlet temperature of say 150^{o}C, and call this point D. See Figure 8.28.

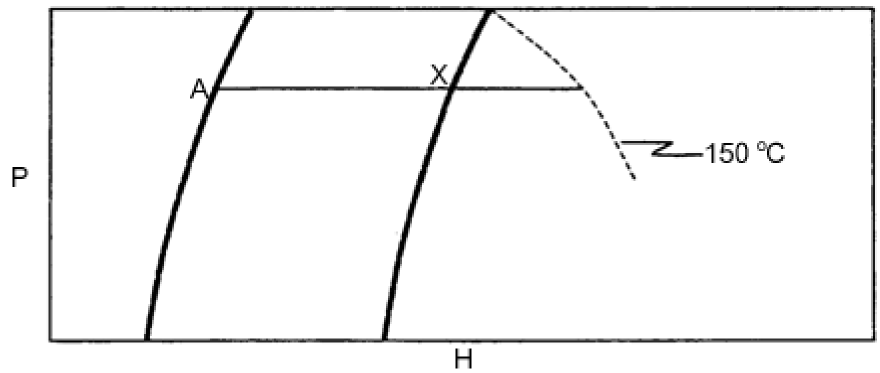


Figure 8.28 P-H diagram indicating the compressor outlet temperature

Draw another horizontal line between the saturated liquid and vapour lines for the evaporating pressure, or temperature. Mark the ends of this line F and C, as follows.

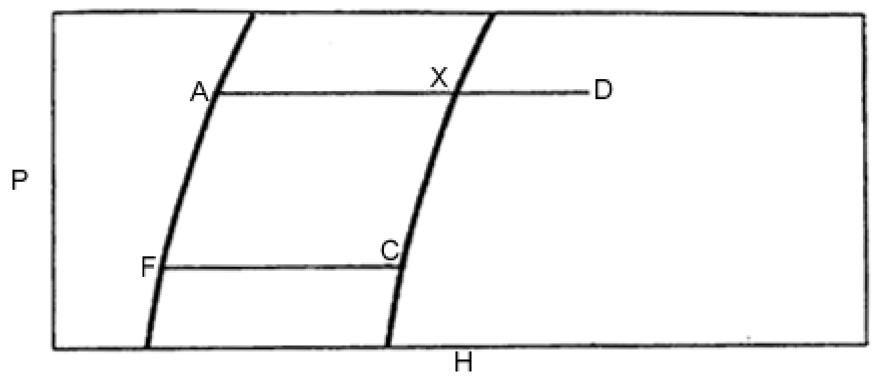


Figure 8.29 P-H diagram with additional horizontal line

Drop a vertical line from A to intersect FC at a point B. Join C and D with a straight line.

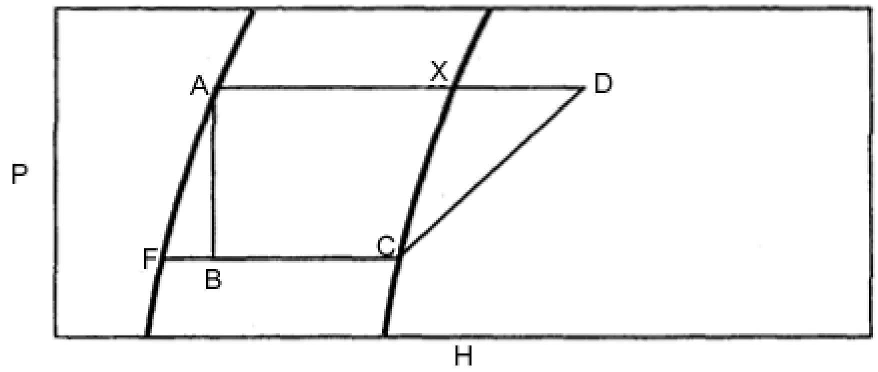


Figure 8.30 P-H diagram with lines to read off value of entropy

Read off the value of entropy at C. Now plot point E on the line XD.

The position of E is fixed by the fact that it has the same entropy as C. Join C and E with a straight dashed line. This is the ideal compression process, and CD the actual one.

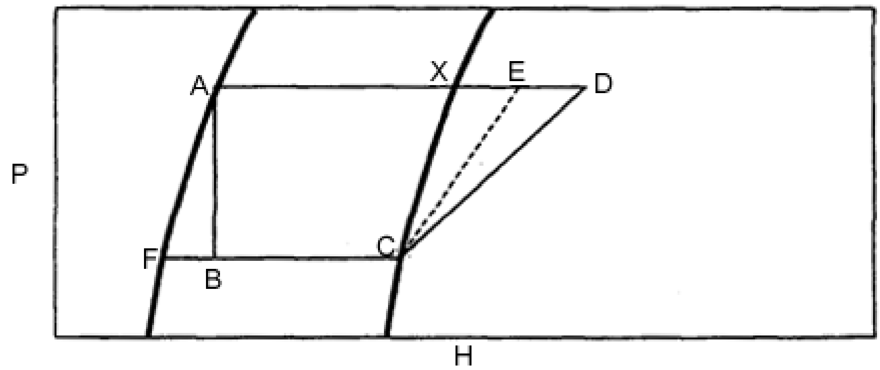


Figure 8.31 P-H diagram which include the ideal and actual compression process

The complete single-stage vapour compression cycle has now been plotted on the P-H diagram.

To help in understanding, the four main components of the refrigeration plant have been sketched in Figure 8.32. The expansion process through the expansion valve is AB; evaporation is from B to C; compression is from C to D; and de-superheating followed by condensation from D to X and X to A.

The refrigerant leaves the condenser at point A as a saturated liquid, and then enters the expansion valve. The function of this valve is to reduce the pressure of the refrigerant from that of condensation to the lower one of evaporation. The refrigerant is then at condition B, which is a wet vapour.

Since there is no work done on or by the refrigerant in passing through the valve, and since the expansion process is essentially adiabatic, it takes place at constant enthalpy. That is, the enthalpy of the refrigerant at B is the same as it is at A.

In the expansion process through the valve, the refrigerant turns partly into vapour. The heat required for the evaporation of the vapour that is present at B comes from the refrigerant itself. This means that its temperature falls. The vapour contained in the wet mixture at state B is called 'flash gas' as mentioned before. About 20% of the refrigerant flashes off typically in the expansion process.

The refrigerant then enters the evaporator at state B and there it draws enough heat from the surroundings to complete the evaporation process, so that the refrigerant leaves the evaporator as a saturated vapour at state C.

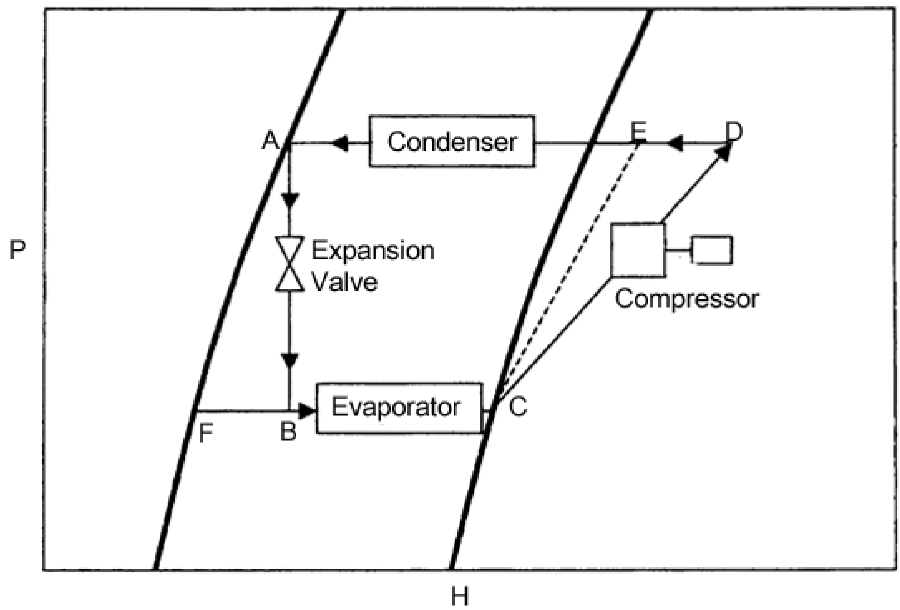


Figure 8.32 Complete P-H diagram with refrigeration components

The **'surroundings'** in the evaporator is of course the water that is being cooled by the refrigeration process.

Point C also represents the state of the refrigerant as it enters the compressor. There it is compressed until it reaches the higher pressure of the cycle, that of the condenser. As already explained, if the compression process were ideal, it would take place at constant entropy along the dashed line CE.

However, the unavoidable inefficiencies in the compression process causes more heat to be added to the vapour than would occur if the process were ideal, and the actual work of compression process takes place from C to D.

At point D the vapour is superheated, and it enters the condenser in that condition. The cooling medium in the condenser (usually water) then removes sufficient heat from the refrigerant that it is first de-superheated from D to X and then condensed from X to A. During de-superheating, the temperature of the vapour drops, but thereafter it is constant during the condensation process. The cycle then repeats itself.

In practice, real refrigeration cycles differ from what we have been describing in a number of respects. Firstly, it is generally arranged that when the refrigerant leaves the evaporator and enters the compressor, it has a few degrees of superheat. This is done by causing the refrigerant to be in the evaporator slightly longer than it needs to, in order to leave in the saturated vapour condition.

The reason for this is to ensure that the refrigerant entering the compressor contains absolutely no liquid droplets, which would in time damage the internal parts of the compressor. See Figure 8.33. To fix point C it is necessary to have the temperature of the vapour at C.

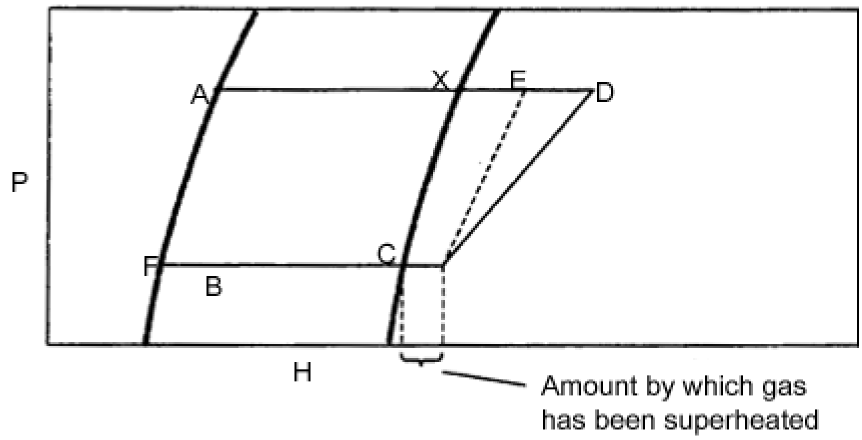


Figure 8.33 P-H diagram displaying the heat process

Similarly, when more heat is removed from the refrigerant than is required to just condense it, it will leave the condenser with a few degrees of sub-cooling. This is usually arranged to ensure that there is not too much vapour in the mixture after expansion at point B. After all, flash gas has already evaporated, and therefore will not contribute to the refrigeration effect in the evaporator.

NB

It must be noted that the heat exchange that takes place in any component of the refrigeration machine can be expressed as kJ/kg or kW.

The heat exchange expressed in kW is just the product of the mass flow of the refrigerant (m_r) and the resultant heat exchange (Enthalpy in kJ/kg) that took place in any part or component of the refrigeration machine.

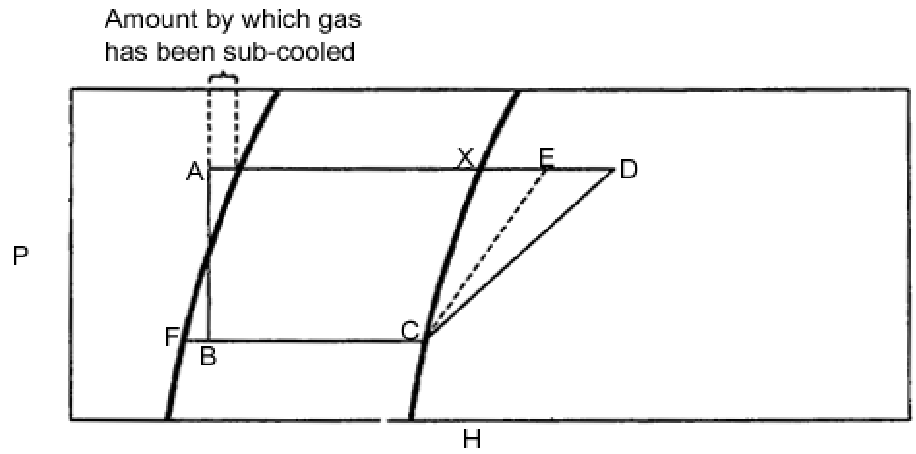


Figure 8.34 P-H diagram displaying cooling process

WORKED EXAMPLE 4

A specific mass of Freon 12 is under a pressure of 745 kPa (absolute) and the temperature that is measured is 5°C. By using a PH diagram for Freon 12, answer the following questions.

All above answers will be described with reference to the PH diagram for Freon 12, Figure 8.35.

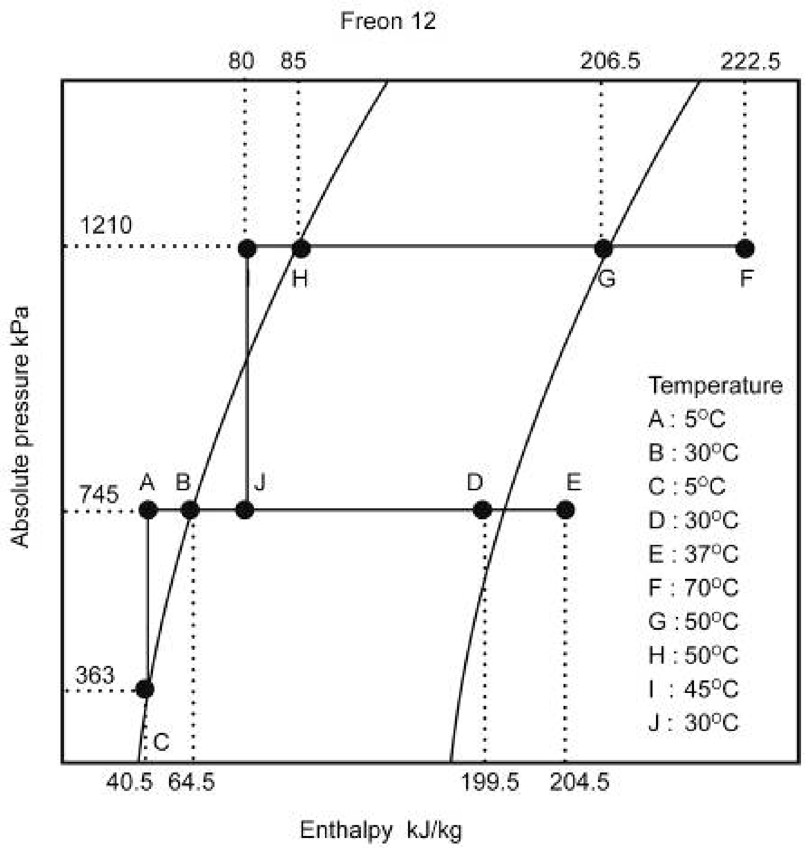


Figure 8.35 Freon 12 P-H diagram plot

Q How much heat must be added to Freon 12 before it starts to evaporate?

A Plot the point of the original phase of the Freon12 and name the point A. This will be the point at 745 kPa and the 5°C or 40.6 kJ/kg enthalpy line, because the temperature line is vertically upwards from the saturated liquid line. A is the liquid or sub cooled phase, which means that the Freon is a liquid. Draw a horizontal line (constant pressure) from A to cut the saturated liquid line at B. This is the point where the Freon will begin to evaporate.

Thus the heat added	=	enthalpy at B - enthalpy at A
	=	H_B - H_A
	=	(64.5 – 40.5) kJ/kg
	=	24kJ/kg

Q Is there any other method to get the agent to evaporate?

A To get the Freon to evaporate without the addition of heat means that the enthalpy remains constant. Draw a vertical line from A to the saturated liquid line at 5°C, at C. The Freon will now start to evaporate. Another method to get the Freon to evaporate is to reduce the pressure. In this case from 745 kPa to a pressure of 363 kPa.

Q From the given starting point of Freon 12, how much heat must be added to fully evaporate the refrigerant and what will the temperature be? Assume that the pressure remains constant.

A Draw a horizontal line from A through B and extend it to cut the saturated vapour line at D. This point D will be where the Freon is fully evaporated and will then be a saturated vapour. The amount of heat required to fully evaporate the Freon from the original position at A is:

Heat needed for evaporation	=	H_D - H_A
	=	(199.5 – 40.5) kJ/kg
	=	159 kJ/kg

The temperature of the Freon will thus be 30°C (the same temperature at B where evaporation started, because evaporation is a latent heat process).

Q From the point where Freon 12 begins to evaporate in question 3 until it is totally evaporated, there is a specific amount of energy required. Determine that amount of heat that needs to be added.

A The amount of heat required to evaporate the Freon (latent heat of evaporation) is calculated as follows:

Latent heat of evaporation $= H_D - H_B$

$= (199.5 – 64.5)$ kJ/kg

$= 135$ kJ/kg

Q After the Freon is fully evaporated, there is an additional 5 kJ/kg added. What type of heat is this and what happens to the Freon (assume constant pressure).

A The enthalpy of Freon at D = H_D = 199.5 kJ/kg

The enthalpy of Freon at the new point E $= H_E$

$= (199.5 + 5)$

$= 204.5$ kJ/kg

Draw a horizontal line from D to cut the 204.5 kJ/kg line at E.

Read the temperature at E = 37.5°C

At point D the Freon of a gas will become superheated if additional heat is added, the temperature will increase from 30 to 37.5°C.

Q Can the Freon be compressed to a pressure of 1 210 kPa without a change in temperature?

A Assume that the Freon stays a gas and draw a horizontal line (at 1 210 kPa) from the saturated vapour line to the right edge of the chart. From point E there is no way that the Freon can obtain a pressure of 1 210 kPa without a change in enthalpy and temperature.

Q Assume that the Freon gas is at a pressure of 1 210 kPa and at a temperature of 70°C. How much heat must be removed from the Freon before condensation starts to take place? How much heat must be removed before the Freon is fully condensed and at what temperature does the gas condense? Assume constant pressure.

A Plot a point F where the 1 210 kPa pressure line cuts the 70°C line. Draw a horizontal line from F to cut the saturated line at G. At this point the Freon begins to condense. The amount of heat that must be removed from the Freon to start condensation is:

Heat required for condensation $= H_F - H_G$

$= (222.5 – 206.5)$

$= 16$ kJ/kg

Draw a horizontal line from G to cut the saturated liquid line at H. At H the Freon is condensed fully to a liquid.

The amount of latent heat that must be removed from the Freon is calculated by:

Latent heat of condensation	=	$H_G - H_H$
	=	(206.5 - 85)
	=	121.5 kJ/kg

The temperature where the Freon condenses can be read from G or H = 50°C.

Q

If a further 5 kJ/kg is removed from the Freon, what will happen? Assume that the pressure is constant.

A

The enthalpy of Freon at H = H_H = 85 kJ/kg

The enthalpy of Freon at new point I = H_I = (85 - 5) is equal to 80 kJ/kg

Draw a horizontal line from H to cut the 80 kJ/kg enthalpy line at E. Read the temperature from I where the line, vertically drawn down, cuts the saturated liquid line, the temperature at I is thus equal to 45°C.

At point H the Freon is a liquid. The removal of heat means that the Freon becomes sub-cooled. The temperature lowers from 50 to 45°C.

If the pressure of the Freon is now reduced from 1 210 kPa to 745 kPa without a change in heat content, what will happen to the temperature and condition of the Freon.

To reduce the pressure from 1 210 kPa to 745 kPa without changing the heat content, means that the enthalpy stays constant. Draw a vertical line from point I to 745 kPa pressure line to cut at J. The temperature of the Freon can now be read to the left or right of point J. The temperature at J is 30°C.

The Freon at point J is in the mixed zone and is thus a mixture of vapour and liquid. Point J is closer to the liquid line than to the vapour line and is therefore more liquid than gas. The percentage of the mixture can be determined by the ratio of the enthalpy of the point between the two lines.

% gas by J	=	$(H_J - H_B / (H_D - H_B) \times 100\%$
	=	(80 – 64.5) / (199.5 – 64.5)
	=	11.5%
% liquid at J	=	$(H_D - H_J / (H_D - H_B) \times 100\%$
	=	(199.5 - 80)/ (199.5 – 64.5)
	=	88.5%

8.5.4 Other Properties of the Pressure Enthalpy Diagram

8.5.4.1 Constant Volume

To the right side of the saturated vapour line of the PH diagram (superheated zone) the Freon is a gas and has a property called constant volume. Constant volume can be linked to the specific volume of **air** that was handled in psychrometry. This gives the volume of a gas per unit mass and therefore the unit is m^3/kg. This property is used to determine the volume flow from, Q = m x v.

WORKED EXAMPLE 5

Assume that 10 kg/s Freon is at a temperature of 15ºC and has an absolute pressure of 57 kPa. What is the volume flow rate of the gas?

Answer

From the Freon 11 chart where the 57 kPa pressure line intersects the 15ºC temperature line, the constant volume can be read off as v = 0.3 m^3/kg.

$$Q = M \times v = 10 \times 0.3 = 3 m^3/s \text{ Freon gas.}$$

NB

Note that the value of the constant volume decreases as the pressure increases (Boyle's law).

8.5.4.2 Entropy

Lines of constant entropy also come in the superheated zone of the PH diagram and the gradient is from the left bottom side to the top right side of the chart. Entropy is a relationship between the heat and the temperature and is given in the unit kJ/kg°C.

Rule

In any compression process where the liquid is compressed, there will be heating. If the process is 100% effective, the entropy of the liquid will not change. As the process is not 100% effective, the entropy will change in the process.

WORKED EXAMPLE 6

Assume that a mass of Freon 12 is at a pressure of 385 kPa and a temperature of 10°C. If the Freon is now compressed to a pressure of 1 210 kPa and a temperature of 70°C; what is the increase in entropy? What is the increase in enthalpy? How effective is the compression process?

Answer

Plot the original condition of the Freon at the point of 385 kPa and the temperature of 10ºC and name it A. Plot B at 1 210 kPa and at a temperature of 70ºC.

Entropy at A, E_A	=	0.7 kJ/kg°C
Entropy at B, E_B	=	0.727 kJ/kg°C
Thus increase in entropy	=	0.027 kJ/kg°C
Enthalpy at A, H_A	=	192.5 kJ/kg
Enthalpy at B, H_B	=	222.5 kJ/kg
Increase in enthalpy	=	30 kJ/kg
Entropy at A, E_A	=	0.7 kJ/kg°C

If the compression were 100% effective the entropy would still have a value of 0.7 kJ/kg°C.

Plot a point C at the cut point of the 0.7 kJ/kg°C entropy line and the 1 210kPa pressure line.

The enthalpy at C, H_C	=	213.3 kJ/kg

For ideal compression the increase in enthalpy would be:

Increase in enthalpy	=	$H_C - H_A$
	=	(213.2 – 192.5)
	=	20.7 kJ/kg

For actual compression the increase in enthalpy would be:

Increase in enthalpy	=	$H_B - H_A$
	=	(222.5 – 192.5)
	=	30.0 kJ/kg

The efficiency of the compression will be determined by the ratio of the ideal increase in enthalpy with the actual increase in enthalpy.

$$\text{Efficiency of compression} = \frac{\text{ideal work of compression}}{\text{actual work of compression}} \times 100\%$$

$$= 20.7/30 \times 100\%$$

$$= 20.7/30 \times 100\%$$

$$= 69\%$$

Figure 8.36 shows how these points are plotted.

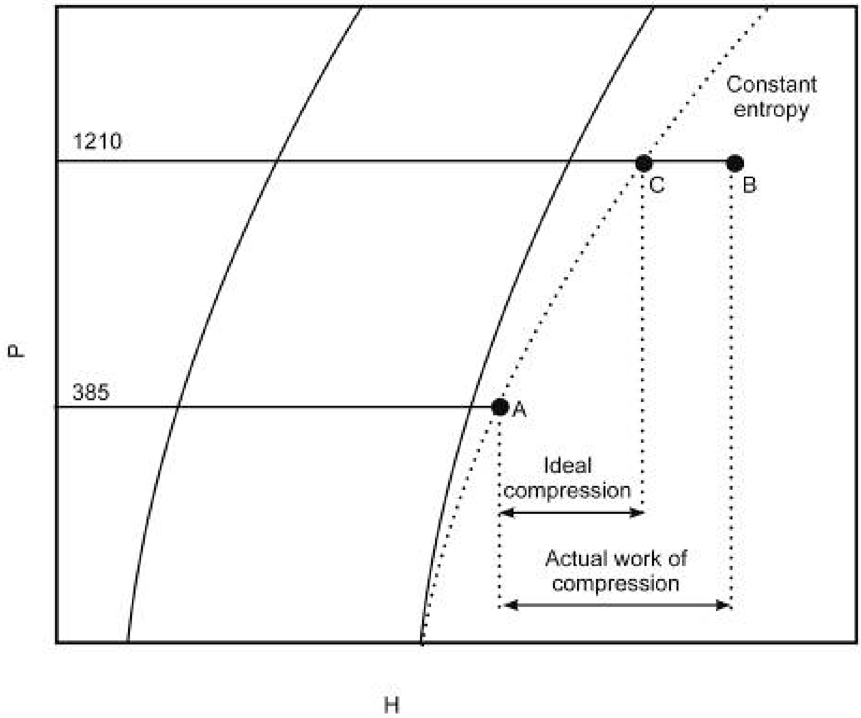


Figure 8.36 Plotting of Entropy points

Note that when the compression takes place the temperature and the enthalpy increases, regardless of whether the compression process is ideal or not. This method of determining the efficiency of the compression is very useful in determining the efficiency of the compressor in the refrigeration system.

8.6 Single Stage Fridge Plant Calculation

8.6.1 Introduction

As mines get deeper and warmer there is pressure on the environmental engineering department to improve the underground working conditions, which results in refrigeration plants underground playing a very important role. Most mines work with a design reject temperature of 28°C and in order to get this, a large amount of capital must be spent on new and improved refrigeration plants.

8.6.2 Definitions: Refrigeration Plant Measurements and Calculations

The following terms are used when the ability of an underground plant must be determined.

Carnot C.O.P (Coefficient of Performance)

The **COP** is the coefficient of performance and for the purposes of standardisation we shall in the future refer to the COP.

The Carnot COP is the best COP for a plant for specific condenser and evaporator temperatures and is expressed as:

$$\text{Carnot C.O.P} = \frac{T_1}{T_2 - T_1} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (8.1)$$

Where:

T_1 = absolute evaporator temperature (Kelvin)

T_2 = absolute condenser temperature (Kelvin)

Overall Compressor C.O.P

The overall COP of a compressor is the ratio between the cooling at the evaporator and the total electrical input to the compressor motor and is expressed as:

$$\text{Overall Compressor C.O.P} = \frac{\text{Cooling at evaporator (kW)}}{\text{compressormotor input power (kW)}} \quad \ldots. \quad (8.2)$$

Actual Compressor C.O.P or cycle COP

The above are defined as the ratio between the cooling at the evaporator and the compressor motor output power (or compressor input power (actual work of compression)) and is expressed as:

$$\text{Actual Compressor C.O.P} = \frac{\text{Cooling at evaporator } (kW)}{\text{Actual work of Compression } (kW)} \quad \ldots\ldots\ldots\ldots \quad (8.3)$$

Overall Plant COP

is the ratio of the cooling at cooling coils and the total electrical input power to the system:

$$\text{Overall Plant C.O.P} = \frac{\text{cooling at coolingcoils (kW)}}{\text{total electrical input power (kW)}} \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (8.4)$$

Overall Plant Cycle Efficiency

It is the ratio between the overall compressor COP and the optimum plant performance (Carnot COP) and is expressed as:

$$\text{Overall Plant Cycle Efficiency} = \frac{\text{Overall Compressor C.O.P}}{\text{Carnot C.O.P}} \times 100\ \% \text{. (8.5)}$$

Actual Plant Cycle Efficiency

It is the ratio between the actual compressor COP and the optimum performance (Carnot COP) and is expressed as:

$$\text{Actual Plant Cycle Efficiency} = \frac{\text{Actual Compressor C.O.P}}{\text{Carnot C.O.P}} \times 100\ \% \text{.. (8.6)}$$

Overall Compressor P.C.R (Power to Cooling Ratio)

P.C.R is known as the electrical usage/power versus the cooling and is the inverse of COP. Overall compressor PCR is the ratio between the compressor motor input power and the cooling at the evaporator:

$$\text{Overall Compressor P.C.R} = \frac{\text{Compressor motor input power (kW)}}{\text{Cooling at evaporator (kW)}} \quad (8.7)$$

Or it can be expressed as the inverse of the overall compressor COP:

$$\text{Overall Compressor P.C.R} = \frac{1}{\text{overall Compressor C.O.P}} \text{ ………… (8.8)}$$

The Nett of Actual Compressor PCR

Actual Compressor PCR is the ratio of the actual work of compression (or compressor motor output power) and the cooling at the evaporator:

$$\text{Actual Compressor P.C.R} = \frac{\text{Actual work of Compression or compressor motor output Power (kW)}}{\text{Cooling at Evaporator (kW)}} \text{(8.9)}$$

Or can be expressed as the inverse of the actual compressor COP:

$$\text{Actual Compressor P.C.R} = \frac{1}{\text{Actual Compressor C.O.P}} \text{ ………….. (8.10)}$$

Overall Plant PCR

This is the relationship between the total electrical input power and the cooing at the cooling coils:

$$\text{Overall plant P.C.R} = \frac{total\ electrical\ input\ power\ (kW)}{cooling\ at\ cooling\ coils\ (kW)} \qquad (8.1)$$

Alternatively:

$$\text{Overall plant P.C.R} = \frac{1}{\text{overall plant C.O.P}} \text{ ……………..…………... (8.12)}$$

Positional Efficiency

This is the ratio expressed as a percentage between the cooling at the working place and the cooling done by the evaporator and is expressed as:

$$\text{Positional Efficiency} = \frac{Cooling\ at\ Cooling\ Coils\ (kW)}{Cooling\ at\ Evaporator} \times 100\% \quad \ldots\ldots (8.13)$$

8.6.3 Typical Performance Figures for Fridge Plants

NB

When the performance figures of a system are calculated, there are typical values that the results can be compared to, in order to see if the plant functions effectively. The list that follows represents typical values that can be used for comparison purposes and should be memorised for comparison purposes based on your own calculation answers.

The list only gives an indication of the values at which a plant operates and must not be taken as actual values (guideline values only). The reason for this is that the plant may be designed for other working conditions.

The values however remain valuable for comparison purposes:

- Carnot C.O.P 6
- Overall Compressor C.O.P 3.5 – 4.0
- Actual Compressor C.O.P 4.0
- Overall Plant C.O.P 2.5
- Overall Cycle Efficiency 60%
- Actual Cycle Efficiency 65%
- Overall Compressor P.C.R 0.25 – 0.3
- Actual Compressor P.C.R 0.25
- Overall Plant P.C.R 0.4

WORKED EXAMPLE 7

The following readings are taken at an underground system, where Freon 11 is used as the cooling agent:

Chilled water flow rate	33.4 ℓ/s
Water at evaporator inlet	18.3ºC
Chilled water at Evaporator outlet	8.4ºC
Chilled water at cooling coil inlet	10.2ºC
Chilled water at cooling coil outlet	17.0ºC
Pump Power at cooling coil outlet	97 kW
Distance between coils and plant	643
Evaporator temperature	6.2 ºC
Evaporator pressure	52 kPa
Airflow volume at cooling coil inlet	38.5 m^3/s
Air temperature at cooling coil inlet	32.3/33.0ºC
Air temperature at cooling coil outlet	27.3/27.3ºC
Barometric pressure at cooling coils	110 kPa
Barometric pressure in the system	104 kPa
Warm water flow rate	72.6 ℓ/s
Warm water entering condenser	36.5ºC
Warm water leaving condenser	42.5ºC
Warm water pump at condenser outlet	52 kW
Condenser temperature	48.9ºC
Condenser pressure	123 kPa (meter pressure)
Compressor electrical volts	6 600 V
Compressor current usage	40.2 A
Compressor motor efficiency	92%
Power factor	0.93
Compressor gas outlet temperature	82 ºC
Air volume entering cooling tower	37.2 m^3/s
Air temperature at tower inlet	31.7/31.8ºC
Air temperature that leaves tower	39.0/39.0ºC
Barometric pressure in tower	105 kPa

Total fan power used in the cool and hot network 170 kW

Determine the following:

i. The Duty
ii. The Carnot C.O.P
iii. The Overall Compressor C.O.P
iv. The Actual Compressor C.O.P
v. The Overall plant Cycle Efficiency
vi. The Actual plant Cycle Efficiency
vii. The Overall plant C.O.P
viii. The Overall compressor P.C.R
ix. The Actual compressor P.C.R
x. The Overall system P.C.R

xi. The plant positional efficiency

xii. The line losses from system to cooling coils in W/m

xiii. The line losses from cooling coils to the system in W/m

xiv. The duty of the cooling tower

xv. The amount of make-up water required at the Cooling tower.

Answer

NB

Before any performance or working parameter can be determined, it is important to establish a **heat balance**. An error margin of 5% is allowed for water measurements taken at the evaporator and condenser. This allowable error margin value increases to 15% for measurements taken at air/water heat exchangers, for example at the cooling coils and the cooling tower. If the error margin figure is higher than the given guidelines, then the reading should be taken again to ensure correctness.

The evaporator duty (q_e) = $m_w \times C_{pw} \times \Delta t_w$

= 33.4 x 4.187 x (18.3 – 8.4)

= 1 384 kW .... (**A**)

The condenser duty (q_c) = $m_w \times C_{pw} \times \Delta t_w$

= 72.6 x 4.187 x (42.5 – 36.5)

= 1 824 kW .... (**B**)

Power at compressor motor input = $\sqrt{3}\ E \times I \times pf$

= $\sqrt{3} \times 6.6 \times 40.2 \times 0.93$

= 427 kW

The efficiency of the compressor motor = 92%

Thus power at compressor inlet = 0.92 x 427

= 393 kW .... (**C**)

This power is now transferred to the cooling agent as heat. The remaining 34 kW is transferred from the motor to the environment. If the compressor were a sealed unit, the entire 427kW would be transferred as heat to the cooling agent.

The **heat balance** as follows (for the plant measurements)

Condenser duty (B) = input power to compressor (C) + duty of evaporator (A)

1 824 $\neq$ [(393 + 1 384) = 1 777].................. (**D**)

Percentage error $= \dfrac{B - D}{B} \times 100\%$

$= \dfrac{1\,824 - 1\,777}{1\,824} \times 100\%$

= 2.6%

This is within the acceptable error margin guideline of 5% and we can now continue with the plant calculations:

i. The plant duty = evaporator duty

= 1384 kW

ii. Carnot C.O.P = $\dfrac{T_1}{T_2 - T_1}$

Where:

T_1 = $T_{absolute\ evaporator}$ = (6.2 + 273.15) = 279.2K

T_2 = $T_{absolute\ condenser}$ = (48.9 + 273.15) = 321.9K

Carnot C.O.P = $\dfrac{279.2}{321.9 - 279.2}$

= 6.54

iii. Total Compressor C.O.P = $\dfrac{\text{Cooling at evaporator (kW)}}{\text{Compressor motor input power (kW)}}$

= 1 384 / 427

= 3.24 / 1 *

= 3.24

* The meaning of the COP is clearly shown.

NB

A COP of 3.24 means that for every 1 kW of electrical power that is supplied to the compressor, there is 3.24 kW of cooling generated at the evaporator. The electrical power to the compressor motor is almost the only electrical power that the mine must pay for in order for the system to operate. The higher the COP, the more effective the system and the better the electrical input power is utilized.

iv.

Actual Compressor C.O.P $= \dfrac{\boldsymbol{Cooling\ at\ Evaporator\ (kW)}}{\boldsymbol{Compressor\ motor\ output\ power\ (kW)}}$

= 1 384 / 393

= 3.52

(This is a low figure compared to guidelines mentioned earlier)

This means that for every 1kW power delivered to the compressor (compressor motor output power is equal to the compressor input power), there is 3.52 kW cooling generated at the evaporator.

$$\text{Overall Cycle Efficiency} = \frac{\text{Overall Compressor C.O.P}}{\text{Carnot C.O.P}} \times 100\ \%$$

$$= \frac{3.24}{6.54} \times 100\%$$

$$= 50\%$$

(This is a low figure compared to guidelines mentioned earlier)

vi.

$$\text{Actual Cycle Efficiency} = \frac{\text{Actual Compressor C.O.P}}{\text{Carnot C.O.P}} \times 100\ \%$$

$= 3.52 / 6.54 \times 100\%$

$= 54\%$ (low)

(This also is a low figure compared to guidelines mentioned earlier)

vii. $$\text{Overall Plant C.O.P} = \frac{\textit{Cooling at cooling coils (kW)}}{\textit{Total electrical input power (kW)}}$$

We must first determine the cooling at the cooling coils:

Air Readings:

$q_a = m_a \times \Delta S_a$

Psychrometric chart

From psychrometric calculations or charts (110 kPa) the following information can be read off as follws:

S_a at cooling coil inlet	=	101.5 kJ/kg
S_a at cooling coil outlet	=	79.3 kJ/kg
ΔS_a (calculated)	=	101.5 – 79.3
	=	22.2 kJ/kg
v (specific volume)	=	0.873 m^3/kg

The mass flow rate air determined as follows:

$$Q = \frac{Q}{v}$$

$$= \frac{38.5}{0.837}$$

$= 46.0$ kg/s

$q_a = m_a \times \Delta S_a$

$= 46.0 \times 22.2$

$= 1\,021$ kW

Heat exchange at the evaporator (water readings at the evaporator):

$$q_w = m_w \times C_{pw} \times \Delta t_w$$
$$= 33.4 \times 4.187 \times (17 - 10.2)$$
$$= 951 \text{ kW}$$

The cooling at the cooling coil is determined by the water and air readings. The heat that is lost by the air should be equal to the heat absorbed by the water.

The two figures differ by 70 kW and the percentage error is less than 15% so we can assume the readings are correct.

NB

Readings taken at the water network are usually **more accurate** than that of the air readings. Measuring of the air temperature is not always possible and in the above instance the water calculation (951 kW) will be considered to be correct.

The Total Electrical Input Power includes the compressor motor input power plus the electrical power that is required for all the pumps and fans that make part of the cooling system

$$\text{Total Electrical Input Power} = (427 + 52 + 97 + 170) \text{ kW}$$
$$= 746 \text{ kW}$$

Now:

$$\text{Overall plant C.O.P} = \frac{\text{Cooling at cooling coils (kW)}}{\text{Total electrical input power (kW)}}$$
$$= \frac{951}{746}$$
$$= 1.28 \text{ (low)}$$

This means that for every 1 kW electrical power to the system as a whole there is 1.28 kW cooling at the cooling coils.

viii.

$$\text{Overall Compressor P.C.R} = \frac{\text{Compressor motor input power (kW)}}{\text{Cooling at evaporator (kW)}}$$
$$= \frac{427}{1\,384}$$
$$= 0.31 \text{ (typical)}$$

This means that for every 1 kW cooling given by the evaporator, 0.31 kW input power must be supplied to the compressor. This is the inverse of overall compressor COP and can also be calculated as follows:

$$\text{Overall Compressor P.C.R} = \frac{1}{\text{Overall Compressor C.O.P}}$$

$$= \frac{1}{3.24}$$

$$= 0.31$$

ix.

$$\text{Actual Compressor P.C.R} = \frac{\text{Compressormotor output Power (kW)}}{\text{Cooling at Evaporator (kW)}}$$

$$= \frac{393}{1\,384}$$

$$= 0.28 \text{ (high)}$$

This means that for every 1 kW that is delivered by the evaporator, 0.28 kW input power at the compressor is required. Remember that the compressor motor output power is the same as the compressor input power.

x.

$$\text{Overall plant P.C.R} = \frac{\text{Ttal electrical input power (kW)}}{\text{Cooling at cooling coils (kW)}}$$

$$= \frac{746}{951}$$

$$= 0.78 \text{ (high)}$$

This means that for every 1 kW cooling that is given by the cooling coils, a total of 0.78 kW electrical input power must be delivered to the system.

xi.

$$\text{Positional Efficiency} = \frac{\text{Cooling at Cooling Coils (kW)}}{\text{Cooling at Evaporator (kW)}} \times 100\%$$

$$= \frac{951}{1\,384}$$

$$= 69\%$$

This figure gives us an idea of how good the system is operating relative to its position. If the system were to be very close to the working place the efficiency would be close to 100%.

xii. Line Losses from the System to the Cooling Coils

Ideally the temperature of the chilled water that leaves the evaporator should be the same as that which enters the cooling cools. In practice this is not possible, as a result of losses that take place. Insulation of chilled water pipes helps reduce the losses, but these can almost never be totally eliminated. In this case the water leaves the evaporator at a temperature of 8.4°C and enters the cooling coils at a temperature of 10.2°C. The heat that is taken up in the pipe can be calculated by:

$$
\begin{aligned}
q_w &= m_w \times C_{pw} \times \Delta t_w \\
&= 33.4 \times 4.187 \times (10.2 - 8.4) \\
&= 252 \text{ kW}
\end{aligned}
$$

Over a distance of 643 m this means that the losses can be expressed as follows:

$$\frac{252}{643} = 392 \text{ W/m}$$

xiii. Line Losses from the Cooling Coil back to the System

NB

The line losses can also be calculated the same as above. It must be noted that the losses should be less, because the temperature difference between the air and water is less. We know that the temperature difference is the driving force for heat transfer.

Temperature of the water at the evaporator inlet = 18.3°C
Temperature of the water that leaves the coils = 17.0°C

$$
\begin{aligned}
q_w &= m_w \times C_{pw} \times \Delta t_w \\
&= 33.4 \times 4.187 \times (18.3 - 17.0) \\
&= 182 \text{ kW}
\end{aligned}
$$

We must however keep in mind that the 182 kW of heat that is absorbed by the **water inclusive is of the heat added by the chilled water pump and that the nett line heat losses** will be the difference between these two figures, therefore the line losses can be determined as follows:

$$
\begin{aligned}
\text{Line losses} &= (182 - 97) \text{ kW} \\
&= 85 \text{ kW}
\end{aligned}
$$

Over a distance of 643 m this means that the losses can be expressed as follows:

$$\frac{252}{643} = 392 \text{ W/m}$$

Psychrometric chart

xiv. To calculate the duty of the cooling tower, we must look at the properties of the air at the inlet and outlet of the cooling tower, and compare them. From the 105 kPa psychrometric chart or calculations:

S_a at cooling tower inlet	=	102.0 kJ/kg
S_a cooling tower outlet	=	145.8 kJ/kg
ΔS_a (calculated)	=	145.8 - 102
	=	43.8 kJ/kg
ν (specific volume)	=	0.874 m^3/kg
Air mass flow rate	=	$\frac{Q}{\nu}$
	=	$\frac{37.2}{0.874}$
	=	42.6 kg/s
out q_a	=	m_a x ΔS_a
	=	42.6 x 43.8
	=	1 866 kW

This figure compares well to the duty of the condenser (1 824 kW), if we consider that there is a 52 kW pump between the condenser and the cooling tower. There is no water temperature for the water entering or leaving the cooling tower. The tower is usually close to the system and therefore the heat losses in the pipe are minimal. In this case we will assume that the temperature of the water that enters the tower is the same as that which leaves. This is normally not the case.

The small amount of heat (52 kW) that is added by the circulation pump to the system can be ignored, because the temperature increase is minimal. The temperature increase as a result of the pump can be determined by:

$$\Delta t = \frac{q}{m_w \; x \; C_{pw}} = 0.17^{o}C$$

Psychrometric chart

xv. The calculation of the amount of make-up water that is required is important (as a result of the loss of water by means of evaporation and or possible dam leakages).

From the 105 kPa psychrometric chart or calculation:

r at cooling tower in	=	29.0 g/kg
r cooling tower out	=	44.4 g/kg
Δr	=	44.4 - 29
	=	15.4 g/kg

We know that the mass flow rate of the air through the tower is 42.6 kg/s (previously calculated). The rate of evaporation can now be determined.

Rate of evaporation at cooling tower	=	m_a x Δr
	=	42.6 x 15.4
	=	656 g/s
	=	0.656 kg/s or ℓ/s.

This is the amount of water that will be lost due to evaporation. We must add this amount of water. The approximate amount of make-up water required in the dam at the cooling tower is **usually double the evaporation rate, whether it is expressed as litres/hour or litres/second**.

Make-up water required	=	0.656 ℓ/s x 2
	=	1.4 ℓ/s

NB

The extra make-up water is required to prevent a build-up of dissolved and suspended solids and because of the possible carry-over of the water droplets by the air velocity and possible leakages from the dam. Remember that the water in the cooling tower sump is pumped back to the condenser.

Rule

Rule of thumb: For every kW of evaporator duty approximately 1.5 litres/hour of water will evaporate at the cooling tower, this amount however needs to be doubled based on our previous assumption mentioned. This means that the amount of water that will have to be added is therefore 3 litres/hour.

In our case the duty of the evaporator is 1 384 kW. The additional water required is:

Make-up water required	=	$\dfrac{\mathbf{1384\ x\ 3}}{\mathbf{60_{min}\ x\ 60_{sec}}}$
	=	1.15
	=	say 1.2 ℓ/s

This compares well to the previous figure calculated to determine the make-up water required.

WORKED EXAMPLE 8

The following measurements were made on a surface cooling plant using R-134a as the refrigerant:

Condensing pressure	Gauge faulty
Condensing temperature	29°C
Evaporating pressure	202 kPa gauge
Machine room barometric pressure	85 kPa
Compressor outlet temperature	50°C
Hot water flow rate	270 ℓ/s
Condenser inlet water temperature	22°C
Condenser exit water temperature	27.5°C

Plot the cycle on the P-H diagram and determine the following:

(i) The Carnot COP
(ii) The cycle or actual compressor COP
(iii) The cycle efficiency
(iv) The compressor efficiency
(v) The percentage flash gas
(vi) The cycle PCR
(vii) The mass flow rate of the refrigerant
(viii) The volumetric flow rate of refrigerant entering the compressor
(ix) The volumetric flow rate of refrigerant leaving the compressor
(x) The plant duty
(xi) The power consumed by the compressor.

Answer

(i) The absolute pressure in the evaporator is:

$$202 + 85 = 287 \text{ kPa or } 85\text{-}(-202) = 287 \text{ kPa}$$

134a is for high pressure machines. The condensing temperature is 29°C. With this information the cycle can be plotted in skeleton form as shown in Figure 8.37.

On the actual P-H chart for R-134a draw the horizontal line FC along the 287 (202 + 85) kPa pressure line. Then draw the horizontal line AX through the wet region coinciding with the 29°C temperature line. Extend this line until it intersects the 50°C temperature line in the superheated region. The cycle can then be completed as previously described. Read off all the important Enthalpy values of the cycle from the chart and enter them onto the skeleton diagram.

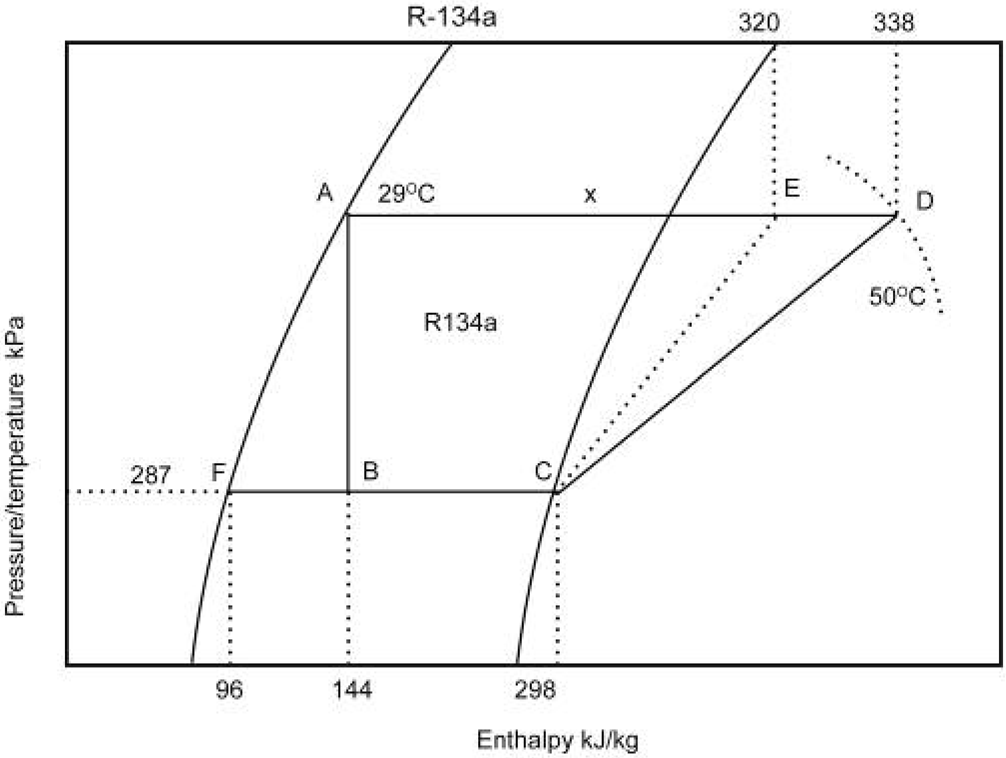


Figure 8.37 Skeleton P-H diagram for R 134a

Carnot COP

This is given by the expression:

$$\text{Carnot C.O.P} = \frac{T_1}{T_2 - T_1}$$

From the P-H diagram for R134 the temperature in the evaporator can be read off as -1ºC.

Where:

$T_1 = T_{\text{absolute evaporator}} = (-1 + 273.15) = 272.2\text{K}$

$T_2 = T_{\text{absolute condenser}} = (29 + 273.15) = 302.15\text{K}$

$$\text{Carnot C.O.P} = \frac{272.15}{302.15 - 272.15}$$

$$= 9.1$$

(ii) Cycle COP or actual compressor COP

$$\text{Actual CompressorC.O.P} = \frac{\text{Cooling at evaporator (kW)}}{\text{Actual work of compression (kW)}}$$

$$= \frac{H_B - H_C}{H_D - H_C}$$

$$= \frac{298 - 144}{338 - 298}$$

$$= 3.85$$

(iii) Cycle efficiency

$$\text{Actual Plant Cycle Efficiency} = \frac{\text{Actual Compressor C.O.P}}{\text{Carnot C.O.P}} \times 100\ \%$$

$$= \frac{3.85}{9.1}$$

$$= 0.42 \text{ or } 42\%$$

(iv) Compressor efficiency
(v)

$$\text{Efficiency of compression} = \frac{\textit{Ideal work of Compression}}{\textit{Actual work of Compression}} \times 100\%$$

$$= \frac{H_E - H_C}{H_D - H_C} \times 100\%$$

$$= \frac{320 - 298}{338 - 298} \times 100\%$$

$$= 0.55 \text{ or } 55\%$$

(vi) Flash Gas

$$\text{Percentage flash gas} = \frac{H_B - H_F}{H_C - H_F} \times 100\%$$

$$= \frac{144 - 96}{298 - 96} \times 100\%$$

$$= 24\%$$

(vii) Cycle PCR or actual compressor PCR

(viii)

$$\text{Actual Compressor P.C.R} = \frac{\text{Actual work of Compression or compressor input Power (kW)}}{\text{Cooling at Evaporator (kW)}}$$

$$= \frac{H_D - H_C}{H_C - H_B}$$

$$= \frac{338 - 298}{298 - 144}$$

$$= 0.24 \text{ or } 24\%$$

This means that for every kilowatt of cooling provided by the compressor, 0.24 kW of power has to be supplied to the compressor.

(vii) Mass flow rate of refrigerant

We can obtain the mass flow rate of the refrigerant from the information given about the condenser. The duty of the condenser can be determined as follows:

Duty of condenser $= m_w \times C_{pw} \times \Delta t_w$

$= 270 \times 4.187 \times (27.5 - 22)$

$= 6\ 218$ say 6.218 MW

NB

The mass flow rate of the water in kg/s is numerically the same as the flow rate in litres per second, since the density of water is approximately 1 kg/litre.

The duty of the condenser is the rate of **heat removal from the refrigerant**. This must be equal to the rate of heat removal from the refrigerant as it flows through the condenser. It shows as follows that:

$$\text{Condenser duty} = m_r \times \Delta H \text{ in kW}$$

$$m_r = \frac{kW}{\Delta H}$$

$$= \frac{kW}{H_D - H_A}$$

$$= \frac{6\ 218}{338 - 144}$$

$$= 32.1 \text{ kg/s}$$

This quantity could also have been obtained if the power input to the compressor or the duty of the evaporator were known. The calculation would proceed in a similar way (because the mass flow rate through the refrigeration machine stays constant – no leaks assumed).

!

If evaporator information were used, the flow rate of the chilled water and its temperature drop as it flows through the evaporator would have to be known. It happens in practice that measurements of water flow rates and temperatures are prone to error, so in determining the flow rate of the refrigerant, it is a **good idea** to use data from all three units, which is the compressor, the evaporator and the condenser.

For the current problem we have been discussing so far, we can only consider the flow rate (kg/s) through the compressor reliable if the measurements made to obtain the condenser duty are trustworthy. We have no way of checking it in this example, as insufficient information has been given.

(viii) Volumetric flow rate **entering** compressor

From the R134a P-H diagram, the density of the refrigerant at point C is read off as 13.5 kg/m^3 and therefore the volume flow of the refrigerant can be determined as follows:

$$\text{Volumetric flow rate } (Q_r) \text{ at C} = \frac{\text{mass flow of refrigerant } (m_r)}{\text{density of refrigerant } (\rho_r)}$$

$$= \frac{32.6}{13.5}$$

$$= 2.4 \text{ m}^3/\text{s}$$

(ix) Volumetric flow rate **leaving** compressor

This can be calculated as before:

$$\text{Volumetric flow rate } (Q_r) \text{ at D} = \frac{\text{mass flow of refrigerant } (m_r)}{\text{density of refrigerant } (\rho_r)}$$

$$= \frac{32.6}{37.0}$$

$$= 0.88 \text{ m}^3/\text{s}$$

Here, the density of the refrigerant at D, 37 kg/m^3, has been read off the P-H diagram at point D. Notice how much smaller the volumetric flow rate is at the exit from the compressor as the result of the compression process.

(x) The plant duty or evaporator duty

NB

The duty of the plant is synonymous with the duty of the evaporator (this is an important point to remember), which is:

$$\text{Plant duty} = m_r \times (H_C - H_B)$$

$$= 32.6 \times (298 - 144)$$

$$= 5\ 020 \text{ kW}$$

Power consumed by compressor

This quantity is determined similarly.

$$\text{Power consumed by compressor} = m_r \times (H_C - H_B)$$

$$= 32.6 \times (338 - 298)$$

$$= 1\ 304 \text{ kW}$$

Note on Fixing Point D

It is not always possible to obtain the correct temperature of the refrigerant vapour after compression (for example poor instrumentation). This means that point D can then not be plotted. However, this can be done in a different way. It can be done as follows:

- Measure the evaporator duty, condenser duty and compressor duty (all in kW), and strike a heat balance, that is, check that the sum of the evaporator duty and the compressor duty equals the condenser duty. The agreement should be within about 5% (of the condenser duty). If not, there is something incorrect about the readings taken to establish the duties, and they should be repeated and checked.

- However, assuming there is a good heat balance, calculate the mass flow rate of the refrigerant through the evaporator as follows:

$$\text{Mass flow rate} = \frac{kW}{\Delta H}$$

$$= \frac{kW_{evaporator}}{H_C - H_B}$$

Now this must be the same as the mass flow rate through the compressor, which is:

$$\text{Mass flow rate} = \frac{kW}{\Delta H}$$

$$= \frac{kW_{compressor}}{H_D - H_C}$$

From this equation, the enthalpy at D can be found, and point D plotted on the P-H diagram at the intersection of the lines for the condensing pressure and the enthalpy of D. Then if need to, the temperature at point D can be read off the chart.

Sometimes the values for the P-H diagram is not obtained from the diagram or chart but from a table and a typical example as seen in Table 8.4 as part of Worked Example 9 .The following Worked Example will explain the application quite clearly.

⌧WORK EXAMPLE 9

The following measurements were made at an underground cooling plant. The refrigerant used is Freon 11.

Evaporator

Water flow rate	28.7 ℓ/s
Inlet water temperature	22.9ºC
Delivery water temperature	7.3ºC

Condenser

Water flow rate	107.1 ℓ/s
Inlet water temperature	37.1ºC
Delivery water temperature	42.3ºC
Barometric pressure	116.5 kPa

Answer the following questions by referring to Table 8.4 which relates to the different temperatures, with reference to the respective absolute pressures:

(i) The duty of the cooling plant in kW?

(ii) What is the input power to the compressor?

(iii) How accurate would you expect the value obtained in (ii) above to be?

(iv) What approximate gauge pressure readings would you expect at the evaporator and condenser of the plant? Table 8.4 gives the absolute saturation pressure of refrigerant 11 at various temperatures.

Table 8.4 Freon 11 (Refrigerant 11) Absolute saturated pressure at various temperatures

Temperature (ºC)	Absolute saturated Pressure (kPa)
0	40.2
10	60.6
20	88.4
30	125.4
40	173.5
50	234.7
60	311.1

Answer

(i) Cooling plant duty = Evaporator Heat Exchange

q_w = $m_w \times C_{pw} \times \Delta t$

= 28.7 x 4.187 x (22.9 – 7.3) ºC

= 1 875 kW

Condenser heat exchange = $m_w \times C_{pw} \times \Delta t$

= 107.1 x 4.187 x (42.3 – 37.1)

= 2 332 kW

(ii) To obtain the heat balance, the condenser heat exchange must be equal to the evaporator heat exchange plus the compressor input power.

Compressor input power = 2 332 – 1875

= 457 kW

(iii) The answer obtained in (ii) will probably be accurate to within 5 % (this is an assumption based on previous knowledge) or 2 332 or roughly 116.6 kW in 457 kW.

NB

(iv) Important to note in this question is that the evaporator and condenser temperatures **were not given** and must therefore **be assumed** (this is a very important point in understanding the question).

NB

The evaporator temperature will be lower than the evaporator water delivery temperature of 7.3ºC

Assume a temperature for the evaporator of 5ºC (always differs from the water with approximately 3-5ºC). Then through interpolation using the values in the table above, the absolute pressure can be determined:

$$P_{abs} = 40.2 + \left(\frac{60.6 - 40.2}{2}\right)$$

$$= 50.4 \text{ kPa}$$

NB

The condenser temperature (read refrigerant temperature) will be higher than the condenser water delivery temperature of 42.3ºC. **Note: Assume** a temperature for the condenser of 45.0ºC. Then through **interpolation** using the values in the table above, the absolute pressure can be determined:

$$P_{abs} = 173.5 + \left(\frac{234.7-173.5}{2}\right)$$

$$= 204.1 \text{ kPa}$$

The gauge at the evaporator will read approximately as follows:

$$= 116.5 - 50.4$$

$$= 66.2 \text{ kPa vacuum (or suction)}$$

The gauge at the condenser will read approximately as follows:

$$= 204.1 - 116.5$$

$$= 87.6 \text{ kPa gauge}$$

SELF STUDY PROBLEM 1

The following readings are taken at an underground system:

Chilled water flow rate	35.1 ℓ/s
Water at evaporator inlet	17.2ºC
Chilled water at evaporator outlet	8.9ºC
Airflow volume at cooling coils inlet	30.4 m^3/s
Temperature of air at coil inlet	31.8/33.9ºC
Temperature of air at coil outlet	27.5/27.5ºC
Barometric pressure at cooling coils	110 kPa
Warm water at condenser inlet	38.4ºC
Warm water at condenser outlet	44.3ºC
Actual compressor C.O.P	3.8

Assume that all heat from the pumps and compressor is given off in the cooling tower.

Determine:

- The plant duty
- The plant positional efficiency
- The water flow rate in the condenser.

(**Answer**: 1220 kW; 57.7%; 62.4 ℓ/s)

SELF STUDY PROBLEM 2

The following readings are taken in an underground refrigeration system where Freon 11 is used:

Condenser pressure	meter faulty
Condenser temperature	45ºC
Evaporator pressure	60.6 kPa vacuum
System barometric pressure	110.0 kPa
Compressor outlet temperature	68ºC
Warm water flow rate	82.1 ℓ/s
Warm water temperature inlet condenser	37.4ºC
Warm water temperature outlet condenser	43.2ºC.

Plot the cycle on the PH diagram and determine (answers shown in brackets):

(i) The Carnot COP (**6.95**)

(ii) The actual compressor COP (**4.42**)

(iii) The cycle efficiency (**63.6%)**

(iv) The percentage flash gas (**18.5%)**

(v) The compressor efficiency (**72.5 %)**

(vi) The actual compressor power to cooling ratio (P.C.R) (**0.226**)

(vii) The mass flow rate of the refrigerant (**10.7 kg/s**)

(viii) The volume flow rate of the cooling agent that enters the compressor (**3.6 m^3/s**)

(ix) Volume flow rate of the cooling agent that leaves the compressor (**0.96 m^3/s**)

(x) The plant duty (**1632 kW**)

(xi) Power consumption by the compressor (**369.2 kW**).

SELF STUDY PROBLEM 3

Figure 8.38 shows measurements made in the intake to a stoping section. The air that comes from B has been cooled through a bulk air cooling system and the efficiency of the fan in the circuit is the overall fan efficiency, determine:

(i) The temperature of the air at A, assuming that there is no change in the moisture content of the air through the fan

(ii) The quantity of fresh air, which should be supplied from B to ensure that the wet-bulb temperature at the stope intake C does not exceed 28.0ºC.

(**Answers**: 30.1/36.2ºC and 18.4 m^3/s)

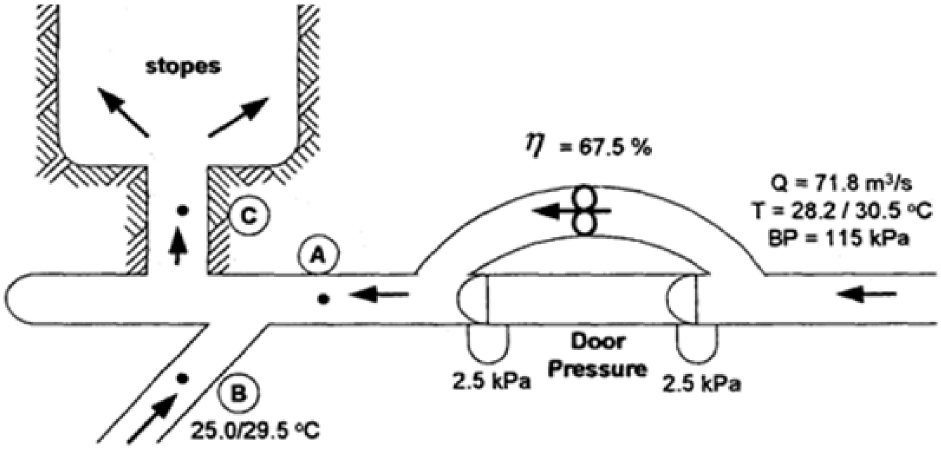


Figure 8.38 A typical stope ventilation layout

SELF STUDY PROBLEM 4

Figure 8.39 shows the layout of an underground cooling plant. Determine the following:

(i) The plant duty
(ii) The plant positional efficiency
(iii) The overall compressor COP
(iv) The amount of heat rejected in the cooling tower
(v) The Carnot COP
(vi) The overall plant cycle efficiency

(**Answers**: 1 700 kW; 81.5%; 2.9; 2 365 kW; 6.9; 42%)

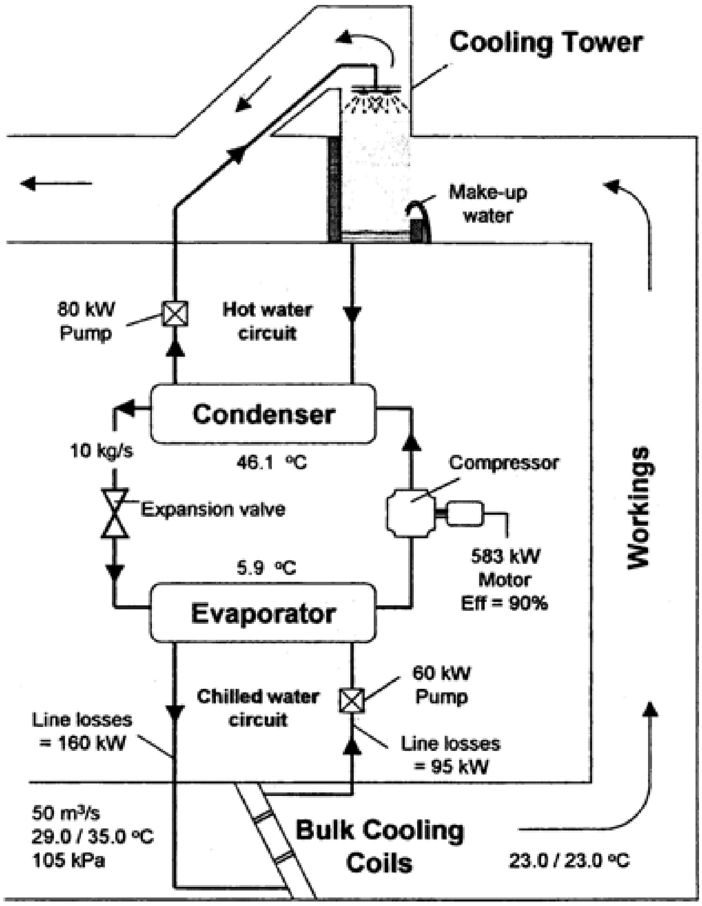


Figure 8.39 Lay-out and measurements from an underground cooling plant

8.7 Multi Stage Refrigeration Systems

8.7.1 Introduction

Until now we have concentrated primarily on single stage vapour compression systems. In practice there are many situations where multistage systems are used. We talk about single stage, two stage or three stage compression. The plotting of the cycle on a pressure enthalpy diagram is more difficult due to the multi stages.

8.7.2 Aspects Relating to Two Stage Compression

8.7.2.1 Cycle Variations

To improve the efficiency of the system, a number of various changes can be made to the vapor pressure – compression cycle, without changing the actual working of the cycle itself. These include economizers with multi stage compressors and sub cooling of the refrigerant when it leaves the condenser.

We will now discuss how the efficiency can be improved.

8.7.2.2 Sub Cooling

Sub cooling of the flowing refrigerant can be obtained by using a medium such as water or air that is at a temperature lower than that of the refrigerant. When sub cooling takes place, the amount of flash gas that is produced at the expansion valve is reduced. This means that the amount of refrigerant that needs to be circulated is reduced (very important point to note).

The overall advantage of sub cooling is to **increase the refrigeration effect** without influencing the compression. Figure 8.40 describes this situation. We can see that the advantage is not that great.

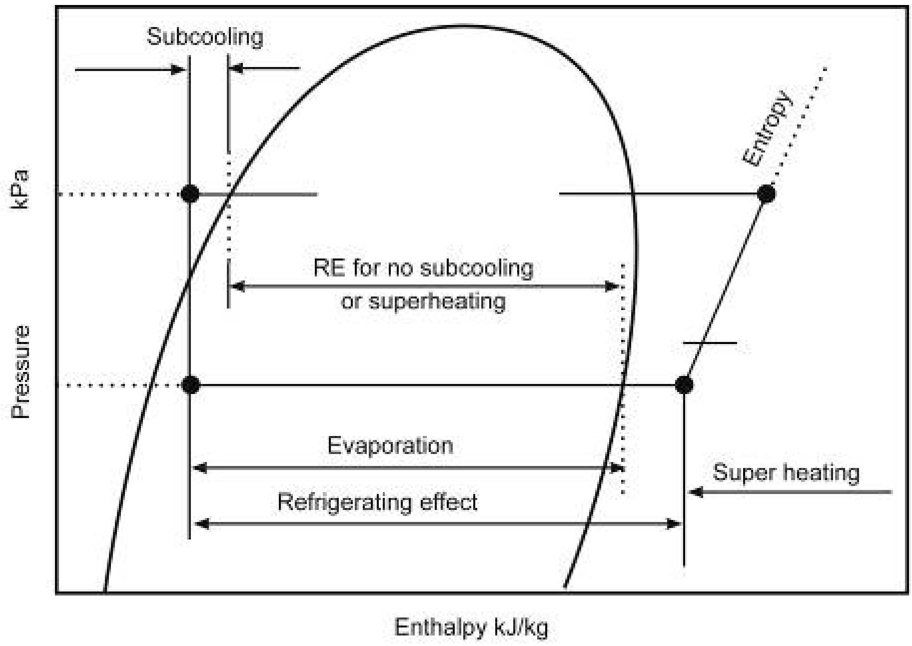


Figure 8.40 Concept of refrigeration and sub-cooling

8.7.2.3 Economisers

The properties of certain refrigerants require that some systems have multi stages in order to handle the pressure change from the evaporator to the condenser. When two or more stages of compression are used, the gas can be cooled after each stage by adding flash gas from the expansion valve.

Therefore the same number of expansion valves is required as there are stages of compression. The refrigeration cycle is the same as that for the basic compression cycle and is shown in Figure 8.41.

The flash gas from the first expansion valve after the condenser is separated from the liquid refrigerant in the economiser and is then piped back to enter the compressor between the first and second stage of compression.

This arrangement results in the hot gas from the 1st stage of compression to be cooled by the flash gas. The liquid refrigerant in the economizer is circulated to the evaporator in the usual manner. The effects of these modifications are illustrated in Figure 8.40.

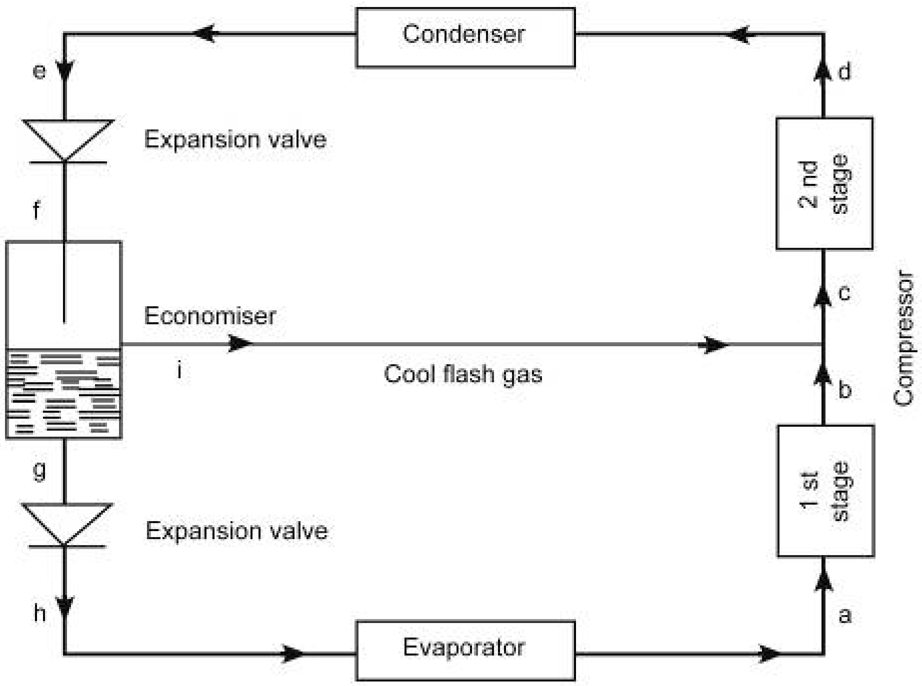


Figure 8.41 Multi stage refrigeration machine

NB

Flash gas entering stage cooling results in an increase in the refrigeration effect. The most important aspect of the economizer is that the flash gas is compressed directly from the first expansion valve at the inter stage pressure to the final condenser pressure rather than from the evaporator pressure. In the basic cycle the liquid refrigerant from the condenser is reduced until all the flash gas is compressed from the evaporator pressure to the condenser pressure.

From Figure 8.42 it is clear that for every 1 kg/s refrigerant that flows from the condenser, x kg/s flash gas forms at the first expansion valve, such that (1 – x) kg/s refrigerant flows to the next expansion valve. The amount of flash gas that is released after each expansion phase can be calculated for any specific pressure or temperature. The amount of input power that is saved by means of having economizers with flash gas between the multistage compressors is shown in Figure 8.43.

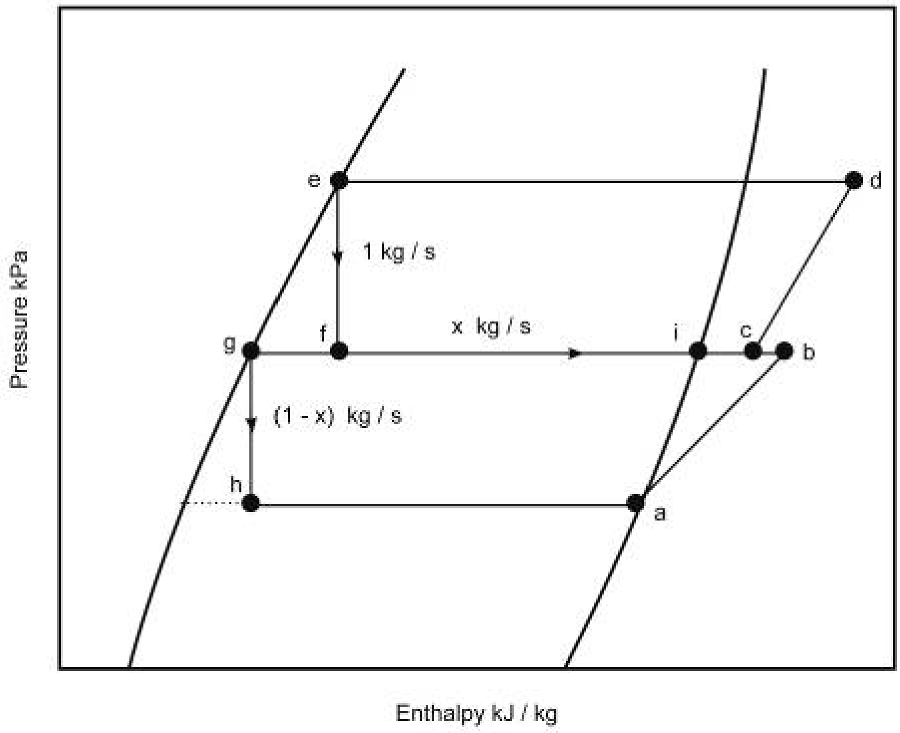


Figure 8.42 Multi stage graphical representation

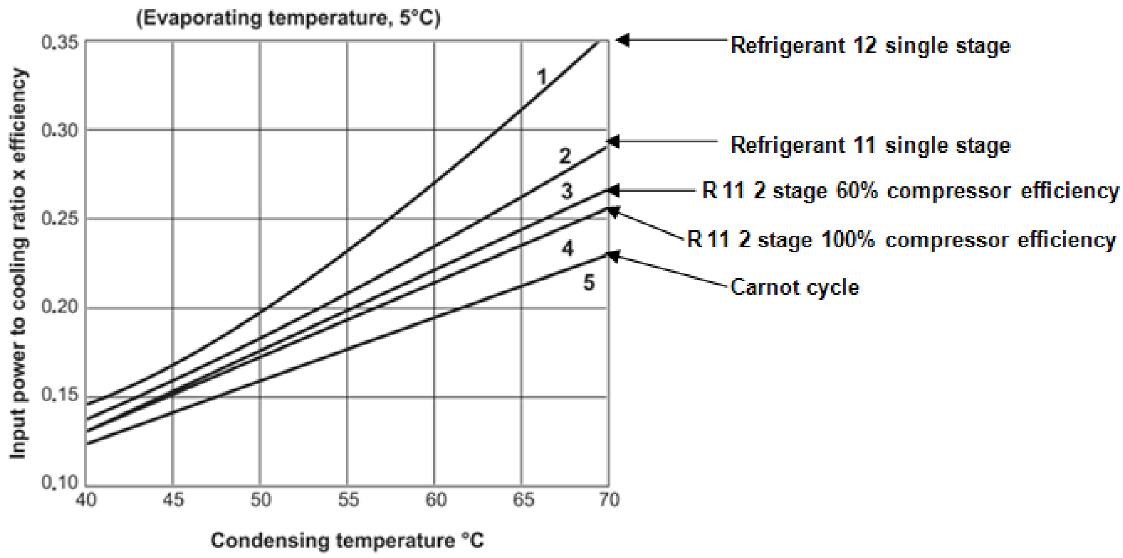


Figure 8.43 Multi stage input power to cooling ratio

The calculation of the various parameters of a multistage system will now be discussed.

WORKED EXAMPLE 10

The following readings were taken at from a two stage compressor system. The refrigerant used is Freon 11.

Evaporator

Water flow rate	26.5 ℓ/s
Water temperature in	23.9ºC
Water temperature out	7.8ºC
Evaporator pressure	50 kPa (absolute pressure).

Condenser

Water flow rate	113.6 ℓ/s
Water temperature in	41.1ºC
Water temperature out	45.9ºC
Condenser pressure	238 kPa (absolute pressure)

Compressor

Pressure between phases	114 kPa
Temperature of gas, Compressor in	5.3ºC
Temperature that leaves Compressor after 1st stage	43.1ºC
Temperature of the gas that leaves the Compressor	78.1ºC

The following needs to be determined:

(i) The performance and duty of the plant
(ii) The refrigerant mass flow
(iii) Calculation of the heat condition of the gas that enters the second stage of compression
(iv) Actual work done by the compressor per unit mass
(v) Power usage of the compressor
(vi) Total input power to the compressor (or compressor motor output power)
(vii) Ideal work done by the compressor per unit mass
(viii) The compressor efficiency
(ix) Percentage flash gas after the first expansion valve.

Answer
The PH Diagram can now be plotted as shown in Figure 8.44 (remember it is a multistage compressor).

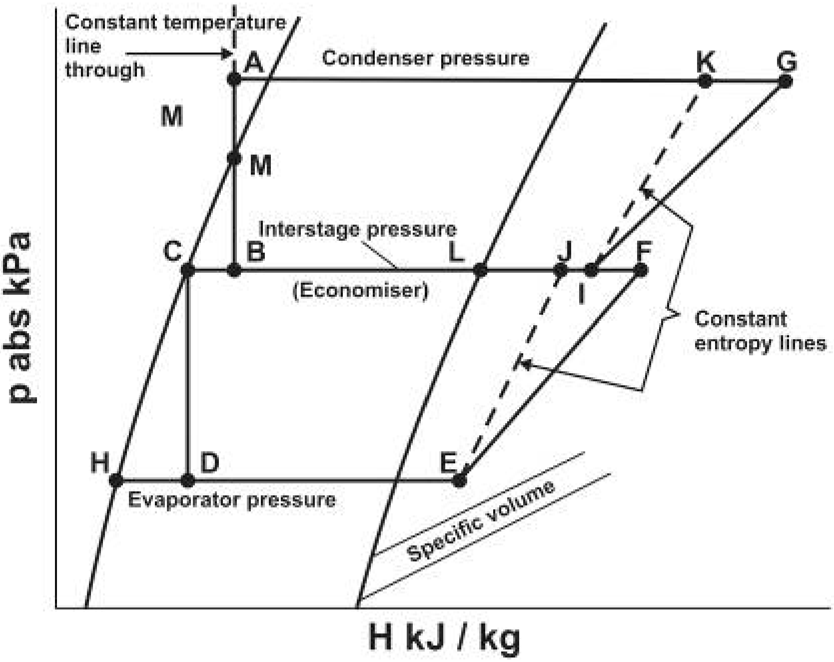


Figure 8.44 Multi stage compression

The three horizontal lines represent the absolute condenser, evaporator and interstage pressure.

G	Measured outlet temperature of the 2nd stage of the compressor
M	Lies on saturated liquid line at the measure condenser outlet temperature
A	Lies on the constant temperature line passing vertically upwards through point M
C,H	Lie on the saturated liquid line.
B,D	Lie vertically beneath M (or A) and C respectively
E	Is at the measured inlet temperature of the 1st stage of compression
F	Is the measured outlet temperature of the 1st stage of compression
J	Lies on a constant entropy line through E
L	Lies on the saturated vapor line at the temperature of the gas in the economizer (this is the flash gas after the first expansion valve)
I	Is the calculated inlet condition to the 2nd stage of the compressor. It lies on the same pressure line calculated inlet condition to stage 2 of the compressor
K	Lies on the constant entropy line through I.

Determining the Performance and the Duty

All enthalpy, temperature and specific volumes can be read directly from the chart.

Evaporator duty (q) = $M_w \times C_{pw} \times t_w$

= $26.5 \times 4.187 \times 16.1$

= 1 786 kW

Condenser duty (q) = $M_w C_{pw}\, t_w$

= $113.6 \times 4.187 \times 4.8$

= 2 283 kW

Refrigeration effect = $H_E - H_D$ = 225.5 – 57.5

= 168 kJ/kg

Heat exchanged at the Condenser

= $H_G - H_A$ = 266.0 – 780

= 188 kJ/kg

Heat content of refrigerant gas leaving economizer:

H_L = 236.5 kJ/kg

Refrigerant mass flow:

= M_{r1} = evaporator duty / refrigeration rate at evaporator

= 1 786 / 168 = 10.6 kg/s

Refrigerant mass flow:

= M_{r2} = Condenser duty / Condenser heat exchange rate

= 2 283 / 188 = 12.1 kg/s

mass flow of 'flashgas':

= M_{r3} = $M_{r2} - M_{r1}$ = 10.6 – 12.1 = 1.5 kg/s

(at economizer)

NB

Note: The mass flow of the refrigerant is only accurate if the duty of the condenser and the evaporator are correct.

Calculation of the heat condition of the gas that enters the 2nd stage of compression.

$$H_I = \frac{(m_{r3} \times H_L) + (m_{rl} \times H_F)}{m_{r2}}$$

$$H_I = \frac{(1.5 \times 236.5) + (10.6 \times 246.3)}{12.1}$$

$$= 245 \text{ kJ/kg}$$

Actual work done by compressor per unit mass

1st stage of Compressor:

$H_F - H_E$ = 246.3 – 225.5 = 20.8 kJ/kg

2nd stage of Compressor:

$H_G - H_I$ = 266.0 – 245.0 = 21.0 kJ/kg

Power usage of the compressor

1st stage

= $M_{r1}(H_F - H_E)$ = 10.6 x 20.8 = 220.5 kW

2nd stage

= $M_{r2}(H_G - H_I)$ = 12.1 x 21.0 = 254.1 kW

Total input power to the compressor:

= 220.5 + 254.1 = 474.6 kW

(Actual work of compression)

Heat Balance

If the above readings are correct then the following is true:

Condenser duty = evaporator duty + total input power to compressor (not motor)

But:

$2\,283 \neq 1\,786 + 474.6$

The percentage error is less than 5% therefore the readings can be considered accurate.

Ideal work done by compressor per unit mass

1st stage of compressor:

$H_J - H_E$ = 241.0 – 225.5 = 15.5 kJ/kg

2nd stage of compressor:

$H_K - H_I$ = 259.0 – 245.0 = 14.0 kJ/kg

Compressor efficiency

$$\text{1st stage of compressor} = \frac{(H_J - H_E)}{H_F - H_E} \times 100$$

$$\text{1st stage of compressor} = \frac{(15.5)}{20.8} x100$$

$$= 75\%$$

$$\text{2nd stage of compressor} = \frac{(H_K - H_I)}{H_G - H_I} x100$$

2nd stage of compressor $= \dfrac{(14.0)}{21.0} x100$

$= 67\%$

Percentage flash gas after 1st expansion valve

% of mass flow through condenser $= \dfrac{m_{r3}}{m_{r2}}$ x 100%

% of mass flow through condenser $= \dfrac{1.5}{12.1} x100\,\%$

$= 12.4\%$

Or as follows:

% of mass flow through condenser $= \dfrac{(H_B - H_C)}{H_L - H_C}$ x 100%

% of mass flow through condenser $= \dfrac{(78 - 57.5)}{236.5 - 57.5}$ x 100%

% of mass flow through condenser $= \dfrac{20.5}{179}$ x 100

$= 11.5\%$

% of mass flow through evaporator $= \dfrac{m_{r3}}{m_{r1}}$ x100%

% of mass flow through evaporator $= \dfrac{1.5}{10.6}$ x 100%

$= 14.2\%$

or as follows:

% of mass flow through evaporator $= \dfrac{(H_B - H_C)}{H_L - H_C}$ x 100%

% of mass flow through evaporator $= \dfrac{(78 - 57.5)}{236.5 - 78}$ x 100%

% of mass flow through evaporator $= \dfrac{20.5}{158.5}$ x 100%

$= 12.9\%$

NB

Note: The two methods that are shown above result in different answers. As we know, the **mass flow rate of the refrigerant is only as accurate as the condenser and evaporator ability**.

$$\text{Carnot C.O.P} = \frac{T_1}{T_2 - T_1}$$

Where:

T_1 = absolute evaporator temperature = 278.45 K

T_2 = absolute condenser temperature = 323.95 K

$$\text{Carnot C.O.P} = \frac{278.45}{323.95 - 278.45}$$

$$= 6.1$$

$$\text{Actual compressor C.O.P} = \frac{\text{Cooling at evaporator (kW)}}{\text{Compressor input power (kW)}}$$

$$= \frac{1\,786}{474.6}$$

$$= 3.8$$

The actual plant efficiency can be calculated as follows:

$$= \frac{\text{actual compressor COP}}{\text{Carnot COP}} \times 100\%$$

$$= 3.8 / 6.1 \times 100\%$$

$$= 62.3\%$$

$$\text{Actual compressor PCR} = \frac{{}^{**}\textit{Compressor}\ \text{input power (kW)}}{\text{evaporator duty (kW)}}$$

$$= \frac{474.6}{1\,786}$$

$$= 0.31 \text{ (typical)}$$

NB

**Remember Compressor input power = Compressor motor output power

Compressor inlet gas volume flow rate

1st stage of compressor:

= $M_{r1} \times v_E$ = 10.6 x 0.34 = 3.60 m^3/s

2nd stage of compressor:

= $M_{r2} \times v_I$ = 12.1 x 0.17 = 2.06 m^3/s

NB

Important Note

If the in-between stage pressure or the temperature of the flash gas in the economizer are not measured, the pressure can be determined by the following formula: Inter stage pressure (abs):

$$\sqrt{P_{\text{abs evaporator}} \times P_{\text{abs condenser}}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (8.14)$$

If the temperature at the outlet of the 1st stage of the compressor is not measured, then the total input power (kW) to the compressor can be determined by:

Input power to Compressor (kW):

$$= \quad m_{r1}(H_G - H_E) + m_{r3}(H_G - H_L) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (8.15)$$

8.8 System Components and Calculations

8.8.1 Introduction

The theoretical part pertaining to the above components have been dealt with before and for the purpose of this chapter, only calculations pertaining to the condenser, cooling tower and the bulk air coolers (spray ponds) will be dealt with.

There are various methods of measuring the performance of cooling towers and spray chambers. We will only look at one method as put forward by Whillier. The performance factors that will be discussed can be applied here. The first and second law of thermodynamics governs the cooling process in a tower or spray chamber and can be stated as follows:

- The first law says that the change in energy in the water must be equal to the change in energy in the air, with add-on for any evaporation. This can be represented by the following Equation:

  $$q = m_w \times C_{pw} \times \Delta t_w = m_a \, \Delta S_a$$

  In this Equation there is only partial provision made for evaporation. The difference is very small and for practical purposes can be ignored.

- The second law applied, says that in a cooling tower where heat is transferred from the condenser water, the water shall not leave the tower at a temperature lower than that of the wet bulb temperature of the intake air, and the air that leaves the tower cannot be higher than that of the water entering the tower.

  NB

  Similarly in an evaporator spray chamber where the air is cooled, the air cannot leave at a temperature lower than that of the water entering the spray chamber. The dry bulb temperature **plays no role** in the heat transfer process. **The temperature only influences the evaporation rate and not the heat transfer process.**

There are various aspects that determine the performance of a cooling tower and will now be discussed briefly.

8.8.2 Aspects Pertaining to (Condenser) Cooling Tower and (Evaporator) Spray Chambers

8.8.2.1 Water Efficiency

The water efficiency of a (condenser) cooling tower or an evaporator spray chamber or pond can be defined as the relationship of the actual change in temperature of the water that goes through the cooling tower or spray chamber and the maximum possible temperature difference. Therefore in a cooling tower where heat is given off from the condenser, the water efficiency can be expressed as:

$$\eta_w \text{ or } N_w = \frac{(t_{wi} - t_{wo})}{(t_{wi} - t_{wbi})} = \frac{\Delta t_w}{(t_{wi} - t_{wbi})} \quad \text{................ (8.16)}$$

Where:

η_w or N_w	=	water efficiency of the cooling tower
t_{wi}	=	temperature water in
t_{wo}	=	temperature water out
t_{wbi}	=	air temperature wet bulb in

If the air is cooled in a spray tunnel or evaporator spray pond, the Equation for water efficiency is as follows:

$$\eta_w \text{ or } N_w = \frac{(t_{wo} - t_{wi})}{(t_{wbi} - t_{wi})} = \frac{\Delta t_w}{(t_{wbi} - t_{wi})} \quad \text{.............. (8.17)}$$

NB

Symbols as defined before. Take note that the temperature of the water that goes into the spray chamber (from the evaporator) enters at a colder temperature than the temperature that exists in the spray chamber (heat extracted from air that flows through the chamber and transferred to the water which then flows back to the evaporator).Refer to Figure 8.13.

8.8.2.2 Air Efficiency of the (Condenser) Cooling Tower

Air arrives at the cooling tower at a wet-bulb temperature lower than the water temperature coming from the condenser. This air has to be heated by the incoming very hot water from the condenser, which is at a certain incoming temperature of the water at the cooling tower. Hence if the tower was perfect, the air could have been heated to the **same temperature** as the incoming temperature of the water to the cooling tower. However, due to inefficiencies in the tower, this is not possible. Refer back to Figure 8.13.

The air efficiency of the condenser cooling tower can be determined by comparing the **actual rise in temperature of the air** to the **maximum possible rise.**

However these ‘rises’ must be expressed in terms of sigma heat content and not temperature, **because temperature is not a linear function of sigma heat when there are changes in the moisture content.**

$$\text{air effciency of tower } \eta_a \text{ or } N_a = \frac{S_{ao} - S_{ai}}{{}^{**}S_{wi} - S_{ai}} \times 100\%$$

Psychrometric chart

**S_{wi} which is the sigma heat content of the incoming water from the condenser, can be obtained from chart no 30 of Psychrometry and Psychrometry charts – 3rd edition Sigma heat value of the air is read off from the 105 kPa psychrometric charts as discussed before.

8.8.2.3 Thermal Capacity Ratio (R)

For a specific installation the water efficiency is dependent on the flow rate and the temperature. The parameter that is used to measure these conditions is the thermal capacity ratio R and is defined as the ratio of the thermal capacity of water with that of air and is expressed as:

$$\textit{Cooling} \text{ tower capacity ratio (R)} = \frac{m_w \times C_{pw}}{m_a \times C'a} \quad \text{................ (8.18)}$$

Where:

$$C'a = \frac{S_{wi} - S_{ai}}{t_{wi} - t_{ai}}$$

R	=	thermal capacity ratio (heat transfer ratio between water and water)
m_w	=	mass flow rate of water
m_a	=	mass flow rate of the air
C_{pw}	=	heat capacity of water
C'_a	=	equivalent specific heat of moist air for a process where the vapour content changes
S	=	Sigma heat

Through manipulation of the Equation 8.18, the value for the thermal capacity ratio R can be determined as follows:

$$R = \frac{m_w \, C_{pw} \, (t_{wi} - t_{ai})}{m_a \, (S_{wi} - S_{ai})}$$

Rule

When the thermal capacity of the water is less than the thermal capacity of the air, the maximum possible value for water efficiency will be the value ‘1’ and if the value of R is greater than ‘1’ then the maximum possible value for water efficiency will be $\frac{1}{R}$

.

NB

Another very important aspect that can be derived from the above is the water-air mass flow ratio and can be expressed as follows:

$$\text{Water-Air ratio } (WAR_o) = \frac{m_w}{m_a}$$

!

This has a major impact on the efficiency of the water tower in terms of having too much air or too much water available for a specific situation and will lead to unnecessary costs if not optimised.

WORKED EXAMPLE 11

100 litres/second of condenser water from a refrigeration plant are to be cooled in a counterflow vertical tower. The following conditions apply:

Air temperature into tower	31ºC saturated
Mean Barometric pressure	115 kPa
Water temperature in	46.0ºC
Water temperature out	34.0ºC
Water/air ratio	0.5

Determine the following:

(i) The leaving air temperature
(ii) The quantity of air required
(iii) The water and air efficiencies of the tower
(iv) The ideal or optimum water/air ratio for these conditions
(v) The water and air efficiencies at optimum water/air ratio
(vi) The quantity of air for optimum conditions.

Answer

Psychrometric chart

(i) From the 115 kPa psychrometric charts the following characteristics of the air:

Apparent specific volume of the air (ν)	0.791 m³/kg
Sigma heat of the air into tower (31ºC)	92.4 kJ/kg

From the water and air heat balance, the following:

$$M_w \times Cpw \times \Delta t = M \times \Delta S$$

$$M_w \times c \times (t_{wi} - t_{wout}) = M_a \times (S_{out} - S_{in})$$

$$\text{but } M_w/M_a = 0.5 \quad \text{(given)}$$

By manipulation of the above formula, S_{out} can be determined as:

$$S_{out} = \frac{m_w \times C_{pw} \times (t_{wi} - t_{wo})}{m_a} + S_{in}$$

$$= [0.5 \times 4.187 \times (46 - 34)] + 92.4$$

$$= 117.5 \text{ kJ/kg}$$

Psychrometric chart

From the 115 kPa psychrometric chart read off the wet-bulb temperature at the sigma heat content of 117.5 kJ/kg.

Wet-bulb temperature = 35.9ºC (air saturated)

(ii) Air quantity required can be determined as follows:

$$M_w/M_a = 0.5 \text{ (given)}$$

$$M_a = M_w / 0.5$$

$$= 100 / 0.5$$

$$= 200 \text{ kg/s}$$

$$\text{but } Q = m \times \nu$$

$$= 200 \times 0.791$$

$$= 158.2 \text{ m}^3\text{/s}$$

(iii) Sigma heat of the water (46ºC) = 189 kJ/kg (using chart 30, Barenbrug at 115 kPa)

The water efficiency can be determined as follows:

$$\text{water efficiency } \eta_w = \frac{(t_{w\,in} - t_{w\,out})}{(t_{w\,in} - t_{wb\,i})} \times 100\%$$

$$= \frac{(46 - 34)}{(46 - 31)} \times 100\%$$

$$= 80\ \%$$

The air efficiency can be determined as follows (**based on Sigma heat difference):**

$$\text{air efficiency } \eta_a = \frac{(S_{a\,out} - S_{a\,in})}{(S_{water\,in} - S_{a\,in})} \times 100\%$$

$$= \frac{(117.5 - 92.4)}{(189 - 92.4)} \times 100\%$$

$$= 26\ \%$$

The air efficiency can also be determined as follows (**based on temperature difference**):

$$\text{air efficiency } \eta_a = \frac{(t_{a\ out} - t_{a\ in})}{(t_{water\ in} - t_{a\ in})} \text{x}100\%$$

$$= \frac{(35.9 - 31.0)}{(46.0 - 31.0)} \text{x}100\%$$

$$= 32.6\ \%$$

(iv) The optimum or theoretically ideal conditions exist when the temperature of the water leaving the cooling tower is equal to the wet-bulb temperature of the air entering the tower and the temperature of the air leaving the tower is equal to the temperature of the water entering the tower. The ideal water/air ratio can now be determined as follows:

$$m_{water} \text{ x c x } \Delta t_{ideal} = m_{air} \text{ x } \Delta S_{ideal}$$

$$\left(\frac{m_{water}}{m_{air}}\right)_{ideal} = \frac{(189 - 92.4)}{(46 - 31) \text{ x } 4.187}$$

$$= 1.538 \text{ say } 1.54 \text{ kg/kg}$$

(v) The optimum efficiency for both air and water will be 100%

(vi) The air quantity required at optimum conditions is as follows:

$$m_{water}/m_{air} = 1.54 \text{ (calculated above)}$$

$$m_{air} = m_{water} / 1.54$$

$$= 100 / 1.54$$

$$= 64.9 \text{ kg/s}$$

$$\text{but } Q = \text{m x } \nu$$

$$= 64.9 \text{ x } 0.791$$

$$= 51.3 \text{ m}^3\text{/s}$$

8.8.2.4 Factor of Merit F

Definition

The theory above is further developed and a term called the **merit factor** is defined, this is to quantify the basic performance characteristics. The factor of merit for a cooling tower is a property that can be compared to the UA value of a heat exchanger. The factor of merit is the value of the water efficiency (η_W) and if the value of the water capacity ratio (R) is equal to 1 then it relates to an ideal cooling tower (which in practice is not possible). At this stage there is no theory available to guess the value of F and it must therefore be determined experimentally. Figure 8.45 shows the relationship between water efficiency, factor of merit and thermal capacity ratio R.

Rule

A perfect cooling tower will therefore have a factor of merit of 1 as mentioned before. The value in practice for a good factor of merit is 0.6. The factor of merit is dependent on the design of the installation. It is relatively **independent** of water and air temperature and barometric pressure and within limits the water- and airflow rates. If the value of F of a tower is available, the duty or performance of the tower under a wide spread of working conditions can be determined. The relationship between these different entities is clearly shown in Figure 8.45.

The following symbols and equations are used in calculating the factor of merit for the tower:

m_w = water mass flow rate (kg/s)

m_a = Air mass flow rate (kg/s)

$$\text{Cooling tower capacity ratio (R)} = \frac{m_w \times C_{pw}}{m_a \times C'a}$$

$$\text{Where} \quad C'a = \frac{S_{wi} - S_{ai}}{t_{wi} - t_{ai}}$$

$$\text{water effciency of tower } N_w = \frac{t_{wi} - t_{wo}}{t_{wi} - t_{wbi}} \times 100\%$$

$$\text{air effciency of tower } N_a = \frac{S_{ao} - S_{ai}}{S_{wi} - S_{ai}} \times 100\%$$

Or

$$N_a = N_w \times R$$

$$\text{Effectiveness of the tower} = E = N_w \quad \text{if } R < 1$$

Or

$$\text{Effectiveness of the tower} = E = N_a \quad \text{if } R > 1$$

$$\text{Factor of merit (F)} = \frac{1}{\left[1+R\left(\frac{1}{E}-1\right)\right]} \quad \text{when } R > 1 \text{ and } E=N_a$$

$$\text{Factor of merit (F)} = \frac{1}{\left[1+\frac{1}{R}\left(\frac{1}{E}-1\right)\right]} \quad \text{when } R < 1 \text{ and } E = N_w$$

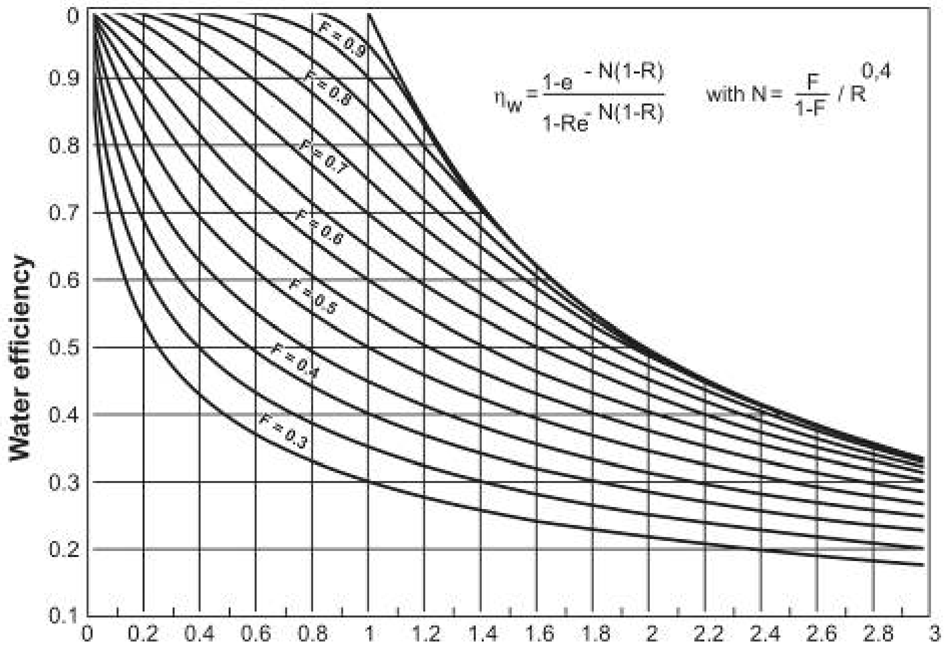


Figure 8.45 Water efficiency, factor of merit and thermal capacity ratio

NB

(Note: e as shown in the diagram in Figure 8.45 is an exponent and must not be read with R, as Re the Reynolds number).

The heat transfer processes that takes place between the refrigerant and the water and the air all have certain characteristics associated with it. To have a common reference guide pertaining to the various temperatures in the whole process (that is refrigerant, water and air temperatures) a standard reference diagram is shown in Figure 8.46.

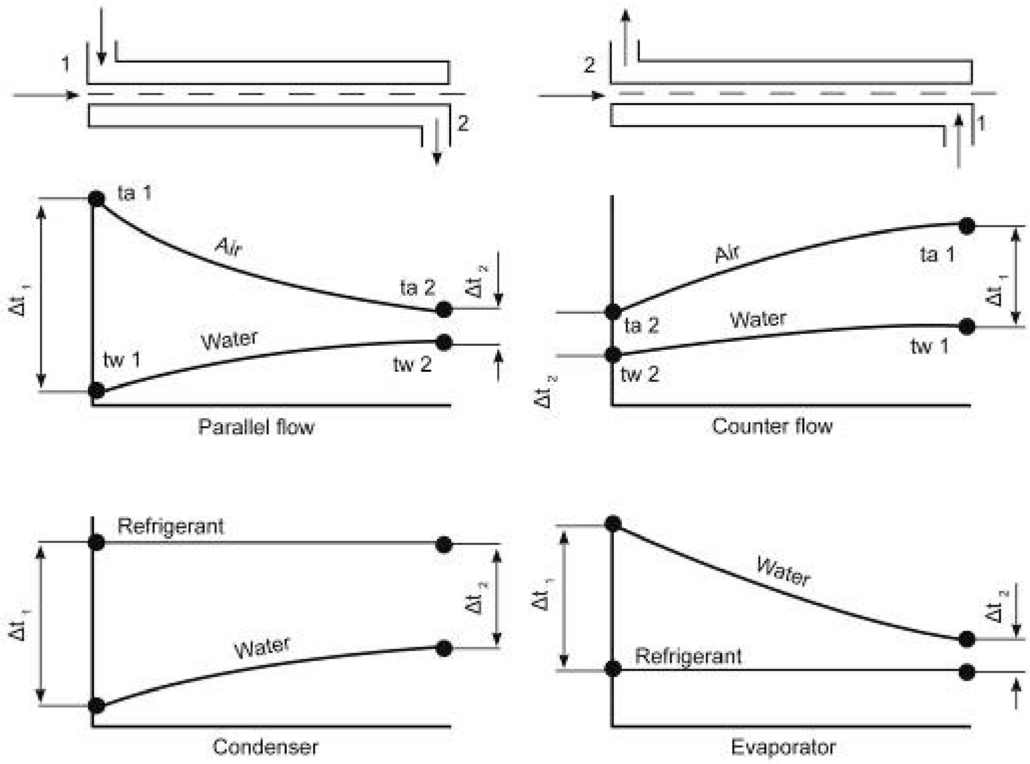


Figure 8.46 Associated temperatures for various types of heat exchangers

The use of the log mean temperature difference (LMTD) will become evident in the sections to follow and is applicable to the condenser and the evaporator.

Condenser

The function of the condenser is to remove the heat of compression and the heat extracted by the refrigerant in the evaporator from the refrigerant gas. The refrigerant gas is condensed back into a liquid in the condenser. The heat transfer in the condenser can be calculated as follows:

$$q = M_w \times C_{pw} \times \Delta t \quad \text{or} \quad q = UA \times LMTD$$

For a condenser the LMTD can be expressed as follows:

$$LMTD = \frac{\Delta t_1 - \Delta t_2}{2.303 \log_{10} \dfrac{\Delta t_1}{\Delta t_2}}$$

And where, Δt_1 and Δt_2 are the differences in the temperature between the water and the refrigerant at the inlet and outlet of the condenser.

$$\frac{1}{UA} = \frac{1}{h_i A_i} + \frac{1}{h_f A_i} + \frac{x_t}{k_t A_m} + \frac{1}{h_o A_o}$$

In the Equation for determining the UA value (W/^{o}C) $\frac{1}{h_i A_i}$ represents the heat transfer due to convection occurring on the inside surface of the tubes, $\frac{x_t}{k_t A_m}$ represents the heat transfer due to conduction through the tubes (k_t, the conductivity of the tube) and $\frac{1}{h_o A_o}$ is the heat transfer due to condensation on the outside surface of the tubes (h_o, the outside heat transfer coefficient). The inside heat transfer coefficient h_i, depends on the water velocity in the tubes, the mean water temperature and the inside diameter of the tubes and can be calculated as follows:

$$h_i \quad = \quad \frac{5\ 680\,(1+0.015t_m)\,[\text{velocity}\ U_w^{0.8}]}{d_t^{0.2}}$$

Where:

t_m	=	average water temperature in ^{o}C
d_t	**=**	**tube diameter in m**
velocity U_w	**=**	**velocity of the water in m/s**

The outside heat transfer co-efficient h_o depends on both the refrigerant and the design of the condenser and can be calculated from careful plant tests if a fouling factor is assumed. The actual values vary considerably but are typically about 2 500 W/m^{2o}C.

The term $\frac{1}{h_f A_i}$ makes an allowance for the build-up of scale and dirt which may occur on the inside surface of the condenser tubes, $\frac{1}{h_f}$ is known as the fouling factor and a high fouling factor can affect the performance of a plant. Condenser tubes should be cleaned regularly to remove scale and dirt. Typical values for the fouling of the condenser tubes are (memorise these as figures as guidelines for use in calculations):

Clean tubes	0.0001 m^{2} ^{o}C/W
Dirty tubes	0.0002 m^{2} ^{o}C/W
Very dirty tubes	0.0005 m^{2} ^{o}C/W

Note: the fouling factor is represented by the formula $\frac{1}{h_f}$, which means that if $\frac{1}{h_f} = 0.0001$ **then** $h_f = \frac{1}{0.0001}$ to be substituted in the formula.

The UA value for different fouling factors can be calculated in a similar matter and a graph of UA against $\frac{1}{h_f}$ can be plotted and is shown in Figure 8.47.

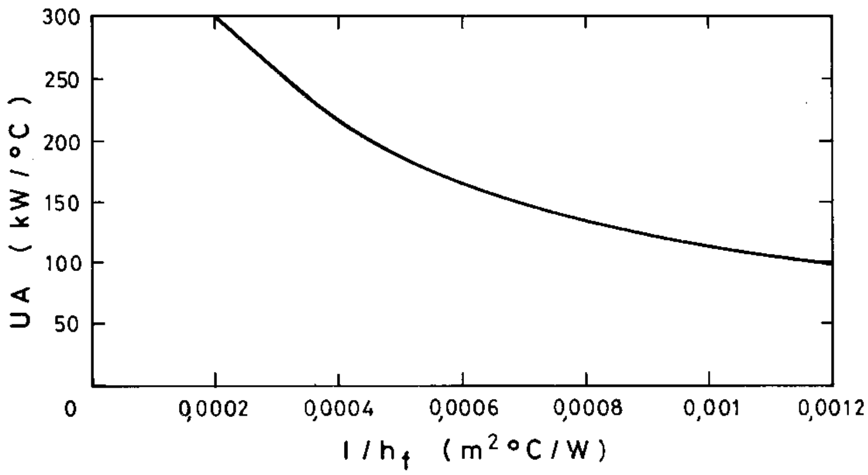


Figure 8.47 UA value versus fouling factor

Cooling Tower and Spray Chamber Calculations

WORKED EXAMPLE 12

The following readings are taken at an underground system, where Freon 11 is used as the cooling agent:

Warm water flow rate	72.6 ℓ/s
Warm water entering condenser	36.5 °C
Warm water leaving condenser	42.5 °C
Condenser Temperature	48.9 °C
Condenser Pressure	123 kPa (meter pressure)
Air volume entering cooling tower	37.2 m^3/s
Air temperature at tower inlet	31.7/31.8°C
Air temperature that leaves tower	39.0/39.0°C
Barometric pressure in tower	105 kPa
Total fan power (cool and hot network)	170 kW

Determine:

(i) Water efficiency of the tower
(ii) Air efficiency of the tower
(iii) Factor of merit of the cooling tower.

Answer

(i) Water efficiency of the cooling tower: Assuming that the line losses between the condenser and the cooling tower are small enough to be ignored, hot water arrives at the tower at a temperature of 42.5ºC. This hot water now has to be cooled by the incoming air (hot return air from workings, but still cooler than the hot water coming from the condenser), which is at a temperature of 31.7/31.8ºC.

If the tower was a perfect one and could transfer all the necessary heat, the water could be cooled from 42.5ºC to 31.7ºC, which is the wet-bulb temperature of the incoming air into the tower. However, because the tower is not a perfect one, the water is actually only cooled from 42.5ºC to 36.5ºC.

From this, the water efficiency of the tower can be determined by comparing the actual cooling of the water to the maximum possible cooling of the water.

$$\text{water effciency of tower } N_w = \frac{t_{wi} - t_{wo}}{t_{wi} - t_{wbi}} \times 100\%$$

$$= \frac{(42.5 - 36.5)^{\circ}C}{(42.5 - 31.7)^{\circ}C}$$

$$= 55.6\%$$

(ii) Air efficiency of the cooling tower: Air arrives at the cooling tower at a wet-bulb temperature of 31.7ºC. This air has to be heated by the incoming very hot water from the condenser, which is set at a temperature of 42.5ºC. Hence if the tower was perfect, the air could have been heated to a maximum temperature of 42.5ºC wet-bulb. However, the air is actually only heated to a temperature of 39.0ºC wet bulb.

The air efficiency of the tower can be determined by comparing the actual rise in temperature of the air to the maximum possible rise. However these ‘rises’ must be expressed in terms of sigma heat content and not temperature, **because temperature is not a linear function of sigma heat when there are changes in the moisture content.**

$$\text{air effciency of tower } N_a = \frac{S_{ao} - S_{ai}}{S_{wi} - S_{ai}} \times 100\%$$

$$= \frac{(145.8 - 102.0)}{(174.0^{**} - 102.0)}$$

$$= 60.8\%$$

However, before this value can be used the R value must be calculated first, R values greater and smaller than 1 have different application values to determine the factor of merit as shown in the examples that follow.

Psychrometric chart

**S_{wi} which is the sigma heat content of the incoming water from the condenser, can be obtained from chart no 30 of Psychrometry and Psychrometry charts – 3rd edition. Sigma heat of the air obtained from the 105 kPa psychrometric charts.

(iii) Factor of merit of the cooling tower

The following symbols and formulae are used in calculating the factor of merit for the tower:

m_w = water mass flow rate (kg/s)

m_a = Air mass flow rate (kg/s)

$$\text{Cooling tower capacity ratio (R)} = \frac{m_w \, x \, C_{pw}}{m_a \, x \, C'_a}$$

$$\text{Where } C'_a = \frac{S_{wi} - S_{ai}}{t_{wi} - t_{ai}}$$

$$\text{water effciency of tower } N_w = \frac{t_{wi} - t_{wo}}{t_{wi} - t_{wbi}} \times 100\%$$

$$\text{air effciency of tower } N_a = \frac{S_{ao} - S_{ai}}{S_{wi} - S_{ai}} \times 100\%$$

Or

N_a = N_w x R

Effectiveness of the tower = E = N_w if R < 1

Or

Effectiveness of the tower = E = N_a if R > 1

$$\text{Factor of merit (F)} = \frac{1}{\left[1+\frac{1}{R}\left(\frac{1}{E}-1\right)\right]} \quad \text{when } R>1 \text{ and } E = N_a = N_w \times R$$

$$\text{Factor of merit (F)} = \frac{1}{\left[1+\frac{1}{R}\left(\frac{1}{E}-1\right)\right]} \quad \text{when } R<1 \text{ and } E = N_w$$

Now, the water mass flow rate, m_w = 72.6 kg/s

And air mass flow rate, m_a = 42.6 kg/s (from Q and ρ values)

$$\text{Cooling tower capacity ratio (R)} = \frac{M_w \times C_{pw}}{M_a \times C'_a}$$

$$\text{where } C'_a = \frac{S_{wi} - S_{ai}}{t_{wi} - t_{ai}}$$

$$= \frac{174.0 - 102.0}{42.5 - 31.7}$$

$$= 6.67$$

$$\text{Cooling tower capacity ratio (R)} = \frac{m_w \times C_{pw}}{m_a \times C'_a}$$

$$= \frac{72.6 \times 4.187}{42.6 \times 6.67}$$

$$= 1.07$$

From the above notes:

N_w = 0.556

N_a = N_w x R (1.07 x 0.556)

= 0.59

NB

Note that the air and water efficiencies are expressed as a ratio to 1 and not 100. The factor of merit for the tower can be determined as follows:

$$\text{Factor of merit (F)} = \frac{1}{\left[1+R\left(\frac{1}{E}-1\right)\right]} \quad \text{when } R>1 \text{ and } E = N_a$$

$$= \frac{1}{\left[1+1.07\left(\frac{1}{0.608}-1\right)\right]}$$

$$= 0.59$$

Charts are more conveniently used to determine the value of F when the values of R, N_w and N_a are known and this will now be dealt with.

WORKED EXAMPLE 13

The following readings are taken during a routine inspection of an underground cooling tower:

Air network	
Air volume flow	213 m^3/s
Barometric Pressure	112.5 kPa
Temperature inlet	30.0 / 34.0°C
Temperature outlet	40.9 / 40.9°C
Water network	
Flow rate	536 ℓ/s
Inlet temperature	43.4°C
Outlet temperature	36.1°C

Calculate the following:

(i) The ability of the cooling tower
(ii) The water efficiency
(iii) The thermal capacity Ratio (R)
(iv) The factor of Merit
(v) Predict the in and outlet temperatures of the water for the above mentioned tower when two new refrigeration plants are installed and the heat that can be removed is increased to 23 250 kW. Assume that the factor of merit, the air and water quantities and the inlet temperatures of the air remains the same.

Answer

(i) It is important to first determine all the relevant psychrometric properties of air. The properties can be given to you in order to speed up the calculation process.

$$S_{ao} = 151.2 \text{ kJ/kg}$$
$$S_{ai} = 89.1 \text{ kJ/kg}$$
$$S_{wi} = 172.0 \text{ kJ/kg}$$
$$t_{wi} = 43.4^{o}C$$
$$t_{wo} = 36.1^{o}C$$
$$t_{wbi} = 30.0^{o}C$$
$$v = 0.813 \text{ m}^3\text{/kg}$$
$$\begin{aligned} m_a &= Q/v \\ &= 213 / 0.813 \\ &= 261.993 \text{ say } 262 \text{ kg/s} \end{aligned}$$

We must first do a heat balance calculation:

For water:

$$q = m_w \times C_{pw} \times \Delta t_w$$
$$= 536 \times 4.187 \times (43.4 - 36.1)$$
$$= 16\ 383 \text{ kW}$$

For air:

$$q = m_a \Delta S_a$$
$$= 262 \times (151.2 - 89.1)$$
$$= 16\ 270 \text{ kW}$$

The heat balance from the duties, as calculated for air and water is done as follows:

(16 383 – 16 270) /16 383 x 100% = 0.7% which is acceptable

(ii) Water efficiency

$$\eta_w = \frac{\Delta t_w}{(t_{wi} - t_{wbi})}$$
$$= \frac{7.3}{13.4}$$
$$= 0.54$$

(iii) Thermal capacity factor R

$$\text{Cooling tower capacity ratio (R)} = \frac{m_w x\ C_{pw}}{m_a\ x\ C'a}$$

$$\text{where } C'a = \frac{S_{wi} - S_{ai}}{t_{wi} - t_{ai}}$$
$$= \frac{172.0 - 89.1}{43.4 - 30}$$
$$= 6.19 \text{ say } 6.2 \text{ kJ/kg}^oC$$

$$\text{Cooling tower capacity ratio (R)} = \frac{m_w x\ C_{pw}}{m_a\ x\ C'a}$$
$$= \frac{536 \times 4.187}{262 \times 6.2}$$
$$= 1.382$$

(iv) Factor of Merit

From Figure 8.45 and with R = 1.382 and $\eta_w = 0.54$ we can read the factor of merit as F = 0.65.

(v) Prediction of water temperature, a trial and error method is used

- Guess the water temperature entering the tower.
- Determine the value for R, determine η_w from Figure 8.45 determine the temperature spectrum Δt_{wi} and calculate t_{wi} and t_{wo}. The procedure must be repeated until the correct values of t_{wi} and t_{wo} are determined.

Let us guess an incoming water temperature of 50°C

Thus:

t_{wi} = 50°C (S_{wi} = 234 kJ/kg, from graph)

t_{ai} = 30°C (S_{ai} = 89.1 kJ/kg)

To determine C'_a, we must remember that sigma values will also change because we are working with a new temperature (therefore a new C'_a).

C'_a = 7.245 kJ/kg°C (check the answer)

R = 1.182 (check the answer)

From Figure 8.45 for

R = 1.182 and F = 0.65 we get η_w = 0.6

Change in water temperature (spectrum)

Definition

We must remember that the definition of **water efficiency** is the change in the actual water temperature difference (t_{wi} - t_{wo}) to the best possible temperature change (t_{wi} - t_{wbi}). We know that the water efficiency can determine the spectrum (t_{wi} - t_{wo}).

$$(t_{wi} - t_{wo}) = \eta_w \times (t_{wi} - t_{wbi})$$
$$= 0.6 \times (50 - 30)$$
$$= 12^oC$$

Duty of the tower

$$= m_w \times C_{pw} \times \Delta t_w$$
$$= 536 \times 4.187 \times 12$$
$$= 26\ 931 \text{ kW}$$

This is higher than the expected 23 250 kW and we must repeat the iteration process to get as close as possible to the original heat exchanged value.

By simple mathematic manipulation and with the duty of the tower equal to the new heat exchanged values (through the iteration process), the various temperatures are:

t_{wi} = 47.9°C and t_{wo} = 37.5°C.

8.8.2.5 Cooling Coil Calculations

Definition

Cooling coils are used to transfer heat from air to chilled water. The duty of the coils can be determined by (the chilled water comes from the evaporator to then cool the hot air):

$$q = m_w \times C_{pw} \times \Delta t_w = m_a \Delta S_a$$

The calculations make use of readings taken from the airway and do not account for the heat transfer as a result of condensation. Heat transfer at the cooling coils is more complicated because heat is transferred through condensation and convection.

An acceptable approximation for mining conditions will be where the intake air into the coil is close to saturated and thus does not cover a wide spectrum of temperatures, and the difference in wet bulb temperature of the air and the water will be considered to be the driving force for the heat transfer. We can therefore say that:
$q = UA \times LMTD$
Where:

$$LMTD = \frac{\Delta t_1 - \Delta t_2}{2.303 \log_{10} \frac{\Delta t_1}{\Delta t_2}}$$

And where Δt_1 and Δt_2 is the difference between the wet-bulb temperature and the water temperature, on each end of the cooling coils (refer to Figure 8.46). The heat exchange is also dependent on whether it is a parallel or counter flow heat exchanger.

The UA of the coil can be determined by:

$$UA = \frac{1}{\frac{1}{h_i A_i} + \frac{1}{h_{fi} A_i} + \frac{x_t}{k_t A_m} + \frac{1}{h_{fo} A_o} + \frac{1}{h_o A_o}}$$

This Equation really serves to define h_o in terms of the approximation mentioned above.

The duty of the coil is calculated with help of observations on the air and water and by using the following Equations:

$$q = m_w \times C_{pw} \times \Delta t_w \text{ and } q = m_a \Delta S_a$$

This is used to determine the correctness of the readings and is calculated with the logarithmic average temperature difference between the air and water. The cooling coil duty is used to calculate the actual UA value.

$$UA = q / LMTD$$

The UA value for the measure air and water flow rates are determined with the assumption that the cooling coil is clean and this means that the inside and outside fouling factor $1/h_{fi}$ en $1/h_{fo}$ is equal to zero, thus:

$$UA = \frac{1}{\frac{1}{h_i A_i} + \frac{1}{h_o A_{oe}}}$$

Where:

A_{oe} = the effective outside area

The x_t / $k_t A_m$ term is also ignored because it is negligibly small, compared to the other terms.

The other two unknown factors h_o and h_i must be determined and this is done by:

$$h_i = \frac{1.4 \times 5\,680 \times (1 + 0.015\, t_m)\, U_w^{0.8}}{d_t^{0.2}} \quad \text{.............. (8.19)}$$

$$h_o = 195 \times \left(\frac{m_a}{A}\right)^{0.549} \quad \text{............................ (8.20)}$$

Where:

U_w = velocity of the water in tube (m/s)

t_m = average water temperature ^oC

d_t = inside diameter of the tube in m

(m_a / A) = mass flow rate of the air per unit area (kg/m^2s)

This step ensures that a comparison can be drawn between the clean UA value and the actual UA value. The two calculated values will be very much the same if the coils are clean and differ greatly if the coils are dirty.

The duty of a clean coil with the same air and water is calculated from the following:

i. The capacity rates ($kW/^oC$) i.e. the heat transfer rate for a unit hange in temperature for air (Ch) and water (Cc) where:

$$C_h = m_a\, x \frac{\Delta S_a}{\Delta t_{wb}} \quad kW/^oC$$

$$C_c = m_w\, x\, C_{pw} \quad kW/^oC$$

The smaller of the two rates is C_{min} and the larger C_{max}

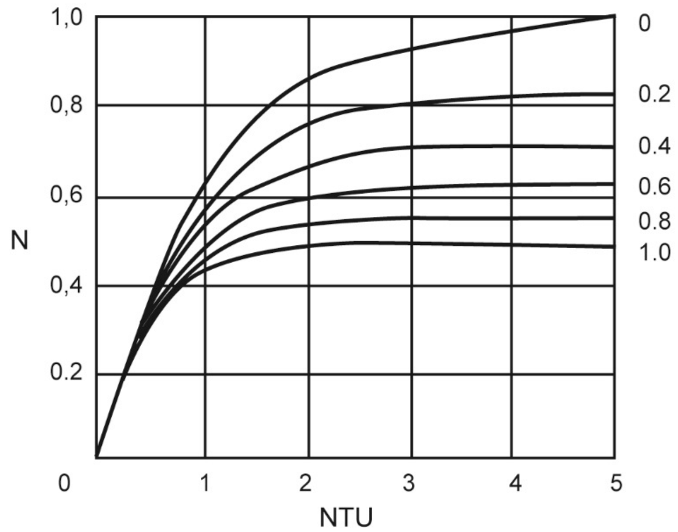
ii. The capacity rate ratio Z is determined by:

$$Z = \frac{C_{min}}{C_{max}}$$

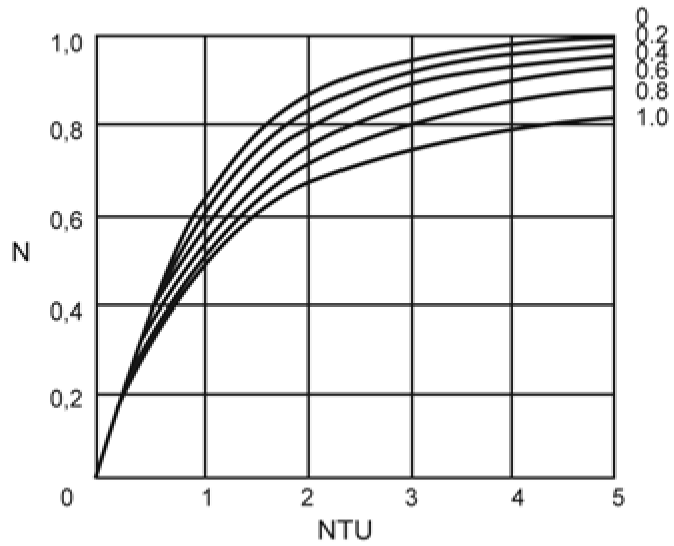
iii. The amount of transfer units can now be determined and this is referred to as NTU (number of transfer units)

$$NTU = \frac{\text{UA of clean coil } (kW/^{o}C)}{C_{min}\ (kW/^{o}C)}$$

iv. The efficiency of the coil (N) is determined by the capacity rate ratio (Z), the amount of NTU's and the type of flow. See Figure 8.48.



Parallel flow



Counter flow

Figure 8.48 Number of transfer units

v. The ability of the cooling coil for the same working conditions is determined by:

$$q = N \times C_{min} \times (t_{wbi} - t_{wi})$$

Figure 8.49 shows the relationship between the UA and the various airflow rates with the different water flow rates. What is clearly visible in the sketch is the UA value of the cooling coil increases as the flow rate of the water increases (this aspect was discussed earlier in detail).

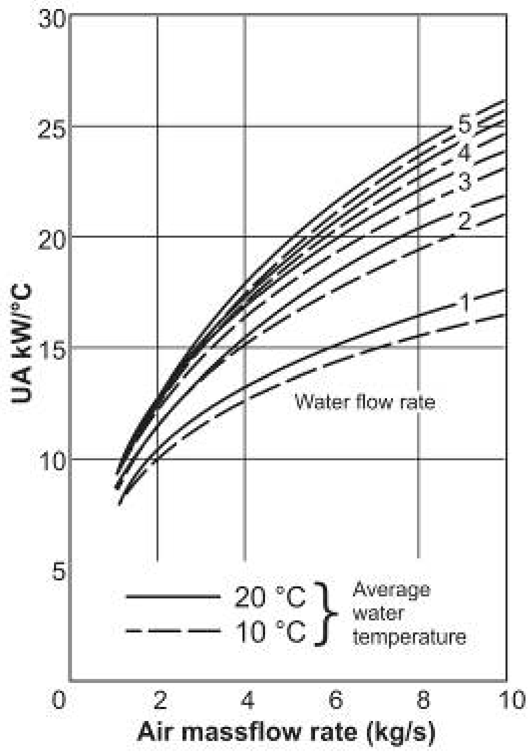


Figure 8.49 UA value for various temperatures and air and water flow rates

Above calculations can be tedious and the duty of the cooling coil can also be determined by using the follow steps:

$$Q = m_w \times C_{pw} \times \Delta t_w$$

The water efficiency can be determined by:

$$\eta_w = \frac{(t_{wo} - t_{wi})}{(t_{wbi} - t_{wi})} = \frac{\Delta t_w}{(t_{wbi} - t_{wi})}$$

Therefore logically the ability of the cooling coil can be determined by:

$$q = m_w \times C_{pw} \times \eta_w \times (t_{wbi} - t_{wi})$$

Ramsden combines the first three terms of the above Equation into a factor K for a specific coil at a specific air and water flow rate.

$$q = K \times (t_{wbi} - t_{wi})$$

Values for K can be obtained for different water and air flow rates by first determining UA, C_{min}, NTU, Z and η_w. Table 8.5 shows different values for K for different flow rates. Figure 8.50 shows it graphically.

Table 8.5 K values for various flow rate of water and air

K Value read off inside table						
Air flow rate (m^3/s)						
Water flow rates (l/s)	1	2	4	6	8	10
1	2.87	3.56	3.91	4.02	4.07	4.1
2	3.58	5.15	6.48	7.05	7.37	7.56
3	3.84	5.85	7.98	9.05	9.72	10.16
4	3.95	6.24	8.93	10.41	11.4	12.09
5	4.03	6.5	9.57	11.4	12.64	13.58

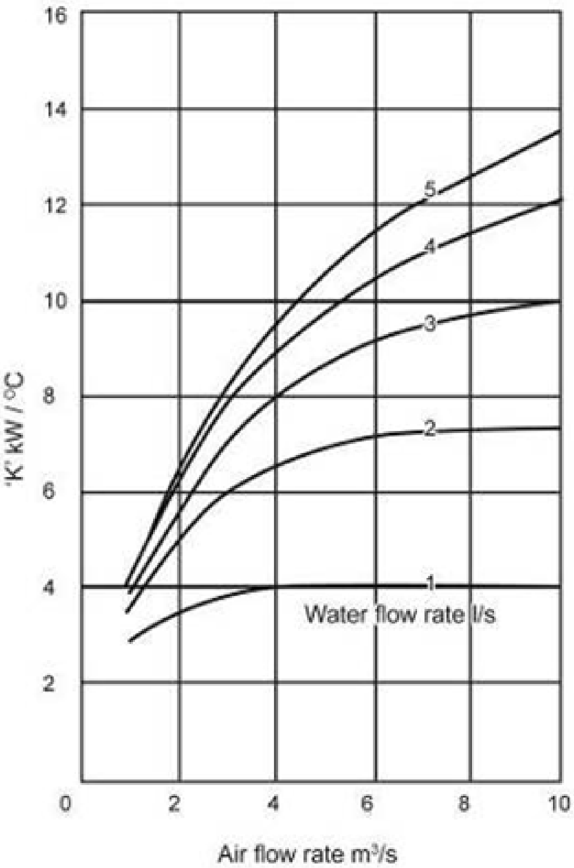


Figure 8.50 'K' value for various air and water flow rates

Chapter 9

PRESSURE SURVEYS

LEARNING OUTCOMES

- Determine pressure differences using different pressure survey methods
- General airflow calculations with Atkinson
- Understand friction factor K.

REFERENCES

Mine Environmental Control (MEC) Certificate, Chamber of Mines, South Africa, Workbook 1.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

9.1 Introduction

Definition

An environmental control department is responsible for the distribution of air to the various working places in a mine. This involves maintaining the desired air quantities in existing sections of the mine and advising on the fans and airways necessary to ventilate new areas efficiently. By carrying out pressure surveys the ventilation officer is able to assess present conditions and obtain information, which is essential for future planning. A **pressure survey** can be defined as the measurement of pressure losses, air densities, air quantities and airway dimensions throughout a mine or part of a mine.

Objectives

Objectives of Pressure Surveys

The results obtained from a pressure survey can be used to:

- Locate areas in which the pressure drop is abnormally high
- Determine the resistance figures for various airways
- Obtain information necessary for future planning requirements (i.e. Types and sizes of airways, fan duties)
- Indicate the total power requirements in different parts of a mine
- Determine whether the power put into the air (air power) is being correctly utilized
- Indicate areas where leakages and/or air recirculation occur.

The student is also reminded about the elementary laws for air flow that were discussed in Chapter 2. Also mentioned before in Chapter 2 of this course, air power is calculated as the product of pressure loss and the airflow volume in kW. The Equation is:

$$\text{Air power needed} = \frac{pQ}{1\,000} \text{ in kW}$$

Where:

p	=	frictional pressure loss (Pa)
Q	=	quantity of air (m^3/s)

9.2 Pressure Survey Instruments

9.2.1 Manometer

A manometer (as shown in Chapter 1) works on the principle that a fluid displacement will take place if there is a pressure difference between two points.

From the well know Equation

$$p = \rho g h$$

We can come to the conclusion that:

$$P_1 - P_2 = p = (\rho_2 - \rho_1)\, gh$$

where the densities differ.

We can therefore say that the pressure difference is proportional to the height difference.

9.2.1.1 Types of Manometers

Self study

The different types of manometer are discussed in detail in VOEE, Chapter 5, pages (79 - 85). The working of these is for self study purposes:

- Vertical U-type
- Vertical reservoir type
- Inclined reservoir type.

Other examples are:

- Hook type
- Micro manometer
- Magnetic manometer
- Liquid manometer.

!

When readings are taken with a manometer, it is possible to have reading errors. The most common sources are:

- Meniscus error (especially with water)
- Surface tension (especially when there is a small delta p)
- Clarity (Clear fluids are hard to see)
- Air bubbles
- Density (The scale of the manometer is set for a specific density)
- Viscosity
- Human error: Scale error:
    - o From producer
    - o Scale not fixed to tube
- Reading error, parallax
- Wrong calculations.

NB

Note: For water 1cm difference is close to 98 - 100 Pa.

9.2.2 Barometers

Remember from previous lessons:

$$P_{absolute} = P_{atmosphere} + P_{meter\ (gauge)}$$

An example of the original barometer is shown in Figure 9.1

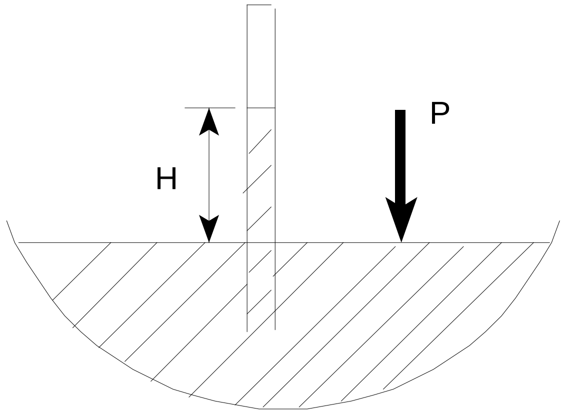


Figure 9.1 Original barometer

Atmospheric Pressure: 760 mm Hg
10.3 m H_2O
101.324 kN/m^2 or kPa

Definition

The **barometer** thus measures the atmospheric pressure. From the height measurement shown above it is clear that it is not practical to use water to measure atmospheric pressure (10.3 m versus 760 mm for mercury).

9.2.2.1 Types of Barometers

- Mercury Barometer
- Aneroid Barometer
- More commonly used nowadays is electronic type measuring instruments. It is however very important that these types of instruments have to be declared intrinsically safe, which means that are not posing a fiery risk. It must therefore not have the potential of causing a spark which may cause a flammable gas explosion underground.

SHE

TYPES OF CORRECTIONS FOR MERCURY BAROMETER

- Calibration correction, error by supplier
- Temperature correction, calibrated at 0°C
- Gravitational Acceleration. $g = 9.79$ m/s^2
- The accuracy of the instrument is close to 7 Pa.

ANEROID BAROMETER

Self study

This barometer does not use a liquid. The full working thereof is described on page 79 of VOEE. In short, the pressure is measured by a sensitive elastic diaphragm and arm combination.

9.3 Thermodynamic Equations

We previously derived the following two Equations (chapter 5):

$$W_f = \int v dP + F + g\int (1+r)dZ + g\int a dZ + \int (1+r)u_a du_a + \frac{1}{2}\int u_a^2 dr + \int a u_w du_w + \frac{1}{2}\int u_w^2 da$$

And the pressure Equation:

$$p_f = (P_2 - P_1) + p + g\int \rho dZ + g\int \frac{a}{\upsilon} dZ + \int \rho u_a du_a + \frac{1}{2}\int \frac{u_a^2}{\upsilon} dr + \int \frac{a u_w}{\upsilon} du_w + \frac{1}{2}\int \frac{u_w^2}{\upsilon} da$$

These two Equations were derived with the assumption that water droplets flow in the same directions as airflow.

The two terms, which signs will change if, the direction of the water droplets changes is:

$$g\int a dZ \quad \text{and} \quad g\int \frac{a}{\upsilon} dZ$$

The above terms can become very simple if the velocity terms are negligibly small. The Equations were derived as follows:

$$p = p_f - (P_2 - P_1) - g\int \rho dZ - g\int \frac{a}{\upsilon} dZ \quad \text{... (9.1)}$$

$$F = W_f - \int \upsilon dP - g\int (1+r)dZ - g\int a dZ \quad \text{.................................... (9.2)}$$

If there is no free moisture in the air, a = 0 and the Equations change to:

$$p = p_f - (P_2 - P_1) - g\int \rho dZ \quad \text{... (9.3)}$$

$$F = W_f - \int \upsilon dP - g\int (1+r)dZ \quad \text{... (9.4)}$$

When there is no fan present in a specific airway, the Equation become:

$$p = -(P_2 - P_1) - g\int \rho dZ \quad \text{.. (9.5)}$$

$$F = -\int \upsilon dP - g\int (1+r)dZ \quad \text{... (9.6)}$$

Equation 9.6 can be further expressed as:

$$F = -\int \upsilon\, dP - g(z_2 - z_1) - g\int r\, dZ \quad \text{... (9.7)}$$

NB

Equation 9.5 and 9.6 **can only be used** under the following conditions:

- Velocity term (difference in velocity between two points) is small enough to ignore
- No water droplets present
- No fan in the airway.

9.4 Friction Equations

In most mine ventilation applications as described before, Atkinson's Equation is used. The two basic Equations for pressure and energy losses, which Atkinson derived, are:

$$p = \frac{kCm^2}{\rho_{std}A^3}\int\frac{(1+r)^2}{\rho}dL \quad \text{.......} \quad (9.8)$$

$$F = \frac{kCm^2}{\rho_{std}A^3}\int\frac{(1+r)^3}{\rho^2}dL \quad \text{.......} \quad (9.9)$$

Where:

k	=	friction factor (Ns^2/m^4)
C	=	airway circumference (m)
m	=	mass flow of air (kg/s)
ρ_{std}	=	Standard air density, kg/m^3 (1.2 kg/m^3)
A	=	Cross sectional area (m^2)
L	=	the length of airway (m)

These two Equations are needed to calculate the friction factor. A more general requirement as mentioned before in Chapter 2 is to determine the resistance of the airway. Airway resistance can be calculated from:

$$R = \frac{KCL}{A^3} \quad \text{.......} \quad (9.10)$$

Where:

R = Airway resistance, m^{-4} (or Ns^2/m^8)

$$p = R\frac{1}{\rho_{mp}}m^2 \quad \text{.......} \quad (9.11)$$

Where:

ρ_{mp} is the average(mean) density for the pressure loss (p) as specified. The average (mean) density is expressed as:

$$\frac{1}{\rho_{mp}} = \frac{1}{L}\int\frac{(1+r)^2}{\rho}dL \quad \text{.......} \quad (9.12)$$

We can also express the friction energy Equation as follows:

$$F = R\frac{1}{\rho_{me}^2}m^2$$

The average (mean) density for the energy as specified is:

$$\frac{1}{\rho_{me}} = \frac{1}{L}\int \frac{(1+r)^3}{\rho^2} dL \quad \text{.. (9.13)}$$

9.5 Pressure Survey Methods

Two types of methods will be discussed, namely the direct and indirect survey methods. Make sure that you understand the difference between the two types of methods and can name and describe some examples of each.

9.5.1 Indirect Methods

9.5.1.1 Barometer Survey (Density Method)

A barometer is set up at either end of an airway. Simultaneous readings of the barometric pressure are made at each point.

The difference between readings gives the difference in barometric pressure, which can be compared with the calculated pressure difference due to elevation and density differences, to determine the pressure loss.

If only one barometer is available, corrections must be made for any possible change in the pressure at the first station while readings were taken at the second station. This is generally done by taking two readings at the first station, one before and one after the reading at the second station. The correct pressure is obtained by linear interpolation.

Another variation in this method is to set up a barometer at a 'control station' and to take readings at regular intervals.

Another barometer(s) is taken to the various other stations and the pressures and times are recorded. By making the necessary time, elevation and density corrections, the pressure loss may be calculated.

In terms of writing the Equations as integrals; it can be done as follows:

$$p = -(P_2 - P_1) - g\int \rho dZ$$

$$F = -\int \nu dP - g(z_2 - z_1) - g\int r dZ$$

From the first Equation the actual pressure increase (P_2 - P_1) and the theoretical pressure increase, - $g\int \rho dZ$.

The biggest problem with the solving of these two Equations is the integrals. Psychrometric calculations of air properties can be determined from two points, this in itself is not really correct, as we **don't know what happens between** the two points.

It is thus logical that the most correct answer is to take a number of readings between the two points, which from a practical point of view, is not always possible. There is a general assumption that the density changes linearly with elevation Z (h). If we apply it as such we can say:

$$\int \rho dZ \quad = \frac{1}{2}(\rho_1 + \rho_2)(z_2 - z_1) \quad \text{.................................... (9.14)}$$

Law

We must assume that the specific volume ν and barometric pressure reveals a polytropic law property, $P\nu^k$ = constant. From Chapter 5 we have established that:

$$\int \nu dP \quad = \frac{k}{k-1}(P_2\nu_2 - P_1\nu_1)$$

$$\text{with } k \quad = \frac{\log\left(\frac{P_2}{P_1}\right)}{\log\left(\frac{\nu_1}{\nu_2}\right)}$$

Rule

When k = 1, then $P\nu$ = Constant for a gas with constant composition the process is then isothermal. If the composition is not constant, such as when there is an increase in moisture content, it is possible to get $P\nu$ = constant approximately proportional for non-isothermal process.

If $P\nu$ = constant, $\int \nu dP$ can then be expressed as:

$$\int \nu dP \quad = P_1\nu_1 \log_e\left(\frac{P_2}{P_1}\right) \quad \text{.. (9.15)}$$

Rule

Although $\int r dZ$ does not have such a great influence on pressure and energy losses as $\int \nu dP$ and $\int \rho dZ$, it is necessary to evaluate the integral. **If just one point's values are known it is safe to assume that the moisture content is linear with respect to elevation**.

References

Refer to example 1 on page 91 of VOEE.

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Sometimes it happens that the incorrect readings are taken and the answers for energy and pressure losses are not representative of what the actual situation is. There exist calculation methods to show these errors, these will be discussed.

If we manipulate Equations 9.8 and Equation 9.9 by representing it as F/p, we can see that $\frac{F}{p} = \nu$. If the two values of the Equation $\frac{F}{p}$ and ν agree, we can accept the results of the pressure survey as correct. If they do not agree, we can assume the results to be incorrect.

The method which is put forward to test if there is an error in energy and/or pressure losses can linearly be expressed as follows:

$$\frac{\left(\frac{F}{p}\right)}{\nu} = x \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (9.16)$$

After algebraic manipulation, we can give the following two Equations for the correct values of energy and pressure losses:

> F' = F (5.002 – 4.035x) and p' = p (3.763 – 2.788x), (F' and p' are the correct values for F and p respectively)

The value of 'x' is thus a relationship between the calculated airway resistance, from the energy loss value, and airway resistance, from the pressure loss. From the above it is clear that the values for 'x' that vary very much from 1 must be deemed suspicious, independent of the pressure survey method or type of airway. This correction method **was derived for deep downcast shafts and must not readily be used for other situations**. Example 2 on page 92 of VOEE explains the situation. Simplistically the density method could also be expressed in terms of the Bernoulli Equation as follows and also gives reasonably accurate results.

References

Self study

$$P_1 + \frac{\rho_1 u_1^2}{2} + \rho_1 g h_1 = P_2 + \frac{\rho_2 u_2^2}{2} + \rho_2 g h_2 + p_L$$

$$\therefore p_L = \left(P_1 + \frac{\rho_1 u_1^2}{2} + \rho_1 g h_1\right) - \left(P_2 + \frac{\rho_2 u_2^2}{2} + \rho_2 g h_2\right)$$

Assuming the change in velocity to be negligible in the shaft or airway, then:

$$\frac{\rho_1 u_1^2}{2} = \frac{\rho_2 u_2^2}{2}$$

And these terms can be ignored.

$$\therefore p_L = (P_1 + \rho_1 g h_1) - (P_2 + \rho_2 g h_2)$$
$$= (P_1 - P_2) + (\rho_1 g h_1 - \rho_2 g h_2)$$

If an accurate mean density (ρ_m) between points '1' and '2' can be obtained,

$$p_L = (P_1 - P_2) + \rho_m g (h_1 - h_2)$$

Rule

Note: Because of the manner in which h_1 and h_2 are measured relative to a datum (see page 10 of EE), the term (h_1 and h_2) is negative if the air flows upwards and positive if the air flows downwards.

This makes it possible to re-write the previous Equation in simpler terms as:

$$\mathbf{p_L = (P_1 - P_2) \pm \rho_m g h} \quad \text{................(9.17)}$$

Where:

h = vertical difference in elevation (m),

Rule

Note: The sign is '+' for air flowing downwards and '-'for air flowing upwards (thumb rule).

Rule

It must also be noted that the value for the pressure loss will always be a positive number (pressure loss cannot be negative). The Equation mentioned above is the basis for pressure survey calculations using the **density method**.

NB

Important points to notice about this Equation are:

- The velocity is assumed to be constant at the two points. If there is a large difference in the velocity at the two points, suitable corrections must be made.
- The pressures at the two points (P_1 and P_2) are usually measured as absolute barometric pressures with a barometer (a barometer generally measures the absolute static pressure).
- Remember that point '1' is the upstream point and point '2' is the downstream point.
- The term 'ρ_m' is the mean density between the two points in question. During the calculation of pressure survey results, the term 'ρ_m' is multiplied by 'g' (9.79 m/s^2) and the difference in elevation, which could be, say 1 000 m, so it is essential that 'ρ_m' is accurate.
- An error in mean density of 0.01 kg/m^3 is multiplied by an amount of 10 000 and leads to an error of 100 Pa in pressure loss.

NB

- The term (h_1 - h_2) is the **vertical difference** in elevation between the two points. It is sometimes necessary to calculate this elevation difference using simple trigonometry.
- If there was no elevation difference, as would be the case in a horizontal airway, the Equation becomes:

$$p_L = P_1 - P_2$$

However, truly horizontal airways are rare in practice, due to the fact that a gradient (normally 1:200) exists for water drainage in most cases.

NB

An important aspect to deal with what has been discussed so far is to understand the difference between the following types of terminology (also note that in these notes the height is sometimes referred to as 'h', or 'z' but all expressed in metre):

- The actual measured pressure difference (normally the difference in measured barometric pressures)
- The theoretical increase in pressure (which ignores the effect of friction on the side walls ($\rho_{average} g h_{vertical}$))
- The pressure loss, which is the difference between the actual pressure increase and the theoretical pressure increase and can also be shown as follows:

$$p_L = (\rho_{average} g h_{vertical}) - (P_2 - P_1)$$

Where:

p_L = (actual) pressure loss

$(\rho_{average} g h_{vertical})$ = theoretical pressure increase ($P_2 - P_1$) = actual pressure increase (measured)

ESTABLISHING THE DIRECTION OF FLOW OF AIR

The direction of flow is not known and sometimes has to be determined. A method of determining this is shown in the Worked Example 1.

WORKED EXAMPLE 1

Assume that instead of the pressure being used up to overcome friction in the shaft it is used up overcoming the resistance of a regulator situated about 1 cm above the lower point. In other words the flow is frictionless above the regulator and the only increase in pressure is that due to the mass of air above the regulator. The average density of the air between points A and B is 1.071 kg/m^3. The height between points A and C is 852 m. Determine the direction of the airflow in the shaft (is it up cast or down cast?).

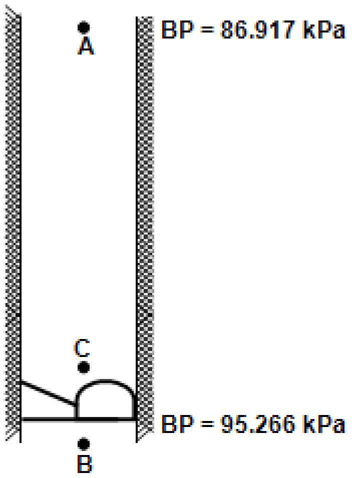

Figure 9.2 Graphical representation explaining airflow direction

Answer

The top of the shaft is the upstream point (point '1') and the bottom of the shaft is the downstream point (point '2'). Air flows downwards so a '+' sign is used in the Equation (application of thumb rule previously mentioned).

Vertical difference in elevation, h	=	852 m
Mean density, ρ_m	=	$\dfrac{1.01 + 1.131}{2}$ kg/m^3
	=	1.071 kg/m^3
Pressure at A (given)	=	86.917 kPa
Theoretical pressure increase A - C	=	ρ_m g h
	=	9.79 x 1.071 x 852
	=	8 933 Pa
∴ Theoretical pressure at C	=	(86.917 + 8.933) kPa
	=	95.850 kPa

Rule

But pressure 1 cm below at B actually is 95.266 kPa, which means that the airflow direction is downcast as theoretical pressure at C is greater than pressure at B (air flows from high pressure to low pressure).

NB

As mentioned earlier, the accuracy of the mean density is very important if accurate pressure loss results are to be obtained. In most instances density varies linearly with depth; however in the case of a shaft pressure survey from surface there are substantial variations in air density at the top 10 m of the shaft because of unstable conditions at the entrance. After 100 m the density again starts to vary linearly with depth. This is shown in the Figure 9.3.

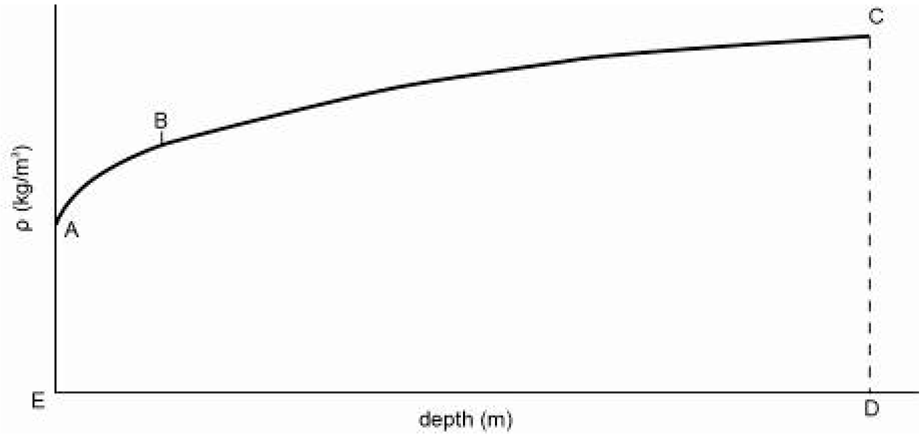

Figure 9.3 Increase in air density with depth

To obtain the true density in the shaft, as many measurements as possible should be taken in the shaft, especially in the first 100 m. These measurements should then be plotted against depth as shown in the graph.

$$\text{Then true mean density} = \frac{\text{total area enclosed by ABCDE}}{\text{total depth of shaft ED}}$$

However in practice it is not always possible to take so many measurements close together during the first 100 m (A - B) and then further apart in the rest of the shaft (B - C). It is for this reason that shaft pressure surveys usually have several sets of observations made in the shaft instead of just the top and bottom measurements, which were used in the last worked example.

TIME CORRECTIONS

The term (P_1 - P_2) is the **actual measured difference** in pressure between the two points. It is assumed when using this term that each pressure was measured at the same time and that there is no error due to pressure variations with time.

If the pressures were measured at different times, **a correction has to be applied to one of the pressures so that both pressures are obtained at the same time.**

If only one barometer is available for the survey the following method/procedure is used.

- Take a pressure reading at point A and the time
- Take a pressure reading at point B and the time
- Return to point A and take a further pressure reading and the time.

WORKED EXAMPLE 2

If the following readings (with one barometer) are obtained, determine the corresponding pressure at the different time.

A:	88.128 kPa	08.10 am
B:	94.385 kPa	09.15 am
A:	88.315 kPa	11.10 am.

Answer

Assuming that pressure changed linearly with time, these pressures can be plotted against time as shown in Figure 9.4

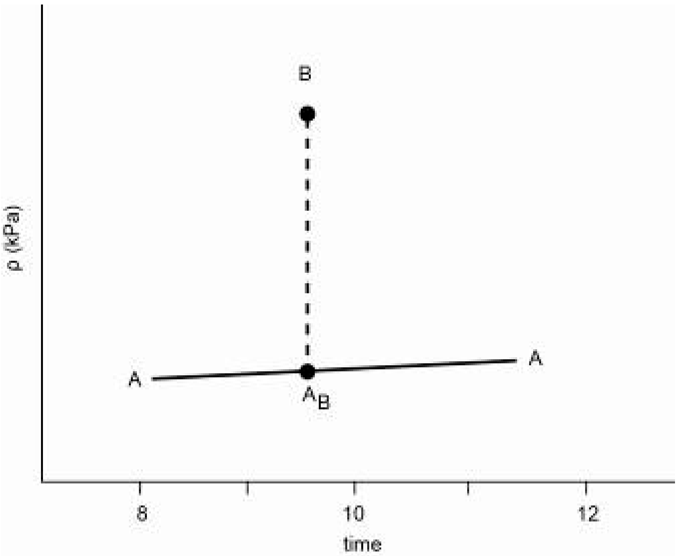

Figure 9.4 Interpolation of pressure readings and time

It can be assumed that A's readings of barometric pressure were increasing with time. When B was read, A's pressure had already increased from that read at 08.10 am. The correct pressure difference between the two points can be obtained by determining what A would have read at the time B's reading was taken (marked as point AB on the graph).

Difference in time between the two readings at A

= 3 hrs (8.10 to 11.10)

= 180 minutes

Difference in pressure between the two readings

= (88 315 - 88 128) Pa

At A = 187 Pa

$\therefore$ A's pressure increased by 187 Pa in 180 minutes

$\therefore$ In 65 minutes (8.10 - 9.15) A's pressure would have increased by:

$$\frac{65}{180} \times 187 = 68 \text{ Pa}$$

$\therefore$ At 9.15 A would have read: $= (88.128 + 0.068)$ kPa
$= 88.196$ kPa

B at 9.15 read 94.385 kPa (given)

$\therefore$ Pressure difference between A and B at 9.15

$$\begin{aligned}(P_1 - P_2) &= (94.385 - 88.196) \text{ kPa} \\ &= 6.189 \text{ kPa} \\ &= 6.189 \text{ Pa}\end{aligned}$$

If two barometers are available, a better method of monitoring changes in barometric pressure during a pressure survey is to have one barometer set up as a 'control barometer; at a convenient point in the mine, and to take readings every five or ten minutes. A second barometer is used for the actual survey and corrections can be accurately determined. The barometer used for the actual survey is known as a 'traverse barometer'.

WORKED EXAMPLE 3

(Density method/Barometer method)

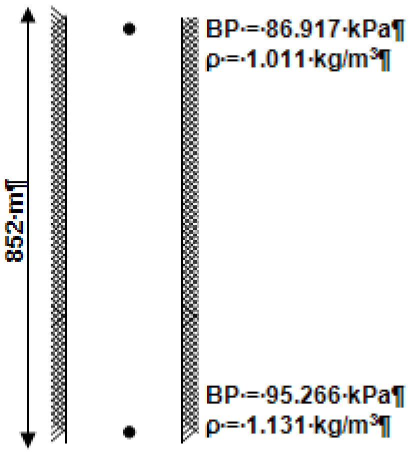

Figure 9.5 Graphical representation of pressures in shaft

The above observations were taken in a section of a downcast shaft. Determine the pressure loss between the two points.

Answer

The top of the shaft is the upstream point (point '1') and the bottom of the shaft is the downstream point (point '2'). Air flows downwards so a '+' sign is used in the Equation.

Vertical difference in elevation, $h = 852$ m

Mean density, $\rho_m = \dfrac{1.01 + 1.131}{2}$ kg/m^3

$= 1.071$ kg/m^3

From the given Equation:

$$
\begin{aligned}
p_L &= (P_1 - P_2) + \rho_m gh \\
&= (86\,917 - 95\,266) + (9.79 \times 1.071 \times 852) \\
&= -8\,349 + 8\,933 \\
&= 584 \text{ Pa}
\end{aligned}
$$

This pressure loss of 584 Pa is at the mean density in the section, namely 1.071 kg/m^3. To express this pressure loss at any other density, pressure varies directly as air density; that is $P \propto \rho$.

Or

$$\frac{P_2}{P_1} = \frac{\rho_2}{\rho_1}$$

For example, express the pressure loss obtained in the last example at an air density of 1.2 kg/m^3.

$$\frac{P_2}{P_1} = \frac{\rho_2}{\rho_1}$$

$$P_2 = P_1 \times \frac{\rho_2}{\rho_1}$$

(Remember point '1' signifies old and '2' signifies new)

$$
\begin{aligned}
P_2 &= 584 \times \frac{1.2}{1.071} \\
&= 654 \text{ Pa}
\end{aligned}
$$

NB

This means that at a higher density the pressure loss will be more and vice versa. An important point to note once again is that **the resistance did not increase**, but that the pressure loss due to an increased air density has increased.

In using the Equation mention above ($p_L = (P_1 - P_2) + \rho_m gh$) two different kinds of pressure increases are obtained and the difference between these two results (as mentioned before) is the actual pressure loss.

The one pressure increase is the actual pressure increase as measured (P_1 - P_2) and the other increase is the theoretical increase in pressure due to the extra mass of air acting on the lower point (ρgh).

NB

This theoretical increase in pressure would exist even if there was no air flowing and is sometimes referred to as a 'no-flow' increase in pressure.

It is not necessary to calculate both increases in pressure at the same time. This can be done as two separate calculations and the results compared with each other.

WORKED EXAMPLE 4

The following figures were obtained when using the density method to determine the pressure loss in a downcast shaft.

Table 9.1 Pressure readings at different air densities

Depth (m)	Pressure (kPa) (Measured)	Air Density (kg/m^3)
0	84.73	1.001 7
305	89.74	1.057 2
914	94.05	1.130 7
1.772	103.54	1.209 7

Determine:

i. The pressure loss in the shaft
ii. The mean air density in the shaft.

Answer

i. The air flows down the shaft hence point '1' will be the higher point (upstream) $\frac{p}{9.79 \text{ x } h}$ and point '2' the lower point (downstream).

p_L = (P_1 - P_2) $\pm$ $\rho_m g$ h ('+' because downcast)

The results can be tabulated as follows:

Table 9.2 Tabulated results from pressure survey

Section	$P_1 - P_2$ (Pa)	ρ_m (kg/m^3)	h (m)	9.79 x ρ_m x h	P_2 (Pa)
1	-3 010	1.0295	305	3 074 Pa	64
2	-6 310	1.0939	609	6 522 Pa	212
3	-9 490	1.1702	858	9 829 Pa	339

Pressure loss in shaft = (64 + 212 + 339) Pa

= 615 Pa

It is not always necessary to calculate the pressure loss in each section as shown above. The theoretical pressure increases (9.79x$\rho_m$xh) shown in the column above could have been added to give a total pressure value of 19 425 Pa and then compared with the actual increase from top to bottom of the shaft:

84 730 - 103 540	=	-18 810 Pa
Theoretical pressure increase	=	19 425 Pa
Therefore pressure loss	=	19 425 - 18 810
	=	615 Pa

ii. Mean density in the shaft : there are several different ways by which a mean density can be obtained from the four densities given (Table 9.1), for example the arithmetic mean of all four, mean of top and bottom, and so forth. However, an accurate mean/average density can be obtained by manipulation of the term for theoretical pressure increase.

$$p = 9.79 \times \rho_m \times h$$

$$\therefore \rho_m = \frac{p}{9.79 \times h}$$

$$= \frac{19\ 425}{9.79 \times 1\ 772}$$

$$= 1.1196 \text{ kg/m}^3$$

Comparison between this mean density and that obtained by arithmetic means, and so forth shows little difference. However, if more observations had been made, in for example 100 m use of:

$$\rho_m = \frac{p}{9.79 \times h}$$ would be far more accurate.

WORKED EXAMPLE 5

The following observations were made during a pressure survey using the 'density' method.

Table 9.3 Pressure survey readings at different time intervals

Control Barometer (CB)		**Traverse Barometer (TB)**		
Time (am)	**kPa**	**Time (am)**	**Station**	**kPa**
10:00	86.720	10:05	A	109.255
10:10	86.728	10:42	B	108.731
10:20	86.742			
10:30	86.754			
10:40	86.763			
10:50	86.768			
11:00	86.774			

Stations A and B are the same elevation, determine the correct pressure loss between A and B.

Answer

It is clear that B **cannot be subtracted from A (the traverse barometer column) because the pressures at A and B were measured at different times**. Examining the control barometer (CB) readings shows that pressures were increasing throughout the mine during the survey period. So the first thing is to determine the amount by which the control readings increased in the time period 10:05 - 10:42. It is then assumed that the difference in the traverse barometer (TB) readings changed by the same amount during this period (linear increase/decrease assumption and then to calculate what the control pressure is expected to be at 10:05).

The **expected** control reading barometric pressure at 10:05 is then ($\frac{5}{10}$ of the difference in the control barometric pressure from 10:00 to 10:10). Remember no pressure was taken by the control barometer at 10:05, but we want to compare the 10:05 traverse barometer meeting with the 10:05 control barometer reading!).

$$\therefore P_{CB} \text{ @ } 10{:}05 = 86.720 + \left[\frac{5}{10} \text{ x } (86.728 - 86.720)\right]$$

$$= (86.720 + 0.004) \text{ kPa}$$

$$= 86.724 \text{ kPa}$$

Similarly,

$$P_{CB} \text{ @ } 10{:}42 = P_{CB} \text{ @ } 10{:}40 + \left[\frac{2}{10} \text{ x } (P \text{@} 10{:}50 - P_{CB} \text{@} 10{:}42)\right]$$

$$= 86.763 + \left[\frac{2}{10} \text{ x } (86.768 - 86.763)\right]$$

$$= 86.763 + (\frac{2}{10} \text{ x } 0.005)$$

$$= 86.764 \text{ kPa}$$

Therefore in the time period from 10:05 to 10:42 the control barometric (CB) pressure increased from 86.724 to 86.764 kPa. The increase in control barometric (CB) pressure:

$$P_{CB} = (86.764 - 86.724) \text{ kPa}$$
$$= 0.04 \text{ kPa}$$

It is now assumed that if the control barometric (CB) pressure increased by 0.04 kPa from 10:05 to 10:42, then the difference in the traverse barometric (TB) pressure would **have increased by the same amount**.

$\therefore$ A's reading at 10:05 = 109.255 kPa

NB

B's reading at 10:05 would be 0.04 kPa **lower** than it read at 10:42 (because **barometric pressure levels were increasing**).

$\therefore$ B's reading at 10:05 would have been = (108.731 – 0.04) kPa
= 108.691 kPa

Because there is no elevation difference between A and B, Equation 9.17 can be written as:

$$p_L = (P_1 - P_2)$$
$$\therefore p_L = P_A \text{ @ } 10{:}05 - P_B \text{ @ } 10{:}05$$
$$= (109.255 - 108.691) \text{ kPa}$$
$$= 0.564 \text{ kPa}$$
$$= 564 \text{ Pa}$$

The same result could have been obtained by converting B to A's time, that is:

B's reading at 10:42 would be 0.04 kPa **higher** than it read at 10:05 (because barometric pressure levels were increasing).

∴ A's reading at 10:42 would have been

	=	(109.255 + 0.04) kPa
	=	109.295 kPa
And p_L	=	P_A @ 10:42 - P_B @ 10:42
	=	109.295 – 108.731
	=	0.564 kPa
	=	564 Pa (as before)

9.5.1.2 Full Volume, Reduced Volume Survey

With this method, wet- and dry-bulb temperatures and barometric pressure readings must be taken in two places at the same time, both at full volume and during period of reduced volume of airflow. The immediate result of a loss in volume flow is a change in barometric pressure, but a change in temperature is also possible. The reduction in air volume is usually obtained by switching off a main fan (hence this method is often called the fan running - fan stopped method).

The advantages of this method are:

Advantages

- All measurements can be taken at fixed spots.
- Corrections due to elevation do not have to be determined.
- Observations can be taken in a short time.
- Few instruments are required and they are simple to use.

The disadvantages of this method are:

Disadvantages

- A main fan usually has to be switched off, thus it cannot be done during the working shift.
- The results are often not reliable and are slightly difficult to calculate.

References

This method of pressure surveying is described on page 142 of EE. The method is derived from the density method and hence some of the terms in the Equation should be familiar. By using the airway pressure loss equation, the full volume flow pressure losses can be expressed as follows (using integrals):

$$p = -(P_2 - P_1) - g\int \rho dZ \qquad (9.18a)$$

and during reduced volume flow conditions expressed as follows ('r' = reduced flow):

$$p_r = -(P_{2r} - P_{1r}) - g\int \rho_r dZ \qquad (9.18b)$$

These two Equations (6a-6b) can be combined to give the following:

$$p - p_r = -[(P_2 - P_{2r}) - (P_1 - P_{1r})] - g\left(\int \rho dZ - \int \rho_r dZ\right) \quad \text{.........} \ (9.19)$$

Advantage

At this stage Equation 9.19 is just a method of determining ($p - p_r$). The advantage of this method is that the barometric pressure readings are at the ends of the airway and these are the different values of the integral terms.

As previously mentioned, barometers are trust worthier to show the difference in pressure compared to measuring absolute pressure. This pressure survey method makes full use of this aspect discussed here.

The difference in friction pressure losses can be expressed in terms of airway resistance, mass flow rate and the average/mean density (as shown before), where 'ρ_{mp}' is mean density at full pressure/volume and 'ρ_{mpr}' is the mean density at reduced pressure/volume:

$$p = R\frac{1}{\rho_{mp}}m^2$$

$$p_r = R\frac{1}{\rho_{mpr}}m^2$$

Combined the Equation will look as follows:

$$p - p_r = R\frac{1}{\rho_{mp}}m^2\left(1 - \frac{\rho_{mp}}{\rho_{mpr}}\frac{m_r^2}{m^2}\right) \quad \text{..........................} \ (9.20)$$

Which simply gives:

$$p - p_r = p\left(1 - \frac{\rho_{mp}}{\rho_{mpr}}\frac{m_r^2}{m^2}\right) \quad \text{....................................} \ (9.21)$$

References

Refer to example 6 on page 101 of VOEE to get an idea of the application of this method.

Simplistically, the method and use can also be expressed as follows:

The Equation applicable to the full volume, reduced volume survey method is simplified as follows:

$$p_f = \frac{\Delta B \pm (\rho_{mf} - \rho_{mr})\, g\, h}{1 - \left(\frac{\rho_{mr}}{\rho_{mf}} \times \frac{Q_R^2}{Q_F^2}\right)} \quad \text{...................................} \ (9.22)$$

Where:

p_f	=	pressure loss for full volume flow (Pa)
g	=	constant for gravitational acceleration (9.79 m/s^2)
h	=	vertical difference in elevation (m)
ρ_{mf}	=	mean air density at full volume flow (kg/m^3)

ρ_{mr} = mean air density at reduced volume flow (kg/m^3)
Q_F = full volume flow (m^3/s)
Q_R = reduced volume flow (m^3/s)
B = the difference between: barometric pressure difference with full volume flow and barometric pressure difference with reduced volume flow (Pa).

That is:

B = (BP_{F1} – BP_{F2}) - (BP_{R1} - BP_{R2})
= (change in BP at '1') - (change in BP at '2')

Subscript '$_F$' indicates full volume flow
Subscript '$_R$' indicates reduced volume flow
Subscript '$_1$' indicates upstream point
Subscript '$_2$' indicates downstream point.

NB

± is used as follows: A plus ('+') sign is used if the air is flowing downwards, and a minus ('-') is used if the air is flowing upwards.

WORKED EXAMPLE 6

The following observations were made during a pressure survey in a downcast shaft using the full and reduced volume method.

Table 9.4 Pressure readings summary using the full volume reduced volume pressure survey method

		Full Volume (150 m^3/s)		**Reduced Volume (28.3 m^3/s)**	
	Elevation (m)	**B.P.(kPa)**	**Density (kg/m^3)**	**B.P.(kPa)**	**Density (kg/m^3)**
1 shaft surface	240	84.074	0.9560	83.80	0.9628
1 shaft 6 level	1 710	98.976	1.1776	99.616	1.7900

B.P. Barometric Pressure

Determine the normal pressure loss in the shaft by using the simplified full volume reduced volume pressure survey Equation.

Answer

$$p_F = \frac{\Delta B \pm (\rho_{mf} - \rho_{mr})\, g\, h}{1 - \left(\frac{\rho_{mr}}{\rho_{mf}} \times \frac{Q_R^2}{Q_F^2}\right)}$$

This Equation can be broken down into smaller terms for simplicity:

Step 1:

$$B = (BP_{F1} - BP_{F2}) - (BP_{R1} - BP_{R2})$$

Because the air flows downwards, point '1' will be 1 shaft surface and point '2' 1 shaft 6 level.

$$\begin{aligned}
\therefore B &= (84.074 – 98.976) - (83.980 – 99.616) \\
&= (-14.902) - (-15.636) \\
&= - 14.902 + 15.636 \\
&= 0.734 \text{ kPa} \\
&= 734 \text{ Pa}
\end{aligned}$$

Step 2:

Air flows downwards, hence a '+' sign is used.

Step 3:

$$\begin{aligned}
g \times h &= 9.79\ (1\,710 - 240) \\
&= 9.79 \times 1\,470 \\
&= 14\,391
\end{aligned}$$

Step 4:

$(\rho_{mf} - \rho_{mr})$

$$\rho_{mf} = \frac{0.9560 + 1.1776}{2} = 1.0668 \text{ kg/m}^3$$

$$\rho_{mr} = \frac{0.9628 + 1.1790}{2} = 1.0709 \text{ kg/m}^3$$

$$\begin{aligned}
\therefore (\rho_{mf} - \rho_{mr}) &= (1.0668 – 1.0709) \\
&= - 0.0041 \text{ kg/m}^3
\end{aligned}$$ **(Note the '-' sign)**

Step 5:

$$\frac{\rho_{mr}}{\rho_{mf}} = \frac{1.0709}{1.0668}$$

$$= 1.00384 \text{ kg/m}^3$$

Step 6:

$$\frac{Q_R^2}{Q_F^2} = \frac{(28.3)^2}{(150)^2}$$

$$= 0.035595 \text{ (ratio)}$$

Step 7:

$$1 - \frac{\rho_{mr}}{\rho_{mf}} \times \frac{{Q_R}^2}{{Q_F}^2}$$

$$= \quad 1 - (1.00384 \times 0.03556)$$

$$= \quad 1 - 0.035\,7$$

$$= \quad 0.9643$$

Now, substituting all the above terms into the original Equation:

$$p_F = \frac{\Delta B \pm (\rho_{mf} - \rho_{mr})\, g\, h}{1 - \left(\frac{\rho_{mr}}{\rho_{mf}} \times \frac{Q_R^2}{Q_F^2}\right)}$$

$$= \frac{734 + (-0.0041) \times 14\,391}{0.9643}$$

$$= 700 \text{ Pa}$$

This is the normal pressure loss in the shaft and is expressed at the true mean density in the shaft. It should be noted that where there is a change in barometric pressure, the relative change in density can be calculated from the Equation:

$$\frac{\rho_1}{BP_1} = \frac{\rho_2}{BP_2}$$

9.5.2 Direct Pressure Survey Methods

9.5.2.1 Trailing Hose Method

The trailing hose method of pressure surveying is simple and easy to use in horizontal airways and gives a direct measurement or indication of the pressure loss. When the airway is inclined or vertical and the densities of the air inside and outside of the tube are not equal, then it is necessary to apply corrections to the readings before the 'true' pressure loss is obtained. This is best explained by the Figure 9.6, which shows a portion of a vertical downcast shaft in which a trailing hose is placed. A manometer attached to the bottom of the tube measures the pressure loss. Refer to Figure 9.6.

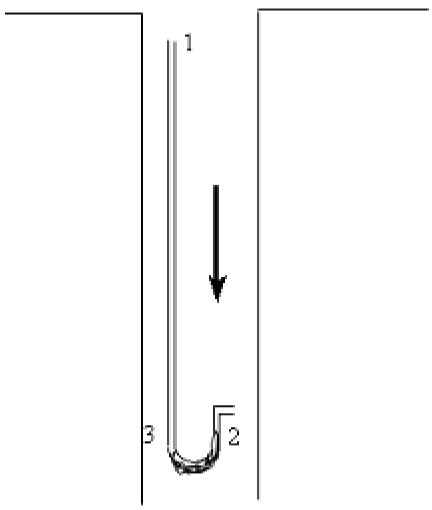

Figure 9.6 Graphical representation of the trailing hose pressure survey

From Figure 9.6 we can derive the following. The pressure Equation can be applied to the shaft and the hose and can be expressed as follows:

$$\text{shaft } p = -(P_2 - P_1) - g\int \rho dZ \quad \text{.......................... (9.23)}$$

$$\text{trailing hose } 0 = -(P_3 - P_1) - g\int \rho_{hose} dZ \quad \text{.................... (9.24)}$$

There is no airflow in the hose and therefore the friction losses are zero. Equation 9.23 and Equation9.24 can be combined to give:

$$\text{p} = (P_3 - P_2) + g\int \rho_{hose} dZ - g\int \rho dZ \quad \text{............................ (9.25)}$$

Because (P_3 - P_2) is the differential pressure difference, as measured by the manometer, it shows that the friction pressure loss is equal to the differential pressure difference plus a correction factor which is equal to the difference in pressure by the air present in the shaft and hose.

If the airway is horizontal, we can see that dZ = 0 and the friction pressure loss will equal the differential pressure difference of the manometer. In all circumstances the density of the air in the hose differs to that of the air in the shaft. **This is as a result of the pressure difference between the two and can also be as a result of moisture in the shaft.**

The measuring of the pressure and temperature at the ends is not a problem and can be done, but the measuring in the hose is a problem. The only practical manner to determine the psychrometric properties of the air in the hose **is to ensure that the air in the hose enters with specific moisture content and the air in the hose has the same dry bulb temperature as the shaft**.

This can be read from the manometer, during a determined time, to ensure that the conditions are the same in the hose and shaft. After the hose is attached to the manometer, one must ensure that one waits long enough to ensure that the dry bulb temperature inside the hose and the shaft is the same.

WORKED EXAMPLE 7

Table 9.5 Trailing hose pressure survey summary

	Elevation (m)	**Pressure (kPa)**	**t_{wb} temperature (ºC)**	**t_{db} temperature (ºC)**	**r (kg/kg)**	**Density (kg/m^3)**
Top	-553.0	86.854	10.0	14.1	0.00723	1.04883
Bottom	-1331.6	94.983	13.9	19.6	0.00822	1.12478

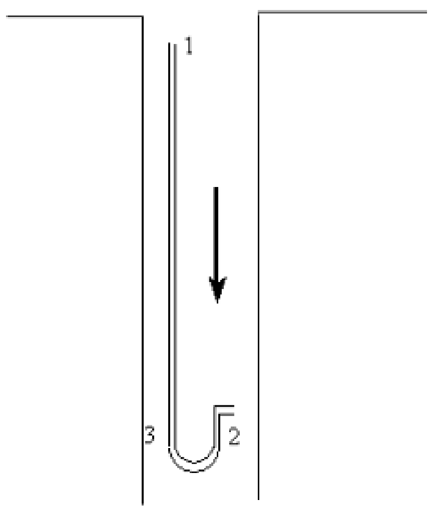

Figure 9.7 Trailing hose method

Note: The values in the table above are of the air in the shaft, of which some is also applicable to the hose. The value of g is given as 9.79 m/s^2 and the pressure reading obtained from the manometer is 376 Pa (this could also have been given as in mm mercury or water and then needed to be calculated accordingly). The friction pressure loss in the shaft needs to be determined.

Answer
It is important that the Psychrometric calculations for air are determined first (calculated or read off from a Psychrometric chart, whichever are applicable). In this example it is given to you, but it can be expected from you to calculate these values yourself or read them from a chart.

We must now determine the Psychrometric properties of the air in the hose. If we assume that the moisture content in the hose remains constant at 0.00723 kg/kg (or 9.23 g/kg), the air density of the air in the shaft is 1.04883 kg/m^3 at the top of the shaft and a dry-bulb temperature of 19.6°C at the bottom, the density of the air in the tube can be calculated as follows (we first have to determine the barometric pressure in the hose).

The barometric pressure of the air in the shaft is as follows:

Pressure in hose = Barometric pressure at bottom + manometer reading
= 94.983 + 0.376
= 95.359 kPa

The density of the air in the hose can then be determined using Psychromtric Equations (assuming that the moisture content stays the same in the hose):

$$\text{Vapour pressure } (P_w) = \frac{r\,P}{0.622 + r}$$

$$= \frac{0.00723 \times 95.359}{0.622 + 0.00723}$$

$$= 1.0957 \text{ kPa}$$

$$\text{Specific volume } (v) = \frac{0.287035 \times (273.15 + t_{db})}{P_{hose} - P_w}$$

$$= \frac{0.287035 \times (273.15 + 19.6)}{95.359 - 1.0957}$$

$$= 0.89143 \text{ m}^3/\text{kg}$$

$$\text{Density } (\rho) = \frac{1 + r}{\nu}$$

$$= \frac{1 + 0.00723}{0.89143}$$

$$= 1.1299 \text{ kg/m}^3$$

Having determined the density in the hose the integrals now need to be solved. The two integrals must be evaluated with the assumption that the density is linear with relation to elevation.

$$\int \rho dZ = \frac{1}{2}(\rho_1 + \rho_2)(z_2 - z_1)$$

$$= \frac{1}{2}(1.04883 + 1.12478)(-1331.6 - (-533.0))$$

$$= -867.922 \text{ kg/m}^2$$

$$\int \rho dZ = \frac{1}{2}(\rho_1 + \rho_2)(Z_2 - Z_1)$$

$$= \frac{1}{2}(1.04883 + 1.1299)(-1331.6 -(-5.33.0))$$

$$= -869.967 \text{ kg/m}^2$$

Therefore:

$$p = 376 + 9.79(-869.967) - 9.79(-869.922)$$

$$= 355.98 \text{ say } 356 \text{ Pa}$$

The trailing hose method is the most accurate method to determine pressure losses, but it is important that all possible errors, such as leakages and correct installation procedures are countered.

Simpler methods (without using integrals) are also used to do the calculations. Figure 9.8 is similar to the one used before, but different footnotes are used however both these methods will lead to the same answer.

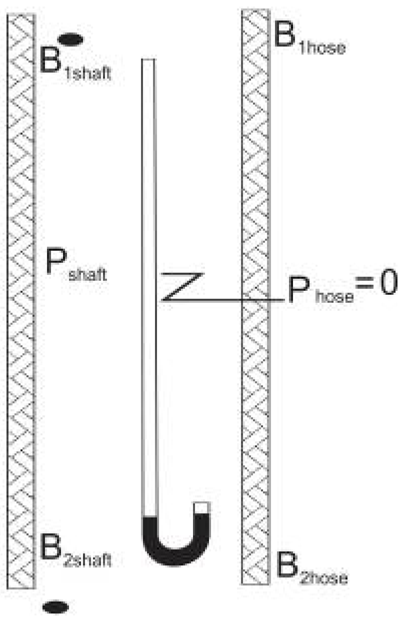

Figure 9.8 Pressure readings in shaft with trailing hose

The absolute pressures at the top and bottom of the section are indicated by B_{1shaft} and B_{2shaft} respectively and the pressure loss in the shaft (P_{shaft}) can be calculated by using the **'density'** method equation.

Therefore:

$$P_{shaft} = \rho_{mshaft}\, g\, h - (B_{2shaft} - B_{1shaft})$$

Where:

h	=	difference in elevation (m)
ρ_{mshaft}	=	mean density of the air in the shaft (kg/m^3)
g	=	gravitational acceleration (m/s^2)

The pressure loss in the trailing hose (P_{hose}) can be calculated in a similar manner.

Therefore:

$$P_{hose} = \rho_{mt}\, g\, h - (B_{2hose} - B_{1hose})$$

Where h and g are as explained above and ρ_{mhose} is the average/mean density of the air in the tube (kg/m^3), B_{1hose} and B_{2hose} are the pressures in the tube at position 1 and 2. However, it will be apparent that $B_{1shaft} = B_{1hose}$ and since there is no air flowing in the tube that $p_{hose} = 0$.

Thus by substituting in the Equation above:

$$P_{hose} = 0 = g\rho_{mhose} - B_{2hose} + B_{1hose}$$

Manipulation of the Equation given above gives the following result for the pressure loss in the shaft:

$$p_{shaft} = (\rho_{ms} - \rho_{mt})\, gh - (B_{2shaft} - B_{2hose})$$

Evaluation of this Equation shows that the term ($B_{2shaft} - B_{2hose}$) is the reading which would be obtained by a manometer connected at the bottom of the tube and the term ($\rho_{mshaft} - \rho_{mhose}$) gh, is the correction needed because of the ‘difference in densities’ between the air inside and outside of the tube. In other words, the manometer reading does not give a direct pressure loss reading.

There are obviously many different positions where the manometer can be situated, although in practice it is most likely that it will be attached either to the top or to the bottom of the tube. Similarly, the shaft may be either an upcast or downcast shaft.

WORKED EXAMPLE 8

The following information was obtained from a trailing hose survey in a vertical downcast shaft. The manometer was attached to the bottom end of the tube. Determine the pressure loss in the shaft.

Table 9.6 Pressure readings from trailing hose pressure survey

	Elevation Above datum (m)	Barometric pressure (Pa)	Shaft air Density (kg/m^3)	Tube air Density (kg/m^3)	Manometer Reading (Pa)
Shaft top	1 723.4	87 770	1.07973	1.07973	-
Shaft bottom	428.0	102 263	1.23322	1.23805	214

Answer

Using the trailing hose Equation:

$$p_{shaft} = (\rho_{mshaft} - \rho_{mhose}) - (B_{2shaft} - B_{2hose})$$

Where:

$$g = 9.79 \text{ m/s}^2$$

$$h = (1\,723.4 - 428.0) = 1295.4 \text{ m}$$

$$\rho_{mshaft} = \frac{1.07973 + 1.23322}{2} = 1.1565 \text{ kg/m}^3$$

$$\rho_{hose} = \frac{1.07973 + 1.238\,05}{2} = 1.1589 \text{ kg/m}^3$$

$$\rho_{mshaft} - \rho_{hose} = 1.1565 - 1.1589 = -0.0024 \text{kg/m}^3$$ (note the minus sign)

$$B_{2shaft} - B_{2hose} = \text{manometer reading} = -214 \text{ Pa}$$

NB

(**Note the minus sign because the air inside the tube is at a higher pressure than the air in the shaft**).

Substituting in Equation

$$p_{shaft} = (9.79 \text{ x } 1295.4 \text{ x } -0.002\,4) - (-214)$$

$$p_{shaft} = (-30.6) + 214$$

$$\therefore p_{shaft} = 183.4 \text{ Pa}$$

If the 'density' method was used to calculate the pressure loss in the shaft, it would be:

$$p_L = (87\,770 - 102\,263) + (9.79 \text{ x } 1\,295.4 \text{ x } 1.156\,5)$$

$$p_L = 173.7 \text{ Pa}$$

NB

The difference between these two readings can be attributed to the fact that the true mean air density in the shaft is not necessarily 1.1565 kg/m^3, but some other value. And as explained earlier, more readings would have to be taken in the shaft to obtain a better value of the mean density of the air in the shaft or some correction factor applied.

In the above example, air was drawn through the tube for a period of time before attaching the manometer. The mean hose air density (ρ_{mhose}) was obtained by working from the assumption that the moisture content inside the hose at the bottom of the shaft was the same as that of the air at the top of the shaft.

Thus to determine ρ_{mhose} the following values were used:

Absolute pressure	=	102 263 + 214
	=	102 477 Pa
Moisture content	=	0.00024 kg/kg (assumed)
Temperature	=	15.17^{o}C
Specific volume	=	0.807887 m^3/kg
Therefore, ρ_{mhose}	=	1.238094 kg/m^3

To use the relationship P α ρ, in this instance would be incorrect as it does not take cognisance of the fact that there is a difference in the moisture content of the air in the shaft, and the air inside the tube. The degree of error in this instance is approximately 7%.

9.5.2.2 Ventilation Pressure by Readings

The difference in pressure over ventilation doors shows the pressure loss in the system and readings of the pressures over the doors can be used to determine the pressure losses. Again we make use of the pressure Equation. Consider the Figure 9.9.

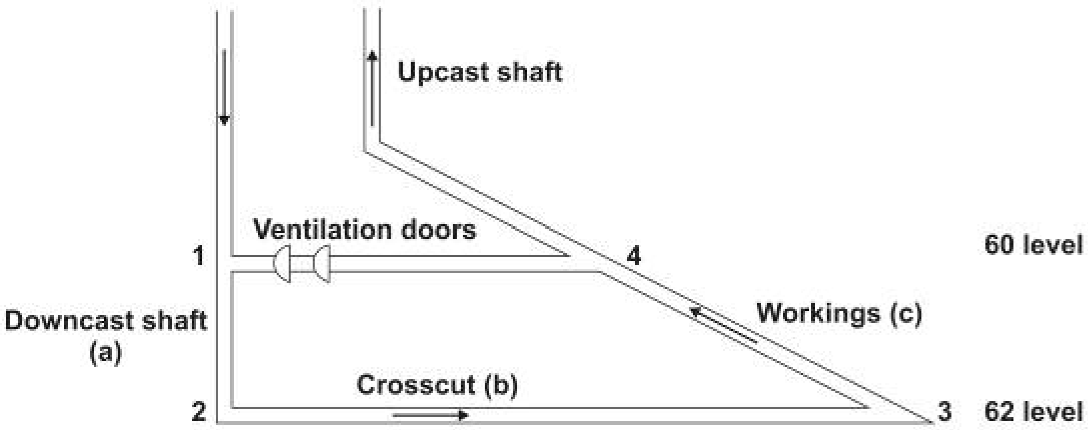

Figure 9.9 Diagrammatic section of airflow through mine

The airway pressure Equation can be written for each of the airways 1 to 4:

$$p_a = -(P_2 - P_1) - g\int \rho_a dZ$$

$$p_b = -(P_3 - P_2) - g\int \rho_b dZ$$

$$p_c = -(P_4 - P_3) - g\int \rho_c dZ$$

If we add the above Equations to each other:

$$p_a + p_b + p_c = (P_1 - P_4) - g(\int \rho_a dZ + \int \rho_b dZ + \int \rho_c dZ)$$

Self study

The term (P_1 - P_4) is the pressure over the two doors on 60 level and the sum of the pressure differences over the network is thus the ventilation door pressure plus a correction factor which is dependant on the density of the air and the elevation of the network. Example problem 7 on page 102, Chapter 6 of VOEE is for self-study purposes.

9.6 Measurement Techniques

9.6.1 Pressure Measuring

NB

Take note that P is the static pressure in all our Equations up to now and that p is the friction pressure loss. When we take readings it is important that the correct pressure is determined.

9.6.2 Temperature Readings

NB

For Psychrometric calculations, it is important that the average temperature for a point in a specific airway is used. The best way is to take a number of readings and to plot this on a graph and the readings will not be place bound.

Self study

Example problem 8 on page 104, Chapter 6 of VOEE is for self-study purposes.

Chapter

10

COMPRESSED AIR FOR MINES

LEARNING OUTCOMES

- Knowledge and understanding of all aspects pertaining to compressed air.

REFERENCES

Fraser, P, 'Saving energy by replacing compressed air with localized hydropower systems: a 'half level' model approach', *Third International Platinum Conference, Platinum in transformation,* The Southern African Institute of Mining and Metallurgy, 2008, p285-292.

Vermaak, JC, 'Compressed air control', *Journal of the Mine Ventilation Society of South Africa,* Jan/March 2002, p12-16.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

10.1 Introduction

Usage

SHE

Compressed air is a very useful and safe source of energy for many mining applications, especially for operating percussion rock drills (pneumatic machinery). Machines driven by compressed air can stand up to rougher handling than electrical machinery. Compressed air, although not very energy friendly, is still extensively used in the South African mining industry, especially hard rock mines, and as a result of these very same useful characteristics of safety and availability, it is also very extensively abused. Research is being done on more efficient electric drilling, but due to the importance of compressed air use on mines it will be dealt with in detail.

10.2 Sources of Compressed Air

Some mines buy compressed air from a central compressor station and pay for it at an agreed price for each unit passing through a meter on the mine property.

Most mines, however, have one or more compressor stations of their own and calculate the cost of an air unit by dividing the total cost of air production by the number of air units metered on the discharged side of the compressors. Compressors fall mainly into two basic categories:

- Positive displacement
- Dynamic type.

References

The positive displacement type as the name suggests, compresses air by drawing a certain volume into an enclosed space and then reducing this volume by mechanical action (screw and reciprocating types). By far the greatest proportion of compressed air used in mines is supplied by compressors of the dynamic type, represented mainly by the centrifugal compressor. Further descriptions and pictures related to the different kinds of compressors can be found in Chapter 7 of VOEE.

The performance of compressors is in many respects comparable to that of fans (that is in terms of efficiencies). A centrifugal compressor has a stall point, its efficiencies vary from very low at free delivery to a maximum near the stall point, and generally the fan laws apply when allowance is made for the compressibility of air.

NB

While a centrifugal compressor (dynamic type) is capable of delivering a large range of air volume over a large range of pressures, it can supply only a particular volume at a particular pressure and at a definite efficiency, unless it's running speed is variable or unless some other type of control is incorporated. The speed of an electrically driven compressor is seldom variable. If, during part of the day, more air is required from it than what it was primarily designed for, **it can supply this increased air quantity only at a reduced pressure and with loss of efficiency**. If, on the other hand, less than the design quantity is required temporarily the compressor is likely to stall and a by-pass valve has to be opened to blow off some of the air to the atmosphere, unless the compressor has inlet vane control. By partially closing the vanes a smaller quantity can be obtained at the desired pressure, but with a considerable reduction in efficiency.

!

The speed of a steam-driven compressor is easily altered by adjusting the supply of steam, but the production of steam in the boilers is not as easily adjusted and therefore every variation in steam requirement must affect the efficiency of the boilers.

Objective

The aim should therefore be to ensure as constant a load as possible.

It may, for example, be possible to arrange that air-consuming processes such as sand-blasting, agitation of mud sumps and the use of air winches and loaders are not conducted during the drilling shift, but considerable variation in air requirements will still be found unavoidable and these variations must be catered for in at least two of the following three ways:

- Several compressors of different capacities must be provided so that judicious combination at these will give approximately the required load at all times of the day.
- Some compressors with variable speed or inlet vane control must be provided to give the desired flexibility when run alone or in conjunction with other compressors.
- Ample storage capacity should be provided to smooth out momentary fluctuations in load. For this reason air receivers are usually installed at the compressor stations and at strategic points along the routes of the air mains. In large mines money is often wasted in this way because the capacity of the receivers is negligible in comparison with the capacity of the great length of large-diameter mains. In a few South African mines, however, benefits have been derived from the provision of very large receivers in the form of isolated rock tunnels with capacities of some thousands of cubic metres.

ML

10.3 Metering of Compressed Air

NB

All mines with pneumatic drilling mining systems need a continuous supply of compressed air. The continuous process of mineral extraction from certain kind of mines needs air for agitation of the contents of certain tanks. Air is sometimes also needed for operating the brakes of hoists. In some instances fans (mostly for emergency use only) and pumps driven by compressed air are also required to run continuously.

Other processes are intermittent. On a producing mine most of the rock drills work for only a limited period during the morning shift, during which period all air fans and pumps are also working. Air-driven loaders and scrapers are operated mostly during the morning and night shifts, while water blasts consume a large quantity of air at the end of the morning shift and at the beginning of the night shift.

Some of these processes require high air pressures whereas others require only relatively low air pressures. Each mine has its own pattern of consumption which again changes as the mine reaches different stages of development.

ML

It is essential, in order to be able to study and control the flow of air to different parts of a mine at all times of the day, to have recording metering devices operating at strategic points.

Most flow meters record the pressure difference across an orifice plate in an air column. This pressure difference is proportional to the square of the air volume flowing through the orifice and is usually recorded on a chart which is graduated to give readings directly proportional to the volume flowing.

Most of these old meters have been changed with electronic flow meters which are much more accurate but also costly.

10.4 Units of Compressed Air

NB

In most instances it is not the amount of air which is available that is of importance**, but the amount of energy which is stored in that air and which can be used for driving machinery**. It is therefore basically correct that, in costing, a figure should be arrived at for the **cost per unit of energy in the air and not a cost per unit mass or volume of air**. A term that is commonly used when we consider compressed air terminology is the term called ‘free air’ expressed in m^3/s or kg/s. The definition of **free air** is as follows:

Definition

It refers to air at a pressure of 83.21 kPa, a temperature of 15.94ºC and a relative humidity of 55%, under these conditions air has a density of almost 1 kg/m^3. This means that a flow rate of 1 m^3/s of free air relates to a mass flow rate of 1kg/s. Due to the large range of pressure changes existing in compressed air systems, it is preferable to use mass flow rates (kg/s) for compressed air (in the mining sense of the word costs are normally related to m^3/s of free air).

ML

References

Mines in most instances produce their own compressed air, however, and find it more convenient to use a unit of volume. On a deep-level gold mine, air compressors typically consume at least 25% the total energy required, compared with up to 40% for a large shallow platinum mine (2009 figures). Given the ESKOM electricity supply constraint and significant energy price increases, alternative more energy efficient technologies, need to be considered. An article written by Peter Fraser of Hydropower Equipment highlights the significance of other drilling sources such as hydropower needs to be considered.

NB

The electric power which is required as indicated above are therefore substantial and capital and maintenance costs can also have a considerable effect to the working cost related to mining. It is therefore an important requirement that compressed air equipment is used at high efficiencies as it contributes largely to costs for mining.

10.5 Air Consumption of Various Appliances

Table 10.1 gives the approximate amount of air consumed by some commonly used appliances and wasted by leakage. These figures, of course, vary with the type and condition of the appliance and with the air pressure. In the last column the annual cost is given of running these appliances continuously assuming an air pressure of 500 kPa and the cost of compressed air as given in Table 10.1.

Table 10.1 Approximate costs of compressed air for different appliances

Appliance	m^3/s Free Air	Cost per Annum for continuous running (2002)	Cost per Annum for continuous running (2010)
Jackhammer	0.06	R2 970	R7 633
Drifter	0.066	R3 285	R8 442
Venturi	0.024	R1 170	R3 007
400 mm Air Fans	0.07	R3 465	R8 905
25 mm water blast	0.23	R11 385	R29 259
Cameron Pump 4x2	0.028	R1 395	R3 585
Cameron Pump 6x3	0.066	R3 285	R8 442
Cameron Pump 8x4	0.123	R6 075	R15 613
Cameron Pump 10x5	0.165	R8 190	R21 048
Cameron Pump 12x7	0.198	R9 900	R25 443
Quimby Pump	0.076	R3 780	R9 715
Turbine Motor 11 kW	0.151	R7 470	R19 198
Turbine Motor 15 KW	0.203	R10 035	R25 790
Mechanical Loader	0.137	R6 750	R17 348
Diamond Drill	0.09	R4 500	R11 565
5 mm leak	0.03	R1 485	R3 816
10 mm leak	0.12	R5 850	R15 035
25 mm leak	0.75	R37 125	R95 411

10.6 Optimum Compressed Air Pressure

The rate of production of mines that uses compressed air extensively for drilling depends to a very large extent on the number and depth of blasting holes drilled daily into the rock. For this a large number of rock drill machines are required. The speed of penetration of a rock drill is dependent to a very large extent on the air pressure supplied to the machine.

In fairly hard ground a typical machine on these fields could, for example, drill:

Rules

- 50 mm per minute with air at 300 kPa pressure
- Or 150 mm per minute with air at 400 kPa pressure
- Or 220 mm per minute with air at 500 kPa pressure.

Specially designed machines can attain still greater rates of penetration if supplied with air at pressures of over 700 kPa; and these machines are also lighter. High drilling pressures can therefore result into more efficient drilling practices, such as less labour and less air used for drilling the same length of hole with high pressures than with low pressures.

In practice, the maximum available drilling pressure is determined by the following:

- Capacities of the compressors
- The pressures that can be withstood safely by the air pipes and hoses
- The loss of pressure due to friction in the pipes connecting the compressors and the machines.

SHE

In some equipment, such as mechanical loaders, pumps, winches riveters and fans, a certain minimum pressure is required for efficient working, but it is usually found that the safety of either the machine itself or of the work being done by it also limits the practical maximum working pressure. For this reason these appliances are often throttled down during the drilling shift. In nearly all cases, with mechanical loaders the possible exception, a pressure of 400 kPa at these appliances is ample.

A pressure of 300 kPa is usually sufficient for all other applications such as Brown tanks, filters, water blasts, hoists, brakes, mud sump agitation, blowing-out of electric motors and assay crushers and so forth.

NB

An important point to note is that there is a natural increase in the gauge pressure of air travelling down a deep mine. This is due to the difference between the density of the air inside the air column and that of the air outside it. Table 10.2 shows how the pressure increases with depth.

Table 10.2 Air Pressures at depth

Depth (m)	Air Pressures (kPa)			
0	400	500	600	700
500	430	524	640	750
1000	460	580	690	800
1500	490	610	730	850
2000	530	650	780	900
2500	560	690	820	950
3000	590	730	870	1010
3500	620	770	910	1060
4000	650	800	960	1110

10.7 Loss of Pressure in Pipes and Hoses

NB

When air flows in a pipe or hose, frictional and shock losses occur just as happens with ventilation currents, and these cause a drop in pressure. D'Arcy's formula for calculating the pressure drop in a compressed air column is very similar to Atkinson's Equation which is used in ventilation calculations. The most significant factor is that the pressure drop is proportional to the square of the air volume flowing. Therefore, doubling the amount of air flowing results in a four-fold increase in the pressure loss, and trebling the air flow increases the pressure loss by a factor of nine ($p \propto Q^2$).

The following tabulations (Table 10.3 and Table 10.4) are useful for calculating the pressure losses in pipes and hoses with various rates of airflow. Table 10.3 shows loss of pressure in air hoses at 500 kPa gauge pressure (for old hoses with many hose couplings, the figures may be doubled) and Table 10.4 shows the amount of m^3/s of free air that would give a pressure drop 2 kPa per 100 m for different pipe/hose diameters.

Table 10.3 Pressure losses associated with different size hoses and lengths

m^3/s free air	**Pressure drop in kPa**			
	25 mm hose		**32 mm hose**	
	30 m	60 m	30 m	60 m
0.05	22	44	5.5	11
0.06	37	74	10	20
0.07	65	130	17.5	35

Table 10.4 Pressure drop in hoses for different pipe diameters

Pipe Diameter	**m^3/s of Free Air that would give a pressure drop of 2 kPa / 100 m**			
mm	At 400 kPa Gauge	At 500 kPa Gauge	At 600 kPa Gauge	At 700 kPa Gauge
25	0.0	0.0	0.0	0.0
51	0.0	0.0	0.0	0.0
102	0.2	0.3	0.3	0.3
152	0.7	0.8	0.9	0.9
203	1.5	1.7	1.8	1.9
254	2.8	3.1	3.4	3.6
305	4.6	5.1	5.5	5.9
356	6.8	7.6	8.2	8.8
406	9.9	11.0	11.9	12.7
457	13.5	15.0	16.2	17.4
508	17.8	19.7	21.3	22.8
559	23.4	25.9	28.1	30.0
610	28.6	31.6	34.2	36.8
660	36.5	40.4	43.6	46.8
711	44.2	48.9	53.0	56.7

Assumptions as used in Table 10.4 above:

1: Used pipes in good condition

2: 1 m^3 of free air–1kg

The advantages of using 32 mm hoses rather than 25 mm hoses on drilling machines, especially when the hoses are long and the machines consume large quantities of air, are clearly illustrated in Table 10.3.

A very simple way of calculating the amount of compressed air that will be delivered by a specific column compared to the guideline pressure loss of 2 kPa per 100 of pipe for a specific diameter of pipe/hose and a specific design gauge pressure is as follows:

- First determine the actual pressure loss per 100 m of pipe
- Read off the correct free m^3/s value for the size pipe in play at the correct gauge pressure
- Then use the following Equation to solve the problem:

$$\left(\frac{\text{actual pressure loss}}{\text{design or guideline pressure loss}}\right)_{/\ 100\ \text{m pipe}} = \left(\frac{Q_{actual}}{Q_{design}}\right)^2 \qquad (10.1)$$

This relates to the very important Equation of Atkinson which implies that the p α Q^2.

It is often found in practice that the hoses supplying certain pieces of equipment driven by compressed air are much too small. The result is that the equipment does not work efficiently. Table 10.5 gives an indication of the size of hose which should be used for certain types of equipment (as guideline).

Table 10.5 Minimum pipe/hose size required for each piece of compressed air equipment

Equipment	Minimum size of hose (mm)
Cameron Pump	38
Large Cameron Pump	51
Quimby Pump	38
Reciprocating Winch 5x7	38
Reciprocating Winch 6x8	51
Turbine Motor 11 kW	51
Turbine Motor 15 kW	51
Mechanical Loader (Small)	51
Diamond Drill	38

Table 10.6 is a guideline for the number of pieces of compressed air driven machinery that should be used for different pipes used to supply compressed air. The values in this table are based on a pressure drop of 2 kPa per 100 m of piping and are therefore suitable for calculating the number of machines that may be used on long columns.

For short columns the load may be increased by 50% if necessary. In determining the number of machines to be supplied from a pipe other equipment must be taken into account. Each compressed air driven fan should be counted as **one compressed air driven machine**, each compressed air driven scraper winch (not commonly used and used for emergencies only, very energy inefficient) **as three compressed air driven machines**, and compressed air pumps according to their sizes.

Table 10.6 Machines supplied for a given pipe diameter

Pipe/hose Diameter (mm)	Compressed air driven machines supplied
51	1
102	5
152	15
203	30
254	55
305	85

10.8 The Reticulation System

NB

The basic facts that have to be known, before the optimum size of a new compressed air pipe can be determined is the final length of the column and the maximum quantity of air that will be required during the drilling shift. The latter is seldom predictable to any degree of accuracy, but a fair estimate can usually be made on the basis of the number of appliances which will be used during the drilling shift and reasonable allowances for leakages and wastage.

As a first estimate of the compressed air requirement of a hard rock mine a useful empirical rule is that 1 m^3 of compressed air is required per second during the peak period of the shift for each 6000 tons of rock broken per month. This estimate has to be adjusted in the light of conditions applying at the particular mine. In practice the reticulation system can be divided into three main parts:

- The main pipe lines which deliver from the compressor stations to the shafts, down the vertical and other shafts (incline or sub vertical) shafts and along the main haulages. The total length of any such column is often of the order of 5 000 to 10 000 metres.
- The branch pipes which lead from the shafts, levels and haulages to the working areas, stopes or development ends. These are seldom more than about 2 000 metres in length.
- The service pipes or hoses which lead from the branches to the different appliances. These are seldom longer than 200 metres.

NB

The main compressed air installations in the past used to be permanent installations with a life of 20 to 40 years. Branch pipes installations are semi-permanent with a life of 5 to 10 years, and service pipe installations are temporary.

Large steel pipes are usually of sizes of 51 mm diameter increments, such as for 254, 305, 356 and 406 mm, and the problem is how to decide when the use of the next larger size pipe becomes justified.

Balancing the cost of the loss in energy due to friction against the capital cost of larger columns, it can be shown that it becomes evident when the pressure drop is of the order of 2 kPa per 100 metre length of column, but because of the permanency of the mains and the fact that future air requirements cannot always be predicted accurately.

It is suggested that the following design guidelines be used for design purposes:

Rules

- 1 kPa pressure drop per 100 m for main compressed air pipe columns
- 2 kPa pressure drop per 100 m for branch pipe compressed air columns
- 40 kPa pressure drop per 100 m for service compressed air pipe/hose columns.

Therefore, when the air requirements have been decided on, the required size of pipe for branch lines can be read off directly from Table 10.6. For mains, the quantity must not exceed 70% of the values given in Table 10.6. For service pipes the load should not be more than four times the value given in Table 10.6.

The tables given here can be used for solving all kinds of problems relating to the reticulation of compressed air, such as the three Worked Examples (1, 2 and 3).

WORKED EXAMPLE 1

A new section of a mine is expected to use twenty pneumatically operated rock drill machines, three compressed air driven fans, one large compressed air driven pump and two air winches. What size of compressed air service pipe is recommended?

Answer

Count the air pump as two units, each air winch as three units (as per previous guideline last paragraph before Table 10.6). The total is 31 compressed air units. From Table 10.6 a 203 mm pipe can cope with 30 machines. If the section is near the shaft and not likely to expand, 203 mm pipe should be used. Otherwise 254 mm should be used. This will be able to accommodate 55 machines but will not cost a lot more in terms of capital expense, but does give the flexibility in terms of higher production needs in future.

WORKED EXAMPLE 2

The pressure drop over 1 500 m of 305 mm compressed air main pipe is 90 kPa. The average pressure is approximately 600 kPa. What is the quantity of air that will be flowing in this column?

Answer

From Table 10.4 it can be found that for a pressure drop of 2 kPa per 100 m of 305 mm pipe at a pressure of 600 kPa, the air quantity flowing is 5.5 m^3/s.

The actual pressure drop per 100 m length of pipe has to be determined and this is:

$$\frac{90}{1500} \text{ which is 0.06 kPa/m or 6 kPa/100 m}$$

Then, by using Equation 10.1 (manipulated):

$$\left(\frac{\text{actual pressure loss}}{\text{design or guideline pressure loss}}\right)_{/\,100\text{ m pipe}} = \left(\frac{Q_{actual}}{Q_{design}}\right)^2$$

$$Q_{actual} = Q_{design} \times \sqrt{\left(\frac{\text{actual pressure loss}}{\text{design or guideline pressure loss}}\right)_{/\,100\text{m pipe}}}$$

$$= 5.5 \times \sqrt{\left(\frac{6}{2}\right)}$$

$$= 9.51 \text{ m}^3\text{/s of free air}$$

WORKED EXAMPLE 3

A large section of a mine served by a 305 mm main uses 4.8 m^3/s during the peak period.

Extension of work on this section is expected to treble the air consumption. The main is 1 600 m long and the average pressure at the beginning of the main compressed air pipe is 500 kPa. The supervisor is asked for his recommendations.

Answer

From Table 10.4 at a gauge pressure of 500 kPa a 305 mm main can carry 5.1 m^3/s for a pressure drop of 2 kPa per 100 m or 32 kPa per 1 600 m the pressure drop at present can be determined as follows:

$$\frac{\text{actual pressure loss}}{\text{design or guideline pressure loss}} = \left(\frac{Q_{actual}}{Q_{design}}\right)^2$$

$$\text{actual pressure loss} = \text{design or guideline pressure loss} \times \left(\frac{Q_{actual}}{Q_{design}}\right)^2$$

$$= 32 \times \left(\frac{4.8}{5.1}\right)^2$$

$$= 28.4 \text{ kPa}$$

If the quantity is trebled to 14.4 m^3/s form the 4.8 m^3/s without adding to the 305 mm main, the pressure drop will become:

$$= \quad 28.4 \times \left(\frac{14.4}{4.8}\right)^2 \quad = \quad 255.6 \text{ say } 256 \text{ kPa}$$

Two alternatives present themselves, namely:

- Replacing the main by a pipe of larger diameter
- Or by installing another column parallel to it.

Rule

In each case the aim should be to reduce the pressure drop to 1 kPa per 100 m or 16 kPa for the 1 600 m. The quantities shown in Table 10.4 correspond to a pressure drop of 2 kPa per 100 m and for half this pressure drop these quantities therefore have to be adapted applying the p α Q^2 rule, the new pressure loss is 0.5 of the original pressure, which means that the new quantity Q_2 can be determined as follows:

$$Q_2 = Q_1 \times \sqrt{\frac{p_2}{p_1}} = Q_1 \times \sqrt{0.5} = Q_1 \times 0.707$$

Table 10.7 can now be determined from values in Table 10.4

Table 10.7 New free air values for reduced pressure loss

Pipe Diameter (mm)	Free air Capacity m^3/s for 1 kPa loss per 100 m
305	3.6
356	5.4
406	7.8
457	10.6
508	13.9
559	18.3

From these figures it is obvious that in order to convey 14.4 m^3/s the 305 mm pipe would either have to be replaced by a 508 mm pipe, or it would have to be supplemented by a 457 mm pipe in parallel with it (3.5 -10.5 = 14.0), although in each case using the next smaller size would not cause a very serious pressure loss. In practice it would probably be found that the difference in cost between a 457 mm pipe and a 508 m pipe would be less than a quarter of the cost of a 305 mm pipe and consequently a 508 mm would be installed and the 305 mm pipe reclaimed for use elsewhere.

Definition

Ring feeds (pipes joining in a circular fashion from a main feed back to the main feed) have both advantages and disadvantages.

Advantages

If the air column feeding one section or level is overloaded while that in a nearby section or level has spare capacity, pressure in the former area can often be improved considerably by cross-connecting the two columns. A further advantage is that, when one of the columns becomes damaged, neither section is completely starved of air. In practice, however, the pressure in both sections usually becomes so low while the one feed is out of commission that neither can do any effective drilling.

Disadvantages

Other disadvantages are that a burst pipe in one section immediately affects both sections, and that an inadvertently closed valve in the one feed can result in low pressures in both sections, which may persist for several days before the cause is discovered. Except in isolated cases, ring-feeding should therefore be used only between major sections of a mine.

On some mines air receivers have been built underground where old tunnels have been sealed off, put under pressure through the use of water and used in times where compressed air was needed in high peak electricity times. Figure 10.1 is an example of such an air receiver used.

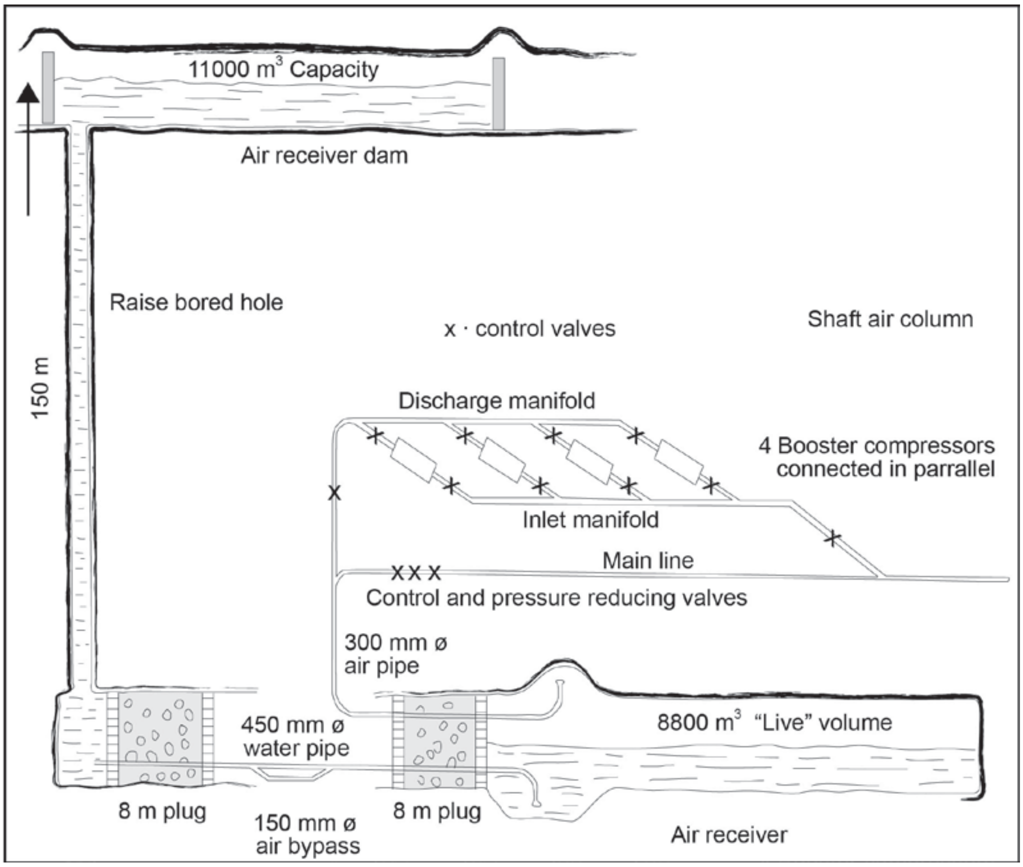

Figure 10.1 Schematic representation of air receiver underground

WORKED EXAMPLE 4

(Additional Compressed Air through air receivers)

A stoped out area with a volume of 15 000 m^3 must be used as an air receiver. It is to be charged to an absolute pressure of 1 500 kPa during the off shift periods. It is intended to draw air from this receiver during the four hour drilling shift until the absolute pressure drops to 700 kPa. What is the rate at which the main air supply can be boosted during this period? Assume an air temperature of 30ºC.

Answer

In order to answer this question the volume of the receiver needs to be expressed as a ratio of the gas constant (287 J/kgK) and temperature of the compressed air (from the ideal gas Equation).

$$m = \frac{PV_r}{RT} = \left(\frac{V_r}{RT}\right) \text{ x } P$$

From the above, when the air temperature is constant, the volume to RT ratio also remains constant for a given receiver. The mass of air (m) present in a receiver is therefore directly proportional to the absolute pressure:

$$\frac{PV_r}{RT} = \frac{15\,000}{287 \text{ x } (273 + 30)}$$
$$= 0.1725 \text{ kg/Pa}$$
$$= 172.5 \text{ kg/kPa}$$

The mass of the air in the receiver for time 1 and 2 must now be determined. The mass of the air in the receiver at the beginning of the shift (m_1)

$$m_1 = \frac{PV_r}{RT} = \left(\frac{V_r}{RT}\right) \text{ x } P$$
$$= 172.5 \text{ kg/kPa x } 1\,500$$
$$= 258\,750 \text{ kg}$$

The mass of the air in the receiver after 4 hours (m_2)

$$m_2 = \frac{PV_r}{RT} = \left(\frac{V_r}{RT}\right) \text{ x } P$$
$$= 172.5 \text{ kg/kPa x } 700$$
$$= 120\,750 \text{ kg}$$

The mass flow rate over 4 hours can therefore be determined:

$$m_2 = \frac{m_1 - m_2}{\text{time (t)}}$$

$$= \frac{258\ 750 - 120\ 750}{4 \times 60 \times 60}$$

$$= 9.58 \text{ kg/s}$$

WORKED EXAMPLE 5

Calculate the mass of air in one air unit for a gauge pressure of 600 kPa.

Answer

The mass of air corresponding to one air unit can be calculated by means of the following Equation:

$$m = \frac{3\ 600 \times 0.641 \times 10^3}{RT_{wr} \times \log_e\left(\frac{P_2}{P_{wr}}\right)} \quad \text{.............................. (10.2)}$$

Where:

R	=	Universal gas constant (287 J/kgK)
T_{wr}	=	mean absolute temperature on the Witwatersrand (289.31 K)
P_{wr}	=	mean absolute pressure on the Witwatersrand (83.21 kPa)
P_2	=	$P_{wr} + P_{meter}$

Substitute the above values in the Equation given:

$$m = \frac{3600 \times 0.641 \times 10^3}{RT_{wr} \times \log_e\left(\frac{P_2}{P_{wr}}\right)}$$

$$= \frac{3600 \times 0.641 \times 10^3}{287 \times 289.31 \times \log_e\left(\frac{83.21 + 600}{83.21}\right)}$$

$$= 13.2 \text{ kg or } 13.2 \text{ m}^3 \text{ of free air}$$

WORKED EXAMPLE 6

A section of a mine is served by a 305 mm diameter main line, which carries a mass flow of 4.8 kg/s of compressed air (or 4.8 m^3/s of free air) at peak periods.

The air in the column is assumed to have a mean temperature of 30ºC and the barometric pressure is 96 kPa. Average gauge pressure is 500 kPa.

Determine the following:

i. If the main is 1 600 m in length, what would the maximum pressure loss be across it?

ii. The pressure loss in kPa/100 m, comment on your answer.

Answer

i. The density of the compressed air must first be determined:

$$\rho = \frac{P \times 10^3}{RT}$$

$$= \frac{(500 + 96) \times 10^3}{287 \times (273 + 30)}$$

$$= 6.85 \text{ kg/m}^3$$

NB

Important to note is that the pressure is the absolute pressure. In this question the resistance factor (R_f) value is not given and must therefore be calculated:

The pressure loss in the pipe can now be determined:

$$R_f = \frac{10^{13.832}}{D^{5.276}}$$

$$= \frac{10^{13.832}}{305^{5.276}}$$

$$= 5.3 \text{ m}^{-5}$$

$$\Delta P = \frac{R_f \times M^2 \times L \times 10^{-3}}{w}$$

$$= \frac{5.3 \times (4.8)^2 \times 1\,600 \times 10^{-3}}{6.85}$$

$$= 28.5 \text{ kPa} \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots(10.3)$$

ii. The pressure loss must now be calculated for 100 m.

$$= \frac{28.5}{1600} \times 100$$

$$= 1.8 \text{ kPa / 100 m}$$

This loss can be considered as too high for a permanent air main.

WORKED EXAMPLE 7

A compressed air fan draws 0.07 kg/s of compressed air from the mains through 20 m of 25 mm diameter plastic hose. If the absolute pressure at the start of the hose is 600 kPa and the air temperature 30ºC.

i. What is the pressure loss through the pipe (show all assumptions)

ii. What is the gauge pressure at the fan; assume a barometric pressure of 100 kPa.

Answer

i. The compressed air density must first be calculated. Note that in this question the absolute pressure is given and should be used as such.

$$\rho = \frac{P \times 10^3}{RT}$$

$$= \frac{(600) \times 10^3}{287 \times (273 + 30)}$$

$$= 6.9 \text{ kg/m}^3$$

The mass flow of compressed air through hoses is low and therefore other pressure losses can be expected. Hoses of 25 mm and 32 mm diameter are in common use and resistance factors for these are normally 2 000 000 m^{-5} and 500 000 m^{-5} (assumed).

The pressure loss can be calculated as follows:

$$\Delta P = \frac{R_f \times M^2 \times L \times 10^{-3}}{w}$$

$$= \frac{2\,000\,000 \times (0.07)^2 \times 20 \times 10^{-3}}{6.9}$$

$$= 28 \text{ kPa}$$

ii. The gauge pressure at the fan
(remember $P_{abs} = P_{atm} + P_{gauge}$)

= (600 – 100) – 28 (100 kPa barometric pressure)

= 472 kPa

WORKED EXAMPLE 8

Ignoring frictional losses, calculate the gauge pressure at the bottom of a 2 000 m vertical shaft under the following conditions:

- Absolute pressure on surface 85 kPa
- Absolute pressure at the shaft bottom 105 kPa
- Temperature of the compressed air 28ºC
- Gauge pressure on surface 400 kPa

Answer

There is a natural increase in the **gauge** pressure when compressed air flows down a shaft. The increase in pressure due to auto compression can be derived from the following Equation:

$$P_e = P_s \times \exp\left(\frac{gh}{RT}\right) \qquad \text{.............................. (10.4)}$$

Where:

P_s = the **absolute** pressure on surface
P_e = the absolute pressure underground exit.
T = Temperature in Kelvin (dry bulb)
h = depth underground
R = Gas constant

The absolute pressure underground must therefore first be determined:

$$P_e = P_s \times \exp\left(\frac{gh}{RT}\right)$$

$$= (400 + 85) \times \exp\left(\frac{9.79 \times 2000}{287 \times (273.15 + 28)}\right)$$

$$= 608 \text{ kPa}$$

The gauge pressure can be determined as follows:

P_{gauge} = 608 - 105
= 503 kPa (an increase of 503 - 400 = 103 kPa)

WORKED EXAMPLE 9

During peak periods 5.1 kg/s of compressed air flows through a 305 mm diameter column in an 800 m deep vertical shaft. The column extends into the haulage and the total length of the compressed air column is 1 000 m. The initial gauge pressure of the compressed air is 500 kPa and the temperature 30ºC.

Barometric pressures at the top and bottom of the shaft are 85 and 94 kPa respectively. The effect of frictional loss and change in elevation on the pressure must be considered. The value for frictional factor R_f is an amount 5.3 m^{-5}.

i. What would the gauge pressure be at the shaft bottom?

ii. What would the gauge pressure be if the elevation difference was zero (horizontal)?

Answer

i. The absolute pressure at the shaft bottom (considering the effect of frictional losses and pressure change due to change in elevation). It is assumed that the temperature remains constant.

$$P_e = P_s\sqrt{\exp\left(\frac{2gh}{RT}\right) - R_f\frac{R^2T^2}{P_s^2 g \times 10^6} \times \frac{(\exp\left(\frac{2gh}{RT}\right) - 1) \times L}{h} \times M^2}$$

$$= (500 + 85)\sqrt{\exp\left(\frac{2 \times 9.79 \times 800}{287 \times (273 + 30)}\right) - 5.3 \times \frac{287^2(273+30)^2}{(500+85)^2 \times 9.79 \times 10^6} \times \frac{(\exp\left(\frac{2 \times 9.79 \times 800}{287 \times (273 + 30)}\right) - 1) \times 1\,000}{h} \times (5.1)^2}$$

$$= 656 \text{ kPa}$$

…… (10.5)

The gauge pressure is therefore:

Gauge pressure	=	656 - 94
	=	562 kPa

ii. When there is no difference in elevation (horizontal), the 'exp' component is then equal to the following Equation:

$$\frac{(\exp\left(\frac{2gh}{RT}\right) - 1) \text{ x } L}{h} = \frac{2gL}{RT}$$

The Equation for the absolute pressure at the exit is then:

$$P_e = P_s\sqrt{1 - 2\,R_f \text{ x } \frac{RTL}{P_s^2 g \text{ x } 10^6} \text{ x } M^2}$$

$$= (500 + 85)\sqrt{1 - 2 \text{ x } 5.3 \text{ x } \frac{287 \text{ x } 303 \text{ x } 1000}{585^2 \text{ x } 9.79 \text{ x } 10^6} \text{ x } (5.1)^2}$$

$$= 583 \text{ kPa}$$

The pressure loss $\Delta P = P_s - P_e = 585 - 583 = 2$ kPa

WORKED EXAMPLE 10

A compressed air driven fan has a duty of 2.1 m^3/s at a pressure of 950 Pa and it uses 0.08 kg/s of compressed air at a gauge pressure of 490 kPa (assume 20 Pa losses from the compressor to the fan). The barometric pressure at the mine is approximately 87 kPa. Assume an electricity cost of 15 c/kWh. The compressor is used 16 hours per day, 365 days per year. The specific volume of the air at the compressor intake is 1 m^3/kg, the motor efficiency is 95% and the compressor has an isothermal efficiency of 65%.

The following must be answered:

i. Define the concept of free air as used in compressed air terminology
ii. The power required to compress 1 kg/s of air to the required pressure to drive the fan
iii. The actual compressor input power to provide compressed air to the fan
iv. The fan efficiency when compressed air is used?
v. The annual power cost for an electric fan (assume fan efficiency as 60% and the fan motor efficiency as 95%) related to the cost of running a compressed air fan, comment on your answer.

Answer

i. Free air refers to air at an absolute pressure of 83 kPa a temperature of 15.94ºC and a relative humidity of 55% and expressed in m^3/s, due to high pressures in compressed air systems, kg/s is used to express flow rates.

ii. An absolute pressure of 87 kPa on surface means the gauge pressure before compression is 0 kPa.

The Equation to determine the power needed to compress the air is as follows:

$$\text{kW} = \frac{P_1\ \nu\ M}{\text{efficiency}_{motor} \times \text{efficiencyisothermal}} \times \log_e\left(\frac{P_c}{P_1}\right)$$

$$= \frac{87 \times 1 \times 1}{0.95 \times 0.65} \times \log_e\left(\frac{(490+20+87)}{87}\right)$$

$$= 271.4 \text{ kW per 1kg/s}$$

.... (10.6)

NB

The most common mistake made is that the wrong pressures are used. In all the calculations related to compressed air problems **the absolute pressures must be used**.

iii. The compressor input power required to provide the air for the fan:

$$\text{kW} = 0.08 \times 271.4$$

$$= 21.7 \text{ kW}$$

vi. The fan efficiency can be determined as follows:

$$\text{Fan efficiency} = \frac{\text{Air power fan}}{\text{input power compressor}} \times 100\%$$

$$= \frac{0.950 \times 2.1}{21.7} \times 100\%$$

$$= 9.2\%$$

v. The annual power cost to operate an electrical fan under normal conditions.

$$\text{Power cost} = \frac{\text{Air power fan x annual power cost}}{\text{efficiency}_{fan} \times \text{efficiency}_{motor}}$$

$$= \frac{0.950 \times 2.1 \times (365 \times 16 \times 0.15)}{0.60 \times 0.95}$$

$$= \text{R } 3\,066 \text{ per annum}$$

The annual power cost to operate a compressed air fan:

Power cost	=	$kW_{\text{compressed air fan}}$ x annual electricity cost
	=	21.7 x R $876_{(365 \times 16 \times 0.15)}$
	=	R 19 009 per annum

This is an increase of almost 6.2 times more than for an electrically driven fan and indicates that the compressed air fan should only be used in emergency situations.

WORKED EXAMPLE 11

Compressed air is used in a development end where the following conditions prevail:

- Air temperature 31.5/31.7ºC
- Barometric pressure 110 kPa
- Compressed air pressure 600 kPa (absolute)

Assume that the compressed air was at the dry-bulb temperature prevailing in the end and that the air was saturated. Determine the cooling effect of the compressed air.

Answer

From the psychrometric charts:

- Sigma heat of air in the end 97.8 kJ/kg (air humid)
- Sigma heat of compressed air 44.0 kJ/kg (air very dry)

The cooling effect of releasing 1kg/s of compressed air can be determined as follows:

kW	=	Sigma heat x mass flow rate of compressed air
	=	(97.8 – 44.0) x 1
	=	53.8 kW.

SELF STUDY

Read the following articles:

References

- ‘Saving energy by replacing compressed air with localized hydropower systems: a ‘half level’ model approach’ by Peter Fraser
- ‘Compressed air control’ by J.C Vermaak.

Chapter

11

MINE GASES AND EXPLOSIONS

LEARNING OUTCOMES

The aim of this part of the course is to familiarise the student with the sources, characteristics and dangers of gases in the underground environment.

Study Objectives: Section 11.1

On completion of this study theme the student will be able to:

- List the characteristics of gasses
- Identify the source and characteristics of general gases found underground
- List gases from mining operations
- List gases from surface mining operations
- Know the reasons for testing for gas underground.

Study Objectives: Section 11.2

At the completion of this chapter, you would have covered the following aspects related to flammable gases:

- History of gas explosions/risks
- Properties of gases
- Methane risks
- Identification/classification of flammable gases (areas, occurrences and composition)
- Influence of flammable gas mixtures
- Flammable gas detection
- Control of gas emissions into workings of a mine
- Typical gas concentration calculation
- Preventing accumulations of flammable gas
- Case studies pertaining to flammable gas explosions
- Controlling of ignition sources.

Study Objectives: Section 11.3

At the completion of this chapter, you would have covered the following aspects related to coal dust explosions:

- History of coal dust explosion/risks
- Mechanism of coal dust explosion
- Coal dust explosion risks
- Co-efficient of explosibility
- Explosibility index
- Coal dust explosion prevention
- Guidelines to stone dusting underground.

REFERENCES

Guidelines for Code of Practice (COP) for Prevention of Coal Dust Explosions, Department of Minerals and Energy, Mine Health and Safety Inspectorate, Section 5, p1-30.

Mine Environmental Control (MEC) Certificate, Chamber of Mines South Africa, Workbook 4.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

11.1 Mine Gases

11.1.1 Introduction

NB

Various gases are found in the underground and surface mining environment. It is important to be able to identify these gases and know the dangers thereof. It is also important to know what the various sources in the underground and surface mining environment are and how to deal with them effectively.

Normal fresh air consists of a mixture of gases, i.e. Nitrogen (N, 78.09%), oxygen (O_2, 20.95%), Carbon dioxide (CO_2, 0.03% or 300 ppm), and argon and other rare gases (0.93%), to make up the 100%.

These gases can also occur in the atmosphere with a mixture of other gases such as methane, nitrous fumes, carbon monoxide, hydrogen and hydrogen sulphides.

11.1.2 List of General Gases Found Underground

Table 11.1 is not complete but contains of the most important gases found underground, their sources, the allowable concentration in fresh air and the specific gravity (SG) thereof.

Definition

The **SG** is the ratio of the density of gas to that of air. A SG value of less than one is an indication that the gas is ‘lighter’ than air and greater than one indicates that the gas is ‘heavier’ than air.

Table 11.1 Characteristics of the most common gases found in the mining industry

GAS	CHEMICAL FORMULA	DANGERS	EXPLOSIVE	SPECIFIC GRAVITY (SG)	Occupational Exposure Limit (OEL)
Nitrogen	N	Inert gas (Oxygen defincіency – Asphyxia)	No	0.97	
Carbon Dioxide	CO_2	Oxygen deficiency – Asphyxia	No	1.53	5000 ppm or 0.5%
Carbon Monoxide	CO	Poisonous and Explosive	Yes 12.5 – 74.2%	0.97	50 ppm or 0.005%
Oxides of Nitrogen (Nitrous fumes)	NO_x	Lungs become waterlogged	No	1.6	3 ppm or 0.0003%
Hydrogen	H_2	Highly explosive	Yes 4% - 74%	0.07	Nil
Hydrogen Sulphide	H_2S	Poisonous. Explosive	Yes 4% - 44%	1.2	10 ppm or 0.001 %
Methane	CH_4	Burns, explodes and oxygen deficiency	Yes 5% - 15%	0.55	14 000 ppm or less than 1.4%
Oxygen	O_2	Oxygen deficiency can result in asphyxia and can be fatal.	No	1.10	More than 19%
Ammonia	NH_3	Colourless. Can smell and irritation of eyes, nose and throat	No	0.6	25 ppm
Hydrogen Chloride	HCl	Poisonous	No	1.27	
Chlorine	C_{12}	Poisonous & cause intense irritation of eyes, nose & throat	No	2.5	0.5 ppm
Sulphur Dioxide	$S0_2$	Poisonous, burns eyes, nose, throat & lungs	No	2.22	2 ppm
Hydrogen cyanide (Prussic acid gas)	HCN	Poisonous. Explosive (5.6 -40%). Can be absorbed through the skin	Yes	0.94	10 ppm
Arsine	ASH_3	Poisonous	No	2.71	0.05 ppm
Sulphuric Acid Mist	H_2SO_4	Poisonous. Irritates eyes, nose, throat & skin	-	-	1 mg/m^3
Mercury Vapour	Hg	Poisonous can be absorbed through the skin	-	-	0.05 mg/m^3
Lead Fumes	Pb	Poisonous	-	-	0.15 mg/m^3

*ppm = parts per million

11.1.3 Sources of Gases Encountered in Mining Operations

Gases found in the mining industry can be subdivided into three main groups namely:

- Poisonous (toxic)
- Flammable
- Inert.

Table 11.2 shows sources of some of the most important gases associated with the various mining operations and from this it can be seen that gases are released into the mine air from basically three sources, namely:

- The strata that is mined
- The mining or mineral processing procedure (details of gases released through mineral processing procedures will not be dealt with in this course)
- Unplanned incidents such as fires (details of gases released with mine fires will not be dealt with in this course).

Table 11.2 Sources of gases in mining operations

Source	Gases
Blasting	Noxious fumes, Carbon monoxide, Carbon dioxide, ammonia, sulphur dioxide (in pyritic ore)
Burning fuses or igniter cord	Nitrous fumes, CO, CO_2
Pilot hole and borehole drilling, fissures, faults	Flammable gas, H_2S, H_2, He, excessive CH_4 causes oxygen deficiency
Diesel engines	CO, CO_2, nitrous fumes, aldehydes, SO_2
Sand fillings	HCN
Poorly-ventilated old workings	Oxygen deficiency, CO, CO_2, CH_4, H_2
Underground fires	Oxygen deficiency, CO, CO_2
After flammable gas explosions	Oxygen deficiency, CO, CO_2
From decaying organic matter	Oxygen deficiency, CO
Water chlorination	Cl_2
Electrostatic precipitators	Ozone, nitrous fumes
Battery charging	Hydrogen
Welding	Nitrous fumes
Stagnant Water	H_2S
Burning PVC-coated electric cables and other plastics	CO, CO_2, HCN, NH_3 and nitrogen oxides
Refrigeration plants	Freon 11 and 12, ammonia

11.1.3.1 Dangerous Gases Found Underground

All gases in some or other way is dangerous in the context of having too much of it (in volume or concentration thereof), but there are some gases that are extremely dangerous that needs to be mentioned, these are:

HYDROCYANIC ACID GAS (HCN)

Hydrogen cyanide (prussic acid) is a colourless gas with a penetrating odour resembling that of bitter almonds. It is slightly lighter than air. The gas can be formed underground in sand fillings if, acid mine water is exposed to the cyanide residue. In practice, its occurrence is extremely rare since sands used for filling contain only minute quantities of cyanide, if any. In addition to being formed underground in sand fillings, Hydrocyanic gas may also be produced by the burning of plastics.

Hydrocyanic acid gas is highly toxic. It is a chemical asphyxiant acting on the nervous system in such a way as to stop the oxidation of the tissues; death results rapidly when HCN is inhaled, even for a short period.

PHOSGENE ($COCl_2$)

Phosgene is an extremely toxic gas and can be formed by the decomposition of Refrigerant gases. In the past it was believed that during a fire phosgene is produced by the burning of plastics. This is not correct. Carbon monoxide, carbon dioxide, hydrochloric or hydro-fluoric acid gas are the main gases produced when plastic burns. Phosgene is heavier than air and has a very distinct smell (very similar to HCN).

GASES RELEASED WITH THE COMBUSTION OF PLASTICS

Plastics are being used to an ever-increasing extent both underground and on the surface. A possible hazard that should be kept in mind when using them is that when they are heated they can give rise to dangerous concentrations of noxious gases.

It is important to note that plastics give off noxious gases when they are heated to 250°C or above irrespective of whether or not oxygen is present. In other words, the plastic polymer itself begins to break down. Active combustion accelerates the process by raiding the temperature.

Examples

The plastics of concern are polyurethane's (for example 'Rigiseal'), nylon (for example instrument tubing, some types of sealing cloth), PVC (for example electrical insulation, some types of brattice cloth) and polyethylene. Small quantities of Teflon, urea formaldehyde ('Bakelite') and acrylonitrile are also used.

All plastics give off copious quantities of carbon dioxide and carbon monoxide when heated. However, in addition, concentrations of hydrogen cyanide (HCN) are evolved from polyurethane (urea-formaldehyde) and acrylonitrile at temperatures above 200°C, the rate of production increasing with temperature. PVC and PTFE are dangerous in that they release hydrochloric or hydro-fluoric acid gases respectively.

Dangerous concentrations of carbon monoxide are also released, although the acid gases poses a greater hazard in the case of these two plastic materials. Polyethylene polypropylenes give rise to CO and CO_2 only when heated, provided that other products used, such as plasticisers or fillers, do not lead to the production of other gases.

Plastics are not unique in this regard. Other materials used underground can when heated, result in the production of noxious gases other than CO and CO_2 for instance, vulcanised rubber or synthetic rubber used in many mines as scatter barricades in stopes evolve H_2S and other sulphur compounds when heated above 200°C. It is therefore necessary to be aware that a potential hazard from these sources exists during a fire.

ALDEHYDES

Definition

Aldehydes are organic compounds that have the general chemical formula CnH_2NO and which are found in the incomplete combustion of a hydrocarbon such as diesel engine fuel. The commonest in mine air is formaldehyde (H.HCO) and acetaldehyde.

Aldehydes have a pungent odour; a concentration in air of 0.00003% can be detected by smell (this concentration is well below toxic level).

SHE

The gases are intensely irritating to the nose, eyes, throat and the skin and act as an irritant poison in the lungs. Concentrations above 0.0025% are difficult to tolerate and are dangerous to health. As aldehydes are so pungent, the normal person would withdraw from them before they reach a dangerous limit. Proper timing of the diesel engine reduces the production of these gases. Adequate ventilation will dilute them.

CHLORINE (CL^2)

This gas is a powerful disinfectant and is used underground to disinfect water supplies. It is a heavy, greenish-yellow gas having a specific gravity of 2.5. It has a choking pungent odour, is a powerful irritant and causes coughing, tightness in the chest vomiting and smarting of the eyes. It will cause oedema (flooding of the lungs), like with nitrous fumes and is rapidly fatal. The OEL for CL_2 is 0.5 ppm or 0.00005%.

Note: Using chemical detector tubes, chemical analysis or specialised electronic detector instruments can test all five gases mentioned above.

11.1.3.2 Gases from the Strata

Gases are released into the working environment through fissures and cracks in the rock strata. There are large reservoirs of natural gases underground and these are connected by means of fissures and cracks to each other and to the surface. If a gas reservoir of this kind is exposed by mining activities, its volumetric flow and pressure may be very high and may remain so for years.

Where this problem occurs, it is common practice to cement and seal off the gas reservoirs on a regular basis. Sometimes the cementation and sealing off of a specific area has the effect of increasing the effusion of gases in another area. Normally, a combination of cementation, sealing off and drainage of gases gives the optimum results.

Another important source of gases underground is fissure water under high pressure. This, results in the gases being dissolved in the fissure water, even those that are not normally soluble. When this fissure water is exposed by mining activities, the gases in solution are released into the atmosphere. Hydrogen sulphide is a gas that often causes this type of problem. It is therefore also important to take the necessary extra measures to dilute the air, for example by having available additional air resources (such as compressed air) at the source, when gas release occurs underground.

11.1.4 Gas Layering

Gases that are released from the rock strata have the tendency to form layers on the hanging wall of the excavation if not dispersed by ventilation. This layering can persist for distances up to 100 metres or more. Generally the layering will move in the same direction of the movement of air, but in some cases can also move against the general direction of flow of air.

Factors that influence layering are:

Rule

- The rate of release of the gas
- The density of the gas
- The inclination of the airway
- The amount of turbulence of the air (the higher the turbulence, the least the chances of layering).

It is therefore important that areas where layering can take place are identified.

For this purpose there is an Equation that can be used to predict whether layering will take place or not, it is as follows:

$$L.N = \frac{U}{\left(g\frac{1-SG}{SG}\frac{R}{D}\right)^{1/3}} \qquad \text{.......................................} \ (11.1)$$

Where:

L.N	=	Layering number
U	=	Velocity of the air (m/s)
R	=	Rate of methane emission (m^3/s)
D	=	Width of the airway (m)
g	=	Gravitational acceleration (9.79 m/s2)
SG	=	Specific gravity of the gas relative to air

Rules

If the layering number **is less than 5**, then the chances are good that **layering will take place** in the specific workplace. For values greater than 8 the chances for layering taking place is quite slim. For values between 5 and 8, other factors may determine whether layering will take place or not. These are:

- The factors such as the inclination of the air way
- The roughness of the hanging wall
- Other physical factors such as shape, size and so forth.

Definition

To the calculation pertaining to layering of the specific gas considered, the rate of gas release must be determined first or estimated. This can be done by measuring the concentration of the specific gas in the air mixture. The best way of getting rid of layering is with the use of a '**jet' fan** (fan in which the exit nozzle has been reduced) that forces air at a high velocity over the layering and in this way gets rid of the layering of the gas. In this way the gas will also be diluted. Where underground air velocities are slow, it increases the chances that layering will take place.

11.1.5 Dealing with Gas Sources Underground

There are several ways of dealing with a large amount of unwanted gases underground, these are:

- Removal at source (through the application of engineering designs to take care of the gases)
- Dilution through the supply of extra fresh air
- Wearing of personal protective clothing such as gas masks
- Administrative procedures such as removal of people from dangerous areas (managing gas concentration below dangerous limits).

The most applicable and extensively used methods are the dilution of the gas within the specific working environment and/or the removal thereof through mechanical means (filters and fans). Dilution and removal of the gases will be discussed in the sections to follow.

11.1.5.1 Dilution and Removal of Gases

One of the most common ways of dealing with gases is the dilution thereof through the supply of extra non contaminated (fresh) air. It is important to understand how to calculate gas concentrations as well as release rates from underground work places and then to be able to calculate the amount of air that will be needed to dilute the gas to within acceptable limits. The section that follows will deal with it.

CALCULATION OF GAS CONCENTRATION AND DILUTION

An Equation that can be used to calculate the percentage of gas in any gas mixture is as follows:

$$\%\ \text{gas} = \frac{\text{total quantity of gas } (q_{gas})}{\text{total quantity of mixture } (q_{mix})} \times 100\%$$

$$= \frac{\text{total quantity of gas } (q_{gas})}{\text{total quantity of mixture } (q_{gas} + q_{fresh\ air})} \times 100\% \quad \text{.....................} \ (11.2)$$

It must be remembered that the following must be kept in mind:

- The quantity of gas and air can be in m^3/s or kg/s provided that the same unit is used for the gas and the air (either m^3/s or kg/s, not both)
- The mixture refers to the sum of the gas and fresh air quantities or mass flow.

WORKED EXAMPLE 1

A force fan delivers 5.2 m^3/s fresh air to a development end. The concentration of methane in the return air from the end is measured as 4.5%. How much **additional** air must be supplied by the fan to ensure that the methane will be diluted to within acceptable standards?

Answer

To answer this question the student must be able to calculate the methane emission rate (through manipulation of Equation 11.2 and the percentage methane has been given as 4.5%) and **also know** that the acceptable methane concentration percentage is 1.4%.

$$\% \text{ gas} = \frac{q_{CH4}}{q_{CH4} + q_{air}} \times 100\%$$

$$4.5\% = \frac{q_{CH4}}{(q_{CH4} + 5.2)} \times 100\%$$

Through simple manipulation of the formula it can be determined that the methane emission rate is 0.245 m^3/s.

The second part of the question can now be done. Let the quantity required by the fan be q_{fan}. By using the above Equation the calculation can be done as follows. Assume a minimum concentration percentage for methane as 1%. This will be sufficient as the minimum percentage acceptable methane concentration is 1.4%. An oversupply of air will take place, which is not wrong (but needs to be managed accordingly, due to the high cost of electricity). This is sometimes necessary due to leakages that take place in the force columns.

$$\% \text{ gas} = \frac{q_{CH4}}{q_{CH4} + q_{air}} \times 100\%$$

$$1.0\% = \frac{0.245}{(0.245 + q_{fan})} \times 100\%$$

Through manipulation of the formula, the value for the air supplied by the fan is 24.3 m^3/s. The question is not finished yet as the question asked how much additional air must be supplied by the fan. The **additional** amount that must be supplied is therefore 19.1 (24.3 – 5.2) m^3/s. Therefore read the question and answer what is asked.

There is also an easier way of calculating the amount of fresh air required for dilution purposes. In using this Equation the maximum allowable concentration (MAC) in ppm. for the specific gas must be known; the normal amount (N) of that gas (ppm.) in air; as well as the rate of emission of the gas (q_{gas}).

$$Q_{fresh} = \frac{q_{gas} \times 10^6}{MAC - N} - q_{gas} \quad \text{.................................} \; (11.3)$$

The reason for the '- q_{gas}' term is that the amount of fresh air needed will then be the resultant answer. If the gas in question does not appear in normal fresh air, then the value of N is zero. There is no carbon monoxide in normal fresh air and therefore the value of N is zero. The value of N for carbon dioxide is 300 ppm.

Managers however can state their own acceptable concentration less than the guideline figures.

This means that if a manager specifies 50 ppm for CO as its OEL more air will have to be supplied (test this statement with a calculation).

In this case the MAC will be 50 ppm and not 100 ppm. Both the terms below the line in the Equation must be in ppm. It is sometimes necessary that a percentage be expressed in ppm. For example 0.2% can be expressed as follows:

0.2 parts per 100

2 parts per 1 000

200 parts per 10 000

2 000 parts per 1 000 000

WORKED EXAMPLE 2

How much fresh air must be pointed at a source of CO gas that emits an amount of 0.0003 m^3/s of carbon monoxide, to ensure that a concentration of 0.005% is not exceeded?

Answer

NB

It is important to establish the percentage of CO in ppm. The concentration of 0.005% is equal to 50 ppm. (required MAC value). The calculation can be done as follows:

$$Q_{fresh} = \frac{q_{gas} \times 10^6}{MAC - N} - q_{gas}$$

$$= \frac{0.0003 \times 10^6}{50 - 0} - 0.0003$$

$$= 5.9997 \text{ say } 6 \ m^3/s$$

WORKED EXAMPLE 3

The following readings were taken in the return air from a fire:

Quantity of air (including gases)	1.4 m^3/s
Carbon dioxide	**0.52 % by volume**
Carbon monoxide	0.15 % by volume
Barometric pressure	95 kPa
Temperature of the fire	92ºC

If 1 kg of timber burns to produce a total of approximately 2 kg of carbon gas (CO plus CO_2), calculate the rate at which the timber is burning.

Answer

NB

The total quantity of the air including gases is 1.4 m^3/s. The Carbon dioxide (CO_2) content is 0.52% by volume. However there is typically 0.03% or 300 ppm by volume of CO_2 present in normal air, the fire therefore produces 0.49% (0.52 – 0.03). This is an important point which is sometimes missed by students. The quantity of the gas can therefore be determined as follows:

$$\text{Quantity } CO_2 = 1.4 \times 0.49$$

$$= 0.00686 \ m^3/s$$

Determine the mass flow of CO_2 by multiplying the SG of CO_2 by the density of the air and in this instance a density of 1.0 kg/m^3 for the mixed air (which includes the gas) would be a reasonable assumption. However, for accuracy the universal gas law must be applied, because the temperature of the fire was given.

The specific Equation can be calculated as follows and then the specific density of the air can be determined as well:

$$\frac{Pv}{T} = R$$

$$v = \frac{RT}{P}$$

$$= \frac{0.2871 \text{ kJ/kg} \times (273 + 92)}{95}$$

$$= 1.1031 \text{ m}^3/kg$$

The density is the inverse of v

$$= \frac{1}{1.1031}$$

$$= 0.907 \text{ kg/m}^3$$

NB

Since CO_2 is approximately 1.55 times heavier than air (**note, this is one of the catches to the question**), and the air density calculated as 0.907 kg/m^3, the mass flow of the CO_2 produced can be established:

The density of the CO_2

$$= SG_{gas} \times \text{density of air}$$

$$= 1.55 \times 0.907$$

$$= 1.4059 \text{ kg/m}^3$$

The mass flow of carbon dioxide from the fire can now be determined as follows:

$$\text{Volume of the } CO_2 = \frac{\text{mass of } CO_2}{\text{density of } CO_2}$$

$$\text{mass flow of } CO_2 = \text{Volume of } CO_2 \; x \text{ density of } CO_2$$

$$= 0.00686 \times 1.4059$$

$$= 0.00964 \text{ kg/s } CO_2$$

Now consider carbon monoxide. In the case of CO where there is none in the general air stream, the following can be said: *"Since CO is approximately 0.97 times lighter than air (**note, this is one of the catches to the question**), and the air density calculated as 0.907 kg/m^3, the mass flow of the CO produced can be established."*

The density of the CO	=	SG_{gas} x density of air
	=	0.97 x 0.907
	=	0.8798 kg/m^3
The quantity of CO	=	1.4 x 15/100
	=	0.0021 m^3/s

$$\textit{Volume} \text{ of the CO} = \frac{\text{mass of CO}}{\text{density of CO}}$$

massflow of CO	=	Volume of CO *x* density of CO
	=	0.0021 x 0.8798
	=	0.00185 kg/s CO

The total mass of carbon gases produced by the fire can be determined as follows:

$$= CO_{\text{mass flow}} + CO_{2\text{ mass flow}}$$

$$= 0.00185 + 0.00964$$

$$= 0.01149 \text{ kg/s}$$

The following ratios were given and the rate at which the timber burns can now be determined:

If 1 kg of timber produces 2 kg of gas, then '?' kg of timber produces 0.01149 kg/s?

$$\text{timber burn rate} = \frac{1 \text{ kg x } 0.00149 \text{ kg/s}}{2 \text{ kg}}$$

$$= 0.00574 \text{ kg/s}$$

WORKED EXAMPLE 4

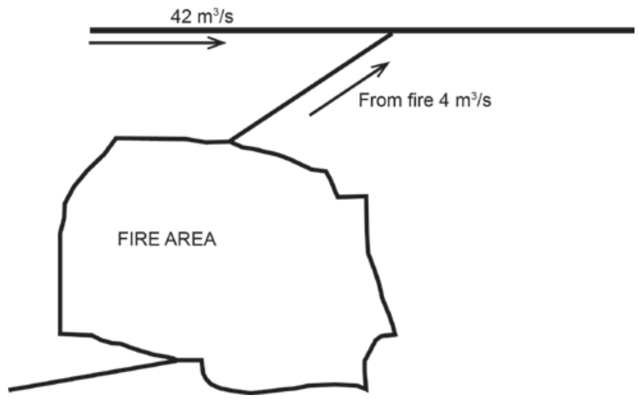

Figure11.1 Diagrammatical representation of gases from fire

Figure 11.1 refers. A volume of 4 m^3/s of fumes from a fire in a stope containing 3.8% CO_2 and 0.32% CO mixes with 42 m^3/s of mine air containing 0.08% CO_2 and no CO. The following needs to be determined:

(i) The concentrations of gases after mixing

(ii) The stope where the fire has been detected is completely sealed off. After sealing, the average air temperature increased from 35ºC to 42ºC (this normally relates to the dry bulb temperature – the sensible heat). If the barometric pressure before sealing was 105 kPa and assuming that there is no leakage of air and no heat flow to or from the rock, what is the approximate pressure difference across the seals? Assume that the barometric pressure outside the fire seal remains constant.

Answer

(i)	Quantity of CO_2 from stope	=	% CO_2 x Q_{stope}
		=	0.038 x 4
		=	0.152 m^3/s
	Quantity of CO_2 in mine air	=	% CO_2 x $Q_{mine\ air}$
		=	0.0008 x 42
		=	0.0336 m^3/s
	Total quantity of CO_2	=	0.152 + 0.0336
		=	0.1856 m^3/s
	Total quantity of mixture	=	4 + 42
		=	46 m^3/s

$$\% \; CO_2 = \frac{q_{CO_2}}{Q_{mix}} \times 100\%$$

$$= \frac{0.1856}{46} \times 100\%$$

$$= 0.403 \; \% \; CO_2 \text{ (or 4 030 p.p.m.)}$$

Quantity of CO from stope	=	% CO x Q_{stope}
	=	0.32 x 4
	=	0.0128 m^3/s
Quantity of CO from mine air	=	NIL
Total quantity of CO	=	0.0128 m^3/s
Total quantity of mixture	=	46 m^3/s

$$\% \; CO = \frac{q_{CO}}{Q_{mix}} x100\%$$

$$= \frac{0.0128}{46} x100\%$$

$$= 0.0278 \; \% \; CO \text{ (or 278 ppm)}$$

(ii) By using the Universal gas law the pressure difference across the seals can be determined:

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Where:

T_1	=	35 + 273	=	308 K
T_2	=	42 + 273	=	315 K

Assume that the volume remains constant (area does not change), from the gas law above P_2 can be determined:

$$P_2 = \frac{P_1 \, T_2}{T_1}$$

$$= \frac{105.4 \times 315}{308}$$

$$= 107.85 \text{ say } 107.9$$

The approximate pressure difference across the seals

$$= P_2 - P_1$$

$$= 107.9 - 105.4$$

$$= 2.5 \text{ kPa}$$

11.1.5.2 Removal of the Gas

Definition

In the underground working environment there are various ways of removing gas from work areas (with specific reference to development ends). The most common used is the **force exhaust method,** which ensures that the gas is immediately removed from the work area by means of an exhaust fan whilst fresh air is delivered to the face by means of a force fan. This fan is normally flame proof. When there is no gas emission in the development end, the air quantity distribution for a typical example is shown in Figure 11.2. No leakages are assumed. Note the various quantities in the diagram.

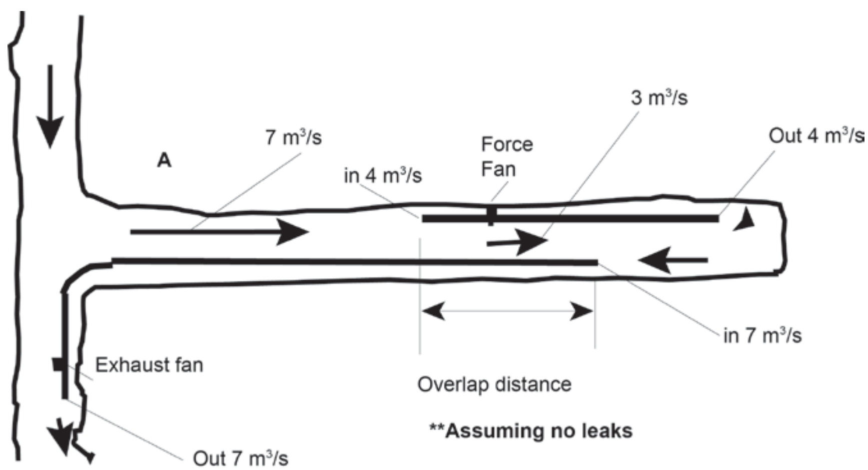

Figure11.2 No gas emission, air quantity distribution in end

NB

However, **when a gas emission is taking place** in such a development end, some major changes take place. The most important is the amount of fresh air that is drawn into the end is now reduced by the amount of gas released in the face.

In Figure 11.3 this fact is clearly shown. Also take note of the effect on the other quantities. The exhaust fan will still handle 7 m^3/s but it will be made up of 6.9 m^3/s of air and 0.1 m^3/s of methane (total of 7 m^3/s). The gas released in the end can also be from a diesel locomotive, but in this case will be gases formed in the engine through air sucked into the engine.

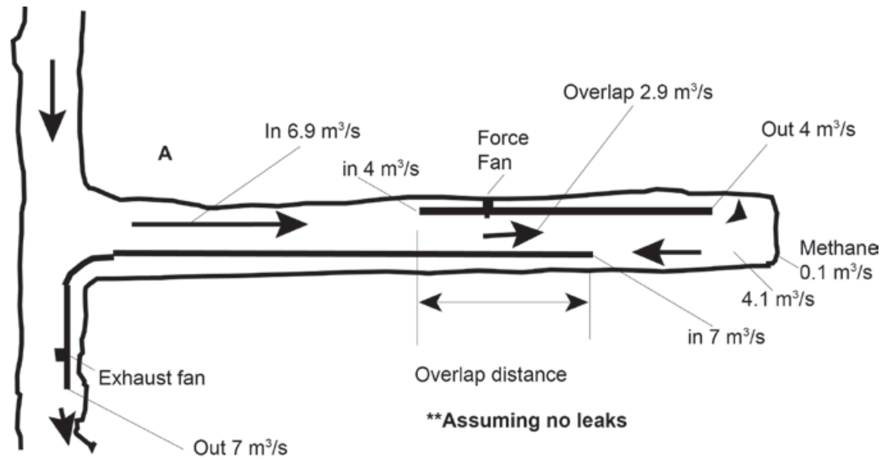

Figure 11.3 The effect of air distribution if gas emission takes place in end with force exhaust columns installed

11.1.6 Recirculation of Air in Development Ends

Definition

An important aspect that also needs to be considered here is the potential of **re-circulation of air**. This happens when the amount of air supplied by the force fan in a force exhaust system delivers more air than the amount of air been exhausted. This can become a serious issue to deal with as there can be a build-up of methane within the system if not enough contaminated air is removed from the workplace (and then also methane) to keep the concentration below the acceptable legal limit of 1.4%. Re-circulation of air can and has been done in certain mines before, but a proper risk assessment needs to be done before this is considered.

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NB

Recirculation of air is never allowed for coal mines.

Figures 11.4, 11.5 and 11.6, shows this effect quite clearly (no circulation (Figures 11.4 and 11.5) and Figure 11.6 which shows recirculation). In this example 6 m^3/s of air is exhausted from the development end and in the example figures the force fan quantity is increased from 2 m^3/s to 4 m^3/s to 12 m^3/s. Recirculation of air can also help to reduce the concentration of a gas on the face due to the larger amount of air being supplied.

!

However there is a concern. If the gas from the end is methane, then through recirculation of air there will be a build up of methane until a steady state condition (i.e. for the concentration of the methane gas in the face) has been reached.

It is therefore important to establish whether this steady state condition (methane gas concentration) will be below the maximum allowable concentration of 1.4%.

It must be noted that through a force exhaust system, fresh air is continuously supplied to the face by the force fan, but that contaminated air is also removed continuously, preventing a continuous build up of methane. A serious situation will however occur if the exhaust fan is stopped, in which case there will be a rapid build up of methane gas. The force fan in re-circulation conditions must be a flameproof fan (a fan in which the electrical switch gear has been sealed off).

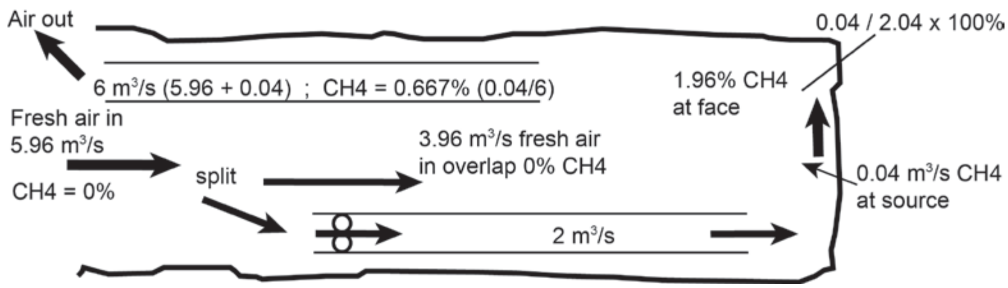


Figure 11.4 Force exhaust system 'low' force fan quantity

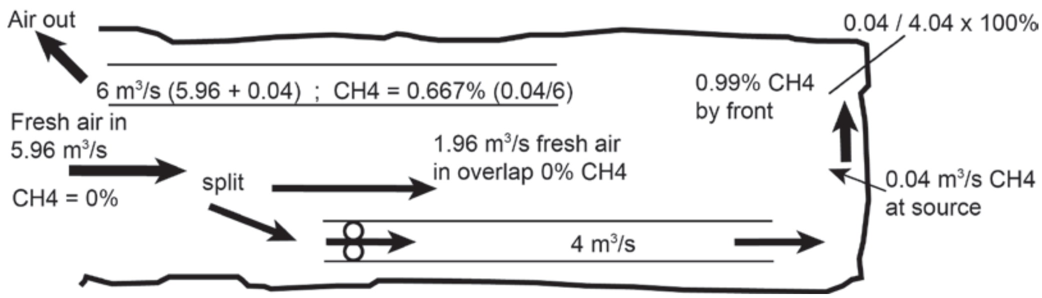


Figure 11.5 Force exhaust system 'higher' force fan quantity

The methane concentration in Figure 11.4 is 1.96%. With the increase in force quantity in Figure 11.5 the methane concentration is reduced to 0.99% (because of the higher flow in air supplied). The quantity of airflow across the overlap will be reduced from, 3.96 m^3/s to 1.96 m^3/s. This will mean a decrease in velocity of air in the overlap and can have the danger of a methane build up in the overlap.

When the force quantity is higher than the exhaust quantity, we have a situation where recirculation of air will take place. The results in Figure 11.6 are the steady state condition and the detail for the calculation is shown in the Excel spread sheet.

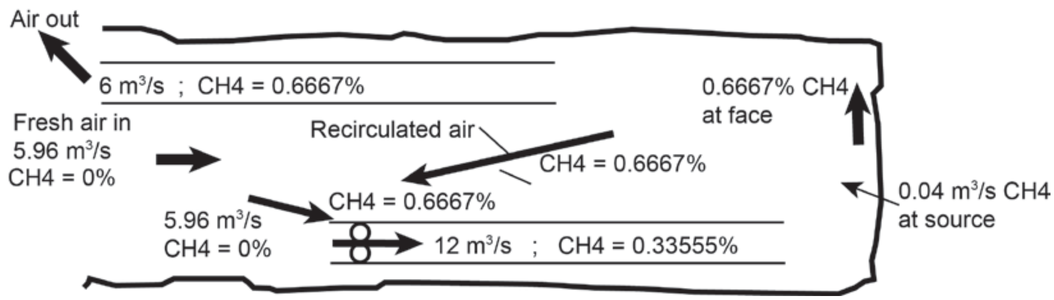


Figure 11.6 Force exhaust system, recirculation of air (steady state)

EXAMPLE QUESTION 5

A fan exhausts 5 m^3/s of air from a development end. A force column and fan delivers fresh air to the face. Methane is released at the face at a rate of 0.05 m^3/s. A typical force fan is available and delivers 7.5 m^3/s. What methane percentage will be present in the following areas? (refer to Figure 11.7)

- The exhaust column (1%)
- The overlap (1%, quantity of methane in overlap is 0.0255 m^3/s)
- Force column (0.34%)
- The face (1%)

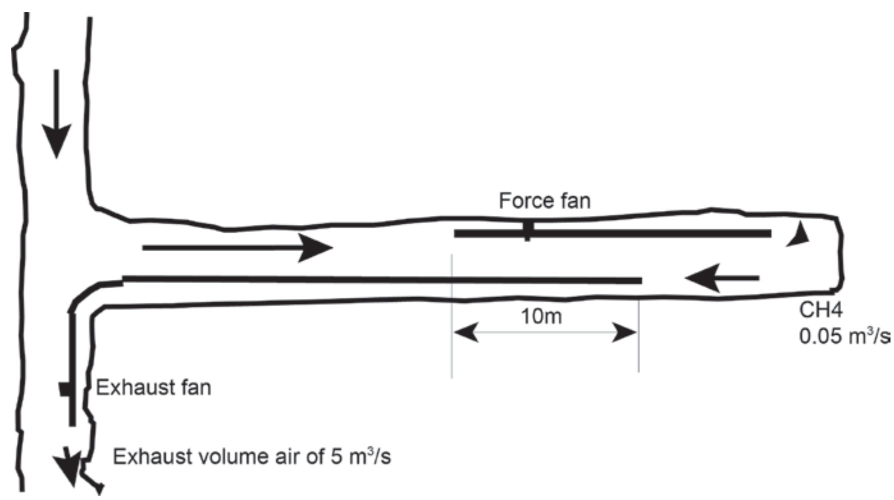


Figure 11.7 Force exhaust ventilation column layout

Gases from strata are normally encountered in the pre developed development tunnels. Generally speaking it is advantageous to ventilate development ends by means of force fans.

NB

It is important to note that the force fan must be situated outside the entrance to the development end (fresh air or through air). If this is not done, re-circulation of contaminated air will take place in the end. This is shown in Figure 11.8.

The advantages of the force column system are therefore the following:

Advantages

- All the gas in the tunnel is concentrated in the return air way. The largest concentration is therefore encountered where the return air joins the through ventilation
- The gas is blown out of the development end
- No additional ventilation equipment is necessary in the end
- Flammable gas is not drawn over the moving components or the motor
- The fan and switchgear can be placed in through ventilation
- The full volume of air is blown over the source and dilutes the gas completely.

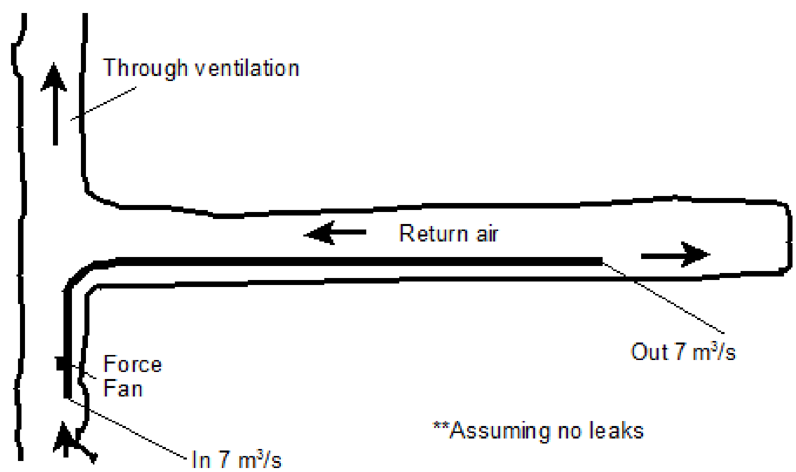


Figure 11.8 Force fan ventilation

References

For more detail on gases being released in underground mines read pages 539-542 of VOEE.

11.1.7 Gases from Surface Mining Operation

References

In mineral processing such as for gold, mercury is used. Mercury is very dangerous and mercury gas can be released into the atmosphere, which is very unhealthy. Hydrogen cyanide is also released in the mineral processing process. For more detail read pages 542-543 of VOEE.

11.1.8 Detection of Gases

The testing for gases is normally done to determine if the air contains dangerous concentrations, which is higher than specified OEL's. It is therefore important to get an immediate indication of the gas concentration so as to have remedial plans immediately in place if necessary.

The purpose of gas detection is therefore:

Objectives

- The detection of inflammable gas
- Determining the oxygen content of air
- Detection of noxious gases
- Sampling of gases
- Control of mine gases.

11.2 Flammable Gas Explosions

11.2.1 Introduction

Methane explosions in the coal mining industry have happened in the past as well but recently have also happened in the hard mining industry with disastrous consequences.

This section will highlight the dangers and severity thereof and other sections in this chapter cover the history of methane explosions in the hard rock mining environment extensively. The student will therefore have a detailed insight into the hazards and risks associated with methane explosions in the hard rock and coal mining environment. For the purpose of this course, all aspects with regards to flammable gas in the hard rock environment will be dealt with in detail.

The earliest recorded methane explosion resulting in the death of a miner occurred in 1621 at Gateshead, England (Duckham, H. and Duckham, B., 1973). As coal gradually became the major supplier of energy over the next several decades in the United Kingdom and Europe, more of these explosions took place. The first recorded major disaster in which 30 people lost their lives also occurred at Gateshead in 1705.

The earliest dust explosion recorded was in an Italian flourmill in 1785. As early as 1803, Buddle (Rice, 1911) pointed out that coal dust had burned and added to the violence of an explosion in an English colliery. This finding was based upon the testimony of witnesses who stated that they had observed 'sparks of burning coal'.

On 9 November 1815 Sir H. Davy read a memoir to the Royal Society of London on the ignition of methane (Humphrey, 1960). He made specific reference to the ignition of methane by electric sparks as well as the possibility that sparks created from a steel mill could act as an ignition source.

The explosions at Jarrow in 1834, Wallsend in 1835 and Springwell in 1837 provided further evidence of the involvement of coal dust in explosions where coked coal dust was observed. In the 1844 Haswell disaster report (Rice, GS, 1911, p. 11), Faraday concluded that "*In considering the extent of the fire from the moment of explosion it is not to be supposed that fire damp was its only fuel* " etc. The industry paid little attention to this view and favoured the view that coal dust did not play any part in a coal mine explosion.

Another ignition source risk, in the form of electricity, was introduced into the mines in 1882.

The measure taken to safeguard against this new risk was that electricity was only allowed to be introduced into areas of good ventilation. Furthermore, the incorporation of flameproof enclosures was intended to reduce the risk of open sparking to an acceptable level. It was initially accepted that low-power electric circuits were safe, but an explosion at Senghenydd in South Wales in 1913 was attributed to a signalling circuit as the ignition source. Following research by Thornton (1919) and Wheeler (1920), the concept of intrinsic safety was developed. By 1920 most of the principles of intrinsically safe circuits were already known.

The introduction of permitted explosives, as early as 1890, which continued until the early part of the 20th century, helped to reduce the risk associated with explosions of this kind.

In the USA coal production expanded rapidly from 160 million tons in 1890 to 270 million tons in 1900. In parallel with this, there was an increase in the death toll from explosions over these 10 years. A total of 1 473 people lost their lives in almost 300 explosions during this time; in 38 of these explosions, 1 006 fatalities were recorded. The number of fatalities increased further to 3 912 for the period 1901 to 1910.

This led to the establishment of the United States Bureau of Mines Bruceton Research Centre to help devise and develop protective measures against explosions associated with coal mining.

Verakis and Nagy (Verakis and Nagy, 1987, p 343) reported on the first people involved with the investigation and the recording of mine explosion accidents.

Although Faraday had drawn attention to the possibility of coal dust extending the flame length, and Galloway, an English inspector, had proved this experimentally in the 1870s, such a view was highly unpopular.

In 1876 Hall (also an English inspector explosions) an explosion at the Seaham Mine in England in 1880 resulted in 164 deaths. As part of the investigation, Abel was tasked with performing moderate-scale tests with coal dust.

Thus the definitive determination of coal dust as an explosion hazard has been credited to England.

Most of the major research initiatives to study and ultimately prevent explosions started around the turn of the century and normally followed some major disaster. The 1906 Courriers disaster in France resulted in 1 099 fatalities. This explosion occurred in a mine where methane had never been encountered before. In the official report it was noted that the explosion was initiated by shot firing.

This disaster shocked the world and brought the issue of preventing these occurrences to the forefront.

This signalled the start of more extensive experimental work. Taffanel started in 1907 at a small surface gallery at Lievin, France, which later expanded into larger galleries (Taffanel, 1913). Throughout the world, the pioneering research in this area was led in France by Taffanel, in Russia by Chennitsyn, in Poland by Czaplinski and Jicinsky, in America by Rice and in Britain by Wheeler and Tideswell.

In the 20th century, many large-scale methane and coal dust explosion galleries were constructed; these include galleries in the USA, the UK, Poland, Germany, Russia, China and South Africa.

Since the 1930s, increasing mechanisation has introduced the risk of frictional ignitions into working headings in which the interaction of cutting tools with quartz bands can result in incentive sparking capable of igniting methane. In recent times the introduction of powerful coal winning machines has increased the number of frictional ignitions. For example, the mine disaster at Glace Bay, Nova Scotia in 1979 claimed 12 lives.

This explosion was attributed to a frictional ignition and the 1993 explosion at Middelbult Colliery in South Africa was also found to be the result of friction at a continuous miner pick.

Many safety measures were researched and developed during the last century. Of these, the following have been instrumental in reducing the frequency and severity of explosions:

- Inerting practices
- Explosion barriers
- Knowledge and understanding of explosions and their prevention
- Development of permitted explosives
- Flame proofing and intrinsically safe electrical apparatus
- Improved ventilation practices
- Improved flammable gas-monitoring devices.

Although intense research attention has been paid to the development of preventive and protection measures against coal mine explosions, numerous disasters still occur.

"Of all the risks inherent in coal mining, the one most feared by coal miners is an explosion. Explosions are not the biggest cause of loss of life. On a statistical basis, explosions may be amongst the less frequent events in mining causing loss of life but, apart from fires and flooding, there is no other cause capable of wiping out the entire workforce below ground at the time."

These words, written by Joseph Dickson, an Inspector of Mines in the Lancashire coalfield, United Kingdom, in 1850 remain equally valid today.

11.2.2 Summary of the History of Explosions

Table 11.3 summarises the history of coal mine fatalities in the UK, USA and Australia until 1996.

The Pike River Mine explosion was a coal mining disaster that began on 19 November 2010 in the Pike River Mine, 46 kilometers northeast of Greymouth, in the West Coast region of New Zealand's South Island.

Table 11.3 History of explosions and associated fatalities

Period	United Kingdom[1]		USA[2]		Australia[3]	
	Number of Explosions	Fatalities	Number of Explosions	Fatalities	Number of Explosions	Fatalities
1850-1860	15	1 069	3	84	-	-
1861-1870	18	1 409	2	86	-	-
1871-1880	20	1 742	12	88	-	-
1881-1890	17	1 122	22	515	4	88
1891-1900	5	661	38	1 006	3	21
1901-1910	9	1 008	113	3 301	3	104
1911-1920	4	682	81	2 144	0	0
1921-1930	3	118	104	2 517	5	108
1931-1940	9	566	38	752	2	6
1941-1950	8	231	37	678	1	4
1951-1960	9	235	19	293	2	4
1961-1970	3	59	35	300	0	0
1971-1980	2	15	5	37	4	46
1981-1990	0	0	12	70	2	12
1991-1996	0	0	44	15	1	11

Notes:

[1] Duckham and Duckham, 1973, Langdon, 1997.

[2] Humphrey, 1960, Richmond et al., 1983, Dobrowski et al., 1996

[3] NSW Dept Mineral Resources Annual Reports, Qld Govt Mining Journals 1900-1925, Qld Mines Dept Annual Reports 1925-1996.

In a study in the UK by Tideswell (1955), the author concluded that the reduction in explosion incidences could be attributed to "*an increasing appreciation of the explosion hazard and an increasing degree of managerial and inspectorial control*". It is clear from Table 11.3 that both the number of explosions and the associated fatalities have decreased as the knowledge and preventive technologies have been developed and implemented in mines. Although coal mining operations in South Africa commenced in 1874 at Molteno in the Eastern Cape, the first recorded explosion occurred in 1891 at Elandslaagte Colliery in Natal.

Since then more than 330 explosions have caused the death of more than 1 000 coal miners (Phillips and Landman, 1993). This number includes the Middelbult explosion of 1993 in which 53 miners lost their lives.

Table 11.4 shows the ten worst explosions in the history of South Africa (Landman, 1992).

Table 11.4 The ten worst explosions in South African history

Colliery	Date	Injuries	Fatalities
Durban Navigation	08/10/1996	0	125
Natal Navigation	16/05/1923	4	78
New Mansfield	31/07/1935	0	78
Hlobane	12/09/1983	8	68
Middelbult	12/09/1944	13	56
Glencoe	13/05/1993	0	53
Burnside	13/02/1908	5	50
Ermelo Mine Services	20/05/1980	0	38
Middelbult	09/04/1987	11	35
Middelbult	12/08/1985	7	34

A summary of the history of explosions over the past century in the South African mining industry is shown in Table 11.5 (Landman, 1992; Phillips and Landman, 1993).

Table 11.5 History of explosions in the South African coal mining industry

Period	Number of Explosions	Injuries	Fatalities
1900-1909	26	52	140
1910-1919	33	57	27
1920-1929	51	63	193
1930-1939	19	28	119
1940-1949	25	72	184
1950-1959	17	53	59
1960-1969	31	31	32
1970-1979	29	41	30
1980-1989	57	73	179
1990-1993	26	34	63

It is evident from the table that South Africa has a long history of mine explosions. Phillips and Landman (Phillips and Landman, 1993) concluded that the huge changes in the number of men at risk, the tonnage produced and the mining methods required further investigation.

See summary of some of the major coal mine explosions (1962 onwards) international disasters with multiple fatalities (> 30 people killed) is given.

The continued occurrence of coal mine explosions in recent times shows that this type of incident is still a major problem in a number of countries. Details of coal mine explosions during the years 1994, 1995, 1996 and 1997 are given in Table 11.6:

- In 1994, 21 explosions resulted in 367 people losing their lives. This equates to a fatality rate of 17.5 per explosion
- In 1995, 19 explosions resulted in 296 fatalities, equating to a fatality rate of 15.6
- In 1996, 14 explosions resulted in 295 fatalities, equating to a fatality rate of 22.7
- In 1997, 33 explosions resulted in 809 fatalities, equating to a fatality rate of 24.5.

Table 11.6 Coal mine explosions and fatal injuries 1994 – 1997

Country	1994		1995		1996		1997	
	Number Killed	Number of Explosions	Number Killed	Number of Explosions	Number Killed	Number of Explosions	Number Killed	Number of Explosions
Australia	12	1						
Bulgaria	4	1					12	2
Chile	20	1						
China	104	6	211	13	251-258	6	630	19
CIS			34	4				
Columbia							16-20	1
Czech Republic					2	1		
India							4	1
Indonesia	5	1						
Iran							30	1
Kazakstan	27	1						
Norway (Russian Mine)							23	1
Pakistan	27	1			10	1		
Philippines	104	1			3-15	1		
Romania							3	1
Russia	5	2			15	2	84	5
South Korea							6	1
Spain			14	1				
Turkey			37	1	5	1		
Ukraine	60	4			6	1		
USA							1	1
Vietnam					3	1		

11.2.3 Analysis of the Trends of Explosions

The previous tables show that there is no clear trend in either the number of explosions or the number of fatalities from these explosions. In two studies, one in the USA (Dobroski, et al. 1996) and another in South Africa (Phillips and Landman, 1993), the researchers tried to determine the trends of explosions in their individual countries. The following sections summarise their findings, indicating the dependence of a number of different variables and the interdependence of changes in the method of extraction.

11.2.3.1 Analysis of Major US Ignition Sources

In the USA a major explosion is classified as one in which five or more fatalities occur. Since 1951 there have been 38 such explosions. No explosions have been recorded since 1994. In a study undertaken in the USA, the fuel source and ignition source of these explosions were identified. The analysis of this is shown in Table 11.7.

Table 11.7 Analysis of major US coal mine explosions by fuel and ignition source (1951-1994) (after Dobroski et al. 1996)

Fuel Source	Numbers Involved	Ignition Source	Numbers Involved
Methane	27	Electrical	19
		Smoking	4
		Flame Safety Lamp	1
		Cutting Torch	1
		Not Determined	2
Coal Dust	6	Explosives	5
		Explosives/Electrical	1
Methane & Coal Dust	3	Explosives	2
		Electrical	1
Distillate Products	1	Fire	1
Not known	1		

Methane was the most frequent fuel source of major explosions and electrical arcing or sparking the most frequent source of ignition. Investigation of the explosions showed that where stone dusting was up to standard, coal dust did not participate in the explosion and did not support flame propagation in the explosion. Where stone dusting was below standard; the presence of coal dust on the floor, ribs and roof assisted in propagation of the explosion far beyond the site of the ignition.

Of the six coal dust explosions, five were caused by explosives and in the other case the ignition source may have been either explosives or electrical. The coal dust explosions generally occurred due to people not complying with the requirements of the legislation. The shock wave generated when the explosive was detonated raised coal dust that had not been wetted or treated with inert dust and was subsequently ignited by the flame from the explosive.

For the three methane and coal dust explosions, both fuel sources were known to be present at the time of ignition. The use of explosives created a hybrid atmosphere in two cases and in the other case both fuel sources were present following a methane outburst.

The one explosion involving distillate products occurred after a fire, when the flame of the fire ignited the distillate products.

After one major explosion, the mine facilities were extensively damaged. There were subsequent explosions and the mine was then sealed at the surface. No underground investigation of the explosion was possible and neither the fuel source nor ignition source were determined. For minor ignitions and explosions in US coal mines, the number of injuries and fatalities are analysed for 10-year periods starting in 1951. These are shown in Table 11.8.

Table 11.8 Injuries and fatalities in minor US coal mine ignitions and explosions (1951-1994)

PERIOD	Number of Ignitions & Explosions	Number Injured	Number Killed
1951 – 1960	242	284	45
1961 – 1970	323	312	46
1971 – 1980	625	89	6
1981 – 1990	891	88	11
1991 – 1994	390	41	7

Although the number of ignitions and explosions increased steadily during this period, there was a reduction in the number of people injured and killed. Table 11.9 shows the ignition sources that accounted for the majority of ignitions and explosions over the period 1951 to 1994.

Table 11.9 The most frequent ignition sources of US coal mine ignitions and explosions (1951-1994)

Period	**The Most Frequent Ignition Sources**		
	Highest	**Second Highest**	**Third Highest**
1951 – 1960	Elect. Arcing	Explosives	Naked Lights
1961 – 1970	Cont. Miner	Elect. Arcing	Smoking
1970 – 1980	Cont. Miner	Cut & Welding	Longwall
1981 – 1990	Cont. Miner	Cut & Welding	Longwall
1991 – 1994	Cont. Miner	Longwall	Cut & Welding

Since 1961 to the present, frictional ignitions at the continuous miner have been the largest single cause of ignitions, followed by cutting and welding and the longwall shearer. The pattern of injuries and fatalities for minor coal mine ignitions and explosions in the USA have also changed over this period.

Although continuous miner operations were responsible for the highest percentage of injuries, they were not the major cause of fatalities.

Electric arcing was the major cause of fatalities over the whole period.

When the risk of injuries and fatalities is analysed for continuous miners and longwalls, it becomes apparent that there have been significant changes over the review periods.

Comparisons of non-fatal injuries and fatalities are shown separately in Table 11.10 and Table 11.11 respectively.

Table 11.10 Comparison of non-fatal injuries for USA coal mines (ignitions and explosions)

Period	Continuous miners			Longwalls		
	Number of Ignitions	Number of Times Injuries Occurred	Frequency of Injury Occurrence (%)	Number of Ignitions	Number of times Injuries Occurred	Frequency of Injury Occurrence (%)
1961-1970	162	62	39	0	0	0
1971-1980	433	25	5.7	37	0	0
1981-1990	479	17	3.5	154	1	<1
1991-1994	260	6	2.3	46	0	0

For the period 1961-1970, an injury or injuries occurred in 39% of ignitions in operations associated with continuous miners. Over the subsequent decades the number of ignitions increased substantially, but the number of times that ignitions caused injuries decreased, as did the frequency of injury occurrence.

NB

The results for longwall operations clearly show that this method of mining has a much lower risk of injury due to ignitions of methane. Part of the increase in the number of longwall ignitions may be accounted for by the greater use of this method of mining.

The steady decrease in injuries is probably attributable to a number of factors, including improved cutting pick design, more effective water sprays, lower cutting speeds, greater awareness of the hazard of blunt picks and better ventilation on the longwall which limits the volume of methane present in the explosive range at the time of ignition. A similar comparison for continuous miner and longwall operations can be made for fatal injuries that occurred over the same period. The analysis is shown in Table 11.11.

Table 11.11 Comparison of fatal injuries for US coal mine ignitions and explosions

Period	Continuous miners			Longwalls		
	Number of Ignitions	Number of times fatalities occurred	% Risk of fatality	Number of Ignitions	Number of times fatalities occurred	% Risk of fatality
1961-1970	162	5	3	0	0	0
1971-1980	433	0	0	37	0	0
1981-1990	479	1	<1	154	0	0
1991-1994	260	0	0	46	0	0

Table 11.11 shows a marked decline in the percentage risk of fatal injuries due to ignitions and minor explosions in continuous miner operations. For longwall faces during the review period, no fatal accidents occurred due to ignitions or minor explosions, this again being the result of the lower risk with longwall operations.

11.2.3.2 Analysis of South African Ignitions

In an attempt to establish a change in the pattern of ignitions, Phillips and Landman (Phillips and Landman, 1993) undertook a comprehensive study in South African mines. They concluded that there was a steady increase in frictional ignitions following the introduction of continuous miners in the early 1970s together with a decrease in the events associated with blasting. In a further attempt to arrive at a more realistic view, a detailed analysis for the period 1984 to 1993 was done. During the analysis period, 51 incidents were reported at an average rate of 5.1 per annum. Of these, 40 were classified as ignitions and 11 as explosions. In these incidents 136 miners lost their lives.

Sources of ignition can be divided into three major categories, namely:

- Frictional
- Flame
- Electric ignitions.

Frictional ignitions include ignitions due to the cutting picks of continuous miners or road headers contacting stone, the cutting picks of coal cutters, the cutting picks of shearers and ignitions due to stone-on-stone or stone-on-steel friction.

Flame ignitions include those due to electric sparks and lightning forming stray currents.

In order to assess the influence of mining method on the occurrence of explosions and ignitions, the data for the period 1982 to 1993 have also been classified according to the location of the incident and the mining method involved. This analysis is presented in Table 11.12.

Table 11.12 Location of incident epicentres in South African underground collieries for the period 1982 to 1993

Location	Ignitions	Explosions	Total	%Total
Face Area				
Bord-and-pillar	25	10	35	55
Longwall development	4	0	4	6
Longwall (face)	3	0	3	5
Abandoned areas				
Pillar extraction goaf	2	3	5	8
Longwall goaf	1	0	1	1.5
Non-face areas				
Intake airways	0	2	2	3
Return airways	0	2	2	3
Shaft areas	0	1	1	2
Unknown	11	0	11	16.5

It is clear from Table 11.12 that 66% of all incidents occurred in the face area. However, many of these were frictional ignitions which were localised and not lethal.

In considering the severity of the problem in each of the three areas of a mine, it is important to consider the number of fatalities caused. Since ignitions are localised events, the data for the 18 explosions during the period 1982 to 1993 have been re-examined. During these years explosions were responsible for 218 fatalities and 93 injuries. The ignition sources and the locations of these explosions are given in Table 11.13.

Table 11.13 Sources of ignition and location of explosions in South African Underground collieries in relation to fatalities for the period 1982 to 1993

Ignition source	**Incidents**	**Injured**	**% Injured**	**Killed**	**% Killed**
CM picks	6	37	40	2	1
Electricity	3	12	13	69	32
Blasting	3	11	12	41	19
Lightning	2	1	1	7	3
Stone on stone	1	11	12	11	5
Spontaneous combustion	1	10	10	0	0
Unknown	2	11	12	88	40
TOTAL	**18**	**93**	**100**	**218**	**100**
Location					
Bord-and-pillar	10	53	57	78	36
Goaf/Abandoned	3	10	11	59	27
Intakes	2	7	7	34	15.5
Returns	2	22	24	46	21
Shafts	1	1	1	1	0.5
TOTAL	**18**	**93**	**100**	**218**	**100**

NB

Philips concluded that although the number of fatalities from explosions in the face area, abandoned areas and out bye areas are similar (78, 59 and 81 respectively), it must be remembered that many of the non-face explosions occur close to working sections and so the face area remains of extreme importance.

11.2.4 Conclusions from Historical Facts

It is evident that the severity of major coal mine explosions in the West has shown a steady decrease. This can be attributed to a gradual growth in the understanding of the propagation of explosions and the development of technologies to prevent and protect against them.

Some of the major advances made in the last century include the development of:

- Flame safety standards
- Intrinsic safety principles
- Use of stone dust
- Implementation of explosion barriers.

Furthermore the development of equipment and technologies has resulted in enormous improvements in productivity and this has reduced the number of people at risk.

It is also known that frictional ignition occurrences have not followed the same trend in producing multi-fatality explosions. Though the frequency is high, the reduction in severity has clearly shown a greatly reduced risk of major coal mine explosions.

While a review of the situation in mature coal mining countries could lead to complacency, this is not true of the lesser-developed coal mining countries. Recent major explosions have occurred in countries where the appreciation of the difficulties associated with proper prevention and protection technologies is not well understood, implemented and regulated.

These major incidents related to loss of life can all be attributed to a failure of compliance to well-accepted standards and human error. Although this statement is true it has also highlighted the need for further development of systems where the need for human interaction and the human interface is reduced.

As an example, the need to maintain passive explosion barriers in a high-quality state to ensure optimum operation characteristics was highlighted in numerous past explosion case studies. To this end, the development of a system or other systems that can reduce the risk of a major explosion, together with the awareness of the risk, are still the major focal points in the world's coal mining industry of today.

As the risk of an underground explosion can never be reduced to zero, research into the development of alternative protective measures and techniques continues. Eisner and Rae (1970) listed a **number of practical limitations of the passive barriers of the day.** They concluded that the effort required to install, maintain and advance the stone dust barrier was a formidable task. They also identified the need to develop a barrier that could be advanced quickly and without dismantling and re-erecting its individual components.

Some of the major findings from the historical facts:

- Explosions in coal mines led to an immense loss of life during the last century.
- The frequency of coal mine explosions in the more advanced countries has declined substantially since the nineteenth century and the early decades of the twentieth century.

- The number of people killed in each explosion has also declined due to the introduction of mechanisation, which in turn has led to improvements in productivity and a reduction in the number of people underground who were exposed to the risk of an explosion.
- New technology, including the use of electricity, introduced new hazards to coal mines but measures were progressively introduced to safely manage these hazards.
- Human factors appear to have been involved in the ignition sources of most major explosions.
- In some cases, ignitions of methane led to the propagation of coal dust explosions for substantial distances through the mine workings and with considerable violence because of an inadequate standard of stone dusting of the mine roadways.
- Coal dust explosions occurred far less frequently than methane explosions.
- Due to the implications of a coal mine explosion, it is doubtful whether a situation would ever be reached where it is considered that the risk has been reduced to acceptable levels.

11.2.5 Properties of Hydro-Carbon Flammable Gases

In the previous section the properties and occurrences of cases generally found in the underground environment have been highlighted. In this section the details pertaining to flammable gases will be dealt with.

NB

Methane, hydrogen and carbon monoxide are flammable gases that may occur in a mine and it is therefore important for us to know more about these gases. Some aspects such as their properties, occurrence and detection, the methanometer, layering numbers, protective measures have been dealt with. The Coward triangle and the use of the USBM diagram in terms of predicting the explosibility of a flammable gas mixture will be dealt with in the sections to follow. Methane is the most important of the three gases and it will therefore be discussed in the most detail. However, attention will also be given to the relationship between the three gases. Other flammable gases are also encountered underground and will also be dealt with in detail in the sections to follow.

Methane has the following basic properties and other more detailed information on the other properties of the hydro carbon gases will also be dealt with later:

- It is colourless
- It is odourless
- It has no taste
- It is lighter than air with a Relative Density (or Specific Gravity) of 0.55 in comparison with air
- It is a flammable gas, but does not sustain combustion
- The ignition temperature is 650ºC and it has a boiling point of 164ºC
- It burns at concentrations between 0 en 5% and when the concentration is higher than 15%, it is explosible between 5 and 15%, reaching maximum explosibility at 9%. At concentrations higher than 15%, it has a smothering effect (displaces O_2).

For the purpose of this course, we will also concentrate on the lesser known family of gases called the hydro carbons. Below follows some of the definitions, as defined by Bates et al. (1980) of all the possible natural hydrocarbon gases expected during underground hard rock mining.

11.2.5.1 Methane (CH_4)

A Colourless, odourless inflammable gas, the simplest paraffin hydrocarbon, formula CH_4. It is the principal constituent of natural gas and is also found associated with crude oil.

11.2.5.2 Ethane (C_2H_6)

A Colourless, odourless, water-insoluble, gaseous paraffin hydrocarbon, formula C_2H_6, which occurs in natural gas or can be produced as a by-product in the cracking of petroleum.

11.2.5.3 Propane (C_3H_8)

An inflammable gaseous hydrocarbon, formula C_3H_8, of the methane series. It occurs naturally in crude petroleum and natural gas. It is also produced by cracking and is primarily used as a fuel and in the making of chemicals.

11.2.5.4 Butane (C_4H_{10})

A gaseous inflammable paraffin hydrocarbon, formula C_4H_{10}, occurring in either of two isometric forms: n-butane, $CH_3CH_2CH_2CH_3$ or isobutane, $CH_3CH(CH_3)_2$. The butanes occur in petroleum and natural gas.

Some of the properties of the gases present are shown in Table 11.14.

Table 11.14 Hydro-Carbon Gas properties

Chemical Properties	Methane CH_4	Ethane C_2H_6	Propane C_3H_8	Butane C_4H_{10}	Hydrogen H_2
Explosive range[1] (%)	5.0 – 15.0	3.0 – 12.4	2.1 – 9.5	1.8 – 8.4	4.0 – 74.0
Minimum ignition temperature[2] (K)	813	788	723	678	not available
Combustion energy (MJ/m^3)	33.95	60.43	86.42	112.41	10.22
Maximum. burning velocity (m/s)	0.33 – 0.43	0.49	0.48	0.45	3.2 – 3.6

***Notes:** a. In air at atmospheric pressure and room temperature
b. At atmospheric pressure

[1] Gibbs and Calcote, J., 1959, Chemical Eng. Data, 4.226
[2] Rose, J. W. and Cooper, J.R., 1977, Technical data of Fuel, Seventh edition
[3] Zabetakis, M. G., 1965, Flammability characteristics of combustible gases and vapours, US Bureau of Mines, Bulletin 627.

Table 11.14 illustrates the major difference between paraffin hydrocarbons and hydrogen. The maximum pressure generated by methane is higher than for hydrogen when equivalent volumes explode. However, the rate of a methane explosion (i.e. the rate of pressure rise over time (dP/dt)) is lower than for hydrogen. The minimum ignition temperature drops from 813 K to 623 K for methane as the pressure increases from 1 atm (atmosphere) to 100 atm.

Definition

Burning velocity may be defined as the velocity, relative to the unburned gas, at which a one-dimensional adiabatic flame propagates normal to itself through a homogeneous gas mixture. Burning velocity is directly related to, inter alia:

- Reaction rate
- Minimum ignition temperature
- Rate of flame travel and pressure rise.

Rule

In a study done by Walters (1992), empirical work indicated that an increase in atmospheric pressure increased the burning rate for various materials tested. He concluded that an increase in air density increased the available oxygen per unit volume of air. In short, 'As air density increases, the burning rate increases'.

11.2.6 Mechanism of a Flammable Gas Explosion

Definition

An **explosion** can be defined as a process in which the rate of heat generation, temperature rise and pressure rise increase tremendously as a result of the impact of the combustion of the gas/air mixture. In a typical methane explosion, the temperature increases up to approximately 2 000°C. The speed of the reaction is so great that it is initially adiabatic. The result is that the pressure in the immediate environment increases very rapidly to reach a maximum and this is then relieved by expansion of the air. Consequently, a shock wave is generated in all directions. In mine workings, this expansion is restricted by airway tunnels, which leads to high air velocities in the specific tunnels.

If the explosion occurs in a development tunnel, the air can only expand in one direction and this can have catastrophic consequences. An explosion in which the air mixture contains 10% methane can reach air velocities of as much as 660 m/s (double the speed of sound). The shock wave is initially the moving boundary between the normal mine air atmosphere and the increased pressure of the expanding explosion. There is therefore a very high concentrated pressure gradient over the shock wave and the result is that everything in the path of the shock wave is 'blown away'.

Moreover, it has the effect of further mixing any other methane concentrations through which it moves, thus setting off a chain of explosions.

11.2.6.1 Gas Explosions

Most explosions at mines have been caused by the ignition of methane. Modern systems and standards in mines deal with and eliminate methane in the ventilating air very effectively. However, the greatest hazard comes from methane released from fissures and the strata, which sometimes contain as much as 90% methane. Dangerous accumulations of methane must therefore be prevented at high places by taking additional ventilation precautions.

In the event of a flammable gas explosion, a number of factors influence the rate of the explosion. The most important factors are:

NB

- Type of fuel
- Concentration
- Volume of fuel
- Strength of the ignition source.

Other factors frequently reported are:

- Pressure
- Temperature.

An increase in temperature normally widens the limits of explosibility. The effect of pressure is neither simple nor uniform. The variability effect arises from the combined influences of physical factors (heat losses) and chemical factors (reaction rates), which may be in opposition to one another.

References

The flammable gases (fuel) available were listed by Cook (1999, Personal Communications). The influence on the flammable limits was also calculated, using the formula derived by Le Chatellier. The major individual flammable gas constituents available were methane, ethane and hydrogen.

$$\frac{p_t}{L_t} = \frac{p_1}{L_1} + \frac{p_2}{L_2} + \frac{p_3}{L_3} + \frac{p_n}{L_n}$$

Where:

p : The individual gas concentrations
L_t : The final or total gas flammability of the mixture (either upper or lower limit)
L : The upper or lower limit of the individual gases in the mixture
P_t : The total % concentration of all the flammable gases.

WORKED EXAMPLE 6

Three gases, Methane, Butane and Hydrogen identified through a chemical gas analysis shows concentrations of 8%, 6% and 10% respectively. The formula to determine the upper or lower explosive range of a cocktail of flammable gases can be determined as follows:

$$\frac{p_t}{L_t} = \frac{p_1}{L_1} + \frac{p_2}{L_2} + \frac{p_3}{L_3} + \frac{p_n}{L_n}$$

Where:

p : The individual gas concentrations
L_t : The final or total gas flammability of the mixture (either upper or lower limit)
L : The upper or lower limit of the individual gases in the mixture
P_t : The total % concentration of all the flammable gases.

Determine the upper and lower limit of the cocktail of the flammable gas and comment on your answer in terms of the risks associated with this mixture.

Answer

The first part of this question is to mention the explosibility range of the gases mentioned. This is as follows (from memorization):

Methane:	5% and 15%
Butane:	1.8% and 8.4%
Hydrogen:	4% and 74%

The lower limit can be calculated as follows:

$$\frac{p_t}{L_t} = \frac{p_1}{L_1} + \frac{p_2}{L_2} + \frac{p_3}{L_3} + \frac{p_n}{L_n}$$

$$\frac{24}{L_{lower}} = \frac{8}{5} + \frac{6}{1.8} + \frac{10}{4}$$

$$L_{lower} = 3.229 \text{ say } 3.23\% \text{ (through manipulation)}$$

The upper limit can be calculated as follows:

$$\frac{p_t}{L_t} = \frac{p_1}{L_1} + \frac{p_2}{L_2} + \frac{p_3}{L_3} + \frac{p_n}{L_n}$$

$$\frac{24}{L_{upper}} = \frac{8}{15} + \frac{6}{8.4} + \frac{10}{74}$$

$$L_{upper} = 17.357 \text{ say } 17.4\% \text{ (through manipulation)}$$

From the results it is obvious that the lower limit has been reduced to lower than that of methane and that the upper limit has been reduced to very close to the upper limit of methane.

11.2.7 Abnormal Release of Methane Gas

Methane is released at a fairly constant rate from the rock strata and controlling it is reasonably simple. However, it is the unpredictable nature of methane that makes it so hazardous.

Various factors that influence the release and retention of methane have been investigated and those that may affect the release of methane are as follows:

- Variations in the flow of ventilating air
- Different mining methods: bord-and-pillar versus longwall mining
- Movement of strata

- Variations in barometric pressure
- Fluidity of the strata.

11.2.8 Ignition Sources

A number of possible ignition sources have been mentioned before (in the historical section) and are mentioned again:

11.2.8.1 Electrical

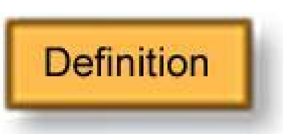
The use of flameproof or non flameproof equipment can be a very much discussed topic when the financial implications of flameproof equipment has to be considered.

When non flame proof equipment are in use, this aspect needs to be brought to the attention of all people concerned and the special precautions that have to be taken when these non flame proof are in use, needs to be looked at in detail.

11.2.8.2 Frictional

- Drilling equipment that can cause sparks on the face when drilling takes place, especially when dry drilling takes place (problems with water feed).
- Temporary steel support units (Camlok) and a hand-held pneumatic drill used can be the source of ignition as well (frictional sparks when rock and metal collide as would have been the case if they had fallen or had been dropped – that could have generated a spark that would have ignited the flammable gas).
- Another possibility that can be a source of ignition is a seismic event that could trigger several other sources of ignition, such as rock on rock sparking.
- The presence of contraband.

11.2.8.3 Methane Drainage Systems



Methane drainage involves ‘bleeding’ methane out of the strata surrounding the working environment. This is done where large quantities of methane have been released into the atmosphere, thus creating unsafe working conditions for the workers. In coal mines, methane is also released from the coal that is being mined.

The basic system used for draining methane is as follows. First, a pattern of boreholes is drilled into the surrounding strata in order to penetrate the methane-bearing fissures. These holes are then connected to a column of extraction pipes which can exhaust into the atmosphere or can in turn be connected to a low-volume pressure fan which exhausts into a safe area.

The gas accumulated in this way can also be used as a source of energy (for heating, etc.).

11.2.8.4 Methane Volume Measurements

Methane volume measurements are done in mines (especially coal mines) as a means of determining the volumetric flow of methane (m^3/s) in a specific district or environment.

The aim of such measurements is therefore to establish whether there are any changes in the rate of methane release and, if so, to determine the extent of the change.

When there is evidence of such a change, one needs to know whether the change is **normal** (change in the release rate concomitant with a change in the production rate) or **abnormal** (factors such as the barometric pressure, fans not being operational and, for example, seals that are inadequate).

An increase in the volumetric flow (m^3/s) of methane may lead to the creation of flammable or explosible methane-air mixtures. It is therefore important that the limits specified in legislation are not exceeded. Moreover, provision must be made during the planning stages for increases, for whatever reason, in the volumetric flow rate of methane.

However, economic aspects also have to be taken into account and unnecessary overprovision must be avoided since this can result in air being wasted.

11.2.9 Explosibility of a Flammable Gas Air Mixture

Various methods do exist to predict the possible explosibility of an air mixture. Gas samples are normally taken from an atmosphere where an accumulation of flammable gas may occur and analysed to ascertain the various gases present in the sample as well as the percentage make-up thereof. The results thereof, is then evaluated to determine whether a mixture is explosible or not. The two methods commonly used to determine whether an air mixture will be explosible or not is the use of the Coward's triangle and the USBM diagram.

These samples are normally taken from sealed off areas such as in the case in coal mines, but the use of the Coward triangle and the USBM diagram will clearly show the effect of the supply of fresh air to an environment in which a methane blower exist as well as the affect of the depletion of fresh air in such an environment.

11.2.9.1 The Coward Triangle

The gases that occur in an air sample can be grouped into three categories, namely:

- Air
- Flammable gases (methane, hydrogen and carbon monoxide)
- Inert gases (surplus nitrogen and carbon dioxide).

The Coward triangle was developed for the sole purpose of predicting the possible explosibility of an air mixture. Gas samples are taken and the results are calculated and plotted on the triangle, which then indicates whether or not the mixture is explosible.

The triangle was compiled for three gases, namely; methane, hydrogen and carbon monoxide.

Examine the graph of the Coward triangle shown in Figure 11.9.

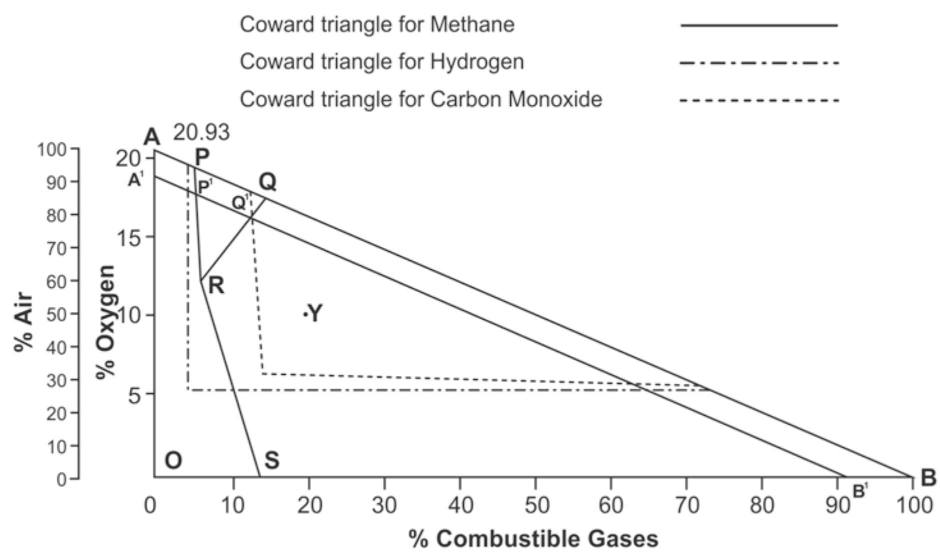


Figure 11.9 The Coward triangle

For the sake of simplicity only the triangle for Methane will be considered for now. From the graph we can make a number of conclusions. Point A represents 100% air and 0% inert gases. Point B represents 0% air, 100% flammable gases and 0% inert gases. Point O represents 0% air, 0% flammable gases and 100% inert gases. A point on the line OA represents a sample containing only air and inert gases (40% air on line OA means 60% inert gases).

A point on the line AB represents a sample containing only air and flammable gases (a point on line AB directly opposite 50% air means 50% air and 50% flammable gases). A point on the line OB represents a sample containing both flammable and inert gases (40% flammable gases on line OB means 40% flammable gases and 60% inert gases).

A point within triangle ABO is a sample containing air, flammable gases and inert gases. Point Y therefore represents a sample containing 50% air, 20% flammable gases and 30% inert gases. If methane is added to the fresh air, the point representing the sample will move away from A along line AB. When point P is reached, the sample contains 95% air and 5% methane, which is below the explosibility limit. A further increase in the methane concentration means that the point will move further along line AB until it reaches point Q. This point represents 86% air and 14% methane and is above the methane explosibility limit.

If we start at A’ (90% air, 10% surplus nitrogen) and then add methane, the point will move down along A’B’ and establish points P’ and O’, which represent the lower and upper limits for air with 10% nitrogen. It is clear that points P and Q are further from each other than points P’ and Q’, which indicates that the explosibility spectrum for P’Q’ is much narrower.

Definition

The addition of extra nitrogen will cause the explosibility spectrum to narrow even further until the two points merge at point R. This point is termed the ‘**nose value’**. All points within the triangle PQR will represent an explosible mixture of air, methane and surplus nitrogen. If other flammable gases (or mixtures) and other inert gases are handled in a similar manner, other triangles similar to PQR can be derived.

If air is added to the atmosphere, the point representing the atmosphere will move from its original position along a straight line in the direction of A (tending towards fresh air). If air is added to an atmosphere represented by a point within square SRQB, the atmosphere will move through the explosible triangle PQR, on its way towards A.

However, if the original point falls within the area encircled by the polygon APRSO, the addition of air will cause the explosibility triangle PQR to be missed out. This area therefore represents air mixtures that will never move through the explosiblity zone when additional air is added.

The addition of an inert gas to the atmosphere will cause the point representing the atmosphere to move away from its original position in a straight line in the direction of point O.

The Coward triangle is therefore a graphic method of representing mixtures of air, flammable gases and inert gases, but the explosibility triangle is based on the fact that methane is the only flammable gas and nitrogen the only inert gas in the mixture.

In South African coal mines, which normally have low methane-release rates, flammable gases in a fiery area are dominated not by a higher methane concentration, but by a complex composition of methane, hydrogen and carbon monoxide. As always, the inert gases are surplus nitrogen, carbon dioxide arising from the use of oxygen, and any inert gas that is added artificially. In this case the Coward triangle will not be applicable.

11.2.9.2 The USBM Diagram

The USBM diagram is an improved version of the Coward triangle and enables us to take into account the percentages of gases such as methane, hydrogen and carbon monoxide (Refer to Figure 11.10). A series of explosibility triangles are drawn for various values of R. R is the relation of methane to total flammable gases. The X and Y coordinates can be determined by using the formula given below the Figure. The point representing the sample can be plotted on the graph.

If the point lies within the applicable triangle on which the R value is based, then the atmosphere at the sampling point is explosible. Each explosibility triangle has a line from its bottom right-hand corner separating atmospheres that will become explosible with the addition of air from those that will not. If these lines are extended downwards, they will converge at the origin of the graph. It is also worth knowing that the Effective Combustible Formula takes into account the fact that hydrogen's contribution to combustion is 25% higher than that of methane and that carbon dioxide's contribution is only 40% of that of methane. Similarly, the Effective Inert Formula takes into account the fact that carbon dioxide's contribution to the ability to suppress combustion is 50% higher than that of nitrogen. The N_2 value used is the surplus N_2 present. The excess of N_2 can also be determined as follows:

$$\text{Excess } N_2 = N_2 - 3.7778\ O_2$$

The use of the above mentioned formula will be described in the example to follow. The oxygen deficiency can also be determined as follows:

$$\text{Oxygen deficiency} = 0.2647\ N_2 - O_2$$

Where:

N_2 and O_2 are the percentage of the gas as determined in the gas sample.

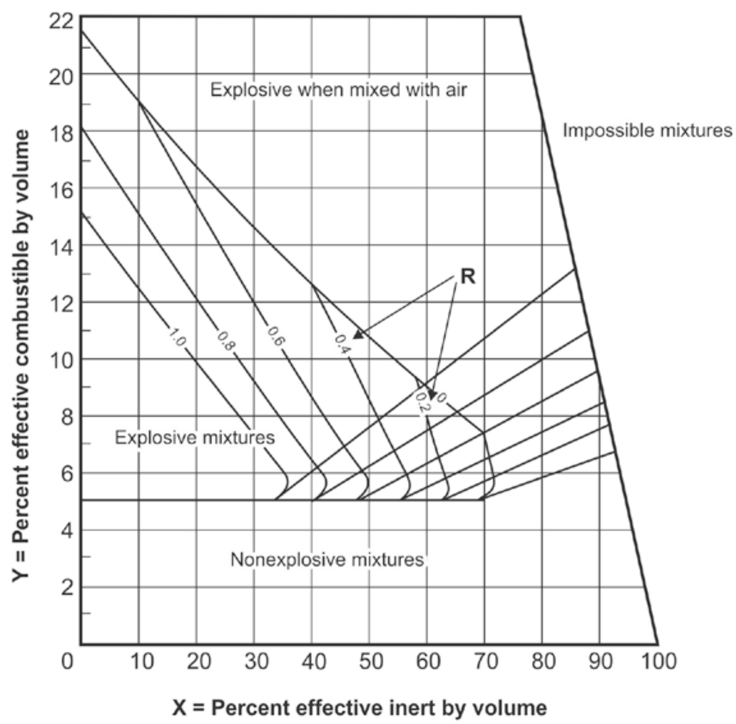


Figure 11.10 USBM diagram to determine explosibility of a flammable gas mixture

Total combustible = D = $CH_4 + H_2 + CO$

R = $CH_4 \div D$

X Co-ordinate Effective Inert = $N_2 - 3.7778\ O_2 + 1.5\ CO_2$

Y Co-ordinate Effective Combustible gases = $CH_4 + I.25\ H_2 + 0.4\ CO$

EXAMPLE QUESTION 7

The laboratory analysis of a gas mixture is as follows:

- O_2 = 12.6%
- N_2 = 57.6%
- CO_2 = 20.0%
- CH_4 = 4.2%
- H_2 = 5.3%
- CO = 0.3%

Calculate the percentages of the gases in question so that you can then plot the values on the Coward and USBM graphs. Comment on the positions of the points plotted on the graphs.

Answer

Calculate the concentration of N_2 in the air:

From relations:

Percentage of $N_{2(\text{normal ratio})}$ in the air = 12.6% / 20.93% x 79.04%
= 47.6%

The surplus N_2 = Lab. result - N_2 in the air = 57.6 – 47.6 = 10%

or by using the formula for excess N_2:

$$\text{Excess } N_2 = N_{2\,(\text{sampled value})} - 3.7778\ O_{2\,(\text{sampled value})}$$
$$= 57.6 - 3.7778 \times 12.6$$
$$= 9.99 \text{ say } 10\%$$

The Coward triangle coordinates are determined as follows:

Air = $N_{2(\text{normal ratio})} + O_{2(\text{sampled})}$
= 47.6% + 12.6%
= 60.2 %

Inert = $N_{2(\text{excess})} + CO_{2(\text{sampled})}$
= 10% + 20%
= 30%

flammable gases = $CH_4 + CO + H_2$
= 4.2% + 0.3 + 5.3
= 9.8%

The total adds up to a 100%.

Plot these and see where the point is lying. Comment on your finding.

For the plot on the USBM graph we need the following calculations:

$$R = \frac{CH_4}{D} = \frac{4.2}{9.8} = 0.43$$

Where:

D = Sum of the % of all the flammable gas (combustibles)

R = The ratio of the methane gas % to the total sample flammable gas.

$$\begin{aligned} X \text{ co-ordinate (effective inert gases)} &= \text{Excess } N_2 + 1.5\ CO_2 \\ &= 10 + (1.5 \text{ x } 20) \\ &= 40 \end{aligned}$$

$$\begin{aligned} Y \text{ co-ordinate (effective flammable gas gases)} &= CH_4 + 1.25\ H_2 + 0.4\ CO \\ &= 4.2 + (1.25 \text{ x } 5.3) + (0.4 \text{ x } 0.3) \\ &= 10.95 \text{ say } 11 \end{aligned}$$

Plot the values on the USBM graph and comment on their positions.

References

There are also three other gas relationships and these are known as the Graham ratio, the Young's ratio and the Willet's ratio, the meaning of each are clearly defined in Chapter 28 of EE.

The Formula applicable to each is as follows:

$$\text{Graham's ratio } (CO/O_2 \text{deficiency}) = \frac{CO}{O_2 \text{ deficiency}} \text{ x } 100\%$$

$$\text{Young's ratio} = \frac{CO_2}{O_2 \text{ deficiency}} \text{ x } 100\%$$

$$\text{Willit's ratio} = \frac{CO_2}{\text{Excess } N_2 + D + CO_2}$$

11.2.10 Gas Sampling

Gas sampling is mainly done if a result is not required immediately and when accuracy is important. For example, if we wanted to check whether the seals constructed to seal off methane accumulations are still operating effectively, it would be advisable to investigate the long-term changes in the gas concentrations in the immediate vicinity of the seal. However, if we simply wanted to know whether it was safe to work in a particular working place, we would use the techniques already described.

11.2.10.1 Procedure for Gas Sampling

Gas sampling involves taking a representative sample of the air in containers and is done mainly by one of four methods:

- Displacement of water
- Displacement of air
- Evacuation
- Compression.

It is important to note that when we take gas samples, we take into account specific aspects such as the density of the gases. For example, we take samples of heavy gases near the floor and samples of light gases against the hanging wall.

11.2.10.2 Characteristics of Detecting Devices

If a specific gas or combination of gases constitutes a hazard to the mine workers, the detection of unsafe concentrations is done mainly by means of sensors. The sensors 'react' to that specific gas and are linked to computers, in this way they can trigger an alarm.

11.2.11 Flammable Gas Detection

Testing for the presence of gas is done mainly to determine whether the air contains hazardous concentrations of gas that are higher than the recommended standard. It is therefore important to get an immediate indication of the gas concentration so that the necessary corrective measures, if needed, can be taken straight away. There is clearly a big difference between gas sampling and gas detection.

The purposes of gas testing are as follows:

- Detection of flammable gases
- Determination of the oxygen content of the air
- Detection of poisonous gases
- Gas sampling
- Control of mine gases.

11.2.11.1 Detection and Testing for Flammable Gases

If a flammable gas is present or suspected to be present (since methane is present in every mine), tests must be carried out in accordance with the code of practice applicable to that mine.

The situations in which gas tests must be carried out are as follows:

- At the beginning of each shift
- After the ventilation has been interrupted for 30 minutes or longer
- Before and after pilot holes are drilled
- Before loading for blasting is started
- After loading is completed and just before blasting starts
- When water, a dyke, a fault or fissure water is encountered or exposed
- At any time when the presence of gas is suspected.

In stope faces or working places in coal mines and fiery mines, or where the presence of flammable gases is suspected, the following must be investigated:

- The return air from the stope or working place
- Faults, dykes, fissures, sockets and high places against the hanging wall
- In advance developed stopes, raise connections and similar places where methane could accumulate
- All electrical switchgear and machinery before they are switched on
- The blasting point, before the blasting wire is connected to the detonator
- All working places adjacent to the stope whether the adjacent working places are connected directly or indirectly.

11.2.11.2 Equipment Used for Methane Testing

Various types of equipment currently exist. The most well-known are the methanometers as well as other electronic equipment, that has to be flameproof for obvious reasons (intrinsically safe).

SELF STUDY

Read the following article by R Henderson: "Choosing the best detection technologies for measuring combustible gas and VOC vapors".

11.2.12 Control of Flammable Gases

Every hazardous gas found in a mine can be classified on the basis of the hazard posed by it and this is normally based on a safe concentration. This is not always as easy as it appears because the Regulations usually indicate the Maximum Allowable Concentrations (quantity), meaning that these concentrations must never be exceeded.

Safe concentrations can also be expressed in terms of time-weighted averages, which means that in specific cases the concentrations at certain stages may indeed exceed the maximum allowable quantity but that they must remain below the limit over an eight-hour shift. In the case of explosible gases, the problem is even more complicated because there is a lower and an upper limit within which the gas may possibly explode.

Below the lower limit there is no danger of an explosion but there may still be a toxic effect, as in the case of carbon monoxide. If the concentration is above the upper limit, there is also no danger of an explosion, but there is a danger of asphyxiation (loss of oxygen in the air).

Objective

Whatever the concentration, the main purpose of ventilation remains the same: to dilute the gases to below their maximum allowable levels.

The control of mine gases (in general) can be dealt with under three main headings (and some mentioned before):

Prevention

- Scrubber box at diesel locomotives
- Explosives that release fewer gases
- Lower gas concentrations
- Avoid human exposure
- Positive pressure.

Removal

- At the source of the gas
- Diesel-driven equipment
- **Methane drainage**.

Dilution

- Good ventilation
- Filters.

11.2.13 Legislation

There are various legal aspects with regard to gas exposure and explosions that you need to be familiar with. In the South African context the following are required.

11.2.13.1 Investigations by the Inspector of Mines

If any person working on a mine has been exposed to gases or a lack of oxygen, specific information must be supplied to the Inspector of Mines and an investigation into the incident must be carried out.

The following information is required:

- Number of persons exposed
- Degree of exposure
- Place of exposure
- Time of exposure
- Possible reasons for exposure
- Reasons for the cause of the exposure
- Preventative measures that had to be taken
- Whether non-compliance with the regulations was the reason for the exposure
- Preventative measures that have been taken to prevent a repetition
- Sketch (plan) of the place where the exposure occurred
- Environmental conditions at the time of the exposure time
- When the Environmental Control Department was informed.

From recent investigations after flammable gas explosions it was felt that for future investigations a detailed methodology should be prepared for an autopsy where the following should be noted:

- Damage to organs
- Flame inhalation
- Amount of burns
- Severity of burns
- All other physical injuries and most likely cause thereof.

NB

It is also important that better forensic evidence should be gathered to answer questions such as:

- Estimated temperature effect
- Evaluation of possible and probable fuel sources
- Investigation of material in a quantitative and qualitative matter.

Due consideration must be given to establish a protocol to be followed in the event of a disaster, detailing contact persons, equipment and the procedure to be followed.

11.2.14 Possible Key Factors Causing Flammable Gas Explosions

Possible shortcomings that can lead to a flammable gas explosion are the following:

ML

- Lack of appreciation of the dangers associated with flammable gas
- Managerial defaults and a lack of the basic knowledge of the reaction and subsequent explosion characteristics.

The major contributor to the most flammable gas explosions is the lack of knowledge. This includes the following:

- Measurement
- Reaction to a positive measurement of flammable gas
- Emergency response procedure (forgetting the critical ‘one hour’ period after an incident occurred)
- Communication (the lack of)
- What flammable gas cocktails are present, how they will react and what the net effect of an ignition could be
- The lack of the installation of secondary early-warning devices
- The lack of formal training and follow-up courses to refresh memories.

EXAMPLE QUESTION 8

The total volume of a sealed-off section is 40 000 m^3 and it contains 70% methane. The pressure inside and outside the sealed-off area is 100 kPa. When an auxiliary fan is switched on close by, it has the effect of lowering the pressure outside the seal by 1.3 kPa.

- If the resistance of the leakage passages around the seal is 180 000 Ns^2/m^8, calculate the rate at which the air/methane mixture will leak out at the seal after the fan has been switched on. (**Answer** 0.09 m^3/s)
- If the air in the airway along the seal has a flow rate to the amount of 25 m^3/s and does not contain any methane, what will be the methane concentration in the air on the downstream side of seal after the fan has been switched on? (**Answer** methane leakage 0.06 m^3/s and gas percentage 0.25%)
- Since the air/methane mixture is flowing out from behind the seal, the pressure inside and outside the seal will actually be the same and this will hold up the return rate of the ventilation. What total volume of air/methane mixture will be displaced before flow ceases? (**Answer** 527 m^3).

SELF STUDY

Read the following article by A Z du Toit: "Opening of sealed-off areas: A practical approach used by Koornfontein mines".

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11.3 Coal Dust Explosions

11.3.1 Introduction

Video

Coal dust explosions have occurred several times in the past. In an underground coal mine a methane explosion normally ignites a coal dust explosion with disastrous consequences (see Figure 11.11) which shows a picture of a simulated coal dust explosion done at Kloppersbos.

Figure 11.11 Simulated coal dust explosion at Kloppersbos

NB

It is therefore important to discuss the various aspects pertaining to coal dust explosions. The danger that coal dust is explosible in a mixture of air is very much dependent of the following factors:

- The concentration of the dust and the presence of methane. The closer the particles are together, the better the chance of an explosion occurring, however when they get too close, the chances are that there will not be enough oxygen to ensure complete combustion.
- The degree of fineness of the coal dust. Any combustible dust that can be picked up into the air can be regarded as explosible, normally the particles sizes are less than 250 m in diameter.
- The type of coal - there are indications that there is a correlation with regards to the explosibility of coal dust and the volatile matter content. The higher the volatile content the greater the chances of an explosion.

- The strength of the ignition source - the energy level of the original ignition determines to what extent the ignition will be propagated.

  Most coal dust explosions have been ignited by methane explosions.

11.3.1.1 Ignition Sensitivity

Definition

Ignition sensitivity can be defined as the minimum temperature or energy amount necessary to create an ignition. In practise, these two terms are not independent and must therefore both be considered. Most dusts also have different ignition sensitivities, depending whether the source is a spark or a hot surface; or whether the dust is in a cloud or a layer.

NB

Another important factor is the minimum dust concentration that will ignite, that is also in its own right dependent on how well the dust has been dispersed, how turbulent the dust cloud is, the dust size distribution and the moisture content of the sample. In the comparison of the ignition energy for flour and methane, it is noted that the ignition energy for methane is 0.3 mJ at a minimum explosible concentration of 36 g/m^3, compared to 50 mJ at a minimum explosible concentration of 50 g/m^3 for flour. The ignition energy for hydrogen is 0.03 mJ at a minimum explosible concentration of 3.6 g/m^3.

11.3.1.2 The Coefficient of Explosibility

Experiments have shown that the explosibility of coal dust can be expressed in terms of a specific explosibility coefficient. Coal dust can be exploded experimentally and with the help of computer detection devices, the maximum and average increase in pressure in the container at the time of explosion can be determined. The formula to determine the co-efficient of explosibility is then expressed as follows:

$$K_{ex} = \sqrt{\text{maximum rate in pressure increase x average rate in pressure increase}} \ \text{bar/s}$$

During this experiment the dust concentrations are changed so as to obtain a different K_{ex} value for different dust concentrations. From these experiments it was found that:

Kex < 70 No explosion hazard

Kex > 95 A definite explosion hazard.

11.3.2 Guidelines for Code of Practice (COP) for Prevention of Coal Dust Explosions

References

The guidelines that are shown in this document are for test and examination purposes also. Make sure that you are familiar with the contents and that you will be able to answer questions applicable to this section. The role of stone dust and other preventative measures are discussed and detail and forms part of this chapter's outcome.

11.3.3 Measures against Coal Dust Explosions

All preventative measures employed should be situated as close as possible to the potential ignition source. This will either prevent the coal dust from participating in an explosion or prevent any further flame propagation.

11.3.3.1 Active On-Board Ignition-Suppression Systems

Objective

The aim of using active suppression systems is to contain the methane flame in the immediate vicinity of any methane ignition. This will prevent a methane explosion which, in turn, could be the ignition source of a coal dust explosion.

Active suppression systems have the following main components:

- Detecting sensors
- Electronic control and self-checking system
- Dust containers
- Flow nozzles.

These individual components are combined into systems, which are mounted on continuous mining machines. When an ignition occurs, its presence is detected by means of the sensor. These sensors are normally sensitive to the ultraviolet light range. An electronic signal from the sensor triggers the suppression system, creating a barrier of flame-suppressing material and containing the flame in the immediate vicinity of its initiation. The flame-suppressing material most frequently used is ammonium phosphate powder, but gases such as NAF S111 may also be considered.

There are two testing facilities for the evaluation of these systems, namely:

- The tunnel at BVS-Derne in Bochum, Germany
- The 20 m tunnel at Kloppersbos, South Africa.

A number of systems have been developed by DMT, a German consulting and engineering company (Faber, 1990a & b), for roadheaders. These have been in underground use since 1989 and are well proven and trusted.

A facility capable of testing such systems against a set protocol (Du Plessis et al., 1996) was built and completed during 1995 at the Kloppersbos Research Facility (part of the CSIR) in South Africa. The facility simulates the various mining configurations encountered in bord-and-pillar mines. To date, two systems have undergone successful evaluations. The first system was developed for low-seam continuous miners (Du Plessis, 1997) and the second for a Dosco 1300H auger-type roadheader (Du Plessis & Van Dijk, 2001) capable of excavating a seam height of up to 4.5 m.

The second system is for use in mining conditions where the roof slopes at an angle of 12^{o} and a double-pass mining method is employed. For this specific evaluation, the maximum height of the roof was 4.5 m. The system was deployed at HBCM mine in the south of France.

11.3.3.2 Inerting of Coal Dust

Several materials for inerting coal dust have been identified and tested by researchers and test facilities worldwide. The first investigation into the use of stone dust as an inerting material for inhibiting coal dust explosions has been attributed to the French investigator, Taffanel.

Over the years, many studies have been conducted to determine the effectiveness of alternative inerting agents. These tests were conducted at all the major research centres, including Tremonia (Germany), Bruceton (USA), the Barbara Experimental Mine (Poland), Buxton (United Kingdom) and the Kloppersbos Research Facility (South Africa).

Most of this work centred on the ability of water, limestone dust and clay slate dust to inhibit the propagation of coal dust explosions and on their limits of effective operation. In recent years, various other methods involving the inhibition of coal dust dispersion as an effective means of preventing coal dust explosions have been investigated. Inert material is applied for two distinctly different purposes in preventing coal dust from participating in an underground explosion, namely as an:

- Inerting agent
- Agent to prevent the dispersion of the dust.

Tests conducted worldwide have led to various general conclusions regarding the effectiveness, or lack of effectiveness, of stone dust and other inerting agents. It is widely held that effectiveness is a function of the:

- Coal dust particle size (Cashdollar & Hertzberg, 1989; Hertzberg et al., 1981)
- Strength of the initiator used (Hertzberg & Cashdollar, 1987)
- Experimental procedure used (Cashdollar & Hertzberg, 1989)
- Presence or absence of a coal dust layer.

The effectiveness is measured in terms of the amount of inert material required to inhibit the propagation of a coal dust explosion. This is normally measured as the percentage TIC (total inert content), which is calculated as the total percentage of inert material on a per mass basis.

The exact means by which stone dust acts as an inerting agent has not been fully explained. However, Cybulski (1975) described it as a combination of the following:

- Stone dust acts as heat sink (adsorption of energy).
- Stone dust screens radiation from the combustion process in between the coal particles.
- The chemical reaction is endothermic, producing carbon dioxide.
- Stone dust particles obstruct the diffusion of oxygen and combustible gases.

Stone dust requirements are calculated using the equations set out below:

$$\%\,Stone\,dust = \frac{(\%TIC - \%\,Ash - \%Water)}{\left\{1 - \left[\frac{\%\,Ash + \%Water}{100}\right]\right\}} \qquad \text{(Michelis, 1991)}$$

Where:

% TIC is the percentage of total incombustible content.

The percentages of ash and water inherently present in the coal dust are taken into account when the calculations are made as shown above. An example of the calculation is given below, where 35 kg of coal dust containing 14.1% ash and 1.6% moisture is to have stone dust added so that the mixture has a TIC of 85%:

$$\% \text{ Stone dust} = \frac{(85 - 14.1 - 1.6)}{\left\{1 - \left[\frac{14.1 + 1.6}{100}\right]\right\}}$$

$$\% \text{ Stone dust} = 82.21\ \%$$

$$\text{Total inert mass} = 35\ \text{kg}/\left[1 - \left(\frac{\%\ \text{Stone dust}}{100}\right)\right]$$

$$= 35\ /\left[1 - \left(\frac{82.21}{100}\right)\right]$$

$$= 35/(1 - 0.8221)$$

$$= \frac{35}{0.1779}$$

$$= 196.7\ \text{kg}$$

Thus the amount of stone dust required to inert 35 kg of coal dust is the difference between the total inert mass and the original coal dust mass as shown below:

$$\begin{aligned}\text{Mass of stone dust} &= \text{Total inert mass} - \text{Coal dust mass}\\ &= 196.7 - 35\\ &= 161.7\ \text{kg}\end{aligned}$$

In South Africa only stone dust is used as an inerting agent. The stone dust used is primarily calcitic or dolomitic in nature. The use of these stone dusts and the extent of the usage are described in the following sections.

11.3.3.3 South African Stone Dust Requirements

Law

The South African requirements for stone dust are set out in the Minerals Act, No. 50 of 1991, Regulation 10:24. These are shown in Table 11.15.

Table 11.15 Specification of stone dust

Item	**Requirement**
Incombustible content	Min. 95%
Free silica	Max. 5%
Size distribution	100% passing 600 mm sieve 50% passing 75 mm sieve

The requirements for the inerting of coal dust were reviewed by a Mining Regulation Advisory Committee (MRAC) task group, which was formed as a result of the recommendations made by the Leon Commission (Leon et al., 1995).

The recommendations set out in the *Guidelines for a Mandatory Code of Practice for Explosion Prevention* (MRAC, 1997) with regard to inertisation requirements included the extent of the application as well as the degree of inertisation required.

EXTENT OF APPLICATION

In order to ensure that the underground workings of a coal mine are adequately protected, all underground areas of a coal mine producing bituminous coal, except those areas extending to the face from and including the last through road and in which the coal dust has been washed from the roof and sides, and the floor is too wet to propagate an explosion, should be stone dusted to within 10 m from all working faces, unless such outbye areas are inaccessible or unsafe to enter. Such areas must be timeously identified by hazard identification and risk assessment and addressed to reduce the hazard.

DEGREE OF INERTISATION

Any coal mine can be divided into different areas, with significantly different potential for experiencing a coal dust explosion, e.g. intake airways far distant from a working face and return airways in the face area.

Considering all available information, the following minimum levels of inertisation by the application of stone dust are required:

- **Intake airways**
  - In the face area, which is defined as anywhere within 180 m of a working face, a minimum percentage by mass of incombustible matter content of 80% must be maintained.
  - Outbye the face area, intake airways must be maintained at a minimum of 65% incombustible matter content. Workshops, substations, battery-charging stations and other similar places where work is done or equipment is maintained, situated in intake air, must nevertheless be maintained at a minimum of 80% incombustible matter content.
- **Return airways**
  - A minimum percentage by mass of incombustible matter content of 80% must be maintained up to a minimum distance of 1 000 m from the face. Beyond this distance, a minimum percentage by mass of incombustible matter content of 65% must be maintained, provided that where approved barriers are installed, the incombustible matter content by mass outbye the face area and outbye the barriers must be maintained at not less than 65%.
  - Return airways in close proximity to sealed areas, or areas in the process of being sealed off, must contain a minimum percentage by mass of incombustible matter content of 80%, unless the area has been properly sealed off.
- **Conveyor belt roads**

  A minimum percentage by mass of incombustible matter content of 80% must be maintained up to a minimum distance of 180 m from the face. Beyond this distance, a minimum percentage by mass of incombustible content of 65% must be maintained.

It should be recognised that conveyor belt roads constitute a particular hazard. Dust is liberated into the ventilation current at transfer points and during the conveying of coal. Spillage from belts also contributes to the problem. In addition, the conveyor structure itself provides an ideal location for the accumulation of coal dust. Stone dusting alone cannot, therefore, adequately protect conveyor belt roads and use must be made of barriers.

- **Abandoned areas prior to being sealed off**

  Before any area is sealed off, the roof, sides and floor must, as far as is reasonably practicable, be stone dusted to ensure a minimum percentage by mass of incombustible matter content of 80%.

  The above recommendations were based on mining practices used in Europe, which were in turn based on extensive research. However, research by Cook (1993) showed that the propagation of a full-scale coal dust explosion can only be stopped if the 80% TIC requirement is adhered to.

  In European Regulations there are two distinctive definitions of dust used: **safe dust** and unsafe dust. The safe dust contains:

  - At least 70% of incombustible if deposited in non-gassy coal fields
  - At least 80% of incombustible if deposited in gassy coal fields
  - Free water in a quantity to prevent dust explosion propagation.

  The only way of protecting against a coal dust explosion occurring is:

  - Llimitation of coal dust generation during extraction and transportation
  - Control of ignition sources (this is common practice in coal mines)
  - Neutralisation of the settled dust by adding incombustible matter, mostly limestone dust.

Rule

In Europe, the general rule is to maintain incombustible matter content at the level of 70%, with some augmentation for gassy seams (80%). In Germany, hygroscopic salts are used to absorb the settling dust (Michelis, 1998).

11.3.3.4 Explosion Barriers

The principle of using explosion barriers to safeguard mines was first noted and evaluated by Taffanel (1910) at the turn of the century. He based it on the fact that the flame of a coal dust explosion is preceded by a blast wave. It was therefore Taffanel who designed the first shelf-based explosion barrier (Taffanel & Le Floch, 1913). Because he lacked the resources that other researchers had later, the barrier failed to stop an explosion in the Clarence mine, France. However, his principle was extensively researched later and formed the basis of a number of shelved stone dust barriers which were developed throughout the world.

In essence, explosion barriers are used to safeguard the rest of the mine outbye the normal production area. Passive barriers are installed to provide supplementary protection against coal dust explosions. To this end, two types of passive barrier have been developed and extensively tested at a number of research institutes. These barriers – the stone dust barrier and the water trough barrier (EN 14591-2, 2007) – have been deployed in mines throughout Europe, Australia and South Africa.

Although tests have indicated certain shortcomings, especially against weak and very strong coal dust explosions, the reduction in the risk of a coal dust explosion propagating throughout a mine when barriers are systematically deployed is well understood and accepted. Passive explosion barriers use the dynamic pressure from the windblast that precedes the flame front to activate and disperse the suppressant material. In the event of a coal dust explosion, the following sequence of events occurs:

- The methane ignites and there is a slow growth in the flame and the size of the fireball, with almost no pressure increase
- Deflection of the methane fireball and pressure from the sides of the gallery create turbulence. At this point, there is almost no time delay between the flame front and the pressure wave
- Coal dust particles start to participate with the methane fireball and become the incendive driver. The dispersion characteristics of the coal dust cloud will determine the initial time delay between the pressure pulse and the progression of the coal dust flame
- From here onwards, the explosion is self-generating and governed only by the amount of fuel readily available.

The time delay between the flame front and the pressure wave is the time available for the passive barrier to disperse enough flame-suppressant material at the correct inert concentration. Thus, passive barriers require a minimum dynamic pressure for activation and a minimum period of time for the dispersal of suppressant material to operate effectively.

The most widely used passive barriers are:

- Polish-type shelf barriers (Figure 11.12)
- German water trough barriers (Figure 11.13)
- South African bagged stone dust barriers (Figure 11.14)

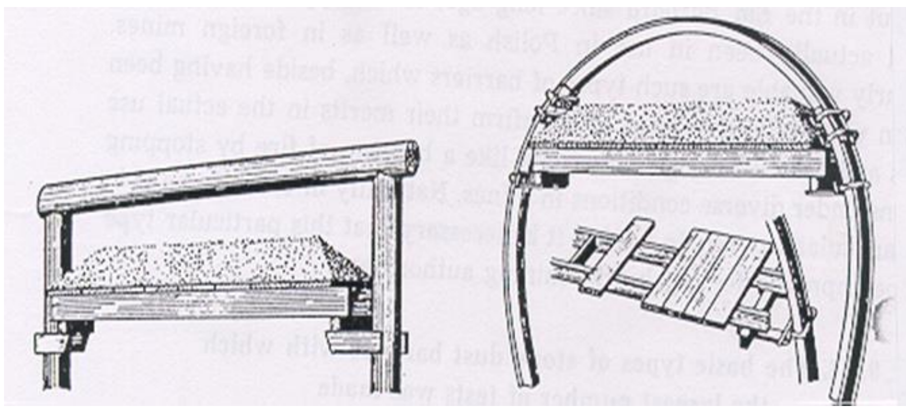


Figure 11.12 Example of a Polish-type shelf barrier

Figure 11.13 Example of a water barrier installed in an underground mine

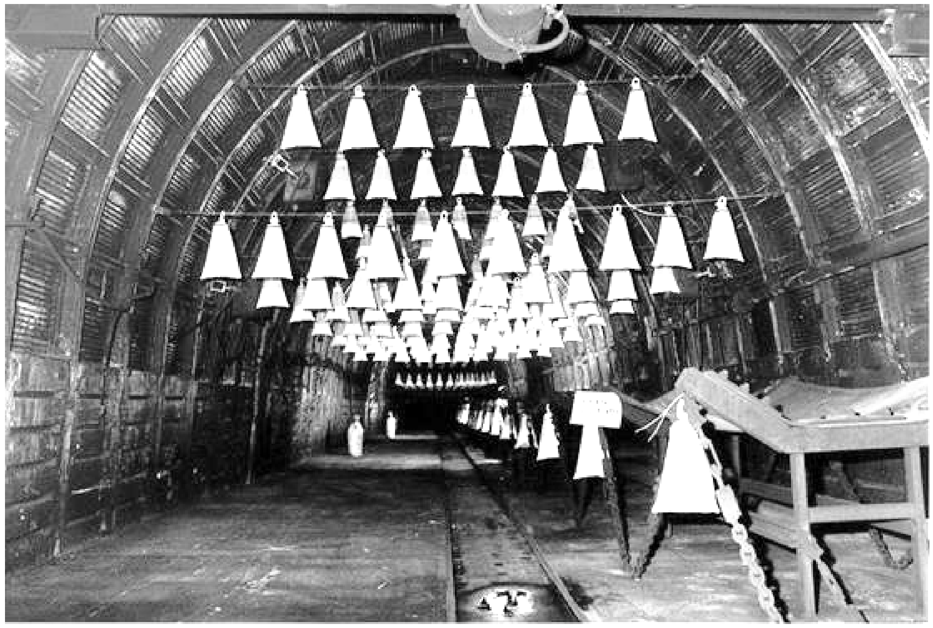


Figure 11.14 Example of a bagged stone dust barrier installed in an underground mine (*Source:* Michaelis et al., 1996)

BAGGED STONE DUST BARRIER

The bagged stone dust barrier system consists of an array of clear plastic bags, containing stone dust, suspended in rows across the width of a mine road. There are two types of approved barrier, which are commonly described as the:

- Distributed barrier
- Concentrated barrier (Du Plessis, 2000).

The design specification remains the same, with the only difference being that the distributed barrier consists of four independent subbarriers and the concentrated barrier is a single entity.

Distributed Barrier

The general requirements for the construction of the distributed bagged stone dust barrier are contained within the *Mandatory Code of Practice for the Prevention of Coal Mine Explosions* issued by the South African Department of Mineral Resources (DMR) in 2002.

The total mass of stone dust to be used in a bagged stone dust barrier is based on either the cross-sectional area of the roadway in which the barrier is to be installed, or on the total excavated volume, between the extremities of the full barrier, of the roadway being protected. The design of the barrier is based on various parameters, as follows.

If M_A is the mass of stone dust based on the cross-sectional area and M_V is the mass based on volume, then M_A must be at least 100 kg/m^2 of the cross-sectional area and M_V must be at least 1 kg of stone dust per cubic metre of roadway volume between the barrier extremities.

The total mass of stone dust to be used in a barrier must be based on the greater of M_A and M_V.

For example, in a roadway 6.6 m wide and 3.5 m high, the cross-sectional area is 23.1 m^2 and M_A would be 23.1 x 100, i.e. 2 310 kg. If in the same roadway the distance between the extremities of the barrier is approximately 120 m, the total volume would be 23.1 x 120.0, i.e. 2 772 m^3; hence M_V would be 2 772 kg. Therefore the greater mass of 2 772 kg would be used.

NB

The layout of the barrier is a critical element in its design and the requirements are as follows. The distributed bagged stone dust barrier consists of four sub-barriers installed over a minimum distance of 100 m of continuous roadway. Three complete sub-barriers must be in position at all times, while the fourth sub-barrier may be in the process of being moved ahead as the section advances. This is only true for advancing sections and the barrier installations in outbye areas must always consist of all four sub-barriers.

The following distances must be maintained in an operational section:

- The first sub-barrier, closest to the last through road, must not be installed closer than 60 m and not further than 120 m from the last through road
- The fourth sub-barrier, furthest from the last through road, must be installed not more than 120 m from the first sub-barrier
- Two intermediate sub-barriers must be equidistant between the first and fourth sub-barriers
- The presence of splits in bord-and-pillar sections must be ignored in determining distances
- The maximum distance between sub-barriers must not exceed 30 m.

Since the concept of bagged stone dust barriers is best illustrated by an example, the following serves as an illustration.

Example

Assume that a bagged stone dust barrier is to be installed in the belt road of a bord-and-pillar section. Assume also that the first row of sub-barriers will be located 100 m from the last through road. The belt road is 3.0 m high and 6.5 m wide.

The distance between the barrier extremities is 120 m and the cross-sectional area is 19.5 m^2. The volume between the extremities of the full barrier is therefore 2 340 m^3. As M_A = 1 950 kg and M_V = 2 340 kg, M_V will be used.

If each bag contains 6 kg of stone dust, altogether 2 340/6 = 390 bags are needed. Based on four sub-barriers, there would be 390/4 = 98 bags per sub-barrier.

Concentrated Barrier

The concentrated barrier was designed for use as an alternative passive barrier system especially for areas where high-speed flame development can occur, for example in longwall sections or outbye areas. The recommended stone dust quantity, M_A, is 100 kg/m^2 and is similar to that for a light Polish shelf explosion barrier. These barriers are by design single entities.

The placement of the barrier should conform to the following design criteria:

Criteria

- The first row of bags must be not nearer than 70 m to the last through road and not further than 120 m in an operational section
- The first row of bags of the second barrier must be no further than 120 m from the last row of bags of the first barrier.

The length of the barrier should be a least 20 m, with a maximum recommended length of 40 m. Furthermore, it is required that at least one fully constructed barrier should be in position at all times, implying leap-frogging from the second barrier in an advancing section.

From this it is clear that should these barriers be installed at distances further than the recommended distances, additional stone dust loading, equivalent to the Polish heavy shelf barrier of 400 kg/m^2 of stone dust, must be considered.

Chapter

12

MINE DUST

LEARNING OUTCOMES

The student will at the end of this module understand the properties and the dangers associated with dust as well as the preventative measures associated with it.

Study Objectives Section 12.1

At the completion of this study theme the student would have covered the following aspects:

- The effects of dust on human beings
- Sources of dust in mines
- Prevention of dust
- Dust filtration
- Dust concentration calculations
- Dust sampling
- Methods of dust sampling
- The Hund Tyndallo meter for measuring dust concentrations.

Study Objectives Section 12.2

At the completion of this chapter, the student would have covered and understood the following aspects related to gravimetric sampling (with specific reference to the gravimetric sampling guideline document):

- Measurement of dust concentrations
- Dust sampling objectives
- Measurement of exposure levels
- Legislation
- TLV concentration calculations
- Air quality indices

- Risk index
- Pollutant index.

REFERENCES

Mine Environmental Control (MEC), Chamber of Mines, South Africa, Workbook 5.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

12.1 Introduction

Dust is formed by reducing materials to a small size. Processes like grinding, crushing, blasting and drilling produce dust particles of sizes varying from very fine to very coarse. The human lungs are protected against coarse dust (as normally encountered in nature), by pre-filters. These filters consist of hair in the nose and constant moistness of the lining of the nose, mouth and windpipe.

SHE

The dust produced by mining operations is a serious hazard as it may cause pneumoconiosis.

Definition

Pneumoconiosis is the term used to describe a large group of diseases of the respiratory organs caused by the inhalation of noxious dust. This general term includes all the specific forms of dust diseases such as silicosis (caused by the inhalation of silica dust), asbestosis (caused by the inhalation of asbestos dust), anthracosis (caused by the inhalation of coal dust), and other forms of pneumoconiosis caused by other noxious dusts.

Any persons suffering from pneumoconiosis contracted as a result of work lawfully carried out at a controlled mine is eligible for pneumoconiosis compensation. A considerable number of cases are at present certified each year in South Africa and the compensation paid by the mines has reached sums of millions of Rand per year. There is no indication at present that the number of cases is decreasing.

SHE

In addition to this high cost to the gold mining industry, there is also a human factor to consider. The health of a person who contracts pneumoconiosis and progresses to a more serious state can become severely impaired.

He may have to stop work, drop from a high salary to a small pension, and would be unable to continue life as a normal healthy person. This not only affects the man concerned but his family as well. Finally the mining industry suffers too because it has to release many men due to this disease at a time when their training and experience is most valuable.

NB

It can be seen therefore, that the dust and pneumoconiosis can play a very important part in the everyday life of persons exposed to a dusty atmosphere. Where dust is produced it must be measured and controlled and these are among the most important duties of the ventilation staff.

12.1.1 New Milestones 2014 and Beyond

Elimination of occupational lung diseases:

- By December 2024, 95% of all exposure measurement results will be below the milestone level for respirable crystalline silica of 0.05 mg/m^3 (these results are individual readings and not average results)
- By December 2024, 95% of all exposure measurement results will be below the milestone level for platinum dust respirable particulate of 1.5 mg/m^3 (<5% crystalline silica), (these results are individual readings and not average results)
- By December 2024, 95% of all exposure measurement results will be below the milestone level for coal dust respirable particulate of 1.5 mg/m^3 (<5% crystalline silica), (these results are individual readings and not average results)
- Using present diagnostic techniques, no new cases of silicosis, pneumoconiosis, coal worker's pneumoconiosis will occur amongst previously unexposed individuals.

NB

'Previous unexposed individuals' are those unexposed to mining dust prior to December 2008 i.e. equivalent to a new person who entered the industry in 2009.

12.2 Properties of Dust

Definition

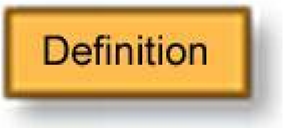
Dust is finely divided solid matter. In mines it is caused by different mining operations such as blasting, drilling, scraping, etc. The name 'dust' is applied whether the particles are airborne or have settled. Dust particles generally have the same chemical and physical properties as the substance from which they are formed, but some properties are altered, for example the rate of solution, which is greatly increased. Coal dust is also a dust and also has detrimental effects to health if not dealt with in a responsible way.

The added danger with regards to coal dust is that it can explode and is normally ignited through a flammable gas that triggered the explosion after being ignited.

12.3 The Effects of Dust on Human Beings

It is generally accepted that large particles of silica of the order of 7 micrometers and larger do not play an important part in causing silicosis. The main reason for this is that during breathing they are effectively filtered from the air by fine hairs in the nose and by sticking to the wet surfaces in the throat, etc. However, they can lead to other diseases such as bronchitis and emphysema.

Fine dust particles of silica below 7 micrometers in diameter do reach the lung and exert some action causing silicosis. Figure 12.1 (the Johannesburg curve) and Figure 12.2 (the British Medical Research Council (BMRC) curve and the American Conference of Governmental Industrial Hygienists (ACGIH) curve), shows that it is only the particles less than 7 micron metres that reaches the lung and that can cause damage. It is obvious then, when measuring the health hazard that more attention is paid to these particles.



Because of the differential deposition of particles of different size in the human lung it has often been suggested that dust sampling should aim to simulate this effect and to collect only what is called the '**Respirable Fraction'** of the dust in the air. Such methods are known as '**Selective Sampling**'.

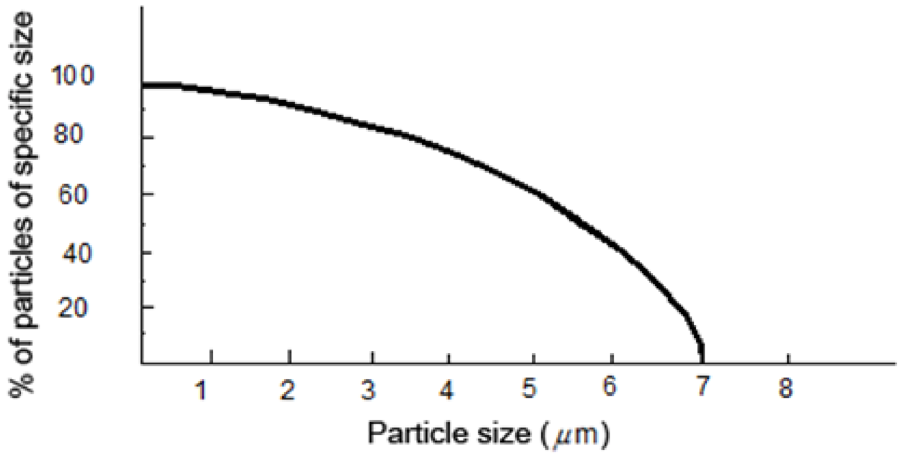


Figure 12.1 The Johannesburg curve (respirable sampling curve)

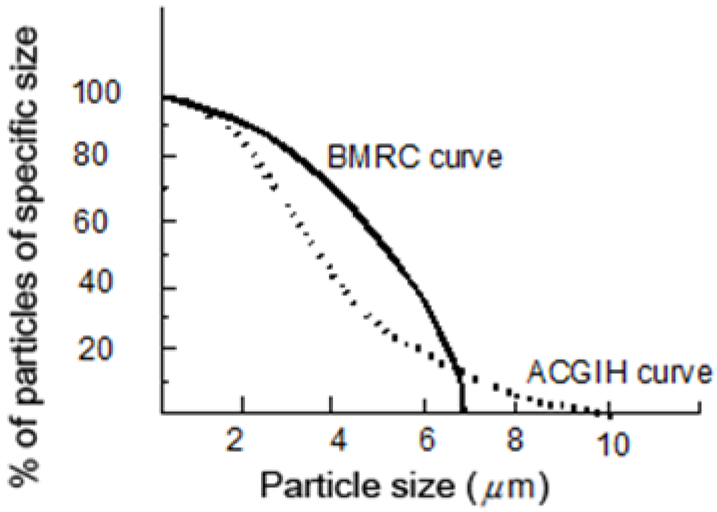


Figure 12.2 The BMRC and CHIG curves for lung penetration by dust particles

NB

Another important property of fine dust is that the particles remain suspended in the air for long periods for example a 0.5 micrometer diameter particle of silica will take three hours to fall through 0.3 m in still air.

(1 micrometer = 1/1 000 millimetre)

12.3.1 Deposition of Inhaled Dust Particles

Dust particles that are breathed in will be deposited in the lungs according to specific mechanisms.

These mechanisms are as follows:

- Gravitational sedimentation as determined by the free falling terminal velocity
- Inertial impaction due to inertia
- Diffusion as applied to microscopic dust particles
- Interception due to the physical size of the dust particle.

Figure 12.3 illustrates these mechanisms.

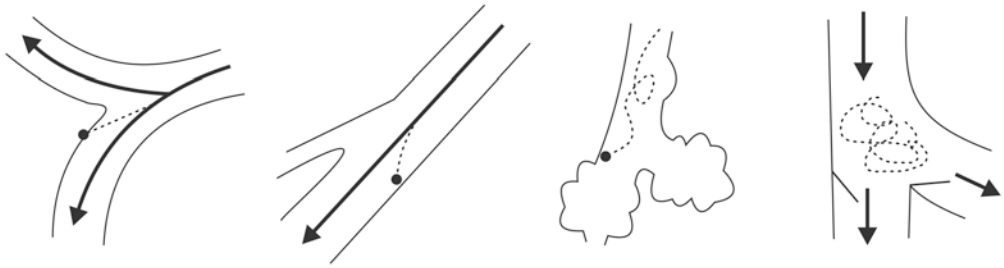


Figure 12.3 Mechanism of deposition of particles in the lung

12.3.2 Health Effects of Inhaled Particles

Depending on the physical, chemical and toxic properties of the inhaled particles, a variety of occupational diseases due to the continuous inhalation thereof can occur. The biological reaction to the different dusts may be non-injurious, slight or serious. Soluble particles may invoke only temporary response with no persistent or serious cell injury, while insoluble particles may irritate the tissue of deposition to a lesser or greater extent, depending on their interaction with the tissue.

Dust related illnesses are a common phenomena on mines (that must be engineered to correction) but there are two aspects that need to be highlighted and this is the difference between non-fibrogenic and fibrogenic dusts.

12.3.2.1 Non-Fibrogenic Dusts

Dusts, which cause little or no reaction of any kind are usually classified as biologically inert dusts

This may however be misleading and it is more correct to refer to these as not causing pulmonary fibrosis, physical impairment, or disease. Any durable inorganic dust will evoke some sort of defence mechanism to effect the expulsion of the inhaled particles (coughing etc.). Carbonaceous material such as coal, carbon black or graphite can be classified as non-fibrogenic or inert.

12.3.2.2 Fibrogenic Dusts

In contrast to ‘inert’ dusts, biologically active dusts such as free crystalline silica and asbestos decrease the active life of cells, resulting in less controlled accumulation of dust in the alveolar region. Human epidemiological studies have indicated that the fibrogenic activity appears to be dose-related. Collageneous pneumoconiosis, that is those caused by fibrogenic dusts, are characterised by the following:

- Permanent alteration or destruction of alveolar architecture
- Collageneous stromal reaction of moderate to maximal degree
- Permanent scarring of the lung.

12.3.3 Prevention of Dust Related Illnesses

The problems encountered with dust related illnesses are researched extensively. The three aspects covered in research studies are:

- Medical research relating to the development, treatment and diagnosis of dust related illnesses
- The techniques of dust sampling and the measurement of respirable dust
- Dust suppression on mines.

12.4 Sources of Dust in Mines

NB

Common sources underground in approximate order of severity are the following.

12.4.1 Blasting

Very high dust concentrations of the order of 100 000 to 250 000 p/ml are produced during blasting. In addition to the dust, blasting fumes are liberated into the air as well.

It is believed that when nitrous fumes are breathed at the same time as dust particles, the toxicity of the particles is increased. Every possible precaution should be taken to prevent exposure of men to dust and fumes produced during blasting.

12.4.2 Drilling

!

Dry drilling, which is forbidden by law, produces high dust concentrations and hence can be very dangerous. Wet drilling can give high dust concentrations if the rock drills are in poor condition, or if dirty water is used.

12.4.3 Blowing-Over in Sinking Shafts

Blowing-over by means of compressed air is carried out to clear away the rock on the footwall, to make space for drilling and to expose misfires. Extremely high dust concentrations can be produced during blowing-over. It is difficult to control this dust and men should wear respirators while engaged in this particular operation.

12.4.4 Scraping

If dry scraping is carried out, high dust concentrations can be produced. In addition to creating dust during the actual operation, scrapers also stir up dust, which has been created by other sources e.g. blasting dust, which has settled out.

12.4.5 Tipping

If the tip is not properly ventilated and particularly if the ore-pass upcast is undesirable dust concentrations can occur. This dust is usually coarse.

12.4.6 Loading Boxes

Again, because of bad or insufficient ventilation and particularly if the ore-ass downcast is undesirable, dust concentrations can occur. The concentration of the dust may reach quite high values if the rock is tipped dry, or if it dries out while in the ore-pass.

12.4.7 Lashing and Mechanical Loading

Hand-lashing (loading by humans with shovels) produces very little dust if the rock is wet. If the rock is dry some visible coarse dust may be produced but the concentration is not very high. Mechanical loaders can produce more dust than hand-lashing especially as it is more difficult keeping all the rock wet when it is removed so rapidly.

Other underground dust sources which usually cause less dust or where the actual operation lasts for a shorter time are:

- Blowing out holes
- Barring down
- Watering down
- Sweepings
- Blow out motors
- Cleaning trucks or material
- Walking, transporting and tramming in haulages
- Changing or cleaning flannel bags.

Dust is also produced at a number of places on surface. These include:

- Crushers and reduction works
- Uranium plants
- Assay crushers
- Drill sharpening shops
- Boilermakers' shops and welding shops
- Dumps.

12.5 Prevention of Dust (Hierarchy of Control)

ML

The basic ways in which high dust exposures can be controlled or prevented are given below. One should remember that 'prevention is better than cure', and this is particularly significant in the case of dust as there is no known cure for pneumoconiosis.

12.5.1 Removal of Personnel

This is the most effective way of preventing exposure to dust. It is mainly applied to blasting dust by the insistence of a minimum re-entry period; by arranging a fixed blasting time for each place so that other workers are not exposed to the blasting dust and fumes and by ensuring that blasting only takes place at the end of the shift and that all workers have already been withdrawn.

When on shift or multi-shift blasting is permitted removal of personnel is impossible and the dust and fumes must either be taken directly to surface or filtered. Other ways in which workers are kept out of dusty air is by arranging that men travel in downcast shafts and by having all underground waiting places in fresh air.

12.5.2 Prevention of Dust at its Source

Every effort should be made to prevent formation of dust at its source or its liberation into the atmosphere. Water is one of the most widely used means of suppressing dust.

The use of wet drilling greatly reduces the liberation of the dust from the rock compared with dry drilling, and is one of the outstanding examples of the suppression of dust at its source. The formation of dust during lashing, scraping and tipping can be greatly reduced if the rock being handled is kept wet.

12.5.3 Dilution by Ventilation

This is essential when the actual suppression of dust at the source has failed. Obviously if the quantity of air passing a given source of dust is doubled the dust concentration from that source will be halved. By delivering enough air the dust concentration can always be reduced to below dangerous limits but it is not always practicable or economical to do this.

12.5.4 Filtration

When dust cannot be controlled by one of the above methods and it has become airborne it can be removed from the air by filtering the air through a suitable filter, which allows the air to flow through it but retains the dust particles. Many types of dust filters are available of which the flannel bag filter is the most commonly known.

12.5.5 Respirators

Respirators are uncomfortable to wear and not practical in hot mines. They should only be considered as a last resort.

12.6 Dust Filtration

Of the dust prevention methods mentioned before, dust filtration forms a very important aspect of the dust control strategy applied on mines. It is therefore necessary to highlight some of the basic parameters associated with it. The purpose of a dust filter is to remove harmful dust from air.

Filtration of the air is required only when other methods of dust suppression, such as watering-down and dilution, have failed. Dust filters have been applied to various dusty processes, including the following:

- Tipping
- Blasting
- Machinery
- General return air
- Surface plants.

An ideal dust filter should:

- Be easy and cheap to install
- Filter out all dust both fine and coarse (i.e. it should be efficient)

- Be unaffected by humidity or moisture, and should not require much maintenance (i.e. it should be reliable)
- Have a low resistance to airflow (that is the filter should be relatively small yet able to filter large volumes of air)
- Be easily maintained and cleaned
- Have a low capital cost (including any excavations required for housing the filter)
- Have a low running cost
- Have a satisfactory efficiency, particularly for fine particles
- Be reliable.

Some of the common types of dust filters in use are:

- Flannel bag filters
- Electrostatic precipitators
- Cyclones
- Wet scrubbers
- Sawdust filters
- 'Pulsaire' dust collectors.

12.6.1 Selection of a Suitable Filter

Points which should be considered are:

- Amount of filtration required. Should only course dust, or course and fine dust be removed?
- How much air must be filtered?
- How much space is available for the filter?
- How long will the filter be used for? If only for a short time, a filter with a low capital cost should be selected, even though the operating costs are high. If the filter is to be used for a long period, a filter with a high initial cost but low running costs will probably be the most economical one.

12.6.2 Flannel Bag Filters (Low Dust Load)

The flannel bag filter is the most commonly used underground. It consists of one or more flannel bags mounted horizontally or vertically. Dusty air from the tip is passed into the bag by means of ventilation piping and a fan. The bags are suspended from horizontal wires by means of eyelets sewn into the material.

The multiple vertical bag filter is used more commonly and consists of a large number (up to 50 or 60) of small bags mounted vertically on a specially constructed base with the air flowing upwards through the bags (Figure 12.4).

Figure 12.5 shows a typical flannel bag arrangement and Figure 12.6 and Figure 12.4 shows how a flannel bag filter is installed for a tipping environment (plan and section view).

The Rabson flannel bag filter has vertical bags installed on a base plate and is attached to a shaking device for cleaning purposes. The bags are 2 m long x 200 mm in diameter.

Another vertical type of flannel bag filter is the Rayner type, which is collapsible and self-cleaning.

12.6.2.1 Requirements for Best Operation

- Air velocity through the filtering material should not exceed 0.1 m/s. Higher velocities result in a decrease in filtering efficiency.
- The air velocity down the actual tip should be 0.5 m/s.
- The bags should not vibrate as this causes the dust to work through the bag. This can be achieved by careful design of the header, i.e. of the fitting to which the bags are fastened.
- Bags should be kept dry. Moisture on bags results in poor filtering efficiencies due to a decrease in electrostatic action, as well as clogging.
- Bags should not become clogged with dust. Although the filtering efficiency of a bag increases as dust builds up inside, if too much dust is collected resistance will increase until very little air will be filtered. It has been found that the bags should normally be cleaned when the pressure loss over them has increased by about 600 Pa.
- There should be no tears or holes in the fabric.
- The fabric should be treated to resist fungal attack.

12.6.2.2 Cleaning of Flannel Bag Filters

Cleaning should be done between shifts. That is when there are few persons in the vicinity, as dust is stirred up during cleaning operations. The large horizontal bags are generally cleaned on surface, having been removed from the plant, transported to surface in dust-proof containers and then cleaned in a bag-cleaning plant.

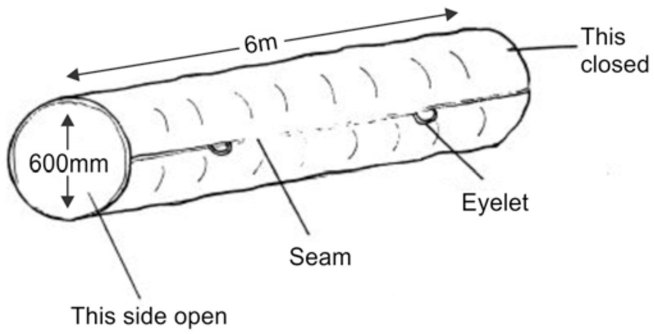


Figure 12.4 Flannel bag

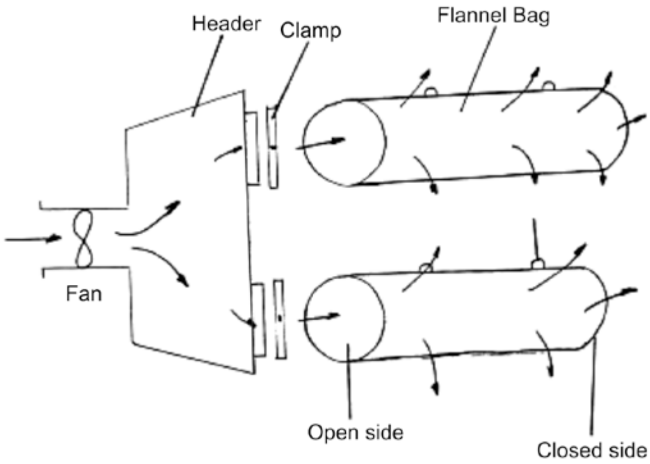


Figure 12.5 Flannel bag filter arrangement

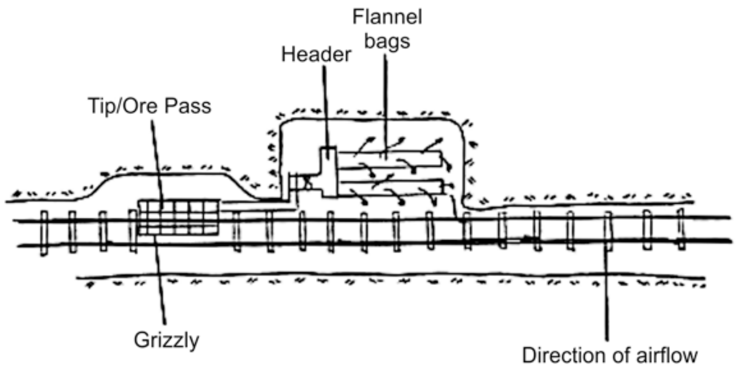


Figure 12.6 Plan view flannel bag installation at tip

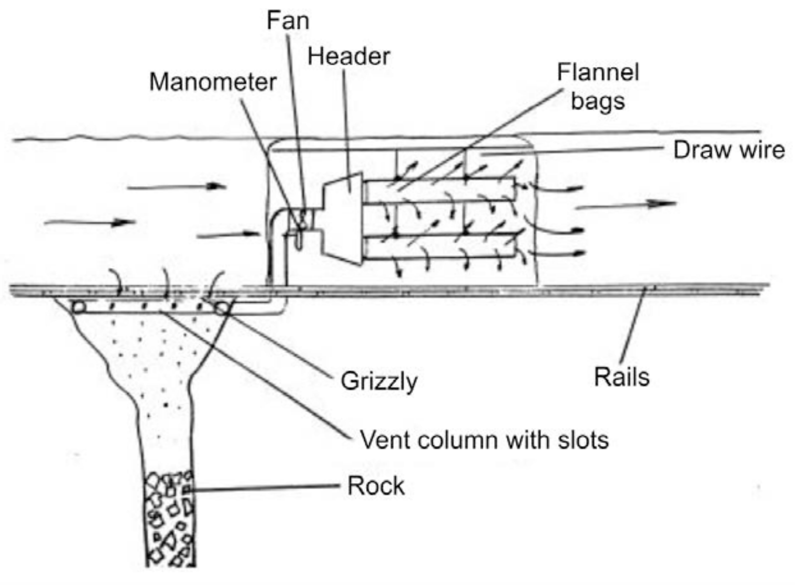


Figure 12.7 Section view flannel bag installation at tip

12.6.2.3 Flannel Bag Calculations for Flannel Bag Filter Arrangements

The next example will clearly show how the calculation with regards to dust filtration can be done and how to establish the number of flannel bags required per installation for a set number of parameters.

WORKED EXAMPLE 1

The length of a flannel bag is given as 6 m and the diameter thereof is 600 mm. The tip size where the flannel bag filter must be installed is 8 m x 2 m. Determine the number of bags that will have to be installed for this installation.

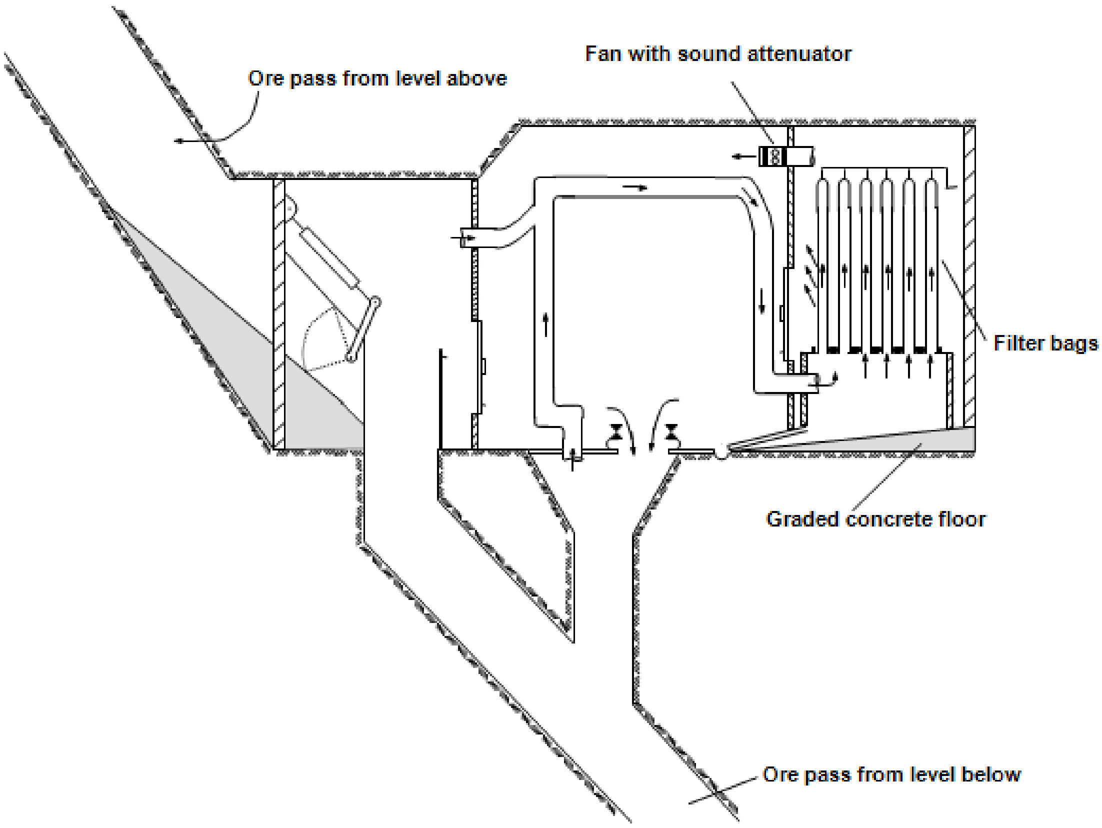


Figure 12.8 Typical section of flannel bag filter underground

Answer

NB

The question must be answered based on certain design parameters mentioned in the notes. Firstly, it is important to calculate the **filter area** of one bag. Given the dimensions above the area can be determined as follows:

$$\begin{aligned}
\text{Filter} &= (\text{the bag circumference x length}) + (\text{end area of the bag}) \\
&= \pi DL + \pi r^2 \\
&= \pi \text{ x } \left(\frac{600}{1\,000} \text{ x } 6\right) + \pi\left(\frac{300}{1000}\right)^2 \\
&= 11.04 + 0.2826 \\
&= 11.586 \text{ m}^2
\end{aligned}$$

The design velocity through the filtering material is 0.1 m/s. The quantity through one filter bag can now be determined:

$$\text{Quantity} = A \times \text{Velocity}$$
$$= 11.586 \times 0.1$$
$$= 1.159\ m^3/s \text{ per bag}$$

The area of the tip can be determined based on the dimensions given:

$$\text{Area of tip} = L \times W$$
$$= 2 \times 8$$
$$= 16\ m^2$$

The downcast air velocity in the tip has been noted as a value of 0.5 m/s, which means that the number of bags can be determined as follows. The volume of air down the tip must first be determined:

$$\text{Quantity} = A \times \text{Velocity}$$
$$= 16 \times 0.5$$
$$= 8.0\ m^3/s \text{ per bag}$$

The number of bags can now be determined:

$$\text{Number of bags} = \frac{Q_{tip}}{Q_{bag}}$$
$$= \frac{8.0}{1.159}$$
$$= 6.9 \text{ say 7 bags}$$

Sufficient dust is left on the bag to ensure a pressure loss of about 100 Pa over the bag when it is again filtered. This is because the filtering efficiency of a perfectly clean bag is only about 50%.

The Rabson filter unit has a shaking device by means of which the filter can be cleaned (mechanical or manual shaking).

The Rayner filter plant is shut down and the bags collapse, causing the dust to settle in trays on the floor of the unit from where it can be removed.

Disadvantages

12.6.2.4 Disadvantages

- Low filtering speed, hence large installations are required if volumes are large
- Efficiency is decreased by moisture
- Frequent inspection required
- Affected by vibration
- Dust produced in cleaning
- Laborious cleaning process, especially if bags are large.

Advantages

12.6.2.5 Advantages of Flannel Bag Filters

- Ease of installation
- High filtering efficiency (95% - 99%)
- Comparatively low capital cost
- No skilled work required for maintenance and cleaning.

12.6.3 Electrostatic Precipitators (High Dust Load)

The filter is made up of a number of cells about 1 m x 0.6 m in frontal area and made up of two sections, namely, the ionising section and the dust collecting section. In the ionising section the dust particles pass through an intense 'ionising field' where they receive an electrostatic charge. A high voltage maintains this ionising field electrically.

The charged dust particles then pass through a series of parallel plates, which is the collecting section. Each alternate plate is earthed while the intermediate plates are maintained at a high voltage. The charged dust particles are repelled by the charged plates onto the earthed plates to which they adhere. An example of the workings of an electrostatic precipitator is shown in Figure 12.9.

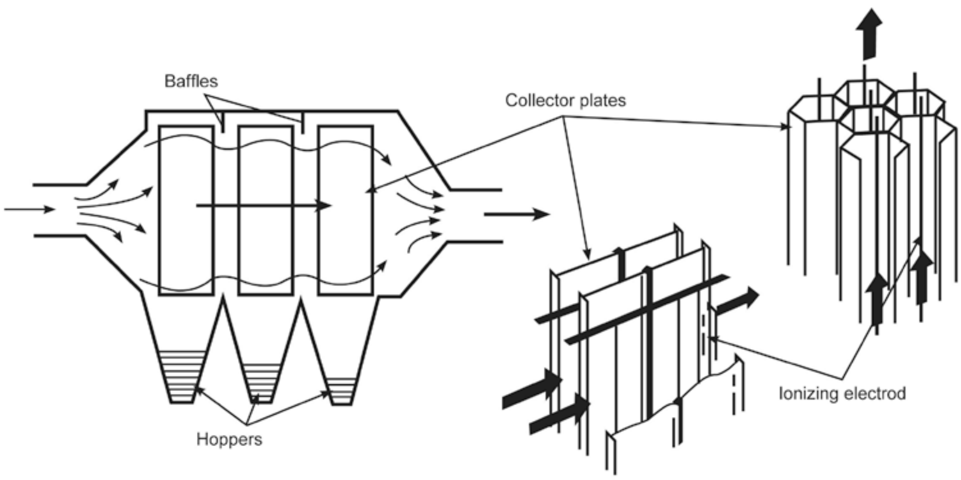


Figure 12.9 Example of electrostatic precipitator

12.6.3.1 Requirements for Best Operation

- The electrical equipment must be correctly maintained
- The air being filtered and the surrounding area should be relatively dry, as moisture causes electric faults, especially with the high voltage in use.

12.6.3.2 Cleaning

The current is first switched off, the plates hosed down with clean water and the sludge deposited in a drain. The plates are allowed to dry before use and may be treated with some substance to increase the adhesion of dust, some of which may otherwise be carried away in the air stream if the air velocity is high.

12.6.3.3 Advantages

Advantages

- High filtering efficiency (almost 100% at 0.5 m/s)
- Lower power cost
- High filtering speeds (from 0.5 to 4 m/s)
- Low resistance to air flow which remains constant irrespective of the period of operation
- Easily cleaned.

12.6.3.4 Disadvantages

Disadvantages

- High capital cost
- Installation requires skilled work
- Requires skilled maintenance
- Moisture may affect electrical circuits
- Danger of electrical shock.

12.6.4 Multi Stage Filter Systems

NB

When designing a multi stage filter system:

- Start with the end in mind, i.e. what conditions do you want at the exhaust of the filter system
- Thus select a final filter that will offer that result
- Then establish how many and the characteristics of the filters to protect that final filter.

All filters have certain attributes:

- They are high efficiency and low dust holding capacity
- Or high dust holding capacity and low efficiency.

Not one filter can provide both. Therefore you need to design around principles of each "stage" to maximize efficiency, dust holding capacity (durability) and practicality.

Application

12.6.4.1 Application

- Tipping Areas
- Crushing
- Transfer Points
- Loading
- Packing.

Advantages

12.6.4.2 Advantages

- High efficiency
- Modular
- Suitable for underground
- Relatively low maintenance
- Installed easily.

Disadvantages

12.6.4.3 Disadvantages

- Initial capital investment
- Requires maintenance
- Moisture may effect some of filter material.

In Figure 12.10 a schematic representation of a multi stage filter system developed by Aircure is shown.

Figure 12.10 Schematic representation of a multi stage filter system

12.6.5 Cyclones

The main mechanism through which particles are filtered is inertial separation. The dust particles enter the cyclone tangentially at the periphery of the cyclone and the swirl is created by removing the clean air at the cyclone axis. Large particles leave the streamlines and migrate to the cyclone wall and then fall into the dust container at the bottom of the cyclone. Smaller particle do not reach the cyclone wall and are removed through the clean air exit.

The efficiency of collection at a given particle size can be improved by both increasing the swirl velocity and reducing the cyclone diameter. An example of the operation of a cyclone is shown in Figure 12.11.

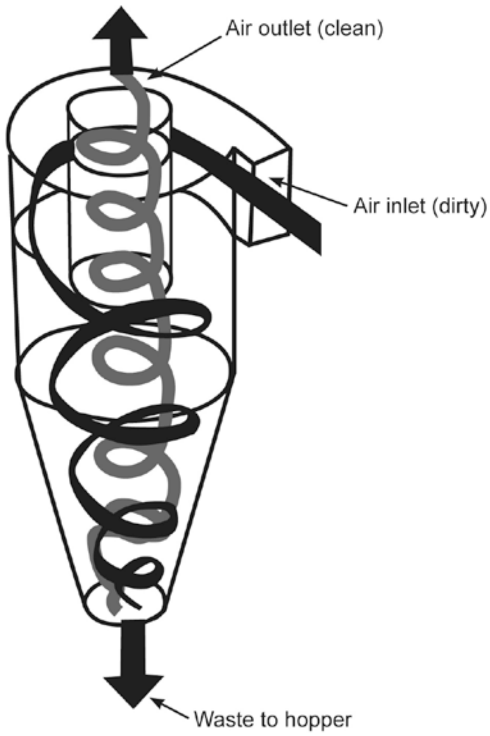


Figure 12.11 Example of operation of a cyclone

12.6.6 Wet Scrubbers

These are normally used in the underground coal mining environment where it is used in conjunction with the continuous miner. It is also used for the filtration of air in crushers on surface. The dust laden air is drawn into a pipe or box in which a fine water spray mist mixes with the dust. The dust particles are then caught in a filter system. The air that goes through the filters also goes through a number of separators where the moisture is removed from the air.

NB

The clean air now exits the scrubber box. It is important that these scrubber boxes and filters are thoroughly maintained; otherwise they will increase the pressure drop across the filter and make it totally inefficient. An example of the working of a dust scrubber is shown in Figure 12.12.

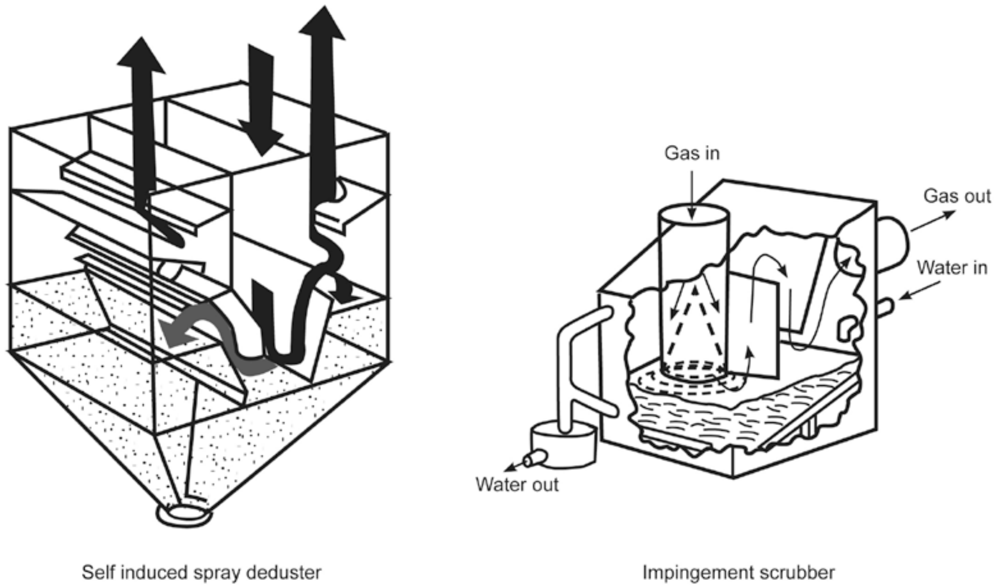


Figure 12.12 An example of the working of a wet scrubber

12.6.7 Other Filters

These include hessian, coconut matting, blanketing, etc. Their filtering efficiencies do not compare very well with those of the filters mentioned above, especially for fine dust. They are, however, used as pre-filters through which coarse dust, which would clog the more efficient filters is removed. A pre-filter of this type is used when blasting dust is filtered.

12.7 Dust Concentration Calculations

The relationship between the amount of air and the dust concentration that is necessary for dilution purposes is as follows:

$$(Q_1 \times C_1) + (Q_2 \times C_2) = (Q_1 + Q_2) \times C_t$$

Where:

Q_1	=	Quantity of air in airway 1
Q_2	=	Quantity of air in airway 2
C_1	=	Dust concentration in airway 1
C_2	=	Dust concentration in airway 2
C_t	=	Total dust concentration in combined airway.

Figure 12.13 will explain the situation better.

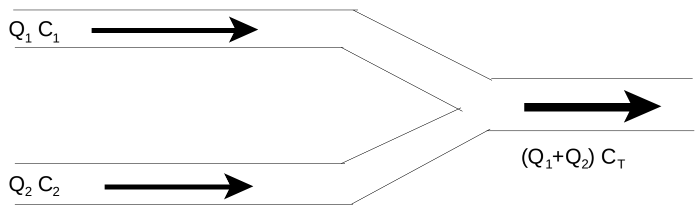


Figure 12.13 Airways joining to give total dust concentration

WORKED EXAMPLE 2

How much fresh air that contains 2mg/m^3, must be mixed with an amount of 11.5 m^3/s air that has a dust concentration of 18 mg/m^3 to ensure that the dust concentration of the mixed air will not be more than 10 mg/m^3?

Answer

Let 'y' be the amount of air that needs to be added.

$$(Q_1 \times C_1) + (Q_2 \times C_2) = (Q_1 + Q_2) \times C_T$$
$$(11.5 \times 18) + (\text{"y"} \times 2) = (11.5 + \text{"y"}) \times 10$$

Through simple manipulation of the formula, the value of 'y' is determined as 11.5 m^3/s.

SELF STUDY EXAMPLE

65 m^3/s of air flows in an intake airway in the direction of the workings. Before the air goes through the workings, 9 m^3/s is drawn off and sent through a filtering system, where the dust concentration after filtration is 3 mg/m^3. 50 m^3/s is sent through a spray chamber, and the air that exists in the spray chamber has a dust concentration of 2 mg/m^3.

The air from the various tunnels then meets again and has a combined dust concentration of 3 mg/m^3. Assume no losses and determine the dust concentration in the air way that has a volume flow rate of 6 m^3/s. See Figure 12.14. (**Answer** 11.33 mg/m^3)

Just as a note of understanding: The dust concentration of the 65 m^3/s of air entering the split cannot be calculated. There is not enough information pertaining to the amount of dust that was filtered in the spray chamber and filtering system. If a percentage efficiency for either was given then the dust count could have been determined before the filter and spray chamber.

The amount of dust that enters the split must obviously be more than the dust amount that exits the split.

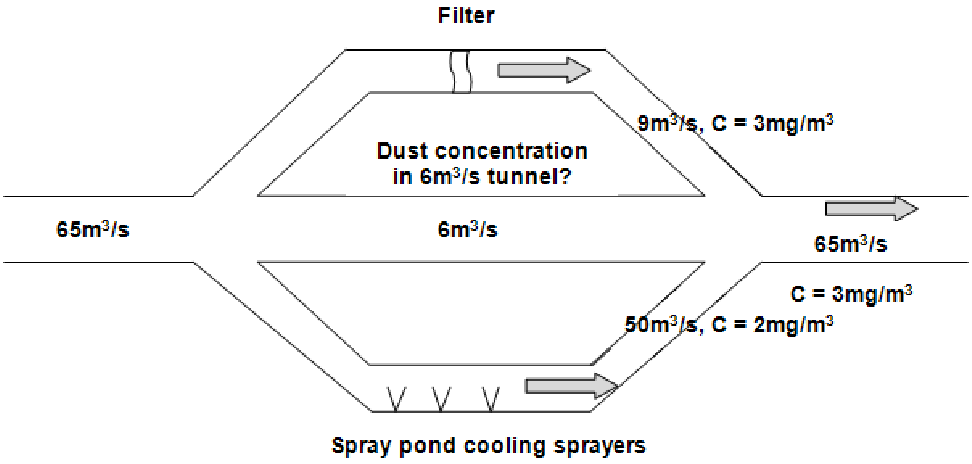


Figure 12.14 Dust concentration layout

12.8 Dust Sampling and Related Technologies

ML

Dust sampling on mines is done for various reasons and the five main reasons are listed as follows:

- To determine the actual levels of dust in a mine and also to determine whether it remains constant or is cyclical. In this way certain trends with regards to dust loads can be highlighted.
- To determine fluctuations in dust concentration levels or to determine the maximum dust levels during a shift.
- To determine how much dust is liberated through certain work procedures such as scraping, loading and drilling and other processes, this will then determine whether new preventative measures should be put in place.
- To determine specific real sources for dust liberation to be able to take the relevant corrective measurements.
- It is the government mining engineers requirement to establish acceptable underground dust concentrations, for example an occupational exposure limit (OEL) of 2 mg/m^3 for silica (SiO_2) content less than 5% and 0.1 mg/m^3 for silica (SiO_2) content more than 5%. In this way a risk assessment can be done.

12.8.1 Requirements for Dust Sampling Equipment

NB

There are various requirements for dust sampling equipment, and these are listed below:

- Must be suitable to determine the specific parameter of the specific dust that is encountered in the work place (for example the silica content)
- Must be accurate
- Automatic control of sample size must be possible
- It must be easy to operate under difficult work conditions
- Must be safe to use even in flammable atmospheres
- Must be robust in construction and must be resistant to moisture and heat
- Must be relatively cheap to use, easy to service and the component parts must be easy to obtain.

12.9 Methods of Sampling Dust

There are basically two types of dust sampling methods, namely:

- Area sampling, sometimes referred to as positional sampling
- Employee sampling methods (gravimetric sampling).

These will now be discussed in detail.

12.9.1 Positional or Area Sampling Methods

Through careful selection of the environment where samples have to be taken, a representative measurement of the dust load in a specific work environment can be obtained by taking only a few samples.

NB

It is important that sampling positions must include points that will include areas where there is an increase in dust load in the air. This will therefore include a position where the air leaves the intake shaft on a specific level to be sampled as well as a position where the air rejoins the return airways.

Typical sampling points include the following:

- Main stations in the downcast shaft
- Positions at the return air side from the loading tips
- Main return airways
- High work areas and anywhere in a stope
- The faces of development ends
- The return air from development tunnels
- Certain worked out areas.

The number of sampling positions that are selected must be chosen in such a way, that the relative number of people that are working in the area must be considered as well. This means that there will be more sampling points in a stope than in a development end. Dust sampling must also be done during the other shift as well, in order to determine the dust levels and difference thereof in different shifts. Real time dust measuring instruments are frequently used to do area sampling. These instuments can include such as the Hund Tyndallo Meter.

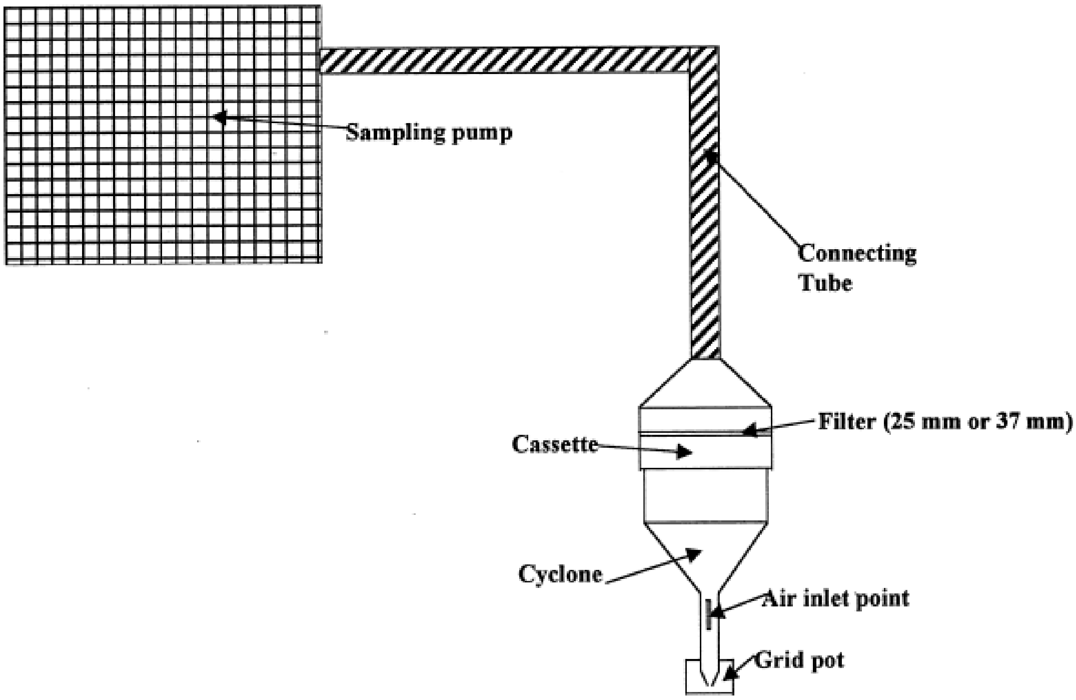
In Table 12.1 the Occupational Exposure Limits for some of the major commodities are shown.

Table 12.1 Occupational Exposure Limits

No	Mineral	OEL
1	Gold mines SiO_2	 0.1
2	Coal mines <5% SiO_2 **>5%** SiO_2	 2.0 0.1
3	Platinum mines <5% SiO_2	 3.0
4	Diamond and other mines <5% SiO_2	 5.0

12.9.2 Gravimetric Sampling

Gravimetric sampling forms a very important part of the ventilation engineer's strategy in the determination of risk areas with regards to unacceptable dust concentrations. Gravimetric sampling is used to allocate personel exposures to workers. Workers wear as sampling train and the measured and calculated concentration is used to evaluate it against the allowable OEL. It is therefore very important that the student has a very sound understanding of all aspects related to gravimetric sampling and the fundamentals associated with it. The gravimetric sampling guideline document will give the student a very thorough exposure to the said requirements pertaining to gravimetric sampling (self study purposes). Figure 12.15 shows the typical components of a 'sampling train' used for gravimetric sampling.



Animation

Figure 12.15 'Sampling train' used for gravimetric sampling

NB

12.9.2.1 Gravimetric Sampling Related Calculations

In the section to follow, various examples (calculations) relating to gravimetric sampling will be dealt with. It is important that the student understand the various principles relating to these calculations.

WORKED EXAMPLE 3

Determine the air quality index for the total population as given in Table 12.2.

Excel

Answer

Table 12.2

Population No	number of persons	Particle concentration (mg/m^3)			Analysis			TLV-TWA	PI	AQI	RISK	PRODUCT
		Respirable		Average	Pollutant	%	mg/m^3 (TWA pollutant)	mg/m^3				
1	12	0.98			Alpha quartz	16						
		0.64										
		1.85										
		1.01			Analysis			TLV-TWA	PI	AQI	RISK	PRODUCT
		0.79										
			Total	Average	Pollutant	%	mg/m^3 (TWA pollutant)	mg/m^3				
2	12		3.40		Arsenic solution	1.7		0.2				
			0.61									
			0.39		Uranium	3.91		0.2				
			1.09									
			1.46		PNOC			10.0				
		Particle concentration (mg/m^3)			Analysis			TLV-TWA	PI	AQI	RISK	PRODUCT
		Respirable		Average	Pollutant	%	mg/m^3(TWA pollutant)	mg/m^3				
3	44	2.11			PNOC			5.0				
		2.00										
		0.91										
		0.73										
		2.08									Product Total	
Total												
Risk =												

Table 12.2: Answer

	(A)	(B)		B/n = (C)		(D)	(CXD/100 = E)	(F)	(E/F=G)	(@SUM G=H)	$(4xH^2=I)$	(IxA=J)
Population No	number of persons	Particle concentration (mg/m^3)			Analysis			TLV-TWA	PI	AQI	RISK	PRODUCT
		Respirable		Average	Pollutant	%	mg/m^3 (TWA pollutant)	mg/m^3				
1	12	0.98		1.05	Alpha quartz	16	0.17	0.10	1.69	1.69	11.38	136.51
		0.64										
		1.85										
		1.01			Analysis			TLV-TWA	PI	AQI	RISK	PRODUCT
		0.79										
			Total	Average	Pollutant	%	mg/m^3(TWA pollutant)	mg/m^3				
2	12		3.40	1.39	Arsenic solution	1.7	0.02	0.2	0.12	0.52	1.09	13.03
			0.61									
			0.39		Uranium	3.91	0.05	0.2	0.27			
			1.09									
			1.46		PNOC	94.39	1.31	10.0	0.13			
		Particle concentration (mg/m^3)			Analysis			TLV-TWA	PI	AQI	RISK	PRODUCT
		Respirable		Average	Pollutant	%	mg/m^3 (TWA pollutant)	mg/m^3				
3	44	2.11		1.57	PNOC	100.00	1.57	5.0	0.31	0.31	0.39	17.26
		2.00										
		0.91										
		0.73										
		2.08										
Total	68										Product Total	166.81
Risk =	2.45											

WORKED EXAMPLE 4

(i) 2 mg of particulates were collected on a filter of a gravimetric sampling device. The flow rate of the sampler was assumed to be 2.2 litres/minute and the sampling time was 4 hours. What is the average dust concentration?

(ii) When calibrating the pump, the flow rate was found to be 1.7 litres/minute. What value does this now give for the concentration?

(iii) What % error was there in the first measurement?

Answer

(i) The sample time must be determined:

Sample time = (4 x 60)
= 240 minutes

The sample volume = Flow rate (litres/minute) x sample time
= 2.2 x 240
= 480 litres of air
= 0.528 m^3

Determine the concentration = Mass / volume
= 2.0 / 0.528
= 3.79 mg/m^3

(ii) The sample $volume_{accurate}$ = Flow rate (litres/minute) x sample time
= 1.7 x 240
= 408 litres of air
= 0.408 m^3

Determine the concentration = Mass / volume
= 2.0 / 0.408
= 4.9 mg/m^3

(ii) The error estimate can now be determined:

$$\text{The \% error} = \frac{4.90 - 3.79}{4.90} \times 100\%$$

$$= \frac{1.11}{4.90} \times 100\%$$

$$= 22.7\ \%$$

It must be noted that this result will be discarded as it resulted in an error measurement.

WORKED EXAMPLE 5

A worker was issued with a gravimetric sampling pump. It's normal operating at 2.2 litres per minute for eight hours and twenty minutes. The following results were obtained from the sample filter:

Table 12.3

Control/blank filters		Sample filter	
pre-filter mass (mg)	**post filter mass (mg)**	**pre-filter mass (mg)**	**post filter mass (mg)**
20.1	20.16	20.12	20.66
20.11	20.18	20.11	20.65
20.09	20.17	20.13	20.67

(i) Determine the sample volume

(ii) Determine the correction factor of the control/blank filter

(iii) Determine the sample mass

(iv) Determine the corrected sample mass

(v) Determine the concentration

(vi) Determine the TWA dust concentration.

Answer

(i) The sample time must be determined:

Sample time = (8 x 60) + 20

= 500 minutes

The sample volume = Flow rate (litres/minute) x sample time

= 2.2 x 500

= 1100 litres of air

= 1.1 m^3

(ii) The correction factor of the control/blank filter must be determined:

Table 12.4

Control/blank filters		
	pre-filter mass (mg)	**post filter mass (mg)**
	20.10	20.16
	20.11	20.18
	20.09	20.17
Totals	**60.30**	**60.51**
Average	**20.10**	**20.17**

The correction factor = post-filter mass-pre-filter mass
= 20.17 - 20.1
= 0.07 mg

Note: In this example the post filter mass was larger and therefore the sign of the correction factor to be applied later will be a 'minus'.

(iii) Determine the sample mass:

Table 12.5

Sample filter		
	pre-filter mass (mg)	**Post-filter mass (mg)**
	20.12	20.66
	20.11	20.65
	20.13	20.67
Totals	**60.36**	**61.98**
Average	**20.12**	**20.66**

The sample mass = post-weight sample mass-pre-weight sample
= 20.66 - 20.12
= 0.54 mg

(iv) Corrected sample mass = Sample mass - correction factor

= 0.54 - 0.07

= 0.47 mg

(v) Determine the concentration = Mass / volume

= 0.47 / 1.1

= 0.43 mg/m^3

(vi) Determine the TWA dust concentration:

$$\text{TWA dust concentration} = \frac{\text{dust concentration x total sample time (minutes)}}{480}$$

$$= \frac{0.43 \times 500}{480}$$

$$= 0.45 \text{ mg/m}^3$$

If in a mine where silica dust is present, the TWA concentration need to be corrected for silica. If the measured silica is 20 % then:

TWA silica dust cone = TWA dust cone x % silica

$$= 0.45 \times \frac{20}{100}$$

$$= 0.09 \text{ mg/m}^3$$

If compared to Table 12.1 where the OEL for silica (S_iO_2) was shown as 0.1 mg/m^2, it shows that the worker exposure is within limit but very close to an over exposure.

Chapter

13

ILLUMINATION

LEARNING OUTCOMES

Study Objectives

- Describe the properties of light
- Can differentiate between luminous intensity, luminous illumination and surface brightness
- Distinguish between various light sources
- Can describe the effect that lighting would have on production in the underground environment.

REFERENCES

Mine Environmental Control (MEC) Certificate, Chamber of Mines, South Africa, Workbook 5.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

13.1 Introduction

Definition

Vision is dependent on light and, lighting systems are designed to provide an environment in which people, through the sense of vision, can function effectively and efficiently. Without light, the important sense of sight is useless, even if the eyes are very good. **Vision** can be defined as the transformation of light energy into biological energy that is converted by the brain. This transformation occurs within the eye through cells known as rods and cones.

The sensation of vision is produced when light enters the eye. The eye pupils increase in size at low illumination levels in order to admit more light into the eyes, and contracts in bright light to admit less light. The amount of light entering the eye falls onto the retina, which is the light-sensing organ and is by far the most important part of the eye. Its main function is to increase the sensitivity of the eyes at low illuminance (illumination) levels through a physiological process.

ML

Good illumination is associated with saving of human energy, reduced accident risk, increased production and, in general, contributes to better utilization of men, material and space.

With deep level mining in South African gold mines, a wide variation in the illumination levels of different working places exists. (± 8 lux to 82 lux). It is thus obvious that poor illumination and any impairment in the ability to dark adapt by an individual may represent a definite safety hazard, not only to the individual but also to his co-workers.

Light is a form of energy and is best described by the comparison of a wave movement, which has the concepts of frequencies and amplitudes associated with it. It therefore exhibits measurable characteristics in terms of the wave length and velocity of propagation, which means that it occupies a place within the electromagnetic spectrum alongside radio waves, x-rays and cosmic rays. The fundamental difference between these various forms of radiant energy is their wavelength and frequency.

The waves can also be absorbed by certain materials and are reflected by others. Sound waves can travel through any medium (except through vacuums). Light waves however can only travel through see-through mediums and can be propagated the furthest and quickest through a vacuum. The reason for this is that the light waves are electromagnetic waves, which do not contain the vibration of physical elements. Light waves are much shorter and also move very much quicker than sound waves. The length of the light wave differs between 400 and 700 nanometres. This is 0.4 to 0.7 micrometres.

13.2 Properties of Light

White light has a mixture of light waves of different wave lengths and when white light is sent through a prism, the light is broken up into a spectrum of different colours. The colour spectrum (as display in Figure 13.1) starts with purple (the shortest wave length) and is followed by, blue, green, yellow, orange, and red (the longest wavelength). Light that enters the eye falls on the retina, which is the light sensor of the body.

The retina is not evenly sensitive for the different light waves. It is most sensitive for light waves with wavelengths of 550 nanometres (yellow to green) and least sensitive for the longest wavelengths of 650 nanometres (red) and 400 nanometres (purple) respectively, which corresponds to frequencies from 8.6×10^{14} Hz and 4.3×10^{14} Hz. Figure 13.1 shows this aspect in detail.

Law

The velocity of light is almost a million times faster than sound. It takes approximately 8 minutes for light from the sun to reach earth. The velocity of any electromagnetic wave in a vacuum is 3×10^{8} m/s. If light were to move at the same speed as sound, it would have meant that sunrise would only occur 14 years after it has actually taken place. Another aspect of light is the light year; this is the distance that air can travel in a year.

Certain characteristics of light are as follows:

- For all practical purposes, light travels in a straight line
- The velocity of light in water is less than in air (factor of ≈ 1.33)
- A ray of light carries energy that is transferred into heat, when it is absorbed by the surface of a body or into electricity when it is absorbed by a thermo coupling
- Light is reflected by a smooth surface where the angle of entry is equal to angle of exit
- Light is refracted (deflected or bent) if it is sent from one medium to another.

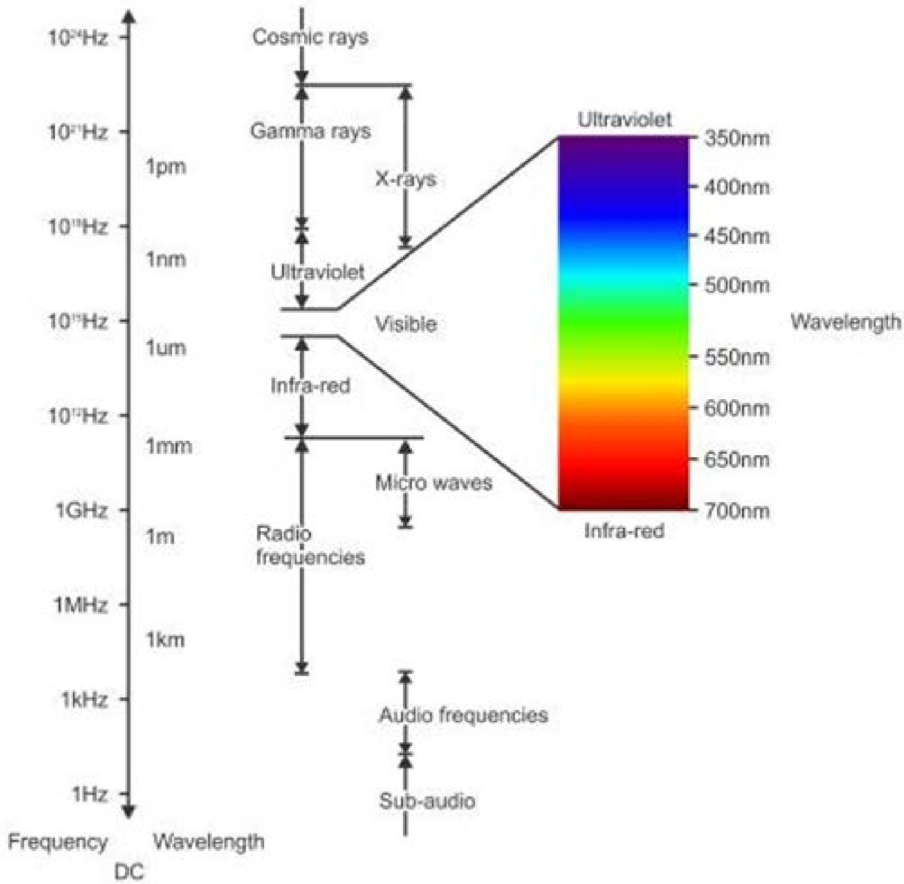


Figure 13.1 A diagram of the electromagnetic wave spectrum

The wave length and frequency of an electromagnetic wave can easily be determined by the use of the following Equation.

$$c = f\lambda \quad \text{...} \quad (13.1)$$

Where:

C	=	the velocity of light (3 x 10^8 m/s)
F	=	the frequency (Hz)
λ	=	the wave length (m)

13.3 Measuring the Properties of Light

Definition

The science of measuring light is called **photometry** and we can define four specific units, namely:

- Luminous intensity
- Luminous flux
- Illumination
- Surface brightness (glare).

13.3.1 Luminous/light Intensity of the Light Source (I) (Candela (cd))

The luminous intensity is the light source measured in candela (cd) in a direction perpendicular to the surface of 1 / 600 000 m^2 of a black body at the temperature of the freezing point of platinum (2 042^o K) and at a pressure of 101.325 kPa. As an example it can be mentioned that:

- A 40 Watt house lamp produces light with a light or luminous intensity of approximately 35 cd
- A 100 Watt house lamp produces light with a light or luminous intensity of approximately 130 cd
- A 40 Watt fluorescent lamp produces light with a light or luminous intensity of approximately 200 cd.

13.3.2 Luminous Flux (lumens)

Definition

Luminous flux is that part of the **total radiant energy per unit time** from a luminous source which produces the sensation of light and is measured in lumens. Lumens is the flux emitted within a unit solid angle of one steradian by a point source having a uniform light intensity of one candela (see Figure 13.2). The energy emitted by a light source is not all visible as light. For example, 70% of the energy that is emitted by a 100 Watt incandescent lamp is in the infrared region and only 10% of that energy is visible light.

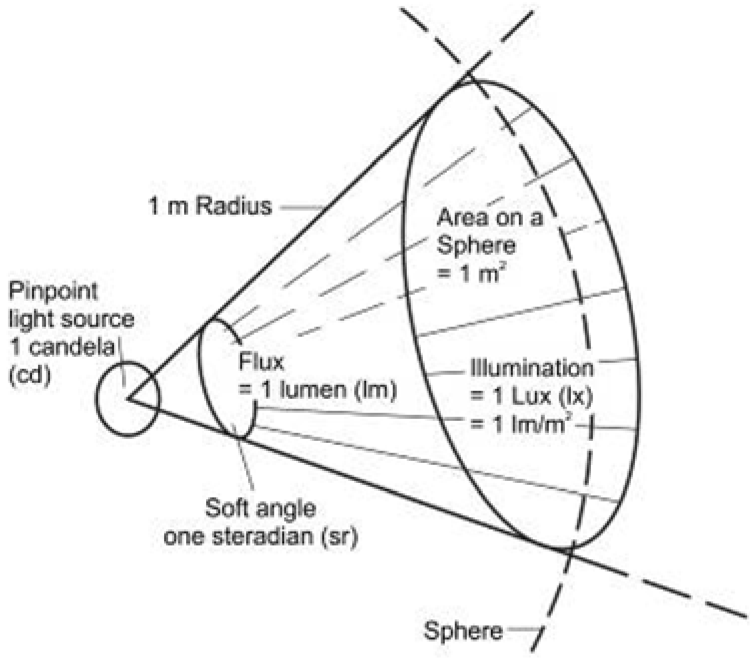


Figure 13.2 Units of light

13.3.3 Illumination (E) (lux = lumen/m^2)

When the intensity of a light source is increased, the luminous flux transmitted to a surface, and the flux on that surface is also increased. In other words, the density of luminous flux on the surface is increased. This is usually referred to as an increase in illumination or illuminance. The unit of illumination is the lux, which is commonly used in routine measurements. It is equal to one lumen/m^2. It can be shown that the illumination on a surface decreases as the square of the distance of the surface from the light source. In other words, if the distance (d) from the light source is doubled, the illumination will fall to a quarter of its initial value. The illumination (E) can also be expressed as a function of the luminous or light intensity (cd) and is shown below:

$$illu\min ation\,(E) \quad = \quad \frac{\text{(I)ntencity of light (candela)}}{d^2}$$

NB

Another important factor which affects the illumination is that the light from a source does not normally fall perpendicularly onto the surface being illuminated, so that the actual area being illuminated when the source is perpendicular is **increased**. A correction factor, which includes the angle of inclination (2), between the normal to the surface and the light rays must be included in any calculation on illumination.

In Figure 13.3, the two lamps L_1 and L_2, have equal intensities, I_1 and I_2 respectively and are situated at points of equal distance, that is R_1=R_2. Using the correction factor for the inclination of source L_2, we have:

$$illu\min ation_{\,2} \quad = \quad \text{illumination}_{\,1} \ \text{x} \ \text{cosine}\,\theta$$

Therefore the illumination for position 2 is always less than for position 1, for any angle where 2∃0º (because cosine 2 always # 1). Refer to Figure 13.3, for a better understanding of the above mentioned.

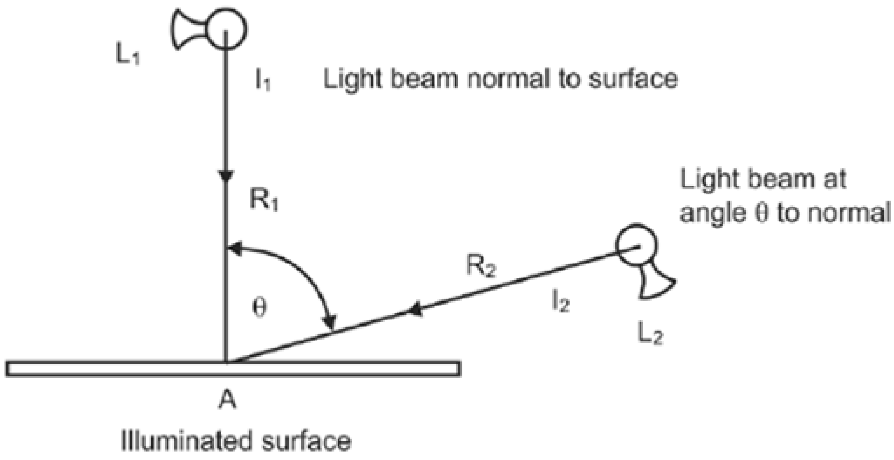


Figure 13.3 The cosine correction factor

13.3.4 Surface Brightness

The surface brightness of an object depends on the intensity of the light striking it and on the amount of the light that is reflected. Thus for a given level of illumination a white surface is brighter than a black surface, because the black surface absorbs more of the light than the white surface does.

Brightness may be either physical or subjective. Physical brightness can be measured and is often called **luminance.** Subjective brightness is that sensed by the eye and varies from individual to individual.

Definitions

Glare is defined as brightness or light intensity that causes discomfort, interference with vision or eye fatigue. When the ability of the eye to distinguish detail is affected by glare, the glare is known as **disability glare.** Glare, which distracts attention or creates a painful sensation, is known as **discomfort glare.**

The relationship between brightness and glare is therefore similar to that between sound and noise, glare and noise both being unwanted. Glare can therefore also be expressed in three forms, namely:

- Relative glare, caused by contrasts in brightness
- Absolute glare, too bright for the eye to adapt to
- Adaptive glare, which is temporary and to which the eyes can adapt.

Factors that influence glare:

- The brightness of the source
- The size of the source
- The position of the source
- The contrast in brightness between the source and its surroundings
- The time of exposure.

13.4 Dark Adaption and Visual Acuity

13.4.1 Dark Adaption

Definition

NB

Dark adaption is the process of change, which occurs in the eyes when the environment changes from bright to dark. This is an important factor in the underground mining environment, as workers move from a brighter to a darker place, there is a period during which their eyes are not fully adapted and they are consequently more prone to accidents. This adapting period to darkness varies greatly from person to person (can take as long as 60 minutes) and indications are that a shortage of vitamin A can extend the period of adaption by several minutes. The only means in reducing this problem of dark adaption in the underground environment are:

- Standardize lighting in a mine to eliminate large variation in underground illumination
- Grade light intensities underground to allow adequate time for the eye to adapt to various light intensities.

13.4.2 Visual Acuity

Definition

Visual acuity can be defined as a measure of the ability to distinguish fine details (sharpness of vision). The most common method to determine visual acuity is the Snellen chart that consists of a row of letters, each row having slightly smaller letters than the row above it.

References

The Snellen E chart can be applied successfully for illiterate adults, people from other language groups and to children of 3 years of age, to determine their visual acuity. For more detail on the application of these charts, candidates can refer to Occupational Hygiene by Schoeman and Schröder, pages 154 to 156.

Figure 13.4 Visual acuity chart

The following factors influence visual acuity:

- Visual acuity improves with improved illumination and is best at levels in excess of 1 000 lux
- It improves with contrast
- It is better with dark symbols on a light background
- It decreases with age.

SHE

It is thus obvious that any impairment in the ability to dark-adapt and, maintain optimal visual acuity under underground mining conditions may represent a definite safety hazard.

13.5 Colour Blindness

This defect can be defined as the partial or total inability to distinguish colour. Colour blindness can be divided into:

- **Partial colour blindness:** is the inability to distinguish the different shades of the compound colours, such as brown and grey.
- **Total colour blindness:** is the inability to distinguish any colour so that only light and darkness can be seen (for example white and black).

- **Red/green colour blindness**: is the inability to distinguish between the primary colours, red, yellow and blue or between the secondary colours, green, orange, brown and purple.

Research determined that the cause of this defect lies in the brain's interpretation of information rather than in the visual organs. Thus, people with this defect or condition should not be employed in certain visual tasks where colour and colour discrimination is a necessity.

13.6 Types of Defective Vision

- **Long-sightedness:** this type of eye cannot clearly see nearby objects
- **Near-sightedness:** this type of eye can see nearby objects very clearly, but objects at a distance are unclear
- **Presbyopia:** is long-sightedness caused by loss of elasticity of the eye lens, occurring especially in middle and old age people - the process of adjusting to near-vision becomes extremely difficult.

13.7 Effects of Good and Poor Illumination

13.7.1 Good Illumination

There is no doubt that good or improved illumination underground and/or on surface have the following beneficial effects:

- Decreases the risk of accidents and reduce errors
- Increases productivity
- Improve working conditions, morale and motivation
- Preserves human energy
- Reduce eye-strain and minor illnesses
- Prevent the accumulation of waste and rubbish
- Better utilisation of men and material
- Improve the ability to move around and provide greater safety.

13.7.2 Poor Illumination

Poor illumination can be an important contributing factor to the following:

- Tiredness: people using their hands and eyes to perform work under poor illumination tires more quickly
- Accidents: causes loss of money, life and incalculable suffering

- Poor productivity:- caused by low morale and motivation in the work force
- Safety hazard: dangerous acts and working conditions cannot be easily noticed or detected
- Causes eye-strain and minor eye illnesses
- Hamper the ability to move around easily
- Sharp increase in the number of errors by workers
- Wastage of material and damage to equipment
- Accumulation of waste, rubbish and old explosives that can pose a fire hazard and unhealthy environments.

SHE

Thus, good illumination can be regarded as important to establish high productivity and a safe and pleasant working environment. Research has also shown that there is a direct correlation between the number of accidents and the type and amount of lighting in a work area underground. Table 13.1 shows this relationship.

Table 13.1 Illumination levels and number of accidents

Illumination (Lux)	Number of accidents as %
20	100
200	68
250	58

From Table 3.1 it is obvious that there is a dramatic decrease in the number of accidents when the lighting is improved. It has also shown that there was an increase in productivity with the new lighting conditions. The light that is given off by a cap lamp is approximately 2-4 lux, which is not always effective enough.

The following are guidelines for illumination quantities for specific work environments underground in South African mines:

- Stations and areas where machines operate 200 lux
- Crossings and dangerous areas 100 lux
- Access ways 20 lux

WORKED EXAMPLE 1

If a light source has an average wave length of 300 nm and a frequency of 10^{15} Hz, what is the approximate velocity of the light?

Answer

$$c = f\lambda$$
$$= 300 \times 10^{-9} \times 10^{15}$$
$$= 300 \times 10^{6}\, m/s$$

WORKED EXAMPLE 2

A light source produces 4κ lumens uniform in all directions. If such a light source is in the middle of a sphere with a radius of 1 m, determine the illumination (E) in lux on the sphere. The area of the sphere is$4\pi r^2$.

Answer

$$1 \text{ lux} = \frac{1 \text{ lumen}}{\text{area}}$$
$$= \frac{4\pi}{4\pi(1)^2}$$
$$= 1 \text{ lumen/m}^2 = 1 \text{ lux}$$

13.8 Light Sources

There are various light sources and will now be discussed in detail. The four most common used light sources are chosen as they are the most frequently used on mines and are of the most effective. These light sources are:

- The filament lamp
- The fluorescent lamp
- The mercury vapour lamp
- The sodium vapour lamp
- Cap lamps for workers.

13.8.1 Filament Lamp

In this type of lamp (Figure 13.5) the light is produced by the heating of a wire to incandescence (white heat) by passing an electric current through it. Only 10% of the energy actually radiated by most filament lamps is in the form of visible light. The greater portion of the energy is in the form of heat.

The life of the lamp and its output are determined primarily by the filament temperature. The higher the temperature the greater the light output but the shorter the life of the lamp. General service lamps have a typical operating life of between 750 and 1 000 hours.

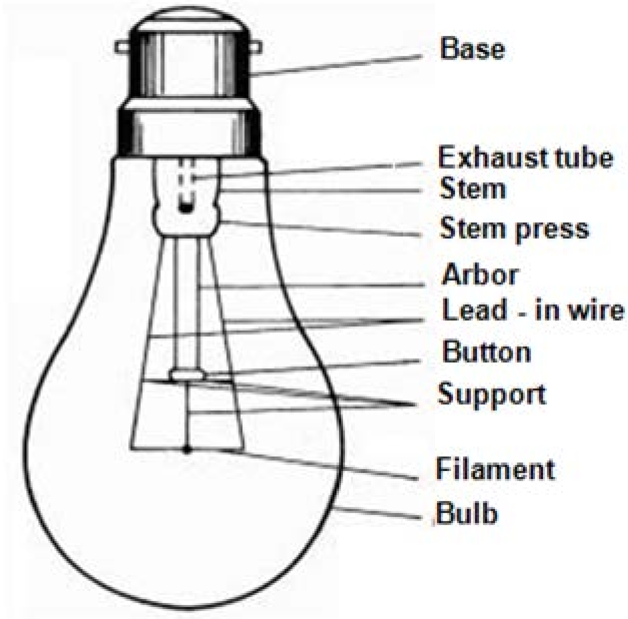


Figure 13.5 Filament lamp

13.8.2 Fluorescent Lamps

This lamp consists of a tubular bulb into each end of which electrodes are sealed. The tube contains a mixture consisting primarily of mercury vapour and argon. The inner wall is coated with fluorescent powder. On applying power to these lamps the electrodes are heated which causes an arc to be struck. The resultant current flow causes radiation to be generated within the gas mixture. The ultra violet energy present is absorbed by the fluorescent powders or phosphors and causes them to radiate light and infrared energy (see Figure 36.5 of EE for an example).

References

The average life span of a fluorescent lamp is about 7 500 hours. The output from fluorescent lamps is high during the first 100 hours of operation and their efficiency in terms of light output is about 22%. Gradual deterioration of the phosphor coating on the inner wall of the tube leads to a decrease in output over time.

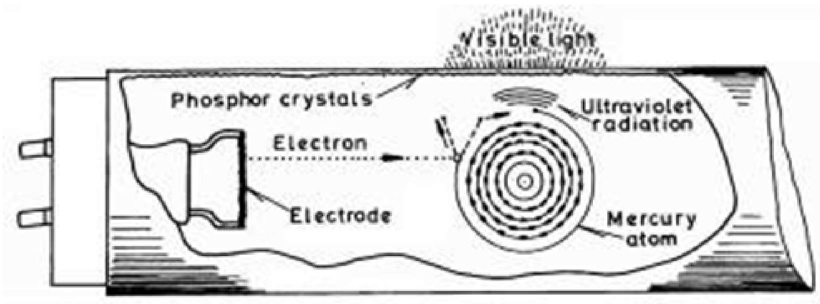


Figure 13.6 Fluorescent lamp

13.8.3 Compact Fluorescent Lamps

Incandescent lamps are being replaced by a device called a 'compact fluorescent lamp' (CFL). A compact fluorescent lamp provides the benefits of fluorescent illumination in a device that can be installed into the screw-in or bayonet sockets used by incandescent lamps (see Figure 13.7). They are, however, incompatible with circuits which dim lights.

Although initially more expensive than incandescent bulbs, compact fluorescent lamps can be as much as 80% more efficient and can last more than ten times longer than filament lamps. For instance, a 25 W compact fluorescent lamp can replace a 100 W incandescent lamp for a saving of 75%.

The Table 13.2 gives typical equivalent CFL ratings for the replacement of common incandescent bulbs.

Table 13.2

Ordinary bulbs	CFLs
40 W	7 – 10 W
60 W	15 –18 W
100 W	20 – 25 W
150 W	32 W

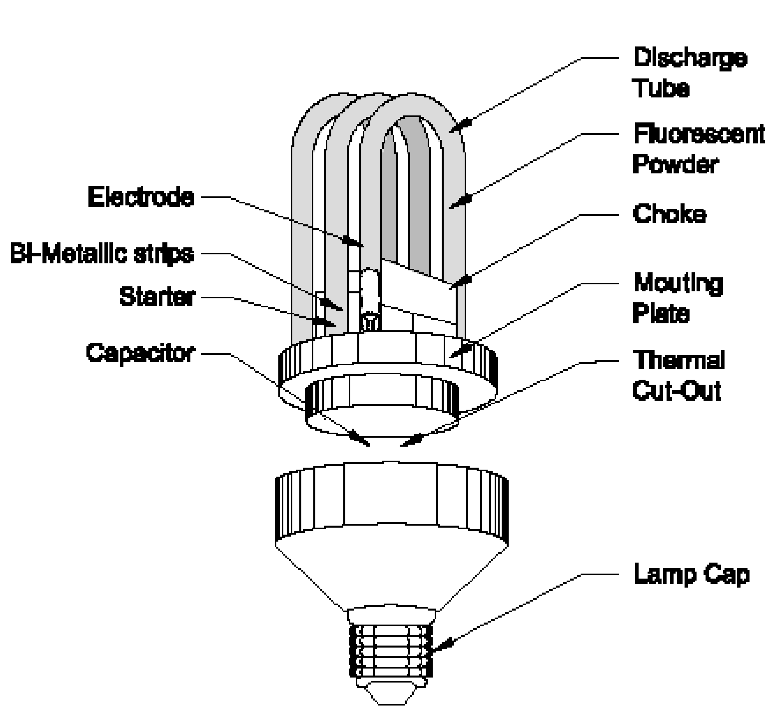


Figure 13.7 Compact florescent lamp

13.8.4 Mercury Vapour Lamps

Mercury vapour lamps belong to the general category known as electric-gas discharge lamps. Light is produced by the passage of an electric current through the mercury vapour.

The basic construction of such a lamp is shown in Figure 13.8. The lamps are normally used in large excavations underground or for night work on surface, as it is very bright and can cover large areas if used in combination with other mercury vapour lamps. Mercury vapour lamps have a very long life span (approximately 24 000 hours, based on a 5 hour burn time per start and operation from the correct ballast which controls the operating current. Output decreases rapidly through the first 100 hours of operation, but then stabilizes.

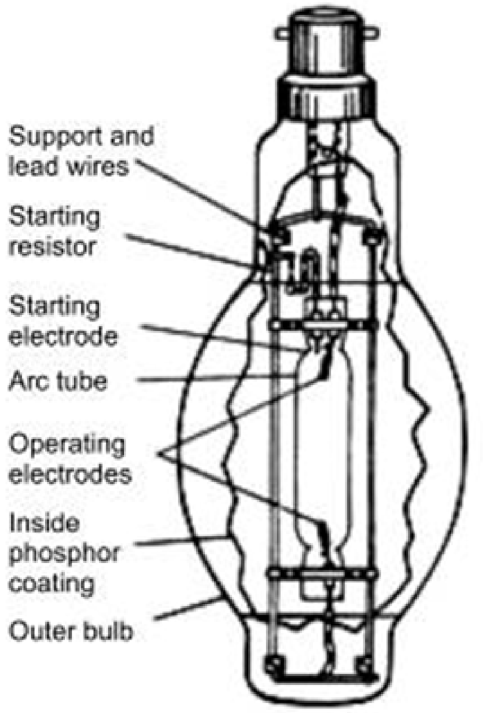


Figure 13.8 Components of a mercury vapour lamp

13.8.5 Sodium Vapour Lamps

Sodium vapour lamps are very similar to mercury lamps, except that the starting gas is neon and the arc is passed through vapourised sodium. The light produced by this type of lamp gives a predominantly yellow colour to the object being illuminated. They are therefore not suitable when accurate colour differentiation is required and their main application is in street lighting.

13.8.6 Cap Lamps

Miner's cap lamps are filament lamps and are basically standard in form as shown in Figure 13.9.

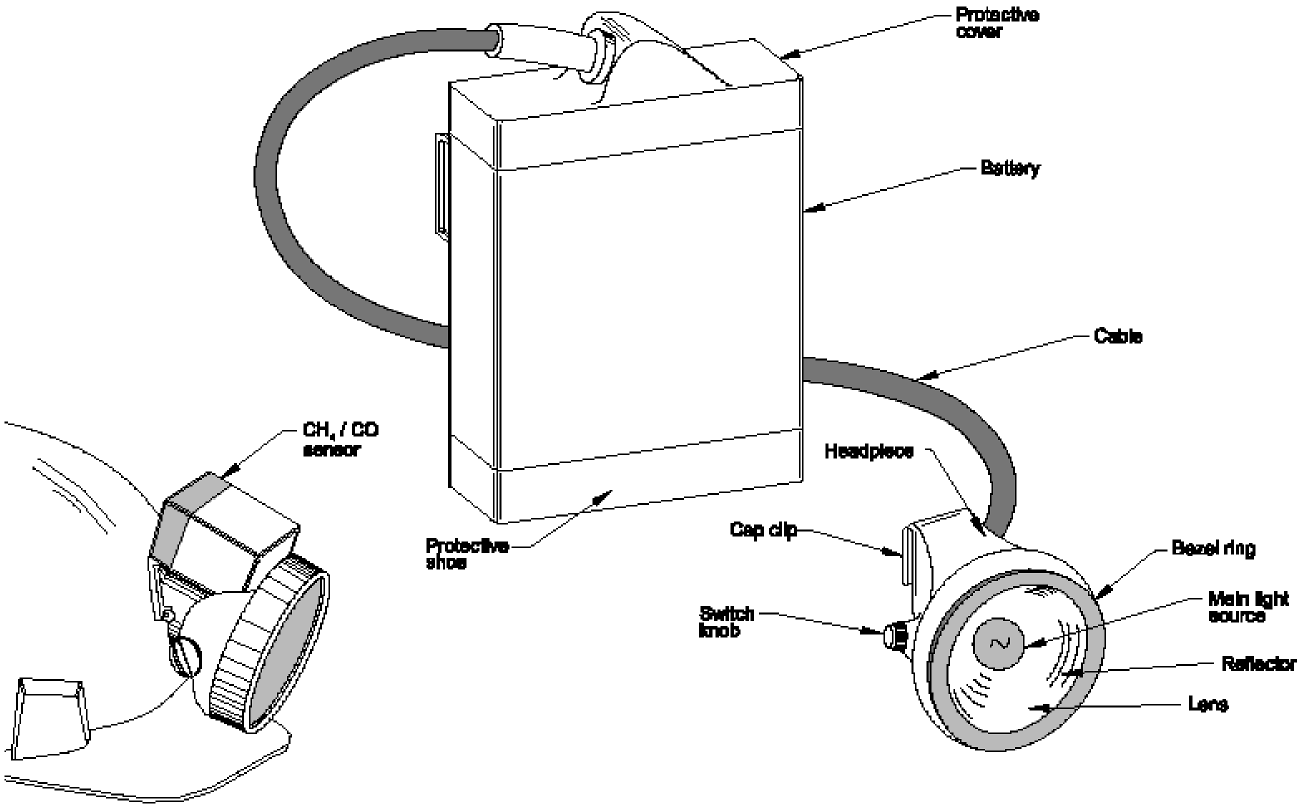


Figure 13.9 Miners cap lamp

The light on the cap lamp is provided by a single or double filament source which draws a specific current normally between 0.67 and 1 A. It is recommended that 0.8 A light sources are used as minimum requirement.

Table 13.3 gives an indication of the different expected burning times and charging times for the various light sources.

Table 13.3 Cap lamp burning times

Light source(A)	Burning time of Lamp (h)	Charging time (h)
0.67	13.0	11.0
0.70	13.0	11.0
0.80	12.3	11.7
1.00	11.5	12.5

Table 13.4 shows the light source versus the light output in lumens ('luminous flux')

Table 13.4 Light source output in Lumens

Light source (A)	Light output (lumens (lm))
0.67	10
0.80	13
1.00	16

13.8.6.1 Batteries of Cap Lamps

These are made up of two lead acid cells connected in series. They produce a nominal output voltage of 4 Volt. The capacity of a fully charged battery is 11 Ah. No leakages are to take place no matter at what angle the battery is carried.

13.9 Recommendations Light Improvement Underground

- Lighting conditions must be standardised
- Existing light sources must be cleaned regularly
- The light intensity must be graded so that from one work area to another the eye must be given enough time to adapt to the new surroundings
- Caplamps must be photometrically tested before use and periodically thereafter
- Glare must be prevented by the use of the proper screens/covers
- Make more use of fluorescent lamps, as they are more effective.

13.10 Light Measuring Instruments

The light meter that is commonly used in the South African mining industry is the luxmeter. The ideal light meter must have the following characteristics:

- Must be of robust construction
- Must be relatively cheap
- Must be accurate and frequently calibrated against test source
- Must be safe to use.

13.11 Illumination Surveys

NB

Illumination surveys are relatively easy to do. The specific position where the measurement is taken is important. Illumination surveys are therefore done to determine the level of illumination in specific work areas to be able to take corrective actions if necessary.

Objective

The information that must be recorded during an illumination survey is the following:

- The light intensity at the specific point being measured
- The type of work being done
- The type and voltage of the lighting installed
- The condition of the lighting (clean or dirty)
- A drawing of the area, including aspects such as the type, colour and measurements of walls, floors and the roof in offices
- The type and number of the lux-meter used.

13.11.1 Frequency of Illumination Surveys

NB

There is no fixed interval, but such surveys are normally done once per year. Surveys can however be repeated with new lighting installed. It is important that illumination surveys be done where there are dangerous work areas. After such an illumination survey, the results must be recorded accordingly and also shown a plan and distributed to the responsible people involved. Each mine has its own procedures, but the minimum requirement is that substandard conditions be dealt with as soon as possible.

13.12 Factors Affecting a Visual Environment

References

(Reference VOEE chapter 36 section 5)

Level of illumination is only one of the factors that determine the quality and hence the safety of a visual environment. Trotter (1982) states, "*Seeing effectively is more complex than merely being dependent on the amount of light being shone at a target*".

In mining, other factors that have been identified as affecting the overall quality of the visual environment include the following:

- Mounting height restrictions and task requirements often place luminaires in the worker's direct line of sight, causing glare
- Mounting positions restrict the size, location and light distribution of luminaires

- Luminaires must meet safety requirements for use in explosive atmospheres
- Inherent vision deficiencies of the workforce population
- Low surface reflectance of rough, darkly coloured rock, which severely limits secondary reflections and indirect lighting
- Suspended dust and water vapour cause back-scattering, thereby reducing apparent illuminance and deposits of dust on brightly painted, potentially hazardous machinery, reduce their reflectance and hence their visibility.Reduction of effective illuminance by protective eye wear.

The range of factors that can influence the visual environment is summarised in Figure 13.10 (ECSC, 1990):

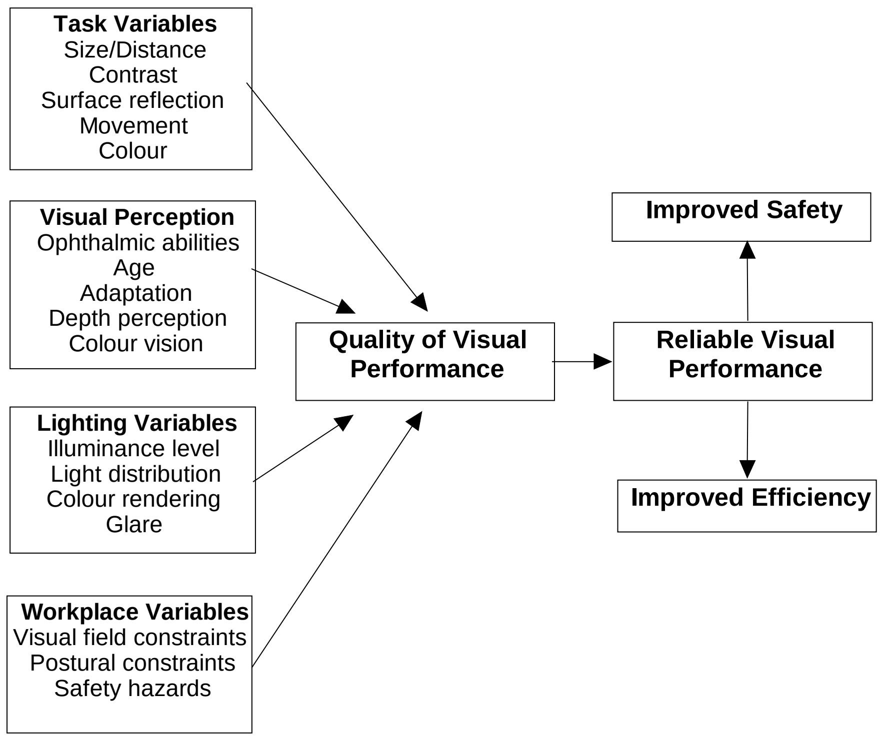


Figure 13.10 Major factors influencing the visual environment and visual performance (ECSC, 1990)

To ensure the quality and reliability of visual performance and hence the safety, all of these factors must be considered. The need for appropriate worker selection to ensure that individuals have sufficient visual abilities to undertake their work safely is a well recognised requirement that should be addressed through programmes of initial screening and periodic testing. However, efforts should be directed at tailoring the visual environment to meet the needs of workers and their tasks, rather than selecting individuals to cope with poor conditions.

13.12.1 Glare

NB

Glare is often a major problem in the mining industry. To reduce this, Baines (1972) suggests that it is better to have more low-power lights with small distances between them than fewer high-power lights further apart. For example, in an underground roadway the former provides even illumination, while the latter produces repetitive transitions between bright glaring light and darkness. Uniform lighting makes for better hazard perception, as dark patches and shadows act to camouflage obstacles and hazards.

Trotter (1982) suggests the following ways to reduce glare:

- Move illuminance sources out of the direct field of view
- Shield sources from direct view
- Use prismatic lenses, filters or cross-polarisers
- Minimise luminance differences between visible sources and background
- Increase background or overall illuminance
- Ensure favourable luminaire positioning relative to work tasks
- Eliminate specular surfaces
- Provide light with the correct level, colour balance and distribution.

It should also be remembered that glare is relative. For example, in an un-illuminated area underground, the cap lamp can give a blinding glare, but at a well-lit shaft bottom it is entirely tolerable, and on surface in daylight it may not even be noticed.

13.12.2 Reflectance

Crooks & Peay (1981) showed that underground work environments differ significantly in their surface luminance and reflectance, output of luminaires, and the nature of visual impairment present. For a given level of illuminance, increasing surface reflectance can enhance the level and distribution of available light. Light-coloured surfaces generally reflect more light than dark-coloured surfaces. Some typical percentages of reflectance in mines, as reported by a range of sources (Best et al., 1981; Baur & Wellbeloved, 1985; Sanders & Peay, 1988), are shown in Table 13.5.

Table 13.5 Reflectance of typical materials in mines

Material	**Reflectance**
Coal	3–15%
Quartz diorite	10–30%
Shale	22–45%
Rust	9%
Fresh Whitewash	65–95%
Faded Whitewash	20–60%
Cement	10–30%

NB

The Table shows that whitewashing a wall can increase its light-reflecting ability considerably. Most underground mines make use of whitewash as a cost-effective means of improving lighting performance, and the application of whitewash in certain locations is a legislated requirement in some countries.

13.12.3 Contrast

Illuminance levels do not fully address the problem of providing a visual environment conducive to safety and efficiency. Detecting the presence of a potential hazard is probably the most common and also the most critical visual task of ensuring safety (for example, the need for drivers to see pedestrians or other obstructions and the need for pedestrians to see slip, trip and fall hazards, etc.).

In such situations, it is frequently the contrast between the visual target and its background that is the most important determinant of reliable hazard detection. Bell (1982) states that the ratio of brightness between the task, the immediate background and the general background should be approximately 10:3:1. Best et al. (1981) state that the ratio between maximum and minimum light levels in any given work area should not exceed 5:1. This is because objects with the same luminance as their surroundings are less likely to be detected, regardless of light level. However, with more light the eye can see more detail, hence less contrast is required.

SHE

In mines, as the light falling on the task and the immediate background are virtually the same, there is in practical terms a trade-off between providing higher light levels and good visual contrast. For example, the use of reflective strips on workers' clothing, tape or paint markings on vehicles or obstructions may provide a more effective solution than the provision of additional or higher-intensity luminaires.

13.12.4 Visibility

Definition

Zones that mine workers need to see to perform their jobs efficiently and safely are generally referred to as '**visual attention areas**'**.**

It is important to distinguish between what *needs to be seen* from *what can be seen.* For example, "…Every year people are killed, equipment is damaged, cables are cut and roof supports are knocked down because the operators of mobile equipment cannot see, due to large blind areas created by the design of such equipment…", Lewis (1986). Poor sight lines are common to a large range of mobile machinery used underground. A study in the United States (US Department of Labour, 1980) concluded that approximately 36% of the fatalities involving mobile equipment in underground coal mines between 1972 and 1979 were directly or indirectly caused by improperly designed operator compartments.

Studies of ergonomics and human errors in the UK coal industry have shown that there is often little consideration given to ergonomics (in particular sight lines) in the design of mining equipment and machinery (Chan et al., 1985; Simpson et al., 1993).

A Health and Safety Executive report (HSE, 1992) concludes that the main area of concern identified with regard to ergonomic aspects of LHD design was poor driver vision. As a result, drivers often encounter problems in judging the sides and corners of a vehicle when manoeuvring, and seeing people and obstacles close to the vehicle. A sight line plot taken from this report is reproduced in Figure 13.11. The grey areas indicate areas where objects less than 1 m high are not visible from various driver's seat positions in two different vehicles, even when they are not loaded. The report also states that between 1986 and 1991, 50% of all major and fatal accidents involving LHDs included cases of pedestrians being run over or struck by moving vehicles, and drivers striking or becoming trapped against the sides of roadways.

A similar conclusion had been reached in an earlier report by the UK Mines Inspectorate (HSE, 1982), suggesting that in over 50% of major injuries caused by mobile machinery, the driver did not detect the presence of the victim.

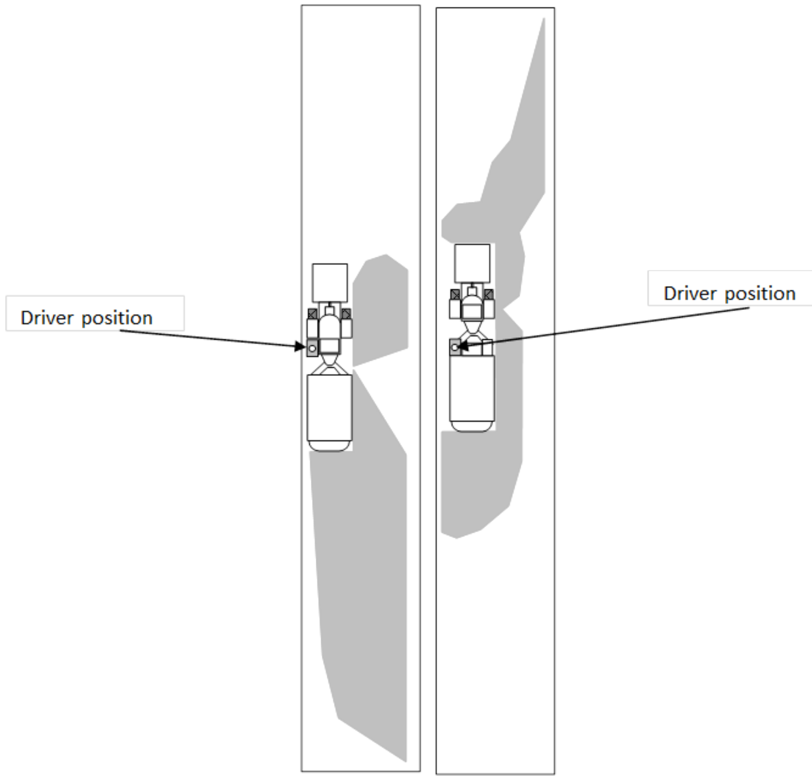


Figure 13.11 Sight line plot for LHD (HSE, 1992)

Similar problems of poor visibility from LHDs in South Africa were identified by van Graan & Schutte (1977). More recent research in South Africa clearly identified similar ergonomic limitations (sight line provision and poor illumination levels) in the design of a range of transport machinery as a prominent latent failure that predisposed towards a considerable number of significant human errors and, therefore, increased accident potential (Simpson et al., 1996).

NB

In order to ensure satisfactory visibility for operators of mining machinery, it is essential to know the areas which they will be required to see to enable safe operation. For mobile machines it is also important to have a means of assessing the operator's sight lines in relation to critical visual targets.

A variety of techniques can be used to evaluate sight lines and produce graphical representations of the results. A number of such techniques were reviewed by Mason et al. (1980) with a view to their use in the mining industry. These can be grouped into the following categories:

- Shadowgraph techniques, which utilise a suspended light at the driver's eye position to cast shadows onto gridded screens that represent the roadway sides. The shadows correspond to areas not visible to the driver in the position being considered, and are recorded photographically. This technique is recommended in ISO 5006-1 (1991) for recording operator sight lines and fields of visibility from surface earth-moving machinery.
- Panoramic photographic techniques in which photos are taken around the machine on surface with the camera at the position of the driver's eye.
- Graphical techniques where sight lines are plotted using projections from general arrangement drawings of the machine.
- Line of sight techniques using a ranging pole around a machine and physically measuring what an operator can and cannot see.

Developments in the field of computer graphics have enabled the construction of a range of more sophisticated computer-based visibility assessment techniques, such as virtual reality. Virtual reality allows the creation of a computer-generated three-dimensional visualisation of the working environment, the movement of all items of equipment and people within it, and their relative behaviour. The computer allows the user to roam around the virtual environment in real time, view it from any angle or position, and interact with any of the objects within.

As an example of the use of virtual reality, a project was undertaken in British coal mines using the technique to examine how operator vision could be improved on LHDs (Denby, 1996; McClarron et al., 1995). A virtual LHD was built with a control panel that allowed the user to drive the vehicle through a range of realistic roadway and lighting conditions, or walk around the underground environment within which the LHD was driven. As the users moved their heads or moved to a different location, a new view of what could be seen from the new position was generated. The model was used to undertake a detailed assessment of what different drivers were able to see directly and using visual aids such as mirrors and CCTV cameras when performing a typical work cycle.

Allowing the workforce to drive the virtual LHD also provided an effective means of raising awareness of the risks associated with working in the presence of such vehicles. In addition, developers of the technique see computer graphics and virtual reality as having important applications in the development of:

- Real-time training simulators
- Accident reconstruction
- Safety awareness training
- Safety planning
- Design and risk analysis.

All of these techniques establish the limits of operators' sight lines, and taking into account minimum requirements for visual attention areas to ensure safety and performance, operator training courses that address sight line restrictions could be developed. In addition, such techniques could aid in the formulation of other risk-control measures, including operational procedures. They could also assist design engineers to identify modifications to improve operator sight lines.

In Table 13.6 the illuminance standards for various countries are shown

Table 13.6 Mine illuminance standards in various countries

	Location and minimum lighting recommendation in lux						
Location	**Shafts**	**Machinery**	**Haulages**	**Loading**	**U/G offices**	**U/G workshops**	**Face**
Range of values	(15-160)	(10-80)	(2.5-21)	(15-160)	(60-270)	(20-400)	(2-15)
Country							
Australia (20-100)	20	20	-	20	100	-	-
Belgium (20-50)	20-50	25	10	20	-	-	-
Canada (20-270)	21-50	50	21	20	270	270	-
Czech Republic (5-20)	15	20	5	20	-	-	5
Germany (15-80)	30-40	40-80	15	40	-	-	-
Hungary (2-100)	40-100	20-50	2-10	40-60	-	20-50	10
Poland (2-100)	30-50	10-50	2-10	15-30	-	30-100	2
United Kingdom (2,5-100)	70	30	2.5	30	60	50-100	-
European Coal & Steel Community (5-90)	40-90	-	5-15	15-80	-	-	-
United States (15)	-	-	-	-	-	-	15
South Africa (20-400)	20-160	10 (at 20m)	20	160	-	400	-

Note: - indicates no official recommendation.

Chapter

14

NOISE

LEARNING OUTCOMES

After completion of the study theme, you should be able to:

- Describe the basic principles of noise
- Calculate Sound intensity, sound energy and sound pressure
- List frequency intervals
- Describe the effect of noise on a human being
- Describe what is meant by acceptable noise limits and noise intervals
- Describe various methods of noise control
- Measurement of noise levels.

REFERENCES

Mine Environmental Control (MEC) Certificate, Chamber of Mines, South Africa, Workbook 5.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

14.1 Noise Induced Hearing Loss (NIHL)

Objective

Noise induced hearing loss (NIHL) has been around the South African mining industry for quite some time now. Compensations paid to employees suffering from NIHL are millions of rand annually and still there are some companies and industries not acknowledging the seriousness of the situation. The purpose of this chapter is therefore to introduce the student to the various hazards that can cause NIHL and also to discuss an overview of preventative measures as well as calculations pertaining to noise/sound calculations.

There are basically two types of deafness or hearing loss, namely:

- Conductive
- Perceptive.

Conductive hearing loss is caused by a disorder in the outer and/or middle ear and prevents the normal amount of sound energy reaching the inner ear. This type of loss may be caused by excessive wax being formed in the auditory canal or by a hardening of the tissues at the foot of the stapes in the oval window. Most hearing impediments of this type are reversible.

Deafness caused by disorders to the inner ear and/or the auditory nerve is known as a **perceptive loss**. This type of loss is permanent and cannot be restored by surgical or other medical means.

As mentioned NIHL can either be conductive or perceptive. Hearing loss due to an explosion (acoustic trauma) is of the conductive type and the eardrum may be ruptured. However, this rarely happens and when it does the level of the sound wave which causes the damage is somewhere around 160-180 dB. Conductive hearing loss can be cured.

Persons exposed to excessive noise for prolonged periods suffer from perceptive loss in hearing. The main disorder occurs in the cochlea and is caused because the vibrations of the fluid in the ear become excessive. This should not be confused with nerve deafness or being completely deaf, which results from a disease of the auditory nerve. Refer to Figure 14.1 for a diagrammatic section of the ear.

14.1.1 New Milestones 2014 and Beyond

Elimination of noise induced hearing loss:

- **Quieting of equipment**: by December 2024, the total operational or process noise emitted by any equipment must not exceed a milestone sound pressure level of 107 dB(A)

NB

This milestone of the sound pressure levels will be verified by initiatives under the COE and MOSH and reviewed in 2016.

- **For the individual**: by December 2016, no employee's Standard Threshold Shift (STS) will exeed 25 dB from the baseline when average at 2000, 3000 and 4000 Hz in one or both ears
- Establish a multi-stakeholder team to consider different compensation systems.

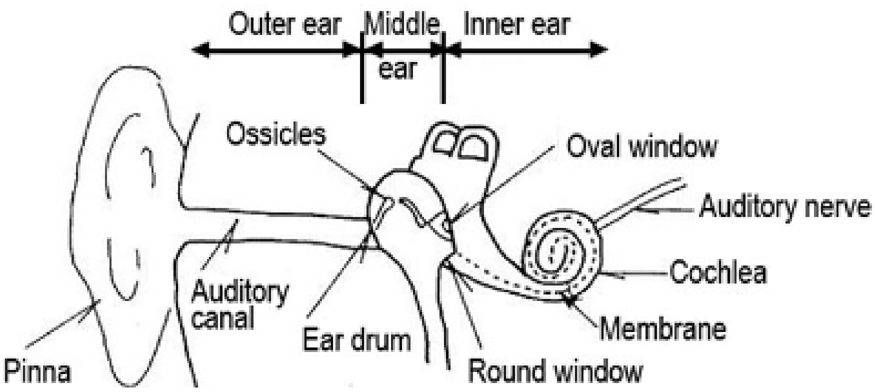


Figure 14.1 The human ear

Definitions

A brief exposure to excessive noise produces a temporary loss in hearing, but hearing returns after a short time if the worker is not exposed to that noise source any longer. This is known as the **temporary threshold shift** (TTS). However, with repeated or prolonged exposure the probability of recovery is reduced and permanent hearing loss can occur. This is known as the **permanent threshold shift** (PTS). Perceptive deafness is incurable but can be prevented. Figure 14.2 shows the relationship between the threshold shift and the noise frequency (this to be discussed in detail later in this chapter.

The main detrimental effects of excessive noise are that the ability to communicate verbally is affected and as mentioned before, hearing loss can occur. There are however other effects of excessive noise such as:

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- Harm to the nervous system
- Increased blood pressure
- Increase heart rate.

The above all cause fatigue. It is also a safety hazard, because danger signals cannot be heard when the noise level is excessive.

The factors causing hearing loss are:

- The noise level
- The frequency of the sound/noise as the ear is frequency sensitive
- The type of noise; as more damage is caused if the energy is concentrated at one frequency, than if it is spread over a band of frequencies
- The duration of the exposure.

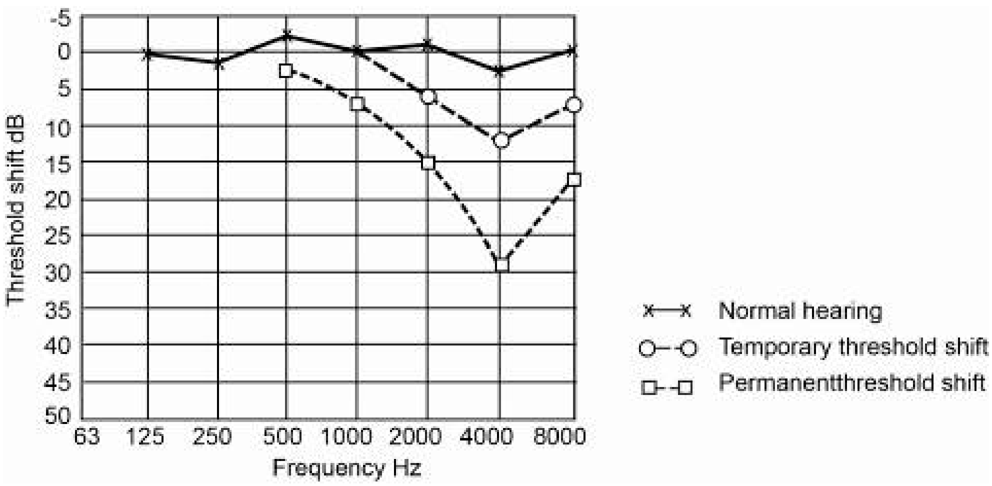

Figure 14.2 Relationship between the threshold shift and the noise frequency

ML

14.1.2 Reducing the Effects of Noise

There are basically three methods to prevent or minimise the effects of noise, namely:

- By controlling the noise through engineering control such as:
  - Insulation
  - Sound absorption in rooms
  - Sound absorption in columns
  - Vibration isolation
  - Control at the source.
- Through administrative measures, such as control of time exposed to noise
- Personal protective equipment.

14.2 Noise Related Definitions and Calculations

Definition

Animation

Click on the link to open

Noise can be defined as unwanted sound and can therefore be detrimental to health. Sound as light is a form of energy and is produced through a wave action similar to that in water. The wave carries energy because the water in the wave is in motion and therefore contains kinetic energy.

The water over which the wave travels remains calm and if we use a cork that floats on the water, it will rise and fall but move forward and backwards to remain on the same spot.

This is an indication that the water is not moving forward, but that kinetic energy is transferred from one mass of water to another.

Each wave therefore carries a definite amount of energy that remains with the wave as it is propagated.

The same energy transfer takes place when billiard balls that are tied together with small springs collide with one another. The energy can only be transferred from one ball to another if there is a collision. The compression of the spring between the two balls has a result that the energy will be transferred to the next ball in a wave fashion and in this way makes the movement of the second ball possible.

Air has both mass and elasticity. At standard temperature and pressure, the density of air is 1.2 kg/m^3. The elasticity or compressibility of air can be demonstrated by the compression of air in a cylinder. As the air is compressed, it offers resistance to the pressure, the same as what happened to the balls mentioned in the example above. Because of these two properties it is possible for sound to be propagated through air. This is the reason why sound cannot be propagated through a vacuum.

14.2.1 Fundamental Aspects of Sound

Sound waves have certain characteristics and can be noted as follows:

- Wavelength
- Cycle
- Frequency
- Velocity/speed
- Intensity
- Sound pressure
- Sound energy level.

14.2.1.1 Wavelength

Wavelength can be defined as the distance (d) between the regions of high or of low-pressure areas and is shown in Figure 14.3.

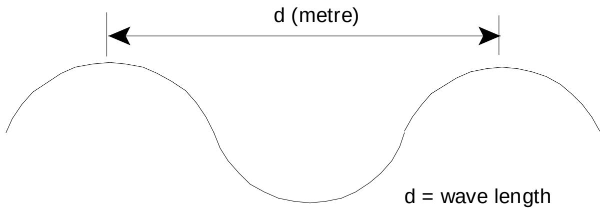

Figure 14.3 Wavelength description

14.2.1.2 Cycle

The travel from the peak position on one wave to the peak position on the next wave is called a **cycle**. One complete vibration of an object therefore corresponds to one complete cycle of a pressure change as shown in Figure 14.4.

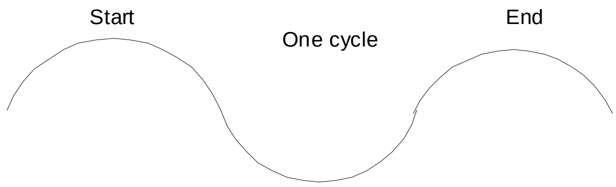

Figure 14.4 A wave cycle

14.2.1.3 Frequency

Definition

Frequency is another property of sound and is described as the number of waves which pass a fixed observation point per unit time, namely second. Obviously, frequency is therefore the number of cycles per second and the unit thereof is Hertz (Hz).

14.2.1.4 Speed/Velocity

Rule

It is obvious that from the description of sound generation that it will take some time for the sound pressure wave to travel between points. In air, at a temperature of 20ºC, the speed of sound is 344 m/s. The speed at which sound travels depends upon factors such as the elasticity, density and temperature of the medium in which it travels. The approximate speeds of sound in various materials vary and are given in the Table 14.1.

Table 14.1 The approximate speeds of sound in various materials

Medium	Average speed of sound (m/s)
Air (1.2 kg/m^3)	344
Lead	1 220
Water	1 410
Cement	3 400
Wood	3 400
Rock	5 000
Steel	5 200

Rule

The ear is only sensitive for frequencies between 20 and 16 000 Hz and the relationship between speed (c), wavelength (λ) and the frequency (f) is given in the following Equation:

$$\lambda = \frac{c}{f} \qquad (14.1)$$

14.2.1.5 Intensity

Sound has already been defined as a succession of pressure waves moving away from a source; in our previous explanation this source was an air piston and billiard balls. Sound waves transmit pressure and since pressure is related to force and energy it is evident that the sound waves transmit energy. In a sound field consisting of waves emanating from a source it is usual to refer to the intensity of a sound at a point.

Definition

Intensity is the amount of energy passing through a unit area per unit time, thus the units of intensity are watts per metre square (W/m^2). There is a relationship between intensity and sound pressure, intensity (I) being proportional to the square of the pressure and shown in the following Equation for sound intensity, where ρ is the density of the medium through which the sound travels and p is the sound pressure in Pascal. Obviously, if the pressure is double the sound intensity will increase four-fold.

$$I = \frac{p^2}{\rho c} \quad \text{...} \quad (14.2)$$

14.2.1.6 Sound Pressure

Definition

Pressure is defined as force per unit area, the unit of pressure being N/m^2. **Sound pressure** is the instantaneous value of the amplitude of the pressure wave. The pressure change follows a cyclic wave, varying from a positive peak value, through zero, to a negative peak value and back again as shown in Figure 14.5.

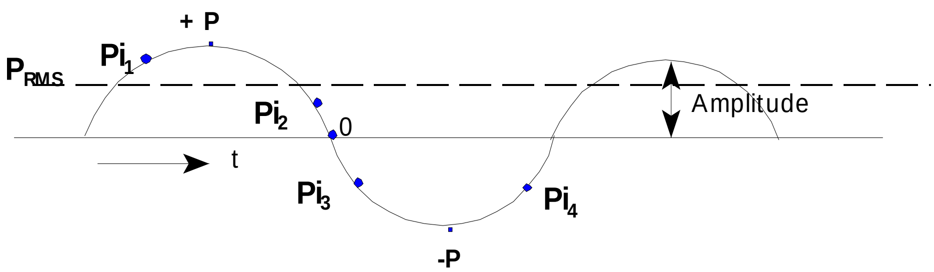

Figure 14.5 Details pertaining to sound pressure

The measure of the sound pressure is the root mean square (RMS) value of the pressure P_i and can be represented as follows:

$$P_{RMS} = \frac{Pi_1^2 + Pi_2^2 + Pi_3^2 + Pi_n^2}{n} \quad \text{......................................} \quad (14.3)$$

This is the pressure measured by the sound level meter. Since sound intensity is proportional to the square of sound pressure, the **sound pressure level (SPL)** can be derived from the Equation shown below:

$$\text{SPL} = \log_{10} \frac{(\text{sound pressure})^2}{(\text{reference pressure})^2} \quad \text{in Bel} \ldots\ldots\ldots\ldots (14.4)$$

The reference pressure is the sound pressure at the threshold of hearing, the value thereof is 2 x 10^{-5} Pa and the frequency thereof is 2 000 Hz. The highest sound pressure that can be heard without pain is 200 Pa. To avoid the excessive use of decimals, it is normally multiplied by 10 and the resultant unit for sound is the decibel as shown in the formula below:

$$\text{SPL} = 10\log_{10} \frac{(\text{sound pressure})^2}{(2 \times 10^{-5})^2} \quad \text{in dB} \ldots\ldots\ldots\ldots\ldots (14.5)$$

NB

By referring to Table 14.2 it can be noted that the sound pressure level of a low voice is 50 dB, whereas for a typical office it is 60 dB. This in fact means that the sound in the office is 10 times greater than that of a low voice, not 20% greater. This is evident when you note that the sound intensity ratios for the 50 and 60 dB levels are 10^5 and 10^6 respectively.

14.2.1.7 Sound Energy Level (SWL)

Definition

The energy associated with sound is known as a **sound power level** and can be simply defined as the nett amount of sound power emitted by a source (such as a fan or motor) to the surrounding medium in the form of sound waves.

If there are no losses in the medium, all of this radiated power must pass through any surface that encloses the source. The larger the enclosing surface, the less sound power per unit area (W/m^2) will pass through any element of the surface. The total sound power is the sum of the products of the intensity through each incremental area. The **sound power level** can be derived as follows:

$$\begin{aligned} \text{SWL} &= 10\log_{10} \frac{\text{sound power(Watt)}}{\text{reference power}} \\ &= 10\log_{10} \frac{\text{sound power}}{10^{-12}} \quad \text{in dB} \end{aligned} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots (14.6)$$

Table 14.2 shows the sound pressure levels related to different working environments.

NB

An important point that needs to be noted now is the fact that the energy that is radiated from a sound/noise source stays the same, but that the pressure related to that sound/noise source is related to the distance from and/or in which way the source has been confined or not. The closer you are to the source the higher the pressure affect on the ear and the further away the lesser the affect.

Table 14.2 Sound pressure levels related to different working environments

Sound pressure (Pa)	**Sound intensity (W/m^2)**	**Sound intensity Ratio Reference (10^{-12} W/m^2)**	**Sound pressure level SPL (dB)**	**Sound source**	**Subjective description**
$2x10^{-5}$	10^{-12}	$10^0=1$	0		Threshold
$2x10^{-4}$	10^{-10}	10^2	20	Broadcast studio low voice	Very quiet
0.0063	10^{-7}	10^5	50		
0.02	10^{-6}	10^6	60	Typical office Conversation	
0.063	10^{-5}	10^7	70		
0.2	10^{-4}	10^8	80	Noisy street traffic Factory	Noisy
2	10^{-2}	10^{10}	100		
20	1	10^{12}	120	Rock drill Pneumatic chipping	Very noisy
63	10	10^{13}	130		
200	100	10^{14}	140	Jet engine at 30m	Intolerable

WORKED EXAMPLE 1

A fan is located in the centre of a large flat, free of obstructions, area. Measurements taken on the surface of an imaginary hemisphere with a radius of 5 m, centred at the fan, indicated a sound pressure of 0.8 Pa. Calculate the sound power level (SWL) on the surface of the hemisphere.

Answer

$$\text{SPL} = 10 \log_{10} \frac{(\text{sound pressure})^2}{(2 \times 10^{-5})^2}$$

$$= 10 \log_{10} \frac{(0.8)^2}{(2 \times 10^{-5})^2}$$

$$= 10 \times 9.2$$

$$= 92 \text{ dB}$$

$$\text{Intensity} = \frac{(\text{sound pressure})^2}{\rho \times c}$$

$$= \frac{(0.8)^2}{1.2 \times 344}$$

$$= 0.00155\ \text{W/m}^2$$

$$\text{The surface is of the hemisphere of radius 5 m} = 2\,\pi\, r^2$$

$$= 2\,\pi\, 5^2$$

$$= 157\ \text{m}^2$$

$$\text{The acoustic power is therefore} = \text{Intensity (I)} \times \text{area}$$

$$= 0.00155 \times 157$$

$$= 0.2435\ \text{Watt}$$

Now:

$$\text{SWL} = 10 \log_{10} \frac{\text{Power}}{10^{-12}}$$

$$= 10 \log_{10} \frac{0.2435}{1 \times 10^{-12}}$$

$$= 10 \times 11.39$$

$$= 114\ \text{dB}$$

NB

Note: Provided that the duty of the fan does not change materially, this sound power level **can be assumed to be constant**.

WORKED EXAMPLE 2

The fan of Worked Example 1 is moved and installed at the bottom of an outside wall of a high building. Calculate the sound pressure level 2.5 m and 5 m from the fan assuming that there are no other obstructions and that the wall and ground do not absorb any sound energy.

Answer

An important part of the facts leading to the answer is that the **acoustic power generated** by the fan **remains constant** at 0.2435 W, unless otherwise stated. The wall of the building and the ground constrain the propagation of the sound waves to a quarter sphere.

(i) For a distance of 2.5 m from the fan the following applies:

Surface area of the ¼ sphere of radius 5 m $= \pi r^2$

$= \pi\ 2.5^2$

$= 19.6\ m^2$

$$\text{Intensity} = \frac{\text{acoustic power}}{\text{area}}$$

$$= \frac{0.2435}{19.6}$$

$$= 0.0124\ W/m^2$$

The pressure at this point can now be determined as follows:

$$\text{Intensity} = \frac{(\text{sound pressure})^2}{\rho \times c}$$

$$\text{pressure} = \sqrt{I\ \rho\ c}$$

$$= \sqrt{0.0124 \times 1.2 \times 344}$$

$$= 2.26\ Pa$$

$$\text{SPL} = 10 \log_{10} \frac{(\text{sound pressure})^2}{(2 \times 10^{-5})^2}$$

$$= 10 \log_{10} \frac{(2.26)^2}{(2 \times 10^{-5})^2}$$

$$= 10 \times 10.11$$

$$= 101\ dB$$

(ii) For a distance of 5 m from the fan the following applies:

Surface area of the ¼ sphere of radius 5 m $= \pi r^2$

$= \pi\ 5^2$

$= 78.5\ m^2$

$$\text{Intensity} = \frac{\text{acoustic power}}{\text{area}}$$

$$= \frac{0.2435}{78.5}$$

$$= 0.0031\ W/m^2$$

The pressure at this point can now be determined as follows:

$$\text{Intensity} = \frac{(\text{sound pressure})^2}{\rho \times c}$$

$$\text{pressure} = \sqrt{I \quad \rho c}$$

$$= \sqrt{0.0031 \times 1.2 \times 344}$$

$$= 1.13 \text{ Pa}$$

$$\text{SPL} = 10 \log_{10} \frac{(\text{sound pressure})^2}{(2 \times 10^{-5})^2}$$

$$= 10 \log_{10} \frac{(1.13)^2}{(2 \times 10^{-5})^2}$$

$$= 10 \times 9.5$$

$$= 95 \text{ dB}$$

NB

This example illustrates the important point mentioned before. The sound power level of a piece of equipment is unlikely to be changed by changing the location, but the sound pressure level depends upon the geometry of the new environment.

WORKED EXAMPLE 3

In Worked Example 1, a second identical fan is installed next to the original fan. Calculate the sound pressure level at 5 m and the sound power level of both fans.

Answer

The total acoustic power will now be double (assuming that the duties of the two fans do not change).

$$\text{Acoustic power for two fans} = 2 \times 0.2435$$

$$= 0.487 \text{ Watt}$$

$$\text{SWL} = 10 \log_{10} \frac{\text{Power}}{10^{-12}}$$

$$= 10 \log_{10} \frac{0.487}{1 \times 10^{-12}}$$

$$= 10 \times 11.69$$

$$= 117 \text{ dB}$$

In Worked Example 1 the area of the hemisphere was calculated as 157 m^2. The sound intensity can be determined as follows:

$$\text{Intensity} = \frac{\text{acoustic power}}{\text{area}}$$
$$= \frac{0.487}{157}$$
$$= 0.0031\,\text{W/m}^2$$

The pressure at this point can now be determined as follows:

$$\text{Intensity} = \frac{(\text{sound pressure})^2}{\rho \times c}$$

$$\text{pressure} = \sqrt{I \quad \rho c}$$
$$= \sqrt{0.0031 \times 1.2 \times 344}$$
$$= 1.13\ \text{Pa}$$

$$\text{SPL} = 10\log_{10} \frac{(\text{sound pressure})^2}{(2 \times 10^{-5})^2}$$

$$= 10\log_{10} \frac{(1.13)^2}{(2 \times 10^{-5})^2}$$
$$= 10 \times 9.5$$
$$= 95\ \text{dB}$$

Both the SPL (95 – 92) and SWL (117 – 114) only increased by 3 dB as a result of the 2nd fan being installed.

WORKED EXAMPLE 4

If the second fan of Worked Example 3 produces a sound pressure level of 86 dB at 5 m when operating by itself, calculate the combined sound pressure level for both fans. Footnotes used are ‘1’ for fan one and ‘2’ for fan two.

Answer

$$SPL_1 = 10\log_{10}\frac{p_1^2}{(2 \times 10^{-5})^2} = 92 \text{ dB}$$

$$\text{therefore} \quad \log_{10}\frac{p_1^2}{(2 \times 10^{-5})^2} = 9.2 \quad \text{and}$$

$$\frac{p_1^2}{(2 \times 10^{-5})^2} = 10^{9.2}$$

$$SPL_2 = 10\log_{10}\frac{p_2^2}{(2 \times 10^{-5})^2} = 86 \text{ dB}$$

$$\text{therefore} \quad \log_{10}\frac{p_2^2}{(2 \times 10^{-5})^2} = 8.6 \quad \text{and}$$

$$\frac{p_2^2}{(2 \times 10^{-5})^2} = 10^{8.6}$$

$$\begin{aligned} SPL_{(1+2)} &= 10\log_{10}\left(\frac{p_1^2}{(2 \times 10^{-5})^2} + \frac{p_2^2}{(2 \times 10^{-5})^2}\right) \\ &= 10\log_{10}(10^{9.2} + 10^{8.6}) \\ &= 10\log_{10}(1.983 \times 10^{9}) \\ &= 10 \times 9.3 \\ &= 93 \text{ dB} \end{aligned}$$

NB

Note: Now the second fan only increased the overall sound pressure level by 1 dB, (93 - 92), although it produced 6 dB less than the first fan.

14.2.1.8 Additional Methods of Determining Sound Pressure Levels

There are other methods to determine the sound pressure levels and will now be discussed in detail. Figure 14.6 shows a graph of the amount of dB's to be added to the highest noise level (SPL) for a specific difference between two noise levels.

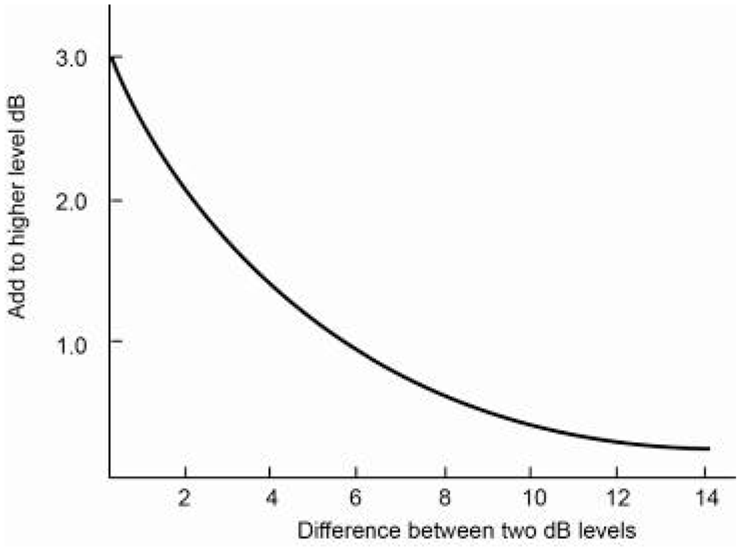

Figure 14.6 Determining resultant noise level

Table 14.3 shows a detailed comparison between the difference in noise levels and the amount of dB's to be added to the highest noise level when they are being experienced simultaneously. The Worked Examples that follow will show the procedure quite clearly.

Table 14.3

Numeric difference between 2 dB levels	Amount to be added to the highest dB level
0	3.0
1	2.5
2	2.0
3	2.0
4	1.5
5	1.0
6	1.0
7	1.0
8	0.5
9	0.5
10 or more	0

WORKED EXAMPLE 5

What is the combined sound pressure level of 3 different sound sources, namely 70, 76, and 85 dB respectively? Make use of the respective table or the graph mentioned before.

NB

Answer

An important aspect to remember in answering this question is to start with the **difference between the two highest noise levels**, **not** the highest and the lowest.

The difference between the two highest levels

= (85 – 76) dB

= 9 dB

By using the table or the graph, the amount to be **added to the highest noise level** is 0.5 dB. This must now be added to the highest value of the two noise levels used to determine the difference. The combined value is therefore:

Combined noise level = (85 + 0.5) dB

= 85.5 dB

The procedure is now repeated by using the 'new' highest level and the remaining low noise level.

The difference between these two levels

= (85.5 – 70) dB

= 15.5 dB

By using the table or the graph, the amount to be **added to the highest noise level** is 0.0 dB. This must now be added to the highest value of the two noise levels used to determine the difference. The combined value is therefore:

Combined noise level = (85.5 + 0.0) dB

= 85.5 dB

Note: The combined level for these three noise sources has been determined as 85.5 dB and that the third noise source did not increase the SPL at all, this is an important point to remember when combinations of noise sources are to be calculated.

14.3 Weighting Networks and Loudness Curves

It was mentioned earlier that the ear is frequency sensitive. In fact it is less sensitive at very low and very high frequencies. The degree of sensitivity in the very high and very low frequencies also depends on the actual sound level. Using a frequency of 1 000 Hz as a base, the sound pressure levels at which sound levels are perceived to be the same are shown in Figure 14.7.

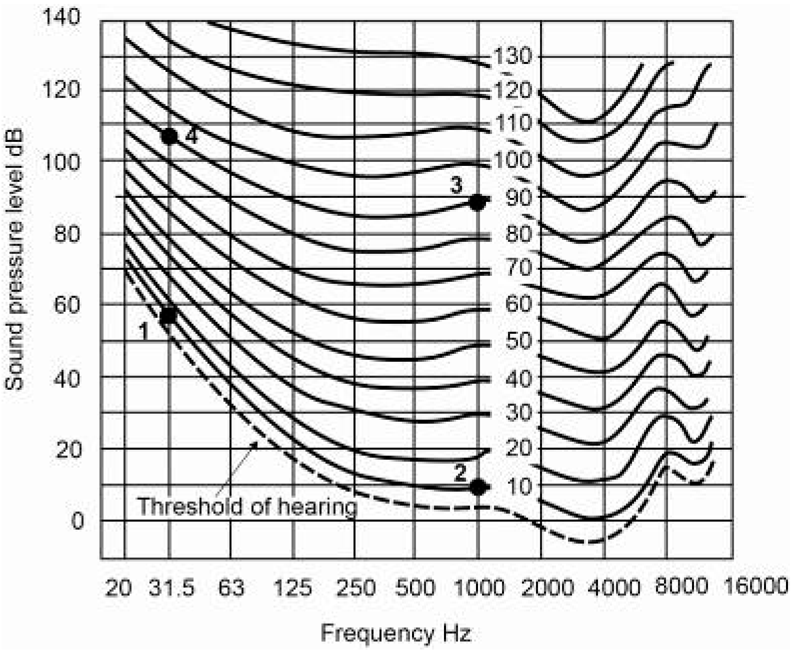

Figure 14.7 Equal loudness contours

The lowest frequency that can be heard by a human being is 20 Hz and the highest varies with conditions such as age and health. Normally the maximum frequency is 16 kHz. On each of the curves shown in Figure 14.7 there are a number of units called phons. The **phon** is the unit for loudness and the measurement of each curve in phons is defined as the value of the sound pressure level of that curve at 1 000 Hz. The dashed curved shows the threshold of hearing. It can be seen that the SPL value at 2 000 Hz is zero and as we already know, is the value of the sound pressure at this point 2 x 10^{-5}.

From this curve a few interesting observations can be made. For sounds with a very low amplitude, (loudness) there is obvious frequency sensitivity. A sound at 30 Hz must for example be 50 dB higher as a sound at 1 000 Hz to be classified as equally loud (points 1 and 2 on the graph). This obvious insensitivity for low frequency sound is dependent on the amplitude (loudness).

If the SPL is 90 dB at 1 000 Hz, then the loudness level at a frequency of 30 Hz and SPL of 108 dB will be perceived as the same (points 3 and 4 on the graph).

NB

The point that is made here is that the ear is not only responding to the frequency, but also to the sound pressure level. It (the ear), weights level of sound according to its frequency content and gives it a certain loudness.

This means that if we want to know how a person judges, or is affected by noise, it is necessary to combine sound pressure level and frequency content into a subjective unit.

14.3.1 Weighting Curves

It was mentioned before that a person judges the loudness of sound according to the sound pressure level and the frequency. In order to combine these two units in a single meaningful index, a method known as weighting curves has been adopted. This principle is incorporated in most, if not all, sound level meters. The purpose of these weighting curves is to reflect in some manner the sensitivity of the ear. In other words, they have to reduce the effect of (strength) of the low frequency components before adding them to the high frequency components.

Definition

Three weighting curves known as '**A**' (suitable for sound levels up to and including 55 dB), '**B**' (suitable for sound levels between 55 and 85 dB) and '**C**' (suitable for sound levels over 85 dB) curves have been accepted internationally. Sound levels measured by means of these curves are expressed as dB (A), dB (B) or dB (C), depending on which curve was used. The shapes of the various curves are shown in Figure 14.8.

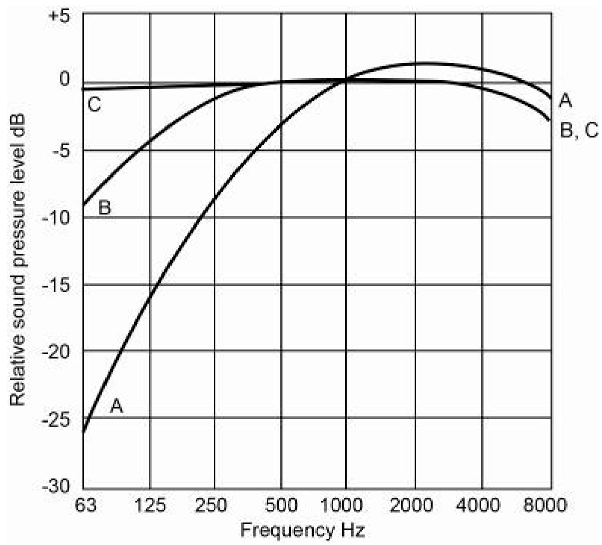

Figure 14.8 Different weighting curves

By comparing these weighting curves with the equal loudness curves described before, it can be seen that there is a certain resemblance between the equal loudness curves and the 'inverted' 'A' curve.

It is because of this resemblance that national and international standards for noise measurements and evaluation recommend that the 'A' weighting curve be used. Weighting curves therefore makes it possible that a single figure can be used to describe the noise level in a specific environment. However, they give no indication of the frequency of the higher noise levels that needs to be controlled.

NB

It must be noted that when excessive noise is measured and noise control measures are required, a more sophisticated set of measurements must be made and these must include sound pressure levels and frequency analysis.

14.4 Acceptable Noise Levels

It is obvious that if persons are exposed to excessive noise for a long period of time during the normal working shift, the risk of hearing damage is considerably greater than if they are exposed for a shorter period (as discussed before). Acceptable noise levels are dependant on the criteria that is used for assessment.

Criteria

The basic criteria, that is used by the environmental engineer is the avoidance of hearing damage.

As with dust and pneumoconiosis, hearing damage depends on:

- The level of noise
- The duration of exposure
- Individual susceptibility.

If the hearing impairment is defined as a hearing loss in excess of 25 dB averaged over the range of frequencies of 500 to 2 000 Hz, then the levels and duration of exposure given in Table 14.4 should not be exceeded. Because of the difference in susceptibility of various individuals it is important to use audiometric testing often if people are exposed at or near these levels.

Table 14.4 The levels and duration of exposure

Duration of exposure (hours/day)	Sound level dB (A)
8	85
4	90
2	95
1	100
0.5	105
0.25	110
0.125	115

NB

A second aspect that is important to mention is the noise levels that influence signal identification and hearing of speech. Here it may be possible that lower levels than that which will cause hearing damage might be needed. Noise that interferes with speech is normally between 500 and 2 000 Hz, levels that are below hearing damage levels. As a basic guideline it can be noted that communication at a distance of 0.5 m and higher than 80 dB environment noise will be difficult. For every doubling of distance, the noise level will have to be reduced by a factor of 5 to sustain the same level of communication.

Rule

Definition

Another aspect to consider is the effect of impact noise and the exposure to it. In this case noise levels vary with time, and where the variations in noise level occur at intervals of more than one second, noise is known as **impact** rather than **continuous noise**. Examples of this kind of noise are hammer work and where breaking of rock on a grizzly by an impact breaker needs to be done.

Examples

Figure 14.9 illustrates the suggested maximum number of impacts that would be permitted for different peak sound pressure levels.

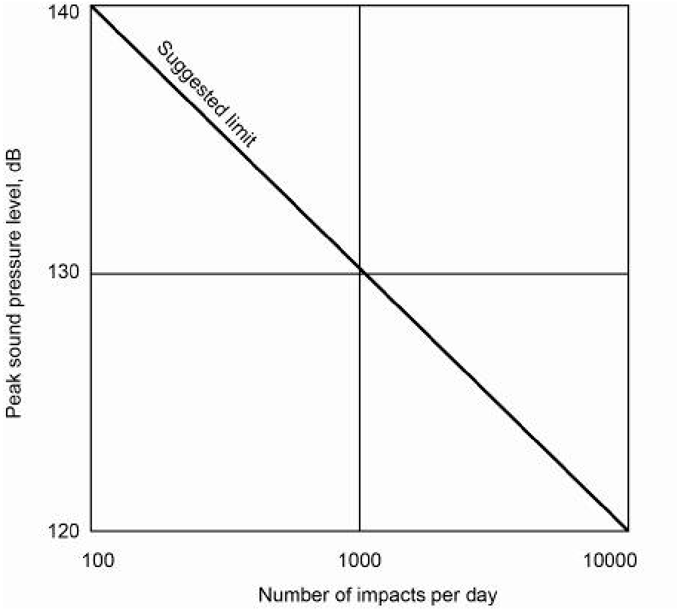

Figure 14.9 Suggested limits for impact noise

Criteria

Another criteria that the mine environmental engineer may be involved with, is comfort levels. This may arise in offices where air conditioning systems operate, or may involve the effects of the noise produced by the mining or metallurgical operations on nearby residential areas. In offices noise ratings between 30 and 50 are acceptable depending on the work being carried out.

Using basic criteria of noise ratings of 25 and 30 dB for inside sleeping and living rooms respectively, these can be modified by up to 30 dB, depending on the duration of the noise. As an example 5 dB may be added if the noise only occurs for 25% of the time. Whereas, 25 dB may be added if, it only occurs for 0.1% of the time (approximately four second in an hour). The type of noise, the time of exposure (day or night) and the type of residential area will also affect the acceptable levels.

14.4.1 Combining Sound Levels (Equivalent Noise Levels (L_{eq}))

Where the exposure to noise varies both in level and duration, an equivalent continuous sound level can be calculated. For an 8-hour workday the fractional exposure factor at each sound level for each time period can be calculated by using different scenarios.

14.4.1.1 L_{eq} for a Steady Noise Level

$$L_{eq} = L_A + C_i$$

Rule

Where: L_{eq} = Equivalent noise level dB (A)

L_A = Measured level of steady noise, dB (A)

C_i = Impulse correction factor which is +10 dB where the noise is of a repetitive nature (example hammering) or when it occurs in single bursts, and +0 dB in all other cases.

14.4.1.2 L_{eq} for a Fluctuating Noise Level

L_{eq} = $L_{A\,(av)} + C_i$

Where : L_{eq} = Equivalent noise level dB(A)

$$L_{eq(av)} = 10 \operatorname{Log}_{10}\left[\frac{1}{100} f_i\, 10^{\frac{L_{Ai}}{10}}\right] \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (14.7)$$

L_{Ai} = Noise level at the mid-point of the I-class, dB (A)

f_i = Duration of the I-class sound level exposure expressed as a percentage of the total analysis time (normalised to a 40 hour total period)

Another Equation to be used:

$$L_{eq} = 10 \operatorname{Log}_{10}\left[f_1 \text{ antilog } \frac{L_1}{10} + f_2 \text{ antilog } \frac{L_2}{10} + \ldots\ldots + f_n \text{ antilog } \frac{L_n}{10} + c_i\right] \ldots\ldots \quad (14.8)$$

Where:

f_1 to f_n = the ratios in relation to 40 hours of duration of exposure to the sound levels L_1 to L_n.

Another formula to be used:

$$f = \frac{t}{8} \text{ antilog } [0.1 \ (L - 90)]$$

$$F = \sum f$$

$$L_{eq} = \frac{\operatorname{Log} F}{0.1} + 90$$

Where:

f = Fractional exposure factor

t = Time period of exposure (hours)

L = Sound level (dB)

F = Total exposure factor

L_{eq} = Equivalent continuous sound level.

NB

Acceptable equivalent continuous sound levels are normally given in dB (A). It was previously noted that the 'C' scale weighting network represents the sensitivity of the ear to sound levels in excess of 85 dB. This appears to be in conflict with the use of the 'A' weighting scale in hearing conservation work. The almost universal use of the 'A' scale weighting stems from the desire to simplify noise measure, particularly when taking into account exposure factors. It was felt that any error involved in using this simplification would be within normal measurement tolerances.

A graph is also available in which the equivalent noise levels are plotted against an exposure factor (D). The Equation for determining the exposure factor is as follows:

$$D = \frac{C_1}{T_1} + \frac{C_2}{T_2} + \ldots\ldots + \frac{C_n}{T_n} \qquad (14.9)$$

Where:

C_1 to C_n = Total time of exposure at noise levels L_1 to L_n

T_1 to T_n = Total time of exposure permitted at noise levels L_1 to L_n

Table 14.5 and Figure 14.10, shows the various permitted values of T for various noise levels and the values of L_{eq} versus exposure factors, respectively.

Table 14.5 Permitted hours for specific noise levels

Allowable exposure time (T) (hours/day)[hours/week]	Noise level dB(A)
8 [40]	85
6 [30]	86
4 [20]	88
3 [15]	89
2 [10]	91
1.5 [7.5]	92
1 [5]	94
0.75 [3.75]	95
0.5 [2.5]	97
0.25 or less [1.25]	100

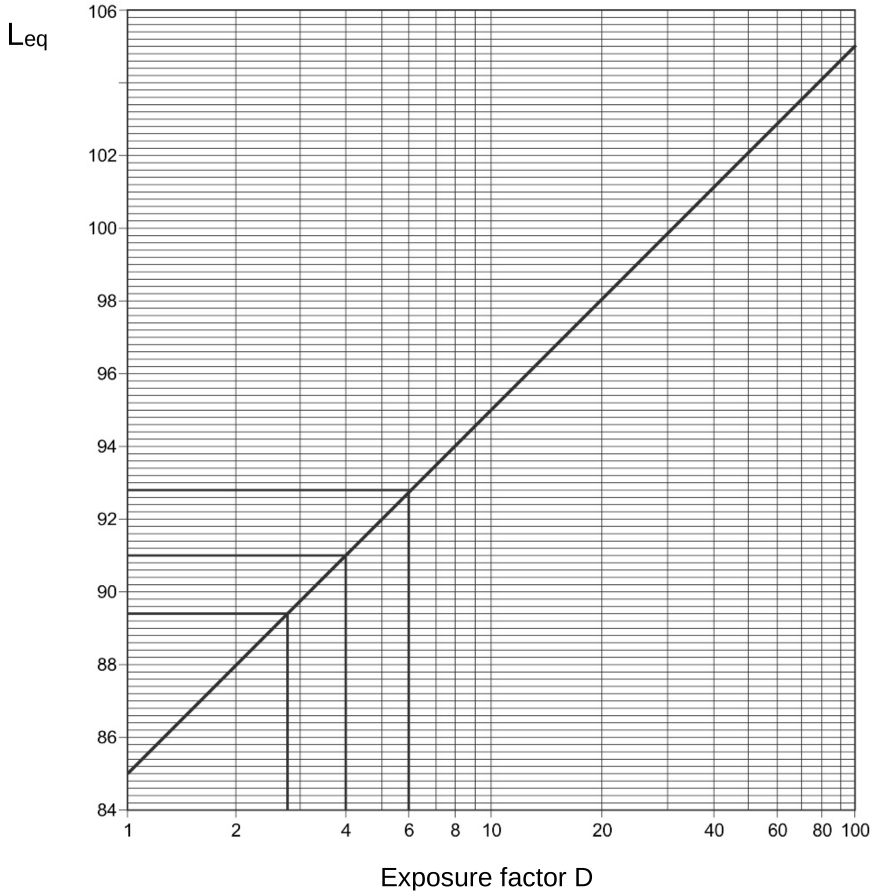

Figure 14.10 Equivalent noise level L_{eq} versus exposure factor

WORKED EXAMPLE 6

A worker is exposed to the following noise levels in a forty-hour work week:

- 85 dB (A) for 4 hours
- 86 dB (A) for 20 hours
- 88 dB (A) for 10 hours
- 94 dB (A) for 6 hours.

Calculate the equivalent noise level (L_{eq}) that this worker is exposed to.

Answer

C_1, C_2, C_3 and C_4 are equal to 4, 20, 10 and 6 hours respectively. By using the values from Table 14.5, the following values for T (allowable hours of exposure per week) can be obtained:

- T1 = 40 hours per week
- T2 = 30 hours per week
- T3 = 20 hours per week
- T4 = 5 hours per week.

The exposure factor must now be determined and can be done as follows:

$$D = \frac{C_1}{T_1} + \frac{C_2}{T_2} + ... + \frac{C_n}{T_n}$$

$$= \frac{4}{40} + \frac{20}{30} + \frac{10}{20} + \frac{6}{5}$$

$$= 2.47$$

From Figure 14.10, the equivalent noise level can be read off as 98 dB. As the maximum allowable noise level is 85 dB, the worker is exposed to excessive noise. From the above it is also obvious that if any of the C/T values is greater than 1, that the worker will be exposed to excessive noise. In other words an exposure factor, D, value > 1 means exposure to excessive noise.

The equivalent noise level could also have been determined mathematically and is normally more correct. There is also a nomogram available which can also be used to determine the values for T, which would then be used as before to determine the exposure factor and subsequent equivalent noise level. The nomogram is shown in Figure 14.11.

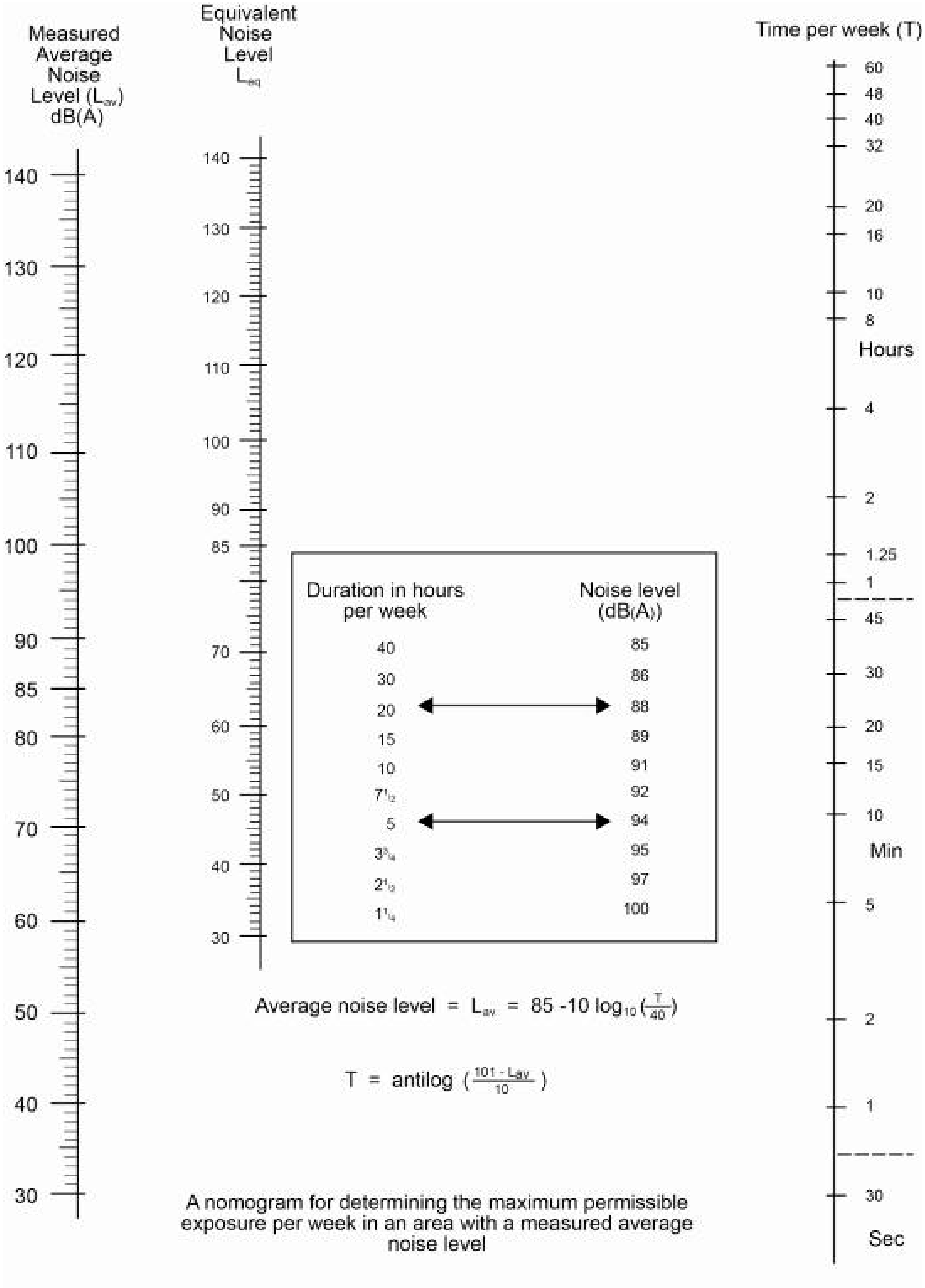

Figure 14.11 Nomogram for determining T levels

14.5 Sound Measuring Instruments

14.5.1 Sound Level Meters

Self study

There are various sound level measuring instruments which are discussed in detail on page 758-760 of VOEE and should be dealt with as self study.

14.5.2 Frequency Analysers

Definition

Sound frequency analysers are usually used in conjunction with sound level meters to form a single measuring unit. Basically a **frequency analyser** does exactly what its name implies. By adjusting the instrument one can measure the sound pressure level at various pre-selected frequencies, which would give you an indication of the frequency that ranges where the machine is producing the most noise. Figure 14.12 shows how the various measured levels at the different frequencies can be plotted. By connecting the various points measured on the graph one can now form a picture of what the frequency distribution looks like.

The machine surveyed here produces its maximum noise in the 500 to 3 000 Hz range (peaking at 90 dB at 1 000 to 2 000 Hz). This information can now be used to select the correct type and thickness of sound damping material to reduce the noise level if so required.

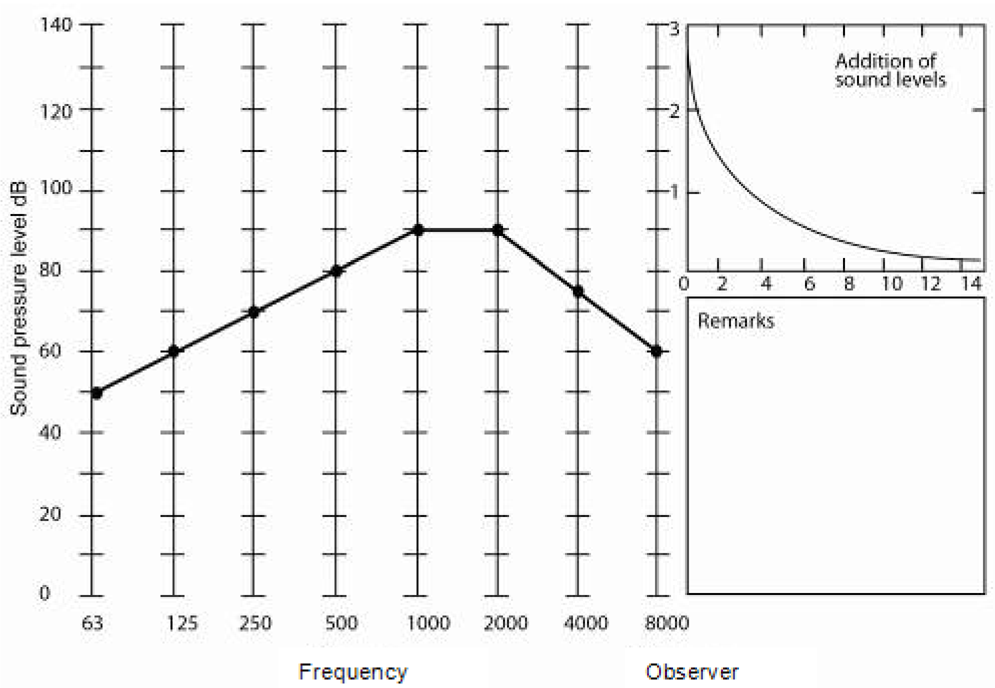

Figure 14.12 Frequency analyser result

14.6 Testing Noise Producing Appliances

In order to determine expected noise levels underground or elsewhere it may be necessary to test a machine before installation. Alternatively, when selecting machinery it is necessary to enquire whether the level of noise generated by it has already been measured by the manufacturer and then to use their results – if any – to predict noise levels elsewhere.

NB

The sound pressure level of a machine depends on the environment in which it is operating. For example, the level of noise generated by a fan in a large workshop will be different from that generated when it is operating in a small development end underground. Because of this aspect one cannot assume that the sound levels at one place will be the same as those generated at another. Fortunately however, sound power levels remain constant no matter where a machine is operating, assuming of course that the machine does not undergo any physical or mechanical changes. In the case of fans the duty of the fan must also not be different. This fact enables sound levels to be predicted for different environments. This is a procedure similar to correcting fan characteristic curves for speed or density changes.

Testing noise-producing appliances generally entails measuring sound pressure levels and then calculating sound power levels. There are many ways of doing this. Four methods will be mentioned but only one, namely duct testing (ventilation pipe column attached to fan), will be discussed in detail.

- Free field testing
- Reverberant field testing
- Semi-reverberant field testing
- Duct testing.

With duct testing all the energy is constrained as it passes along a rigid duct. This method is a combination of the free-field end reverberant methods and is particularly suitable for testing fans. In fact, many existing fan test sites would require only minor modifications to be acceptable for this purpose.

The fan is positioned at the intake to the duct and a sound measuring station is installed some distance along the duct. Sound pressure levels and frequency measurements are made at a number of points at this station and the average readings obtained.

The power level is then obtained by using the following formula:

$$SWL = SPL + 10 \,\text{Log}\, A$$

Where:

SWL	=	Sound power level (dB)
SPL	=	Sound pressure level (dB)
A	=	Cross sectional area of the duct at the measuring station (m^2)

When measuring is carried out using this method, a windshield should be fitted to the microphone to prevent aerodynamic pressure fluctuations at the microphone diaphragm.

It is also necessary to prevent reflected energy from giving a high reading and therefore an acoustical expander (a type of evaseè) and a small chamber, known as an ‘anechoic termination’ is fitted after the measuring station. Figure 14.13 shows the detail pertaining to such a noise testing arrangement for fans.

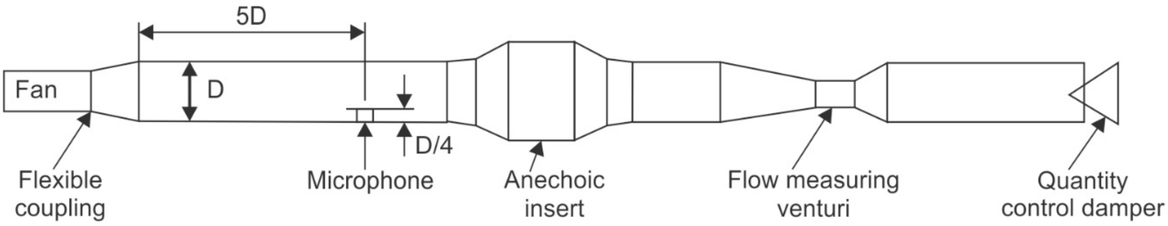

Figure 14.13 Noise testing for fans

14.7 Noise Level Predictions

There are various techniques that exist whereby noise levels produced by machinery may be predicted. Some of these will now be discussed.

14.7.1 Prediction of Noise Levels at Fans

There are three main sources of noise at fans, namely:

- Vortex noise caused by turbulence of the air over the blades
- Periodic variations in pressure due to rotation of the blades (blade noise are normally not the main source of noise)
- Mechanical vibrations can also lead to noise (fans that has not been serviced for a long time, will cause problems).

In the design of a fan, the following formulae can be applied to determine the various energy levels (sound power levels) for different parameters such as the kW rating of the fan and the quantity (Q, m^3/s) of air delivered by the fan as well as the **static pressure** (P, kPa) of the fan.

$$SWL = 97 + 10\,\text{Log kW} + 10\,\text{Log P} \quad \text{dB}$$
$$SWL = 100 + 10\,\text{Log Q} + 20\,\text{Log P} \quad \text{dB}$$
$$SWL = 95 + 10\,\text{Log kW} - 10\,\text{Log Q} \quad \text{dB}$$

Where:

SWL is the sound energy level of the fan in the octave frequency band 31.5 – 8 000Hz.

For axial flow fans that are normally used in development ends, the following formula can be used to determine the sound power or energy level.

$$SWL = 100 + 10\,\text{Log}\,QP^2 \quad \text{dB}$$

where

$$P = \text{Total pressure in kPa}$$

14.7.2 Prediction of Noise Levels at Rock Drills

There are various sources of noise in a rock drill, some of these are:

- The exhaust noise
- Noise as a result of the rotation of the drill steel
- Noise as a result of the working components in the rock drill.

The formula that can be used to predict the sound power level of a rock drill is as follows:

$$SWL = 140 + 10\,\text{Log}\;Q \quad \text{dB}$$

where

$$Q = \text{The free air consumption in } m^3/s$$

NB

It must be remembered that hydraulic rock drills have overall sound power levels between 5 and 8 dB lower than the equivalent pneumatic rock drill. The frequency spectrum will be very similar with a slightly larger correction factor to be applied in the low frequencies and a smaller correction in the high frequencies.

14.7.3 General Noise Predictions

For the prediction of SWL at the exhaust from diesel machinery

$$\text{SWL} = 110 + 10 \text{ Log kW dB}$$

where

$$\text{kW} = \text{Rated kW of the machine}$$

For the prediction of SWL at the engine of diesel machinery

$$\text{SWL} = 100 + 8 \text{ Log kW dB}$$

where

$$\text{kW} = \text{Rated kW of the machine}$$

NB

The values of the above only applicable to machinery with rated capacity **less than 300 kW.**

14.8 Routine Measurements

Routine measurements are normally done to ensure that workers are not exposed to a high risk of noise exposure. This is usually carried out using a basic sound level meter, displaying the levels according to a weighting network (usually A scale).

NB

Sound levels are taken either on a predetermined grid or where the operators would normally work. For correct measurements it must be ensured that a slow response time should be selected. The next example will explain the procedure.

WORKED EXAMPLE 7

A frequency analysis of a hydraulic rock drill is given in the table below. The overall level measured before and after the frequency analysis was 113 dB.

Mid-band frequency (Hz)	63	125	250	500	1 000	2 000	4 000	8 000
SPL (dB)	99	101	106	107	107	103	95	88

Step 1 – The two highest SPL values are in this case 107 and 107 dB. The difference between these two values is zero, and from Table 14.3 or Figure 14.4, 3 dB must be added to the highest value SPL. In this case it does not matter which).

Mid-band frequency (Hz)	63	125	250	500	1 000	2 000	4 000	8 000
SPL (dB)	99	101	106	107	107	103	95	88

|0 +3|

110

STEP 2 – The next highest level is 106 dB, the difference is now 110-106 = 4 and according to Table 14.3, 1.5 dB must be added to the highest level SPL.

Mid-band frequency	63	125	250	500	1 000	2 000	4 000	8 000
SPL (dB)	99	101	106	107	107	103	95	88

```
             101    106    107
              |      |      |
              |      |_0  +3|
              |     110
              |      |
              |_4  +1.5|
                111.5
```

STEP 3 – The process is repeated until such time that a zero correction must be applied; this will then be the final iteration.

Mid-band frequency	63	125	250	500	1 000	2 000	4 000	8 000
SPL (dB)	99	101	106	107	107	103	95	88

```
      |      |      |      |      |
      |      |      |_0  +3|      |
      |      |     110            |
      |      |      |             |
      |      |_4  +1.5|           |
      |        111.5              |
      |             |_8.5    +0.5_|
      |                  112
      |___11        + 0___|
             112 (FINAL)
```

The total pressure level that is measured in this way is therefore 112 dB and compares within 2 dB with the overall measurement that is measured before and after and is therefore acceptable.

WORKED EXAMPLE 8

Calculate the expected dB (A) level for the frequency spectrum of example 7. Given that the correction factor for dB (A) is as follows:

Mid-band frequency	63	125	250	500	1 000	2 000	4 000	8 000
A-Scale correction (dB)	-26	- 16	- 9	- 3	0	+ 1	+ 1	- 1

The sound pressure levels and mid-band frequency as follows:

Mid-band frequency	63	125	250	500	1 000	2 000	4 000	8 000
SPL (dB)	99	101	106	107	107	103	95	88

NB

Answer

What is important is that the correction **must first be applied** to the original set of SPL readings. The rest of the example is then done the same as before.

Mid-band frequency	63	125	250	500	1 000	2 000	4 000	8 000
SPL (dB)	99	101	106	107	107	103	95	88
A-Scale correction (dB)	-26	- 16	- 9	- 3	0	+ 1	+ 1	- 1
Corrected level (dB)	73	85	97	104	107	104	96	87

```
 97    104    107        104
 |      |      |          |
 |      |      |_3____+2__|
 |      |               109
 |      |_5____+1.0_____|
 |           110.0
 |             |
 |             |
 13_______+ 0  |
       110 (FINAL)
```

14.9 General Sound Surveys

With these types of surveys, sound pressure levels are measured at the noise sources and at persons that work close to these sources. Sound pressure levels must be measured at the source, the ear position of persons that work close to the source and at relevant intervals from the source so as to identify risk zones.

The following are of the most common data taken during such a survey:

- Sound pressure levels (SPL) at the source, at ear level and at relevant intervals (+- 5 m from the source of the sound). The ‘A’ weighted scale to be used throughout.
- Background noise levels.
- A description of the area where the measurement is taken. Show the conditions and type of the floor, wall and roof as well as applicable dimensions.
- The type of machinery that causes the noise as well as the distance from the source where the noise is measured.
- The time and date when measurements were taken, as well as the name of the person who took the measurements as well as the numbers of the equipment that was used.
- In some cases it may be necessary to measure the time of exposure.

Chapter

15

BASIC VENTILATION PLANNING CONCEPTS

LEARNING OUTCOMES

After completion of the study theme, you should be able to:

- Describe a systematic method of planning the environmental control requirements of a colliery or metalliferous mine.
- List what you consider to be satisfactory environmental conditions for a deep mine or colliery and give your reasons for selecting these standards.
- List and discuss a minimum of eight factors which would have to be considered when making long-term ventilation plans for a new colliery or gold mine and the important decisions which would be made after considering these factors.
- Compare the merits of using various parameters, e.g. cubic metres per second per kiloton per month, when planning ventilation requirements for a mine.
- Discuss the factors which must be considered when it is necessary to provide additional air through a mine or colliery by either using more fan power or excavating additional airways.
- Discuss the effects of mechanisation on mine ventilation planning of collieries (bord-and-pillar mining and longwall systems) or metalliferous mines.

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Webber-Youngman, RCW, 'Controlled recirculation as a strategy to reduce air supply costs' - An extract from the M.Eng. thesis, Determining the physical and economic impact of environmental design criteria for ultra-deep mines, Nov 2000.

15.1 Introduction

You have now reached the module that is basically putting together all the information that you have gathered thus far. Chapter 16 to follow is all about the economics associated with choice of system required and will show you how to choose between different proposed ventilation layouts and/or systems.

It is therefore important to have a very good understanding of all the principles associated with Mine Environmental Engineering, with specific reference to air supply, refrigeration and how to deal with all the contaminants associated with it. Planning the ventilation and cooling requirements for an underground mine (deep and shallow) or a section of a mine therefore requires a good fundamental basis.

NB

At this stage you should also be capable of using books and technical articles as resource material. Most of the resource material in this module will be of that nature. You are advised to read and research all the resources you are referred to as it forms part of the assessment procedure.

To create an acceptable underground environment, ventilation and cooling estimates therefore need to be made. These estimates need to be made early in the life of the mine to ensure that acceptable conditions are maintained. A systematic approach is thus required. This chapter will discuss ventilation and cooling planning for mines over the life of a mine. It is important to note that air is required for the following reasons underground:

- Oxygen
- Reduce and remove dust
- Remove heat from the mine.

15.2 Systematic Planning Approach

When you are involved in any planning project, a specific procedure needs to be followed in planning the optimised layout for the mine. What is therefore needed is a planned and sequenced approach. This approach is graphically shown in Figure 15.1.

Click on a block in the process to go to that part in the document where the step is discussed.

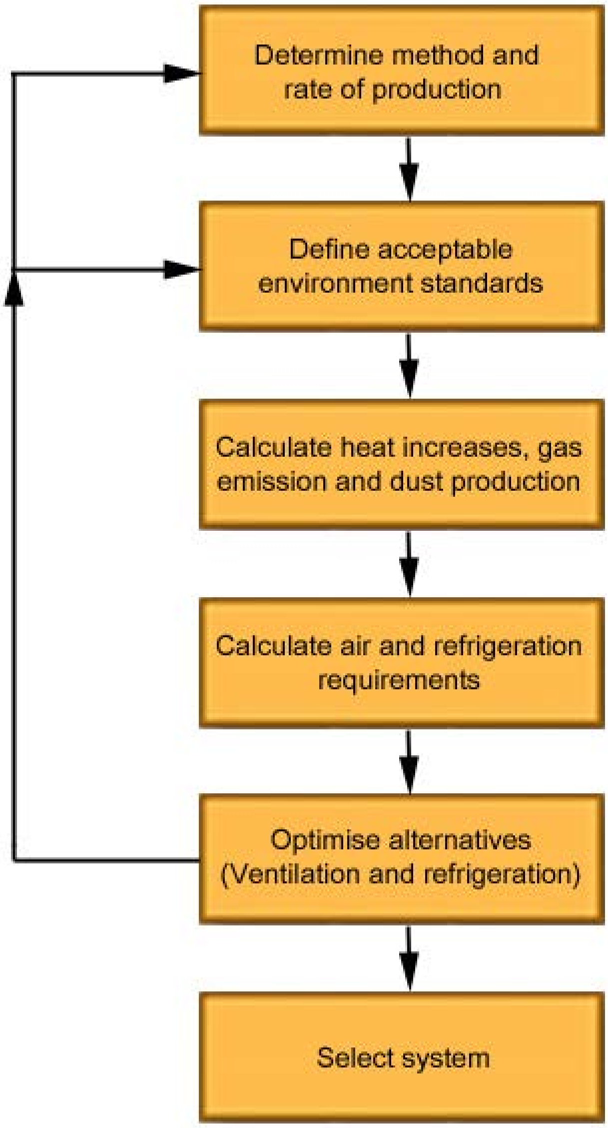


Figure 15.1 A systematic approach to mine ventilation planning

Each of these aspects forms a very important part of the procedure and will now be discussed in detail.

Process

Click on the button to open the process and display where you are in the process.

15.2.1 Determine Method and Rate of Production

Although the method of mining and the rate of production are usually pre-determined, the Environmental Engineer can provide useful input, especially where the shape of the working places (development ends and stopes) and the positioning of airways are concerned. Obviously, rock engineering aspects and geological formations also have an influence, and for this reason a multidisciplinary approach in planning is best.

!

A word of caution at this stage is that although shaft sizes, production rates and so forth are pre-determined you should always bear possible future expansion in mind. A typical ‘contingency factor’ which is sometimes planned for, is that the mine or section, will eventually produce ore at a rate at least 15% to 20% more than originally planned for.

Process

Click on the button to open the process and display where you are in the process.

15.2.2 Define Acceptable Environmental Standards

Before planning can proceed any further, standards have to be established to provide a basis. Factors that should be determined at this stage are:

15.2.2.1 Dust

Depending on the type of ore being mined, certain standards exist for the various types of dust being produced. Your selected standard must be compared to present legal limits (OEL’s), and also to accept threshold limit values (TLV’s).

Self study

You should study Tables 13.2 and 13.3, pages 245, 247 and 248, in VOEE, for guidance in this matter.

HARD ROCK MINES (SUCH GOLD AND PLATINUM)

The reasons for selecting pre determined dust concentration values as guidelines for planning ventilation requirements are as follows:

- Mines pay a levy, which is used to fund compensation payments to workers with diseases that are compensatable. This levy is based on a ‘pollutant index’ and also discussed in the chapter on dust which included the gravimetric sampling guideline document.
- A continued low dust figure results in a healthier work force and large financial savings to the mine, which easily offset the expenses of additional dust suppression techniques. It is important that the student remembers to apply dust OEL values applicable to a specific mine (as shown in the guideline document as well).

NB

BASE METAL MINES

For base metal mines there are also specific pollutant OEL's specific to that specific mining environment and these also needs to be considered and incorporated in the ventilation requirements for the mine. The various toxicity levels of the various types of dust encountered in the particular mine therefore needs to be considered (you should note, however, that because of factors such as gas and the possible explosibility of sulphide dust, where applicable, the air quantities flowing through base metal mines usually far exceed the requirements for dust suppression).

COAL MINES

Coal mines also have their own set of OEL's applicable to dust and these also needs to be incorporated in the ventilation planning requirements for the mine. South African coal mines are not as deep and hot as some coal mines in Europe and therefore refrigeration requirements for coal mines in South African mines are not as applicable as for the deep hot coal mines in Europe.

Obviously, because modern collieries are highly mechanised, the dust production rates on coal faces are very high. Because of this, dilution of the dust by the airflow alone is usually not adequate, and water spray systems and dust extraction / filter units have been designed to cope with this problem. Read the following by Anglo Coal; Planning guideline for ventilation planning: Coal Mines.

15.2.2.2 Gas

Maximum allowable levels for the various types of gases do exist and were discussed in the chapter on gases and should service as a basis of approach. For example, the proposed OEL for methane in a flammable mine is 1.4%, but you will often find that mine apply a value of 1 % by volume, which would mean that more air will have to be supplied to keep the levels below a specific pre-determined value. Dilution of gases is planned for in a similar way. Where inadequacies exist in the ventilation system, it should be addressed through a risk assessment and the corrective actions put in place to deal with the situation.

15.2.2.3 Heat, Temperature and Humidity

An important aspect that needs to be mentioned here is air cooling power (ACP) which is an indicator of the ability of the air to cool down the body of a worker, it is made of two components namely:

- Wet-bulb air temperature
- Velocity of the air.

This is also known as one of several heat stress indices such as temperature, wet kata and specific cooling power (SCP) and so forth.

The Chamber of Mines Research Organisation conducted the most comprehensive investigations of heat stress in mining. This resulted in a concept which is fundamentally sound, from both Engineering and a physiological perspective, and defines the balance between metabolic heat (M) and the collective cooling effects of radiation (R), convection (C), and evaporation (E), as expressed in the following Equation (also known as thermal equilibrium):

$$M = R + C + E \quad \text{......} \quad (15.1)$$

SHE

The units of these parameters are normalised to watts of heat transferred per square metre of skin surface area. Respiratory cooling, conductive heat transfer and mechanical work output are assumed to be negligible. The summation of R + C + E can be regarded as a measure of ACP. Provided this value remains equal to or greater than M, workers will maintain thermal equilibrium. If, on the other hand, metabolic heat production exceeds ACP, the skin temperature of the worker will rise. This may be sufficient to increase ACP until a heat balance is achieved, otherwise the worker's body core temperature will increase. If the latter situation develops and is not alleviated, the worker will exhibit progressive symptoms of heat strain.

The Equations that lead to the calculation of ACP are convenient for incorporating into computer programs but cumbersome for manual calculations. For this reason, nomograms and tables are available (Stewart, 1989b) for specified values of t_{wb} (wet-bulb).

References

Stewart, J., 1989b. Fundamentals of human heat stress. In: J. Burrows, R. Hemp, W.Holding and R.M. Stroh (Eds.), Environmental Engineering in South African Mines: 495-533. Johannesburg, The Mine Ventilation Society of South Africa.

NB

Figure 15.2 shows that air velocities of approximately 0.5 m/s and 1.5 m/s are required to achieve an ACP of 300 W/m^2 at wet-bulb temperatures of 27 and 29°C, respectively. It is important to note that increasing air velocity from 0.5 m/s to 1.5 m/s at a constant wet-bulb temperature will increase ACP by approximately 20%. In contrast, decreasing wet-bulb temperature from 31ºC to 25°C for a constant air velocity of 0.5 m/s will increase ACP by nearly 60%, and would provide an even greater increase in ACP for a velocity of 1.5 m/s.

When designing an environmental control system, ACP can be used to determine the required reject wet-bulb temperature for areas with high air velocities. However, ACP does not lend itself to subsequent monitoring and control in the underground work situation, as it cannot be measured with a single instrument.

Non-environmental co-determinants of ACP, such as skin temperature, sweat rate, work rate, etc. are difficult to measure and, thus, reduce the practicability of ACP for monitoring and control purposes.

Accordingly, the use of ACP should be limited to design applications, with wet-bulb temperature and air velocity (both of which are readily measured with reasonable accuracy) being used for monitoring specific areas and workplaces.

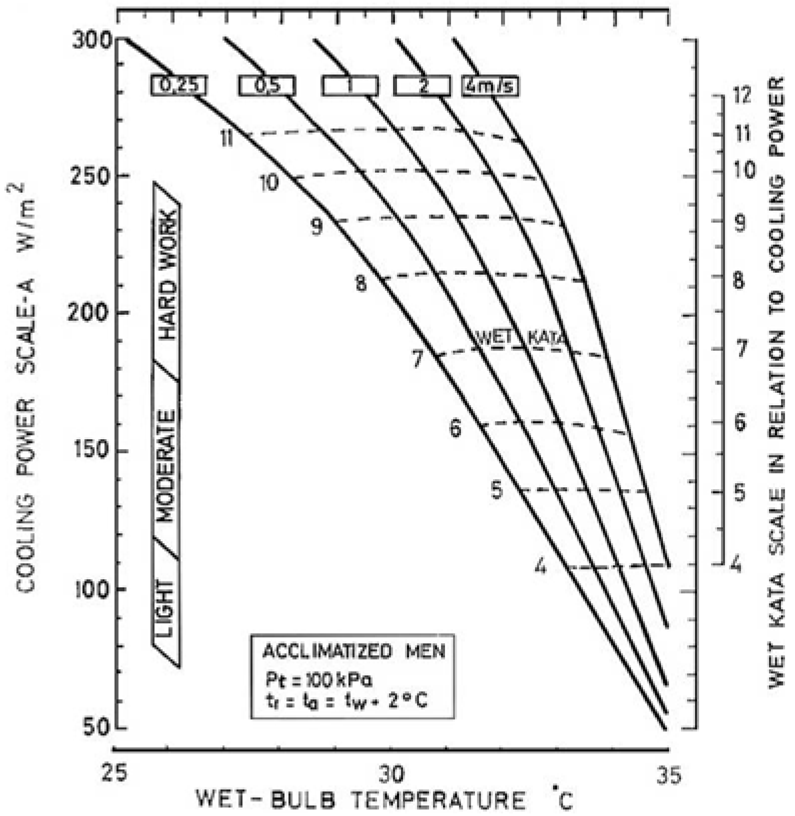


Figure 15.2 Environmental design parameters in relation to Air Cooling Power (ACP)

Definition

This more limited definition is sometimes termed the **Specific Cooling Power** (SCP) or A-Scale cooling power, to distinguish it from general ACP.

Rule

For design purposes, it has been recommended that ACP should not be less than 300 W/m^2 (MacPhearson,1984)

References

McPherson, M.J., 1984. Mine ventilation planning in the 1980s. International Journal of Mine Engineering, 2: 191-202.

Although lower design and control limits for ACP are in use, most notably in Australia, these are within the context of high levels of mechanisation and air-conditioned operator cabins.

The ventilation engineer can use calculations of SCP to determine those combinations of temperature and air velocity that most economically provide the required cooling power or rate of cooling.

Values for SCP can also be used in conjunction with estimates of metabolic rate to determine the environmental conditions necessary to maintain thermal equilibrium for a person performing a specific task.

NB

Ignoring the negligible amount of energy leaving the body in forms other than heat, a person working in a particular environment will be able to maintain equilibrium for as long as the cooling power of that environment equals or exceeds the metabolic rate associated with the task being performed. An acclimatised worker will be able to maintain equilibrium with a skin temperature of 35°C and experience only mild heat strain if the SCP equals or exceeds the metabolic rate. The metabolic rates typical of light, moderate and hard work are 90, 180 and 270 W/m^2, respectively.

An analysis of past environmental performance in various GENMIN mines indicated that the average SCP in their deeper and hotter mines was 293 W/m^2, with values ranging from 241 to 318 W/m^2 (Dumka, 1988).

These related to stopes with air velocities ranging from 0.73 to 1.24 m/s around an average of 0.93 m/s. Dumka made the following recommendations for the design of workplace thermal environments:

Rules

- A minimum SCP of 300 W/m^2
- A range of average stope velocities from 0.5 to 1.2 m/s
- New mine designs should be based on the combinations of wet-bulb temperature and air velocity indicated in Table 15.1

SHE

- Consideration should be given to the fact that the combined conditions of 0.5 m/s for air velocity and 27°C for wet-bulb temperature provide the greatest safety factor, in terms of catering for ventilation breakdowns or interruptions.

Table 15.1 Recommended combinations of wet-bulb temperature and air velocity

Stope face air velocity (m/s)	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2
Wet-bulb temperature (°C)	27.0	27.5	27.8	28.1	28.4	28.6	28.8	29.0

Rules

Heat, temperature and humidity have been dealt with extensively in prior chapters of this course. Some recommended values and reasons for planning the requirements for heat and temperature mentioned above is as follows:

Reject temperature	:	27.5°C wet-bulb
Air cooling power (ACP)		
Light work	:	115 W/m^2
Moderate work	:	180 W/m^2
Hard work	:	240 W/m^2
Wet Kata	:	not less than 10

The reasons for selecting these values are:

i) **Safety**: It has been shown that the chances of an average person suffering from heat stroke at temperatures below 27.5°C wet-bulb are less than one in a million.

ii) **Productivity**: It has been shown that heat is a factor involved in the capacity of a person to perform work, and the figures quoted above ensure a reasonable potential for work.

iii) **Cost benefits**: Because of the fact that acclimatisation (which is an expensive procedure) is not necessary at temperatures below 27.5°C wet-bulb, and the productivity levels involved can be maintained, there are definite cost advantages to planning to the above parameters.

15.2.2.4 Noise

A code of practice for 'The Implementation and Control of a Hearing Conservation Programme in the South African Mining industry' is available. This code offers some assistance when planning for noise abatement and control measures.

References

In the meantime you should refer to the modules on noise, and Chapter 37, 'Noise Control' in 'VOEE, in this regard.

Rule

The noise limit planned for should be equivalent to 85 dB (A) exposure during an 8 hour shift or 40 working hour week.

Obviously these levels present difficulties where machinery such as rock drills and auxiliary fans are concerned, and where the above level cannot be attained, hearing protection should be provided.

The reasons for recommending 85 dB (A) L_{eq} as limit are to:

- Avoid permanent hearing loss and possible claims for compensation at a later stage.
- Improve communication, safety and health.

15.2.2.5 Illumination

Chapter 13 refers to acceptable illumination levels and you should read this again if necessary. Some guideline figures are given below:

- Station and working places, 100 lux
- Junctions and hazardous areas, 50 lux
- Travelling ways, waiting places, 20 lux

The benefits of adequate illumination levels are improved safety, productivity and worker moral.

15.3 Calculation of Heat Loads, Gas Emissions and Dust

Click on the button to open the process and display where you are in the process.

15.3.1 Heat

The determination of the heat load in a deep mine has been dealt with in the chapter on heat and should be reviewed again.

15.3.2 Gas Emissions

Data concerning gas emissions and the types of gas to be expected in a proposed mine, or section of a mine, can often be obtained from prospect borehole samples. Alternatively other mines in the area will be able to provide this type of information.

In the case of coal mines actual coal samples can be extracted from boreholes and the coal degassed. The volume of gas liberated compared to the mass of the coal sample will give an indication of the expected gas emission rate during mining operations. In the case of uranium bearing ores, sampling values obtained can be used to calculate expected random gas emission rates. Chapter 11 on gases has provided extensive background on dust calculation and dust elimination practices and should be reviewed again.

15.3.3 Dust Production

The method of mining and the mineral being mined will, to a large extent determine the amount of dust that may be expected.

References

Experience plays a large role in this case, and you will have to draw from existing data to obtain the information you require. Chapter 12 of these notes and Chapter 15, 'Sources and Methods of Dust Control' in VOEE deals with this subject to some extent, and you should again read the relevant section.

Process

Click on the button to open the process and display where you are in the process.

15.3.4 Calculate Air and Refrigeration Requirements

These can be done through the use of advanced simulation packages such as VUMA (Ventilation of Underground Mine Atmospheres or Ventsim). You can also refer to available graphs and charts to obtain this data. Factors, which play a very important role here, are:

- Depth of workings (to determine residual cooling power of the air)
- Air travel distances
- Shaft and airway sizes (hence, velocities)
- Average surface air temperatures
- Total heat loads
- Expected water usage
- Pre-determined acceptable environmental standards
- Expected dust and gas production underground
- Production rates and methods
- Minimum face velocities.

When all these factors have been taken into account, the airflow requirements must be balanced with the heat, gas and dust loads and if required, refrigeration techniques or additional air volumes considered. A practical example will be referred to at a later stage.

Process

Click on the button to open the process and display where you are in the process.

15.3.5 Optimise Alternatives

At this stage you will have a good idea of what you have planned for the new mine / extension. Some relevant questions that needs to be asked can and may include the following:

- What if you should install ice plants versus conventional water chillers?
- What if the anticipated chilled water usage does not meet the actual usage, or if the actual usage is significantly less than anticipated?
- What if more gas is encountered than anticipated, or maximum allowable concentrations change?

NB

- What are the implications of a larger or smaller overall air volume?
- Will you have to acclimatise workers, and if so, how many?
- What are the economic implications of more or less refrigeration / air volume circulated?

15.4 Factors to be considered in Long Term Planning

15.4.1 Estimating Air Quantities

NB

In the planning of mine ventilation systems it is customary to express the air requirements in m^3/s per ton of rock broken per month. This normally takes into account the amount of air needed to dilute contaminants such as dust and gases and also to remove heat.

On the Central Rand at a depth of 2 650 m, where the virgin rock temperature is about 43.3ºC, the air wet-bulb temperature in summer at the downcast station is about 27.8ºC. In the Orange Free State the same virgin rock temperature is obtained at a depth of approximately 1 650 m, there the down cast station wet-bulb temperature in midsummer is only about 23.9ºC.

Obviously, therefore, for the same virgin rock temperature the air in the Orange Free State can absorb much more heat than the same amount circulated in a Central Rand Mine.

!

Although no method, whatever relationship is adopted, can ever give an accurate estimation of the air requirements because of the great variety of mining conditions. Heat has become such an important item in our deep mines that many people are sceptic whether we will be able to mine profitably at depths due to the high costs associated with refrigeration for deep mines.

The effects of heat generation in mines can be counteracted mainly in two ways:

1. By air circulation through the miner
2. By refrigeration.

Which of the two methods is the most economical depends upon many factors, and probably no hard and fast rule can be laid down.

Objective

A major factor in deciding which method to adopt is probably the life of the mine. It may be economical for short-life mines to resort sooner to refrigeration rather that to sink costly shafts. On the other hand, for long-life mines it may sometimes be more economical to resort to the circulation of larger quantities of air and to defer refrigeration to a later stage. Whether one chooses air circulation or refrigeration the objective of either is to remove heat from the mine.

The correct approach to deep mine ventilation is, therefore, to estimate the amount of heat energy that is liberated from the surrounding rock and to remove this energy either by air circulation or refrigeration.

If we plot the heat production per ton broken per month for various methods of mining against the virgin rock temperatures and at the same time the amount of heat that can be removed by the air, for various volumes per ton broken, then we can see whether some of this heat must be removed by refrigeration or not.

15.4.2 General Ventilation Planning Guidelines for a New Mine

There are several factors to be kept in mind as ventilation planning guidelines for a new mine, these are:

(i) **Natural Rock Temperatures** or VRT

(dealt with in the chapter on heat).

(ii) **Total air requirements**

Ventilating air can be obtained from two sources, namely, from the surface atmosphere direct or from cooling plants which reconditions used air for subsequent re-use.

In calculating total air requirements it is fortunately not critically important to know in advance the proportions to which the air will be derived from these two sources, and this simplification has considerably facilitated the subsequent calculation of the economics of fresh surface air versus reconditioned air.

(iii) **Estimating refrigeration requirements**

(iv) **Basic Assumptions of certain requirements to be made, such as**:

- Tonnage broken in tons/month/shaft (reef and waste)
- Total underground air requirements
- Virgin rock temperatures
- Refrigeration requirements
- Capital cost of main fans
- Capital cost of booster fans
- Capital cost of refrigeration plant u/g and/or on surface
- Electricity cost (continuous operation)
- Refrigeration plant performance numbers such as COP
- Efficiency of main fans (to assume overall efficiency of fans as 70%)
- Efficiency of booster fans (to assume overall efficiency of fans as 70%)

- Interest rates (economics to be discussed in detail in Chapter 16)
- Capital amortisation over 16 years.

(v) **Main fans**

The main exhaust fans at the top of each shaft (s) will need to be assumed (normally parallel placement) and the air density to be measured. Fan motor sizes will have to be included as well as well as running speed of motor which will have a direct impact on the fan speed. Provision also needs to be made for the deposition and drainage of water and for self-closing doors at the fans.

(vi) **Streamlining of buntons in downcast shafts**

Much work has been done in recent years, both locally and overseas, on the streamlining of shaft equipment, especially the buntons (part of shaft steal work) and its effects on resistance to airflow. Shaft buntons are normally streamlined in one way or another. The final co-efficient of friction for a shaft equipped with such buntons are normally less and will have a major impact on the costs associated with air supply to the mine.

J d V Lambrechts did investigations on a mine and found that if the shaft buntons were planned rounded at the ends would have a significant impact on the pressure losses associated in the shaft. The figures listed in Table 15.2 shows a comparison of pressure and air power losses in a 9.6 m diameter downcast shaft carrying approximately 700 kg/s of air:

Table 15.2 Comparison of pressure losses associated with different types of buntons

Shape	K-factor	Pressure Loss (Pa)	Air Power (kW)
Conventional ‘I’-beam	0.0393	3 027	1 894
Rectangular	0.0302	2 326	1 455
Skittle shape	0.0199	1 533	959
Rounded shape	0.016	1 233	771
Diamond hexagonal	0.0147	1 132	708

As can be seen from these figures, installing rounded buntons instead of I-beams in this particular shaft can derive a saving of approximately 59%.

(vii) **Up-cast Shaft**

NB

These are normally one continuous chimney and must be kept as dry as possible due to the damaging effect of water on the surface fans.

It is therefore important that water traps/water collectors should be installed to withdraw water from the air before it reaches the main fans (normally centrifugal fans).

(viii) **Refrigeration**

Siting of fridge plants is a very important economic consideration. Underground and surface fridge plants have very distinctive properties associated with each, with the condensing temperature for a similar plant to be used for underground or on surface, to be much higher for underground plants.

The cooling towers for condenser water-cooling should include consideration of distance from plant and also of such size that it will be able to accommodate future increases in heat loads expected for the mine.

NB

Chilled water use for secondary cooling as well as cooling coil placements for underground work places become a very important point to consider.

Pipe sizes for service water and refrigeration become important aspects to consider for the effective supply thereof. Cascading dam sizes needs to be considered for specific levels and for closed systems, the design of pressure releasing stations needs to be done as well. Reticulation of water has been done in detail in Chapter 7 of this course.

(ix) **Development Ventilation**

The consideration of spot coolers (cooling cars), types of fans to be used as well as extraction strategies, such as exhaust, force or force exhaust.

(x) **Stope Ventilation**

Mining (and ventilation) will be done on a predetermined method and the advantages and disadvantages need to be considered quite well before a decision is made. The air can be supplied in various ways such as for up dip, down dip, breast and long wall mining configurations and also scattered.

(xi) **Dry Mining**

Several advantages pertaining to dry mining has been discussed before and needs to be revisited. Whilst appreciating the 'ventilation' advantages accruing from very dry intake airways and the controlled use of water, particularly in stoping.

(xii) **Cooling of large underground machinery**

Temptation does exist some times to short-circuit some air to the up-cast shaft after it has ventilated pumping, plant and hoisting machinery, but it is not always necessary to discard air which has been used in this way as the air might still have some heat removal capacity and in this way can be lost or wasted.

15.4.3 Additional Planning Parameters

There are various additional parameters used by the Environmental Engineer when planning a new mine. Some of these are as follows:

15.4.3.1 Air tonnage Ratio (J de V Lambrechts)

Figure 15.3 shows how the estimated air quantity can be determined for specific virgin rock temperature.

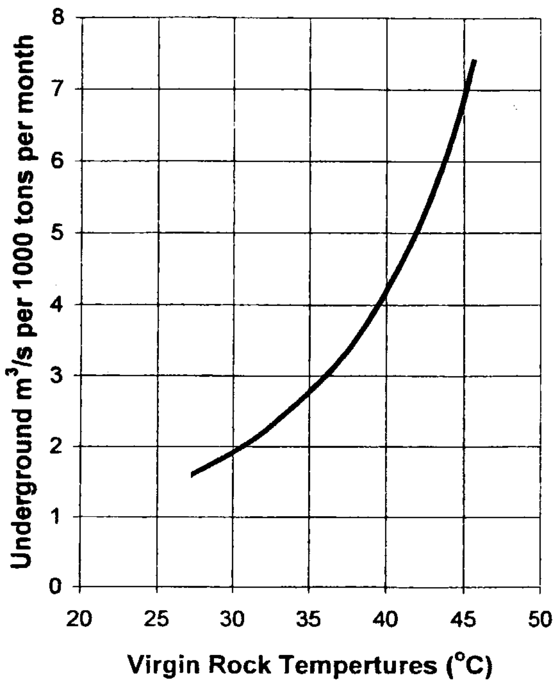


Figure15.3 Estimation of air flow quantities from VRT's

The advantages of using this figure are that it is based on actual measured data, and that it is simple and easy to use.

The disadvantages, however, are that the figure obtained is a ‘wide’ figure applying to a group of mines (in other words the line is fitted to scattered data), and because the curve is exponential, small changes in the VRT over 40°C results in large changes in air volumes circulated which makes it difficult to accurately estimate the air requirements.

As a matter of fact, once a certain pre-determined depth is passed the air volume circulated could in fact be reduced.

15.4.3.2 Heat Production per ton Mined

The use of the Barenbrug and R Stroh graphs have been discussed in detail in Chapter 6 on heat.

WORKED EXAMPLE 1

An underground pump chamber is to be ventilated by taking air from a main intake airway. Two schemes are being considered.

- **Scheme 1**: To pass the air from the pump chamber direct to return, leaving the remainder of the intake air to ventilate the working section

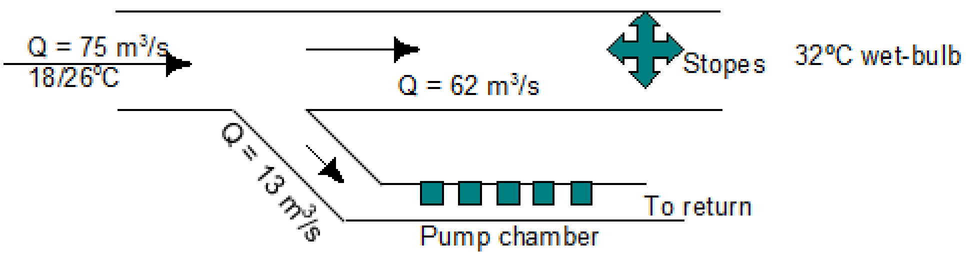


Figure 15 3

- **Scheme 2**: To allow the air, after ventilating the pump chamber, to rejoin the intake air, the working section being ventilated by the mixture of the two air streams.

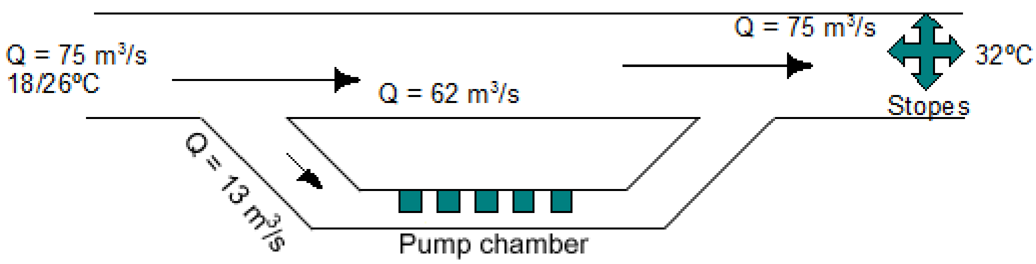


Figure 15.4

The total intake air quantity is 75 m^3/s and the temperature at the entrance to the pump chamber is 18/26ºC. The quantity of air passing through the pump chamber is 13 m^3/s and the total power of the pumps is 1 000 kW of which 70% appears in the air stream in the form of heat.

The barometric pressure is 100 kPa. If the limiting return wet-bulb temperature from the section is 32ºC and the barometric pressure in the return system is also 100 kPa:

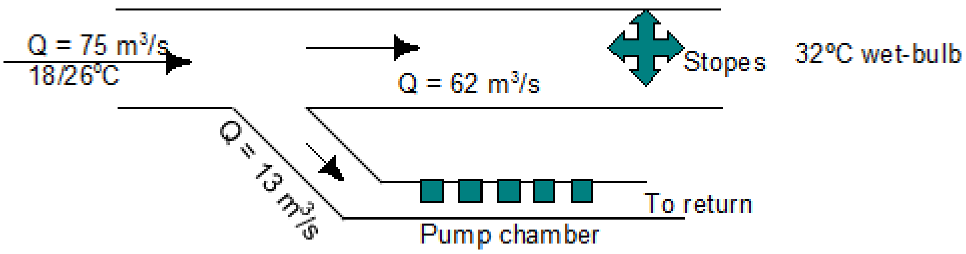
(i) What is the heat absorption capacity (kW) of the air available to the working section in the case of Scheme 1?

(ii) What is the heat absorption capacity (kW) of the air available to the working section in the case of Scheme 2?

(iii) Which scheme would you recommend?

Answer

(i) **Scheme 1**



Psychrometric chart

Figure 15.5

Using the 100 kPa psychrometric chart, the characteristics of the air can be determined:

The specific volume of the air in main airway	=	0.873 m^3/kg
The quantity of air in the main airway	=	75 m^3/s
The quantity of air flowing to the pump chamber	=	13 m^3/s
The quantity of air flowing to the section	=	62 m^3/s (75 – 13)

$$\text{Mass flow of the air} = \frac{Q}{v}$$

$$= \frac{62}{0.873}$$

$$= 71 \text{ kg/s}$$

Sigma heat of air leaving stopes = 107.2 kJ/kg

Sigma heat of air in main airway = 50.2 kJ/kg

Heat absorption capacity of the air (kJ/kg) = ΔS
= (107.2 – 50.2)
= 57 kJ/kg

Heat absorption capacity of the air (kW) = m x ΔS
= 71 x 57
= 4 050 kW

(ii) **Scheme 2**

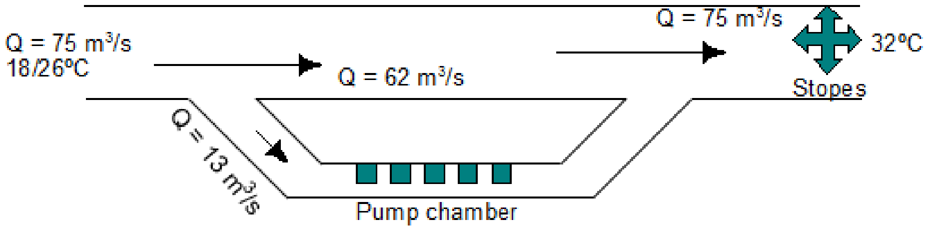


Figure 15 6

Psychrometric chart

Using the 100 kPa psychrometric chart, the characteristics of the air can be determined:

The specific volume of the air in main airway = 0.873 m^3/kg

The quantity of air in the main airway = 75 m^3/s

The quantity of air flowing to the pump chamber = 13 m^3/s

The quantity of air flowing to the section = 75 m^3/s

$$\text{mass flow of the air} = \frac{Q}{v} = \frac{75}{0.873} = 85.9 \text{ kg/s}$$

Sigma heat of air leaving stopes = 107.2 kJ/kg

Sigma heat of air in main airway = 50.2 kJ/kg

Heat absorption capacity of the air (kJ/kg) = ΔS
= (107.2 – 50.2)
= 57 kJ/kg

Heat absorption capacity of the air (kW) = m x ΔS
= 85.9 x 57
= 4 900 kW

Heat added by the pumps = 1 000 x 0.70
= 700 kW

Heat absorption capacity of the air is = 4 900 – 700

= 4 200 kW

(iii) Scheme 2 would be the better scheme as the air is able to absorb an extra 150 kW of heat.

In the example that will now be discussed we use data from existing mines. The parameters that are considered are VRT, design reject temperature and the distance from the shaft.

WORKED EXAMPLE 2

(i) When we need to determine the amount of air required to ventilate zone 1, we can make use of Figure 15.7 in order to obtain a reasonable estimate. For a depth of 1

(ii) 250 m, the amount of air required is approximately 6 kg/s per kton per month if the design reject temperature is 27°C. This is for workings that are 2 500 m from the shaft. **Remember this is only applicable to zone 1.**

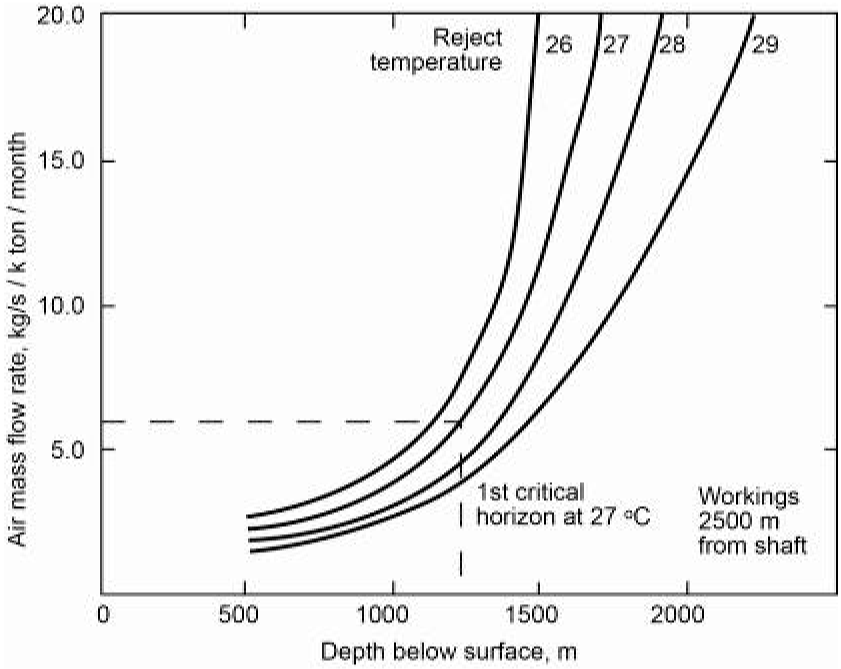


Figure 15.7 Estimation of air flow requirements

(ii) We need to determine the heat load for mine. Figure 15.7 gives the heat produced in kW per kton per month versus the VRT for a Free State mine. The distance from the shaft is also considered. **It is important to remember that the reject temperature correction factor must be used.** Let us assume a distance of 2 500 away from shaft. Also assume VRT of 60°C. Then read off graph the value of heat production as 280 kW/kton. We need to apply correction factor.

From small scale in the left hand top corner and reject temperature of 27°C, read off percentage correction factor as +10%. Take the value as read off 280 kw/kton and multiply it with 10% (1.1). The **answer** for the heat load is now 308 kW/kton.

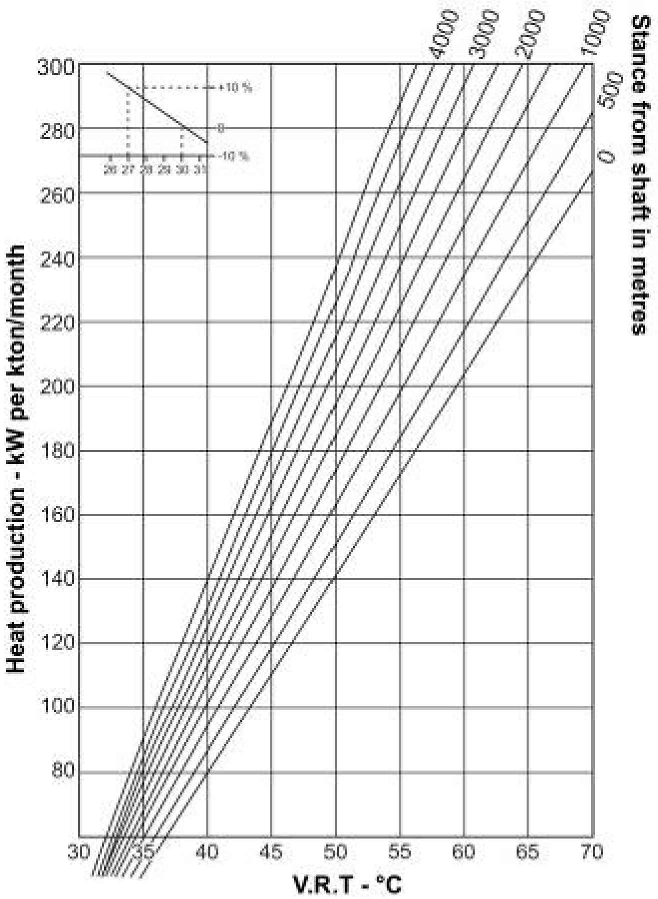


Figure 15.8 Estimation heat production per kton per month

15.5 Fan Power/Additional Airways

When the air volume circulating through a mine has to be increased, there are usually two alternatives available, namely:

- Increase fan power
- Excavating additional airways.

Before you can consider any of these alternatives, you need to obtain detailed information on the air flow circuit (that is the overall resistance and individual airway resistances).

This data is obtained by means of a pressure survey of the mine as was mentioned before in Chapter 9.

Once the data has been obtained from which you can determine what additional fan power or airways are required to meet the new requirements of the mine, you can make a meaningful comparison.

This comparison would include the economic evaluation of the project. In other words, you will have to determine what, for example, it would cost to develop new airways or sliping (cut away of rock) of existing airways, and what the ultimate effect on the fan performance would be. Any additional power costs will also have to be obtained.

Secondly, you will have to determine what the additional fan power requirements would be when installing a new fan or fans. Included in this figure will be the purchasing plus any additional maintenance costs plus erection / excavation costs for the new fan/s.

Other practical considerations, which must be considered, are:

- **Life of the project / section** You have to determine how long the additional air will be required for, as this will have an influence on which scheme you select.
- **Availability of labour / material** Should you wish to have additional airways developed or existing airways slipped, you will have to determine whether there is sufficient labour and material available in the section or on the mine to do the necessary work.
- **Rock Mechanics** In the case of new airways being selected you would also have to determine whether problems might occur regarding the stability and life expectancy of the airways due to poor ground conditions.

15.6 The Effects of Mechanisation

The ultimate result of technological progress is that mines have become more and more mechanised. The implications of mechanisation is increased production rates, resulting in increased dust production, and where gas is involved, increased gas production (whether from the mineral being mined, e.g. coal, or from the exhaust gases in the case of LHD's).

This means that the Environmental Engineer will have to take the increased dust and, where applicable, gas production rates into account when planning a mine.

He would have to know what the anticipated dust / gas production rates in relation to the mineral production rates will be, based on this he will then have to ascertain the air volumes required to effectively dilute and remove all the contaminants.

Another factor, which is of importance with increased mechanisation, is the additional heat that would be liberated into the environment by these machines. This factor, obviously, also needs to be investigated and catered for when required.

Obviously, should attempts at designing rock cutters succeed another resulting factor would be a reduction in average stoping width. This in turn will lead to smaller air requirements needed as the volume of the areas that needs to be supplied with air will become less.

Other benefits would be:

Advantages

- In a mining cycle where explosives are eliminated, it would be possible to install heat exchangers right on the face.
- The number of tramming and travelling ways would be reduced (as the amount of ore to be handled and the number of workers required would be greatly reduced).
- The smaller air volumes referred to above would also result in smaller shafts, fans, return air airway sizes, and so forth.

Other technological advances, such as the placement of tailings in a cement/tailing grout mixture and filled into a bag to form a support system for use underground, has led to the following ventilation benefits:

- Improved air utilization, hence reduced are requirements
- Reduced heat flow from worked out areas resulting in a smaller heat load on the heat removal mechanism (whether air or refrigeration).

In collieries, the placement of ash filling is beneficial in the environmental control context in that it assists in reducing the occurrence of spontaneous combustion and reduces leakage between intakes and returns (where this is practically possible).

In summary it can be said that technological advances and mechanisation of mines have certain very definite advantages, and also some disadvantages.

15.6.1 Planning the Ventilation Requirements for a Mechanized Gold Mine

Let's consider a mine where 80 000 tons per month are mined with mechanised equipment and the total capacity of the diesel equipment is 4 000 kW. A few important aspects that will influence the planning of the ventilation requirements are:

- Diesel fumes
- Air losses to mined out areas
- Ventilation of ramps

- System choice series/parallel
- Dust control.

15.6.1.1 Diesel Fumes

Law

By law NO_x levels should not exceed 5 ppm. The air factor, from Figure 15.9, shows approximately 0.1 $m^3/s/kW$ as the minimum air requirement (through experience it has been proved that this is quite a high factor and can be much less, say 0.06 m^3/s/rated kW).

We can now question what the influence of availability of machines will have on minimum air requirements. Considering the availability of the machines we can express the minimum air flow requirements as follows:

= availability x utilisation x rated kW x 0.1 (or any specified factor)

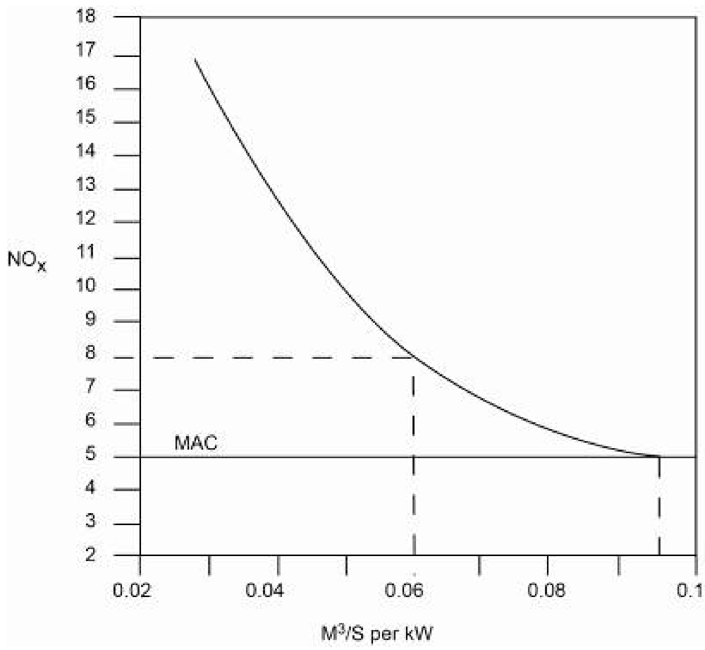


Figure 15.9 Determining air quantities for nitrous fumes dilution

SELF STUDY

Read the article by CJ Pretorius: "Trends from South African historical diesel particulate matter data".

15.6.1.2 Losses to Mined Out Areas

Losses to mined-out areas are typically in the order of 20 – 40% and can also be determined by the assumption that each seal will leak between 1 and 2 m^3/s. If the number of seals are known, we can determine the losses as follows:

Total air volume required = volume flow (from kW) + losses

Where:

Losses = 25% of volume flow (from kW value)

Or:

Losses = number of seals x $1.5m^3/s$

15.6.1.3 Ventilation of Access Roads

If we assume that the largest truck is 180 kW and the total number of roads is 7, then the amount of air that is required for the number of roads can be calculated as follows:

= 180 kW x $0.1m^3/s$/rated kW x 7 roads

= 126 m^3/s

The total volume of air now required can be determined as follows:

= Volume flow + losses + road volume

The above can simply be expressed as $m^3/s/kW$ (total volume flow per total kW) or m^3/s/ton mined.

15.6.1.4 Dust Control

There are a few places where ventilation is used as a measure for dust control and these are in the working place where loading is taking place after blasting. Dust is also further controlled by means of filters and with the help of water sprayers.

Read the following for self study purposes: Lonmin Platinum-Karee Mine; Higher production, lowerProfile: Mining's Evolution.

15.7 Planning the Ventilation and Refrigeration Requirements of a Deep Hot Gold Mine

Parameters that should be considered:

- Mining Factors
    - Production rate per month
    - Type of mining
    - Maximum distance from the shaft
    - Average depth
    - VRT (virgin rock temperature)
    - Expected service water
    - Fissure water.
- Standard ventilation requirements

- o Design reject temperature
- o Air velocity in working, volume flow in development end
- o Surface temperature
- o Air mass flow rate kg/s.

NB

15.7.1 The Influence of Depth and Air Quantity on Refrigeration Requirements

The amount of cooling required is primarily dependant on the depth underground, the Virgin rock temperature (VRT) and the cooling or heat absorption capacity of the available air. The cooling capacity of the air is the difference between the sigma heat content of the air at the station temperature and the sigma heat of the air at the design reject temperature. The station temperature is influenced by the effect of auto-compression at different depths.

It is thus clear that with an increase in depth, the higher the temperature of the air that arrives at the station and the lower the cooling capacity of the air (based on a specific reject temperature) will be.

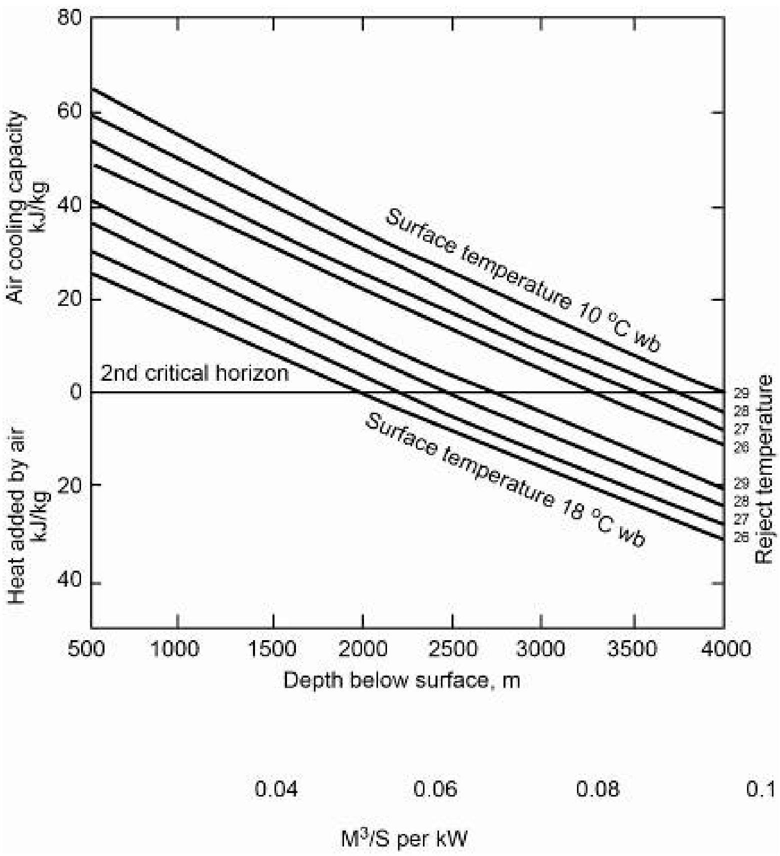


Figure 15.10 Estimation heat added by air various depths

Figure 15.10 makes it possible to read off the cooling capacity of the air for various depths and reject temperatures. Two groups of lines are shown; one group for the average surface temperature (18ºC) of the air without cooling and the other group for air that is cooled to 10°C.

15.7.2 Cooling Zones

The cooling of a deep mine can be divided into three basic zones. Figure 15.11 shows these zones.

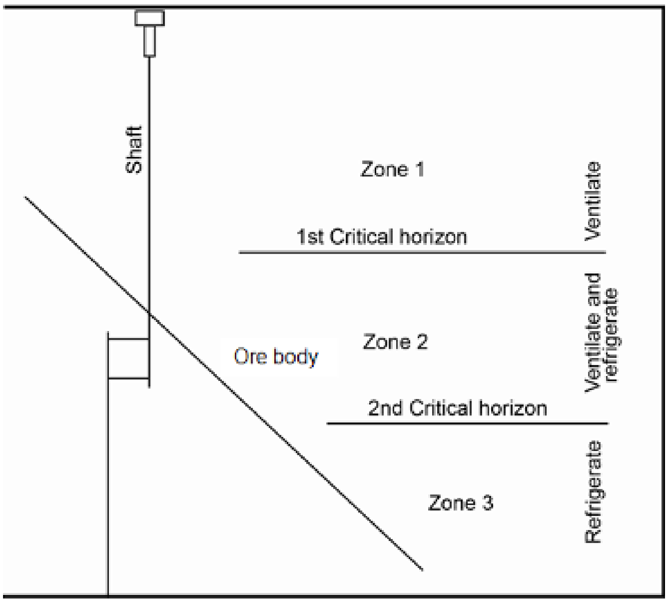


Figure 15.11 Different cooling zones

Zone 1:	1st critical horizon – workings are only cooled by ventilated air
Zone 2:	2nd critical horizon – workings are cooled by a combination of ventilated air and cooled air.
Zone 3:	In this zone the downcast air has no cooling effect and all the heat must be removed by cooled air.

WORKED EXAMPLE 3

The following design specifications are given for a specific working place in a Free State mine. Determine the following:

(i) The mine's heat load.
(ii) Heat transferred as a result of fissure water
(iii) The total heat load of the mine
(iv) Ventilation requirements for the mine

Production rate per month	150 000 ton
Type of mining	semi concentrated with dip of 20°

	80% of the ore body is mined
VRT	47.5°C
Distance of workings from station	2 500 m
Design Reject temperature	27°C
Expected fissure water	15 l/s
Service water used	1 ton of water / ton of rock mined
Surface air temperature	18°C
Temperature bulk volume air down shaft	10°C

Answer

(i) The mine heat load can be determined as follows:

From Figure 15.8 we read off the heat produced per 1 000 tons per month as 176 kW. The design reject temperature is 27°C and the correction factor is 12% (from graph). If we add 12% to 176 kW, the heat load is 197 per 1000 ton per month.

The total heat load of the mine (through rock breaking activities) can be determined with the following two Equations:

= Heat produced (kW/kton/month) x production rate

= 197 x 150

= 29 550 kW

The fissure water also adds heat:

Fissure water heat load = m x c x Δt

= 15 x 4.187 x (47.5 – 27.0)

= 1 288 kW

Total heat load is the sum of the above:

= 29 550 + 1 288

= 30 838 kW

(ii) Ventilation Requirements

The actual calculations for the ventilation requirements have not been done in detail but the procedure is clearly shown. Ventilated air is required for the removal of gasses and dust. The air is also used to transport heat to the coolers. The higher the air velocity, the higher the cooling effect. A typical figure for air mass flow in a Free State gold mine is between **3 and 6 kg/s** for every kton mined. For our example:

Air required = production rate (kton) x kg/s

NB

The wastage of air as a result of leakages is a big problem on the mine. Small mines normally control air better than large ones.

Cooling Capacity of Downcast Air

The cooling capacity is determined by using Figure 15.10

Total cooling capacity (theoretical):

= mass flow x air cooling capacity (from graph)

As a result of poor placement of the refrigeration system we must apply a positional factor of 80% (assumed), therefore the nett cooling capacity can be determined as follows:

= Theoretical cooling capacity x factor.

When air is not cooled on surface the cooling capacity (or the ability to remove heat) changes considerably. This is clearly shown in Figure 15.10 (take readings from the 18°C group lines).

Cooling Requirements

The air is cooled on surface and the cooling capacity of the system is determined by:

$$\text{Cooling capacity of plant} = M_a \times \Delta S$$

There is a sigma heat graph for various wet-bulb temperatures and different barometric pressures. The various sigma heat values can be read off from the graph and thus the cooling capacity of the plant can be determined.

To determine the cooling requirements for a distance (say x) from the shaft:

Cooling required = total heat load – net air cooling capacity

The plant duty of the surface fridge plant if we exclude losses can be determined as follows:

Duty of plant = underground cooling required + surface cooling

NB

If the workings are beneath critical zone 2, the downcast air will no longer have a cooling effect and the amount of cooling at surface plant must be increased**.**

Distribution of Cooling

If we assume that the service water is cooled to a specific temperature (y) the delta t is ($t_{\text{reject design}} - y$). The service water will therefore also have a cooling effect. We must first determine the amount of service water used in a month in ℓ/s as was done in Chapter 7 of this course. Usually there are 24 shifts and the water usage is approximately 1 ton/ton of rock produced.

$$\text{Water usage (l/s)} = \frac{(\text{ton/month} \times 1\,000)_{\text{to convert to kg's}}}{(24 \times 24 \times 3\,600)_{\text{to convert to seconds}}}$$

The cooling capacity of the water, as we already know, is calculated by:

$$q = m \times C \times \Delta t$$

The underground cooling required can therefore be determined as follows:

$$= \text{cooling required U/G} - \text{cooling by service water}$$

The water flow rate for air cooling is:

From $q = m \times C \times \Delta t$

$$m_{water} = \frac{q}{C_{p\ water} \quad \Delta t}$$

The temperature that is used to obtain the Δt is smaller (design temperature minus 5°C (assumption)), because the cooling is not 100% effective.

WORKED EXAMPLE 4

At a mine in the West Witwatersrand region, it has been decided to deepen an existing operational shaft, so as to access a lower lying ore body (thus extending the operational life of the shaft). Access to this portion of the ore body will be through a newly sunk shaft, serving five working levels, each spaced 100 m apart, will be driven from the shaft to the reef horizon. The reef will be extracted using a series of single sided long walls. The planning parameters for this project will be:

Tonnage broken (tons/month)	110 000
Ratio reef tons to development tons	5:1
Average depth below surface	2 800 m
Dip of the ore body	26º
Stoping width	1.8 m
Average stoping face advance	8 m / month
Expected VRT	45ºC
Density of rock to be mined	2 740 kg/m^3
Heat production rate of area to be mined	180 W / ton mined
Chilled service water arrival temperature	12ºC
Return service water temperature	24ºC
Planned water consumption	2 ton/ton mined
Production month consist of	23 days

In addition 25% of the total downcast volume will be used to ventilate shaft infrastructure (including ventilation door leakages). All development will be ventilated using established through ventilation to stoping areas.

Given the following environmental requirements:

Face air velocity	> 1 m/s
Minimum stope air utilisation (AU)	
(air on face supplied at stope entrance)	80%
Maximum ventilation control distance from the face	10 m

Calculate the following:

(i) Total quantity of ventilation required for the ventilation of the stoping areas

(ii) For logistical reasons, the newly sunk shaft has a diameter of 7 m, will this shaft be adequately sized to carry the required volume of ventilation, briefly defend your answer

(iii) Determine the heat load of this production area

(iv) Calculate the cooling load provided through the usage of chilled water

(v) How much additional cooling must be provided to overcome the heat load of this area

(vi) Briefly describe how you might deploy this additional cooling.

Answer

(i) The total amount of reef tons that must be mined must be determined

Total tonnage reef to waste is 5:1 (given)

Reef tons = Total tons x 5/6 (reef tons to total tons ratio)
= 110 000 x 5/6
= 92 000 tons per month

The longwall backlength needs to be determined (using the vertical height between deepest and shallowest level):

$$\text{Longwall back length} = \frac{400}{\sin 26^{o}} = 915 \text{ m}$$

Tons per longwall per month calculated as follows:

= length x face advance/m x SW x rock density
= 915 x 8 x 1.8 x 2.74
= 36 000 tons/month

Number of stoping raises

$$= \frac{92\ 000}{36\ 000}$$

$$= 2.55 \text{ say 3 raises}$$

(ii) The air quantity that needs to be circulated per stope can now be determined face velocity of 1 m/s is given:

Air volume before AU = velocity x SW x strike distance

= 1 x 1.8 x 10

= 18 m^3/s

But the AUI needs to be incorporated

$$= \frac{18}{0.8}$$

$$= 22.5 \text{ say } 25\ m^3/s$$

Three stope lines will therefore be needed

= 25 x 3

= 75 m^3/s

The new shaft area to be determined:

$$\text{Shaft area} = \frac{\pi D^2}{4}$$

$$= \frac{\pi (7)^2}{4}$$

$$= 41.9\ m^2$$

Assume the downcast volume in the shaft to be value 'a'. The leakage/shaft infrastructure will be 0.25 'a' say we call this 'b' (0.25 'a' = 'b', therefore 'a' – 'b' 75, the amount needed for the raises will be 75 m^3/s (calculated above)). The amount of air to be used:

The total downcast volume required

'a' – 0.25 'a' = 75

0.75 'a' = 75

'a' = 100 m^3/s

The shaft air velocity for the 7 m shaft needs to be determined:

$$U = \frac{\text{air quantity required}}{\text{Area}}$$

$$= \frac{100}{41.9}$$

$$= 2.39 \text{ say } 2{,}4 \text{ m/s}$$

The down cast velocity will be well within design specifications and will have a small influence on the total resistance of the mine.

(iii) The heat load in stope faces = 110 000 x 180
= 19 800 kW

NB

(iv) The flow rate of the service water needs to be expressed in litres/second. It is important to first establish the total amount in water used per month. The usage of water has been expressed as 2 ton water per ton of rock mined.

Tons of water per month	=	110 000 x 2
	=	220 000 tons of water/month
Expressed as litres/month	=	220 000 x 1000
	=	220 Mega litres per month

$$\text{flow rate in litres/second} = \frac{220\ 000\ 000}{23 \times 24 \times 60 \times 60}$$

$$= 110.7 \text{ say } 111\ \ell/s$$

(v) Cooling from service water = m x c x Δt
= 111 x 4.187 x (24 – 12)
= 5 577 kW

The additional cooling can be required as follows:

= heat load – service water cooling
= 19 800 – 5 577
= 14 250 kW

(vi) Additional cooling might be best deployed through placement of shaft BAC systems, in this particular shaft 50% of the cooling load based at the shaft BAC's, with the remaining cooling split between level BAC's (2 of) sited closer to the reef on the main air intake levels and approximately 14 mobile coolers (rated at 250 kW each).

15.8 Planning of Ventilation and Cooling Requirements for a Mechanized Base Metal Mine

Examples of massive ore bodies are:

- Okiep
- Black Mountain
- Cullinan.

Usually these mines are not deep and therefore heat is not a problem. Heat is usually as a result of diesel operated machinery.

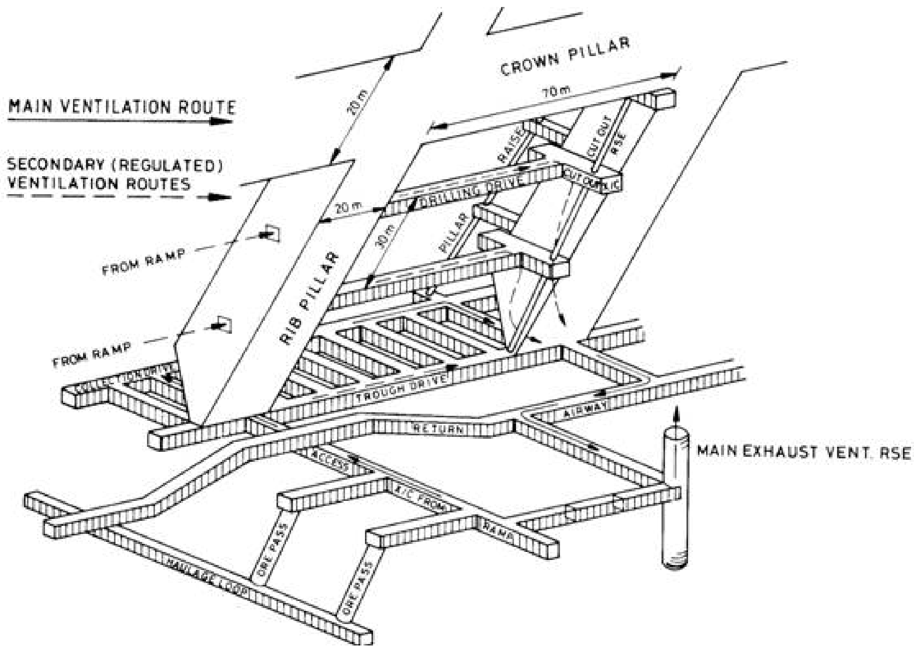


Figure 15.12 Massive mine ventilation layout

Figure 15.12 shows the layout for a typical massive ore body. From the figure it is very clear that all the tunnels in the ore body are developed in advance and fan drilling is done.

15.8.1 Methods and Production Rates

This method of mining is called sublevel open stoping. The ore body is near vertical (70° to 90°). The average width of the ore body is 9.7 m and delivers primarily copper and zinc. The ventilation network is of such a nature that the downcast air goes in with the cross tunnels and is distributed by the ramps, then to the development places and out with the return air way to the upcast shaft. The workings usually consist of a collection drive and 3 lower level reef drives development tunnels. Air comes from the ramp.

15.8.2 Ventilation Standards

Most of the contaminants are diesel gasses, dust and sulfide dust. It is not easy to design for this because diesel fume concentrations change constantly. The principle that is used for design purposes is that 80 m^3/s air must be supplied per MW electrical energy available.

The nitrous fumes and carbon monoxide from diesel machinery are exceptionally dangerous and we must consider this when designing the ventilation (similar as was discussed in section 15.6.1.1).

15.9 Coal Mine Ventilation Planning (Board and Pillar)

There are various methods that are used to mine coal. The most commonly used is board and pillar. Coal mining is mostly mechanized, making use of coal cutters, shuttle cars and mechanical loaders. Conveyors are used to transport the coal out of the mine. Aspects that must be considered when planning for a coal mine are the following:

- Dust
- Blasting
- Gasses
- Spontaneous combustion.

Heat in general is not a problem in coal mines, because most of the machines are electrical and do not have the same heat load as diesel machines. The amount of ventilation required to suppress dust cannot be determined easily because the concentrations are constantly changing. The same argument applies for blasting fumes (can be determined with some effort). A maximum for nitrous fumes, by law, is specified. Non-flammable gasses are often encountered underground, but there is no specific standard for the ventilation that is required for these.

The two major dangers are (discussed in detail in Chapter 12 of this course):

- Coal dust explosions
- Spontaneous combustion (dealt with in 3rd year of mining course).

Ventilation requirements are greatly influenced if the decision is made to ventilate mined out areas. Why would we ventilate mined out areas? These areas are normally fully controlled by maintaining good working conditions and good gas monitoring systems. In a long wall system the ventilation of the mined out areas is dependant on the chosen ventilation system and the amount of ventilation air has a direct influence on the possibility of spontaneous combustion of coal in these mined out areas.

15.9.1 Quantity of Air Required

!

The ventilation of a large number of tunnels in a mechanized mine results in problems. To ventilate these tunnels force or exhaust fans must be used and when the tunnel is advanced twice its width it is classified as a dead end and must then be ventilated by means of a fan or other means.

Guideline

There is a guideline that states that for every m^2 of development tunnel mined, there must be at least 0.2 $m^3/s/m^2$ of face mined. The guideline for air velocity in the last through road of a board and pillar layout for coal mining is at least 0.4 m/s, but for design purposes it should be an average of at least 1.0 m/s. A figure of between 1.0 and 1.5 is therefore advisable. Air volumes required at the face are dependant on the through road measurements, air control measurements, number of intake roads and the ventilation system used.

References

Read the following article by Anglo Coal; Planning guideline for ventilation planning: Coal Mines.

WORKED EXAMPLE 5

The ventilation requirements for a coal mine must be determined. The following aspects must be considered:

Mining Method	Board and Pillar
Road Sizes	3 x 7 m
Air velocity over conveyor	2 m/s
Air velocity – intake road	2.5 m/s
Air velocity – Return road	5 m/s
Standard Panel length	1.2 km
Design velocity in last through road	1.5 m/s
Production ton/continuous miner/month (low figure)	60 000
Production tons per year	4 000 000
Number of roads per section (assumption)	5

Answer

Let's follow an analytical approach:

Total area per road = 3 x 7 = 21m^2

Air flow volume on conveyor/belt road (Q_B)	=	21 x 2
	=	42 m^3/s
Air flow volume return road (Q_{RAW})	=	21 x 5
	=	105 m^3/s
Air flow volume intake road (Q_{INTAKE})	=	21 x 2.5
	=	53 m^3/s
Minimum volume air in last road (Q_R)	=	21 x 1.5
	=	31.5m^3/s
Total tons/month	=	4 000 000/12
	=	333 333 tons/month
Sections required	=	333 333/60 000
	=	5.6 (say 6)
Minimum air flow volume (Q_P)	=	31.5 m^3/s
Air flow volume for 6 sections	=	6 x 31.5
	=	189 m^3/s
Assume air utilization	=	90%
Air volume required (Q_A)	=	189/0.9
	=	210 m^3/s

The total amount of air that must eventually be delivered by means of secondary or even tertiary development to 6 sections is 210 m^3/s (no losses or extra volumes of air considered). The total volume of air that will be needed for secondary development is 210 m^3/s.

If we want to determine the total amount of secondary roads that must be developed:

Number of intake air roads required as follows:

$$= \frac{(Q_A - Q_B)}{Q_{in}}$$

$$= \frac{(210 - 42)}{53}$$

$$= 3.16$$

Say 4 roads, excluding belt road.

The maximum volume flow for 1 return airway is an amount of 105 m^3/s calculated before and the maximum volume of air with 4 intake air roads and 1 conveyor air intake road must then be determined as follows:

$$= (4 \times 53) + (1 \times 42)$$

$$= 254 \text{ m}^3/\text{s}$$

The number of return air roads can be determined as follows:

$$\text{Return air roads (RAW's)} = \frac{254}{105}$$

$$= 2.4 \text{ (say 3roads)}$$

The total number of secondary roads that must be developed to supply the amount of air is calculated as follows (that is total intake and return roads):

$$= 3 + 5 = 8$$

But the number of development roads must be odd and therefore we choose 9 roads.

Total number of air intake roads is now:

	=	6
Air flow volume possible	=	(5 x 53) + (1 x 42)
	=	307 m^3/s
Total return roads (as calculated)	=	3
Possible return airflow volume	=	3 x 105
	=	315 m^3/s

The volume of air delivered cannot be less than the volume of air extracted and we therefore work with a possible airflow volume of 315 m^3/s.

To determine the volume flow rate for primary development, we must also consider the workshop requirements.

Volume flow for workshops (assume)

$$= \quad 100\ m^3/s$$

Quantity of air down shaft (Q_{shaft})

$$= \quad 254 + 100$$

$$= \quad 354\ m^3/s$$

Number of air intake roads for primary development:

$$= \quad \frac{(Q_{shaft} - Q_B)}{Q_{in}}$$

$$= \quad \frac{(354 - 42)}{53}$$

$$= \quad 5.89$$

Say 6 roads, excluding belt road.

Total primary air intake roads $\quad = \quad 6 + 1$

$= \quad$ 7 roads that have a potential to deliver 360m^3/s ((6 x 53) + (1 x 42)). The return air roads should be able to handle the quantity of air in the intake roads.

Number return air roads $\quad = \quad \frac{360}{105}$

$= \quad$ 3.42 (say 4 roads)

It is clear the total number of primary roads will be 11.

15.9.2 Different Aspects of Fans in Coal Mining

NB

It is important to note that the main fan installation must be on surface.

Q

What do you think is the advantage of an extraction fan to that of a force fan?

A

When an exhaust fan stops, the barometric pressure is increased underground. Contaminants in the strata stay in equilibrium and do not enter the work place. Remember methane in the rock strata can enter the mine if the barometric pressure drops (just before rain, why?).

15.9.3 Shaft Size Dimensions and Pressure Losses in Coal Ventilation Planning

The fan pressure is primarily dependant on the mining layout as well as the air velocity in the airways. There are three ways to move air in the system and that is by:

- Downcast shafts
- Inlet and outlet through roads
- Up cast shafts.

The pressure losses and quantities associated with these three aspects will now be discussed.

15.9.3.1 Downcast Shaft

NB

It is important that the air velocity in the conveyor road does not exceed 2.5 m/s, to minimize the creation of dust. This means that the incline shaft will not be able to handle the entire amount of air and a downcast shaft will be required. The shafts are normally circular and are sunk as air is required in specific areas.

A design air velocity for a downcast shaft is 8 – 10 m/s and the area can be determined as follows:

$$\text{Area} = (360\ m^3/s) / (8\ m/s)$$
$$= 45\ m^2 \text{ (air flow volume from example).}$$

The diameter is thus 7.6 m (check answer).

Normally the shafts are not fully equipped and hence the friction losses are minimal. The friction pressure losses will be about a third to that of the velocity pressure loss (thumb rule). Inlet and outlet losses will dominate and the total friction pressure losses of $\frac{2.5\ \rho\ U^2}{2}$ can be expected.

The amount of friction pressure loss is therefore (air velocity amount of 8 m/s):

$$\frac{2.5 \times 1\ (8)^2}{2} = 80\ \text{Pa}$$

15.9.3.2 Pressure Losses in Tunnels

Pressure losses do not depend only on the friction pressure losses calculated from Atkinson’s Equation, for a major part of the pressure loss occurs as a result of the expansion and contraction that occurs at each cross connection between pillars. Pressure survey results show that for design purposes there is a 1 velocity pressure loss for each pillar. If we assume that pillars are 10 m x 10 m and the road is 6 m wide, one velocity pressure will be lost per 16 m, or 62.5 Pa per kilometre of through road.

The pressure loss for inlet through roads is therefore:

$$\frac{62.5 \times 1_{density} \times (2.5_{m/s})^2}{2} = 195 \text{ Pa}$$

Higher velocities can be used for return airways (3 – 3.5 m/s). Return airway losses are thus between 280 and 380 Pa per km. Each return air roadway can carry between 50 and 60 m^3/s. Adequate allowance must be made for pressure losses at air crossings and at dyke intersections.

15.9.3.3 Upcast Shafts

As coal mining is relatively dry, condensation and water are not really a problem in shallow mine's upcast shafts. The design velocity for an upcast shaft is 10-12 m/s.

$$\text{Shaft friction losses} = \frac{2.5 \rho U^2}{2}$$

The area of the shaft = 360/10 m^2 and from this D = 6.8 m.

NB

It is also important to remember that high methane readings will result in additional air quantities required.

15.10 Other Planning Aspects for Shafts

Until now it has been clear that ventilation planning is very dependant on certain aspects and these are summarized again below:

- U/G environmental condition
- Depth of the mining operation
- Type of mining
- Development program
- Shaft sizes (air velocity downcast 8-10 m/s, upcast 16-20 m/s)
- Also to consider maximum design velocities for intake and return airways (RAW's)
- Economic factors; Double shaft vs. single.

The shafts that are typically used for ventilation purposes are single divided shafts or twin shaft systems that are approximately 100 m apart. The distance depends on the size of the ore body and the number of shafts available.

What is the advantage of twin shafts? (Not just ventilation). The answer is obvious:

- Fast development
- Equipped with service winder provides a second outlet.

Figures 15.13 and Figure 15.14, shows typical aspects of a shared shaft that is used for ventilation purposes.

The dividing wall is normally made of 300 mm thick, 1.5 m high and 10 m long pre cast concrete slabs. The joints are sealed with epoxy cement.

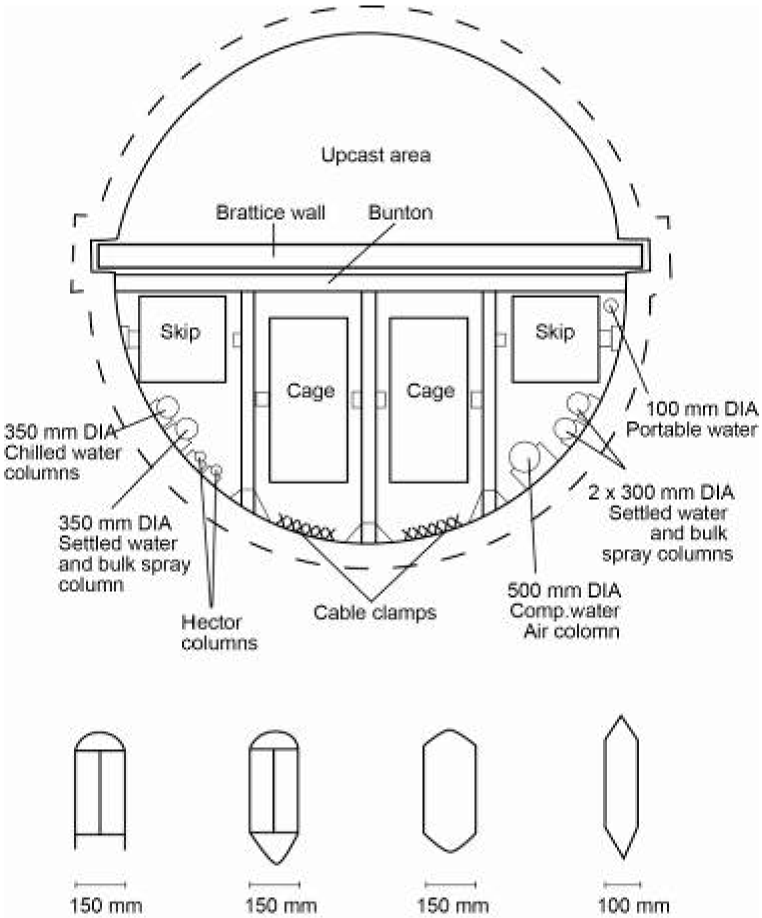


Figure 15.13 Typical shaft plan view of divided shaft

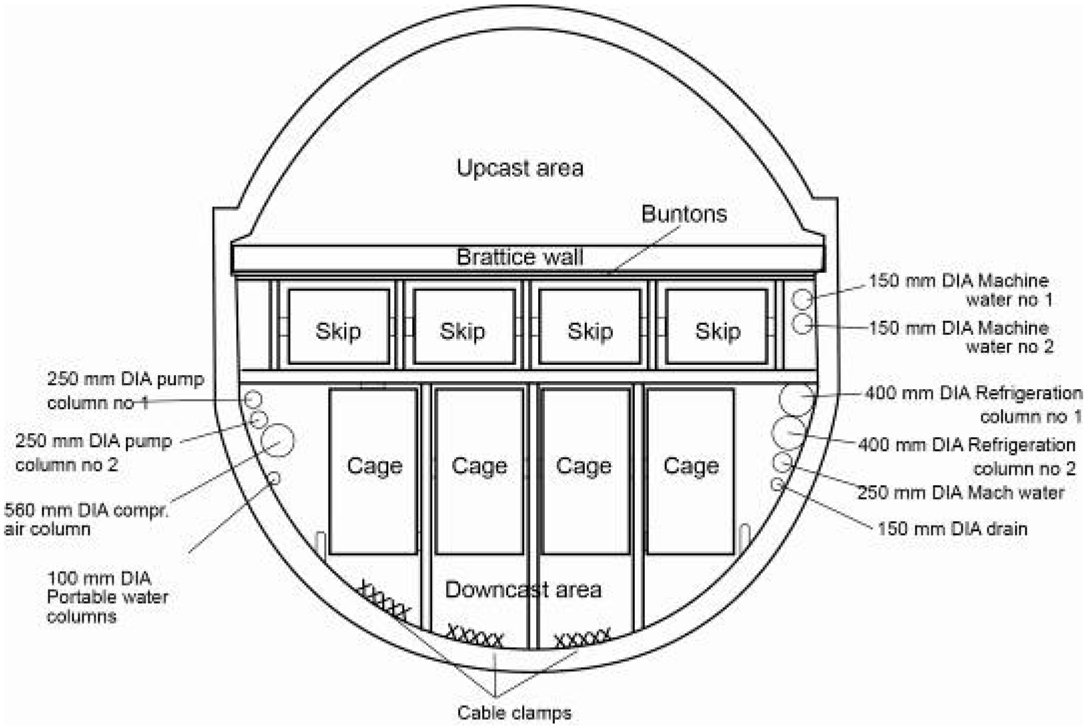


Figure 15.14 Another example of divided shaft

15.10.1 Typical Velocities in Shafts and Airways

For upcast shafts the velocity varies between 18 and 22 m/s (deep hard rock mines), and for downcast shafts between 10 and 12 m/s. For deep gold mines the up cast shaft velocity must be greater than 12 m/s. This is to ensure that water droplets are carried out of the mine. In gold mines the velocity of the air in the main intake airways is between 6 and 8 m/s.

15.11 Fan Planning for Mines

15.11.1 Main Fans

The volume flow rate for various shafts today is often more than 1 000 m^3/s. Electrical motors up to 1 400 kW are used to drive these fans. The size of the fan can vary from 25 m^3/s at 400 Pa to 350 m^3/s at 9 kPa. The fans are usually connected in parallel and should one of the fans fail, there should still be approximately 70% of the air available. This loss in air can be handled for a short period of time.

Centrifugal fans are usually used for surface applications because they operate at a high working pressure. This fan can be very efficient, operating at 85% of the design operating point, but the efficiency decreases should the operating point move away from the designed operating point. These loses can be reduced by installing an inlet guide vane control.

15.11.2 Booster Fans

NB

The function of the fan is to increase the volume flow if it is placed in semi series (not series) with the main stream. Booster fans usually handle between 60 and 90 m^3/s at pressures up to 3 kPa. These are usually axial fans.

15.12 Planning for Losses

When planning a ventilation system, the following losses must be considered:

- Shaft system: up to 5% of the main fan Q
- Divided shafts: up to 5% of Q for cement panels and 10% of steel panels
- Intake air roads: up to 15% of main fan Q
- Stopes: 50% of the air provided to the stope and 25% where backfill is used.

15.13 Recirculation of Air

Investigations have indicated that recirculation of air on a large scale is a viable means of effectively ventilating stopes as well as in larger areas of a mine. This principle involves feeding a particular area with a limited amount of air while fans are used to create controlled recirculation and hence, improved velocities in the area. This is shown in Figure 15.15.

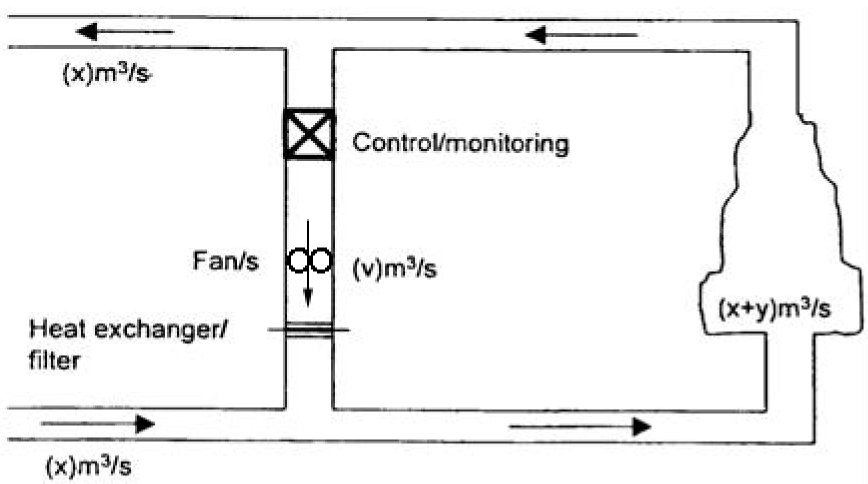


Figure 15.15 Basic principle of recirculation of air

Obviously, there are several factors to consider where re-circulation is involved, such as:

- Maximum anticipated dust and gas loads
- Methods of filtering / removing contaminants

- Contingency planning in case of a main fan breakdown or the occurrence of a fire in the area.

Another layout is also shown in Figure 15.16 and is a more detailed representation of recirculation of air as employed in a hard rock mine.

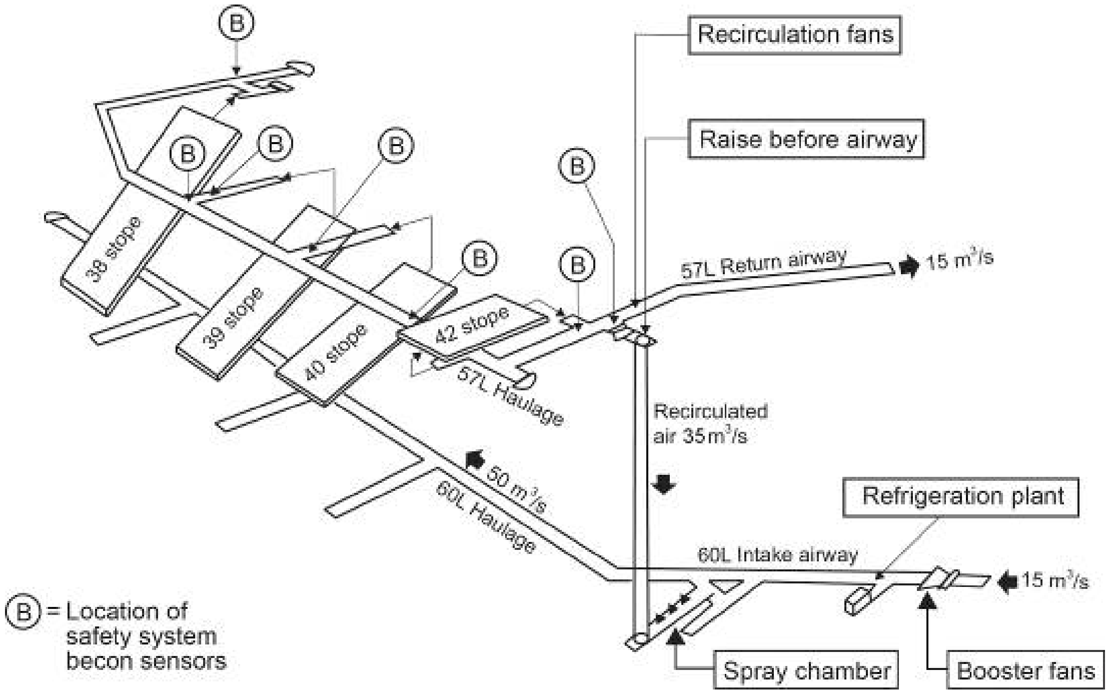


Figure 15.16 Simplified 3D drawing of a recirculation of air layout

A chapter on the recirculation of air as part of a M.Eng study that was written by RCW Webber-Youngman should be read and studied for self study purposes.

Read the following article by M Biffi: “Aligning modern ventilation engineering with risk management”.

15.14 Computer Simulation Packages

VUMA is an air and heat flow simulation program that enables the mine ventilation engineer to simulate various layouts underground to establish which layout is the most productive and profitable in terms of long term capital and running costs as well as the optimum placement of fans and refrigeration units underground so as to ensure a safe, healthy and productive working environment.

Gold Fields mine has developed an ice cooling strategy for their South Deep project and this design was done with the use of Ventilation simulation packages. The power point presentation (courtesy Gold Fields South Africa) has been included as part of the study material to give the student an idea of the magnitude of numbers applicable to ventilation and refrigeration planning, that is capital costs, layouts, air supply quantities as well as refrigeration requirements, with specific reference to ice cooling to be employed.

Chapter

16

MINE VENTILATION ECONOMICS

LEARNING OUTCOMES

At the completion of this chapter, the student would have covered and understood the following aspects related to mine ventilation economics:

- Define simple interest, compound interest, present value and sinking fund.
- State the approximate costs of a ventilation door, a large main fan, a large motor, the annual operating costs of a large motor, the equipment for the environmental control staff on this mine, developing an airway, sliping an excavation, lengths of 570 mm and 760 mm ventilation duct and the monthly ventilation costs of a large colliery or mine.
- Determine the most economic of two or more ventilation schemes with particular regard to fans duct sizes and improving environmental conditions and compare the merits of these schemes.

REFERENCES

Mine Environmental Control (MEC) Certificate, Chamber of Mines, South Africa, Workbook 3.

Mine Ventilation Economics VOEE, Chapter 33, p673-699.

Du Plessis, J.J.L., 2014 (editor). Ventilation and Occupational Environment Engineering in Mines. 3rd Edition. Mine Ventilation Society of South Africa. ISBN 978-0-620-61172-5 (The Blue Book).

16.1 Definitions

You should study each definition below together with the discussion in the textbook. This will assist you in getting a clearer picture of the subject.

Definitions

16.1.1 Interest

Interest is money paid for the use of money, which is borrowed and is usually expressed as a certain percent per year (or rate of interest per year). The original sum of money borrowed or lent is the principal (P).

16.1.2 Simple Interest

Simple interest is interest paid on a sum of money borrowed, but is always calculated on the base of the original amount borrowed, i.e. interest is payable on the original principal throughout the period of the loan for example, if you borrow R1 000 at 30% p.a. simple interest, the interest is:

$$\frac{\textbf{R 30 x 1000}}{\textbf{100}} = \textbf{R 300 p.a.}$$

Disadvantage

This means that if you pay the loan off over three years (36 months), the interest amounts to R 900 and you will be paying interest on the original principal even when making your final payment. This is the major disadvantage of borrowing money on hire purchase. The total amount of money repaid for the above loan is R1 900 or R52.78 per month. If the loan had been on a reducing balance basis (such as housing loans are), interest is only payable on the balance owing and the amount repaid would be R1 206.36 or R33.51 per month.

16.1.3 Compound Interest

Compound interest results when interest is paid on the total amount of money owing, including the principal and any accumulated interest. For example if you invest R1 000 with the bank at 20% interest (compound annually):

$$\text{After 1 year amount} = 1\,000 + \left(\mathbf{1000 \times \frac{20}{100}}\right) = \text{R1 200}$$

$$\text{After 2 year amount} = 1\,200 + \left(\mathbf{1200 \times \frac{20}{100}}\right) = \text{R1 440}$$

$$\text{After 3 year amount} = 1\,440 + \left(\mathbf{1440 \times \frac{20}{100}}\right) = \text{R1 728}$$

and so on.

As can be seen in the above calculation, the bank pays you interest on your principal of R1 000 for the first year (R200), and then pays interest on the R1 000 + R200 for the second year, i.e. you are paid interest on capital and accumulated interest.

This amount can also be determined by the Equation:

$$F \text{ (or A)} = P\left(1 + \frac{i}{100}\right)^{n} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(16.1)}$$

Where:

F	=	Total amount at end of time period
P	=	Principal invested
i	=	Interest rate (%)
n	=	Number of years in time period.

Investing R1000 in the bank at 20% p.a. for 3 years yields:

$$F = 1000 \times \left(1 + \frac{20}{100}\right)^{3}$$

$$= \text{R1 728}$$

16.1.4 Annuity

Definitions

Very simply, an **annuity** is a regular yearly payment for a certain period of time. There are many kinds of annuity, but the scope of these notes is restricted to simple annuities, which are a series of annual payments of equal amounts, starting one year hence.

16.1.5 Present Value

Present value is today's equivalent of a future lump sum payment. For example, if you were due to receive a sum of R 5 000 in 10 years time and the interest rate is 20% but you wanted the sum now, you would obviously receive less than R5 000. The sum to be received now can be calculated by manipulation of the Equation given in 16.1.

$$F \text{ (or A)} = P\left(1 + \frac{i}{100}\right)^{n}$$

$$P = \frac{F \text{ (or A)}}{\left(1 + \frac{i}{100}\right)^{n}}$$

$$\text{So} \quad P = \frac{5\,000}{\left(1 + \frac{20}{100}\right)^{10}}$$

$$= \text{R 808}$$

In other words, if R808 is invested at 20% interest for 10 years, it will accumulate to R5 000. However, the present value of the R5 000 is R808.

16.1.6 Present Value of an Annuity (PV)

Definition

The **present value** of an annuity is today’s equivalent of future periodic payments made at yearly intervals and starting one year hence.

NB

This is an extremely important concept and is used in almost all ventilation economic evaluations.

There are various tables from which the present value of an annuity (PV) can be obtained. These tables usually give the PV of an annuity of R1 per annum. The PV can also be determined from the Equation:

$$PV = A \times \left[\frac{1 - \dfrac{1}{\left(1 + \dfrac{i}{100}\right)^n}}{\dfrac{i}{100}} \right] \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (16.2)$$

Where:

PV	=	Present value of the annuity an amount of annuity
A	=	Amount of annuity
i	=	Interest rate p.a.

Obviously if the annuity is R1 then ‘a’ falls away from the Equation.

For example determine the PV of R1 per year for 4 years at 20% interest (A = 1).

$$PV = \frac{1 - \dfrac{1}{\left(1 + \dfrac{20}{100}\right)^4}}{\dfrac{20}{100}}$$

$$= \frac{1 - \dfrac{1}{(1.28)^4}}{0.2}$$

$$= 2.59$$

R2.59 is thus the present value of R1 per year for 4 years at 20% interest.

This means that if you invested R2.59 in the bank at 20 % it would be sufficient to make annual withdrawals of R1 for the next 4 years.

This is shown in the calculation below:

Table 16.1 Calculation explaining previous example

2.59		Invested at 20%
0.52		Interest for 1^{st} year
3.11		**Total**
-1.00	**1**	First withdrawl
2.11		Balance
0.42		Interest for 2^{nd} year
2.53		**Total**
-1.00	**2**	Second withdrawl
1.53		Balance
0.31		Interest for 3^{rd} year
1.84		**Total**
-1.00	**3**	Third withdrawl
0.84		Balance
0.17		Interest for 4^{th} year
1.00		**Total**
-1.00	**4**	Fourth withdrawl
0.00		Balance

This calculation shows clearly how the investment of R2.59 now is sufficient to cover withdrawals of R1 per year for the next four years.

This is extremely useful when budgeting for future costs, for example if one requires a small 10 kW fan to run for four years and the power costs are R1 500 per year, one can then invest an amount of:

R (2.59 x 1 500) = R3 885 and therefore:

At an interest rate of 20%, it will be sufficient to withdraw R1 500 per year for the next 4 years (R2.59 is only enough to withdraw R1 per year, so to draw R1 500 per year one must invest R1 500 x R2.59. The figure of R1 500 is the amount of the annuity, ‘a’ which is given in the formula for PV).

The amount of R3 885 is termed the ‘present value of power costs’ or ‘PV of power costs’. To invest the R3 885 now to cover the complete power costs for 4 years is equivalent to simply pay the four payments of R1 500 as they arise.

The concept of PV is applied to all future costs when evaluating various schemes. Future costs include such items as power, repairs, maintenance operating costs, spares and so forth.

NB

When making the evaluation, each of these items must first be converted to a yearly cost (annuity) before it can be multiplied by a PV, for example the above mentioned fan will need repairs every 10 months and the repair bill will be R250.

Then annual repair costs $= R\,(250 \times \frac{12}{10})$

$= 300$ per annum

PV of repair costs for 4 years at 20% interest

= (annual cost x PV)

= R (300 x 2.59)

= R777

The investment of R777 will now be sufficient to cover their repair costs for the next four years. Manipulation of the formula for PV can also give the required repayments on a loan. The formula is manipulated to:

$$a = \frac{\text{sum borrowed}}{\text{PV of R1/year}}$$

For example if you borrowed R25 000 from the bank for a house at 20% interest for 20 years, the monthly repayments can be determined as follows:

$$\therefore \text{Annual repayment, } a = \frac{R2\,500}{4.870}$$

$$= R5\,133$$

$$\therefore \text{Monthly repayment} = \frac{5\,133}{12}$$

$$= R\,427$$

16.1.7 Sinking Fund

Definition

A **sinking fund** is the accumulated amount of yearly deposits of equal amounts at a certain rate of interest.

This accumulated amount (S) can be determined from:

$$S = \frac{\left(1 + \frac{i}{100}\right)^n - 1}{\frac{i}{100}} \times a \text{ .... (16.3)}$$

Where:

S = Accumulated amount
i = Interest rate (%)
n = Number of years
a = Annuity which is invested

For example, if one invests R1 per year for 5 years at 20 % interest the accumulated amount at the end of that period will be:

$$S = \frac{\left(1 + \frac{20}{100}\right)^5 - 1}{\frac{20}{100}} \times 1$$

$$= \frac{(1.2)^5 - 1}{\frac{20}{100}} \times 1$$

$$= \frac{2.48 - 1}{0.2} \times 1$$

$$= R7.44$$

This means that if R1 is invested at the end of every year (starting one year hence). At 10% for 5 years, the accumulated amount at the end of that period will be R6.11 (this is a new answer to the above calculation but now using the interest rate as 10% and not 20%).

References

It is not necessary to use the above formula to determine S, as there are tables available such as the one on page 474 of EE, which enables the value of S to be read off. These tables usually give the accumulated amount if R1 per year is invested. Obviously if the amount is R40 per year the accumulated amount is 40 times greater than that in the table.

References

Page 203 of Mine Ventilation Notes for Beginners' - 3rd edition, by W.L. le Roux, shows how sinking funds can be used to solve economic problems. These notes, however, will concentrate on the PV approach.

16.1.8 Amortization

Definition

Amortization means the 'wiping out of a debt' or 'paying off a debt'. For example, in it was shown that to pay off a loan to the amount of R25 000 at 20% over 20 years required a monthly payment of R427/month.

It therefore requires a payment of R427/month to 'amortize' the debt of R25 0000. Also a time period of 20 years is required to 'amortize' the debt of R25 000.

16.1.9 Direct and Indirect Costs

Example

Direct costs **are directly attributed** to the cost objective. An example is the development of an airway where the quantity of explosives used is the direct cost. Other direct costs are the wages of the development team, drill steel, pipes, tracks etc.

Indirect costs are **cost elements that are associated with more than one cost objective**, but are not directly attributable to any of them. In the example of the development of an airway the costs of the management team are normally an indirect cost.

Definitions

16.1.10 Variable and Fixed Costs

Variable costs are costs that vary directly and proportionately with ‘size’. **Fixed costs** do not vary with size. The pipes and tracks required for development are examples of fixed costs.

16.2 Costs

The approximate capital and running costs of various ventilation appliances need to be identified for the list below. These numbers are not given as they change on a yearly basis. It is expected of the students to relate these numbers to their ‘own’ mines. This section therefore only relates to the types of costs that are applicable in ventilation planning.

- **Large Cooling Plant**

Capital cost Surface plant	R? / rated kW
Capital cost underground plant	R? / rated kW
Running cost	R? / rated kW cooling per annum.

- **Small Cooling Plant (Spot-cooler)**

Capital cost	R? / rated kW of cooling

- **Insulation of Chilled Water Pipe**
  R ? per metre 25 mm thick for a 1 560 mm diameter pipe (depends on type of insulation)

- **Dust Filter Plant**
  Approximately R? / m^3/s of air filtered - includes all costs.

- **Ventilation Doors**
  An air lock complete with frames and doors costs R?. A single 2 m x 2 m door costs R1 100.

- **Large Main Fan**

Capital cost	R? / rated kW
Running cost power	R? / rated kW (normally electric cost)
Maintenance cost	R? / rated kW

- **Large Motor**
  R? per kW

- **Fans**

570 mm electric fan	11 kW - R?
	19 kW - R?
760 mm electric fan	-22 kW - R?
	45 kW - R?

- **Ventilation Pipes**

570 mm diameter	R? per metre
760 mm diameter	R? per metre

- **Ventilation Equipment for an Observer**

Anemometer (Lambrecht's)	R?
Rods (1.2m)	R?
Whirling hygrometer	R?
Wet kata and flask	R?
Manometer	R?
Stopwatch	R?
Tape(Linen)	R?
A Fuess aneroid barometer costs	R?

- **Airways Development**
  To develop 1 metre of 3 m x 3 m haulage costs R?/m

- **Sliping**
  To slipe 1 m^3 of rock costs R?

- **Costs of Ventilating a Large Deep Mine**
  R? per ton of rock broken and this can vary considerably from mine to mine.

16.3 Cost and Economics (including Economic Evaluation)

NB

There are several financial evaluation methods that can be used to compare different projects. It is however important to note that each has its own unique applications and this section will deal with the most important ones used for project evaluations. In economic studies it is not enough to know what a machine or system costs in the first instance, in other words, what it will cost to buy and thus own. Similarly, what it will cost to run or operate a machine does not give any complete answer. What is most important is that the sum of owning and operating costs should be a minimum.

To a major extent, the method of financial evaluation will depend on the scope of the project and the policy of the company. Most of the problems encountered will be alternative choice situations and if the consequences can be satisfactorily measured in financial terms, the time value of money should be taken into consideration. The concept of present value is common to all the normally used methods and the differences between simple present value and discounted cash flow is a matter of sophistication related to the number of factors taken into account in the evaluation.

16.3.1 Total Owning Cost (T.O.C)

Definition

Total owning cost is the total amount of money it costs to 'own' a scheme at today's value. This sum includes any capital costs involved in the scheme plus the present value of any future costs involved in the scheme. The total owning cost can be determined as follows:

= PV of capital costs + PV of running costs.

The term 'PV of capital costs' is the same as just 'capital costs'. This is today the equivalent of, today's money, spent now. The term 'PV of running costs,' includes all the future cost items. These items must all be converted to a yearly cost and then multiplied by the PV to obtain a 'PV of running costs'. Before the economic merits of any two schemes can be compared, the T.O.C for each scheme must be obtained.

For example, if a mine is considering buying a booster fan to run for 10 years and has a choice in either buying a fan from Company A or Company B. There is no use in selecting the fan with the cheaper capital cost, as it may be the more expensive fan to run over the period of ten years. To compare the economic merits of the two fans it is necessary to determine the total owning cost of **each** fan and then to compare these two owning costs.

NB

Obviously, the one with the cheaper total owning cost would be the one to select, but in certain cases where the total owning costs of two schemes are much the same, other factors such as delivery time, guarantee period, maker's reputation, etc., play an important role in the final selection.

Before any schemes can be compared, the total owning costs for each of the schemes must be obtained. It should be remembered that the final selection of any scheme does not depend entirely upon its economic merits, i.e. the cheaper scheme is not necessarily the better scheme. The point is that practical consideration often plays a large part in the final selection of any scheme.

For example, a shaft on a mine may have an **economic diameter** of 7.0 m from ventilation point of view, but the chosen diameter may be 8.0 m simply because of the importance of adequate hoisting capacity.

There are countless different examples of the application of economics and the following three worked examples illustrate how the concept is applied in the economic evaluation of various schemes.

Excel

In order to be able to do these calculations the student has to be able to read off the PV values from the Tables mentioned before.

Attached for the convenience of the students are PV tables that can be used.

WORKED EXAMPLE 1

A comprehensive survey of fan repair costs has revealed the following information concerning 570 mm diameter electric fans:

	Fan A	**Fan B**
Average cost per repair	R2 400	R2 900
Average time between repairs	10 months	15 months
Cost of new fan	R10 850	R13 100

Both fans have rated duties of 6.5 m^3/s of air at a pressure of 2 400 Pa. At this duty 'fan A' has an overall efficiency of 70% and 'fan B' an overall efficiency of 78%. Each fan is expected to run for 8 years and the interest rate is 20%. The electricity costs is 75 c per unit (remember a unit is expressed as c/kWh). Which fan would be more economical?

Answer

From present value tables, the present value of R1 per year for 8 years at 20% = R3.837.

Fan A

Capital cost = R10 850

Air power, W_a = $\frac{p \times Q}{1\ 000}$

= $\frac{(2\ 400 \times 6.5)}{1\ 000}$

= 15.6 kW

At 70% efficiency:

Input power = $\left(15.6 \times \frac{100}{70}\right)$kW

= 22.29 kW

Cost of power = 75c per kWh

Assume that the fan runs continuously for 24 hours per day, 365 days per year.

Cost of power p.a. = $R\left(\frac{75}{100} \times 24 \times 365\right)$ per kW p.a

= R6570 kW p.a.

$\therefore$ Cost of input power p.a. = R22.29 x 6570

= R146 445.3 pa

PV of power costs = R (146 445.3 x 3.837)

= R561 910.62

Repair costs = 2 400/10 months

Repair costs p.a.	=	$R\left(2\,400 \times \frac{12}{10}\right)$
	=	R2 880 p.a.
PV of repair costs	=	R (2 880 x 3.837)
	=	R11 051
Now, PV of capital costs	=	R10 850
PV of running costs	=	R (562 910.62 + 11 051)
	=	R572 961.62
Total owning costs	=	PV running + PV capital
	=	R583 811.62

Fan B

Capital Cost	=	R13 100
Air power, W_a	=	15.6 kW (same duty as fan A)
At 78% efficiency:		
Input power	=	$\left(15.6 \times \frac{100}{78}\right)$ kW
	=	20.0 kW
Cost of power p.a.	=	R (20 x 6570)
	=	R131 400 pa
PV of power costs	=	R (131 400 x 3. 837)
	=	R504 181.8
Repair costs	=	R2900/15 months
Repair costs p.a.	=	$R\left(2\,900 \times \frac{12}{15}\right)$
	=	R2 320 p.a.
PV of repair costs	=	R (2 320 x 3.837)
	=	R8 900

Now:

PV of capital costs	=	R13 100
PV of running costs	=	R (504 181.8 + 8 900)
	=	R513 081.8
Total owning costs	=	PV running + PV capital
	=	R 513 081.8 + 13 100)
	=	R526 181.8

The T.O.C. of each fan can now be compared and it can be seen that it is more economical to buy fan B.

Note:

- PV of capital cost is the same as capital cost, as the cost of the fan must be paid now, not in the future.
- Power costs and repair costs are future costs hence these costs are first obtained **per year** and then multiplied by the PV to see how much money in theory should be set aside **now** to cover these future costs.
- Because the duty of the two fans is the same (6.5 m^3/s @ 2 400), the air power of 15.6 kW was not recalculated for fan B, i.e. it was used as a **constant**. When using any constant in a series of economic evaluations it is extremely important to **make sure that the value used initially for the constant is correct**, otherwise each successive evaluation will be incorrect.
- The calculation of PV of power costs was done step by step in this example. It can also be done as one calculation for example Fan A, PV of electricity costs:

$$= \quad R\left(W_a \text{ x } \frac{100}{\text{eff}} \text{ x electricity p.a, x PV}\right)$$

$$= \quad R\left(\frac{2\,400 \text{ x } 6.5}{100} \text{ x } \frac{100}{70} \text{ x } 6570 \text{ x } 3.837\right)$$

$$= \quad R561\ 802$$

  (The slight difference is because each step was rounded off in the worked example)

- Answers to this kind of economic evaluation should only be given to the nearest R1, R10, R100, or even R1 000, depending on the total sum involved. (The nearest R1 000 would have been sufficient in the worked example where sums of R18 000 - R19 000 were involved, this is about 0.5 per cent of the total sum). The reason for this is that one is more concerned about the difference in cost between the schemes, than the actual cost. Also, once a scheme has been selected the mine accountant will take care of the individual Rand and cents.
- Again it must be emphasised that the Total Owning Cost of each scheme must be obtained before any selection of scheme can be made.

WORKED EXAMPLE 2

A downcast shaft is 1 324 m deep. The air quantity at mean shaft density is 315 m^3/s and the pressure loss in the shaft is 1 745 Pa. The shaft has a K-factor of 0.021 Ns^2/m^4. The main fans have an overall efficiency of 73% and power costs are R1 500 per kW per year. It is desired to increase the air quantity to 410 m^3/s by increasing the speed of the main fans. At the same time consideration is given to streamlining the shaft buntons. Two kinds of streamlined buntons are available:

Type 'A': reduces the K-factor of the shaft to 00014 Ns^2/m^4 and will cost R520 per metre depth of shaft.

Type 'B': reduces the K-factor of the shaft to 0.011 Ns^2/m^4 and will cost R711 per metre depth of shaft.

The life of the mine is 18 years and the interest rate is 20%. Which of the following schemes would you recommend?

- **Scheme 'A'** Install type 'A' streamlined buntons
- **Scheme 'B'** Install type 'B' streamlined buntons
- **Scheme 'C'** Leave the shaft as it is.

NB

The main fans can be adjusted to suit any of the above three schemes.

Answer

From tables, the present value of R1 p.a. for 18 years at 20%
= R4.812.

Scheme 'A' (Installing type 'A' streamlined buntons)
= R1 324 x 520)
= R688 500

Pressure required for present quantity of 315 m^3/s

Pressure required for new quantity of 410 m^3/s:

$$P_2 = P_1 \times \left(\frac{Q_2}{Q_1}\right)^2$$

$$= 1\,745 \times \left(\frac{410}{315}\right)^2$$

$$= 2\,956 \text{ Pa}$$

This would be the pressure required if the K-factor had remained constant, but the K-factor has decreased from 0.021 Ns^2/m^4 to an amount of 0.014 Ns^2/m^4 because of the streamlining. Hence the pressure requirement decreases proportionally.

Pressure required	=	$2\,956 \times \frac{0.014}{0.021}$
	=	1 971 Pa

Then PV of power costs

$$= R\left(\frac{p \times Q}{1\,000} \times \frac{100}{eff} \times \text{elec cost p.a.} \times PV\right)$$

$$= R\left(\frac{1\,971 \times 410}{1\,000} \times \frac{100}{73} \times 1\,500 \times 4.812\right)$$

$$= R7\,990\,326$$

Total owning costs	=	PV capital + PV running
	=	R (688 500 + 6 944 211)
	=	R8 678 826

Scheme 'B' (Installing Type 'B' streamlined buntons)

Capital cost of streamlining the buntons	=	R (1 321 x 711)
	=	R941 400

Assuming that the overall efficiency of the main fans remains constant, the PV of power costs will vary directly with the K-factor.

Therefore, at a K-factor of 0.011 Ns^2/m^4, PV of power costs.

$$= R\left(6\,944\,211 \times \frac{0.011}{0.014}\right)$$

$$= R5\,456\,166$$

T.O.C	=	PV capital + PV running
	=	R (941 400 + 5 456 166)
	=	R8 931 726

Scheme 'C' (Leaving the shaft as it is)

Again assuming that the overall efficiency of the main fans remains constant, the PV of power costs will vary directly with the K-factor.

At a K-factor of 0.014 Ns^2/m^4 PV power = R6 944 211

Therefore, at a K-factor of 0.021 Ns^2/m^4 PV of power costs

$$= R6\ 944\ 211 \times \frac{0.021}{0.014}$$

= R10 416 320

T.O.C = PV capital + PV running

= R (0 + 10 416 320)

= R11 978 515

Conclusion

Table 16.2 Summary of findings

SCHEME		T.O.C
A	Use type 'A' streamlining	R 8 678 826
B	Use type 'B' streamlining	R 8 931 726
C	Leave shaft as it is	R 11 978 515

Hence Scheme 'B' would be the more economical scheme

Note

NB

!

The objective behind this example is to increase the quantity of air down the shaft from 315 m^3/s to 410 m^3/s. One of the most common mistakes made in this type of exercise is to forget to change the pressure requirement for the increased quantity, i.e. to pass 315 m^3/s of air down the shaft requires a pressure of 1745 Pa, but to pass 410 m^3/s of air requires a pressure of 1956 Pa. This mistake is not due to lack of knowledge of economics, but a lack of knowledge of the elementary laws of air flow.

The PV of power costs was only calculated in full in scheme 'A'. For scheme, 'B' and 'C' the PV of power costs was determined by proportion of the respective K-factors as all other factors remained constant. Again, this makes use of a 'constant' and the constant should be double-checked to ensure that it is correct, or else all three evaluations are incorrect.

Note that the various costs have been rounded off to the nearest R100 in the calculations. The answers to each scheme could have been rounded off to the nearest R1 000 without affecting the final selection.

WORKED EXAMPLE 3

A mine wishes to purchase 620 m of ventilation ducting to install in an abandoned development end. The column will handle 11 m^3/s of air for a period of 6 years and the interest is 20 %.

The installed cost of the ducting is $\frac{D_{mm}^2}{1.290}$ cents per metre

where D is the pipe diameter in mm. Electric power costs are R260 per kW per annum (please take note that this is an example only and approximately 5 to 6 times lower than the current electricity price, 2010 value). Assume that the mine already has a fan capable of the required duty and assume a value of 65% for the overall efficiency of the fan. The ducting can be re-sold for 10% of its capital installed cost at the end of its life.

(i) Assuming that all the duct sizes under consideration have a K-factor of 0.003 5 Ns^2/m^4, determine the most economic size of duct to use in the end.

(ii) Plot a graph showing how capital, running and total owning costs vary with duct diameter.

Answer

(i) From tables, the present value of R1 per annum for 6 years at 20% is R3.326.

PV of Capital Costs

PV of Capital Costs

Capital installed cost = $\frac{D_{mm}^2}{1.290}$ c/m (where $D_{mm} = D_m \times 1\ 000$)

Duct Length = 620 m

$\therefore$ Capital installed cost = $\left(620 \times \frac{D_{mm}^2}{1.290}\right)\frac{c}{m} \times m$

$= R\left(620 \times \frac{D_{mm}^2}{1.290} \times \frac{1}{100}\right)$

$= R\,(0.004\ 806 \times (D_m \times 1\ 000)^2)$

(D now converted to m)

$= R\,(4\ 806\ D_m^2)$ (D in m)

But the ducting can be re-sold at the end of its life (after 6 years) for 10% of its capital installed cost. Today's value of this 10% of capital is obviously less and can be determined from the formula for the present value of a future lump sum.

The formula is:

$$P = \frac{A}{\left(1 + \frac{i}{100}\right)^n}$$

The amount (A) to be received in 6 year's time for re-selling the ducting is:

$$= R\left(\frac{10}{100} \text{ x } 4\,806\right) D^2$$

$$P = \frac{A}{\left(1 + \frac{i}{100}\right)^n}$$

$$= \frac{480.6\ D^2}{(1.1)^6}$$

$$= R271.3\ D^2$$

Deducting this value from the capital installed cost, (similar to cash discount) gives:

$$\text{Capital installed cost} = R\ (4\,806\ D^2 – 27\,303\ D^2)$$

$$= R\ (4\,534.7)\ D^2$$

PV of running costs

The only running costs for the piping are the power costs.

PV of power cost

$$= R\left(\text{air power x } \frac{100}{\text{fan eff.}} \text{ x electricity cost p.a. x PV}\right)$$

$$= R\left(\text{air power x } \frac{100}{\text{fan eff.}} \text{ x electricity cost p.a. x PV}\right)$$

Substituting $\frac{KCLQ^2}{A^3}$ for p in the Equation gives:

PV of power cost

$$= R\left(\frac{KCLQ^2}{A^3} \text{ x } \frac{Q}{1\,000} \text{ x } \frac{100}{65} \text{ x } 260 \text{ x } 3.33\right)$$

$$= R\left(\frac{0.003\,5 \text{ x C x } 620 \text{ x } 11 \text{ x } 11}{A^3} \text{ x } \frac{11}{1\,000} \text{ x } \frac{100}{65} \text{ x } 260 \text{ x } 3.33\right)$$

$$= R\left(3\,847 \text{ x } \frac{C}{A^3}\right)$$

Therefore:

$\frac{C}{A^3}$ is the only term, which will change as the diameter is varied.

Now:

$$\frac{C}{A^3} = \frac{\pi D}{\left(\frac{\pi D^2}{4}\right)^3}$$

$$= \frac{\pi D}{\left(\frac{\pi^3 D^6}{64}\right)}$$

$$= \frac{64 D}{\pi^2 D^6} = \frac{64}{\pi^2 D^5}$$

$$= \frac{6.48}{D^5}$$

$$\text{PV of power cost} = R3\,847 \times \frac{6.48}{D^5}$$

$$= \frac{R(24\,929)}{D^5}$$

Total Owning Cost

$$\text{T.O.C.} = \text{PV of capital + PV or running}$$

$$= R\left(4\,634.7\,D^2 + \frac{24\,929}{D^5} \quad \text{(Both 'D' in m)}\right)$$

This formula for T.O.C can now be applied to any diameter of duct to determine its owning cost. Assuming a duct diameter of 900 mm to begin with, and using increments of 100 mm, the following results can be obtained in Table 16.3.

Table 16.3 Tabulation of results for different duct sizes

D (m)	**PV capital** R(4 534.7 D^2)	**PV running** $R\left(\frac{24\,929}{D^5}\right)$	**T.O.C** (R)
0.9	R3 670	R42 217	R45 887
1.0	R4 530	R24 929	R29 459
1.1	R5 490	R15 479	R20 969
1.2	R6 530	R10 018	R16 548
1.3	R7 660	R6 714	R14 374
1.4	R8 890	R4 635	R13 525
*1.5	R10 200	R3 283	R13 483
1.6	R11 610	R2 377	R13 987
1.7	R13 110	R1 756	R14 866
1.8	R14 690	R1 319	R16 009

* From the above table, it can be seen that the 1.5 m diameter would be the most economical diameter.

ii) The cost can now be plotted against diameter as shown in Figure 16.1.

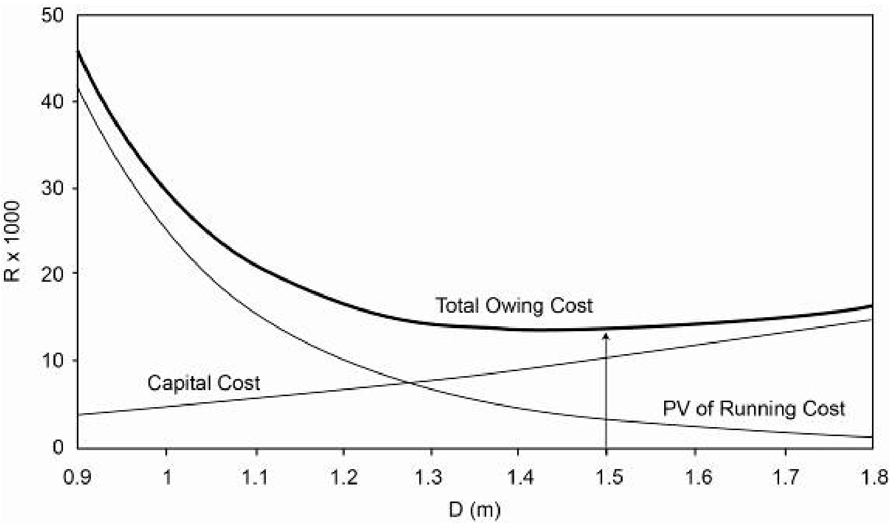


Figure 16.1 Graphical representation of economic diameter for shaft

The answer can also be read off the graph as 1.5 m.

Note:

1. To avoid repetition of the same lengthy calculations for each diameter of duct, extensive use was made of constants in this answer, that is a formula for T.O.C. was derived which depended only on the duct diameter.
2. Any ‘conveyor’ of air, be it a duct, airway or shaft, has a certain most economic size. The reason for this is well illustrated in the graph. As the diameter (size) increases the capital cost also logically **increases**, but as the diameter increases the running costs **decrease** because less pressure (and hence less power) is required to pass the air through these larger sizes. Somewhere in between there is a ‘minimum’ point (the lowest point on the T.O.C. curve) from which the most economical size can be determined.
3. The actual most economical diameter of 1.5 m could be determined more accurately by assuming smaller increments in size. However, as can be seen by the T.O.C. figures in the table (and the graph) it would not really matter whether the mine chose a duct diameter anywhere between 1.4 and 1.6 m as there is little difference in cost. This ‘band’ between 1.4 m and 1.6 m on the T.O.C. curve (where the curve ‘flattens out’) is sometimes called the ‘tolerance’. (See examples on page 690 of ‘VOEE).

References

Once this ‘tolerance’ has been established, the duct diameter chosen would be one that best suits the practical situation, i.e. for a duct the 1.3 m diameter might be chosen for lack of space. On the other hand, the 1.5 m diameter might be chosen because it is manufactured as a standard size.

4. The concept of resale value is a little difficult to follow in the worked example because of all the other factors involved in the example. The following example will attempt to clarify it.

WORKED EXAMPLE 4

Assume a length of ducting has a life of 8 years, the interest rate is 20%, the capital installed cost is R10 000, and the ducting has a resale value of 15% of capital installed cost at the end of its life. Determine the actual capital installed cost.

Answer

$$\text{Capital cost to purchase ducting} = R10\,000$$

$$\text{Resale value in 8 year's time} = R\left(\frac{15}{100} \text{ x } 10\,000\right)$$

$$= R1\,500$$

This means that R1 500 will be paid back to the mine in 8 years time. For the mine to be paid back now means the amount paid back will be less than R1 500. The amount is actually the present value of the amount of R1 500 in 8 year's time.

$$A = p\left(1+\frac{i}{100}\right)$$

$$\therefore \quad p = \frac{A}{\left(1+\frac{i}{100}\right)^n}$$

So:

$$p = \frac{1\,500}{(1.2)^8}$$

Hence the sum of R1 500 is 8 years time is worth R348 now.

$\therefore$ Actual capital installed cost = R (10 000 - 348)

= R 9 652

5. In assuming the various diameters to consider, practical experience plays a key part, for example it would have been ridiculous to assume a 570 mm diameter for this example as the pressure required to carry 11 m^3/s of air through 620 m (from Worked Example 3) of 570 mm diameter duct would be quite large (28 300 Pa in fact).

Note:

References

Example 15, page 688, 'VOEE deals with a similar type problem. You should also study this example to ensure that you are able to solve this type of problem. Section 9, pages 695 to 696, deals with the comparison of ventilation versus cooling. You should also study this section in detail to ensure that you understand the principles involved.

16.4 Inflation, Depreciation, Taxation, NPV, IRR and Discount Rates

To be able to properly understand the application of the above mentioned parameters in the context of evaluating projects, it is important to be able to distinguish the characteristics of each. The following section will deal with some of the most important issues applicable to each.

16.4.1 Inflation

The effect of inflation is normally accounted for in the interest rate in that as inflation increases, so does the interest rate.

The real rates of interest, that is the actual rate adjusted for inflation, are normally very low and in the general economy these are between 2 and 4%. Where interest and inflation rates are known, the present value of an annual amount can be obtained with the use of the following Equation:

$$\text{Present value} = \frac{1 - \dfrac{(1 + i^*)^n}{(1 + i)^n}}{\dfrac{(1 + i)}{(1 + i^*)} - 1} \qquad (16.4)$$

Where i^* = the inflation rate

i = the interest rate

16.4.2 Depreciation and Taxation

When equipment is bought, it depreciates in value over a specific period of time. The effect of this and how taxation is influenced by it will become clear in Worked Example 5.

WORKED EXAMPLE 5

A machine is bought for R50 000 and costs R30 000 per year in operating costs. The estimated life span is 15 years and interest is reckoned at 10% compound. Another machine can be bought for R70 000 and the operating costs will be reduced to R25 000 per year. Briefly discuss the significance of the numbers mentioned in the context of depreciation and taxation.

Answer

The problem can be restated by asking the following question:

Does the saving in operating cost of R5000 per year that is the amount of (R30 000 – R25 000) for 15 years justify the additional capital outlay to the amount of R20 000 (R70 000 – R50 000)? The significance of these numbers is that, if the R70 000 machine was bought, the mine would have made a greater profit.

This profit is taxable and the actual saving would be less than the actual R5 000 indicated.

Businesses are however allowed to write off or depreciate capital expenditure against profits before tax. This means that tax is only payable on the profits in excess of the capital expenditure. Alternatively, the machine may be a replacement and considered an operating cost rather than a capital cost. Again, this is written off against profits before tax.

Taxation rates vary throughout the world and can be either a constant proportion of the profits or a variable rate dependent on the level of profits.

WORKED EXAMPLE 6

What is the present value of a profit of R5 000 before tax if the expenditure was R20 000? Assume a life of 15 years, interest at 10%, straight line depreciation and a 35% tax rate.

Answer

Depreciation allowance

R 20 000 $\div$15	=	R1 333 per year

Profit subject to tax is therefore:

R 5 000 – R 1333	=	R3 667 per year
Tax to be paid	=	R3 667 x 0.35
	=	R1 283 per year

Cash flow to the company is (Profit – Tax payable + depreciation) per year amount, therefore:

Cash flow	=	R3 667 - R1 283 + R1 333
	=	R3 717

The present value of R3 717 per year for 15 years at 10% interest is calculated as follows:

- The present value of R1 at 10% interest in 15 years is read off from Table or calculated as R7.606
- The present value of R3 717 for same period and term is therefore an amount of R28 272 (R3 717 x 7.606). This is more than the amount of R20 000 initially expended and therefore the **expenditure is justified**.

WORKED EXAMPLE 7

What is the present value of a profit of R5 000 before tax if the expenditure was R20 000? Assume a life of 15 years, interest at 10%, allowing depreciation according to profits and use 35% as the tax rate.

Answer

The total expenditure (additional)	=	R20 000
The total saving per annum	=	R5 000

This means that the total allowance amount will be applicable for

$$\frac{\text{R 20 000}}{\text{R 5 000}} = \text{4 years}$$

This means that the for the first 4 years the profit subject to tax = R0

Tax flow to the company will therefore be = R5 000

During the last 11 years there was no depreciation allowance and therefore the profit subject to tax will be R5 000.

The cash flow to company	=	R5 000 x (1 – 0.35)
	=	R3 250 per year

The present value for R 1 per year for **4 years** at 10% is R3.1699

The present value of the profits after tax is as follows:

R5 000 x 3.1699	=	R15 850
Plus R 3 250 x (7.606 – 3.1699)	=	R14 417
	=	**R30 267**

This means that the expenditure was also justified. This brings into consideration a term called the nett present value (NPV).

16.4.3 Net Present Value

Definition

The **net present value** is the present value of the savings or profit less the original investment. For worked example 6 the NPV is an amount of R8 272 (R28 272 – R20 000), the NPV for Worked Example 7 is an amount R10 267 (R30 267 – R20 000). The greater of the two numbers is therefore the better option to consider. If the present value is calculated using the company's minimum discount or interest rate and the NPV is positive, then the investment is justified.

16.4.4 Internal Rate of Return (IRR)

Definition

The **IRR** is the discount rate that equates to the present value of the expected future cash flows to the initial investment. In other words, the IRR is the discount rate at which the NPV value is zero.

16.4.5 Discount Rate or Interest Rate

Company policy normally determines what the acceptable discount or interest rates are. Mining is normally considered a fairly high risk investment, which results in interest rates above the average in the economy.

Most mining companies require between 15 and 20% return on an investment excluding inflation effects and between 20 and 30% including inflation. The actual discount rate will depend on the financial position and goals of the company and the state of the economy in general.